

Report: Core vs Divergent Metabolism + Clinical + Survival

Core patients: 99 | Divergent: 11

Metabolic variables analyzed: 248 | Significant ($p_{\text{adj}} < 0.05$): 1

FBA solutions: FBA, pFBA, L1w, L2, L2w

Structure: Clinical → TNBC → Metabolism → Integration → Survival

Effect size: Cliff's δ | ≥ 0.147 small · ≥ 0.330 medium · ≥ 0.474 large

SECTION I

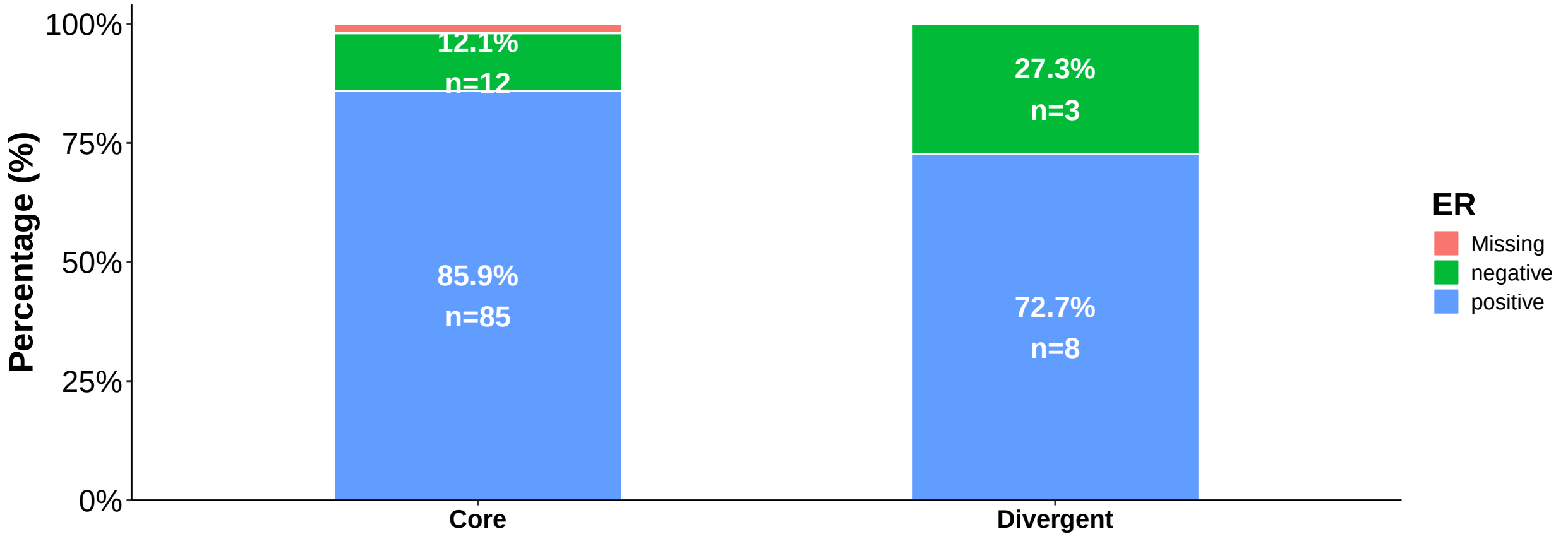
Clinical Differences: Core vs Divergent

Macro-categories: Tumor Biology | Stage | Treatment History | Patient Factors

Categorical Clinical – A. Tumor Biology

ER

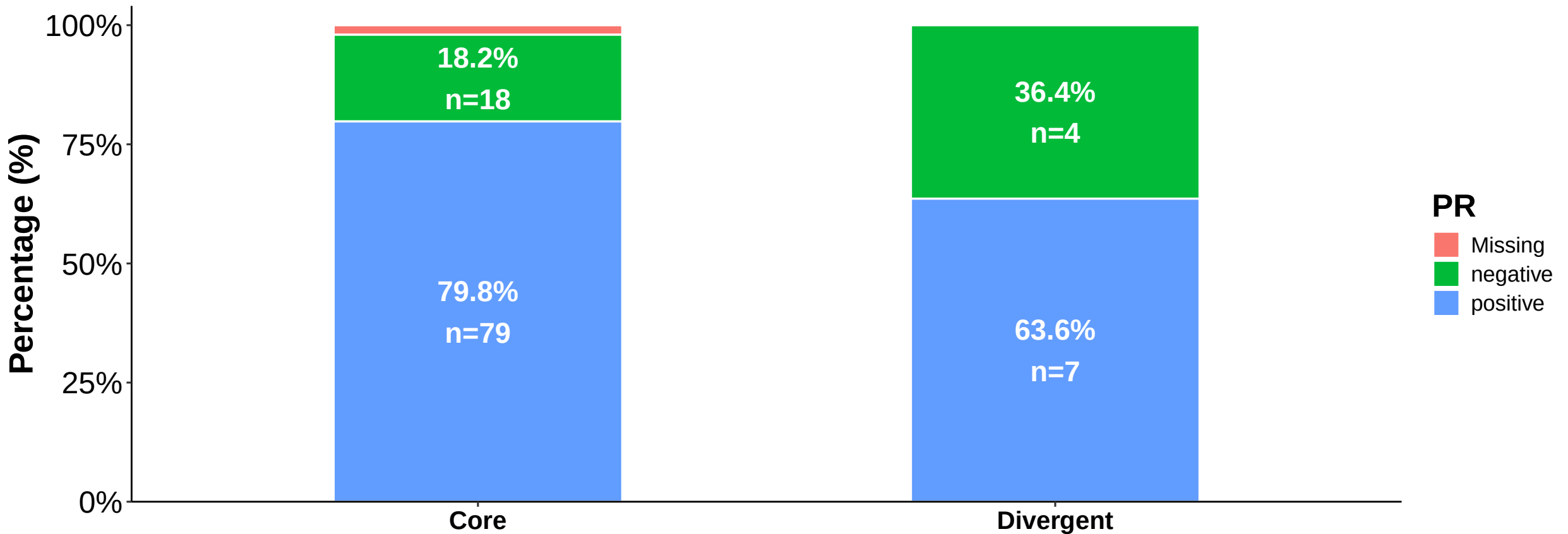
Chi² p: 0.352



Group

PR

Chi² p: 0.335

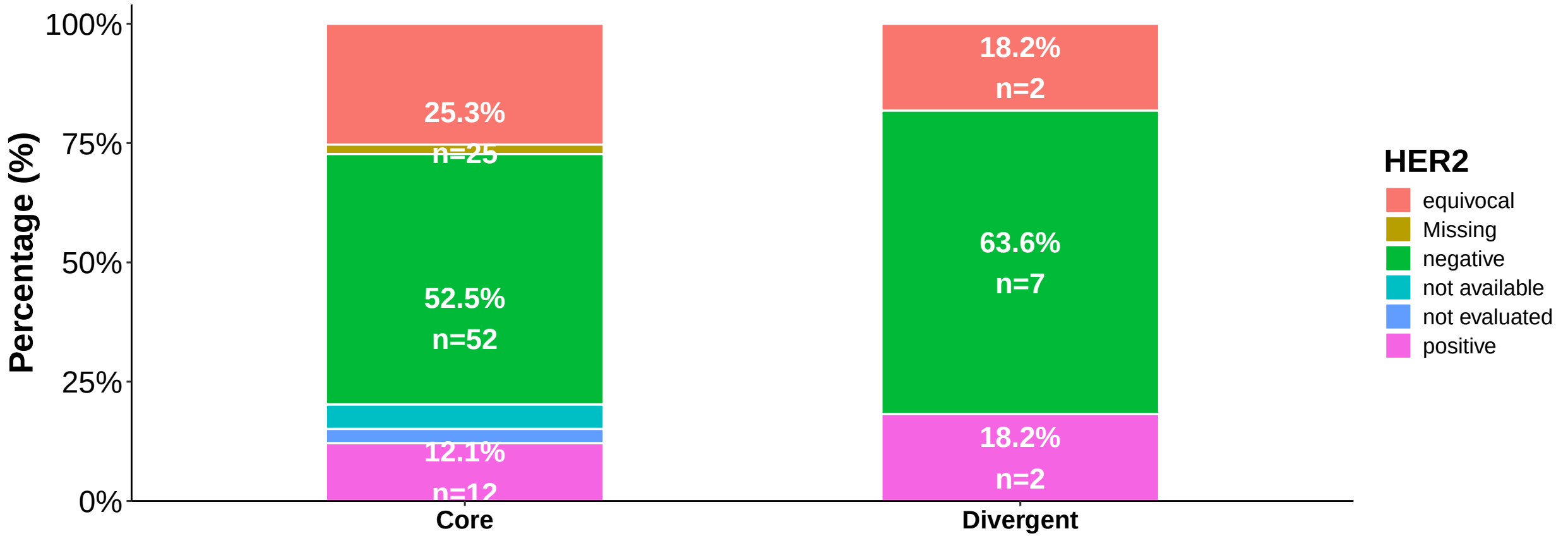


Group

Categorical Clinical – . A. Tumor Biology

HER2

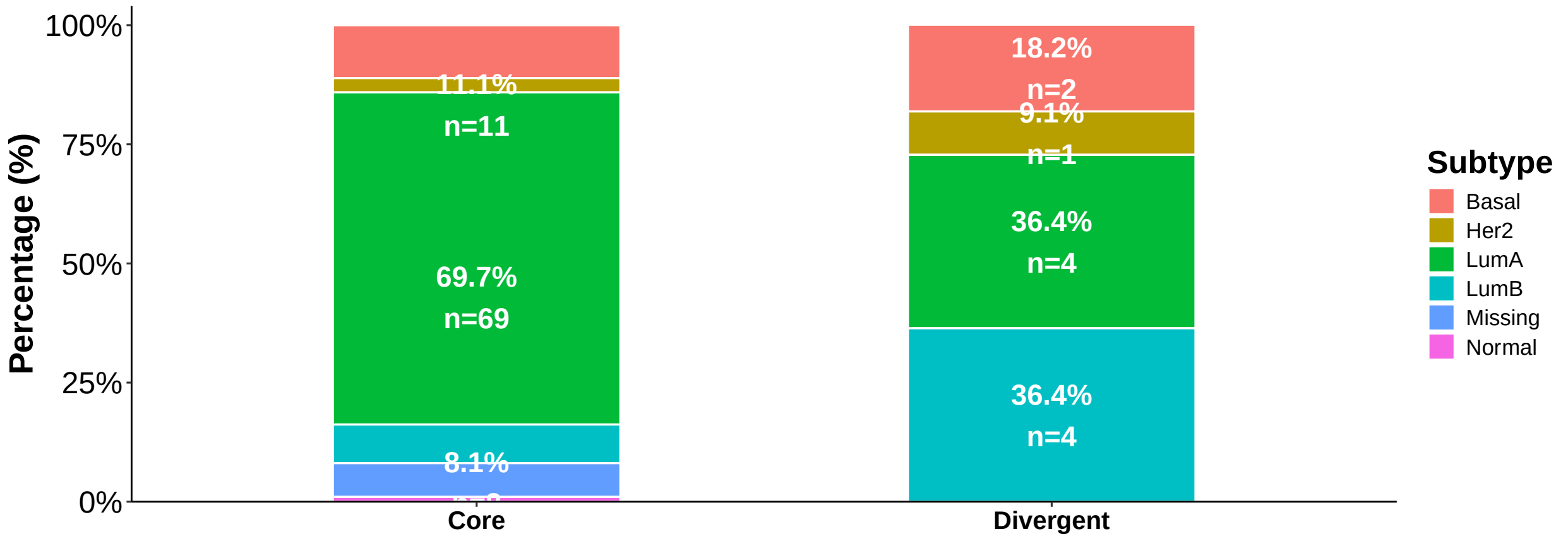
Chi² p: 0.873



Group

Subtype

Chi² p: 0.0471

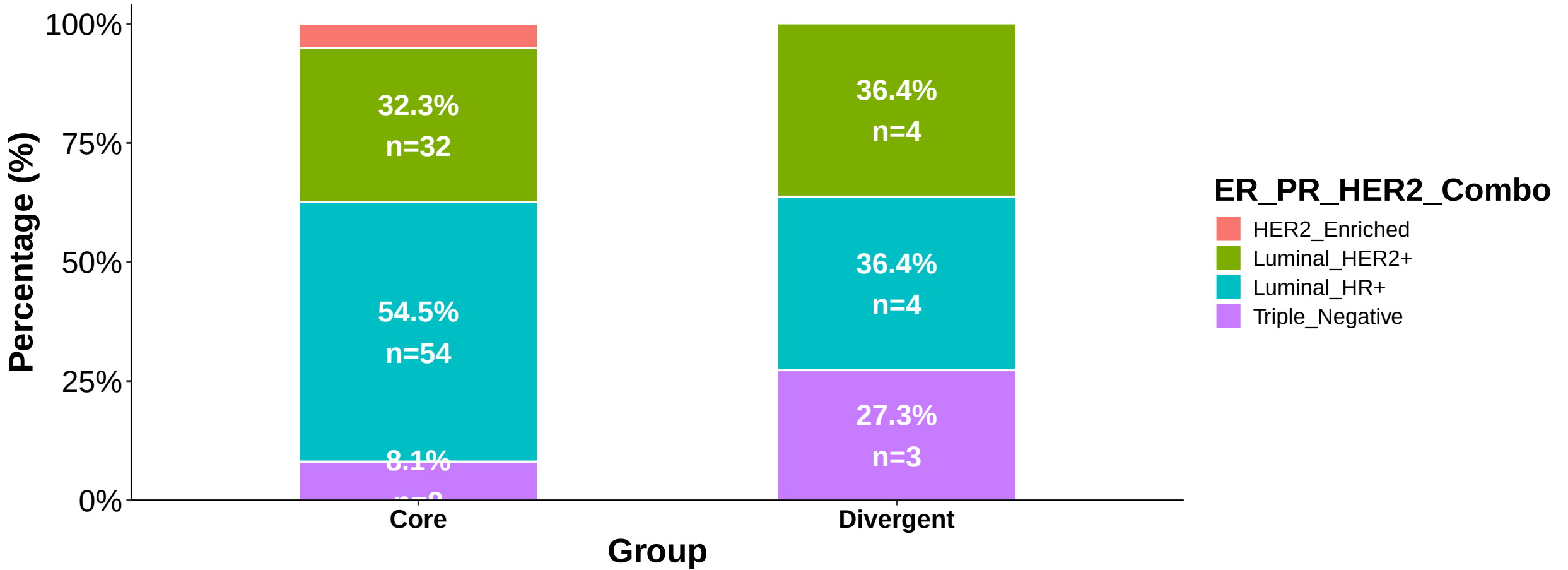


Group

Categorical Clinical – A. Tumor Biology

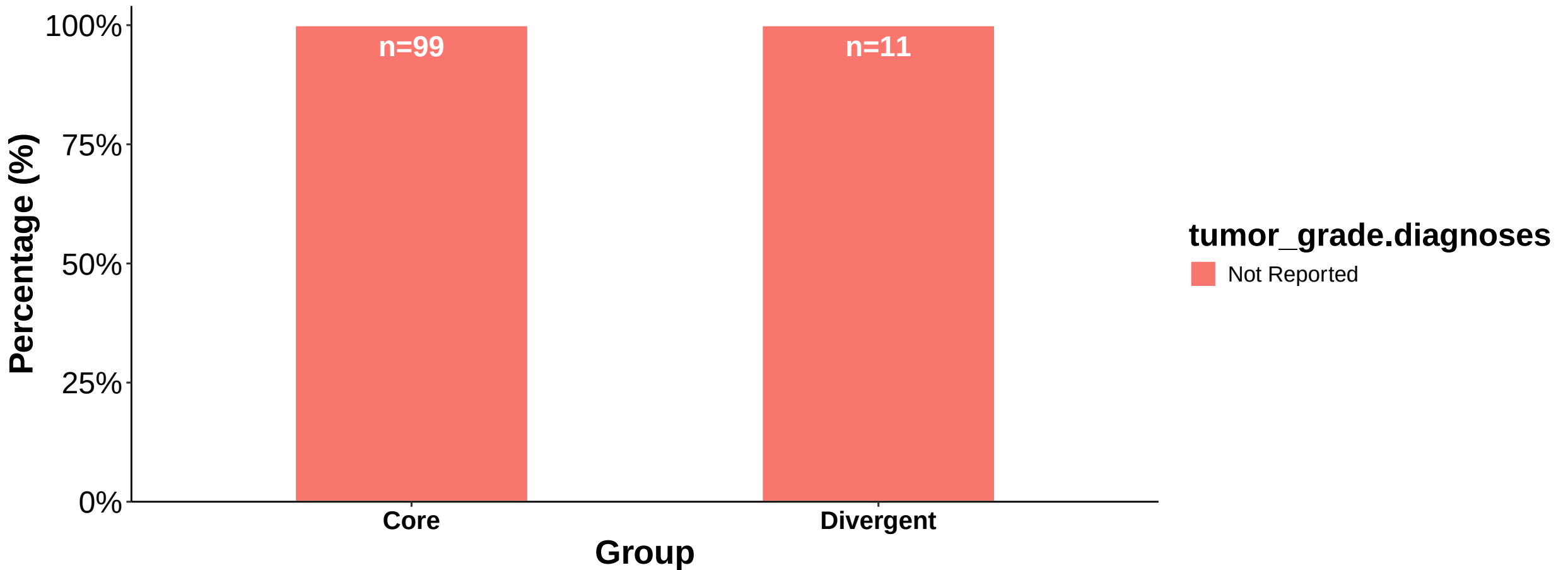
ER_PR_HER2_Combo

Chi² p: 0.181



tumor_grade.diagnoses

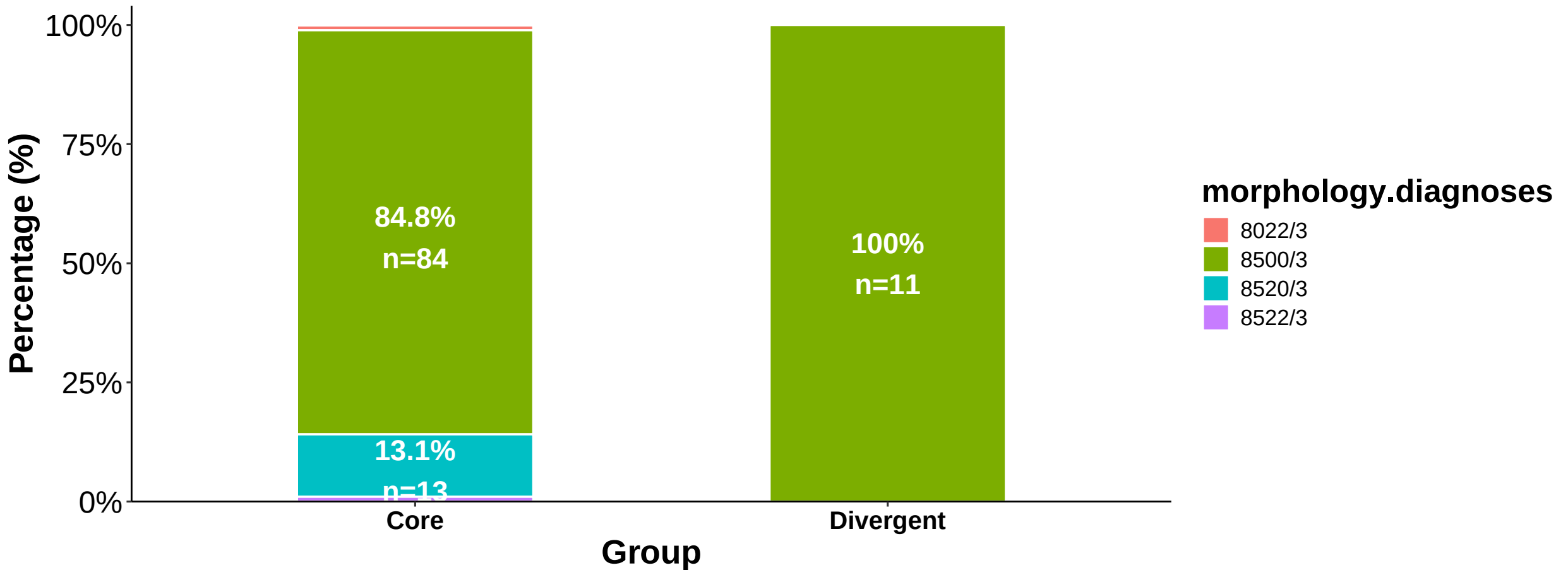
Chi² p: 4.84e-17



Categorical Clinical – . A. Tumor Biology

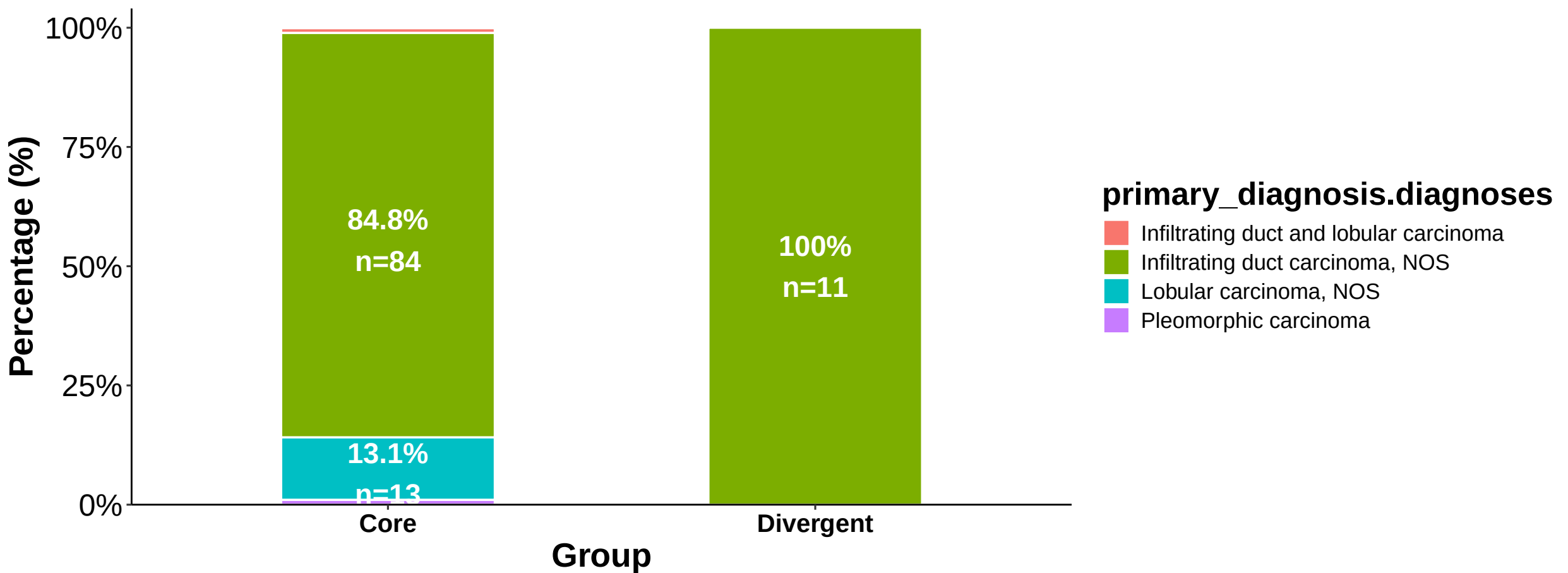
morphology.diagnoses

Chi² p: 0.587



primary_diagnosis.diagnoses

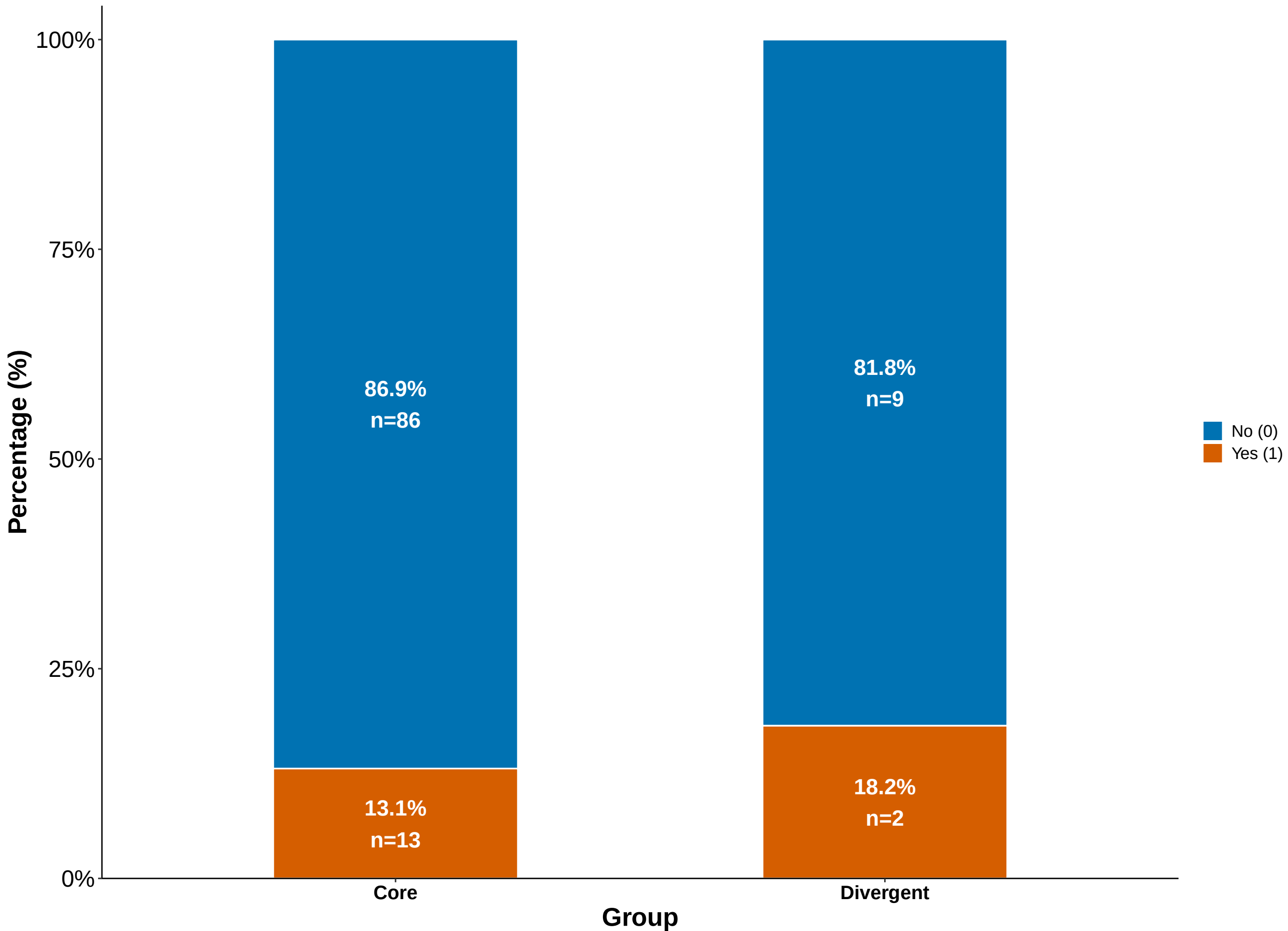
Chi² p: 0.587



Numeric Clinical – . B. Stage and Tumor Burden

Metastasis_Flag

p-adj: 0.762 | . = 0.05 (Negligible) |



Cliff's .: Clinical: . B. Stage and Tumor Burden

Blue = Core > Divergent | Orange = Divergent > Core | * p_adj<0.05

Cliff's .
Div vs Core1.00.5.00.51.0



Metastasis_Flag

ns .=0.05

Cliff's . – Clinical: .B. Stage and Tumor Burden

Cliff's . + .% median | . higher in Core · . higher in Divergent

Metastasis_Flag

ns .=0.05 (NA%)

-0.6

Large

Medium

Small

0

Small

Medium

Large

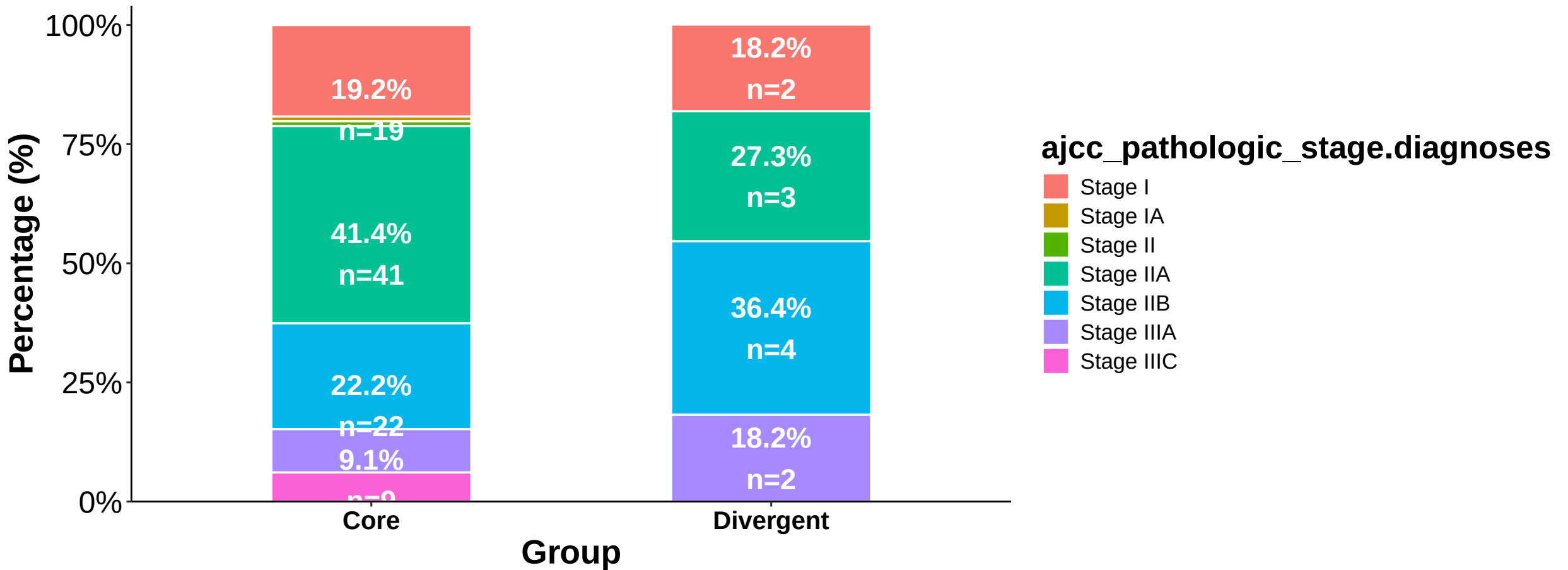
0.6

Cliff's delta [Core vs Divergent]

Categorical Clinical – . B. Stage and Tumor Burden

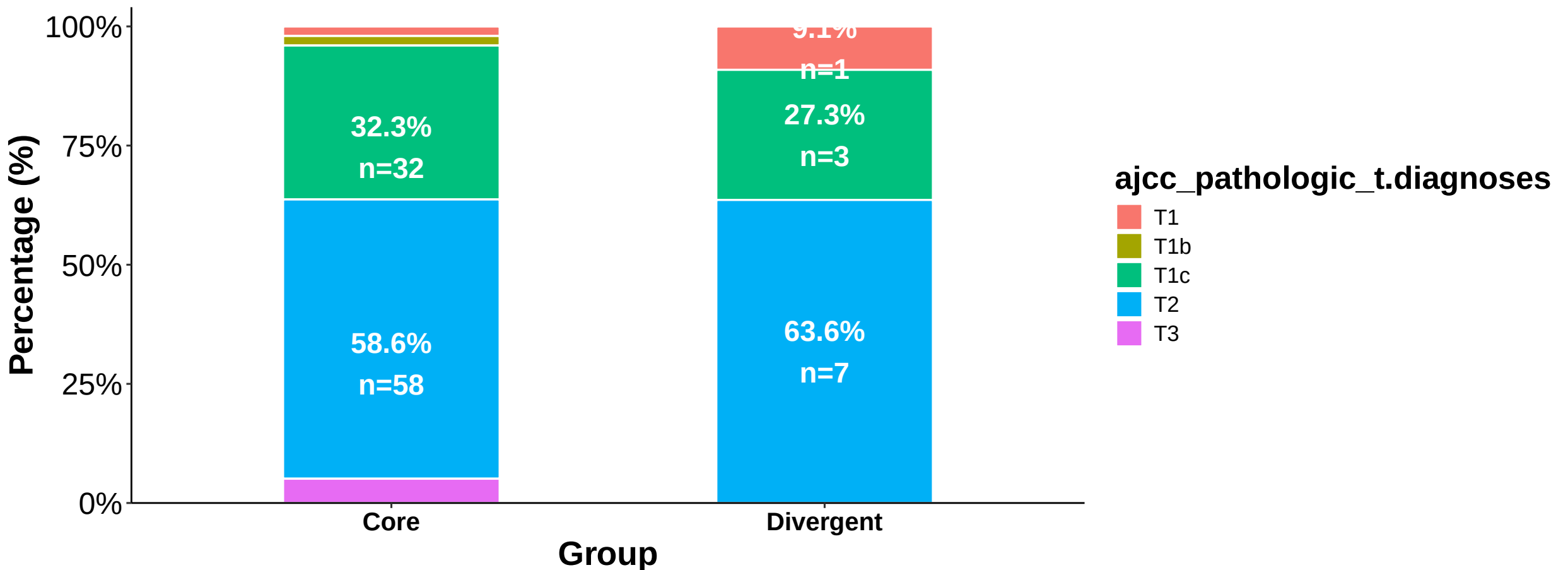
ajcc_pathologic_stage.diagnoses

Chi² p: 0.803



ajcc_pathologic_t.diagnoses

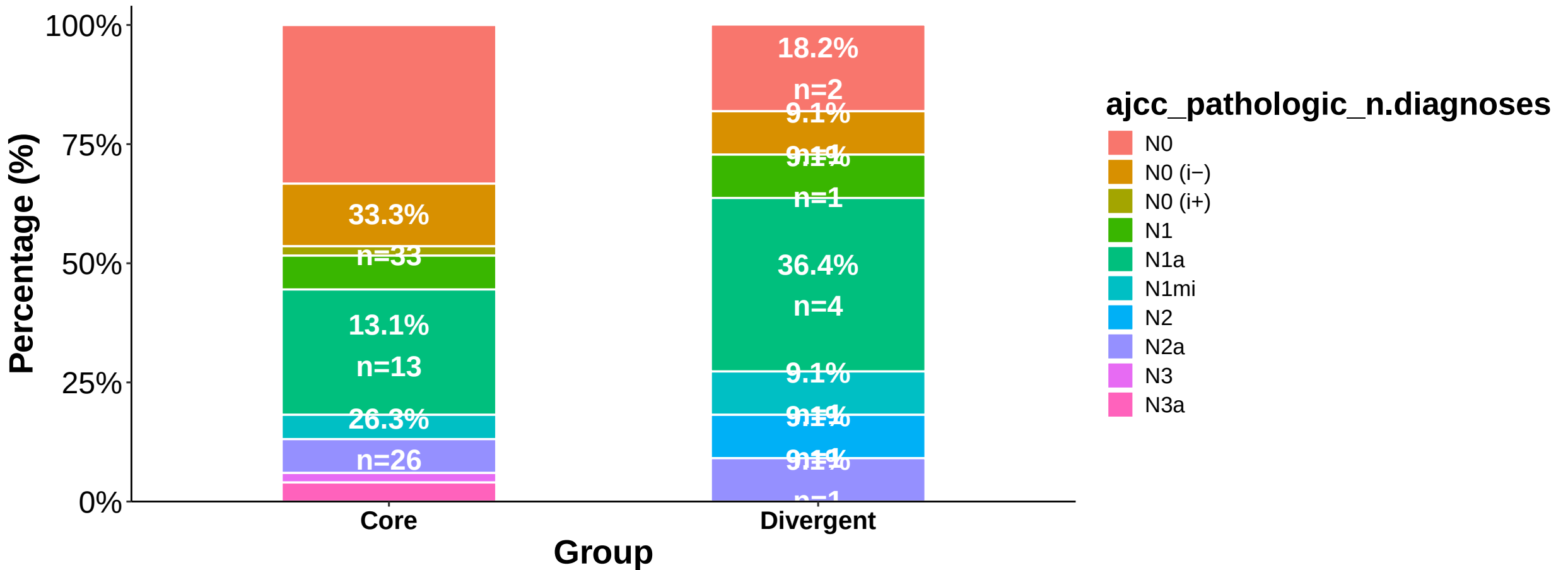
Chi² p: 0.607



Categorical Clinical – . B. Stage and Tumor Burden

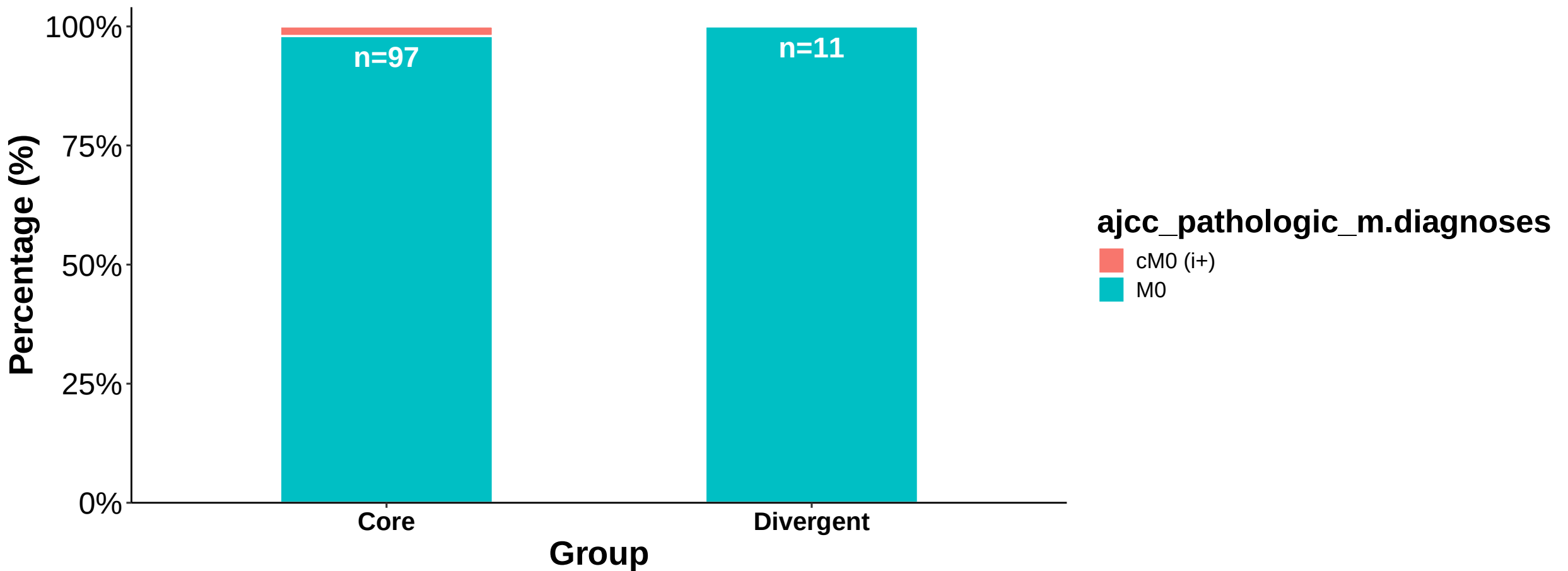
ajcc_pathologic_n.diagnoses

Chi² p: 0.242



ajcc_pathologic_m.diagnoses

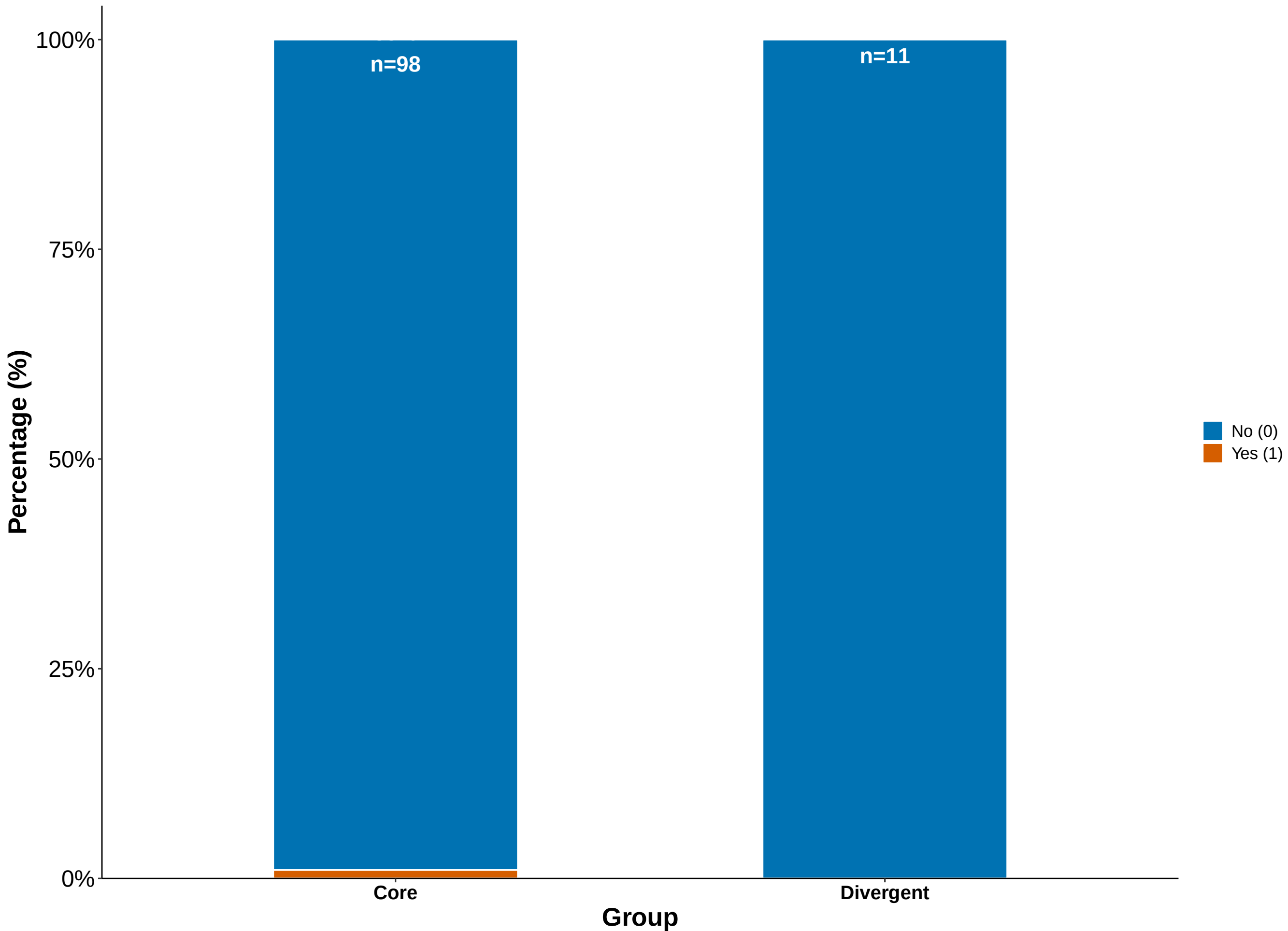
Chi² p: 1



Numeric Clinical – . C. Treatment History

Prior_Treatment_Flag

p-adj: 0.762 | .=-0.01 (Negligible) |



Cliff's .: Clinical: . C. Treatment History

Blue = Core > Divergent | Orange = Divergent > Core | * p_adj<0.05

Cliff's .
Div vs Core



-1.0 -0.5 0.0 0.5 1.0

Prior_Treatment_Flag

ns .=-0.01

Cliff's . – Clinical: . C. Treatment History

Cliff's . + .% median | . higher in Core · . higher in Divergent

Prior_Treatment_Flag

ns .=-0.01 (NA%)

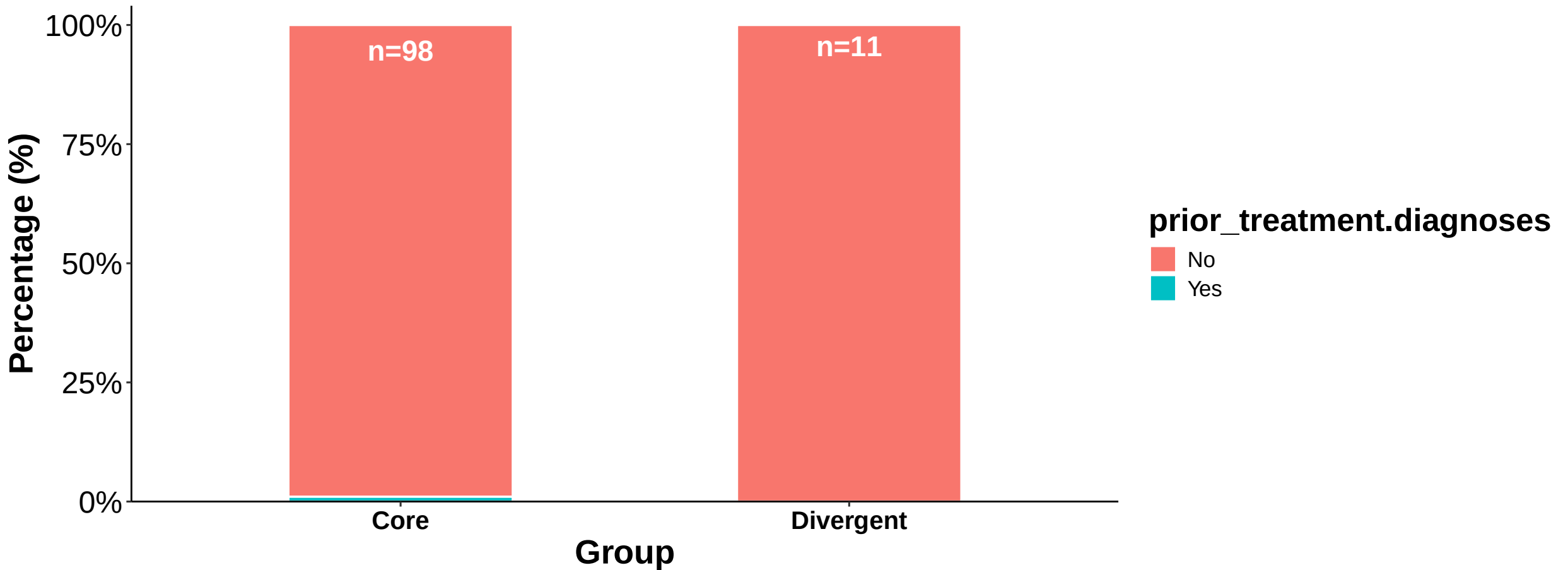
-0.6 Large Medium Small 0 Small Medium Large 0.6

Cliff's delta [Core vs Divergent]

Categorical Clinical – .C. Treatment History

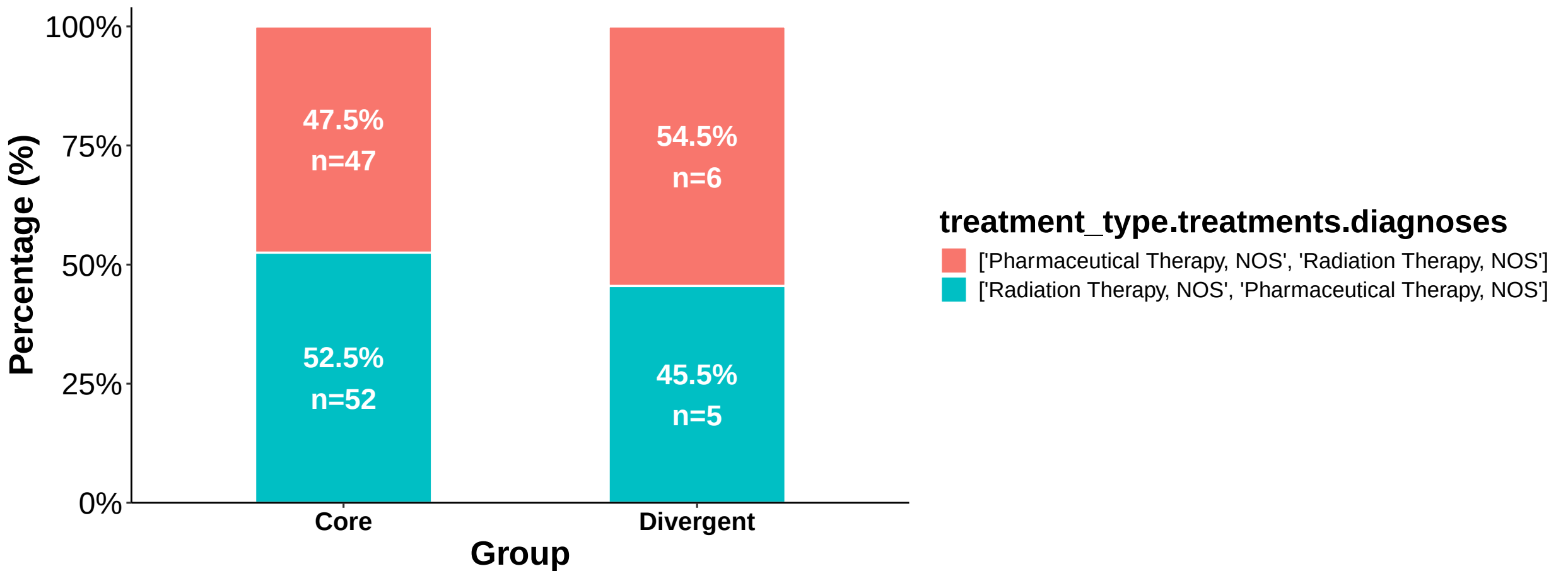
prior_treatment.diagnoses

Chi² p: 1



treatment_type.treatments.diagnoses

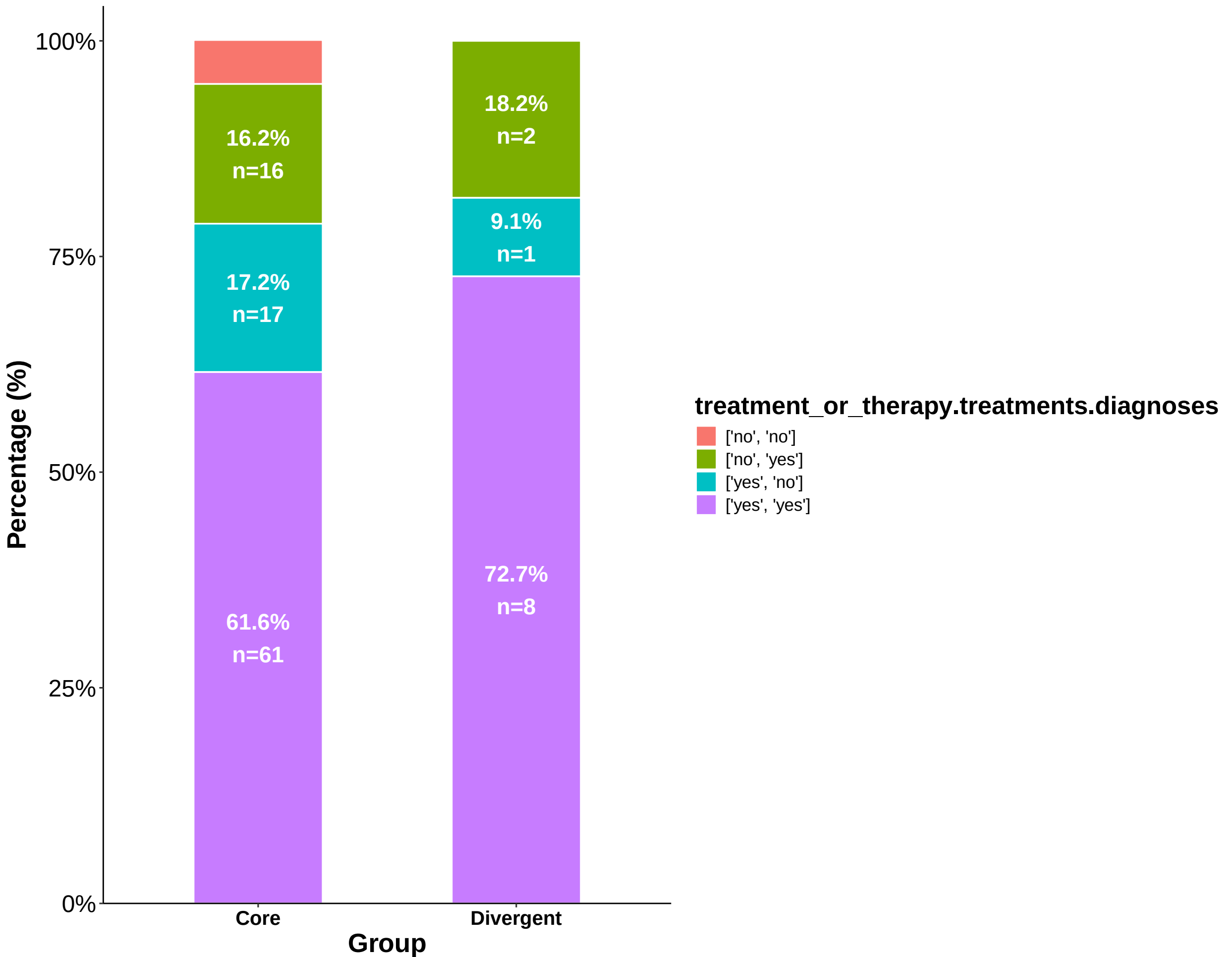
Chi² p: 0.899



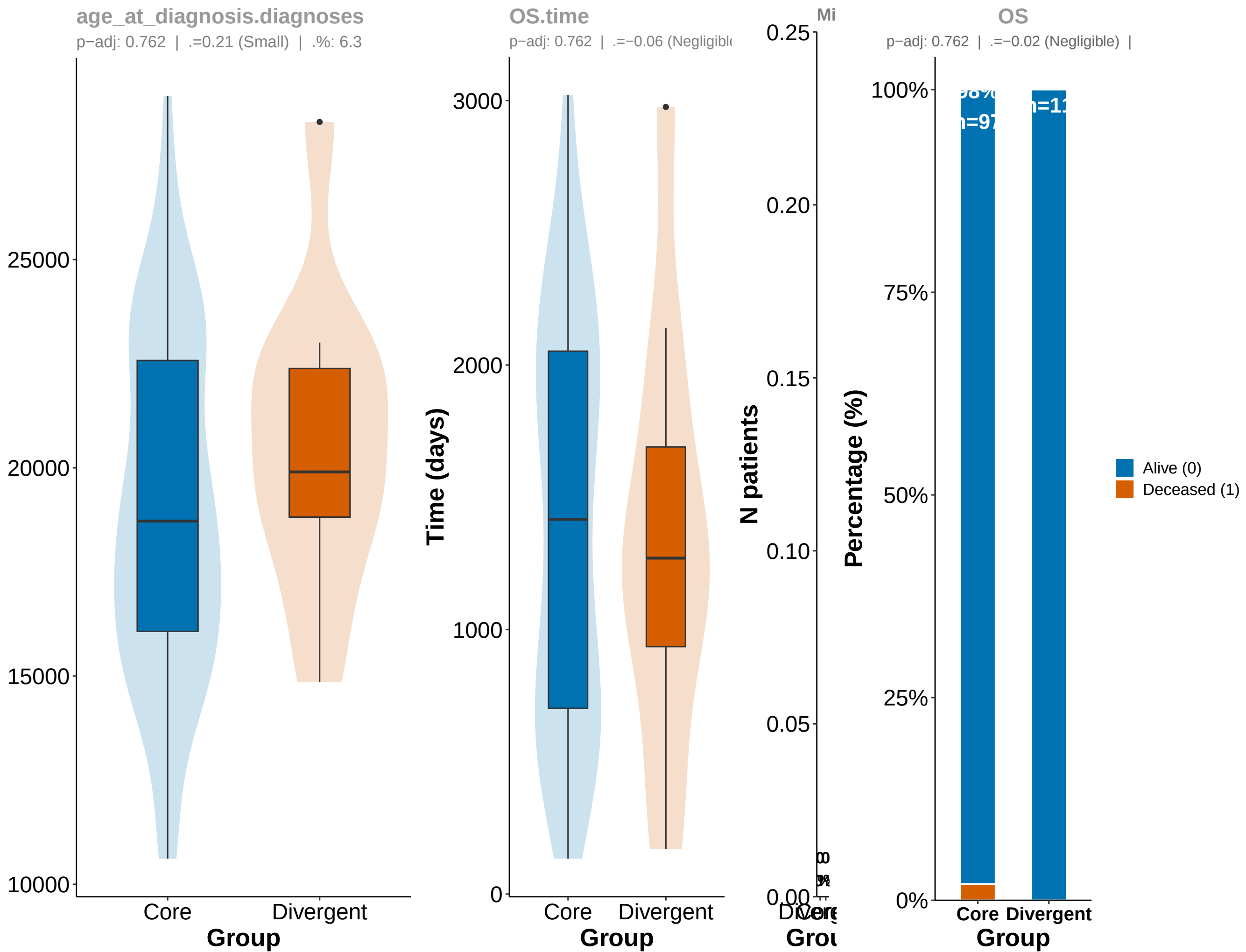
Categorical Clinical – .C. Treatment History

treatment_or_therapy.treatments.diagnoses

Chi² p: 0.76



Numeric Clinical – .D. Patient Factors



Cliff's .: Clinical: . D. Patient Factors

Blue = Core > Divergent | Orange = Divergent > Core | * p_adj<0.05

Cliff's .
Div vs Core1.00.5.00.51.0



age_at_diagnosis.diagnoses

ns .=-0.21

OS

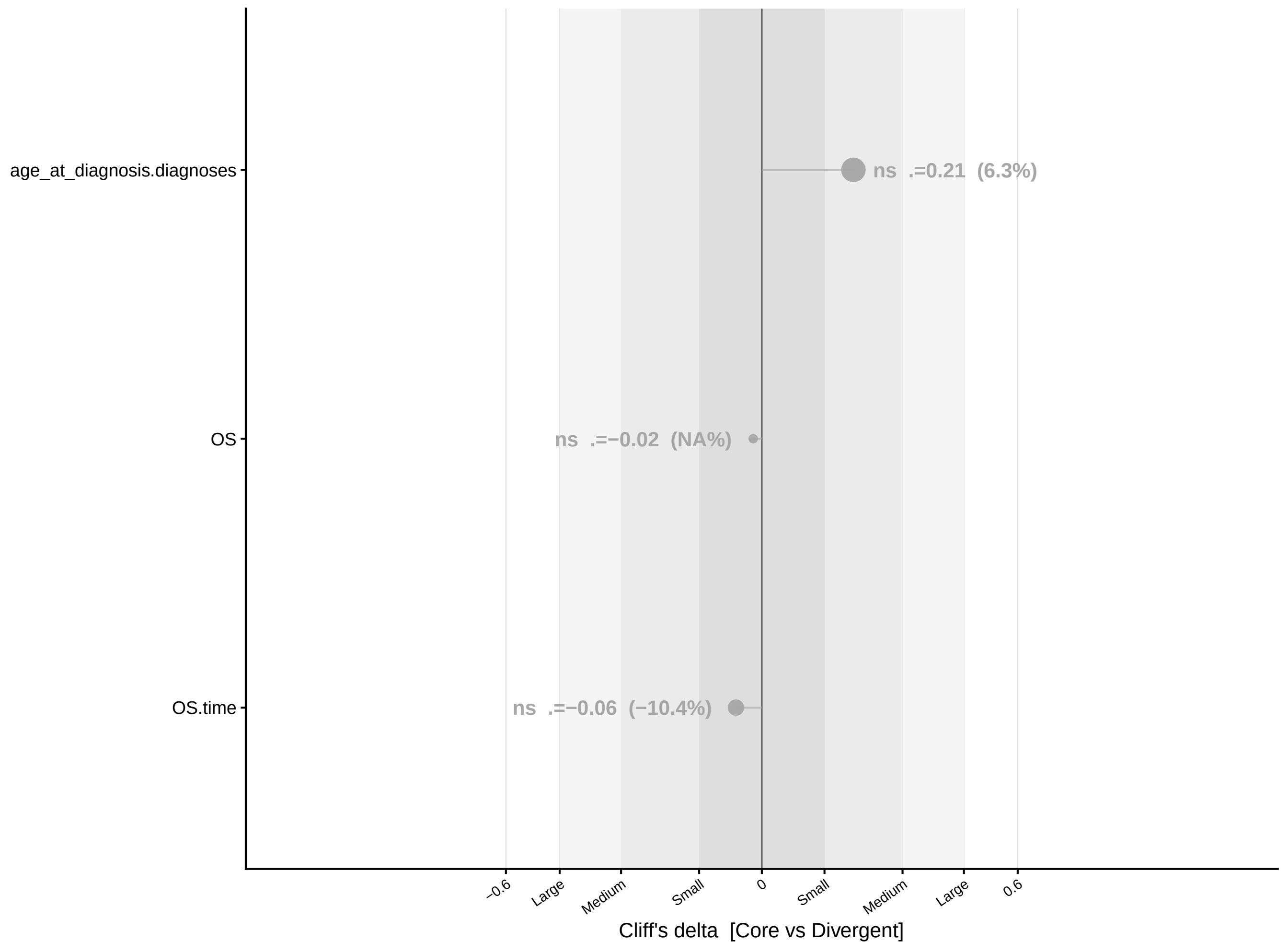
ns .=-0.02

OS.time

ns .=-0.06

Cliff's Δ – Clinical: D. Patient Factors

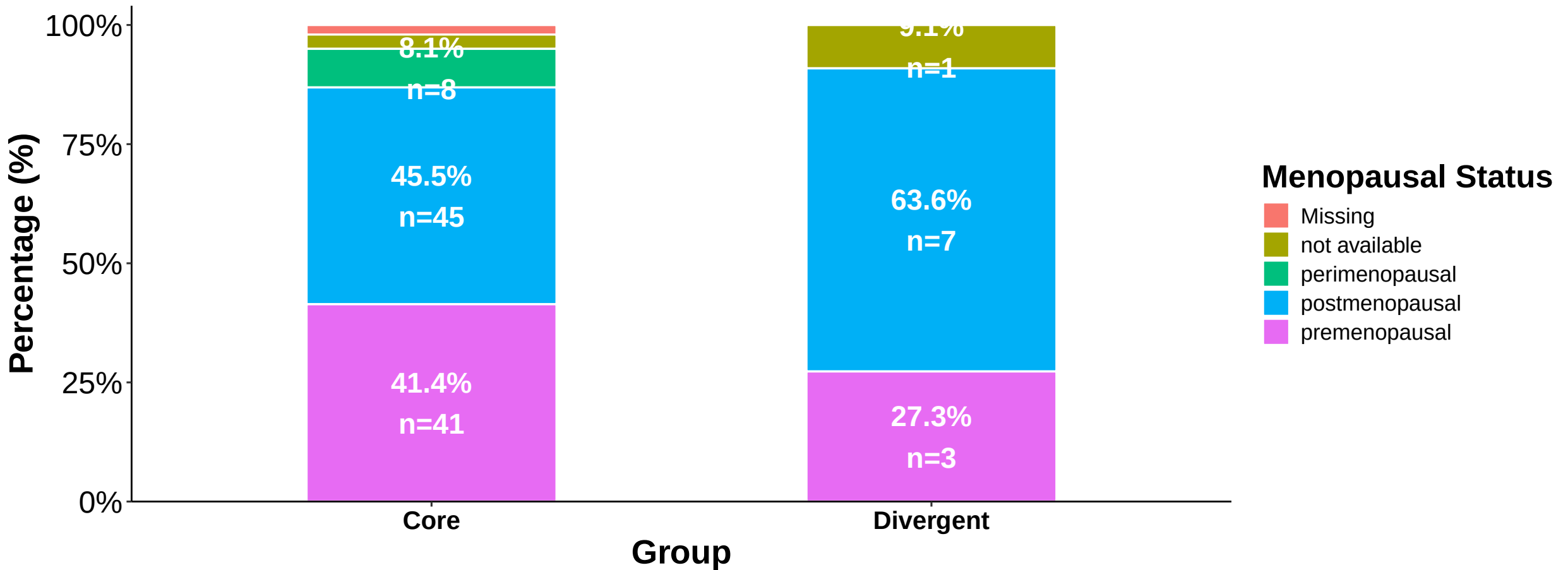
Cliff's Δ + Δ% median | Δ higher in Core Δ higher in Divergent



Categorical Clinical – . D. Patient Factors

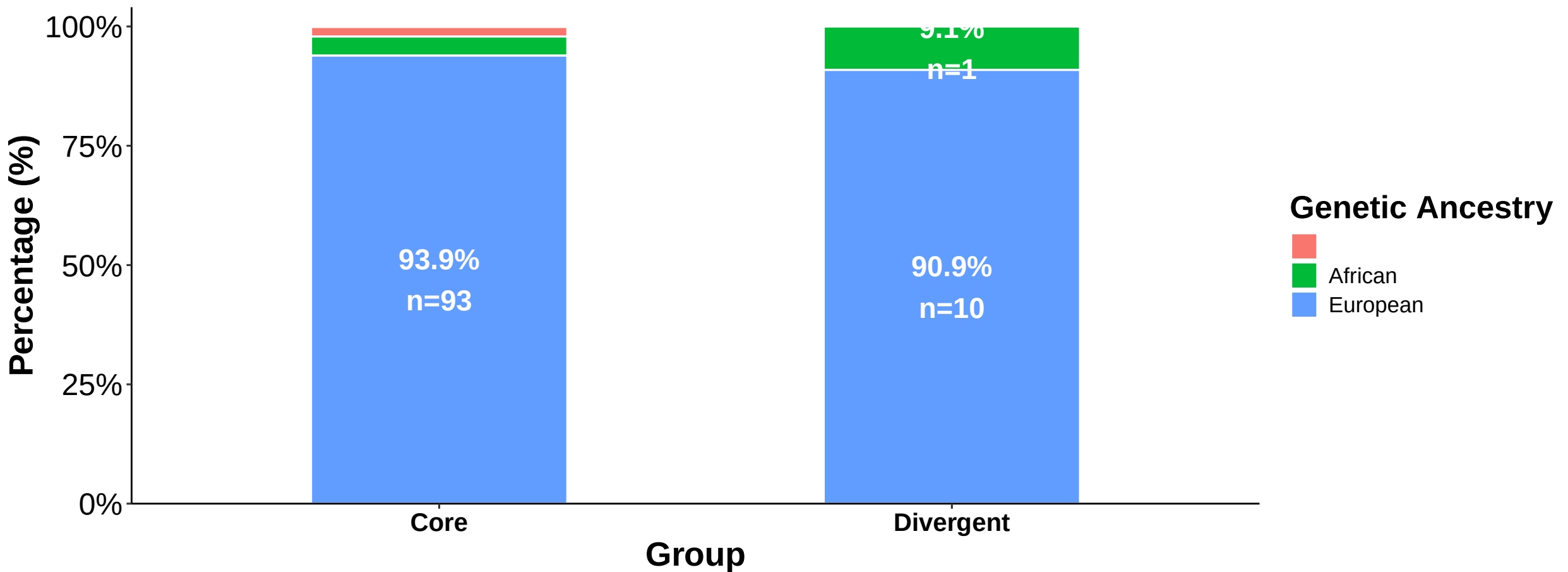
Menopausal Status

Chi² p: 0.509



Genetic Ancestry

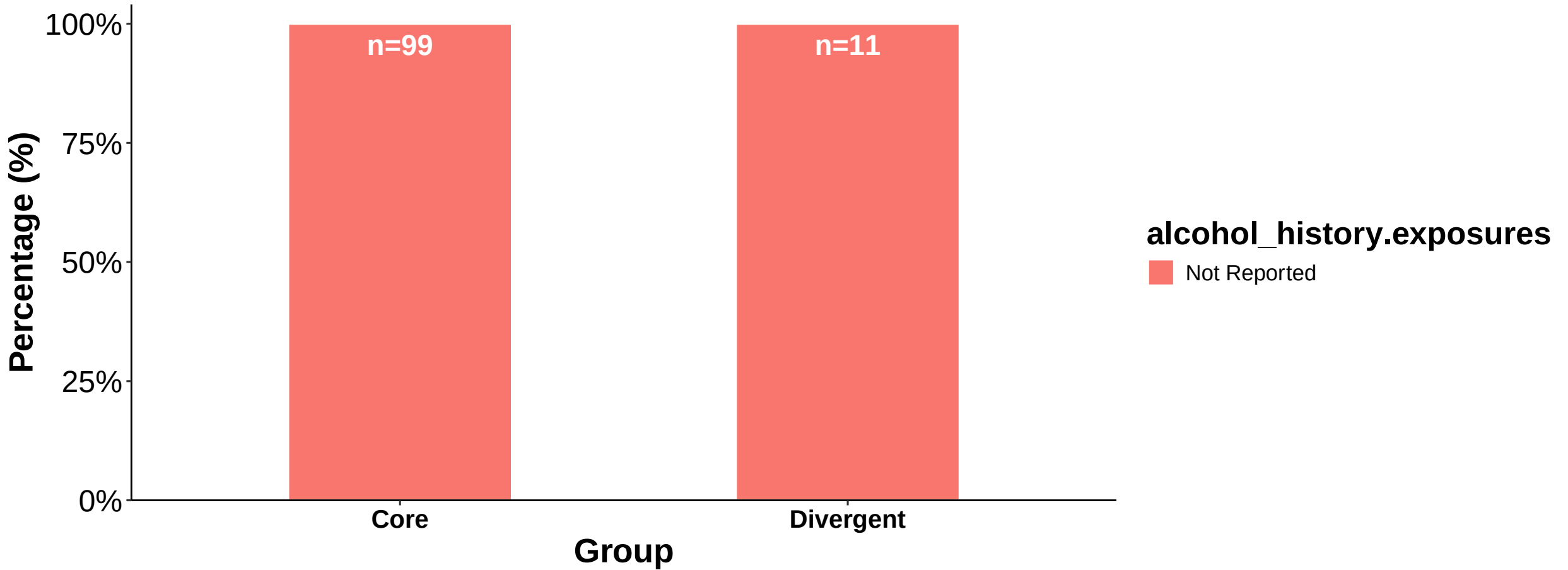
Chi² p: 0.675



Categorical Clinical – . D. Patient Factors

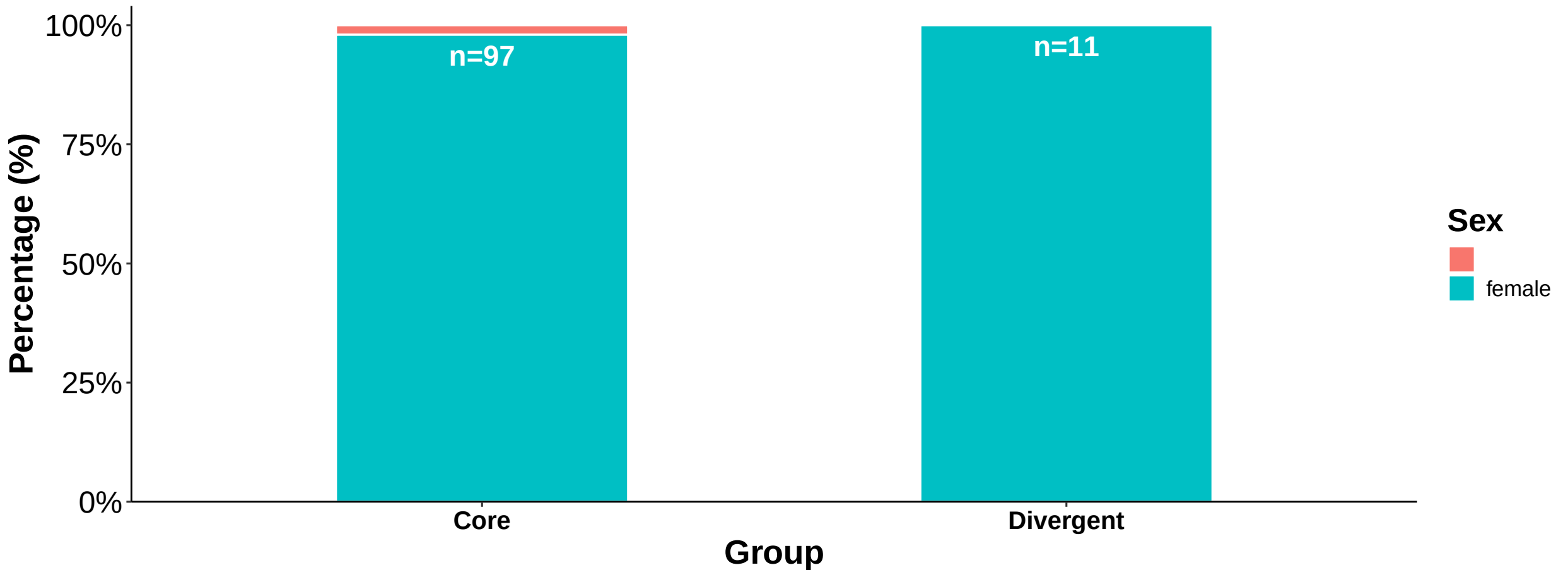
alcohol_history.exposures

Chi² p: 4.84e-17



Sex

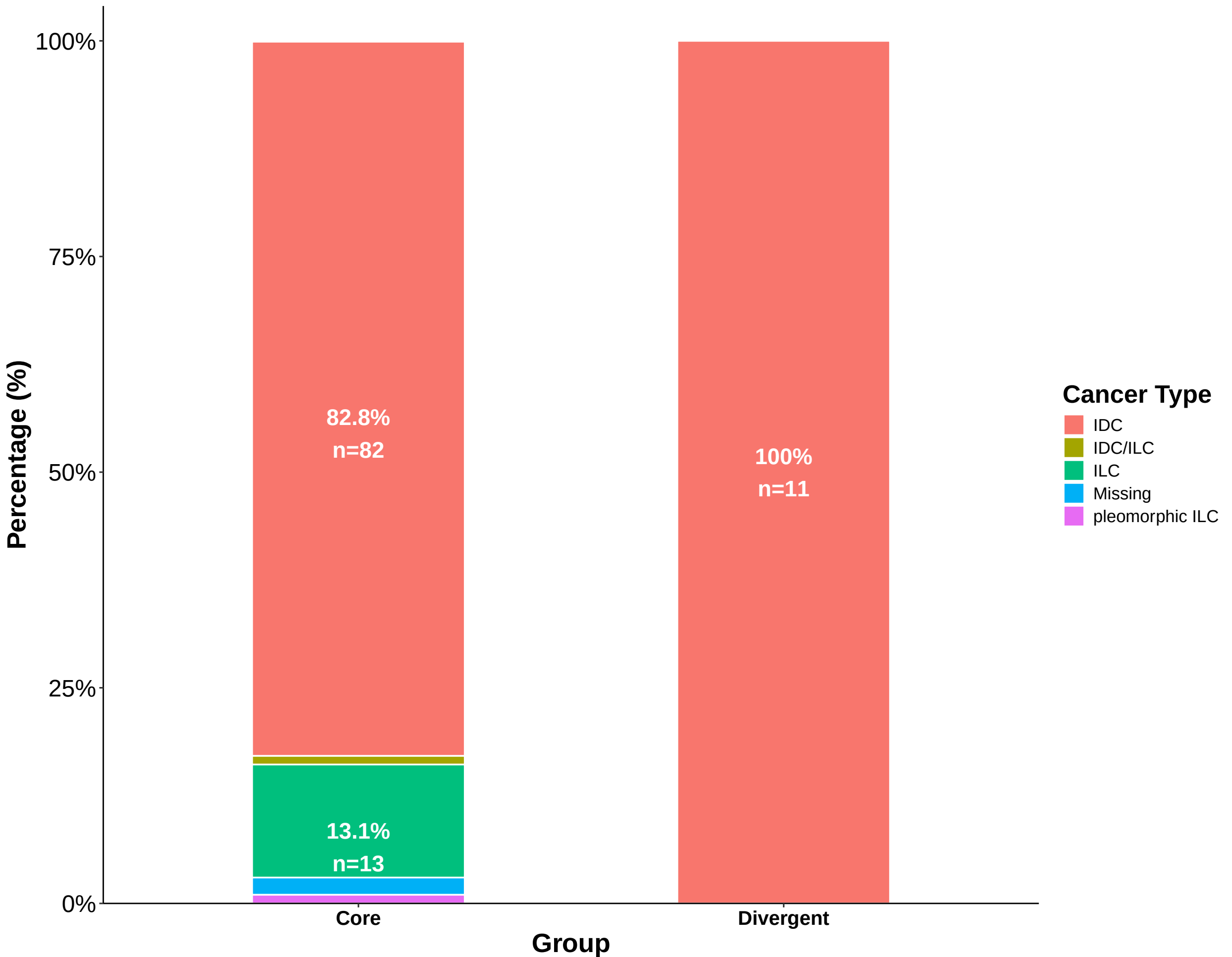
Chi² p: 1



Categorical Clinical – . D. Patient Factors

Cancer Type

Chi² p: 0.693



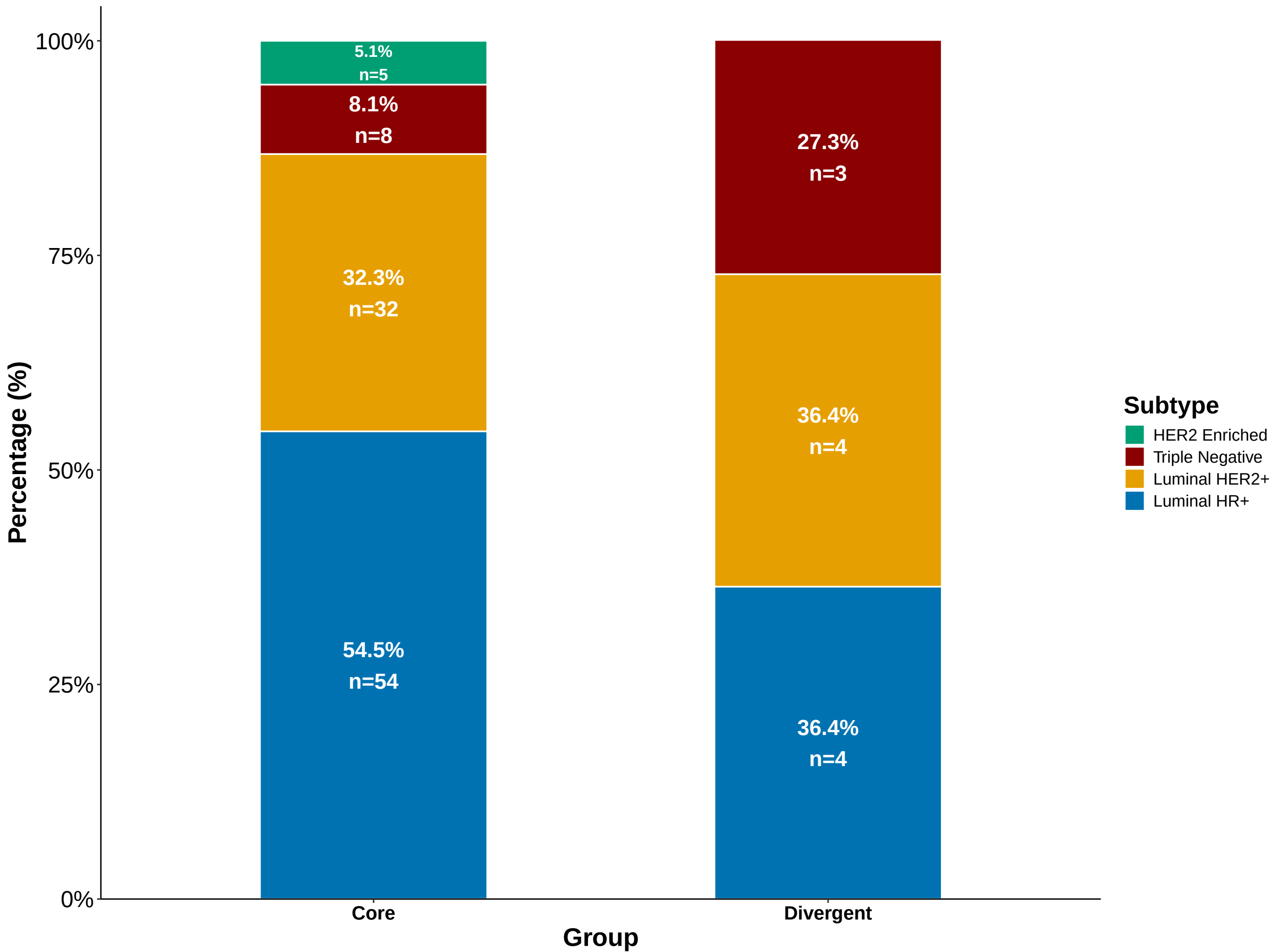
SECTION I–B

Molecular Subtype Analysis: Core vs Divergent

Source: ER_PR_HER2_Combo | Triple Negative · Luminal HR+ · Luminal HER2+ · HER2 Enriched · Missing

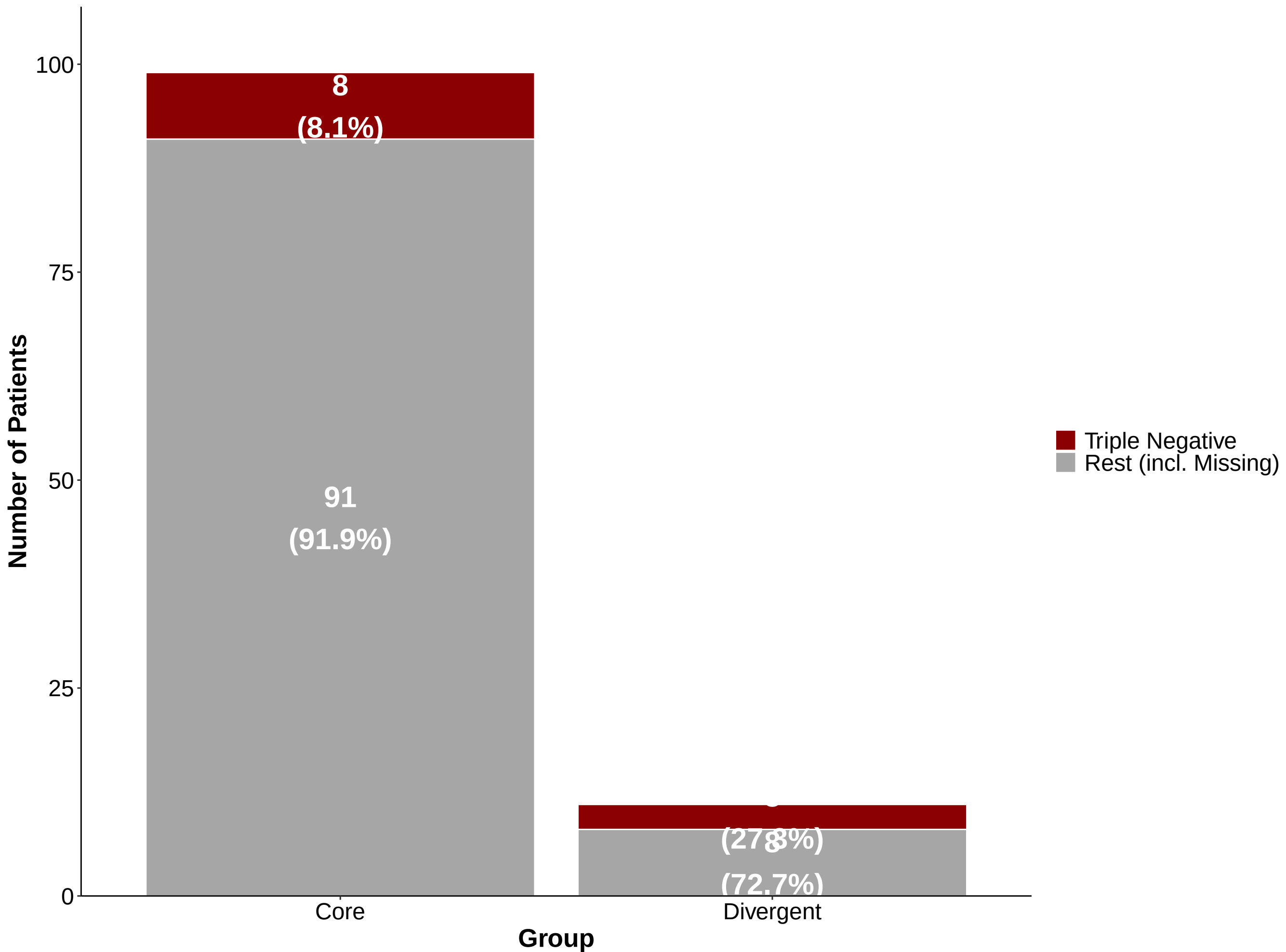
Molecular Subtypes: Proportion (%)

Chi² p = 0.181



Triple Negative vs Rest – Core vs Divergent

Chi² p = 0.181



Triple Negative Summary (source: ER_PR_HER2_Combo):

Core: 8 TNBC out of 99 total (8.1%)

Divergent: 3 TNBC out of 11 total (27.3%)

Chi² test subtypes vs Group: Chi² p = 0.181

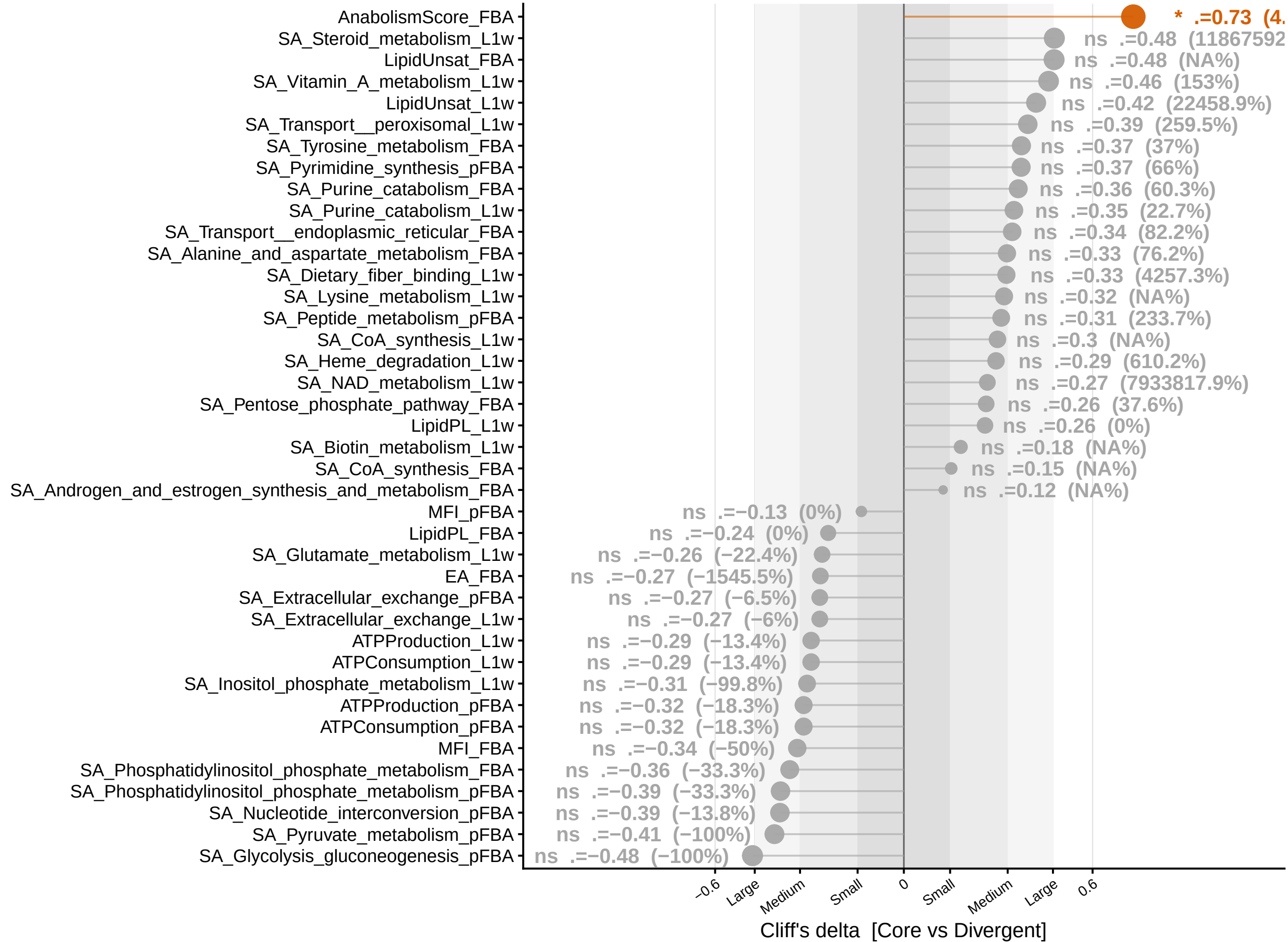
SECTION II

Metabolic Differences: Core vs Divergent

5 functional blocks: Energy | Redox | Lipids | Dependencies | Systemic Reprogramming

Top 40 Metabolic Variables – Cliff's .

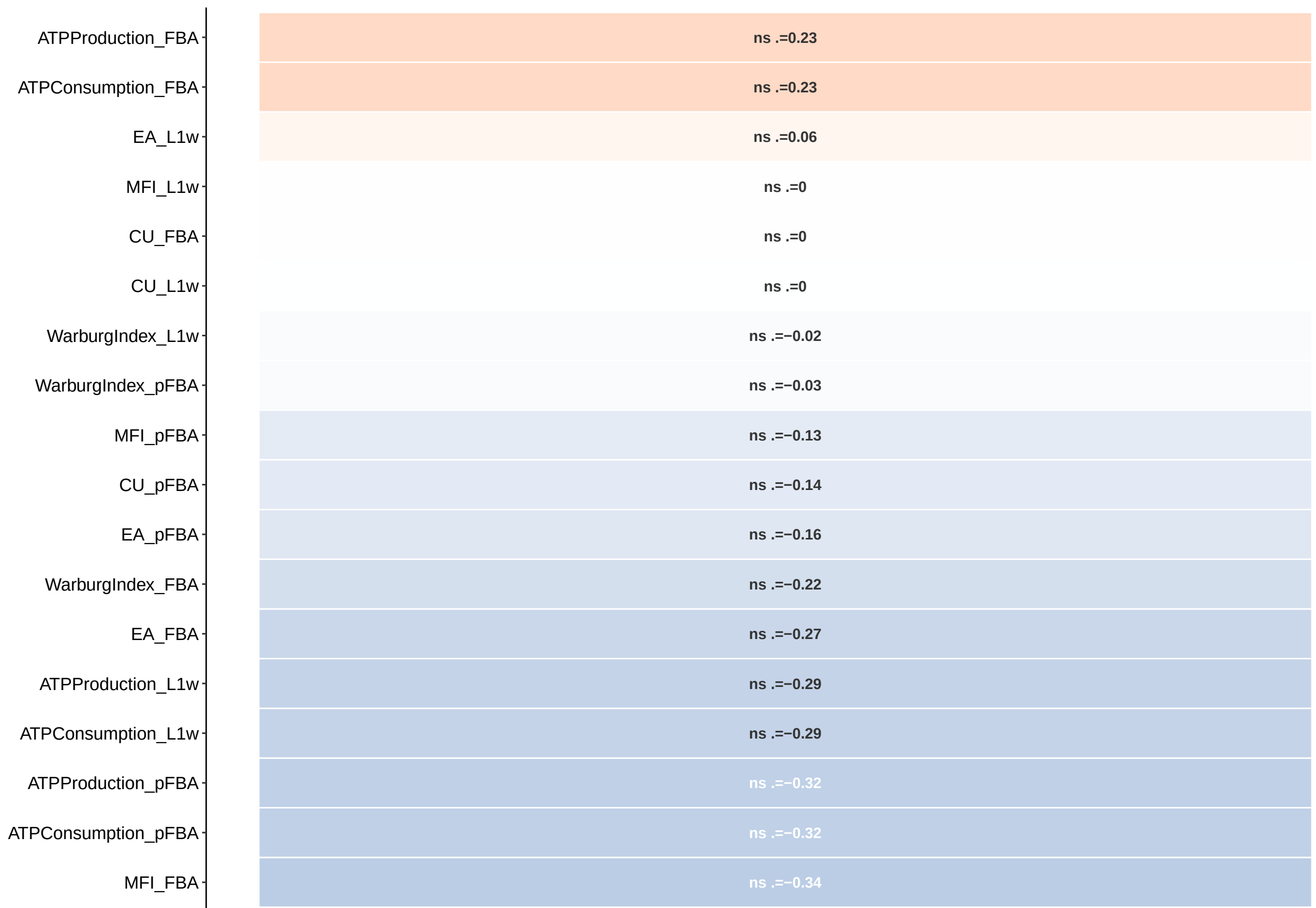
Cliff's . + .% median | . higher in Core · . higher in Divergent



Cliff's .: . 1. Energy and Bioenergetics

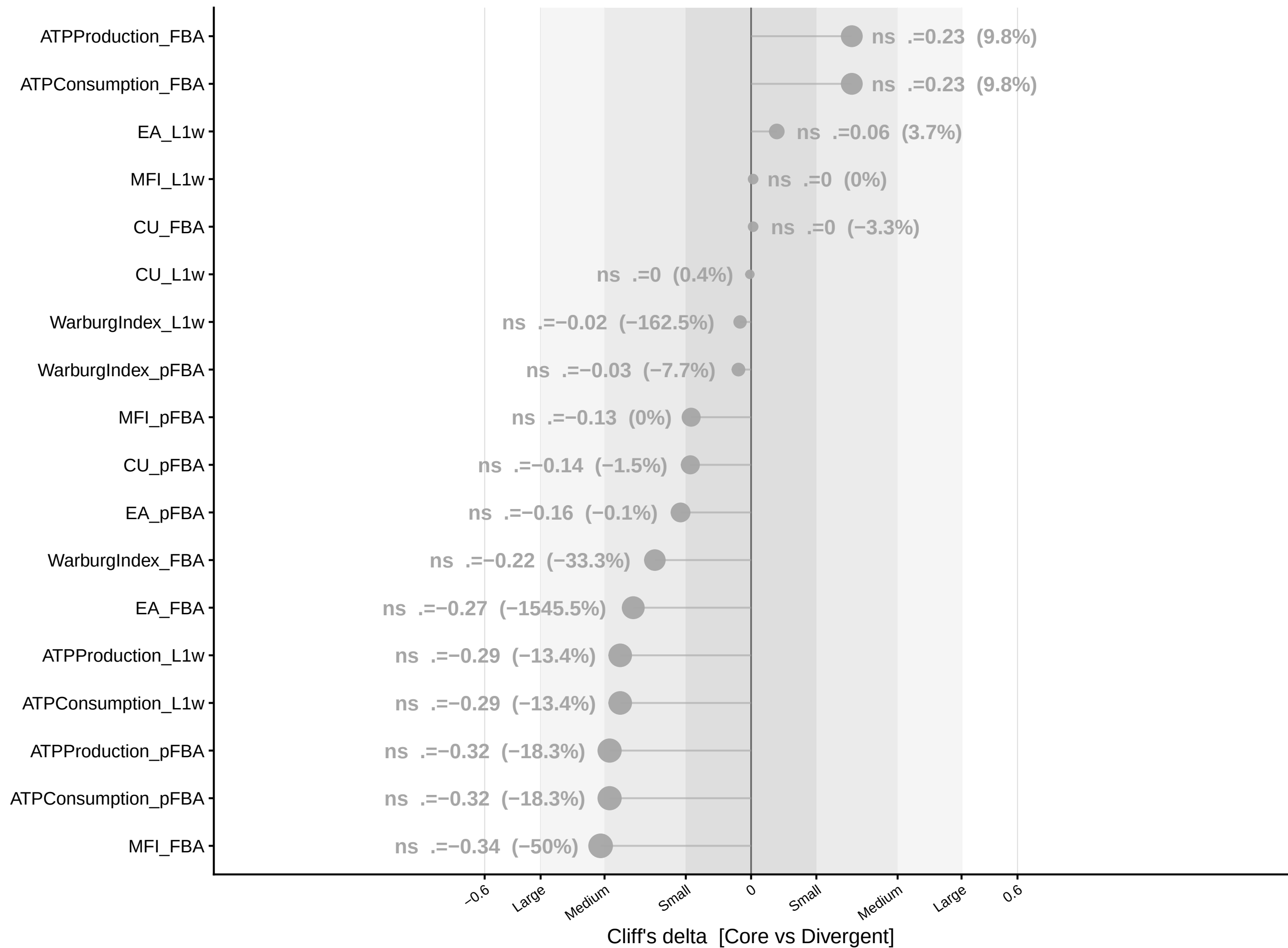
Blue = Core > Divergent | Orange = Divergent > Core | * p_adj<0.05

Cliff's .
Div vs Core1.00.50.00.51.0



Cliff's . – . 1. Energy and Bioenergetics

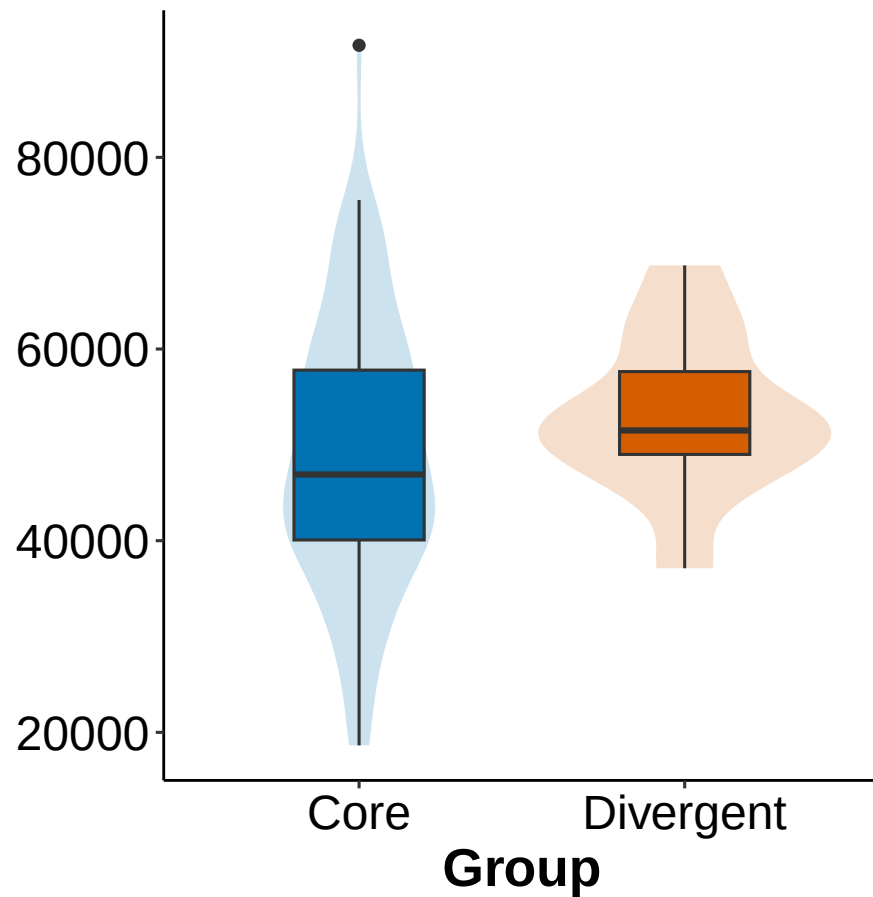
Cliff's . + .% median | . higher in Core · . higher in Divergent



Metabolism – . 1. Energy and Bioenergetics

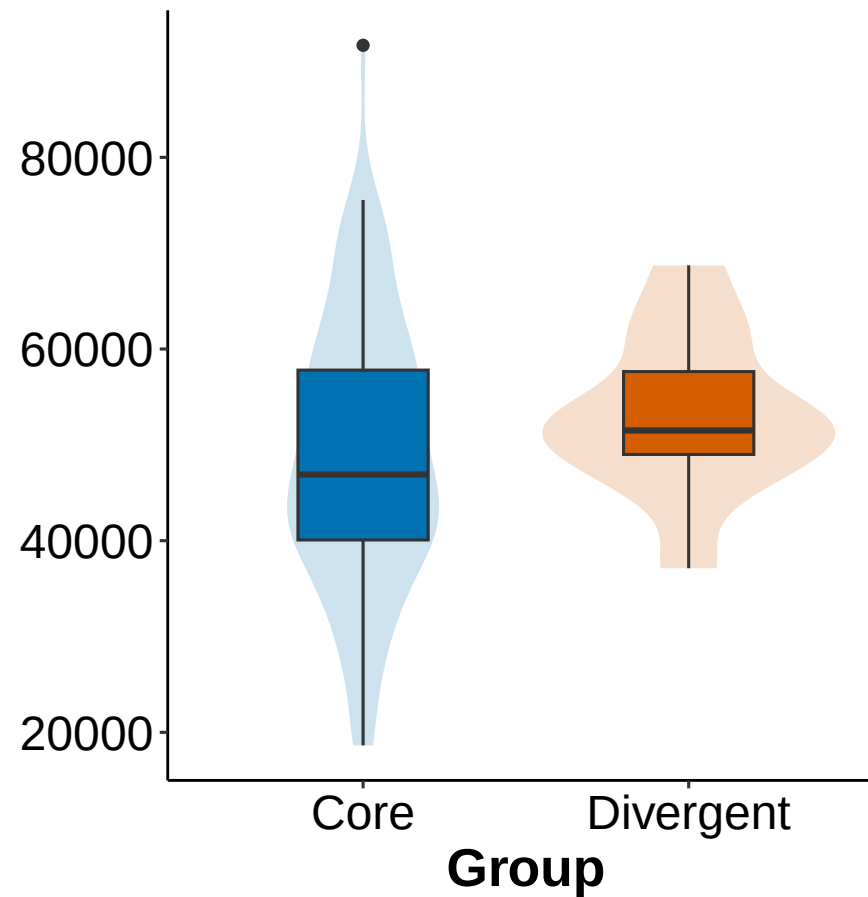
ATPConsumption_FBA

p-adj: 0.885 | .:=0.23 (Small) | .%: 9.8



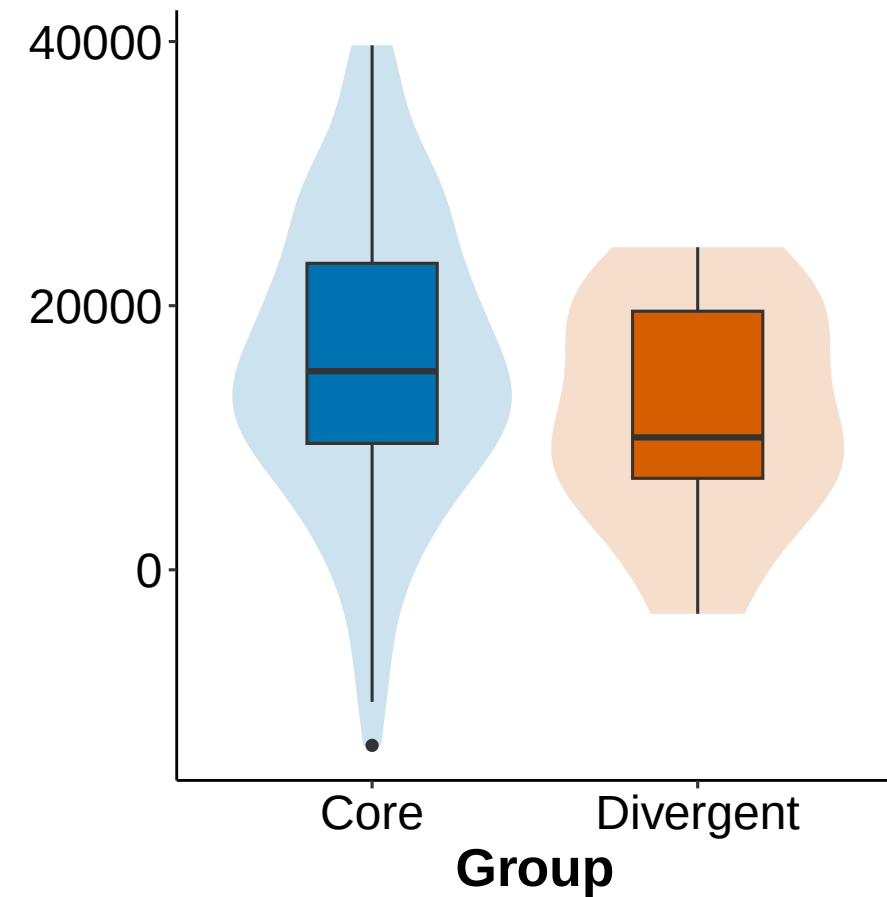
ATPProduction_FBA

p-adj: 0.885 | .:=0.23 (Small) | .%: 9.8



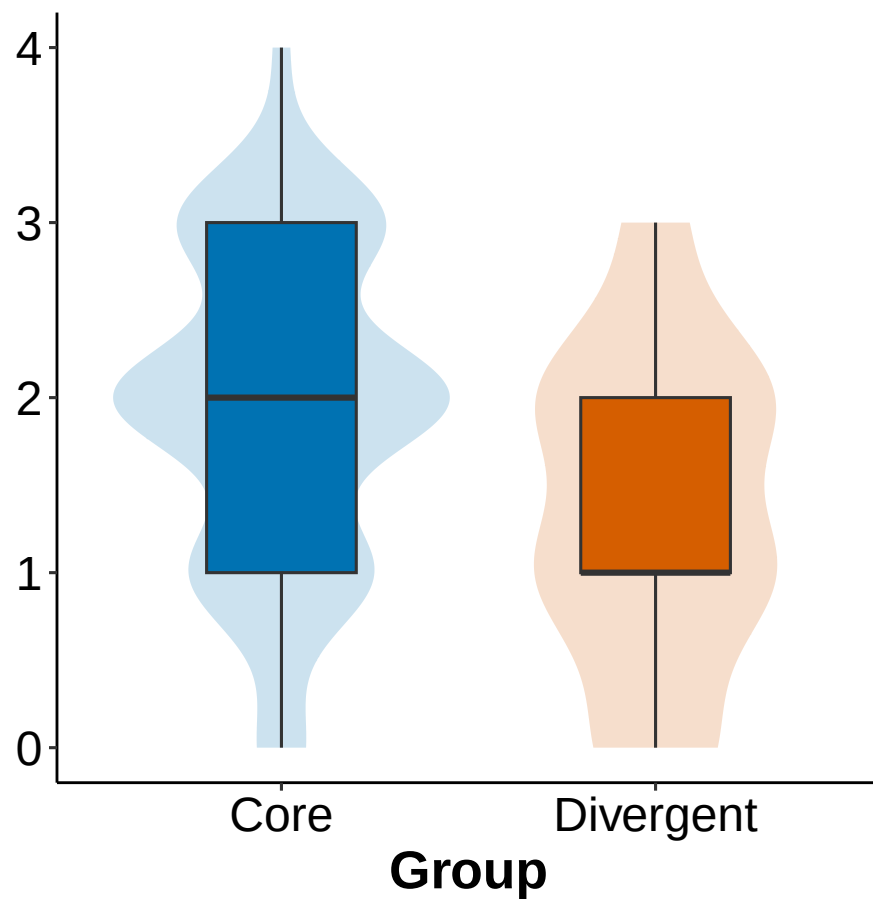
WarburgIndex_FBA

p-adj: 0.885 | .:= -0.22 (Small) | .%: -33.3



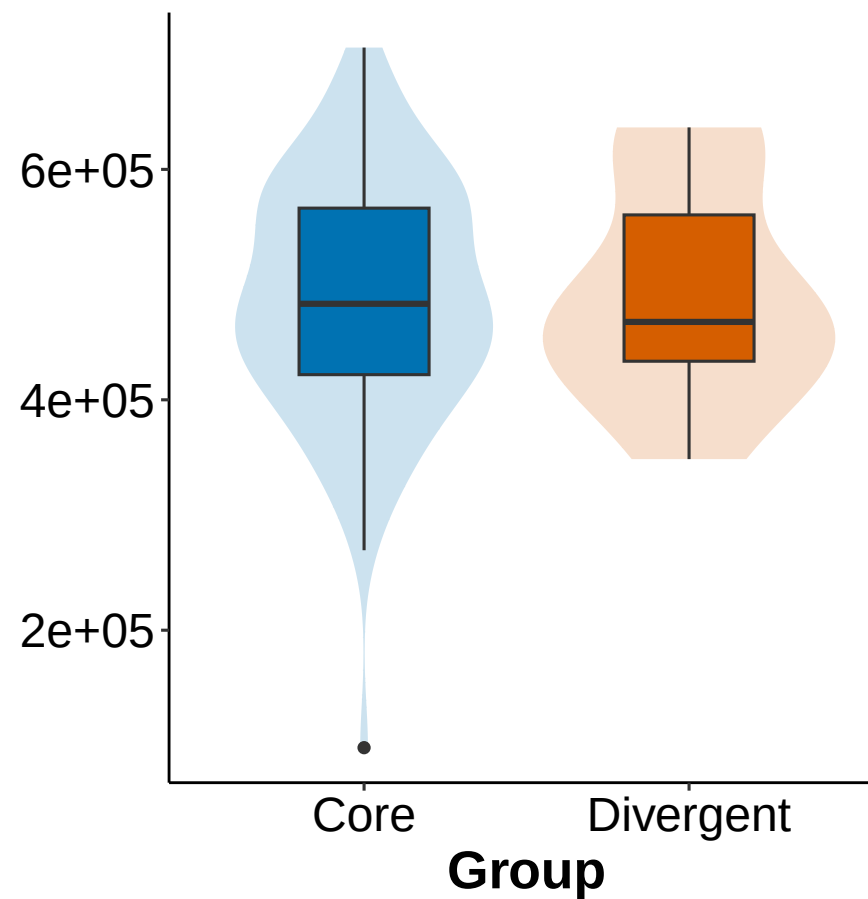
MFI_FBA

p-adj: 0.843 | .:= -0.34 (Medium) | .%: -50



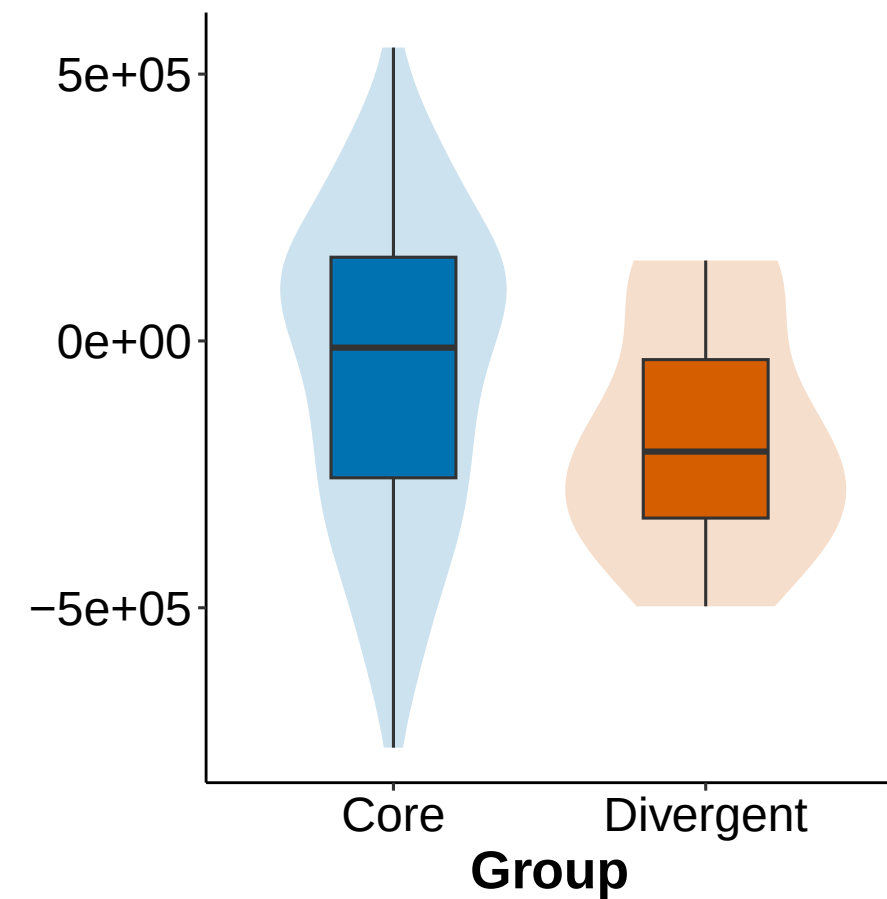
CU_FBA

p-adj: 0.992 | .:=0 (Negligible) | .%: -3.3



EA_FBA

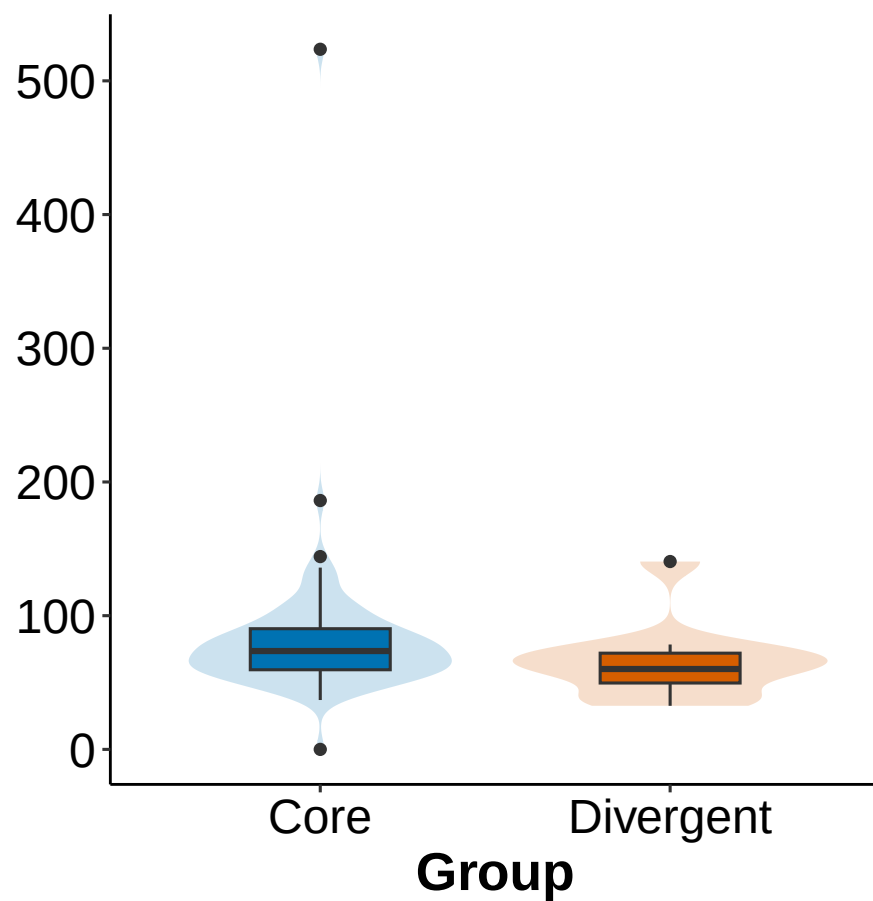
p-adj: 0.872 | .:= -0.27 (Small) | .%: -1545.5



Metabolism – . 1. Energy and Bioenergetics

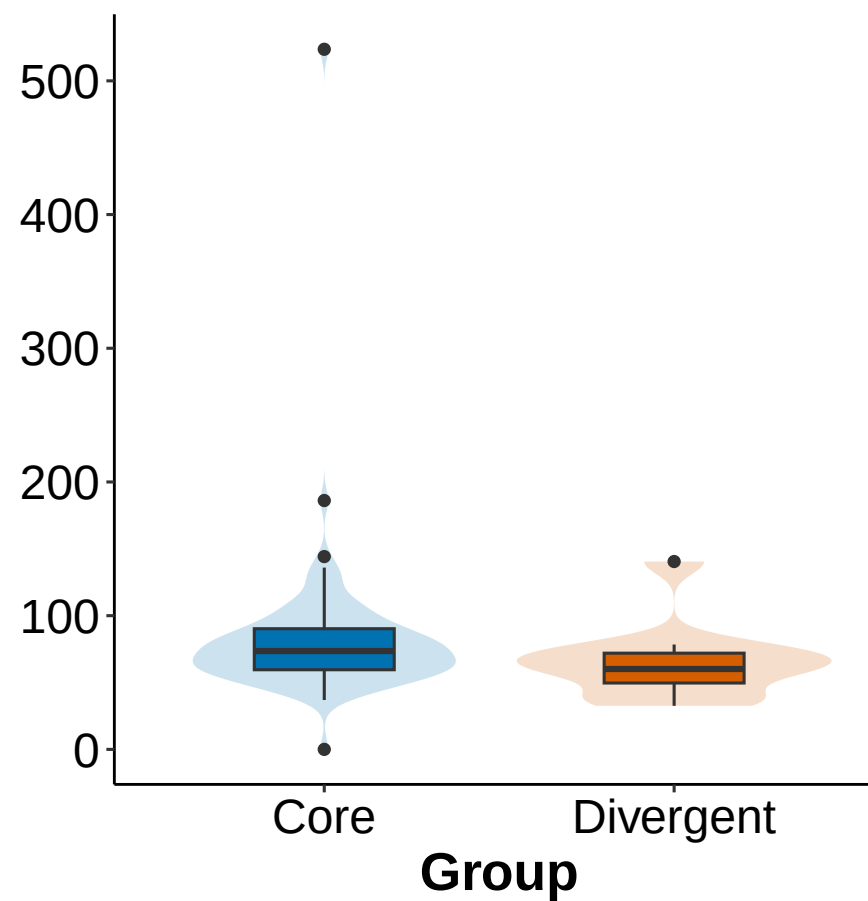
ATPConsumption_pFBA

p-adj: 0.843 | .=-0.32 (Small) | .%: -18.3



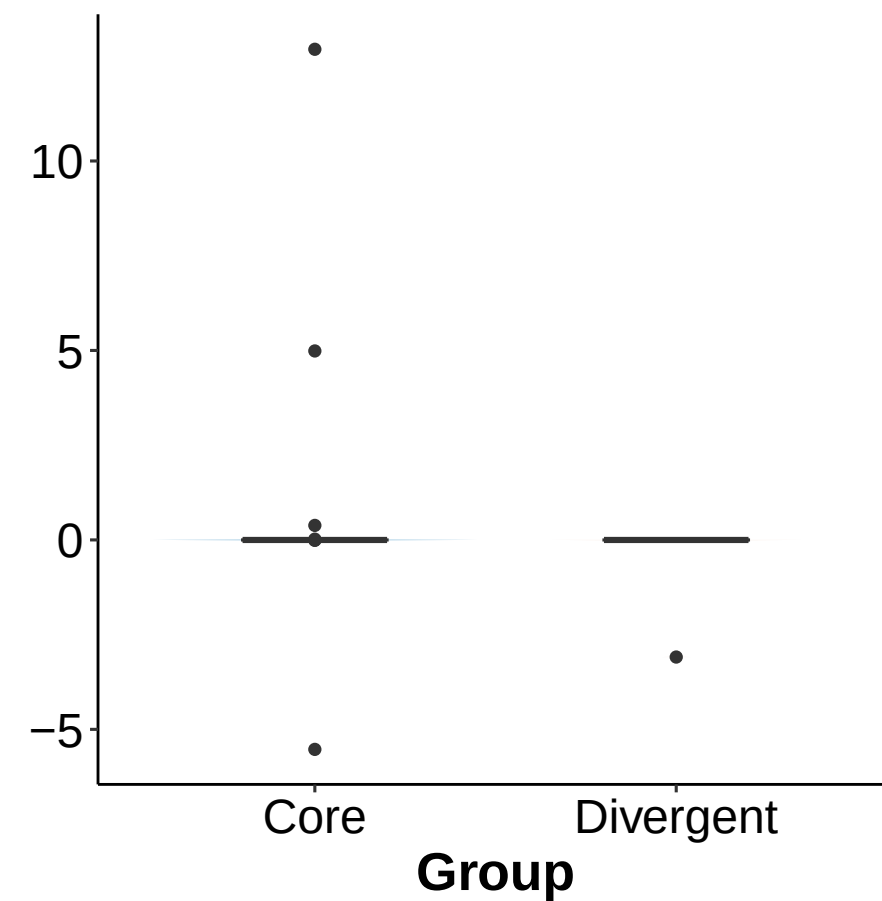
ATPProduction_pFBA

p-adj: 0.843 | .=-0.32 (Small) | .%: -18.3



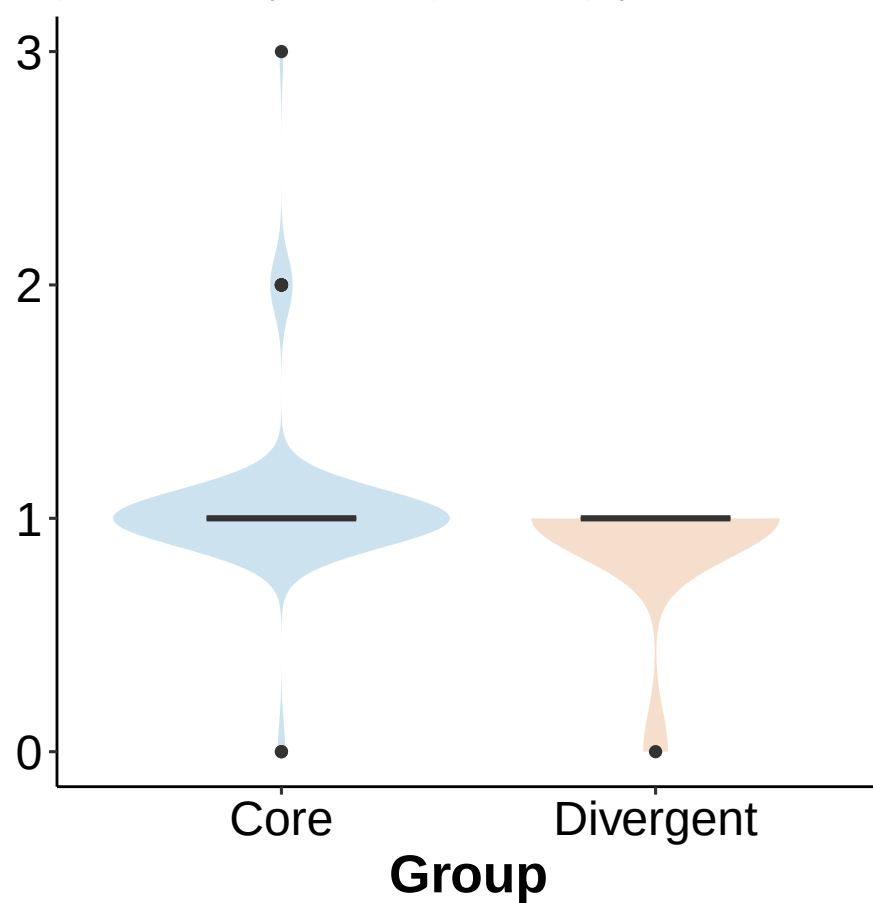
WarburgIndex_pFBA

p-adj: 0.954 | .=-0.03 (Negligible) | .%: -7.7



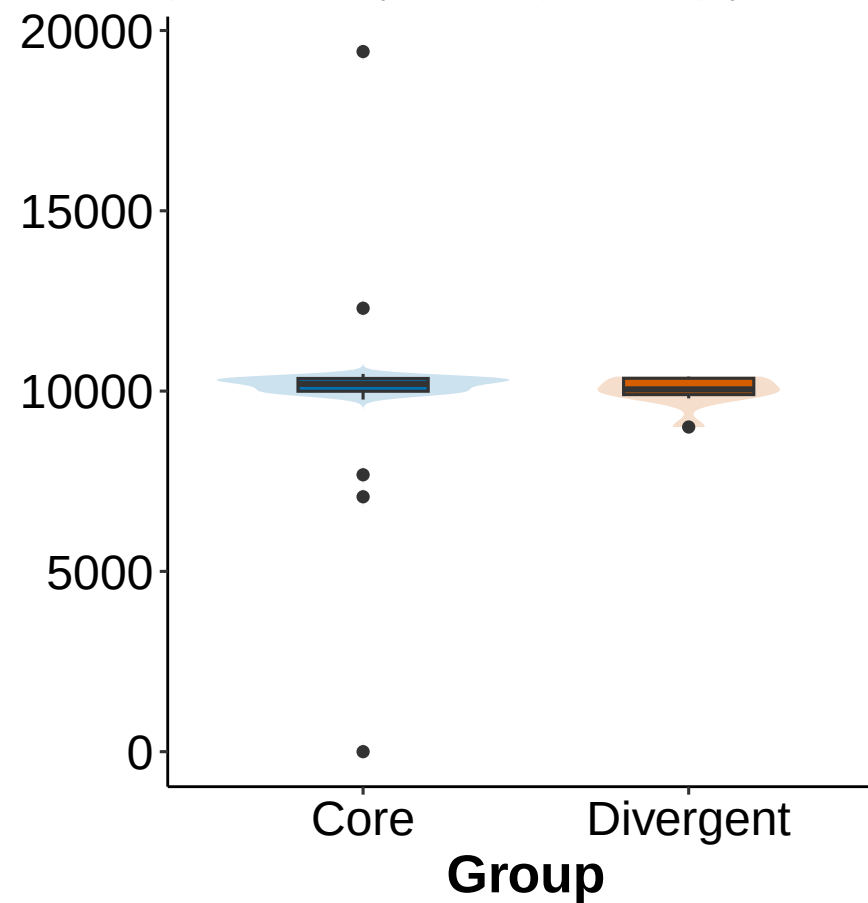
MFI_pFBA

p-adj: 0.872 | .=-0.13 (Negligible) | .%: 0



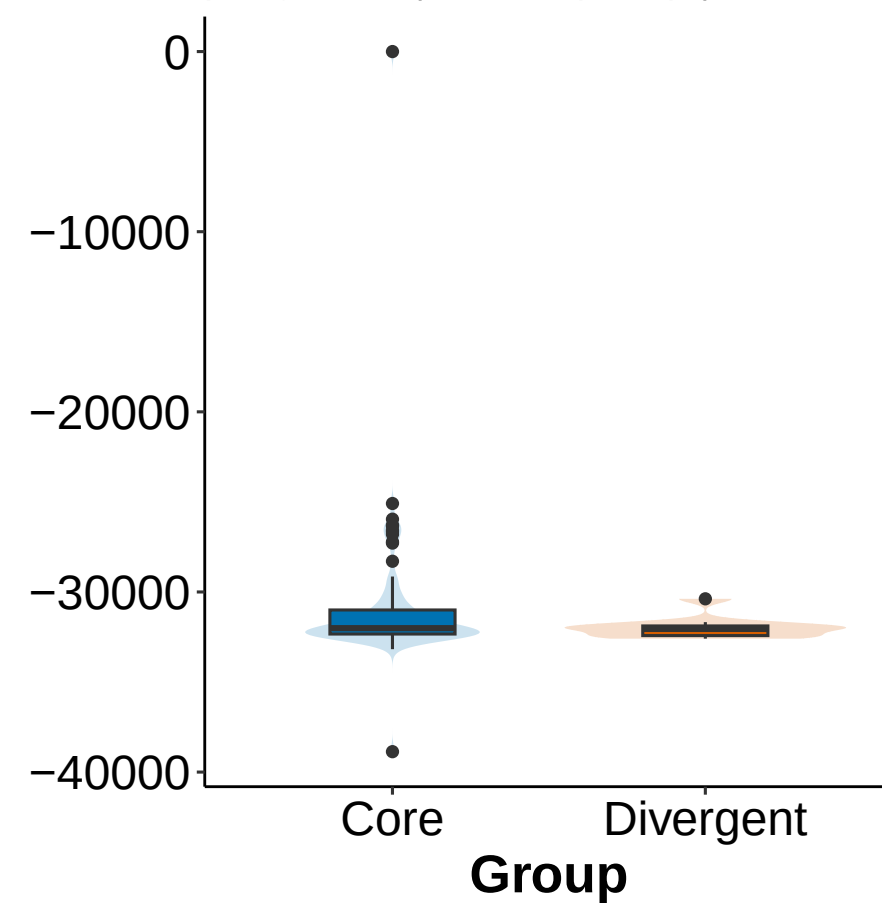
CU_pFBA

p-adj: 0.896 | .=-0.14 (Negligible) | .%: -1.5



EA_pFBA

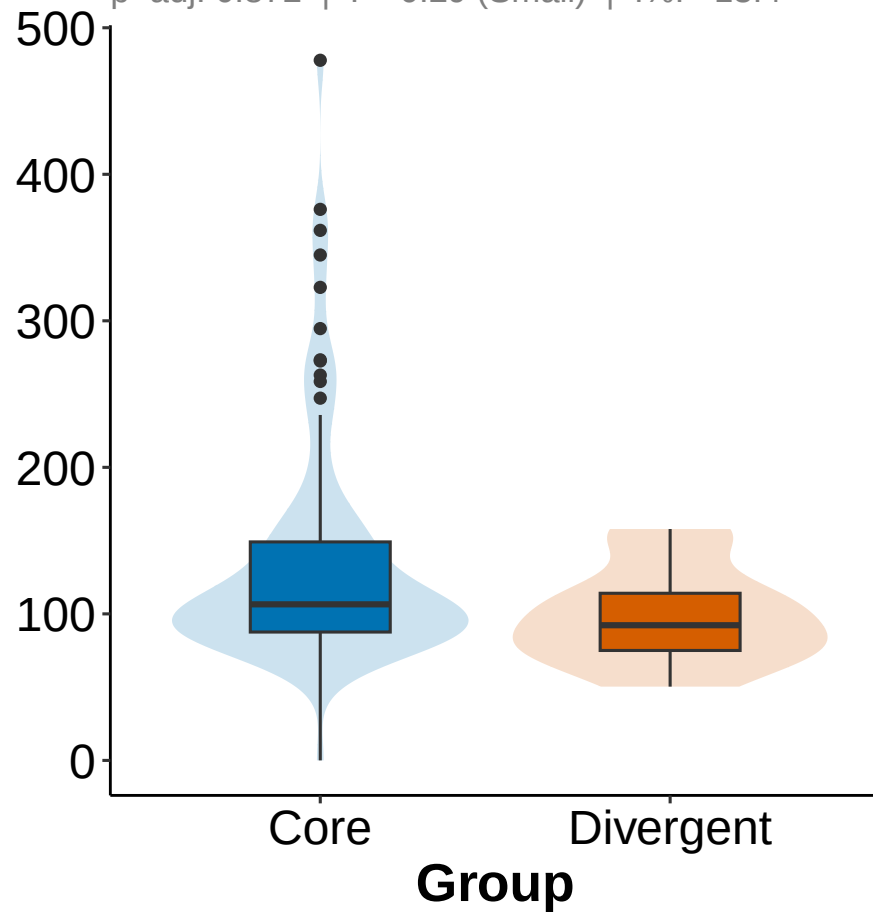
p-adj: 0.885 | .=-0.16 (Small) | .%: -0.1



Metabolism – . 1. Energy and Bioenergetics

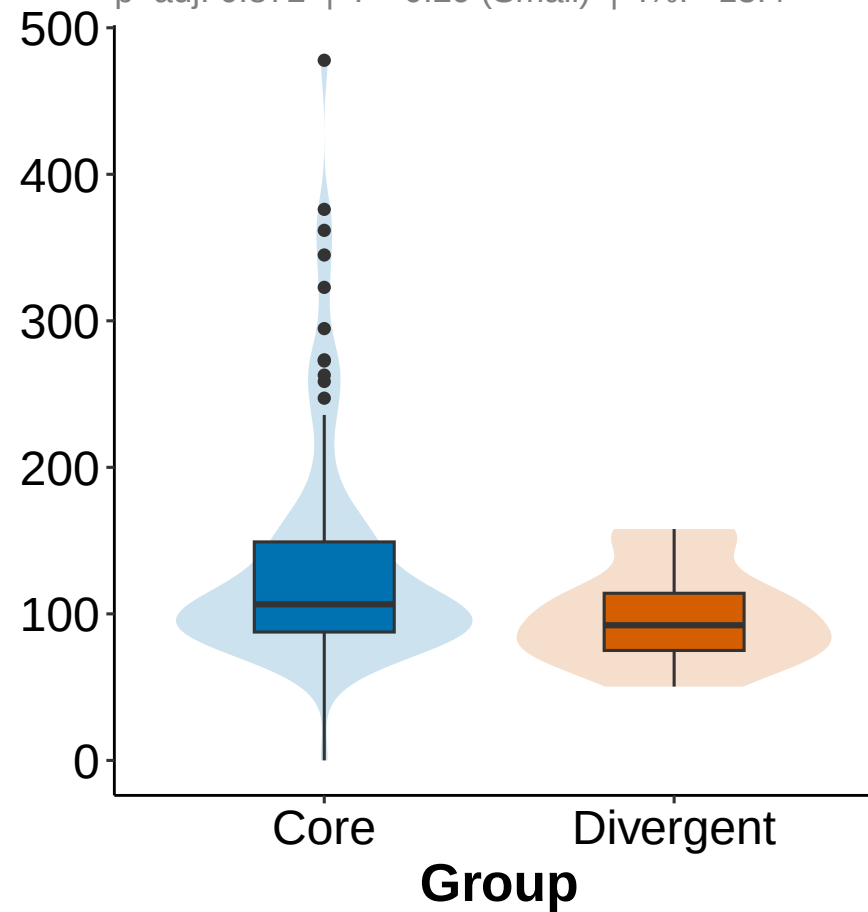
ATPConsumption_L1w

p-adj: 0.872 | .=-0.29 (Small) | .%: -13.4



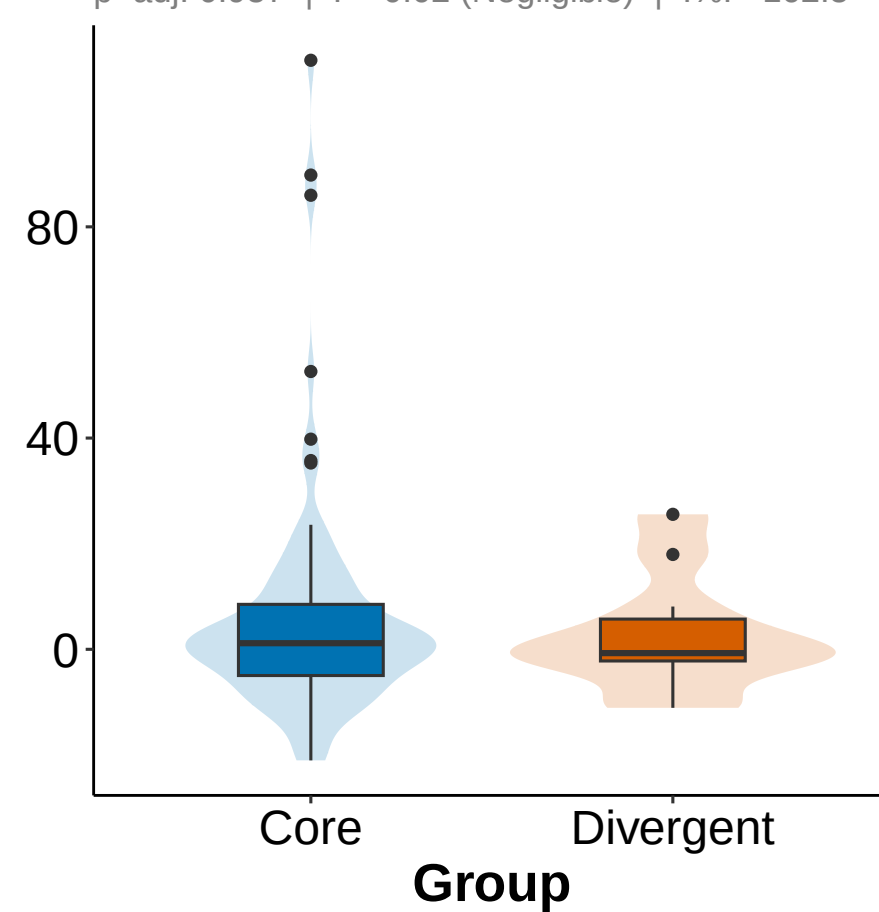
ATPProduction_L1w

p-adj: 0.872 | .=-0.29 (Small) | .%: -13.4



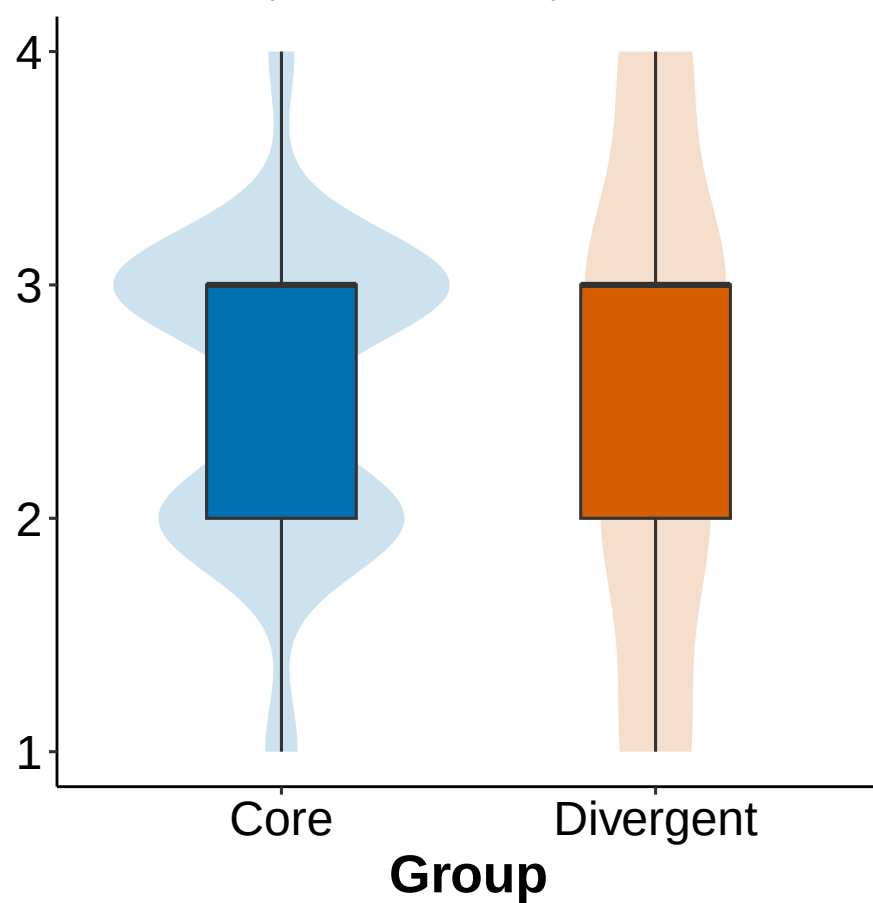
WarburgIndex_L1w

p-adj: 0.957 | .=-0.02 (Negligible) | .%: -162.5



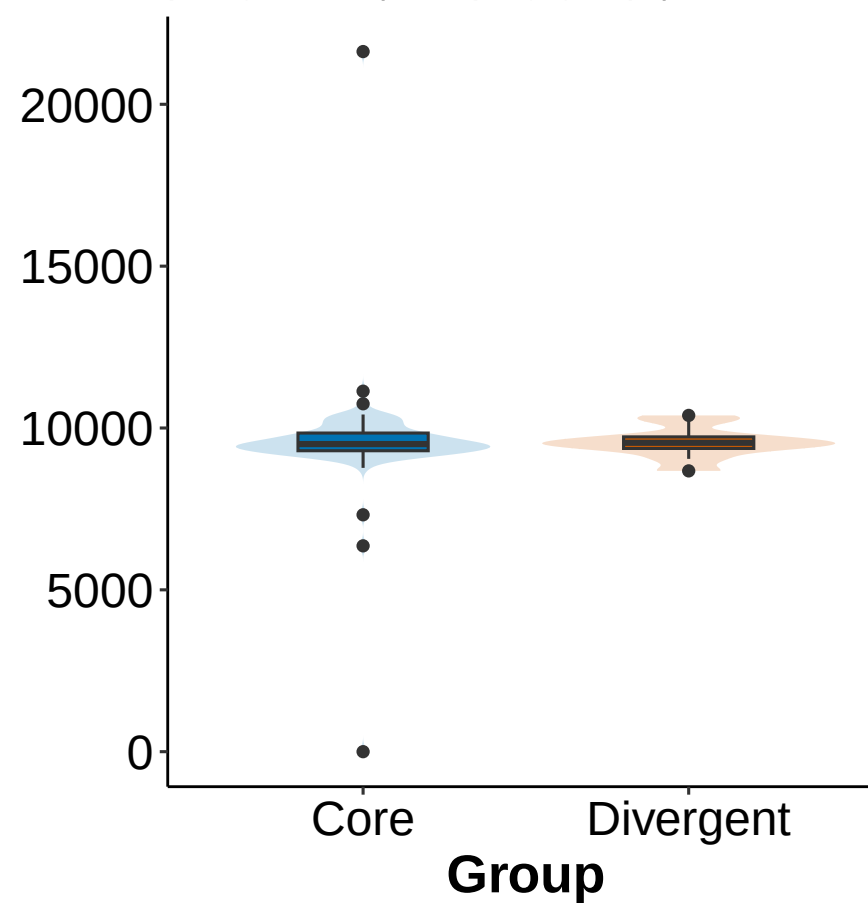
MFI_L1w

p-adj: 0.992 | .=0 (Negligible) | .%: 0



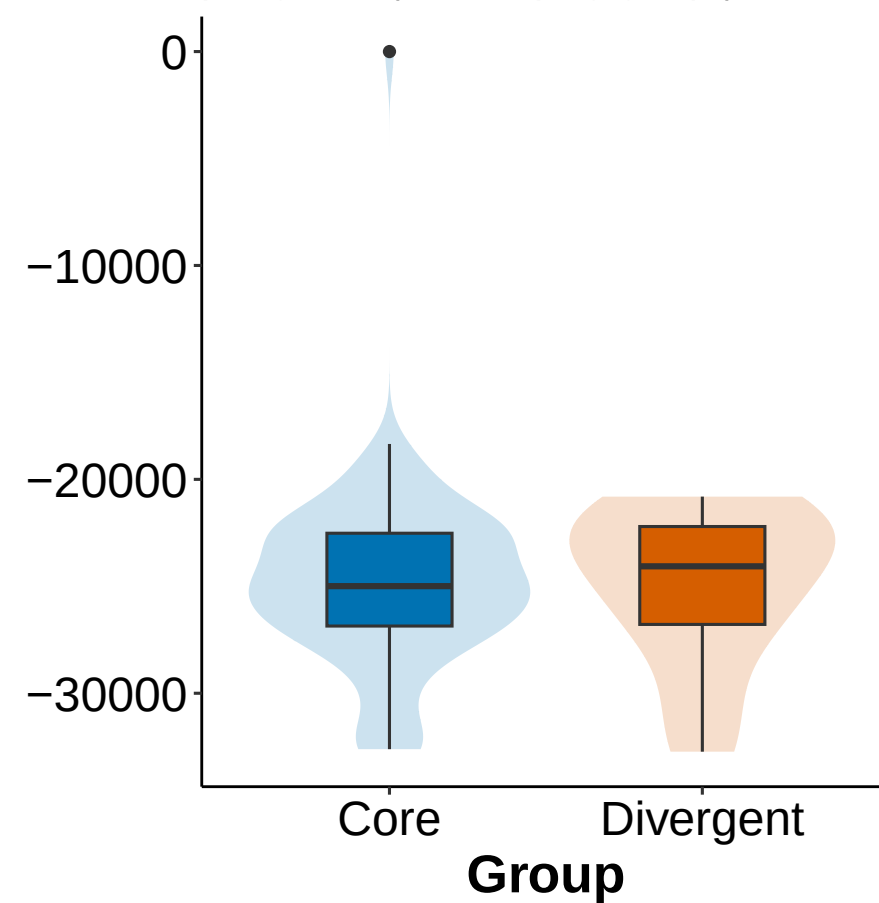
CU_L1w

p-adj: 0.992 | .=0 (Negligible) | .%: 0.4



EA_L1w

p-adj: 0.95 | .=0.06 (Negligible) | .%: 3.7



Cliff's .: . 2. Redox State and Anabolism

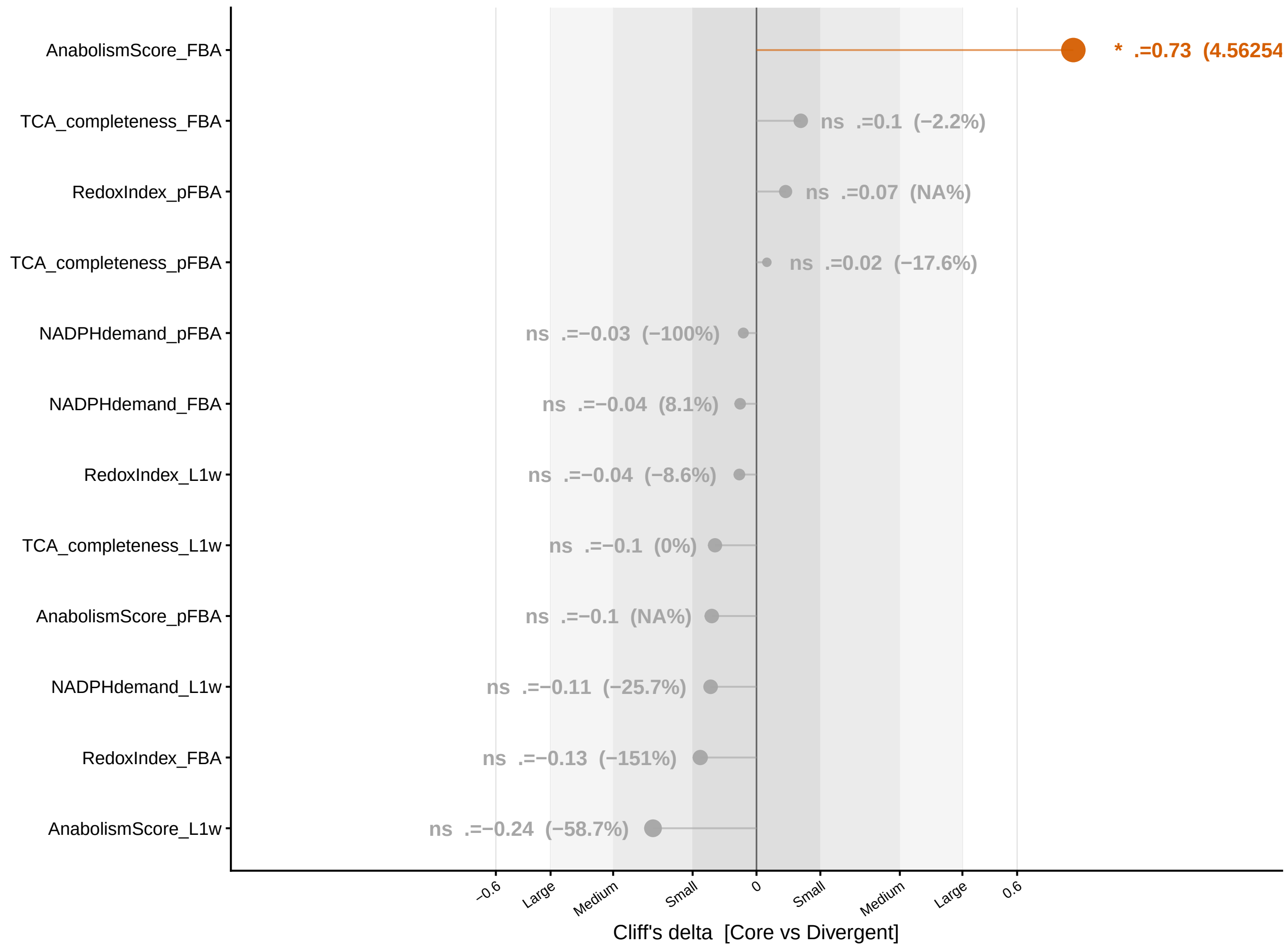
Blue = Core > Divergent | Orange = Divergent > Core | * p_adj<0.05



AnabolismScore_FBA	* .=0.73
TCA_completeness_FBA	ns .=0.1
RedoxIndex_pFBA	ns .=0.07
TCA_completeness_pFBA	ns .=0.02
NADPHdemand_pFBA	ns .=-0.03
NADPHdemand_FBA	ns .=-0.04
RedoxIndex_L1w	ns .=-0.04
TCA_completeness_L1w	ns .=-0.1
AnabolismScore_pFBA	ns .=-0.1
NADPHdemand_L1w	ns .=-0.11
RedoxIndex_FBA	ns .=-0.13
AnabolismScore_L1w	ns .=-0.24

Cliff's . – . 2. Redox State and Anabolism

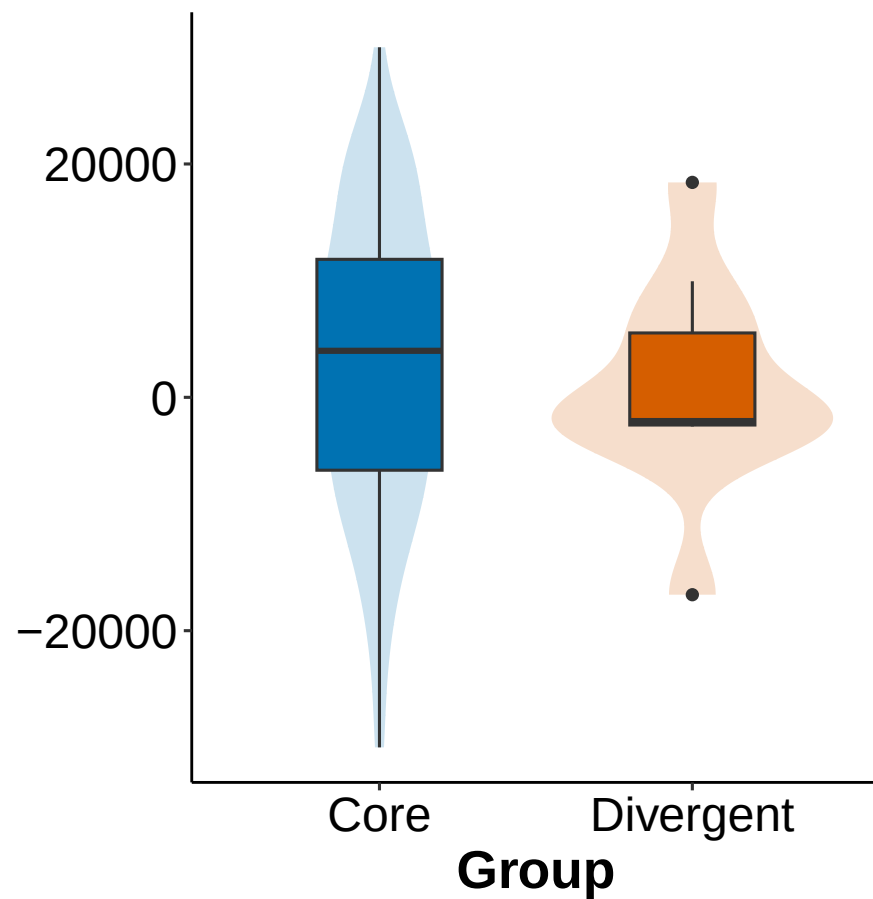
Cliff's . + .% median | . higher in Core · . higher in Divergent



Metabolism – . 2. RedoxState and Anabolism

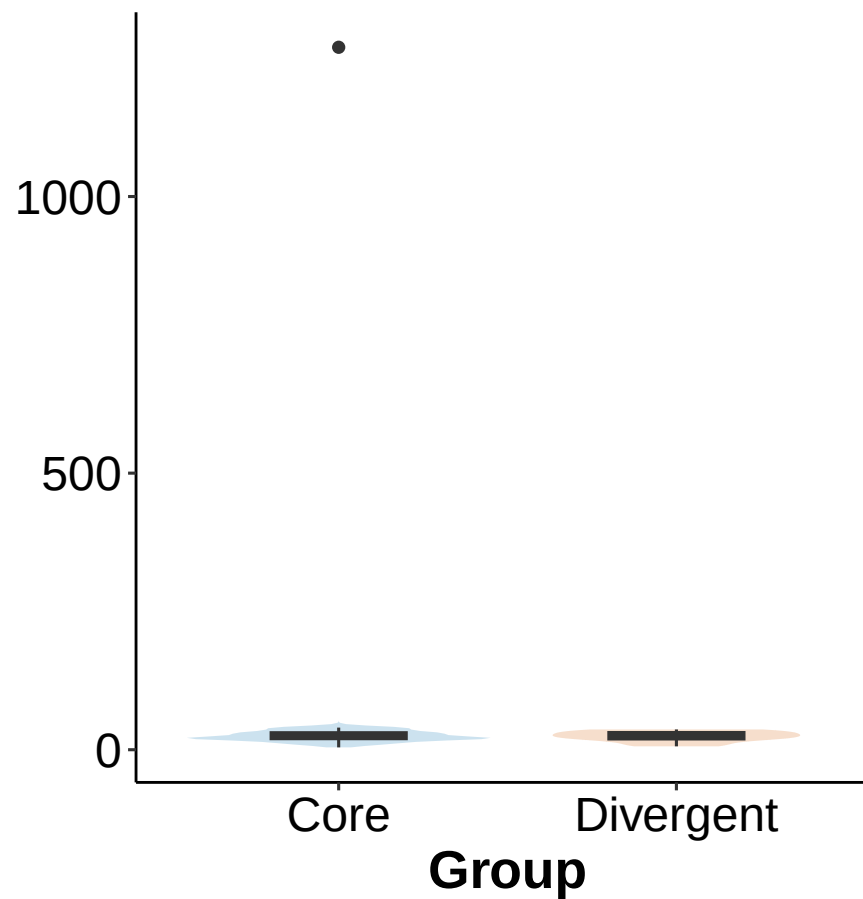
RedoxIndex_FBA

p-adj: 0.917 | .=-0.13 (Negligible) | .%: -15



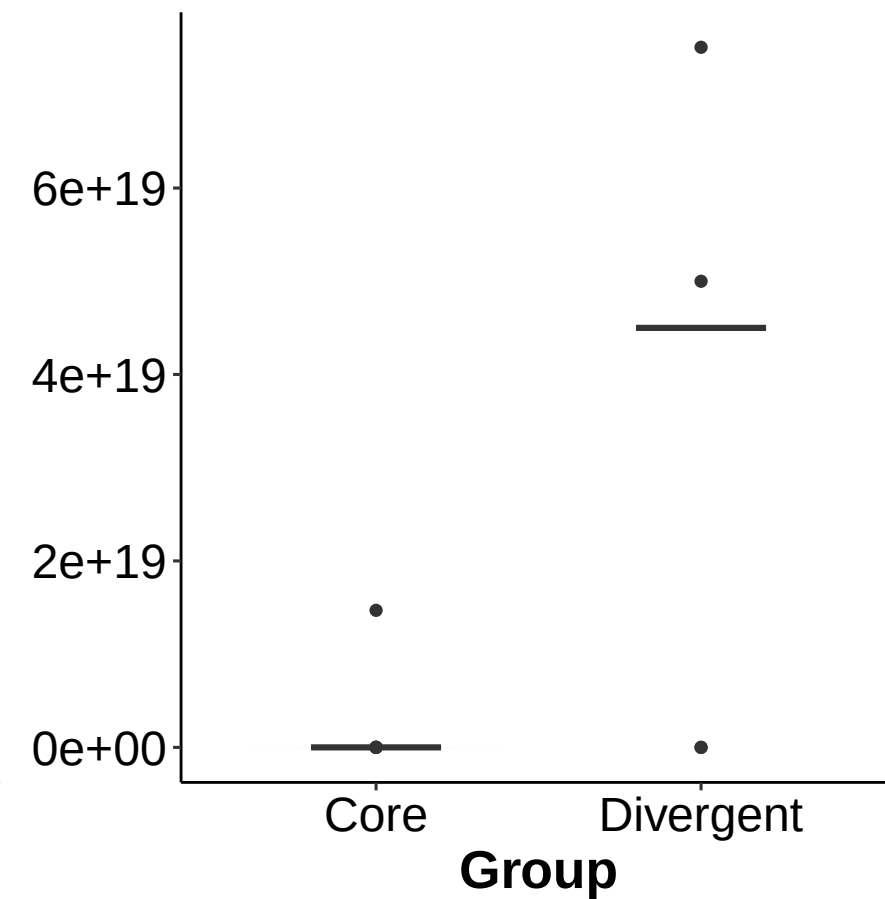
NADPHdemand_FBA

p-adj: 0.954 | .=-0.04 (Negligible) | .%: 8.1



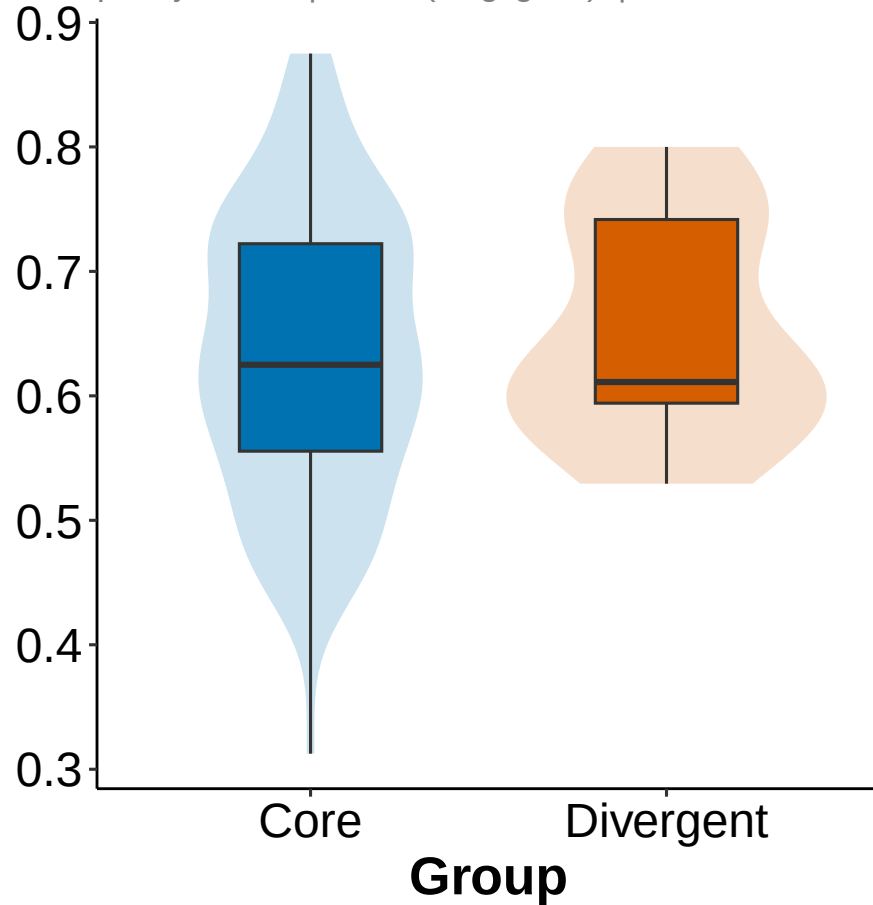
AnabolismScore_FBA

p-adj: 0.0481 | .=0.73 (Large) | .%: 4.562543



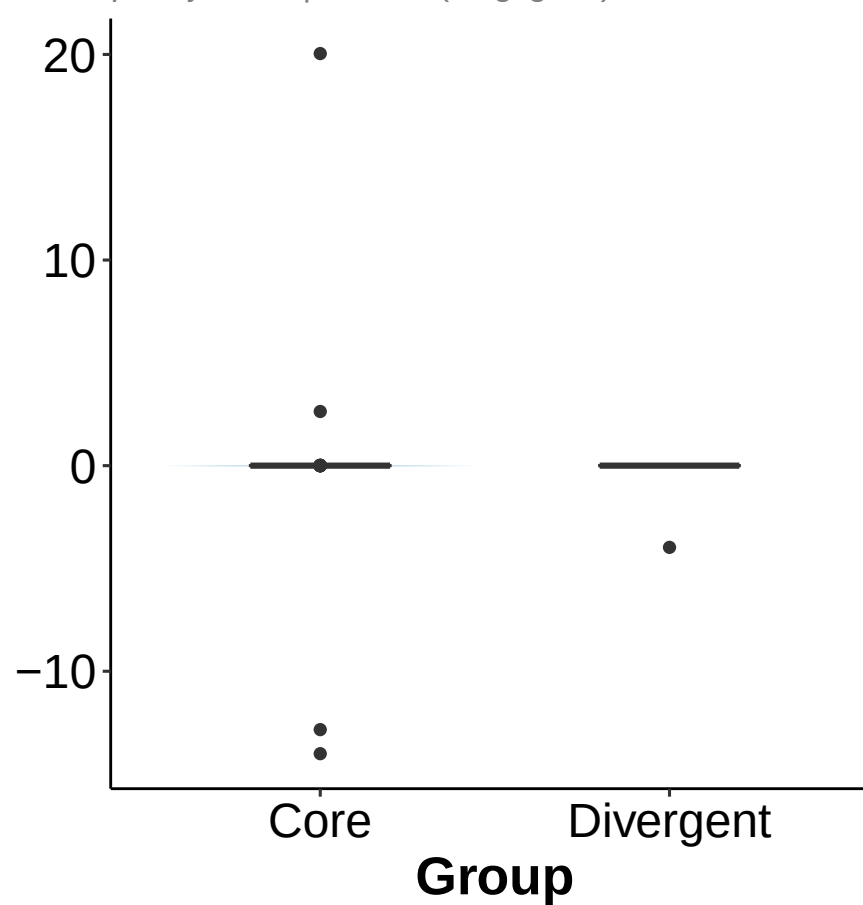
TCA_completeness_FBA

p-adj: 0.928 | .=0.1 (Negligible) | .%: -2.2



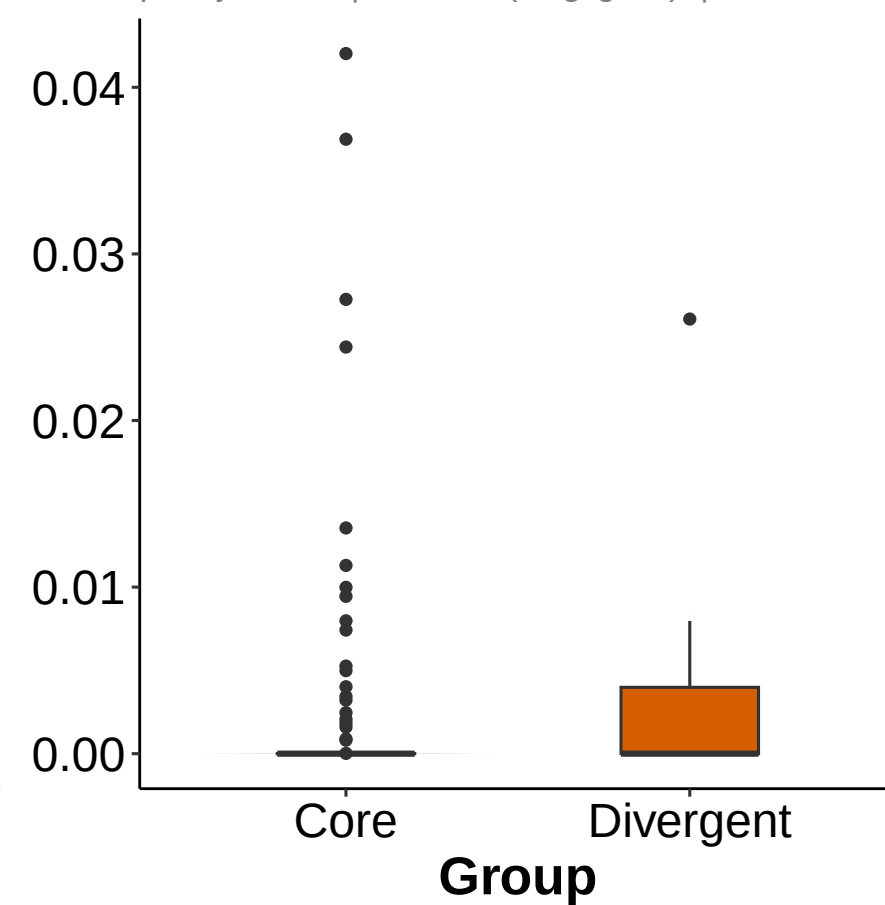
RedoxIndex_pFBA

p-adj: 0.95 | .=0.07 (Negligible)



NADPHdemand_pFBA

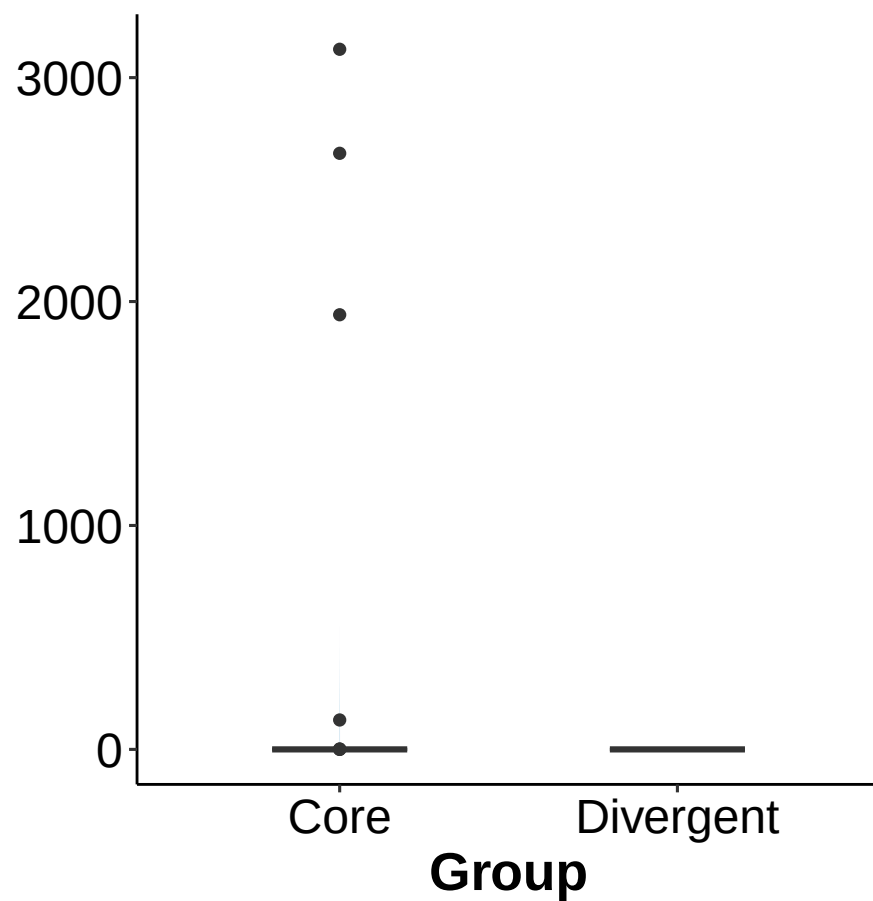
p-adj: 0.954 | .=-0.03 (Negligible) | .%: -100



Metabolism – . 2. RedoxState and Anabolism

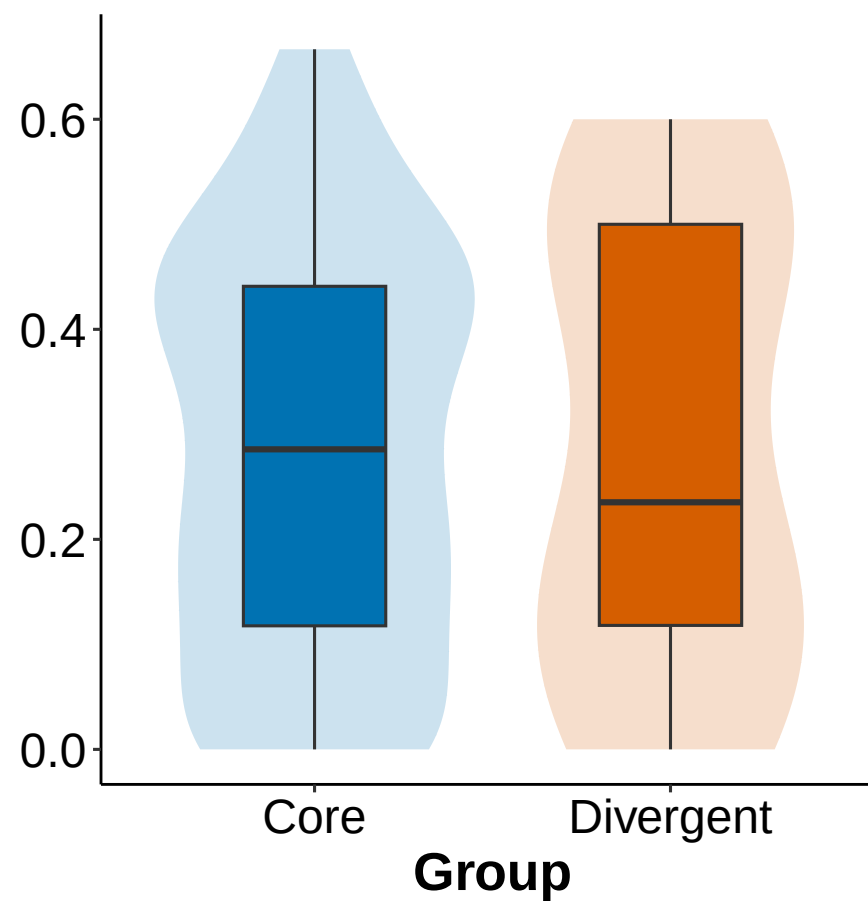
AnabolismScore_pFBA

p-adj: 0.885 | .=-0.1 (Negligible)



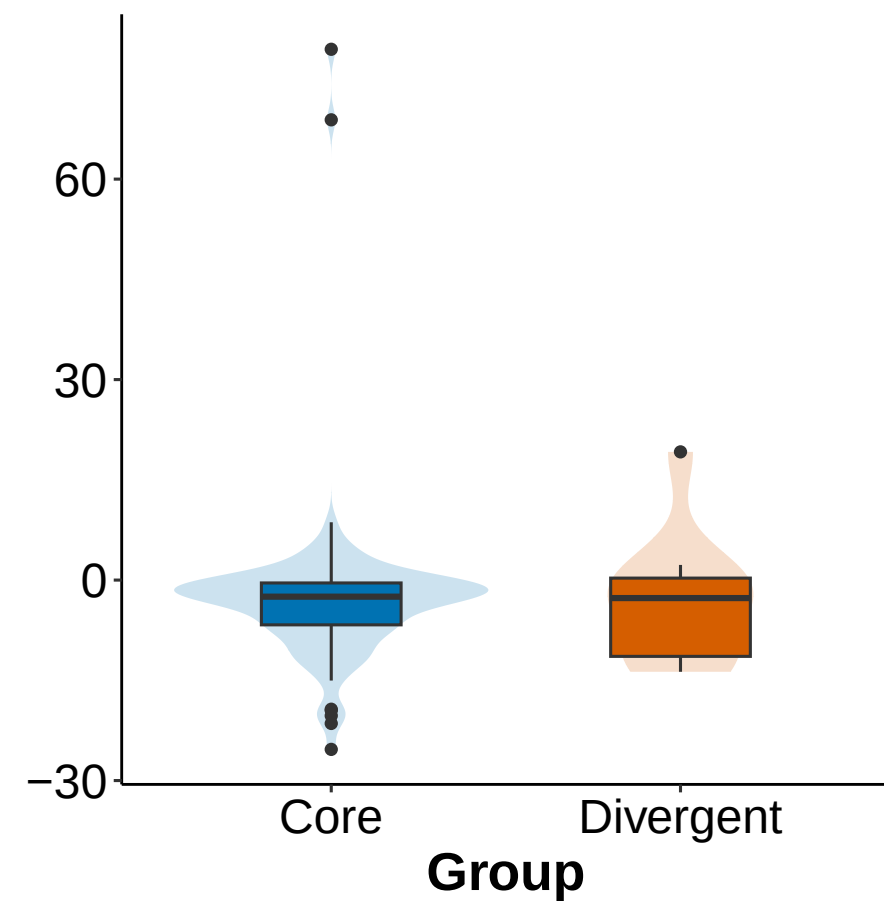
TCA_completeness_pFBA

p-adj: 0.957 | .:=0.02 (Negligible) | .%: -17.6



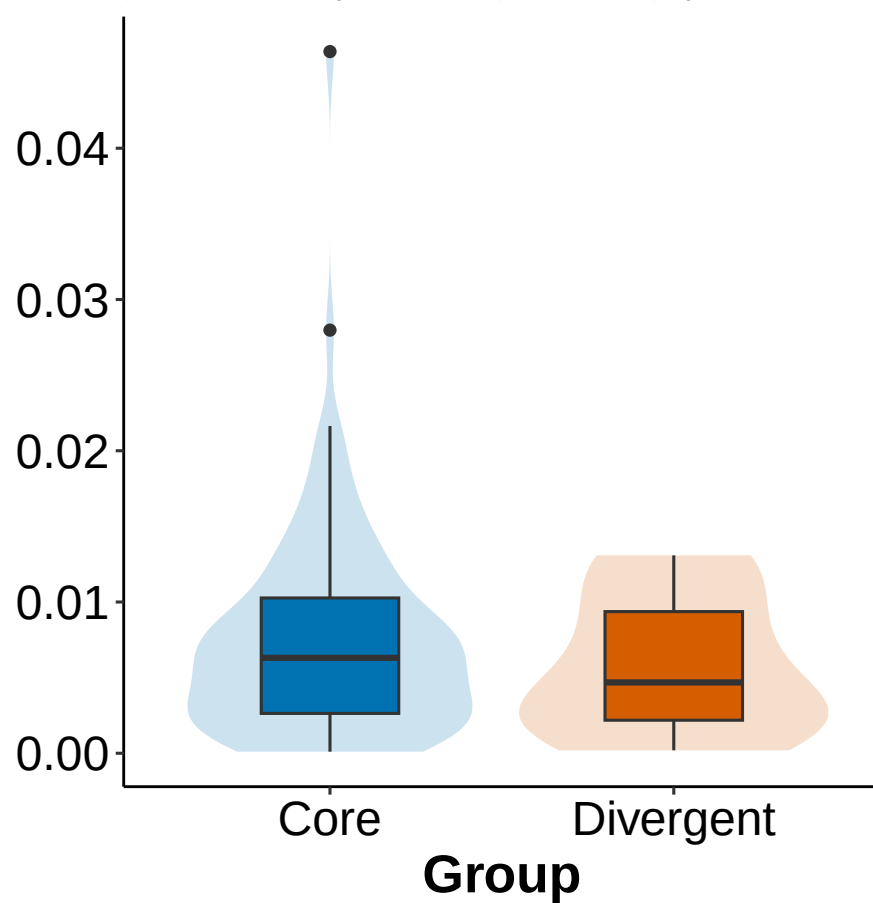
RedoxIndex_L1w

p-adj: 0.954 | .=-0.04 (Negligible) | .%: -8.6



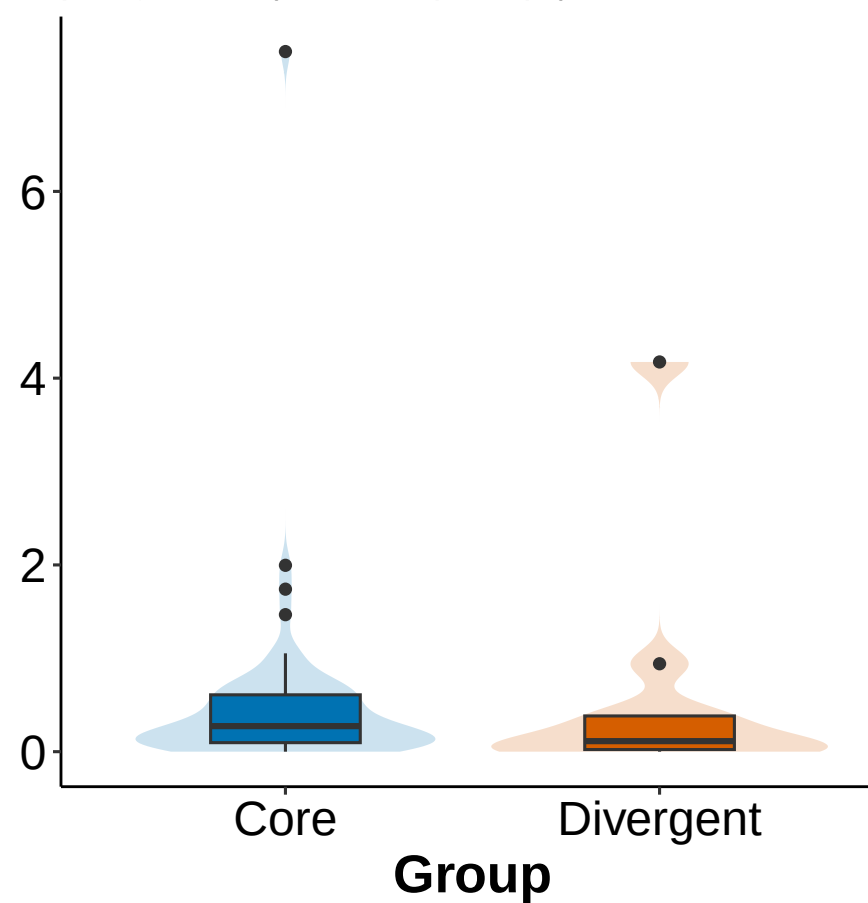
NADPHdemand_L1w

p-adj: 0.927 | .=-0.11 (Negligible) | .%: -25.7



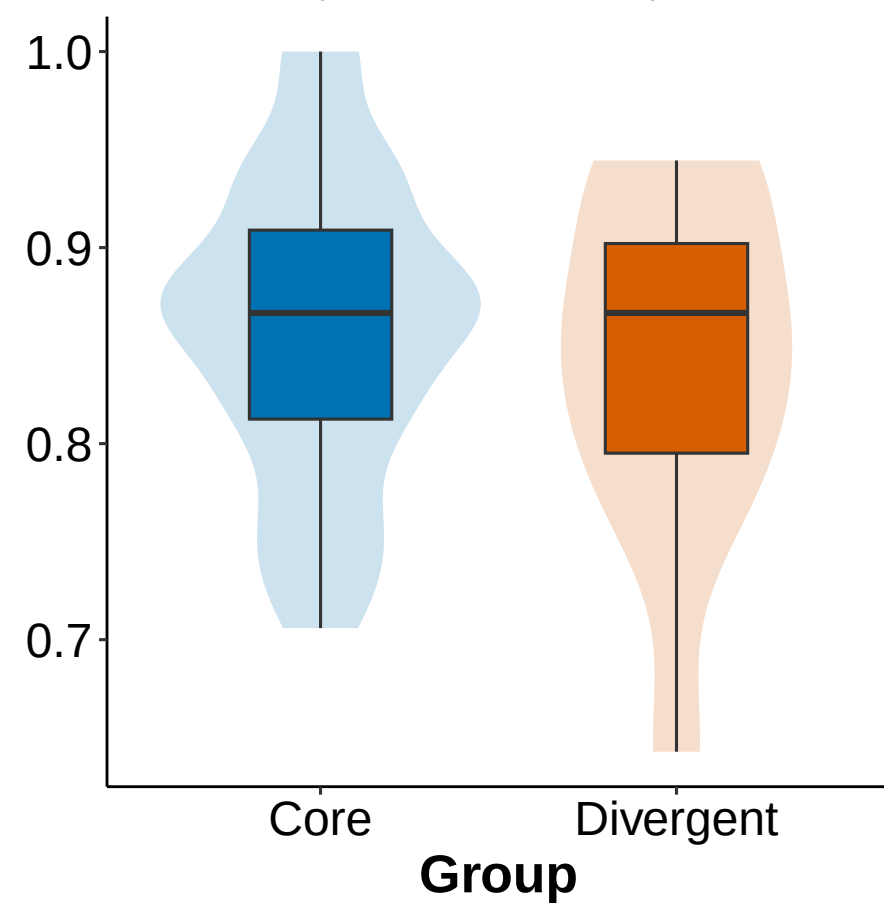
AnabolismScore_L1w

p-adj: 0.885 | .=-0.24 (Small) | .%: -58.7



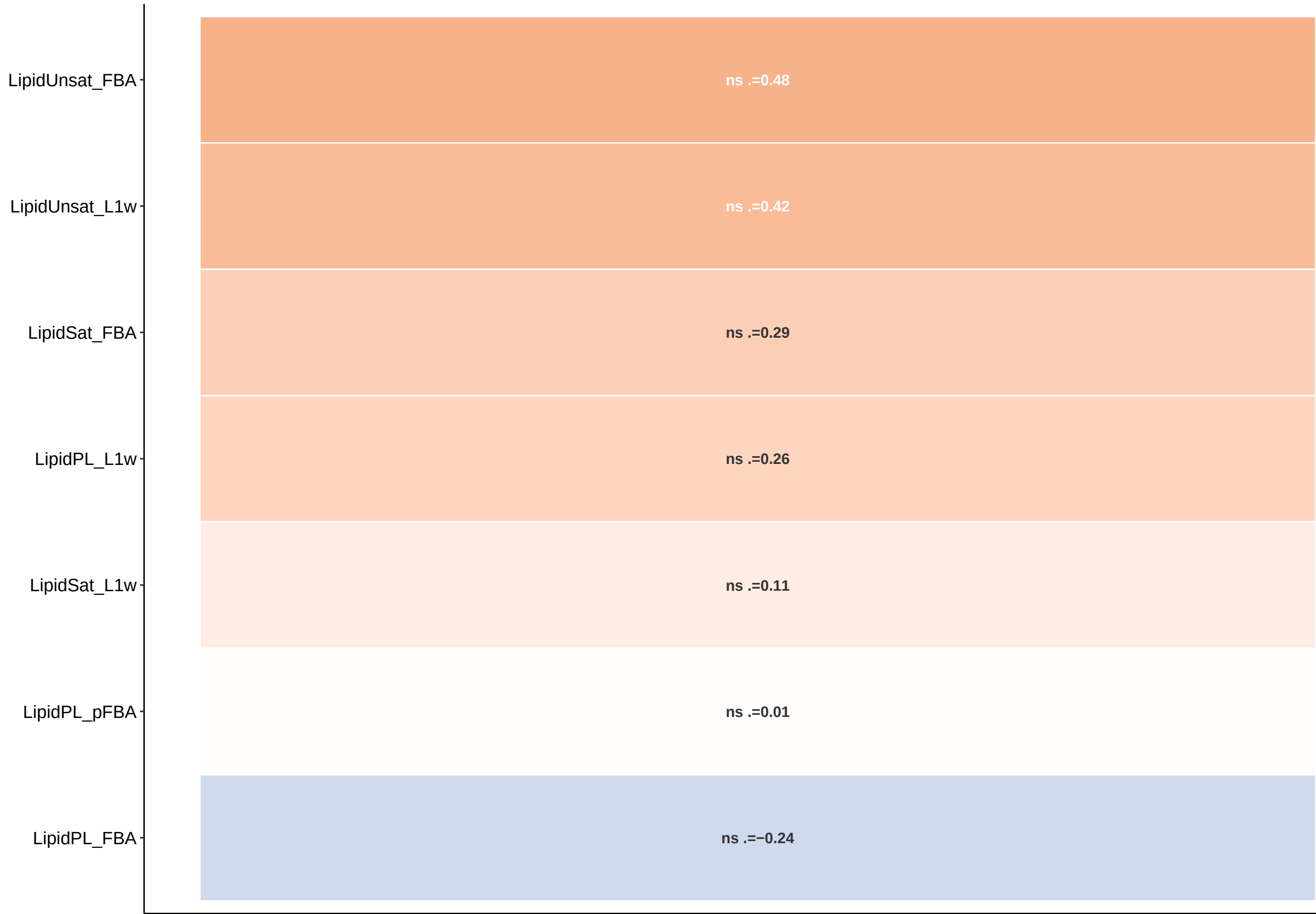
TCA_completeness_L1w

p-adj: 0.942 | .=-0.1 (Negligible) | .%: 0



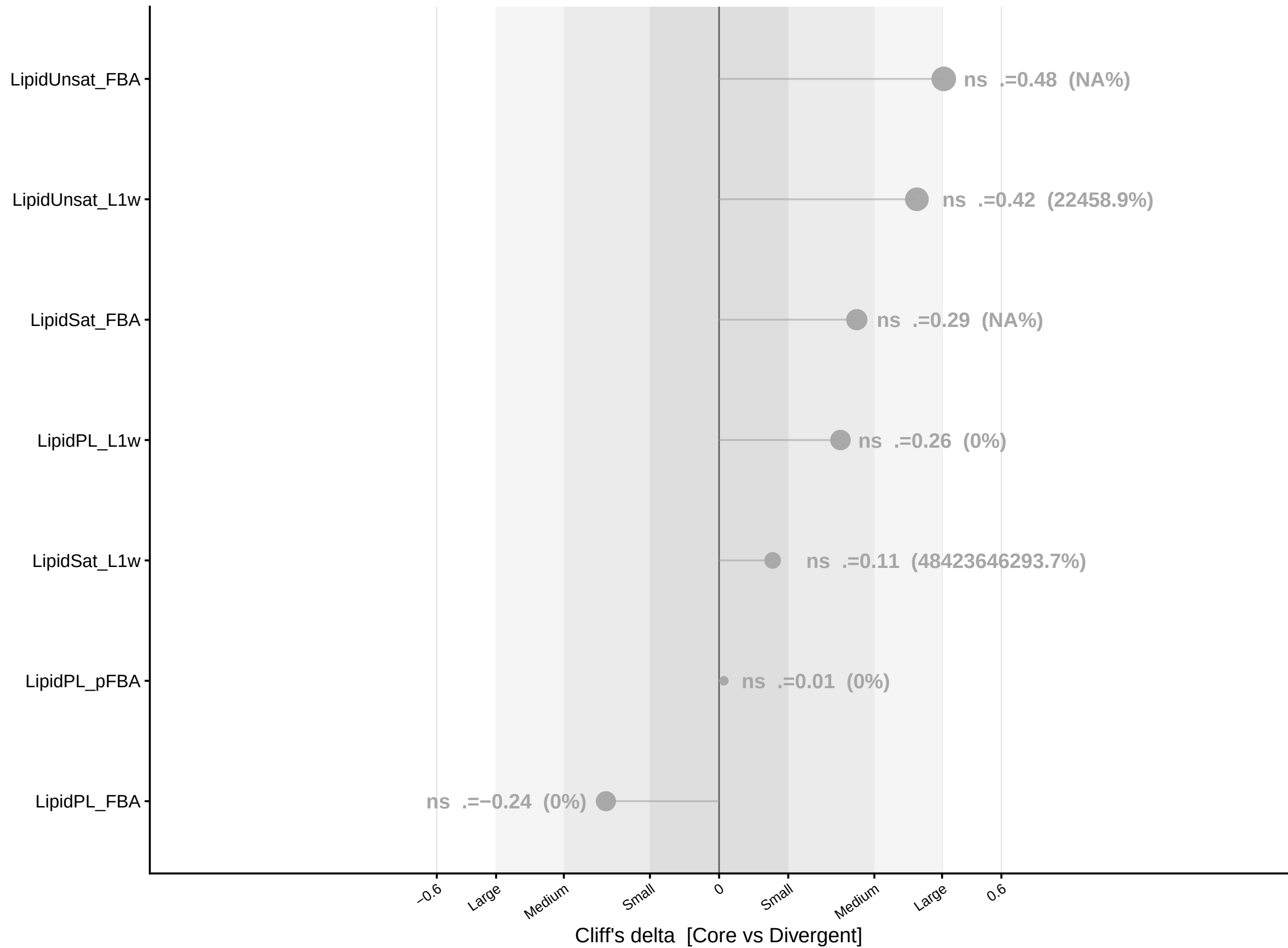
Cliff's .: . 3. Lipid Metabolism

Blue = Core > Divergent | Orange = Divergent > Core | * p_adj<0.05

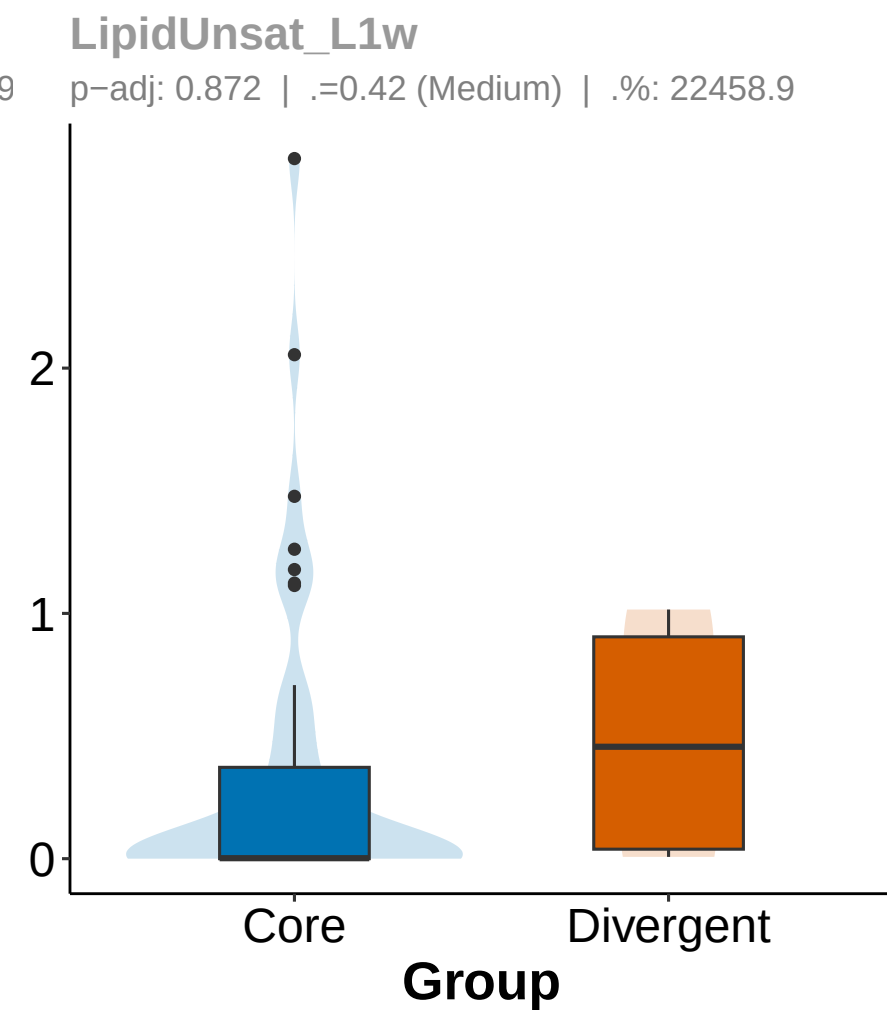
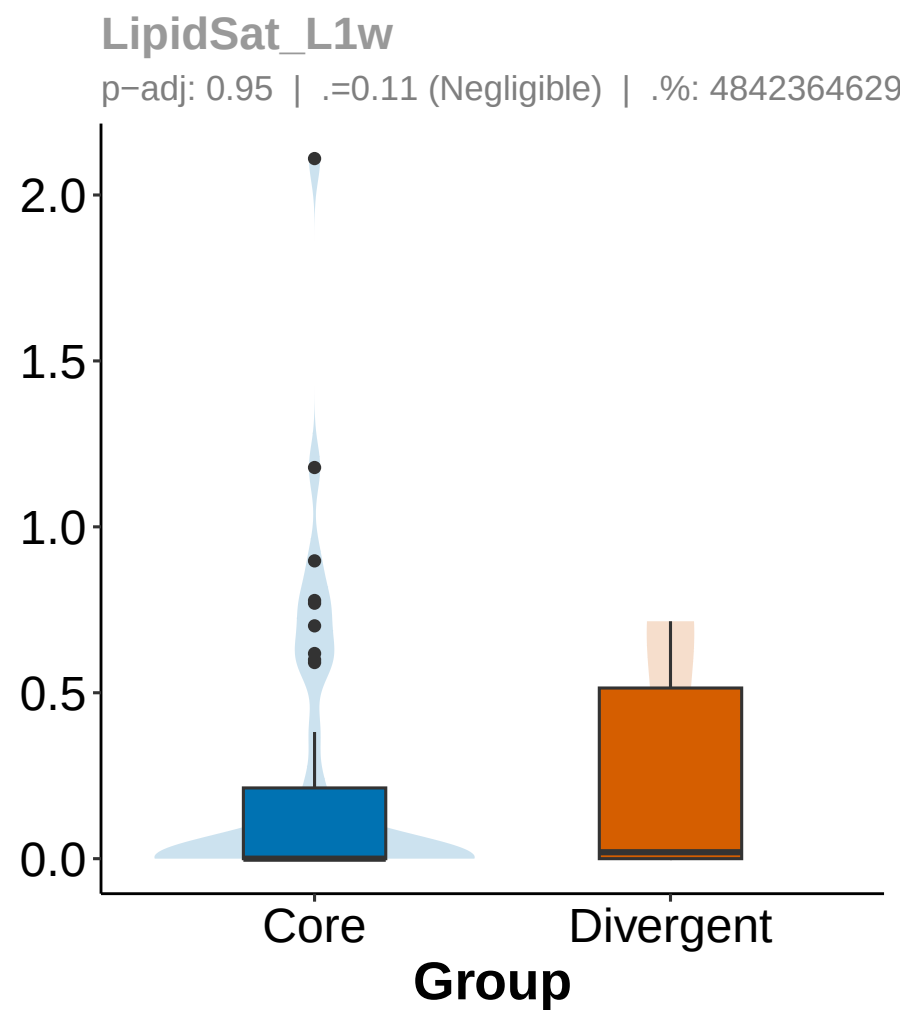
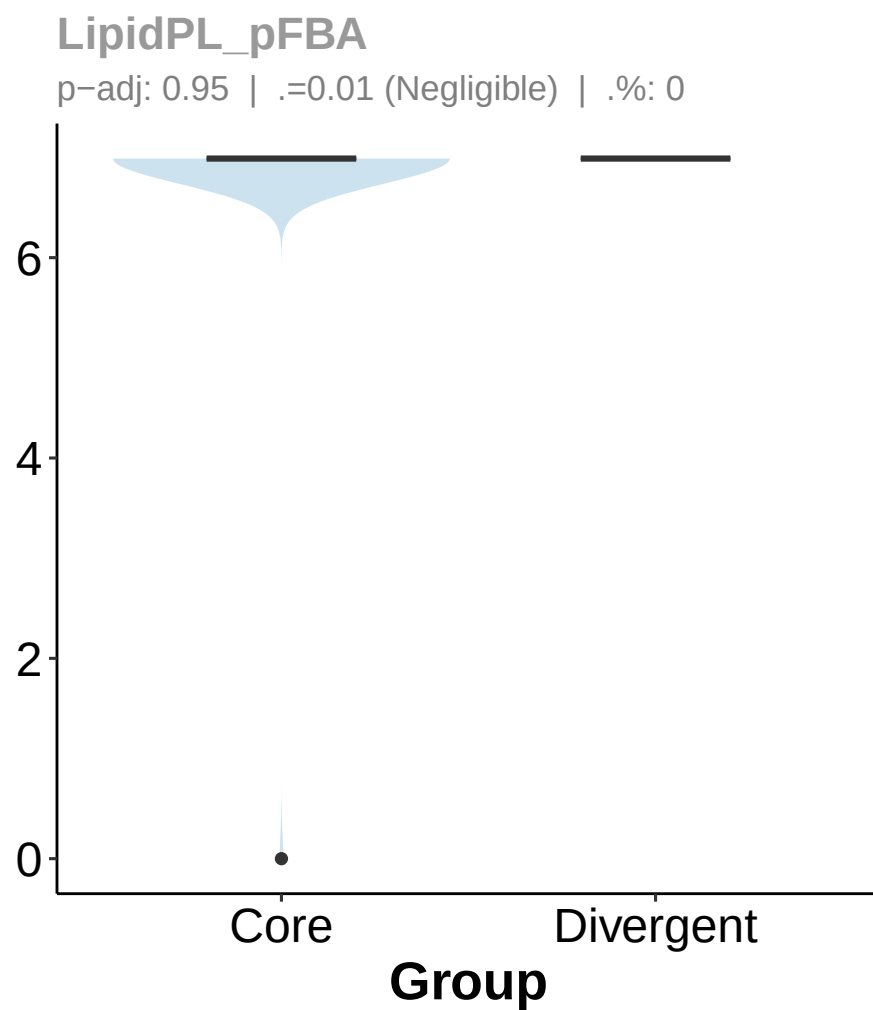
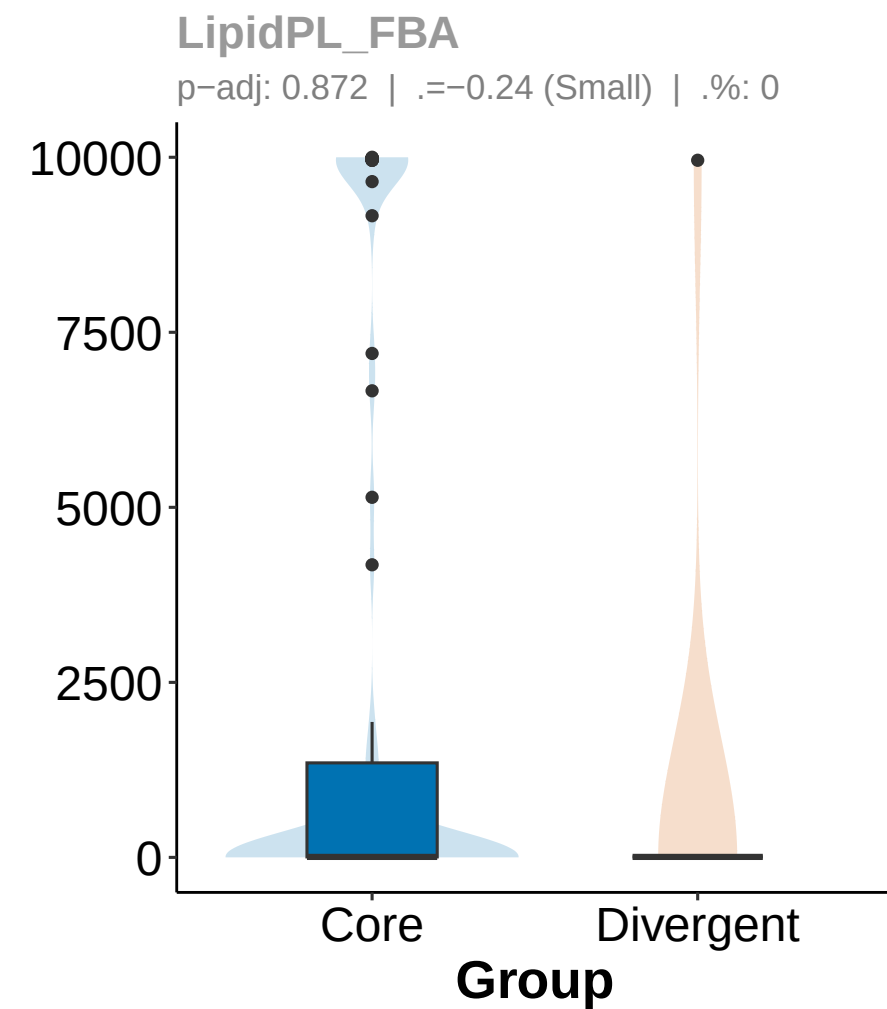
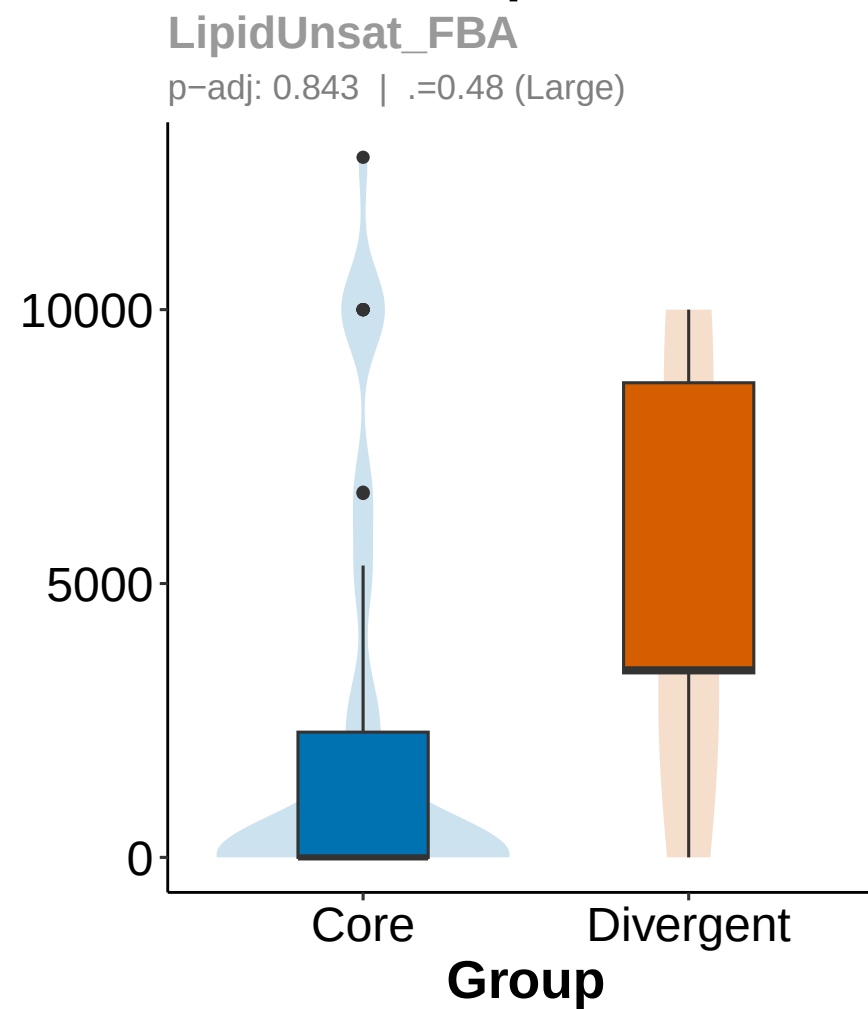
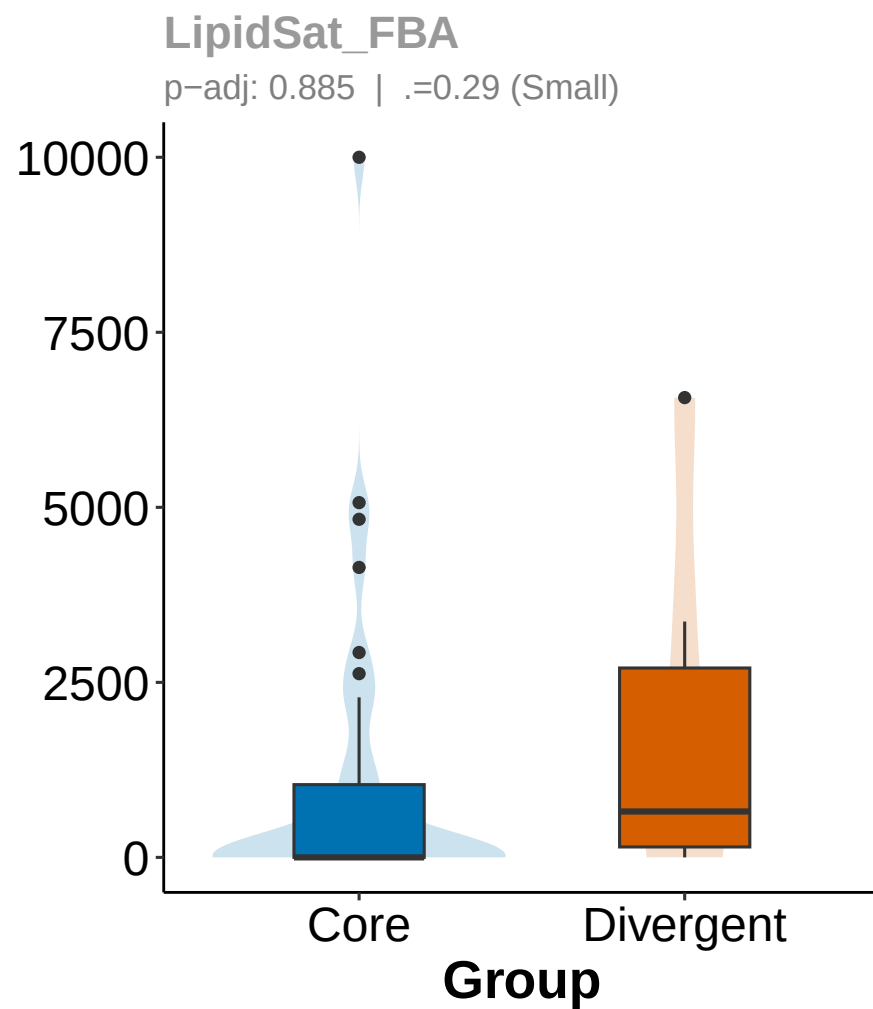


Cliff's . – . 3. Lipid Metabolism

Cliff's . + .% median | . higher in Core · . higher in Divergent



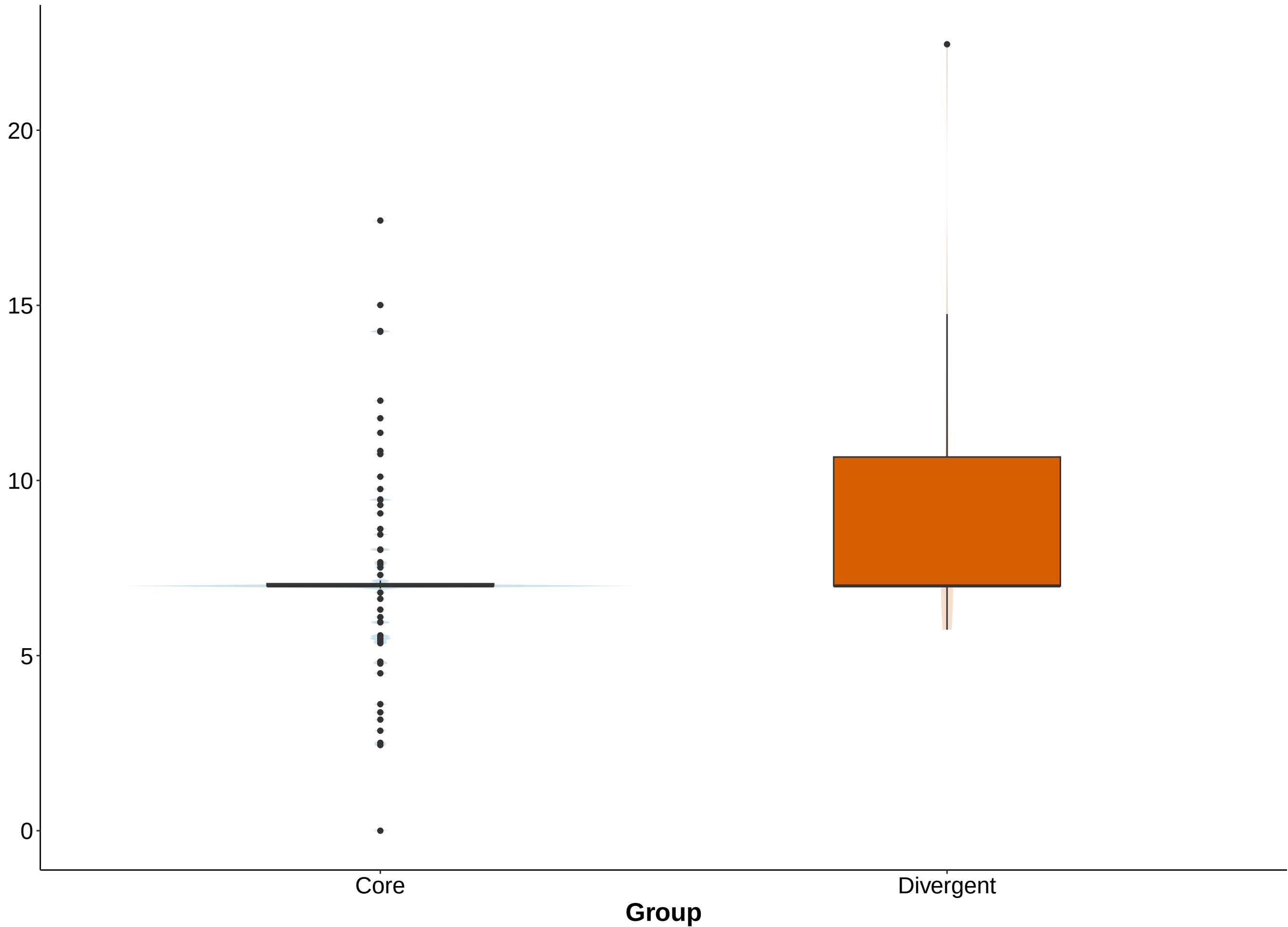
Metabolism – . 3. Lipid Metabolism



Metabolism - . 3. Lipid Metabolism

LipidPL_L1w

p-adj: 0.872 | . = 0.26 (Small) | .%: 0



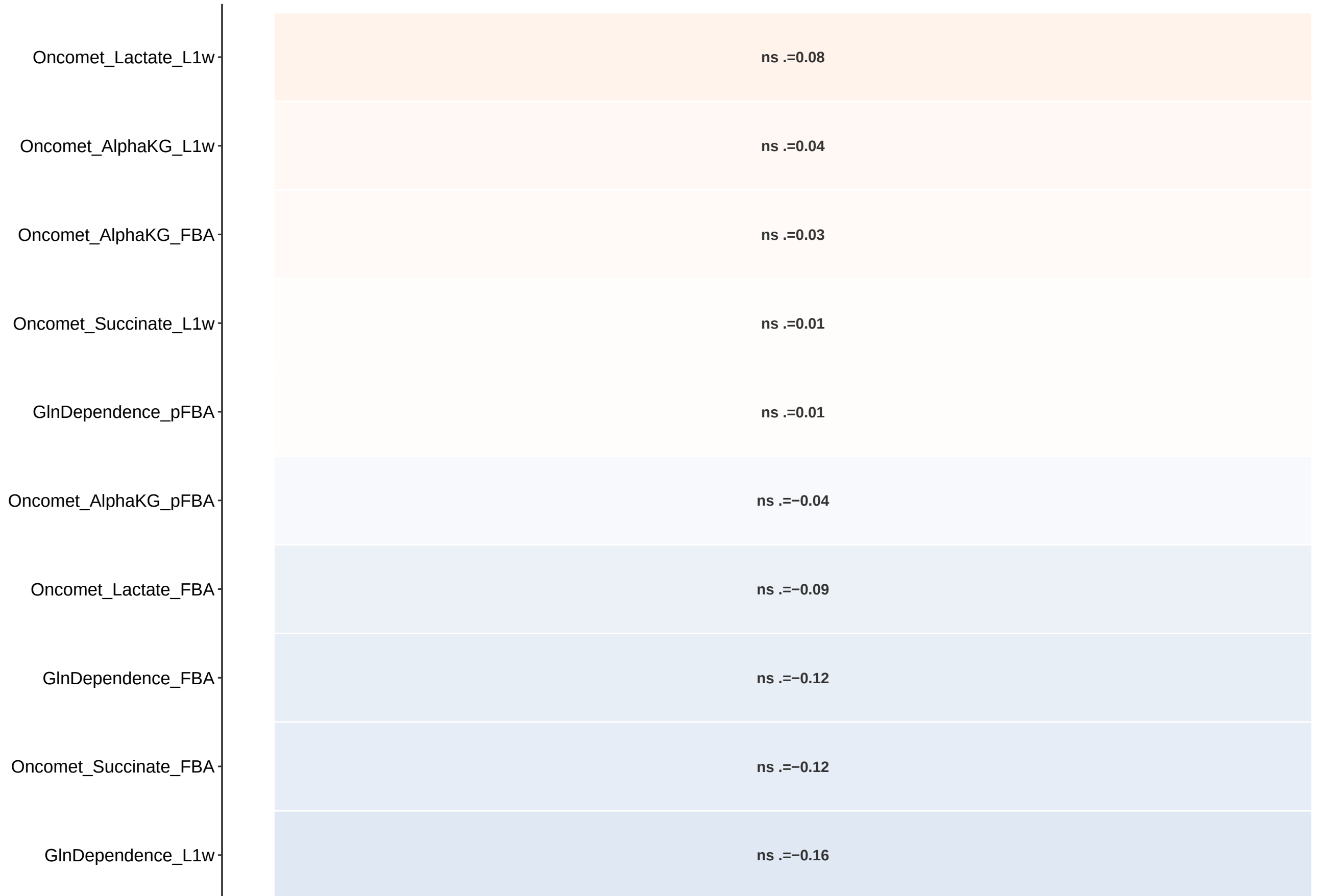
Cliff's .: . 4. Specific Metabolic Dependencies

Blue = Core > Divergent | Orange = Divergent > Core | * p_adj<0.05

Cliff's .
Div vs Core

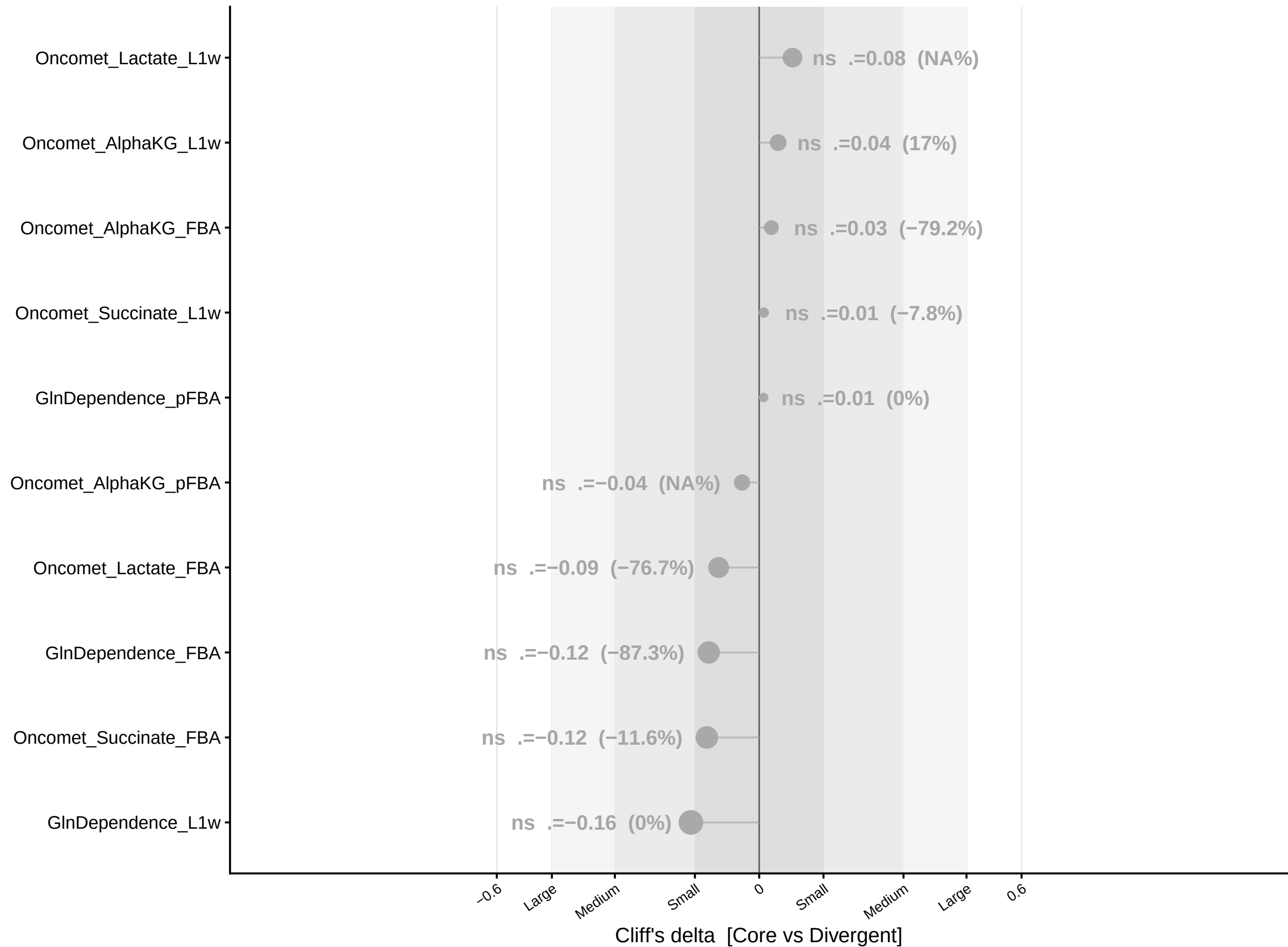


1.0 0.5 0.0 0.5 1.0



Cliff's . – . 4. Specific Metabolic Dependencies

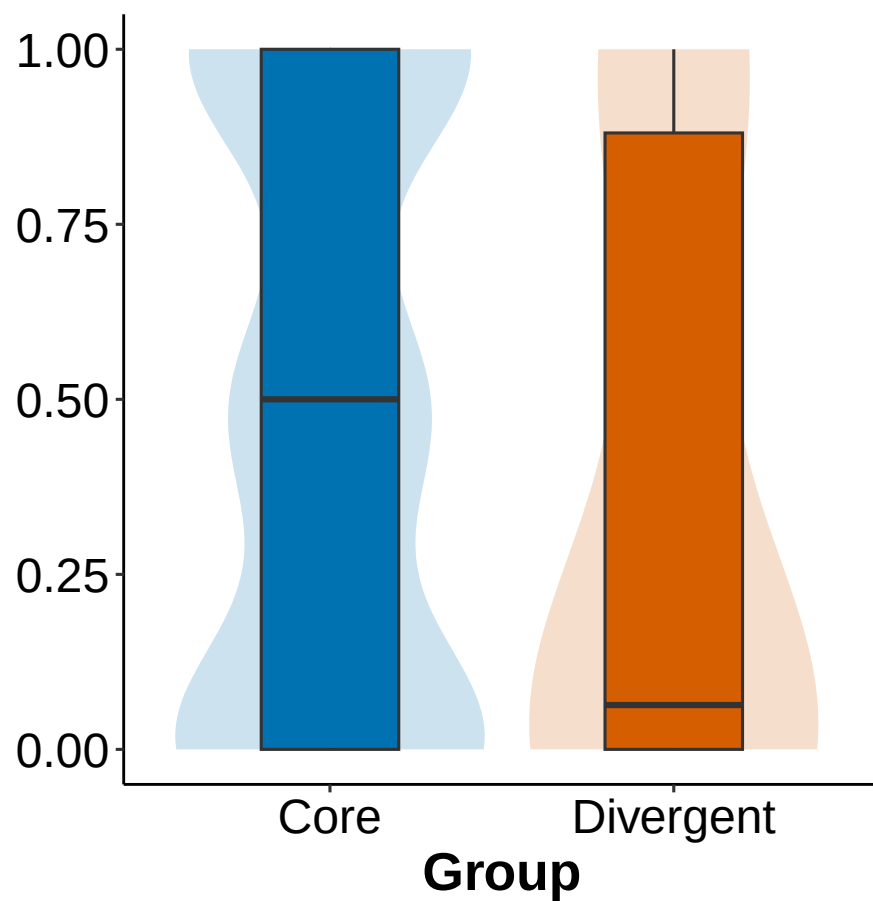
Cliff's . + .% median | . higher in Core · . higher in Divergent



Metabolism – . 4. Specific Metabolic Dependencies

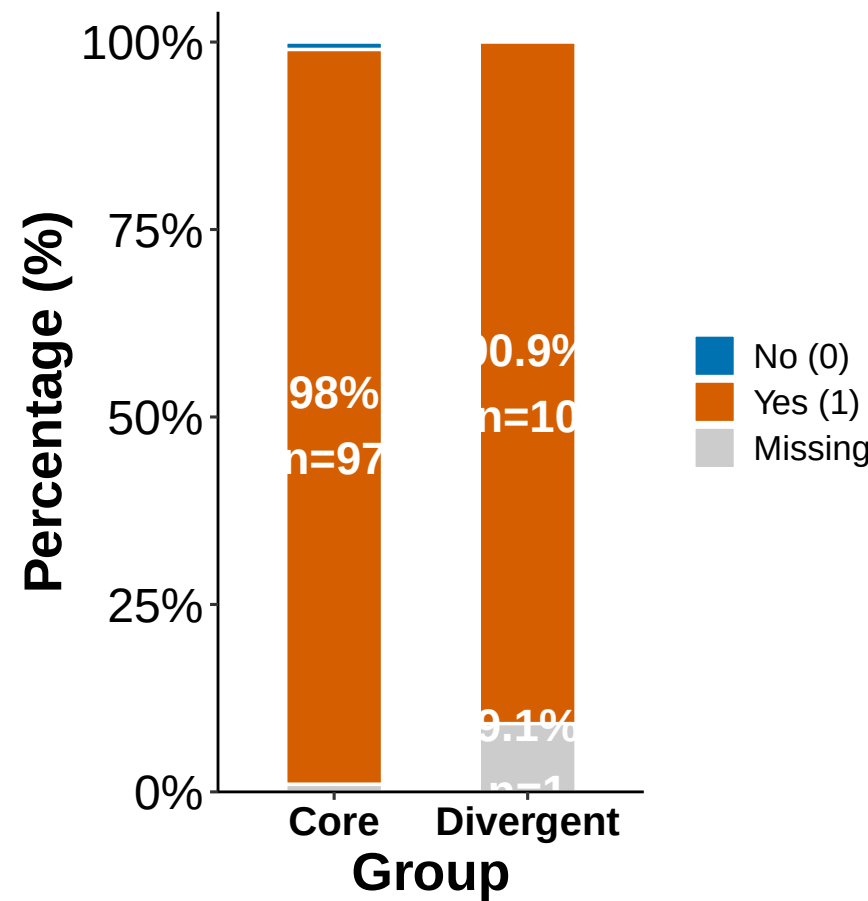
GlnDependence_FBA

p-adj: 0.917 | .=-0.12 (Negligible) | .%: -87.3



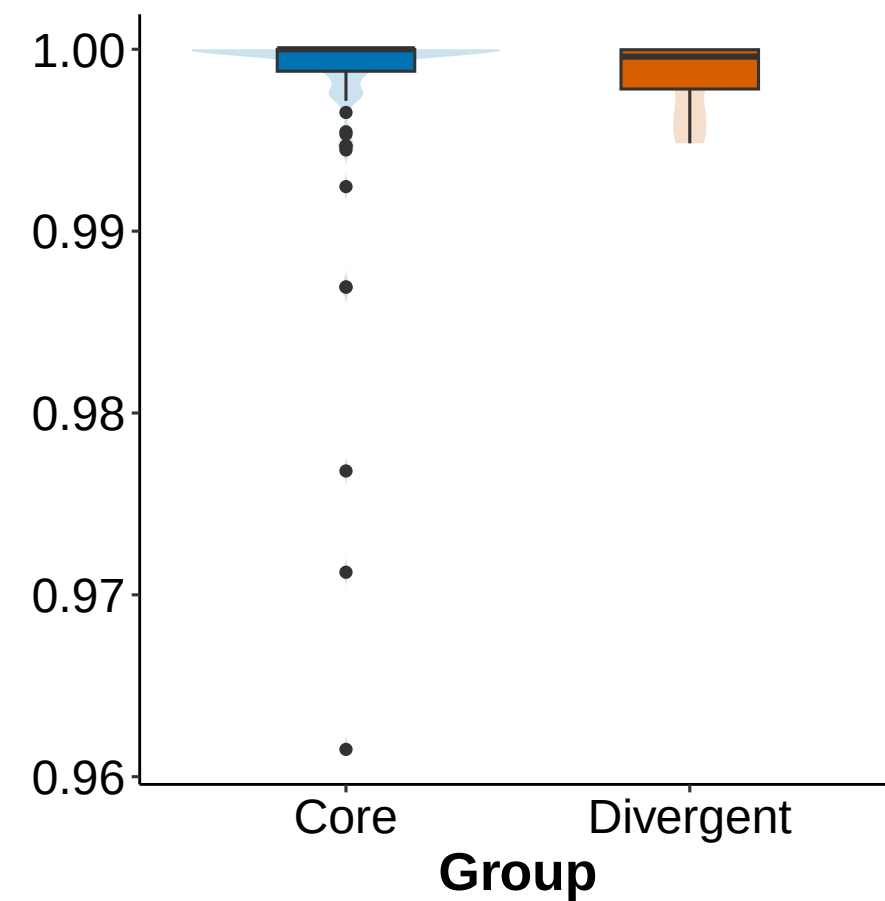
GlnDependence_pFBA

p-adj: 0.95 | .=-0.01 (Negligible) | .%: 0



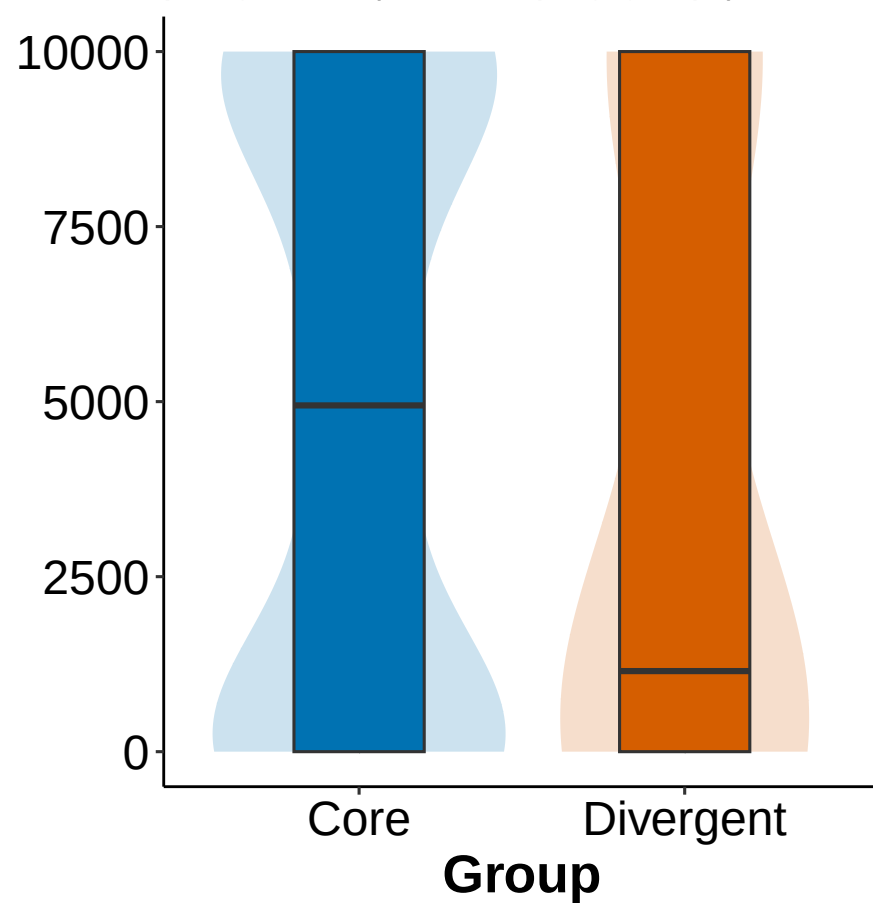
GlnDependence_L1w

p-adj: 0.885 | .=-0.16 (Small) | .%: 0



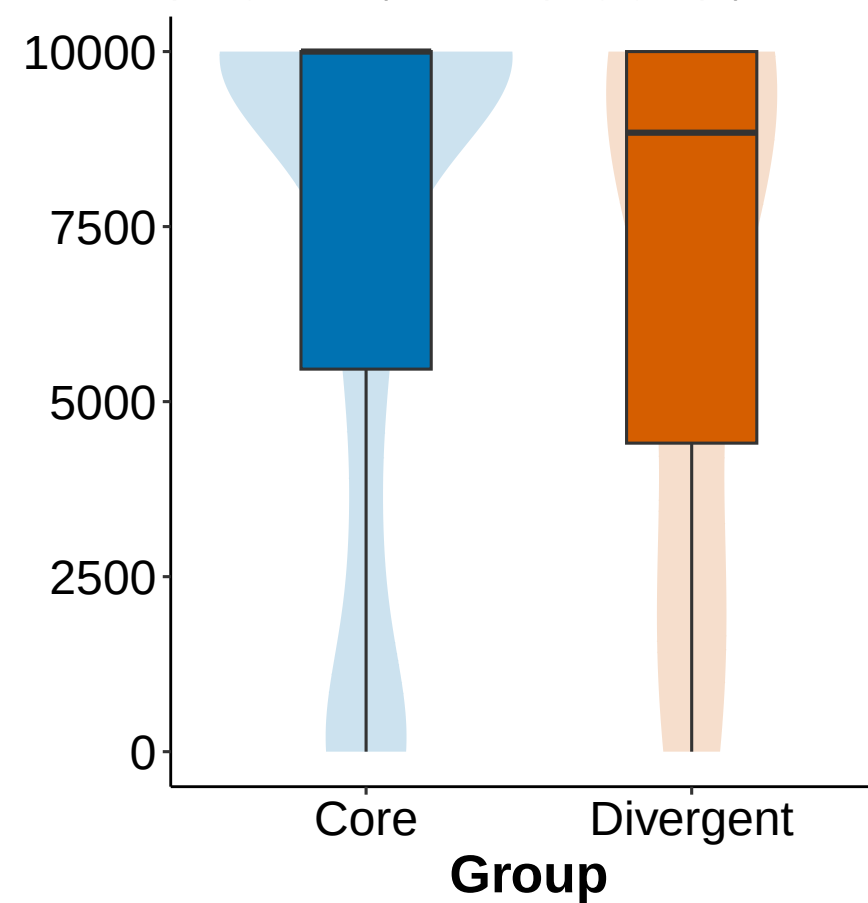
Oncomet_Lactate_FBA

p-adj: 0.942 | .=-0.09 (Negligible) | .%: -76.7



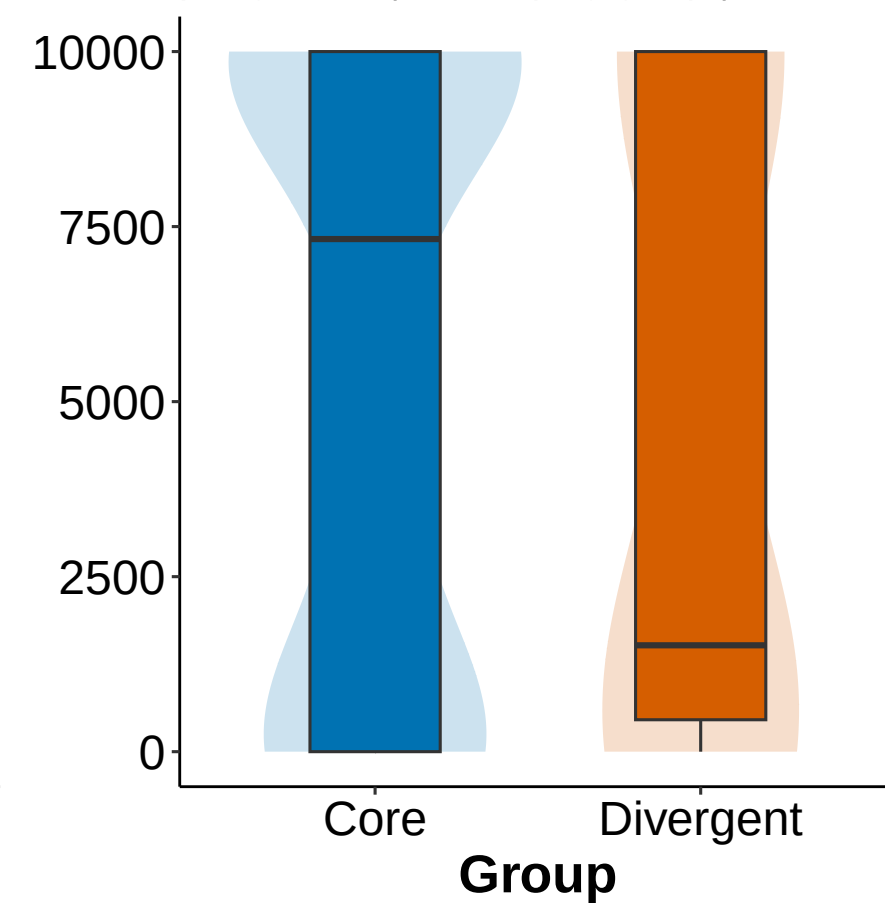
Oncomet_Succinate_FBA

p-adj: 0.917 | .=-0.12 (Negligible) | .%: -11.6

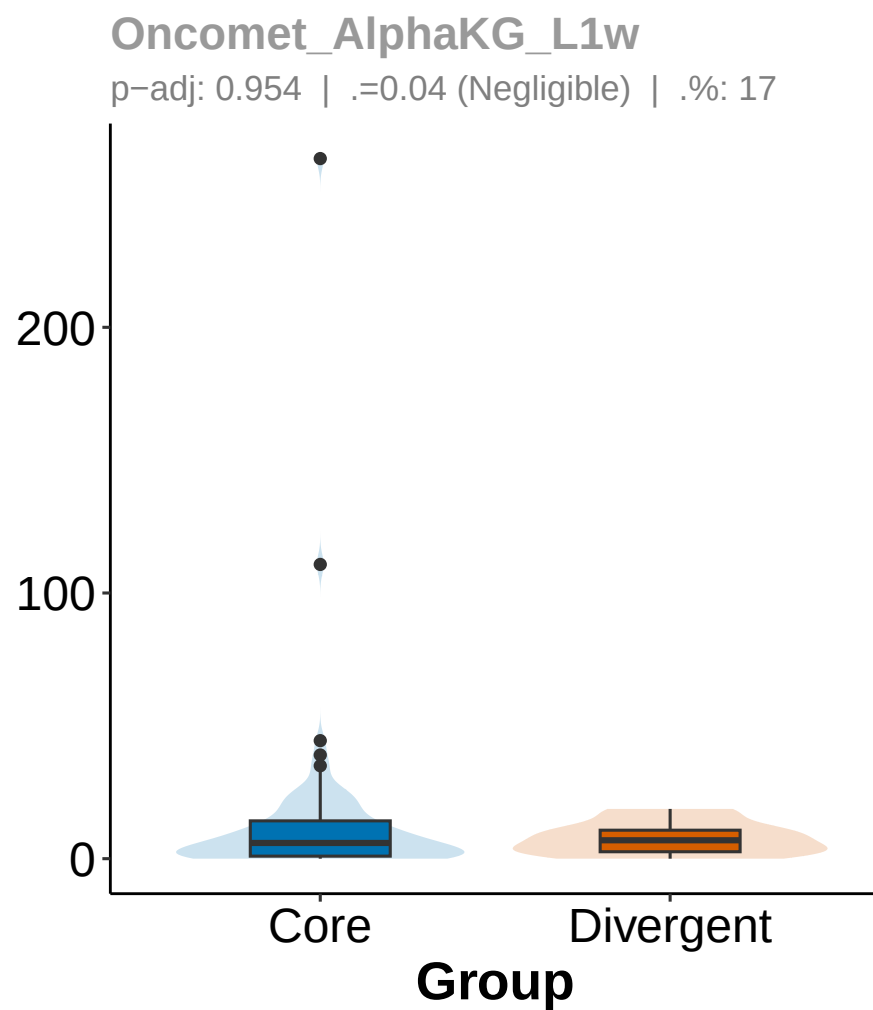
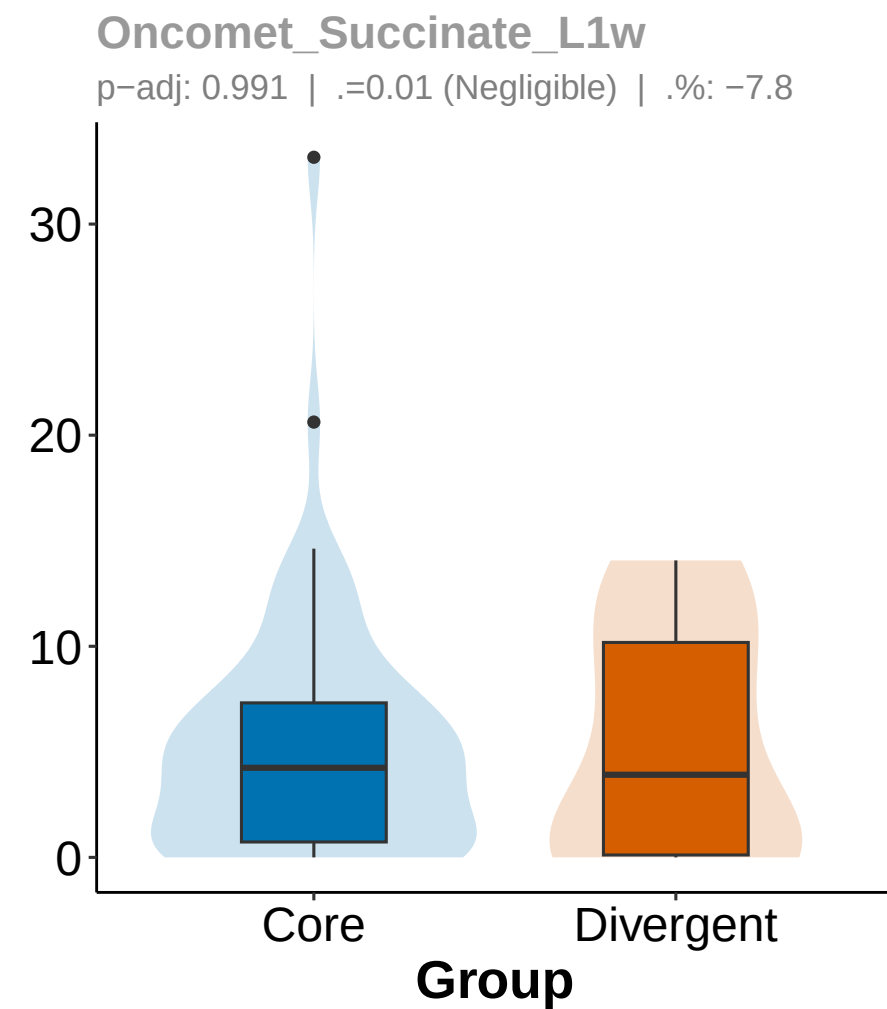
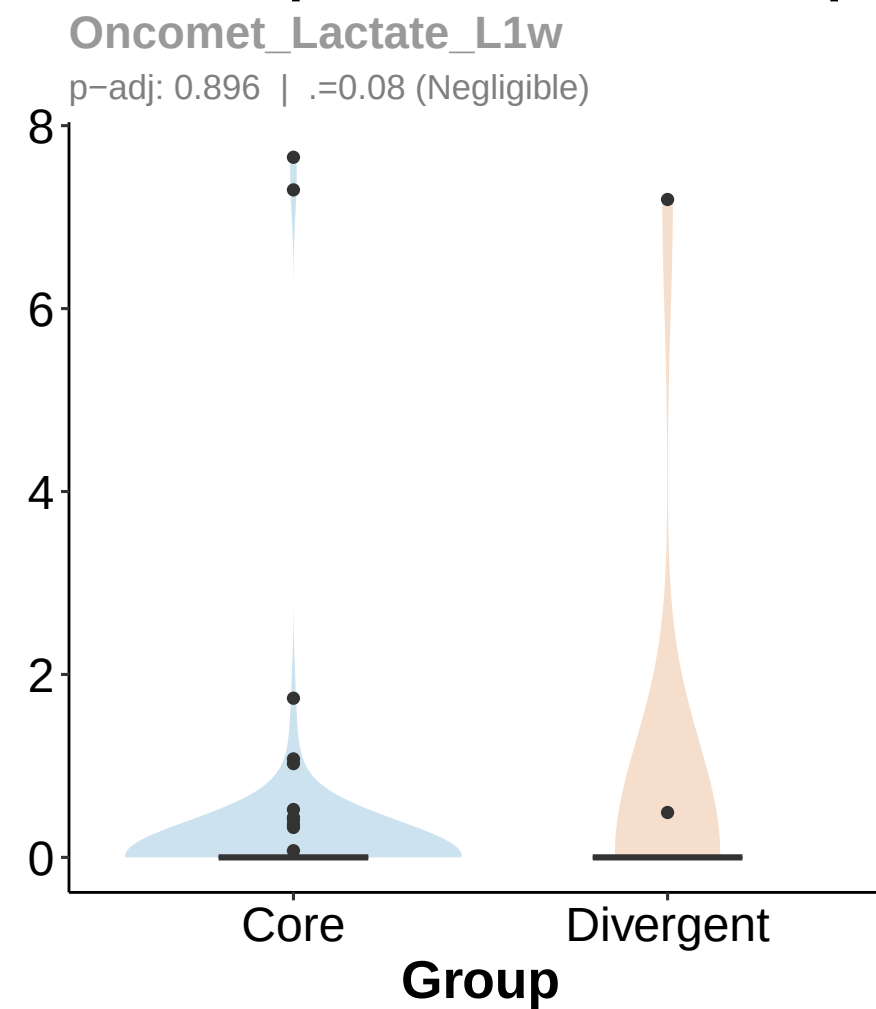
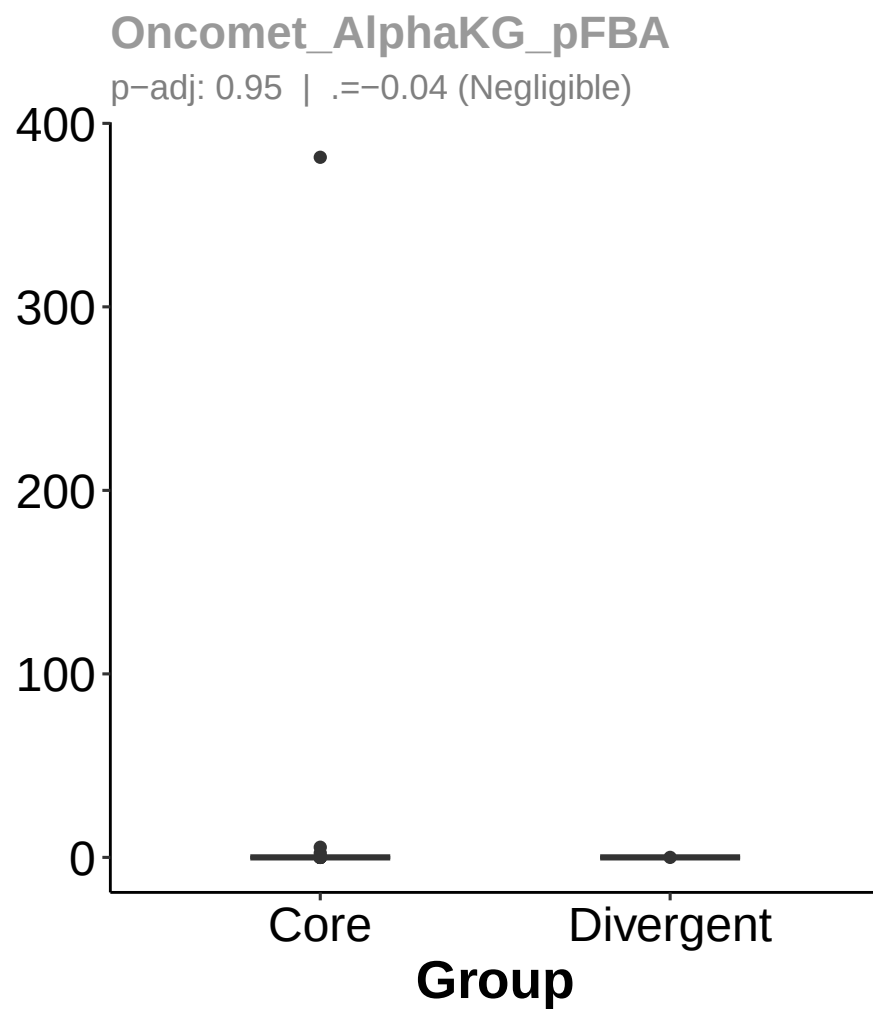


Oncomet_AlphaKG_FBA

p-adj: 0.954 | .=-0.03 (Negligible) | .%: -79.2



Metabolism – . 4. Specific Metabolic Dependencies



Cliff's .: . 5. Systemic Reprogramming (SA_*)

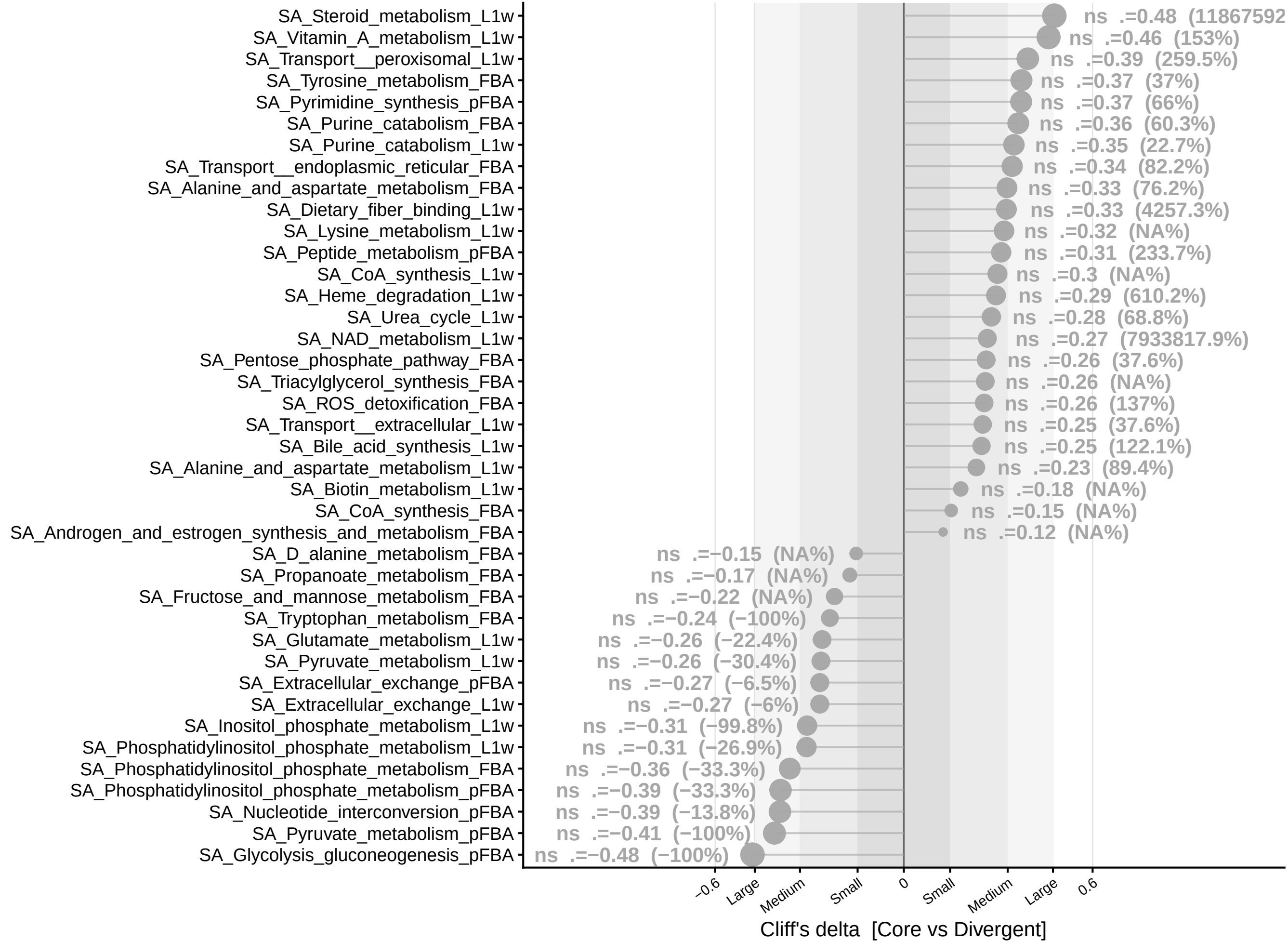
Blue = Core > Divergent | Orange = Divergent > Core | * p_adj<0.05



SA_Steroid_metabolism_L1w	ns .=0.48
SA_Vitamin_A_metabolism_L1w	ns .=0.46
SA_Transport__peroxisomal_L1w	ns .=0.39
SA_Tyrosine_metabolism_FBA	ns .=0.37
SA_Pyrimidine_synthesis_pFBA	ns .=0.37
SA_Purine_catabolism_FBA	ns .=0.36
SA_Purine_catabolism_L1w	ns .=0.35
SA_Transport__endoplasmic_reticular_FBA	ns .=0.34
SA_Alanine_and_aspartate_metabolism_FBA	ns .=0.33
SA_Dietary_fiber_binding_L1w	ns .=0.33
SA_Lysine_metabolism_L1w	ns .=0.32
SA_Peptide_metabolism_pFBA	ns .=0.31
SA_CoA_synthesis_L1w	ns .=0.3
SA_Heme_degradation_L1w	ns .=0.29
SA_Urea_cycle_L1w	ns .=0.28
SA_NAD_metabolism_L1w	ns .=0.27
SA_Pentose_phosphate_pathway_FBA	ns .=0.26
SA_Triacylglycerol_synthesis_FBA	ns .=0.26
SA_ROS_detoxification_FBA	ns .=0.26
SA_Transport__extracellular_L1w	ns .=0.25
SA_Bile_acid_synthesis_L1w	ns .=0.25
SA_Alanine_and_aspartate_metabolism_L1w	ns .=0.23
SA_Biotin_metabolism_L1w	ns .=0.18
SA_CoA_synthesis_FBA	ns .=0.15
SA_Androgen_and_estrogen_synthesis_and_metabolism_FBA	ns .=0.12
SA_D_alanine_metabolism_FBA	ns .=-0.15
SA_Propanoate_metabolism_FBA	ns .=-0.17
SA_Fructose_and_mannose_metabolism_FBA	ns .=-0.22
SA_Tryptophan_metabolism_FBA	ns .=-0.24
SA_Glutamate_metabolism_L1w	ns .=-0.26
SA_Pyruvate_metabolism_L1w	ns .=-0.26
SA_Extracellular_exchange_pFBA	ns .=-0.27
SA_Extracellular_exchange_L1w	ns .=-0.27
SA_Inositol_phosphate_metabolism_L1w	ns .=-0.31
SA_Phosphatidylinositol_phosphate_metabolism_L1w	ns .=-0.31
SA_Phosphatidylinositol_phosphate_metabolism_FBA	ns .=-0.36
SA_Phosphatidylinositol_phosphate_metabolism_pFBA	ns .=-0.39
SA_Nucleotide_interconversion_pFBA	ns .=-0.39
SA_Pyruvate_metabolism_pFBA	ns .=-0.41
SA_Glycolysis_gluconeogenesis_pFBA	ns .=-0.48

Cliff's . – . 5. Systemic Reprogramming (SA_*)

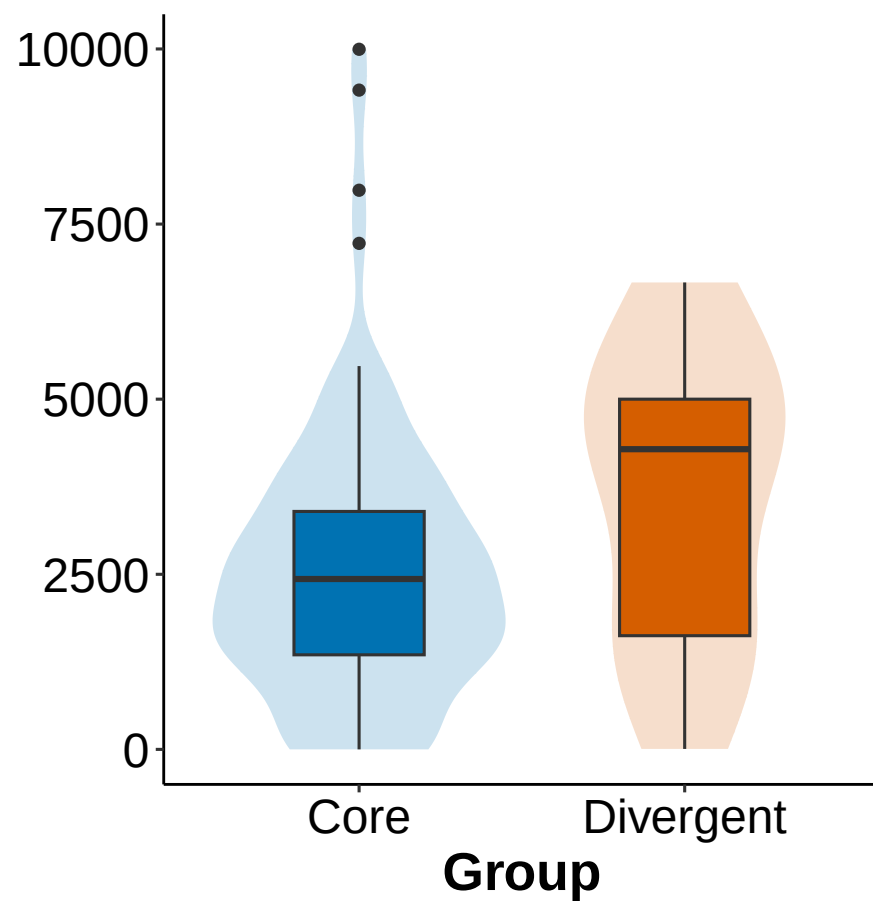
Cliff's . + .% median | . higher in Core · . higher in Divergent



Metabolism – . 5. Systemic Reprogramming (SA_*)

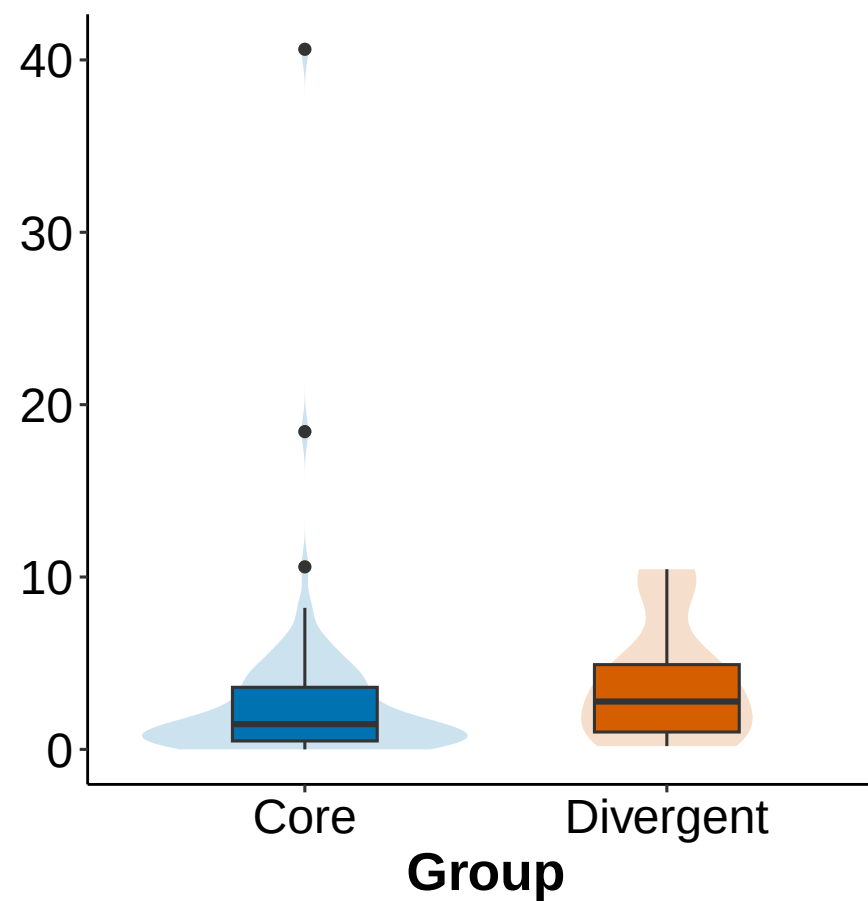
SA_Alanine_and_aspartate_metabolism

p-adj: 0.843 | .= 0.33 (Small) | .%: 76.2



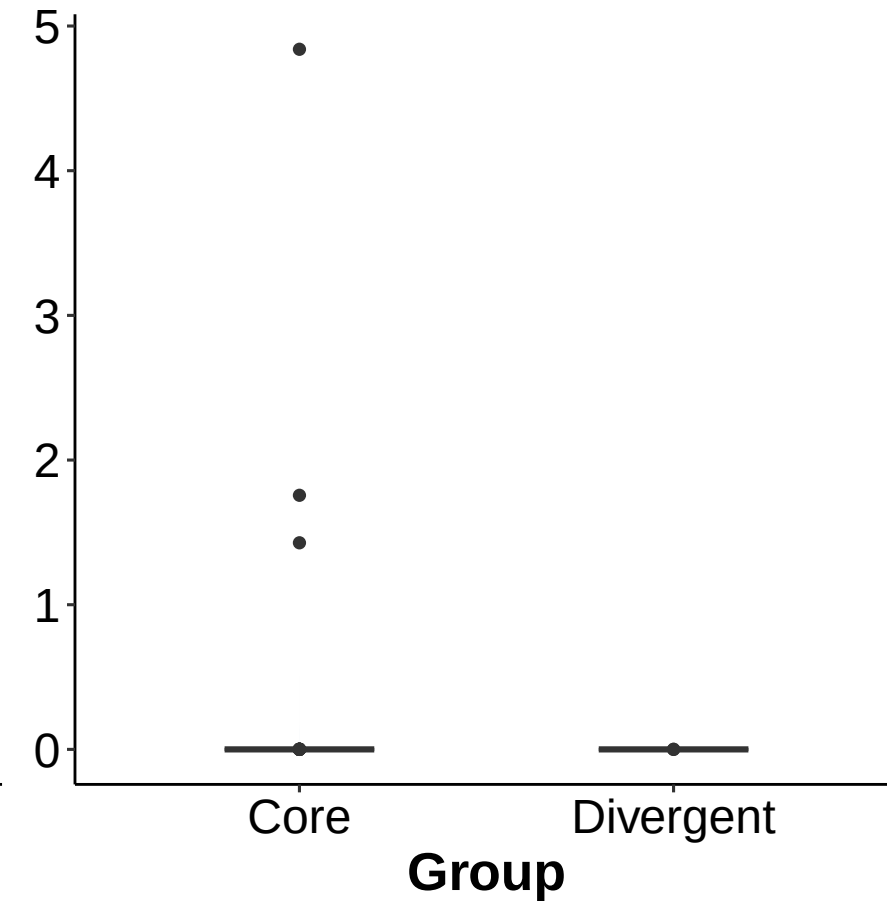
SA_Alanine_and_aspartate_metabolism

p-adj: 0.885 | .= 0.23 (Small) | .%: 89.4



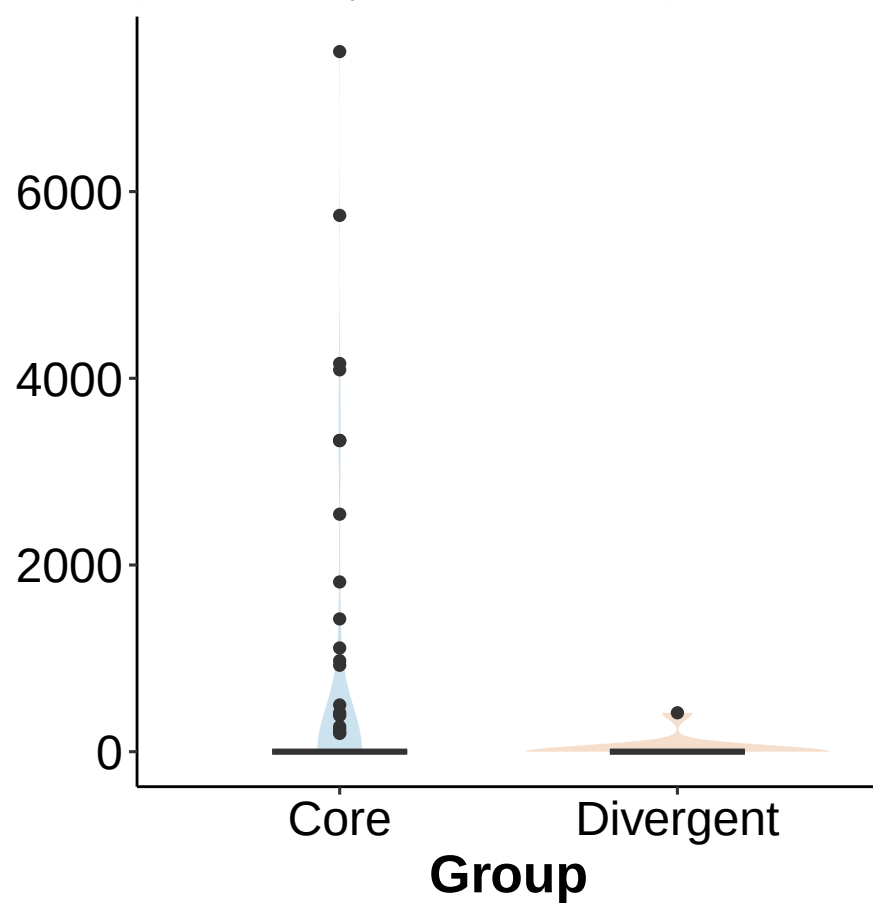
SA_Alanine_and_aspartate_metabolism

p-adj: 0.95 | .= 0.05 (Negligible)



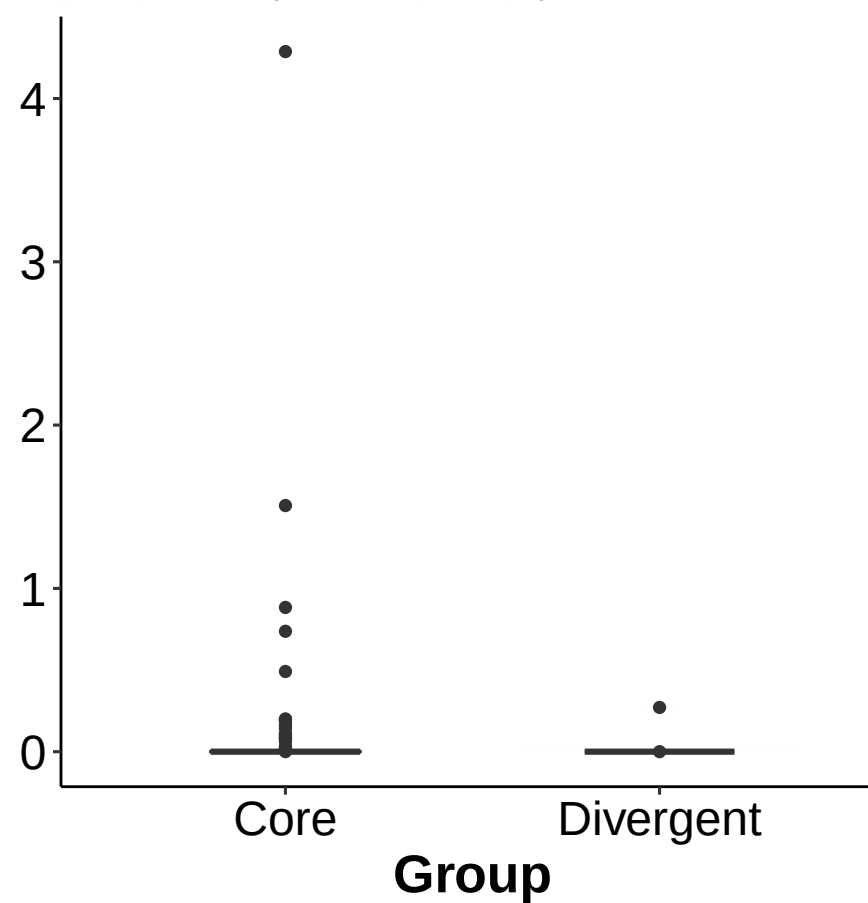
SA_Aminosugar_metabolism_FBA

p-adj: 0.885 | .= -0.12 (Negligible)



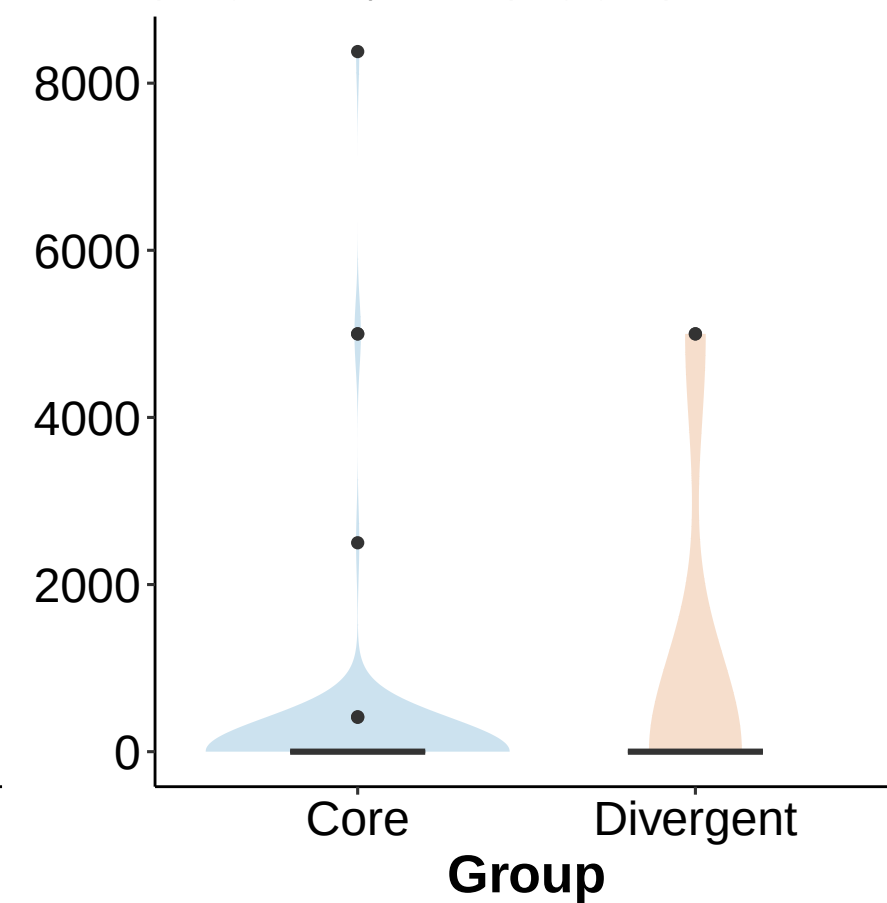
SA_Aminosugar_metabolism_L1w

p-adj: 0.885 | .= -0.2 (Small) | .%: -40.3



SA_Androgen_and_estrogen_synthesis

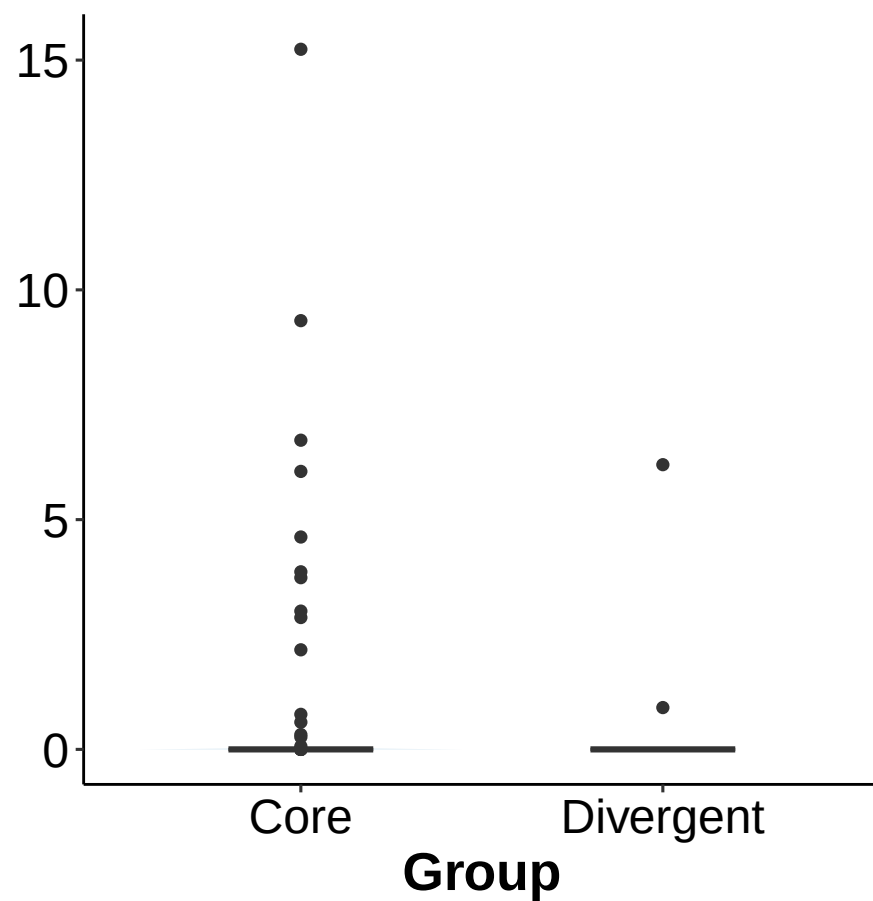
p-adj: 0.872 | .= 0.12 (Negligible)



Metabolism – . 5. Systemic Reprogramming (SA_*)

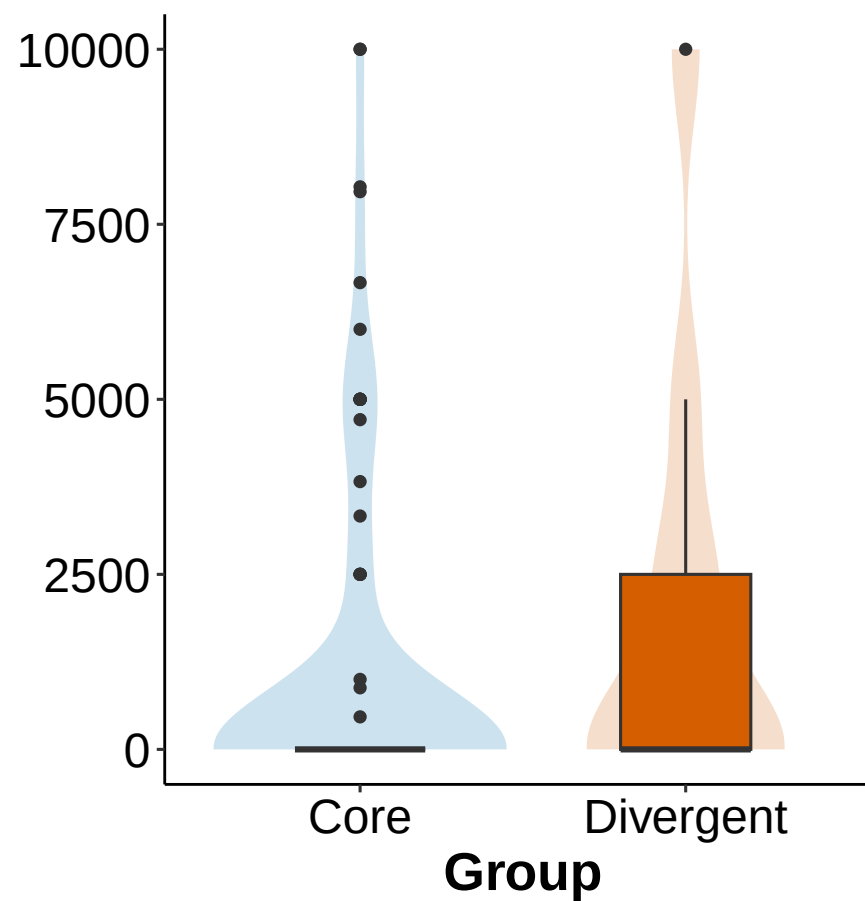
SA_Androgen_and_estrogen_synthes

p-adj: 0.885 | .:=0.15 (Small)



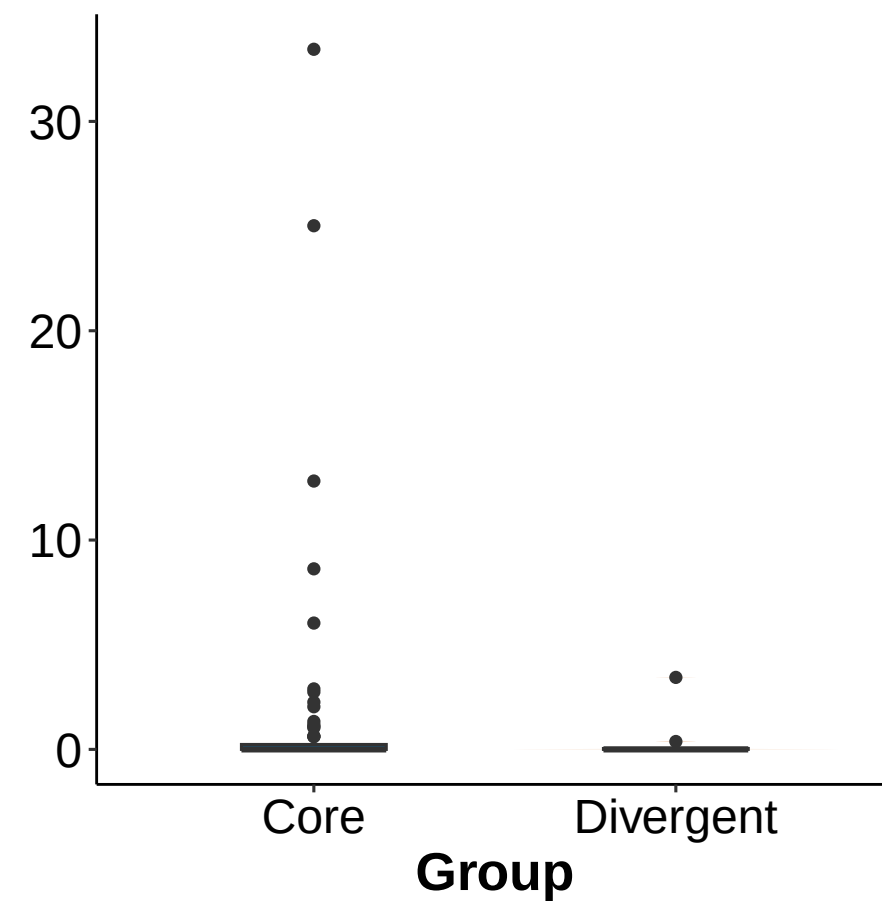
SA_Arachidonic_acid_metabolism

p-adj: 0.885 | .:=0.12 (Negligible)



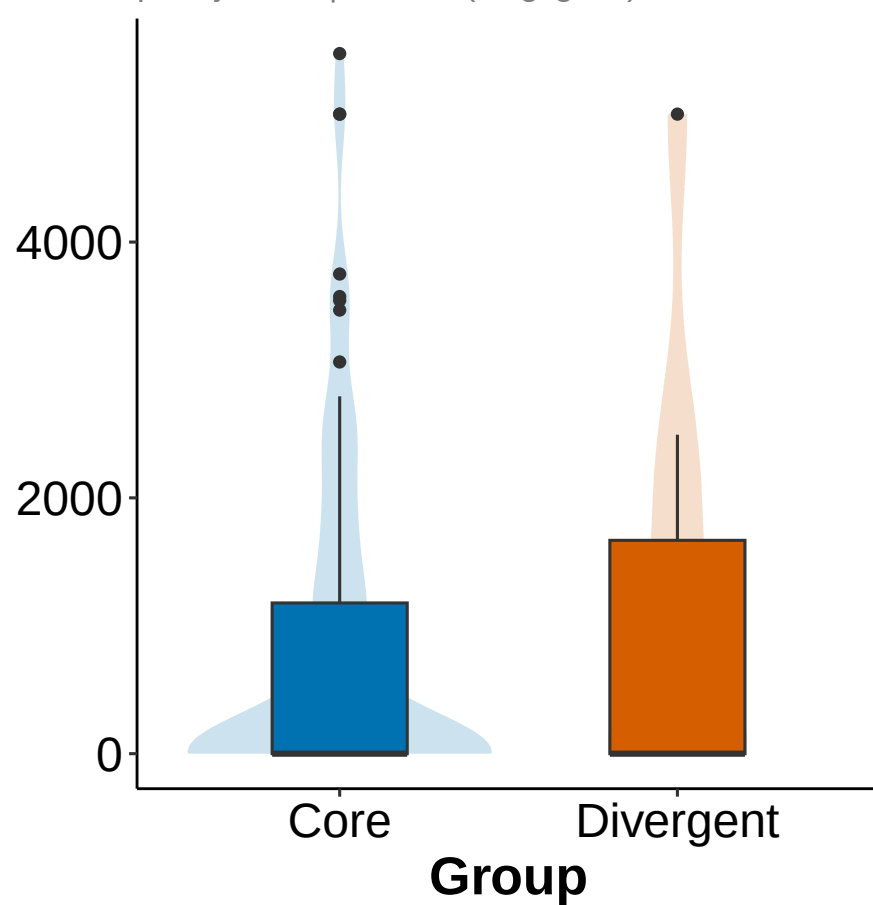
SA_Arachidonic_acid_metabolism_L1

p-adj: 0.885 | .:=−0.17 (Small) | .%: −100



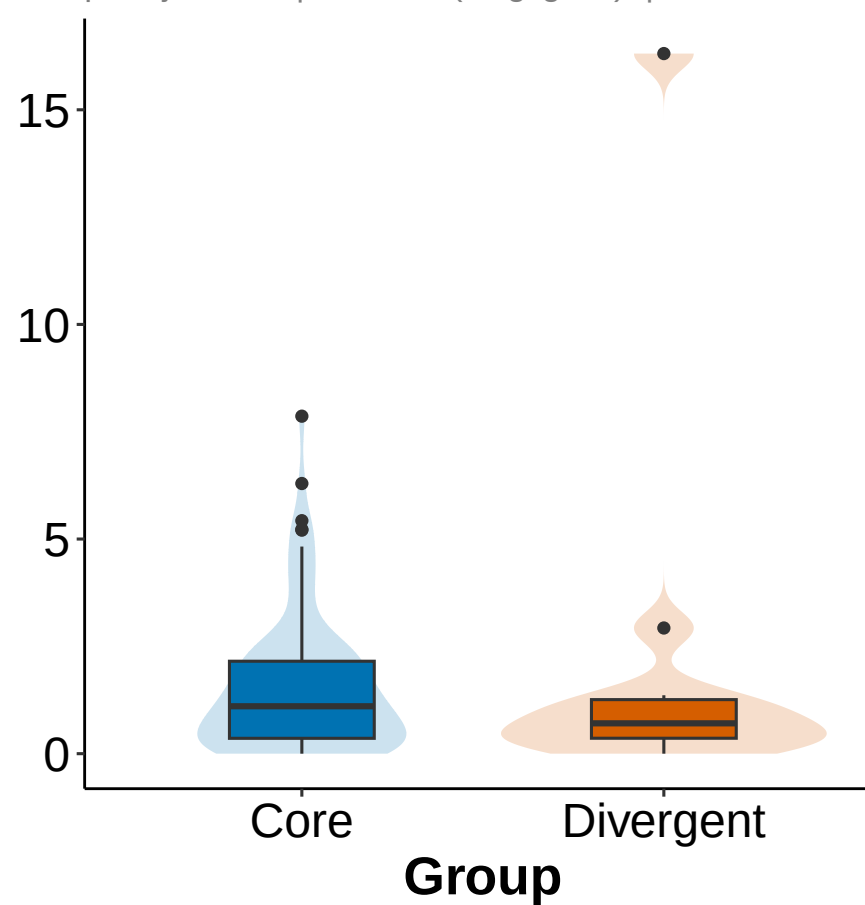
SA_Arginine_and_proline_metaboli

p-adj: 0.95 | .:=0.06 (Negligible)



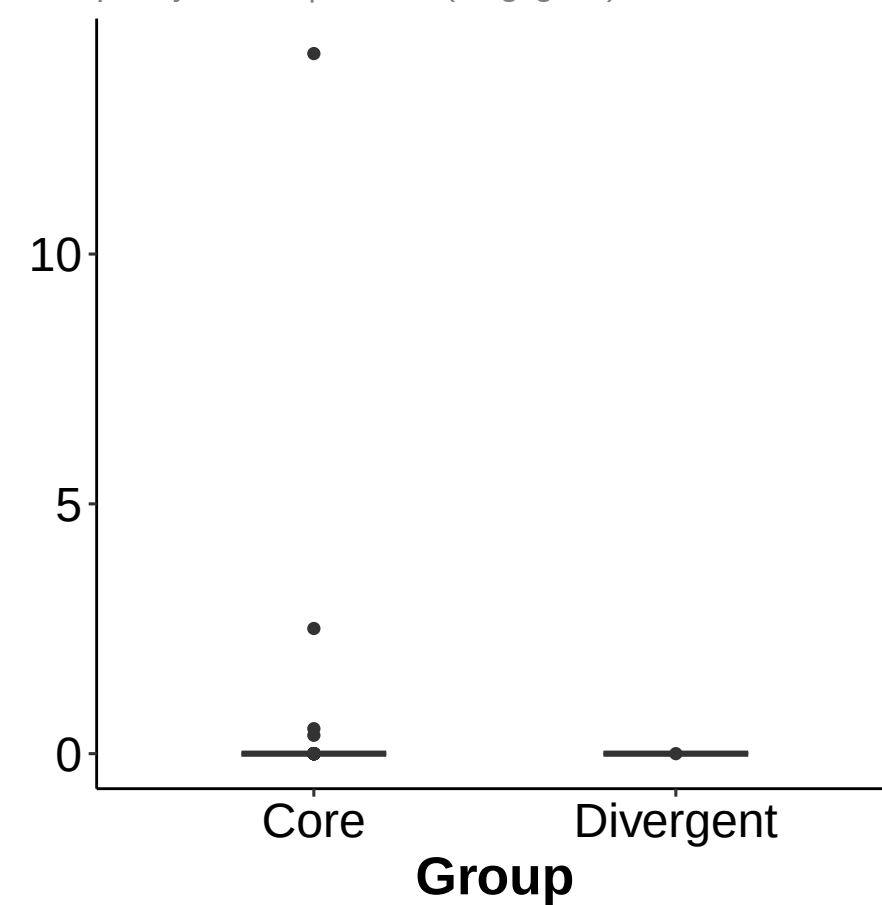
SA_Arginine_and_proline_metabolism

p-adj: 0.917 | .:=−0.12 (Negligible) | .%: −36



SA_Arginine_and_proline_metabolism

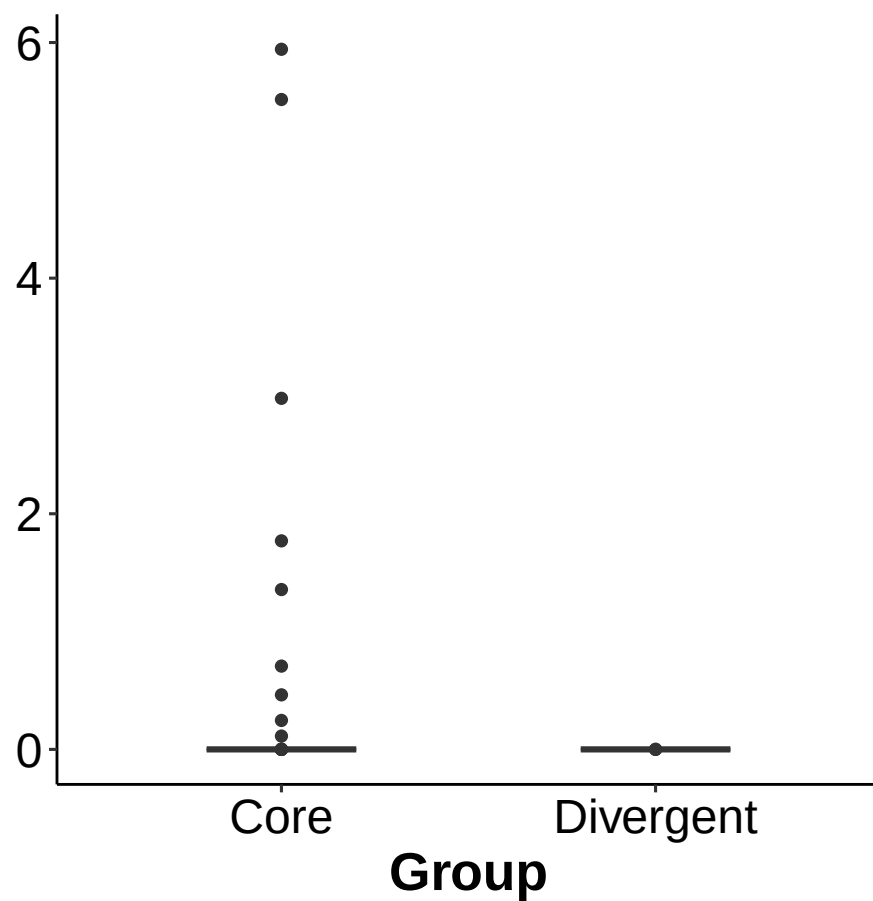
p-adj: 0.896 | .:=−0.1 (Negligible)



Metabolism – . 5. Systemic Reprogramming (SA_*)

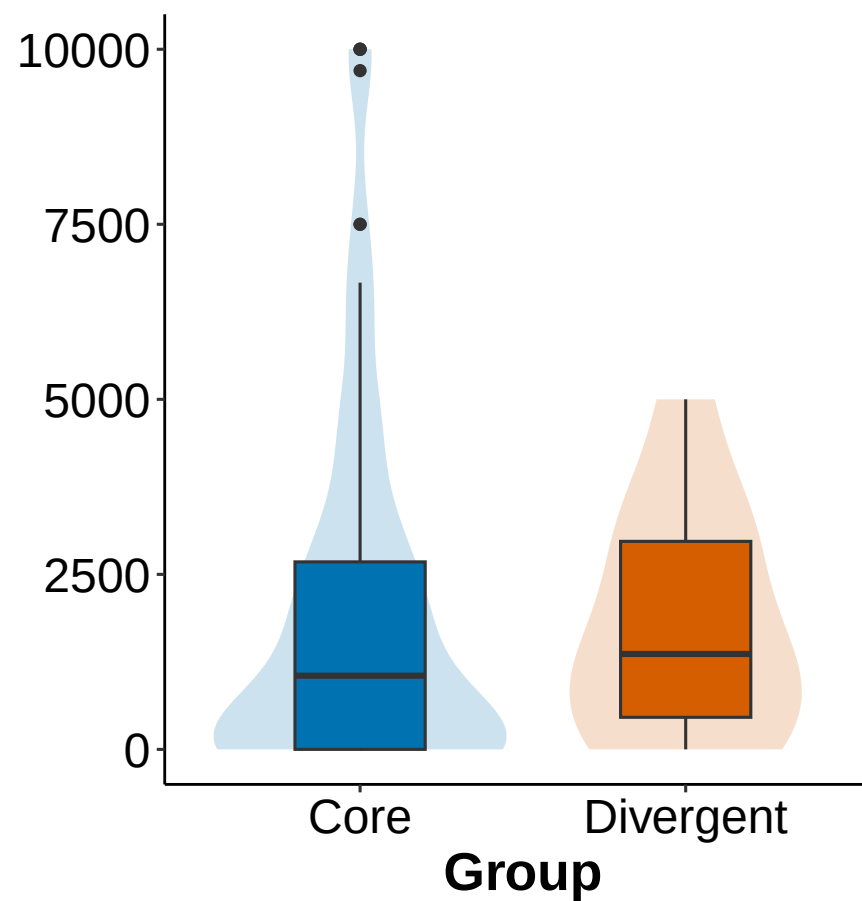
SA_Beta_Alanine_metabolism_L1w

p-adj: 0.954 | .:=0.03 (Negligible)



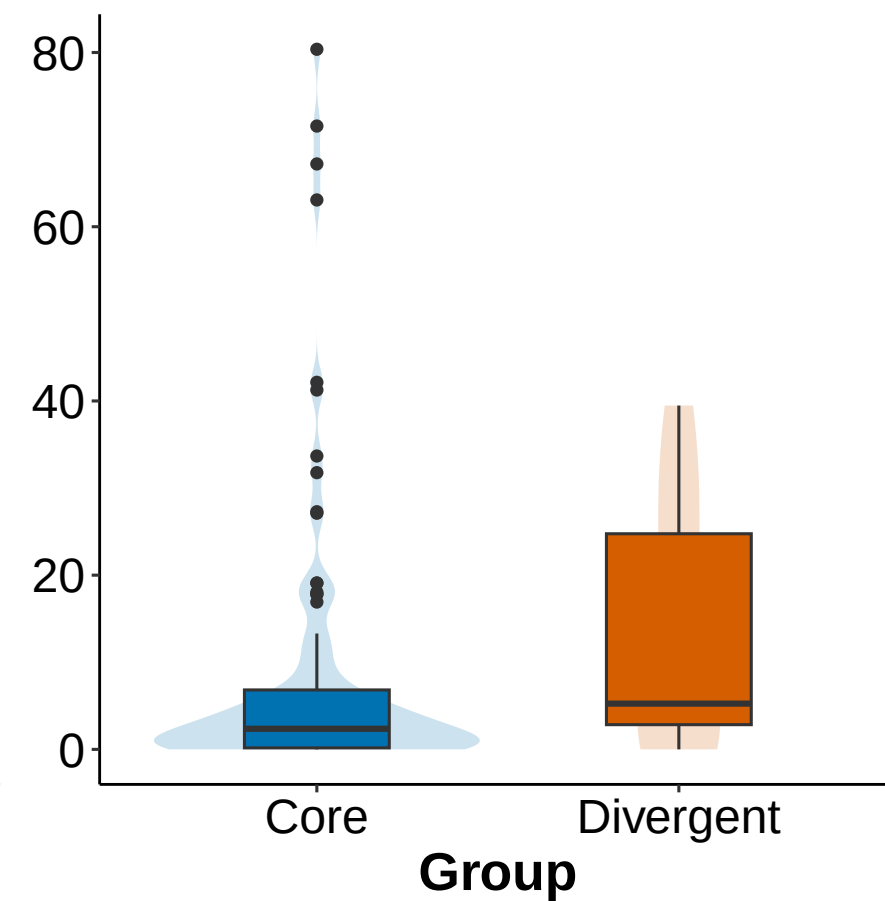
SA_Bile_acid_synthesis_FBA

p-adj: 0.917 | .:=0.12 (Negligible) | .%: 29.5



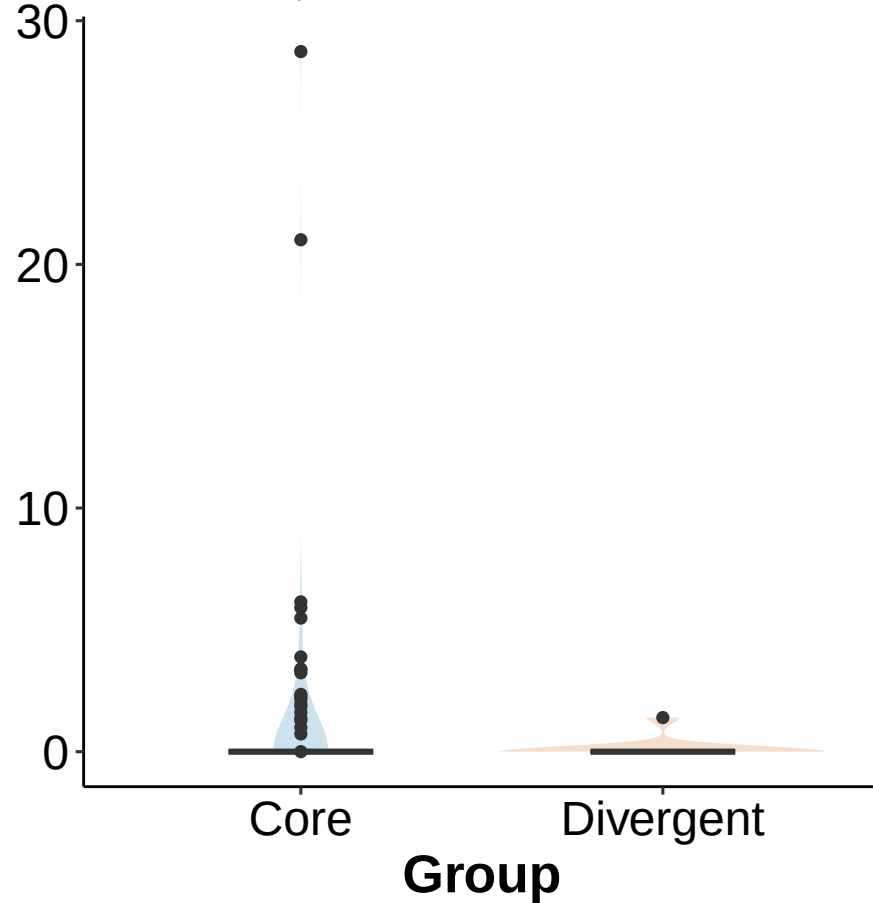
SA_Bile_acid_synthesis_L1w

p-adj: 0.885 | .:=0.25 (Small) | .%: 122.1



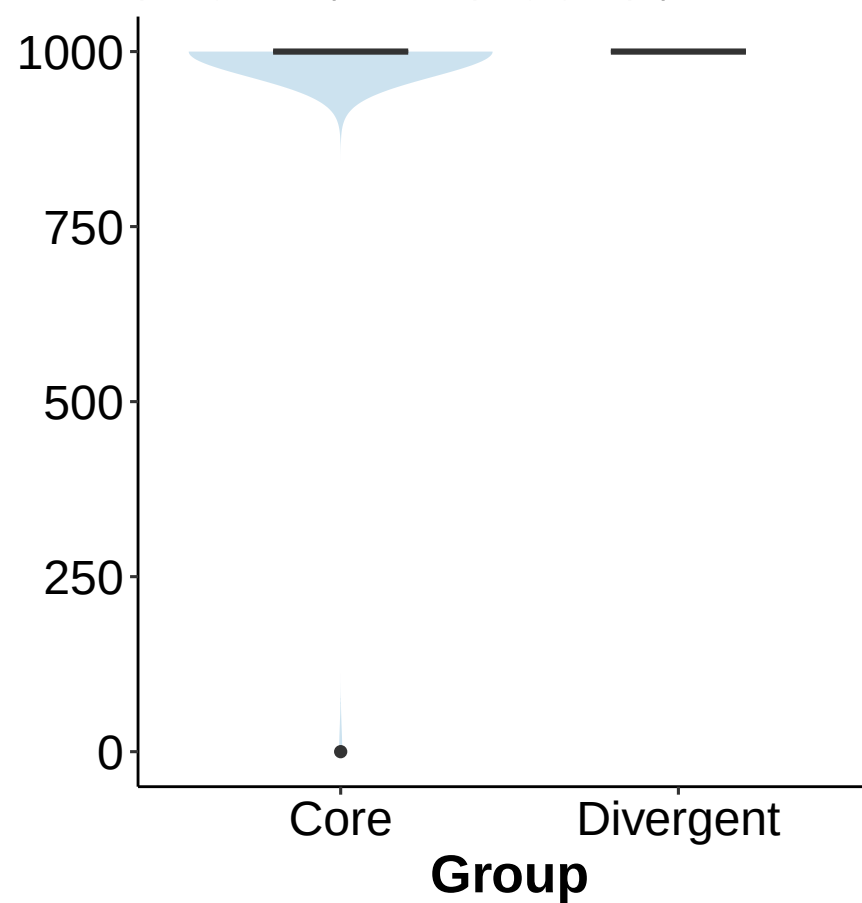
SA_Bile_acid_synthesis_pFBA

p-adj: 0.885 | .:=−0.12 (Negligible)



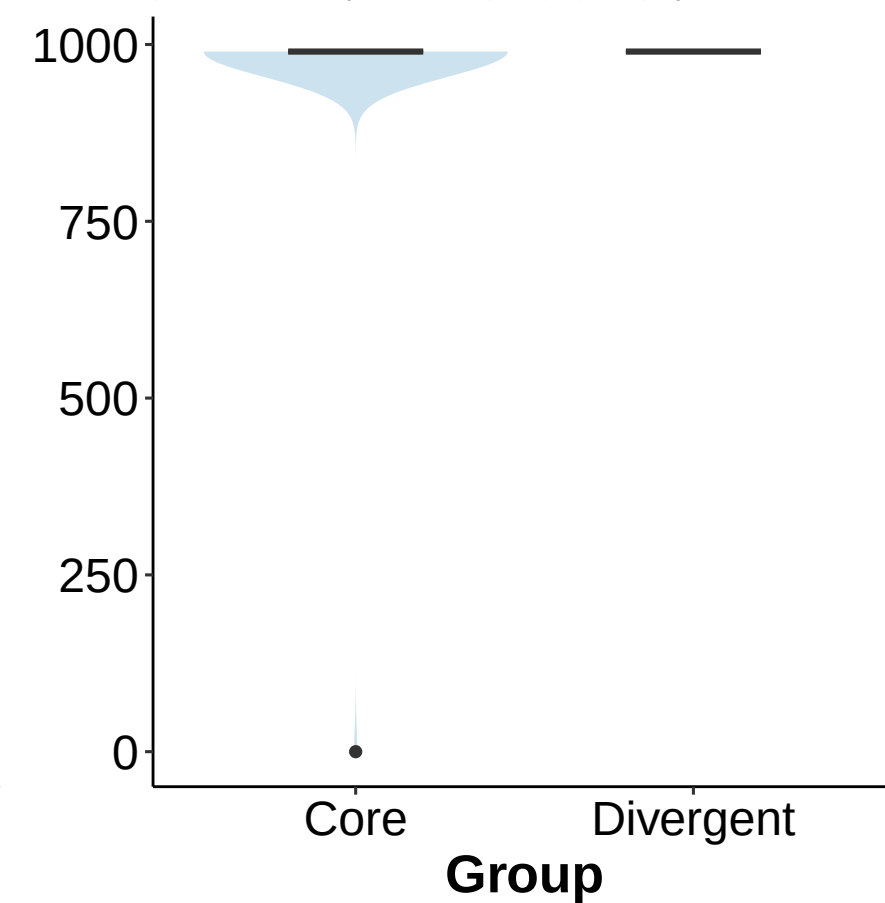
SA_Biomass_reactions_FBA

p-adj: 0.95 | .:=0.01 (Negligible) | .%: 0

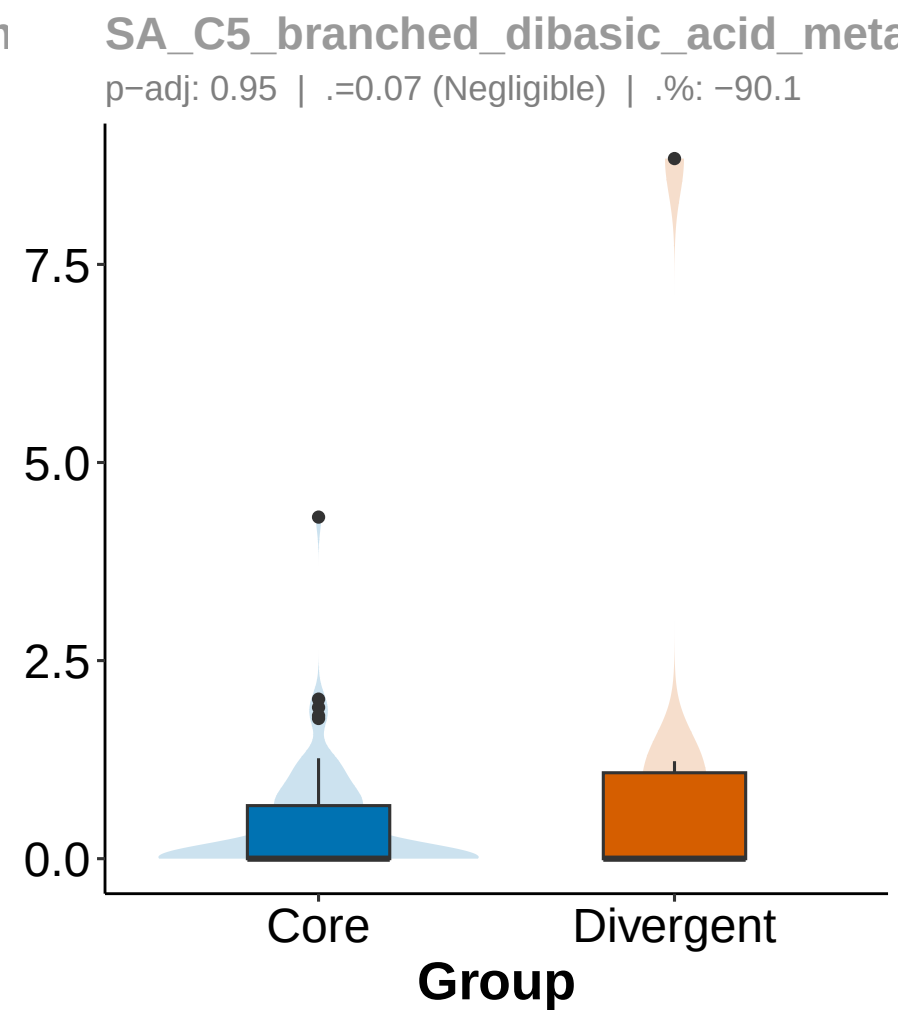
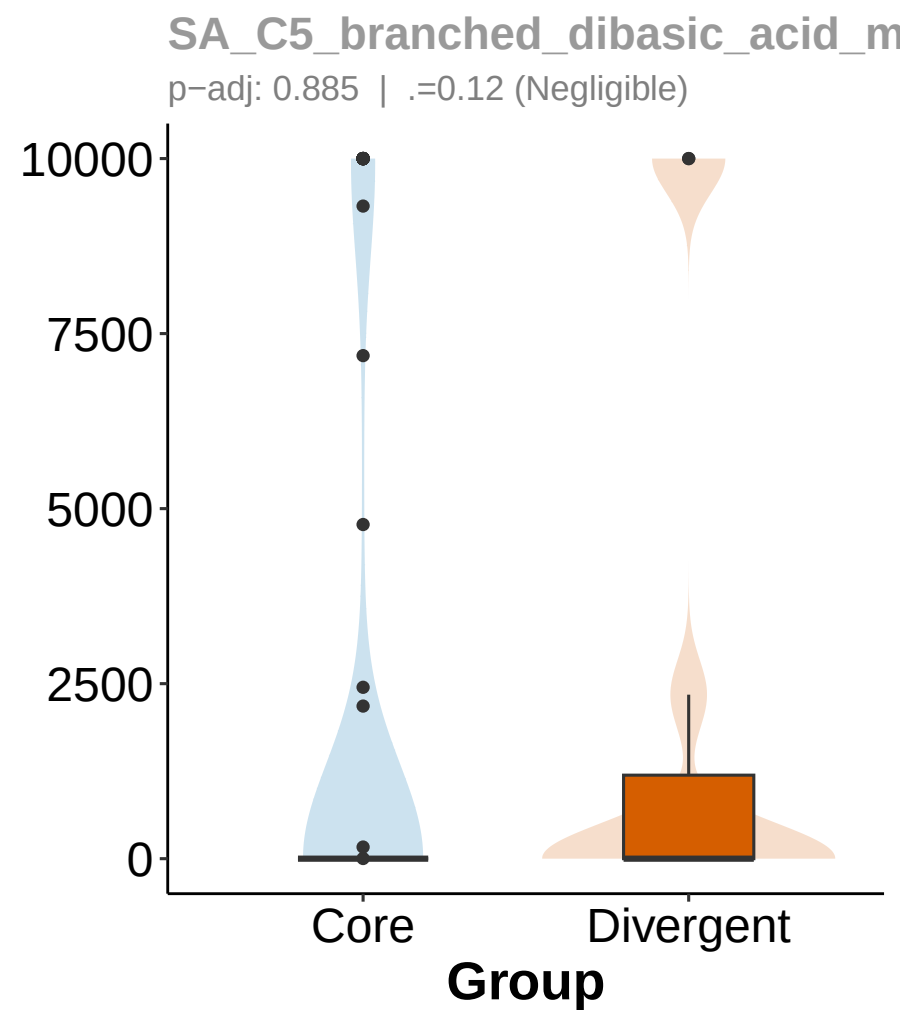
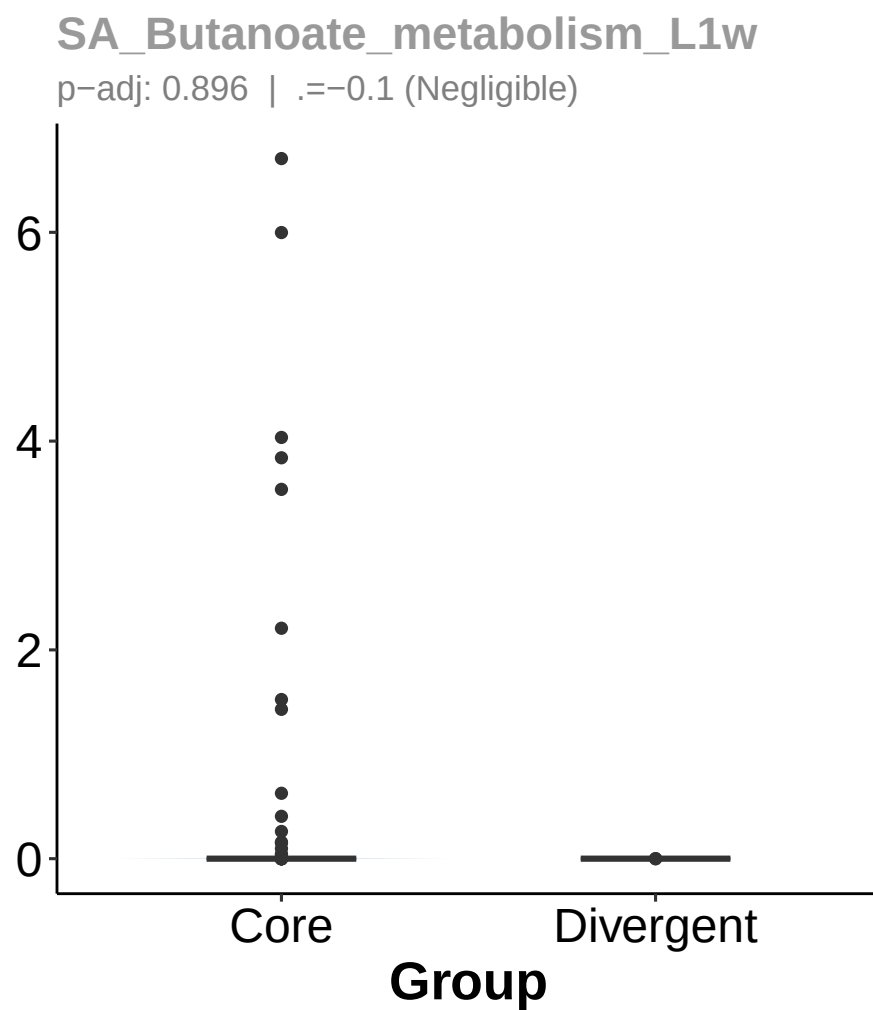
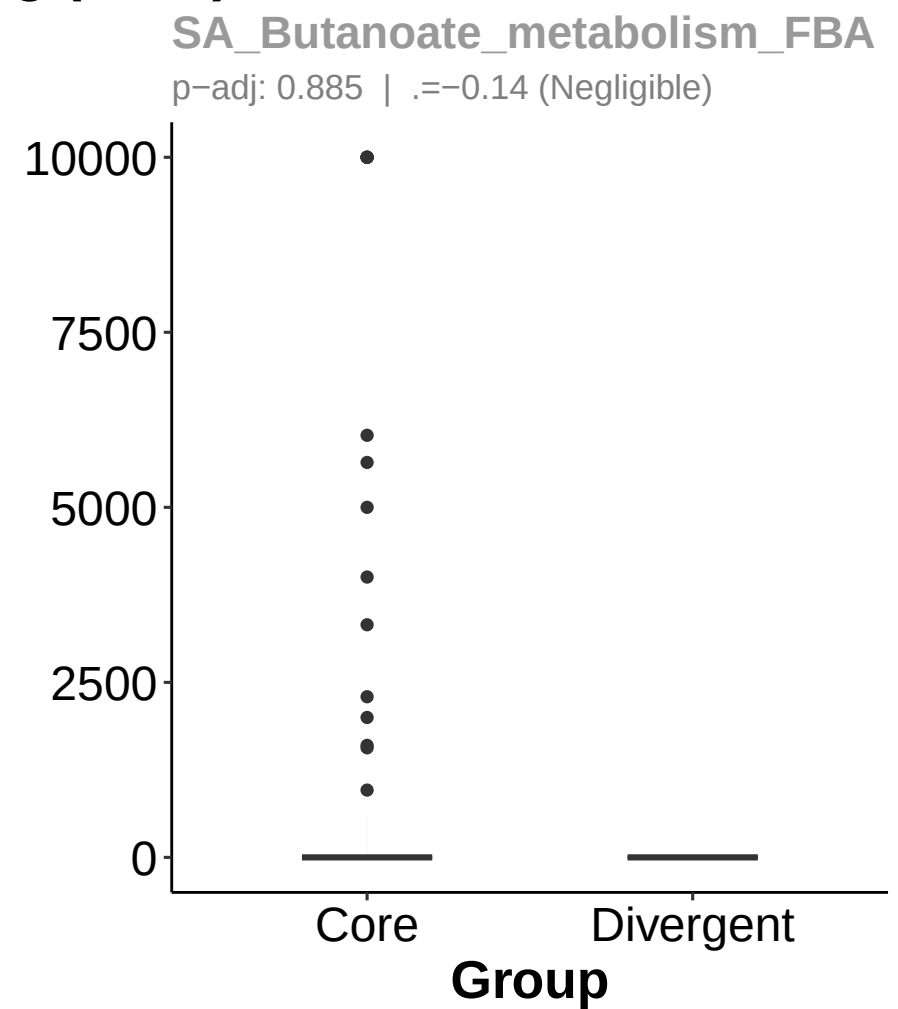
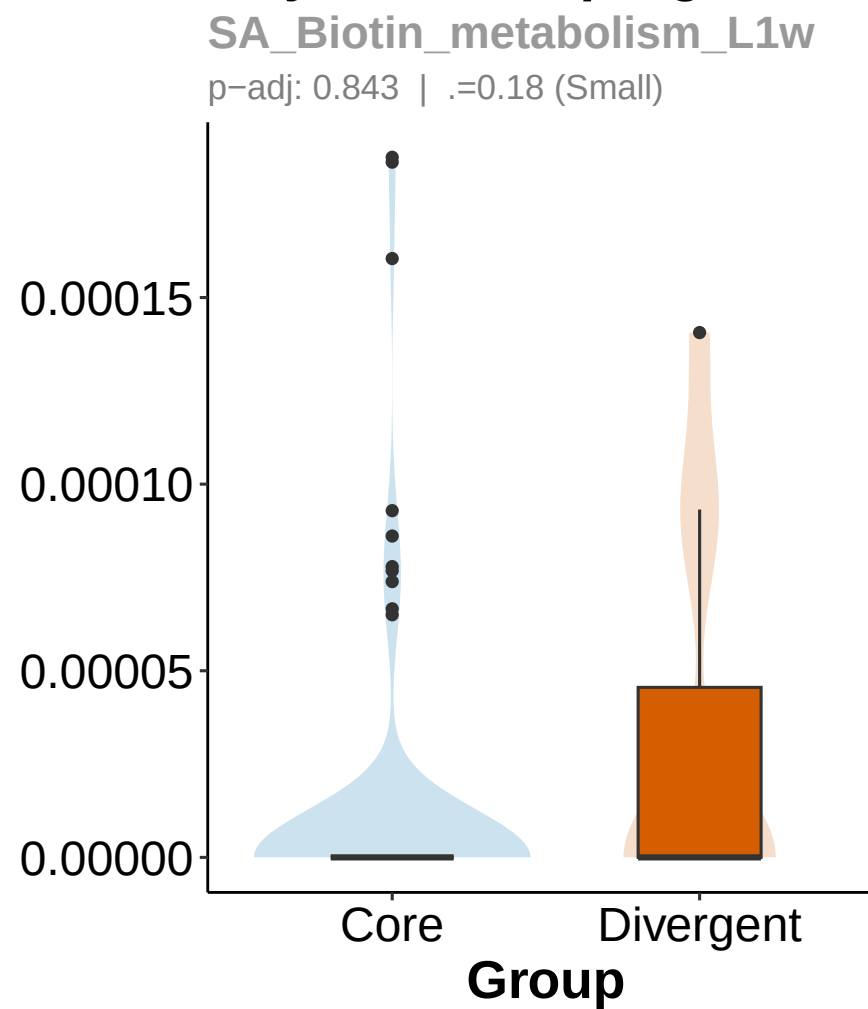
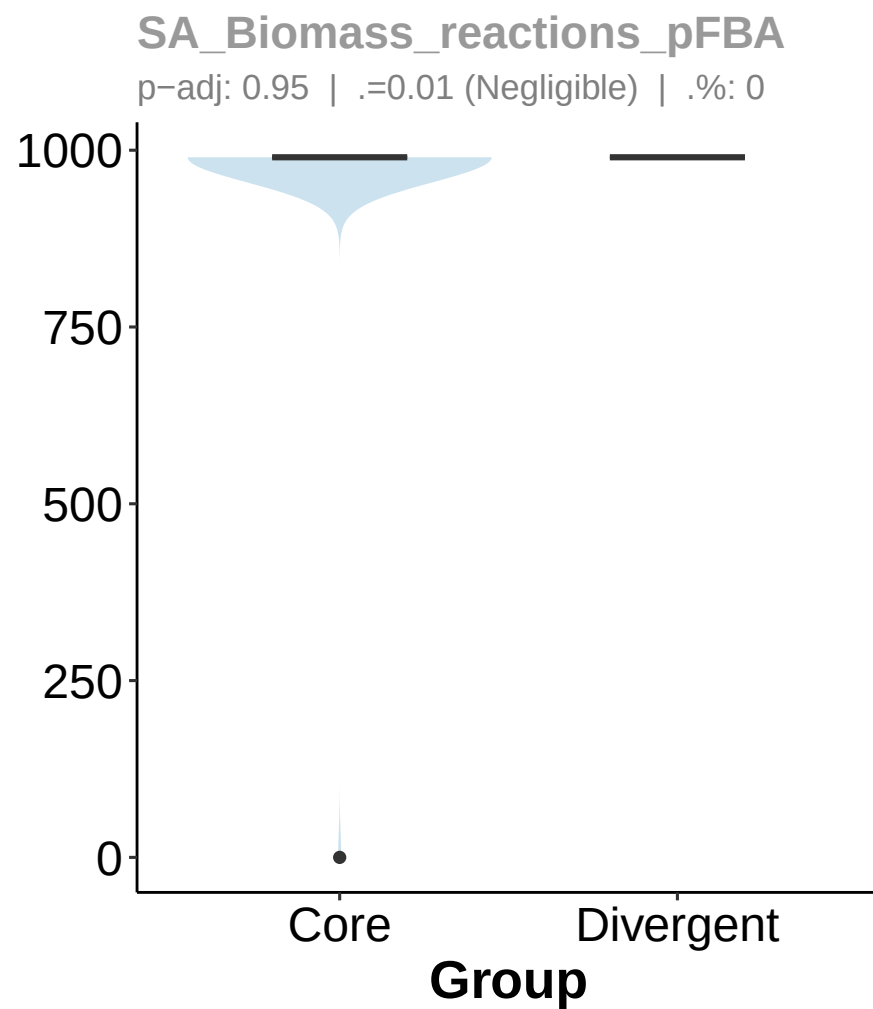


SA_Biomass_reactions_L1w

p-adj: 0.95 | .:=0.01 (Negligible) | .%: 0



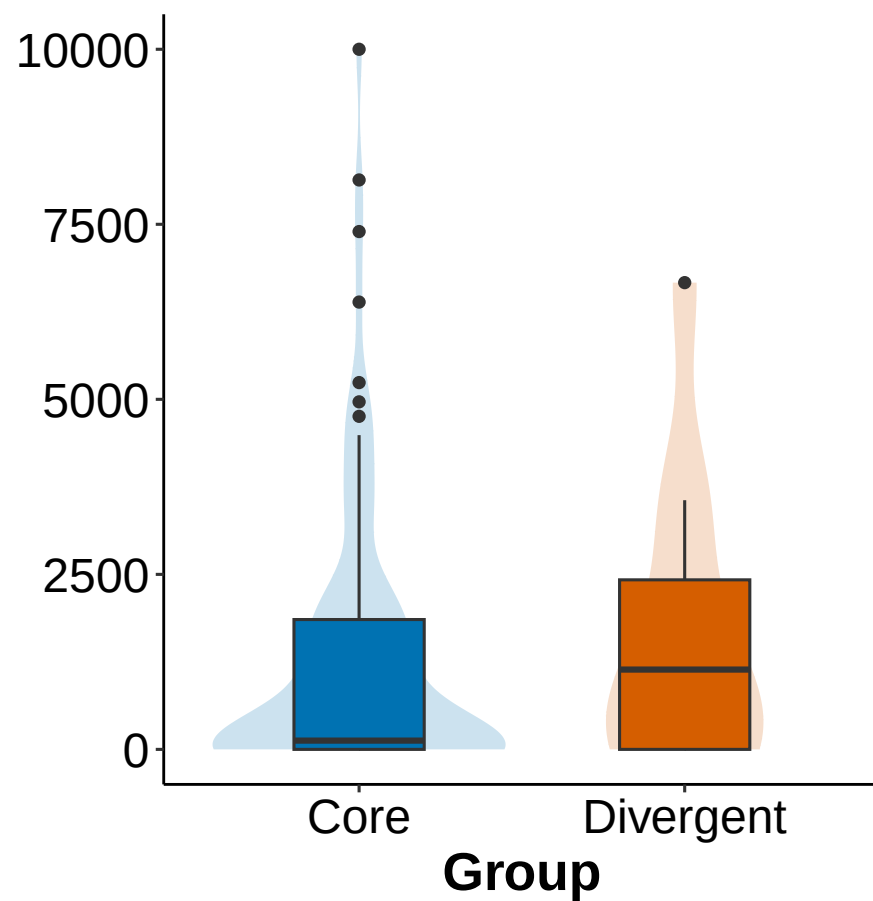
Metabolism – 5. Systemic Reprogramming (SA_*)



Metabolism – . 5. Systemic Reprogramming (SA_*)

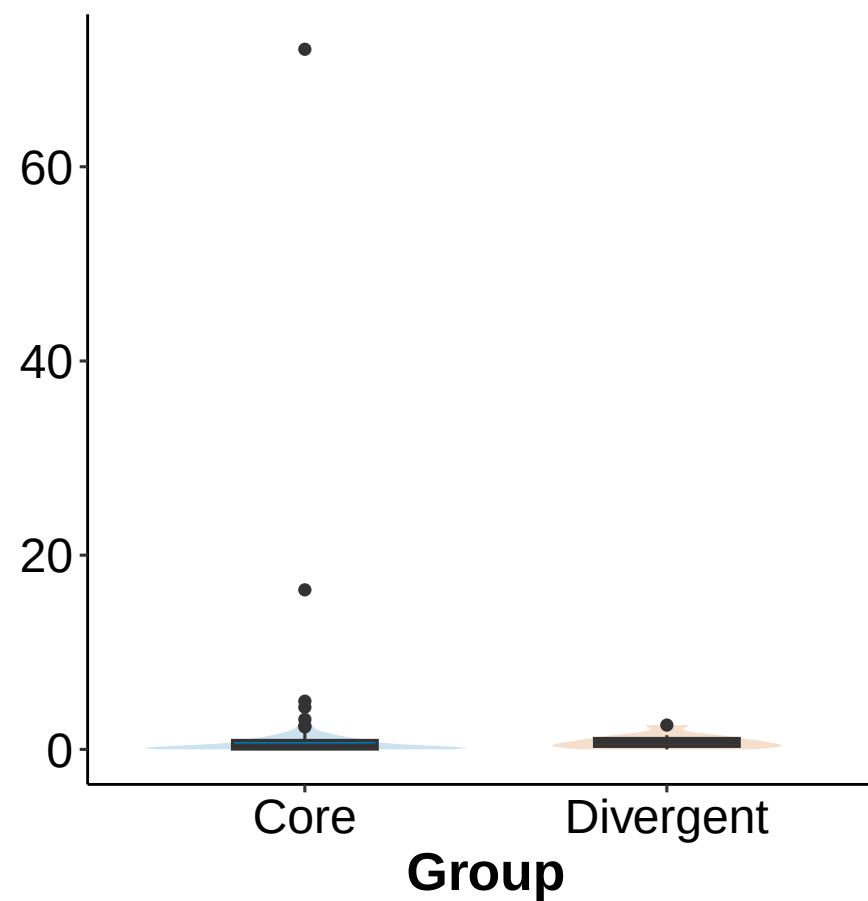
SA_Cholesterol_metabolism_FBA

p-adj: 0.933 | .=0.1 (Negligible) | .%: 808.5



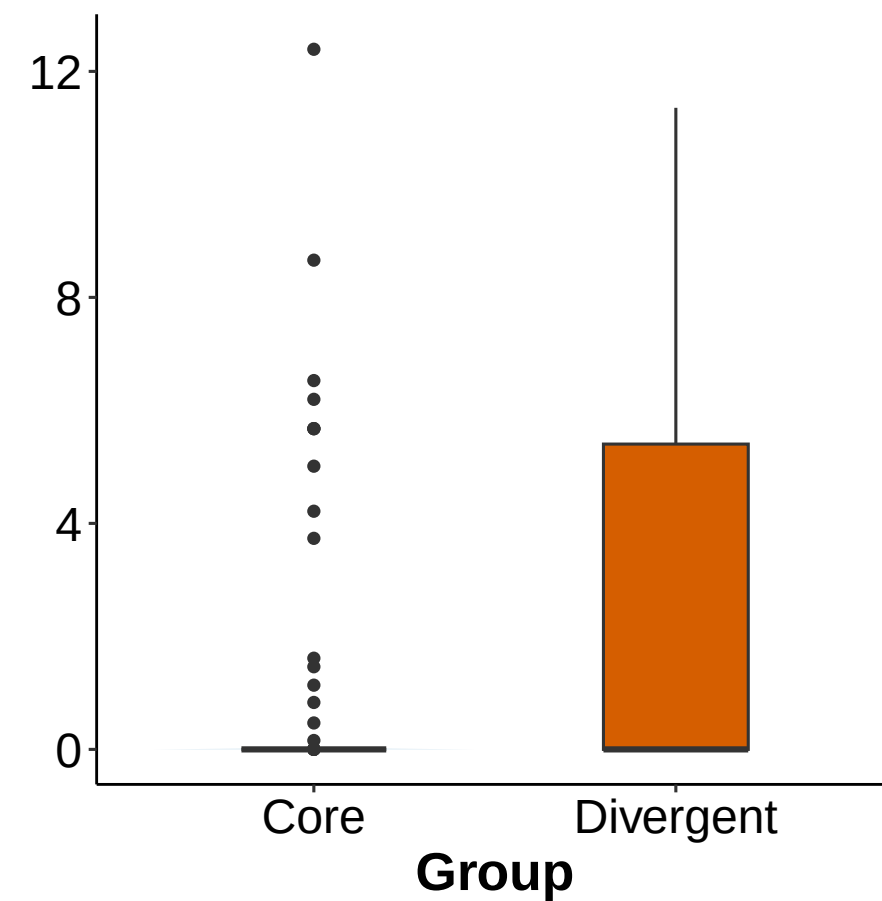
SA_Cholesterol_metabolism_L1w

p-adj: 0.885 | .=0.21 (Small) | .%: 188.7



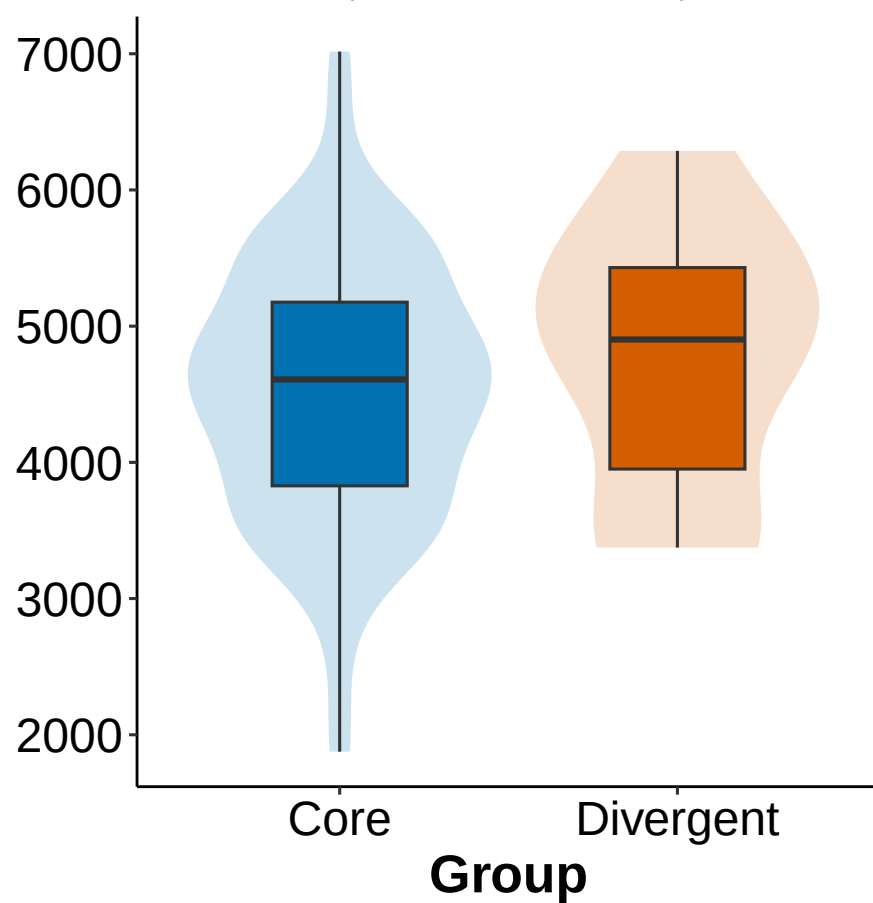
SA_Cholesterol_metabolism_pFBA

p-adj: 0.885 | .=0.15 (Small)



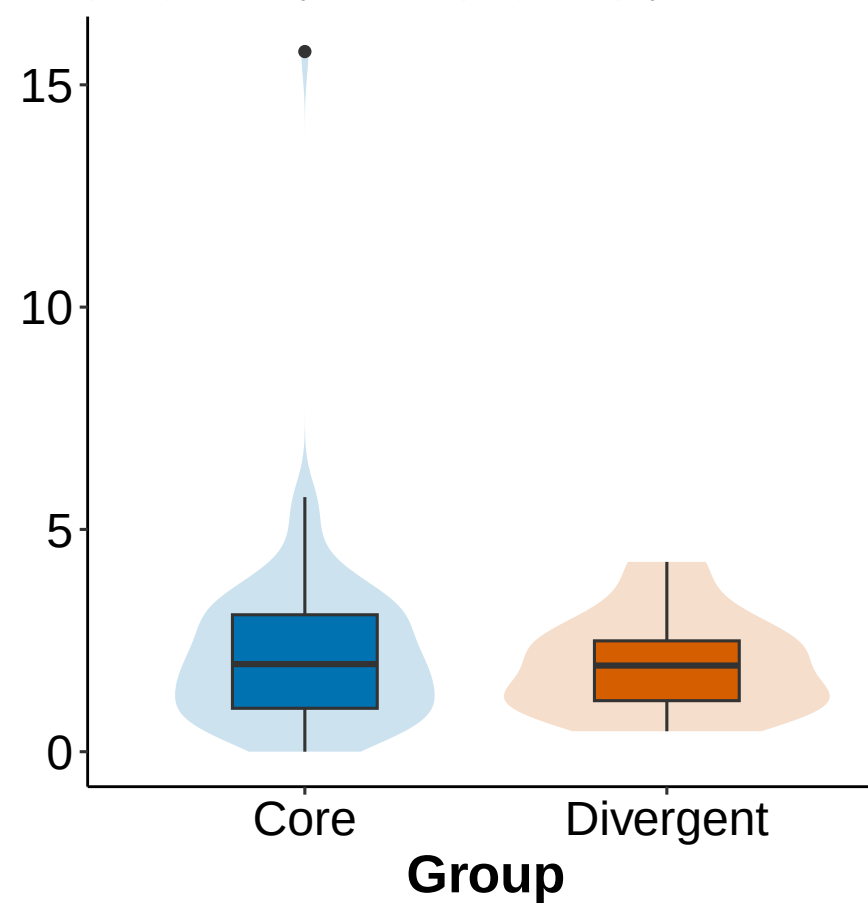
SA_Citric_acid_cycle_FBA

p-adj: 0.896 | .=0.14 (Negligible) | .%: 6.4



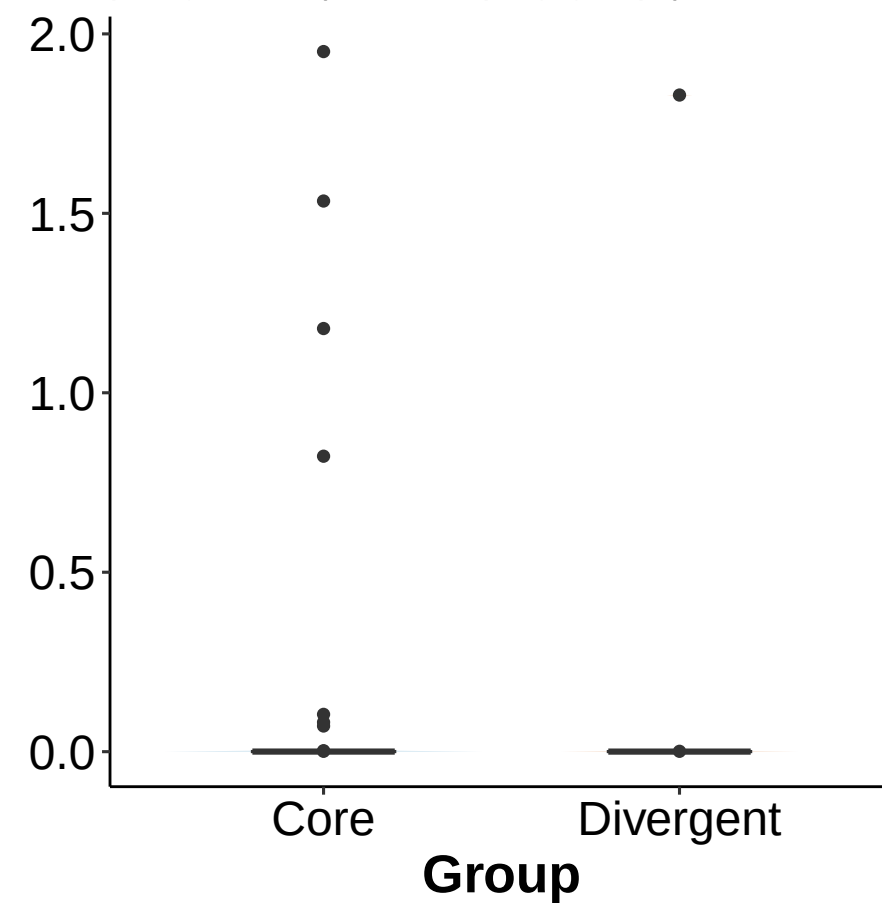
SA_Citric_acid_cycle_L1w

p-adj: 0.954 | .=-0.03 (Negligible) | .%: -1.7



SA_Citric_acid_cycle_pFBA

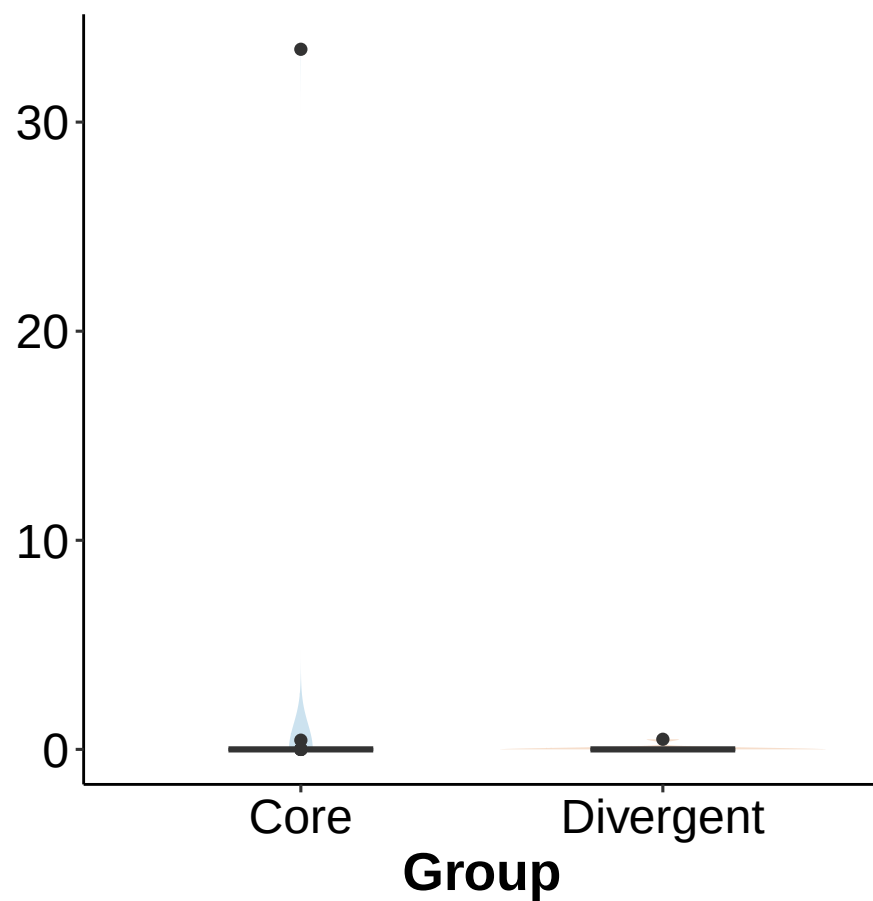
p-adj: 0.945 | .=-0.09 (Negligible) | .%: -24.4



Metabolism – 5. Systemic Reprogramming (SA_*)

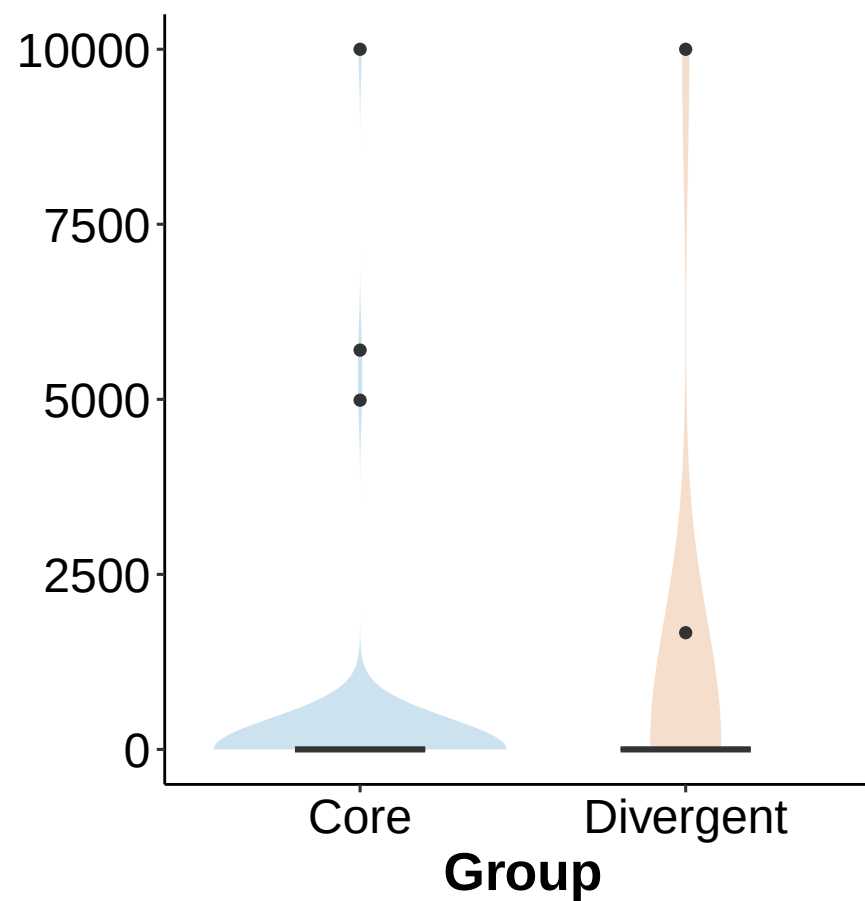
SA_CoA_catabolism_L1w

p-adj: 0.95 | . = 0.02 (Negligible)



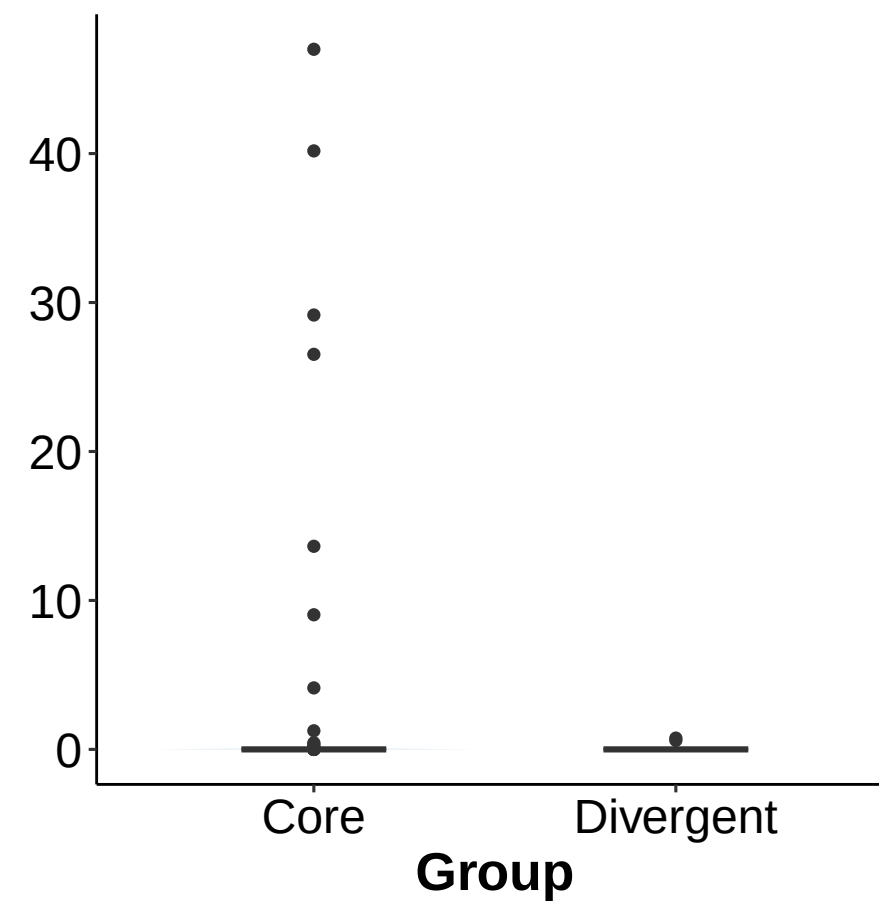
SA_CoA_synthesis_FBA

p-adj: 0.843 | . = 0.15 (Small)



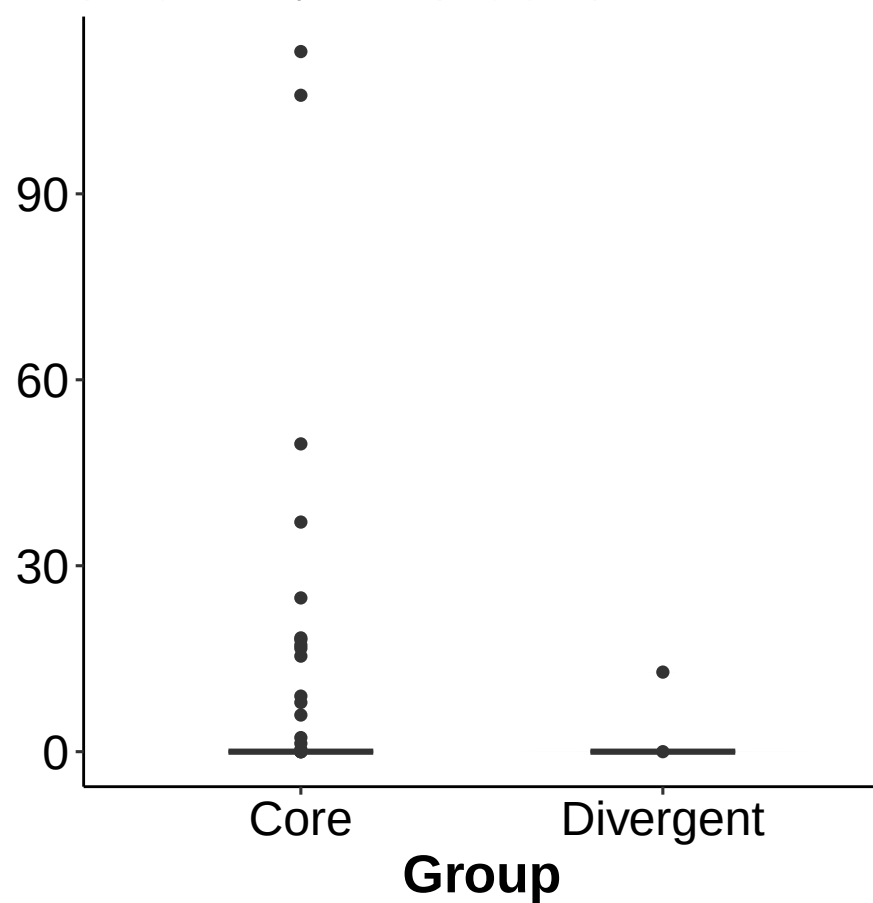
SA_CoA_synthesis_L1w

p-adj: 0.843 | . = 0.3 (Small)



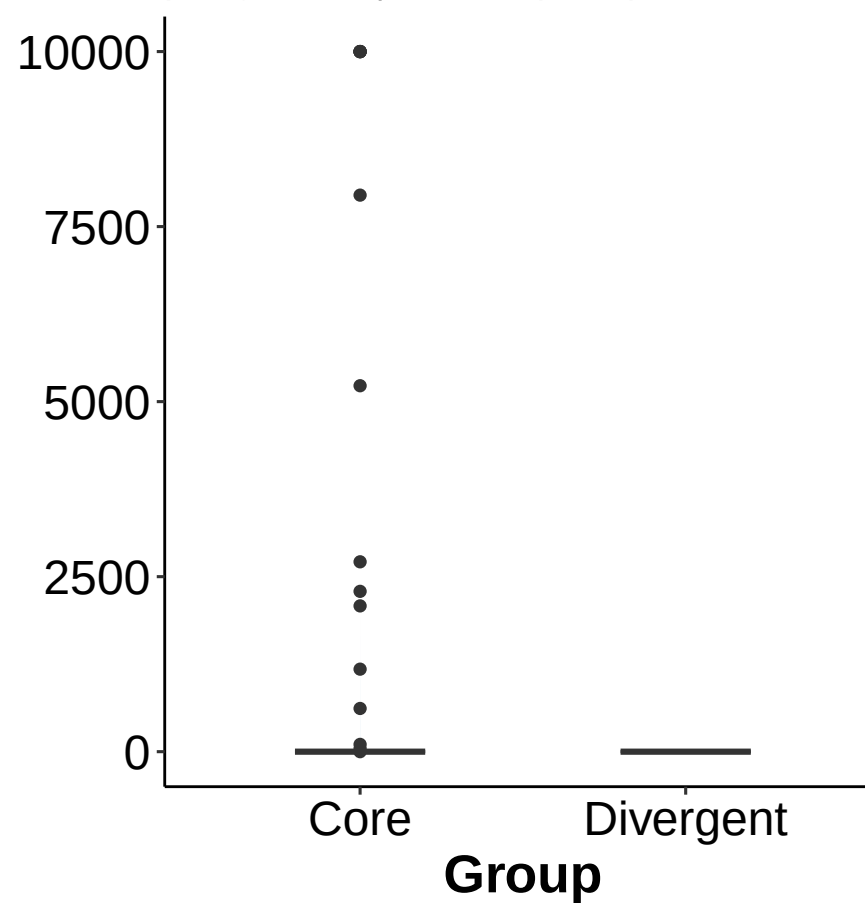
SA_Cytochrome_metabolism_L1w

p-adj: 0.954 | . = 0.02 (Negligible)



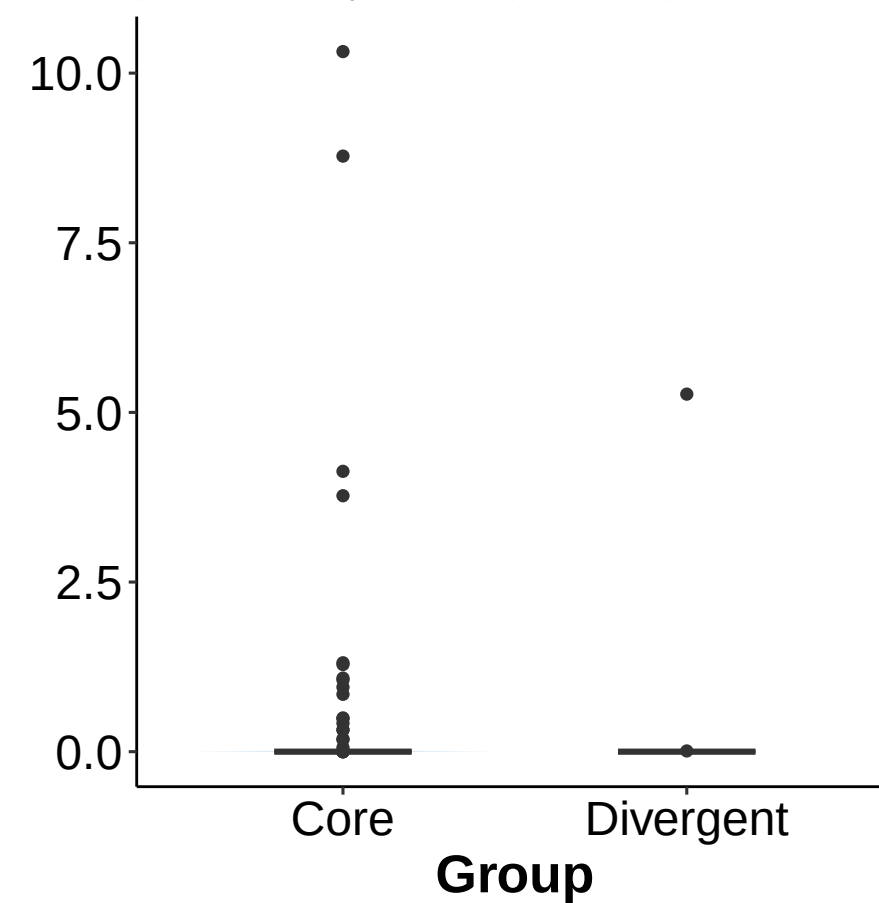
SA_D_alanine_metabolism_FBA

p-adj: 0.885 | . = -0.15 (Small)



SA_D_alanine_metabolism_L1w

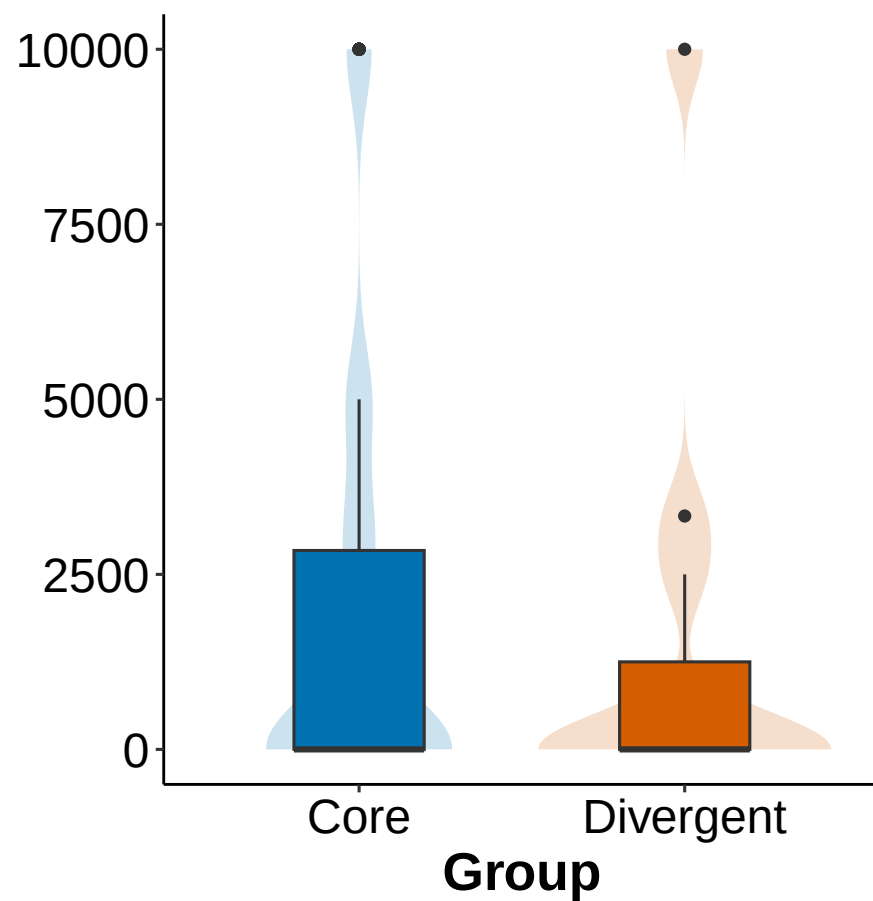
p-adj: 0.896 | . = -0.11 (Negligible)



Metabolism – . 5. Systemic Reprogramming (SA_*)

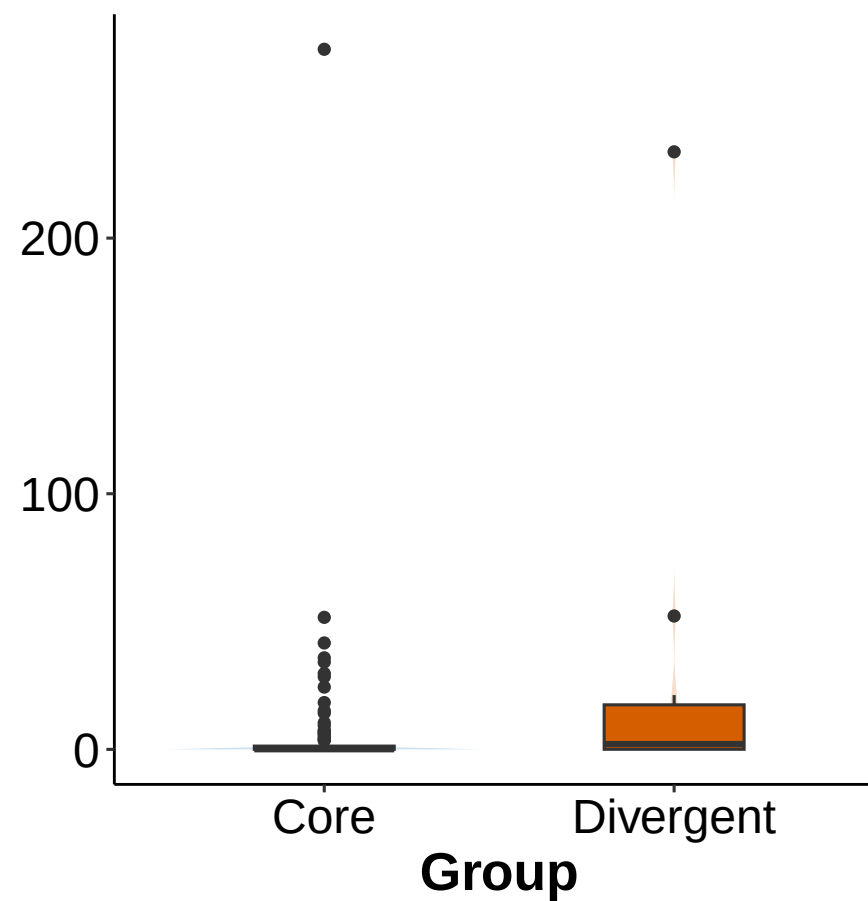
SA_Dietary_fiber_binding_FBA

p-adj: 0.95 | .=-0.07 (Negligible)



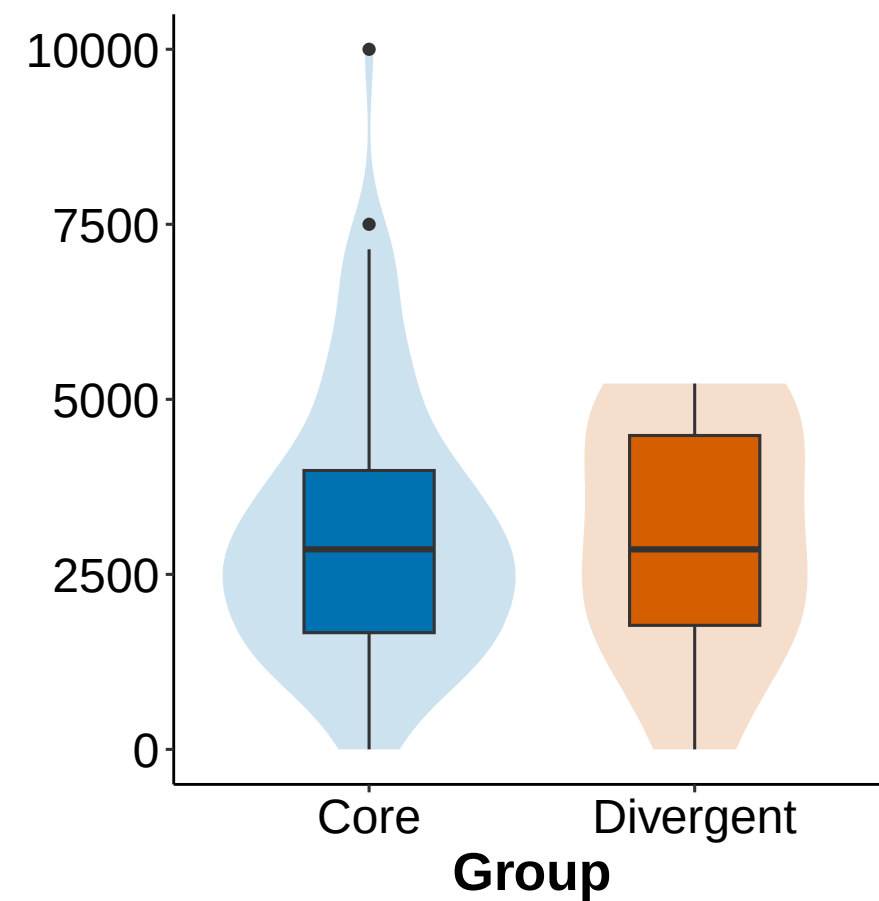
SA_Dietary_fiber_binding_L1w

p-adj: 0.843 | .:=0.33 (Small) | .%: 4257.3



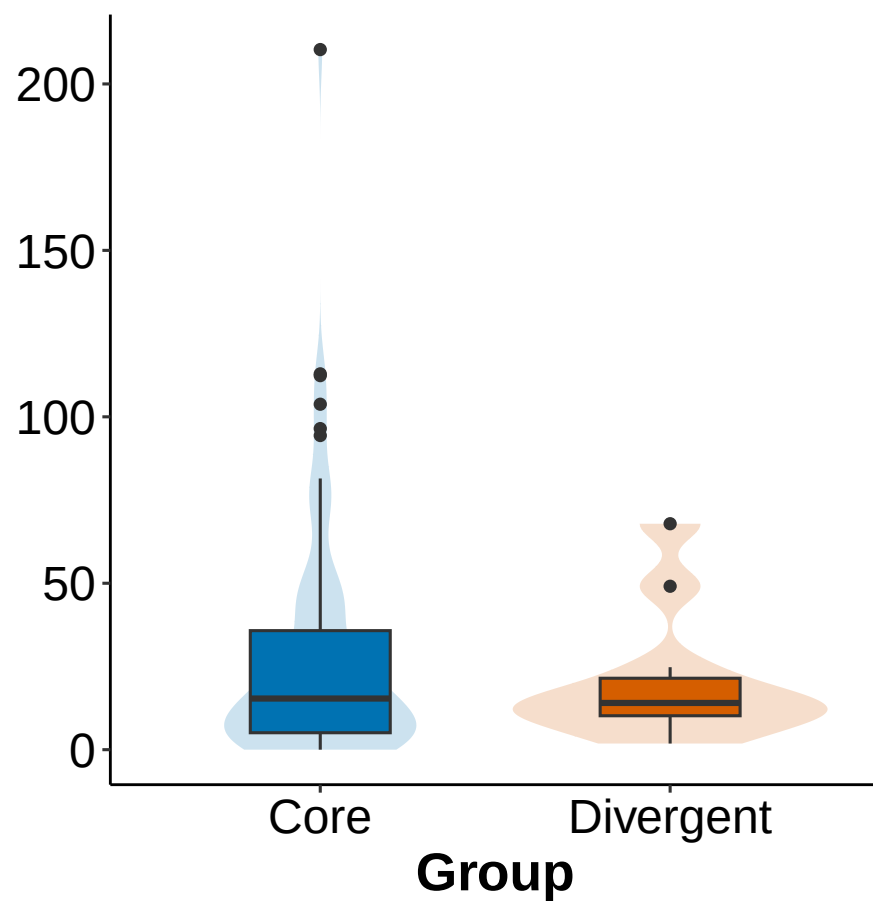
SA_Drug_metabolism_FBA

p-adj: 0.954 | .:=0.04 (Negligible) | .%: 0



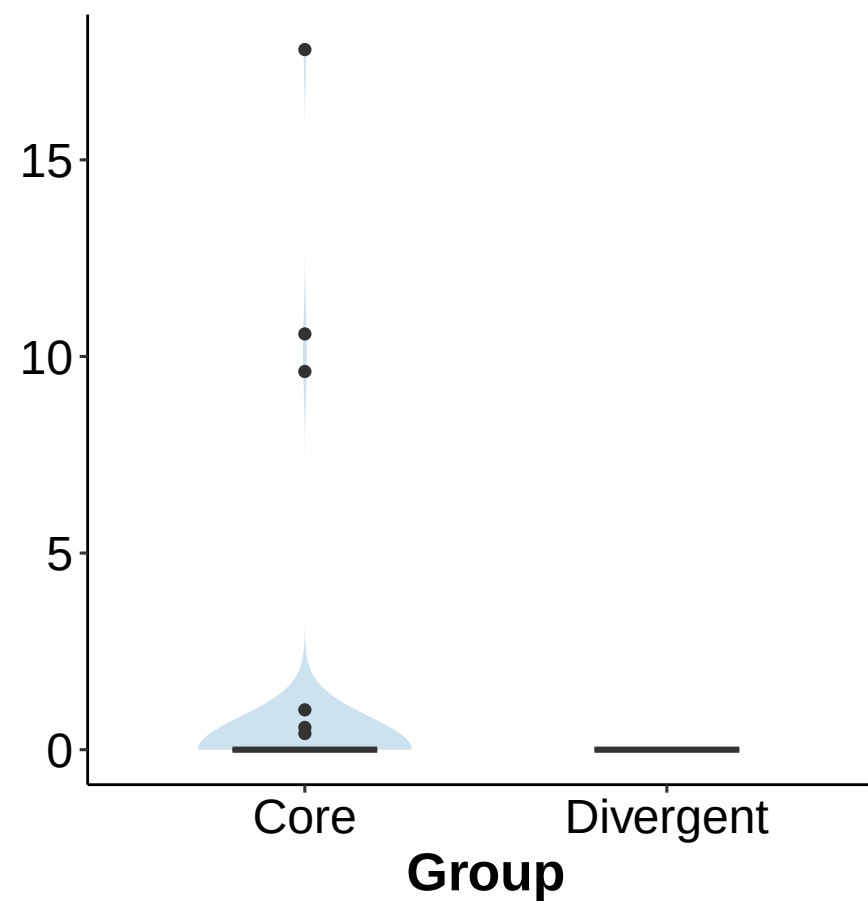
SA_Drug_metabolism_L1w

p-adj: 0.954 | .:=0.04 (Negligible) | .%: -8.6



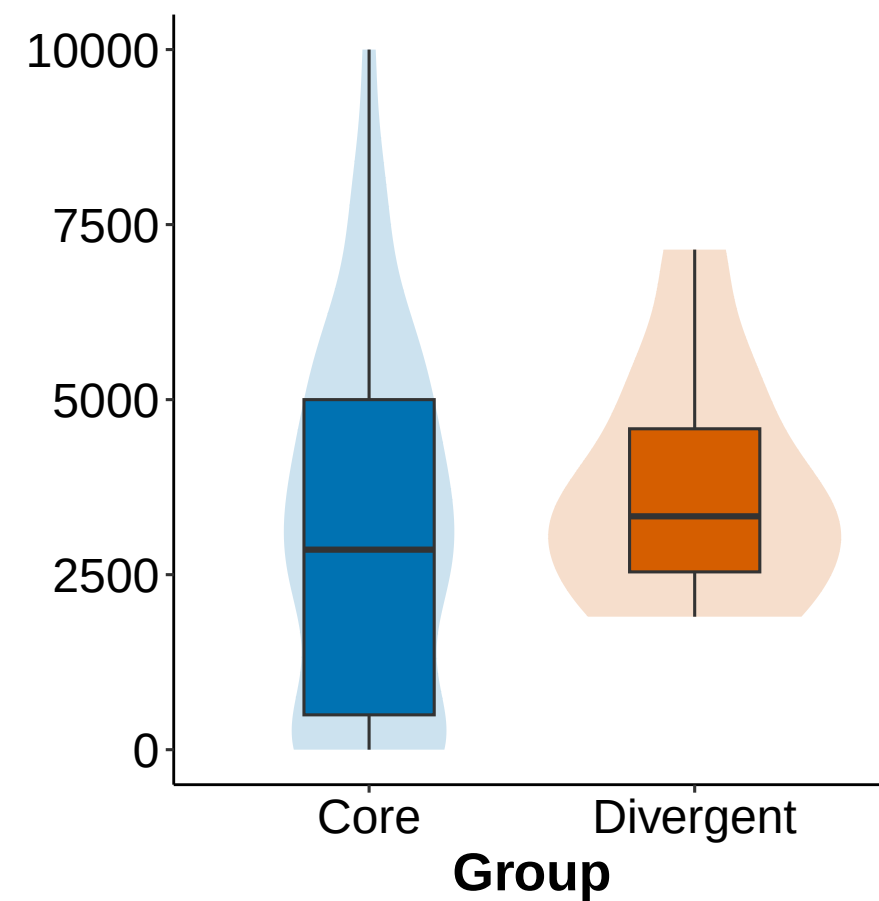
SA_Drug_metabolism_pFBA

p-adj: 0.885 | .=-0.06 (Negligible)

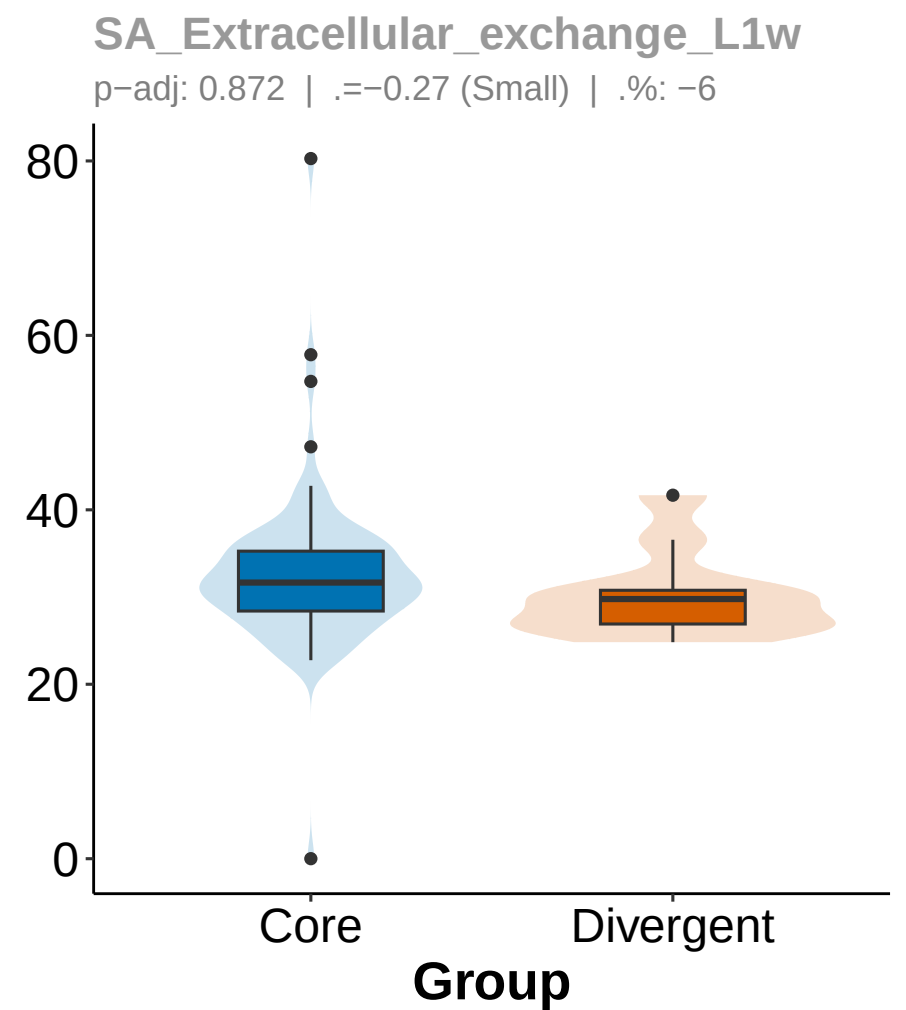
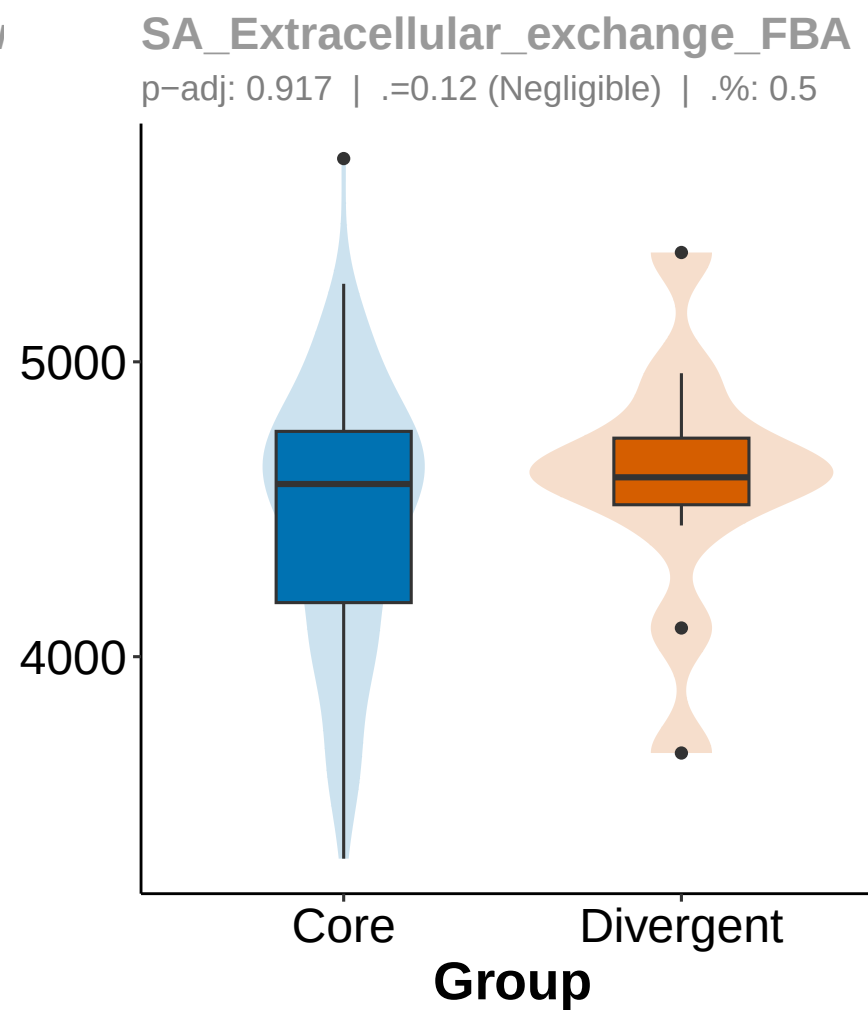
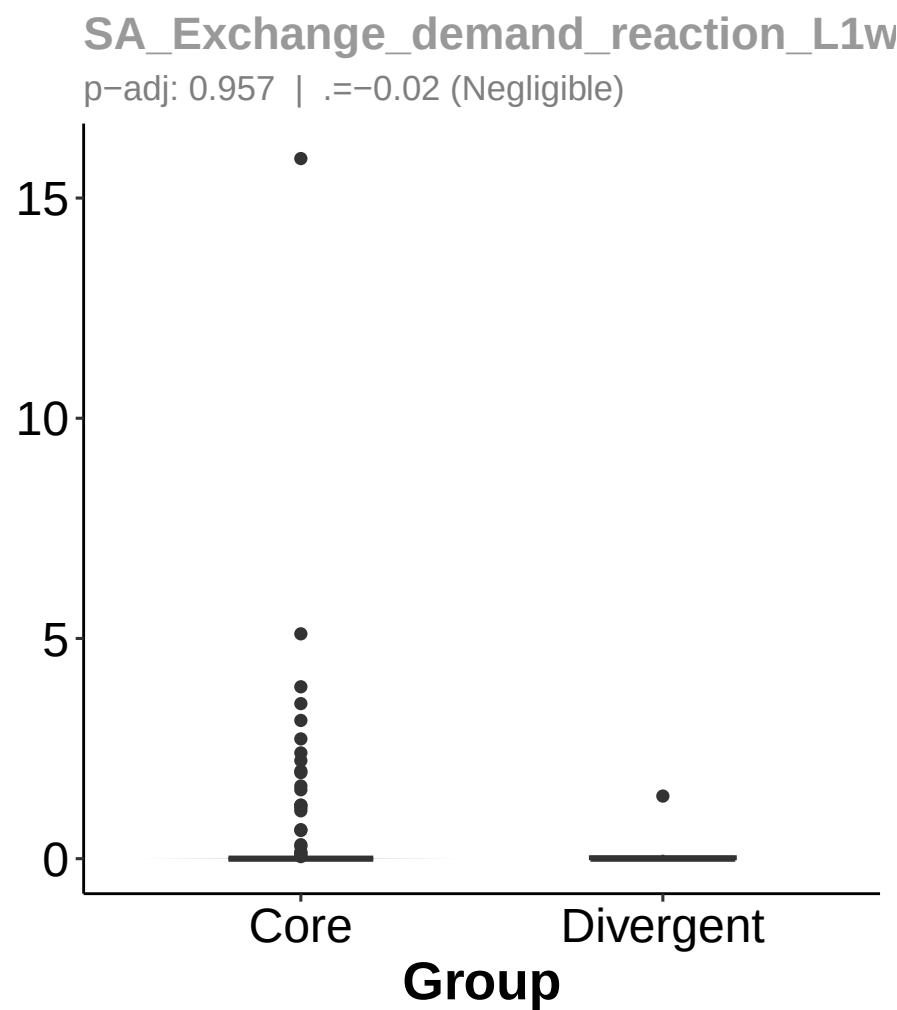
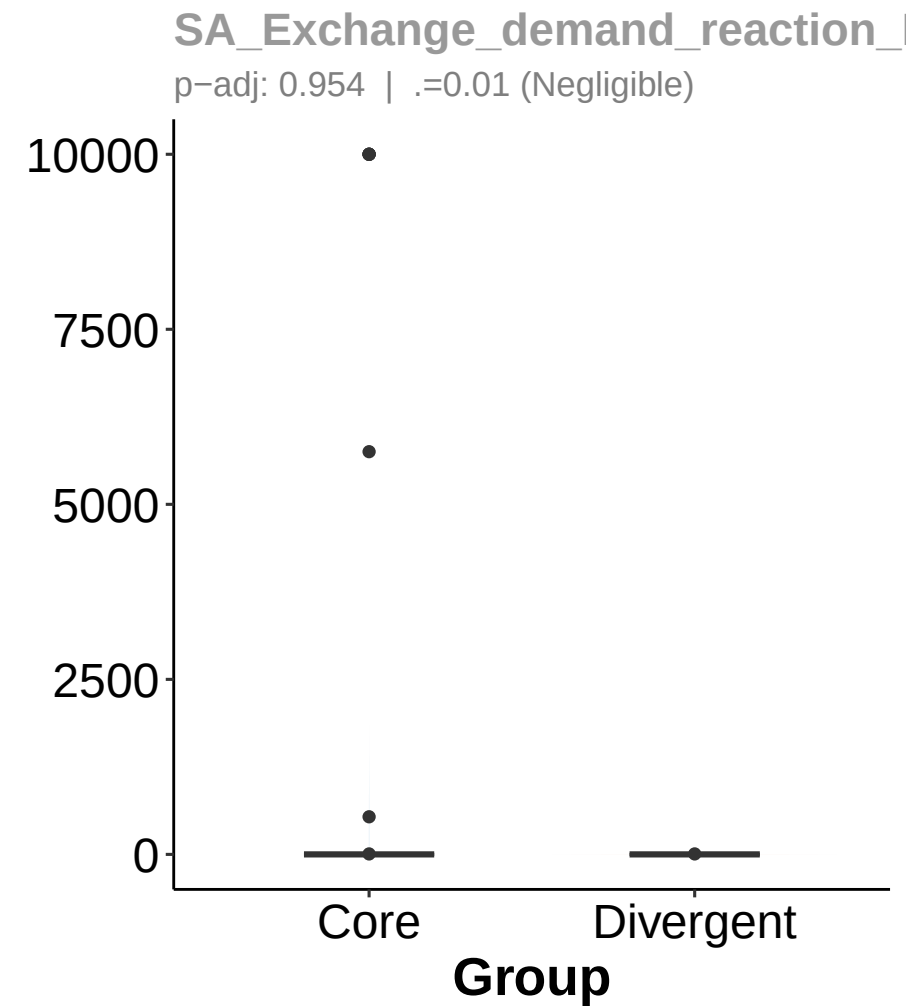
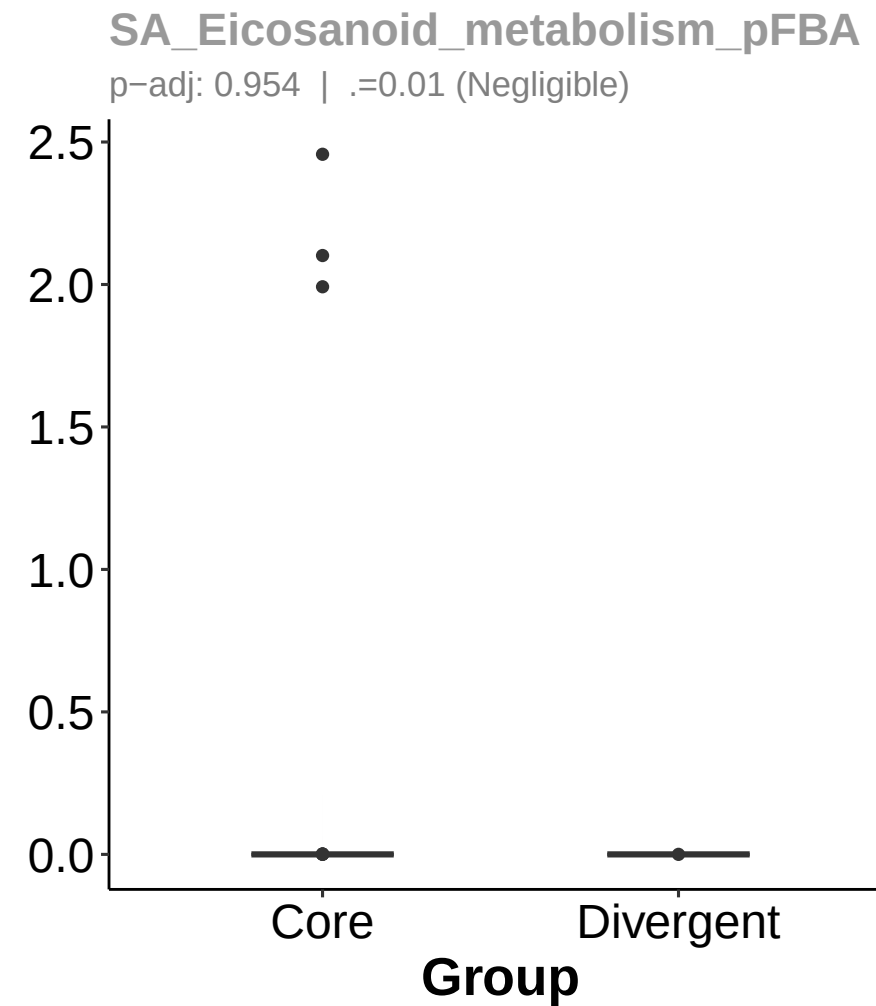
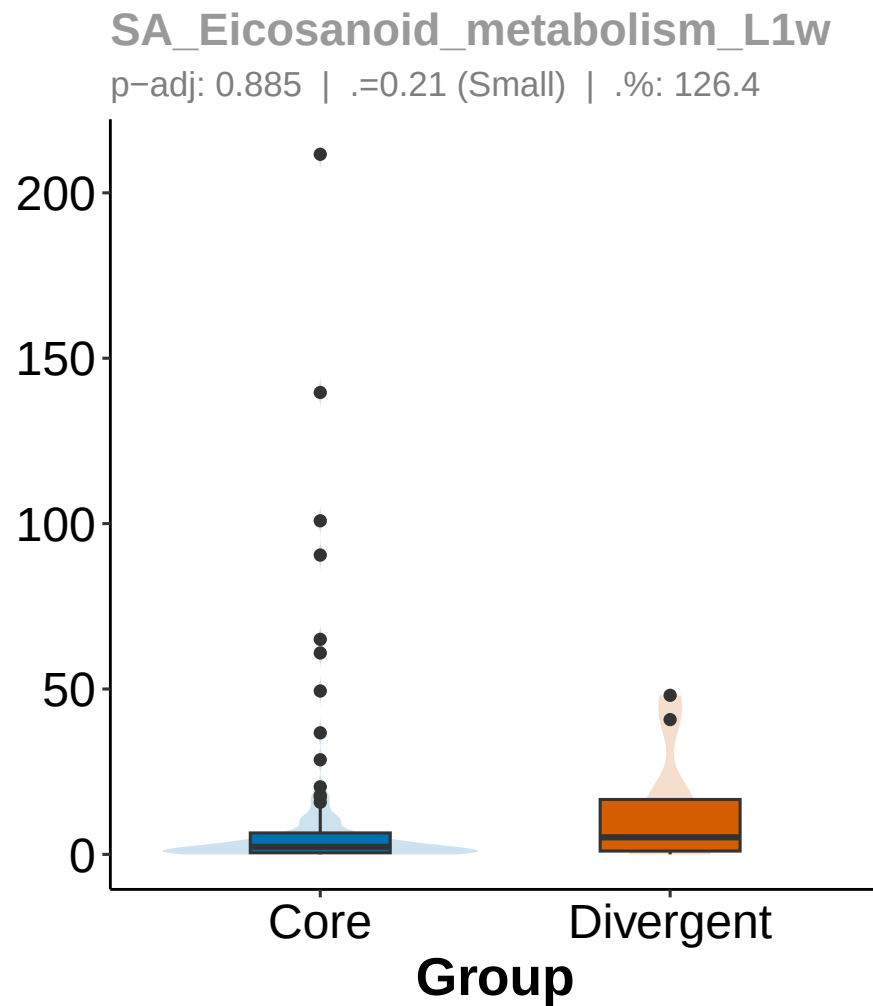


SA_Eicosanoid_metabolism_FBA

p-adj: 0.885 | .:=0.17 (Small) | .%: 16.7



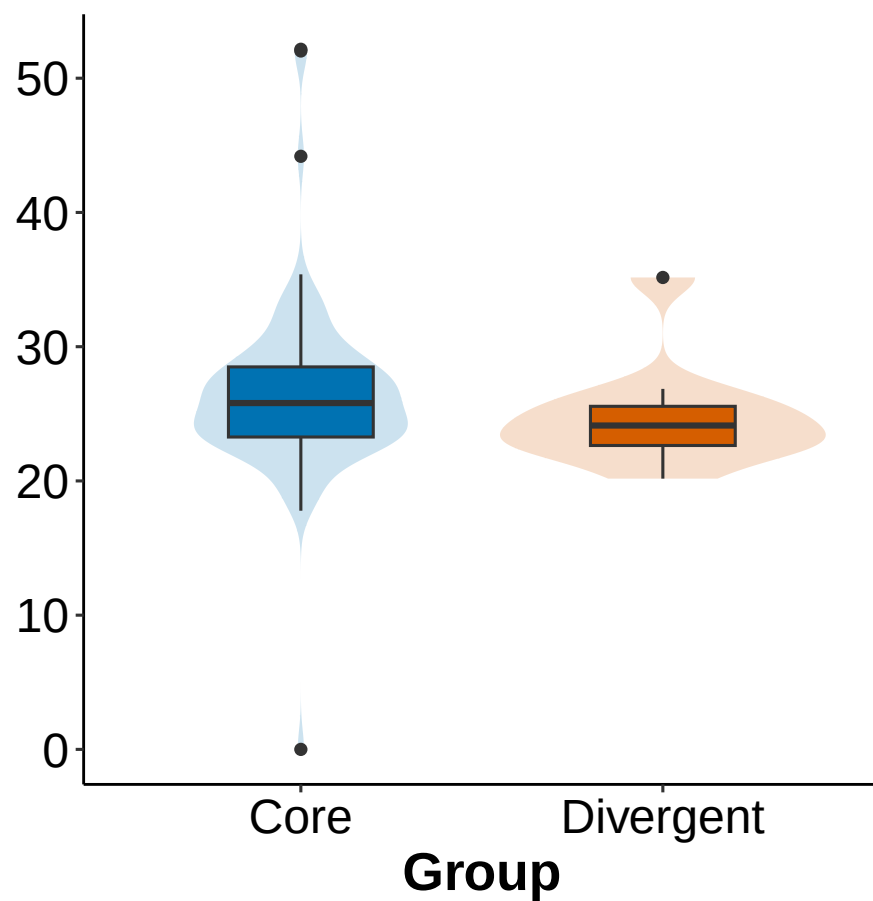
Metabolism – . 5. Systemic Reprogramming (SA_*)



Metabolism – . 5. Systemic Reprogramming (SA_*)

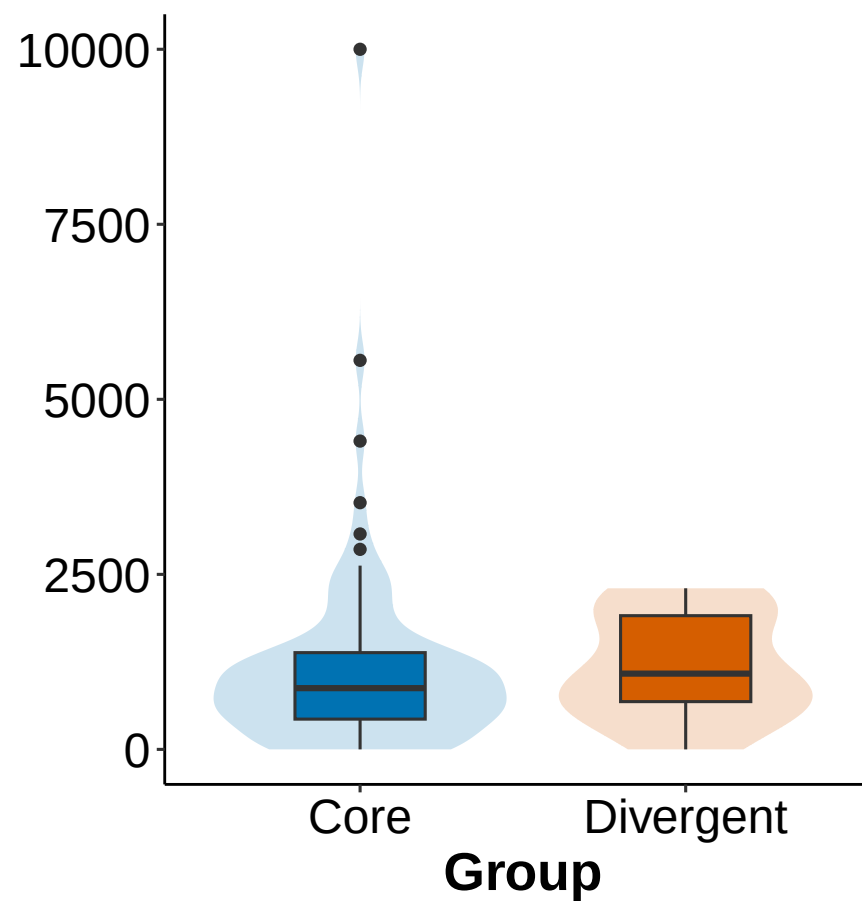
SA_Extracellular_exchange_pFBA

p-adj: 0.872 | .=-0.27 (Small) | .%: -6.5



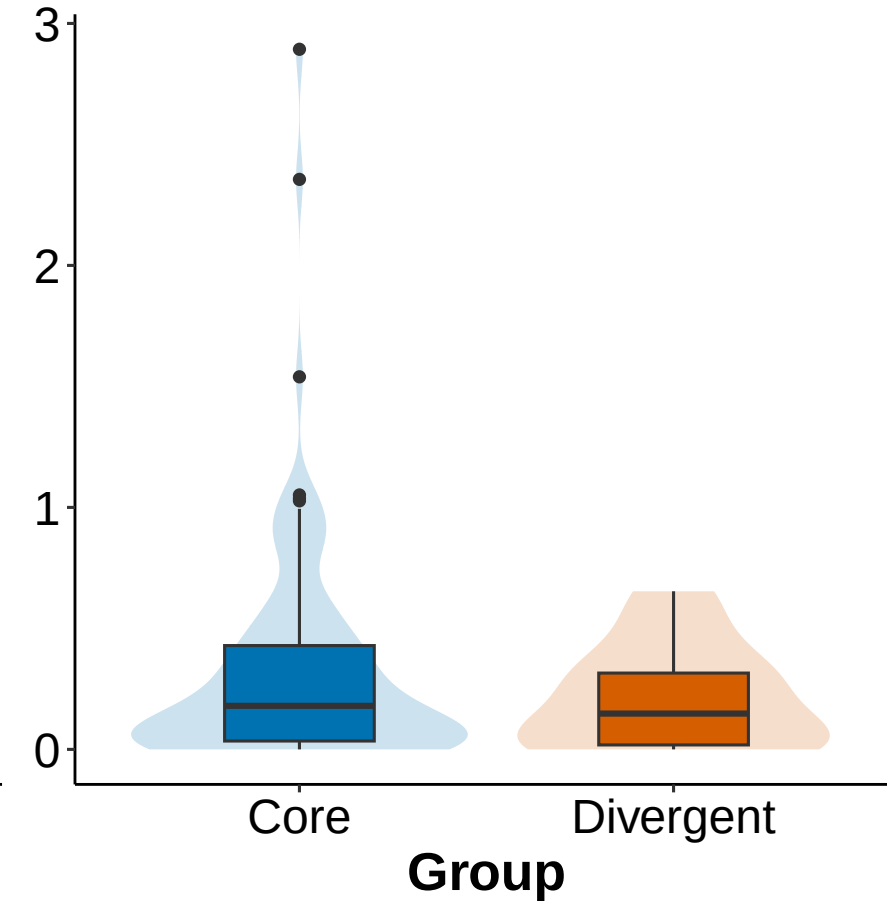
SA_Fatty_acid_oxidation_FBA

p-adj: 0.896 | .=-0.14 (Negligible) | .%: 23.7



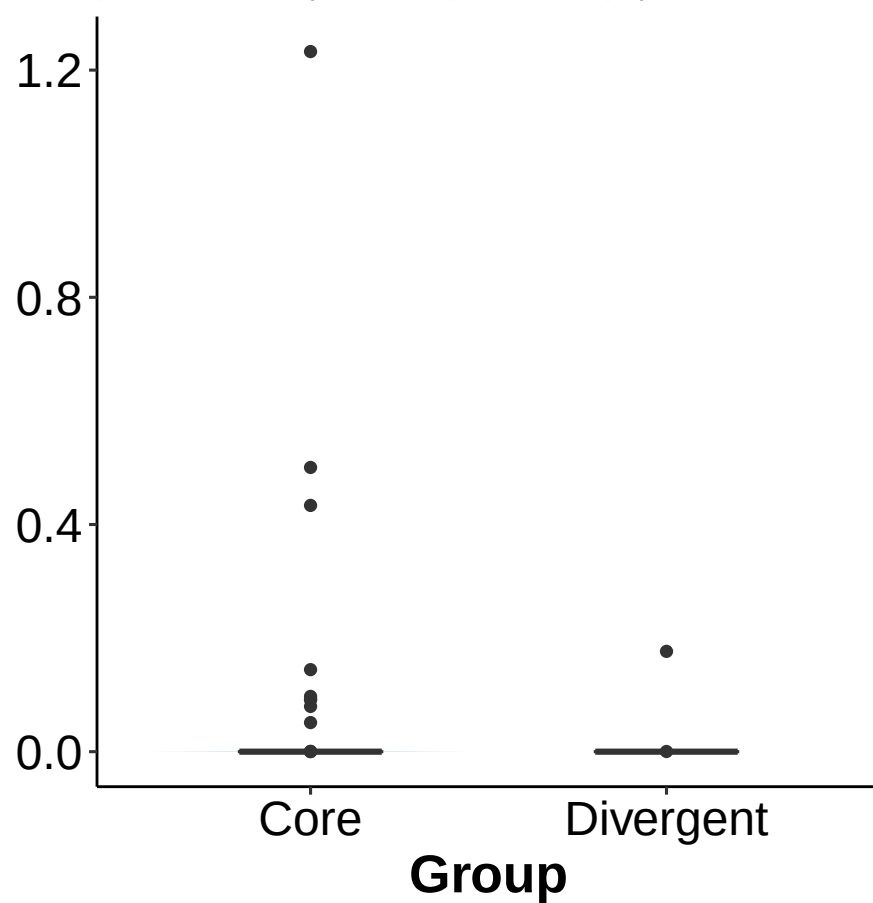
SA_Fatty_acid_oxidation_L1w

p-adj: 0.928 | .=-0.1 (Negligible) | .%: -17.8



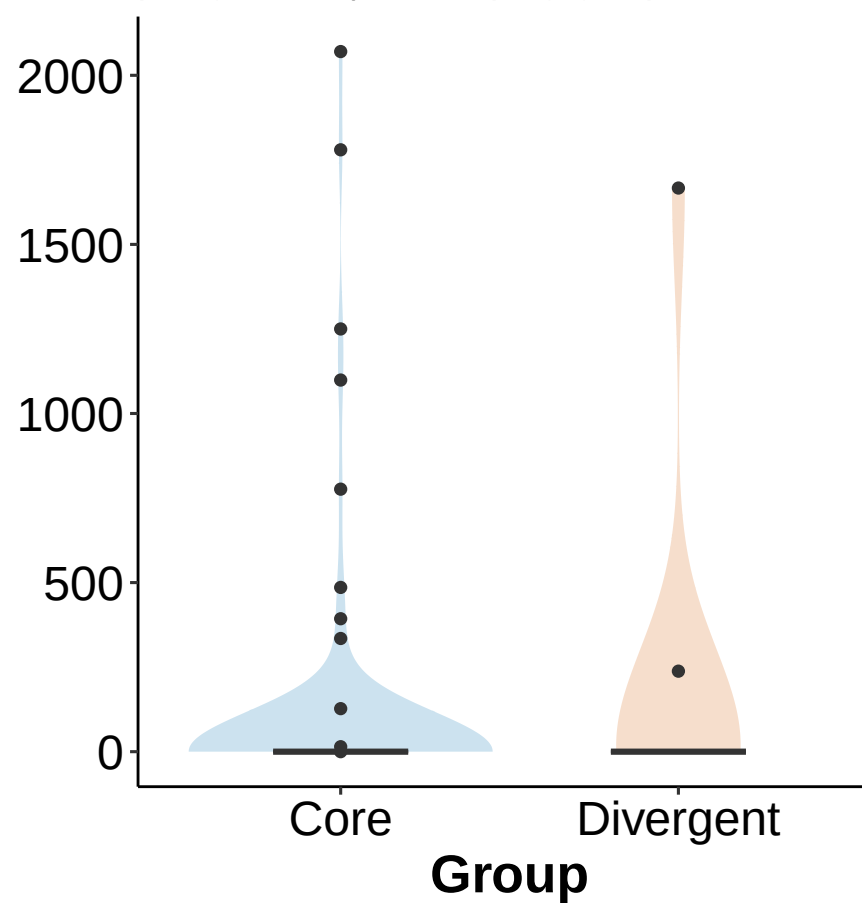
SA_Fatty_acid_oxidation_pFBA

p-adj: 0.896 | .=-0.13 (Negligible) | .%: 126



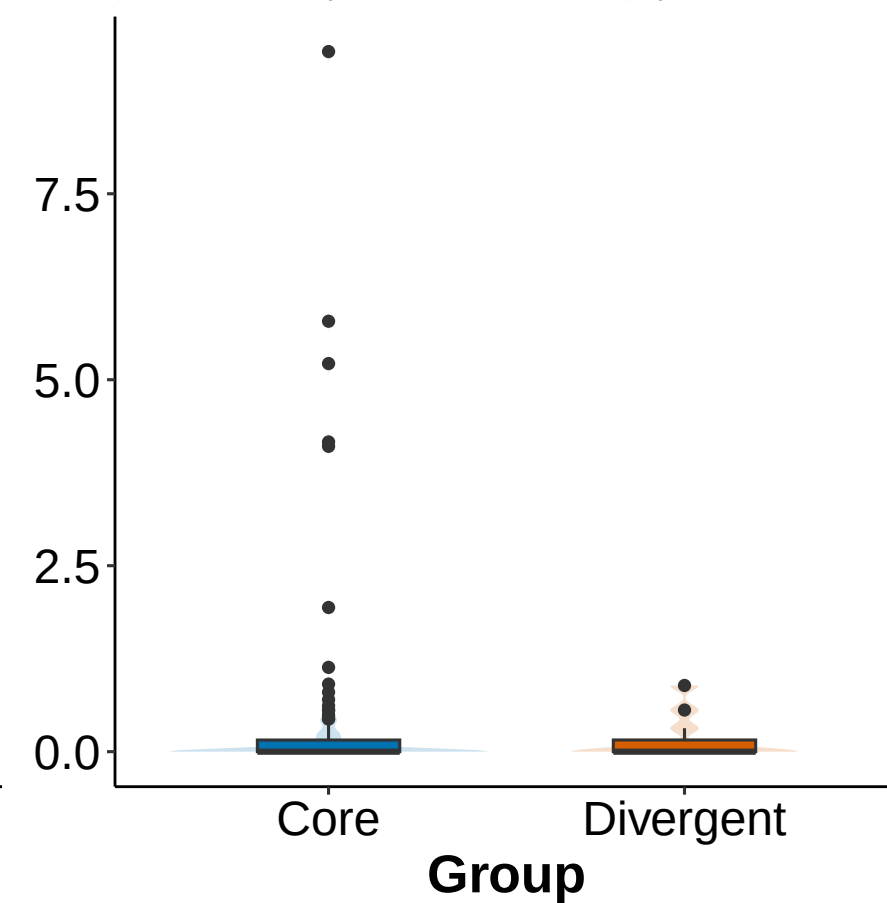
SA_Fatty_acid_synthesis_FBA

p-adj: 0.917 | .=-0.07 (Negligible)



SA_Fatty_acid_synthesis_L1w

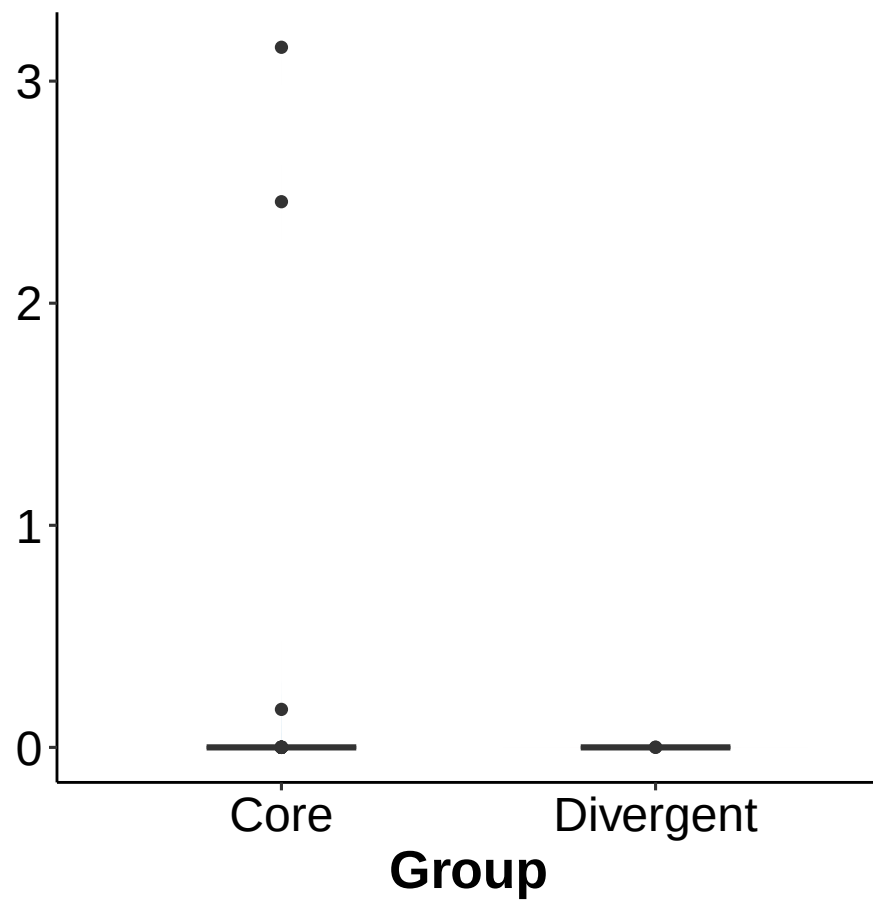
p-adj: 0.942 | .=-0.1 (Negligible) | .%: -78.5



Metabolism – . 5. Systemic Reprogramming (SA_*)

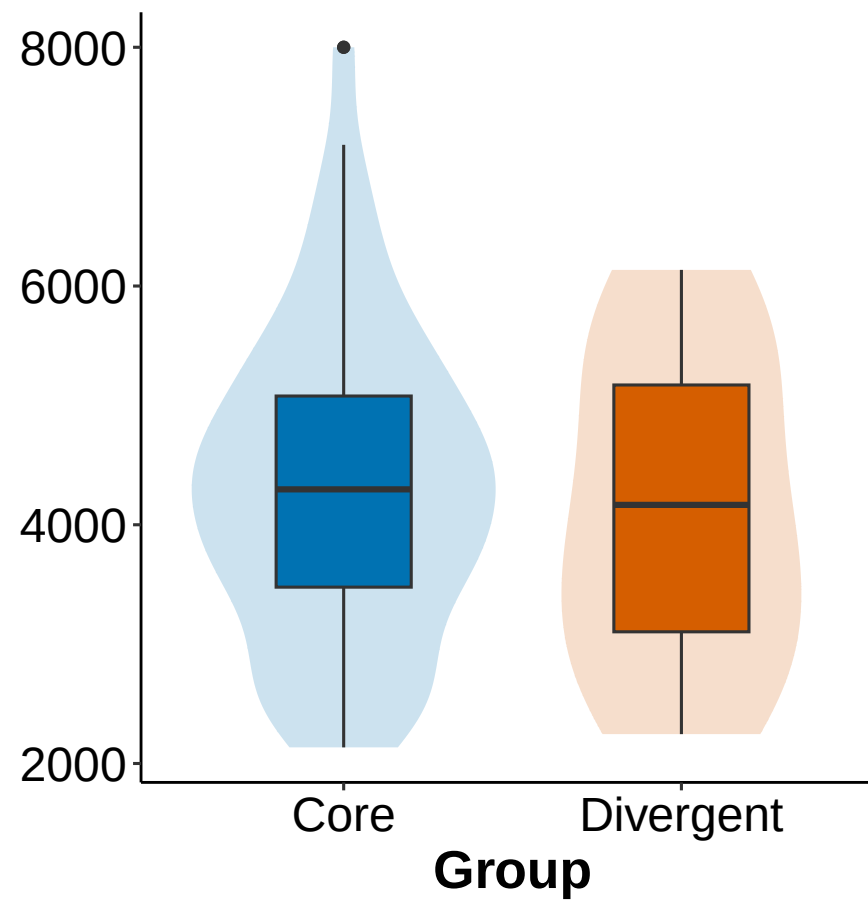
SA_Fatty_acid_synthesis_pFBA

p-adj: 0.95 | .:=0.04 (Negligible)



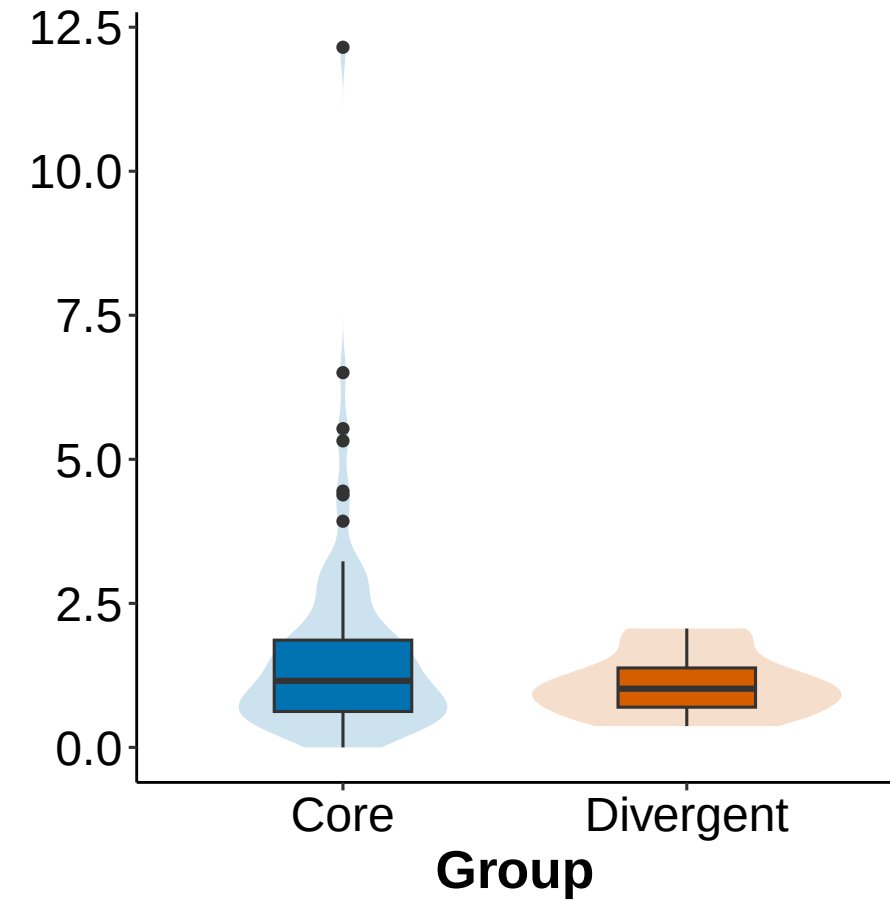
SA_Folate_metabolism_FBA

p-adj: 0.95 | .:=−0.09 (Negligible) | .%: −3



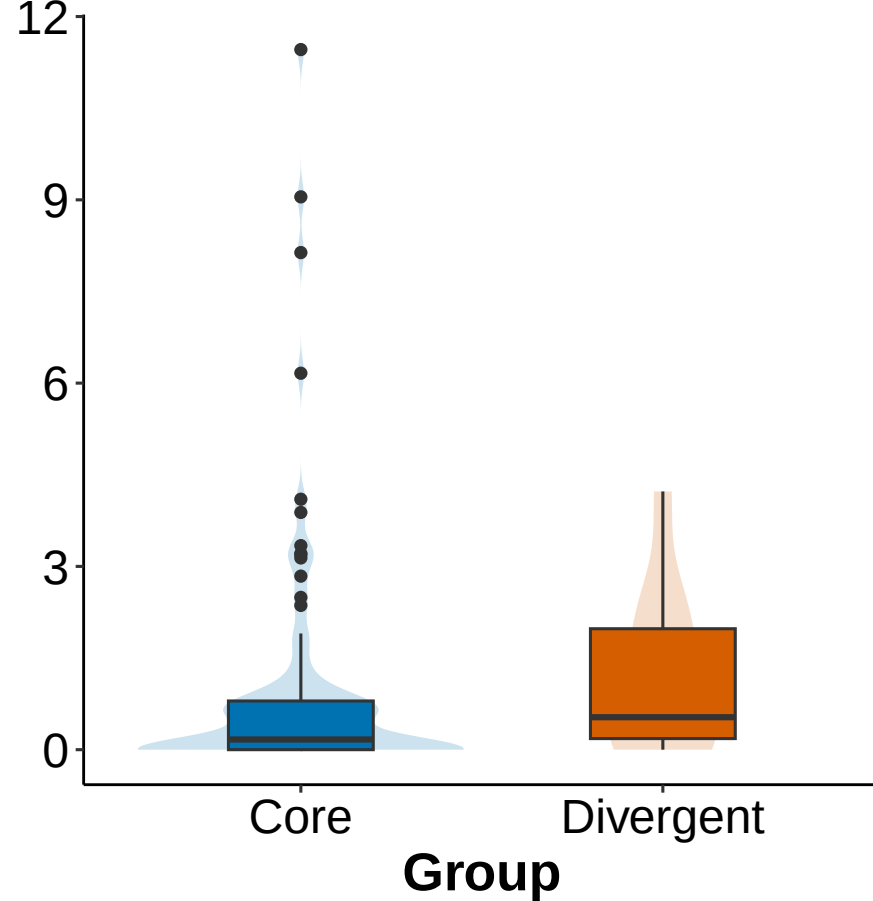
SA_Folate_metabolism_L1w

p-adj: 0.917 | .:=−0.11 (Negligible) | .%: −11.8



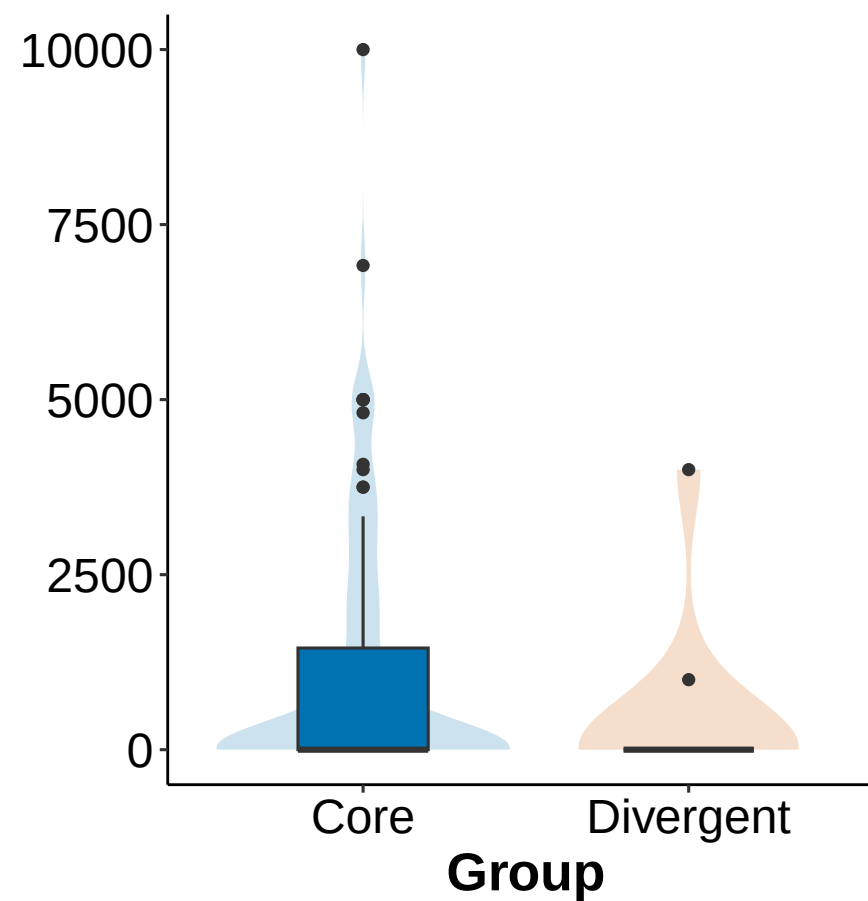
SA_Folate_metabolism_pFBA

p-adj: 0.885 | .:=0.23 (Small) | .%: 218.7



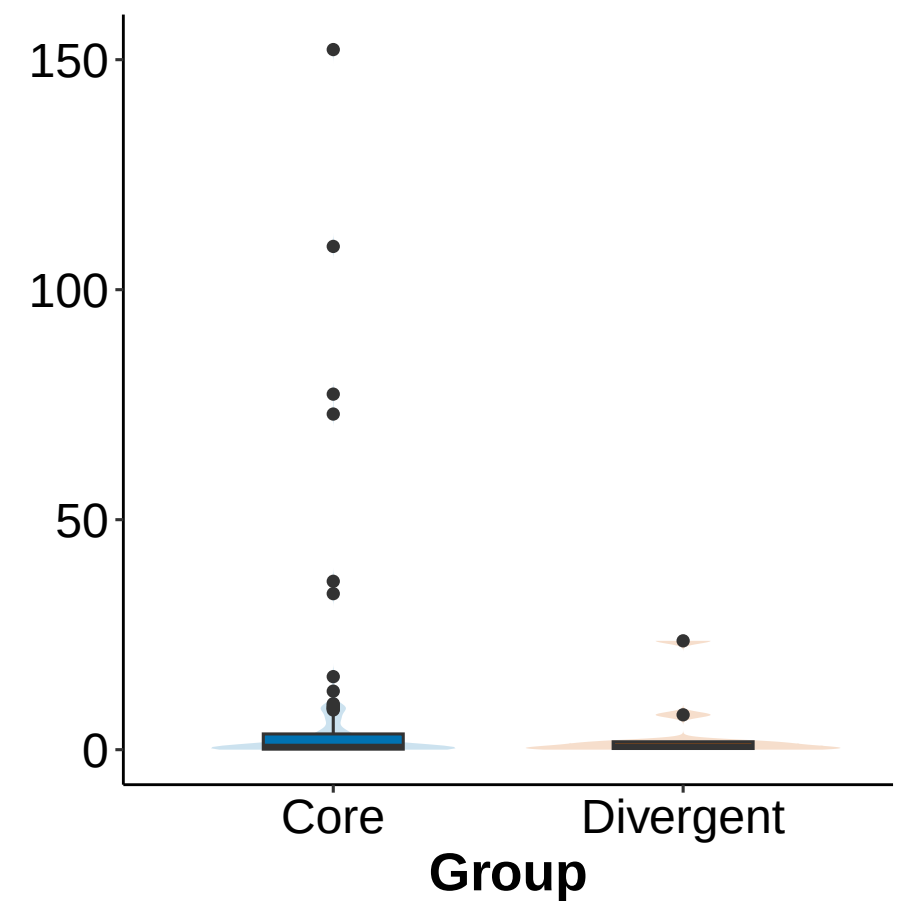
SA_Fructose_and_mannose_metabo

p-adj: 0.885 | .:=−0.22 (Small)



SA_Fructose_and_mannose_metabo

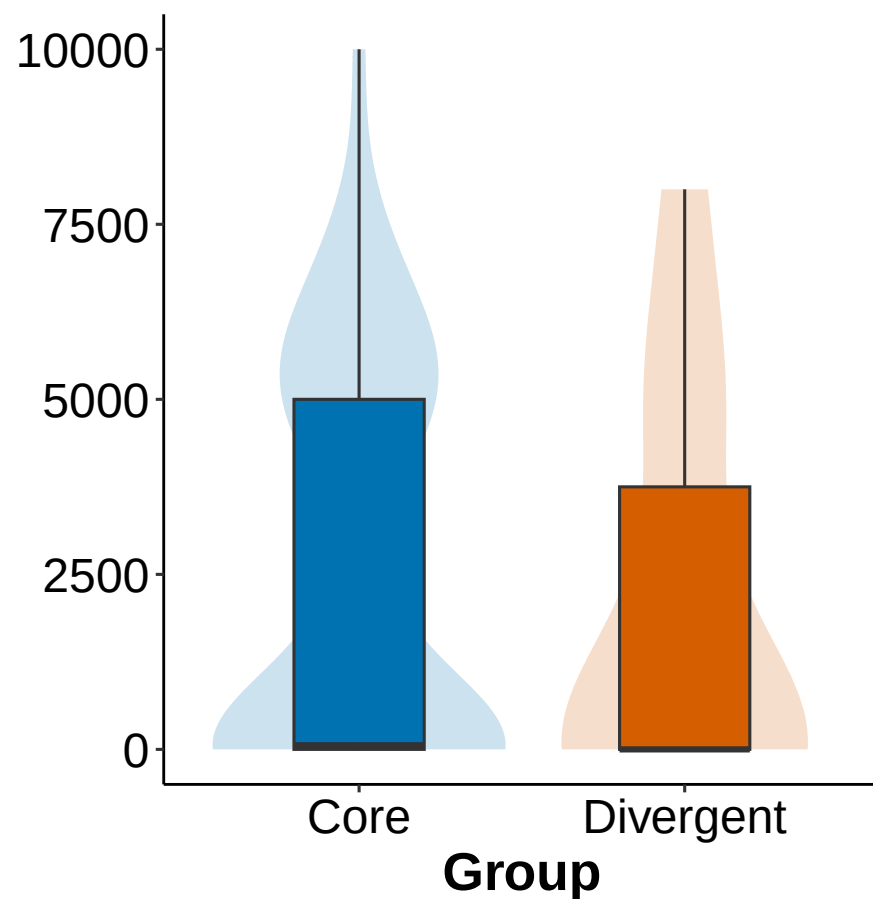
p-adj: 0.959 | .:=0.02 (Negligible) | .%: 7.9



Metabolism – . 5. Systemic Reprogramming (SA_*)

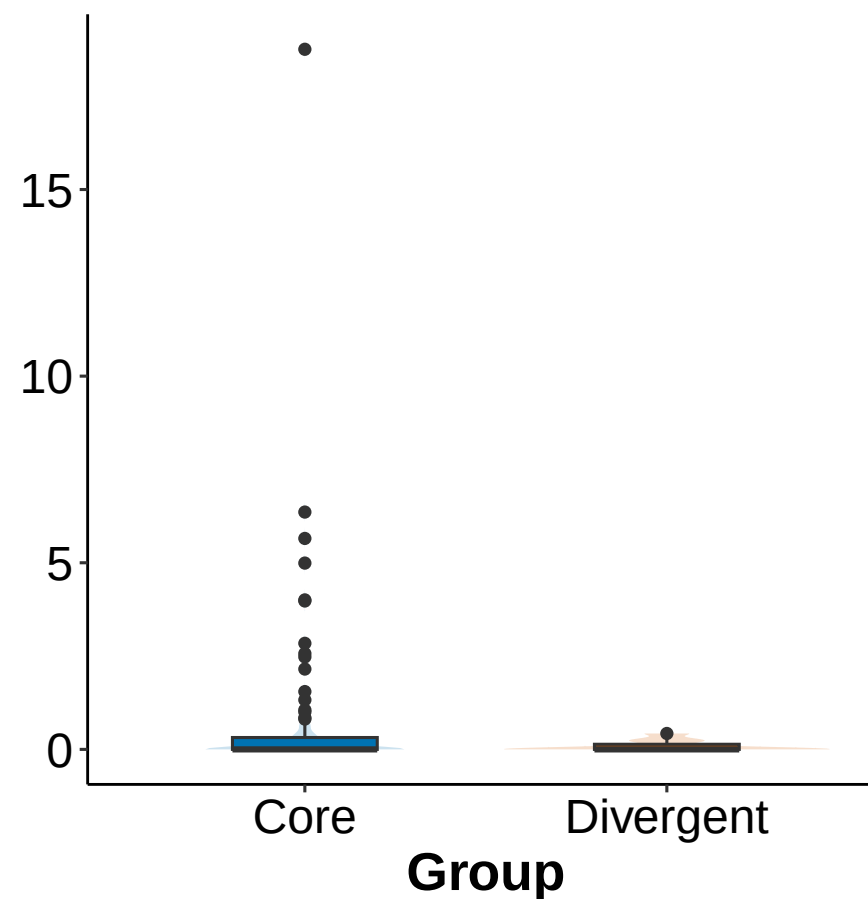
SA_Galactose_metabolism_FBA

p-adj: 0.896 | .=-0.13 (Negligible) | .%: -100



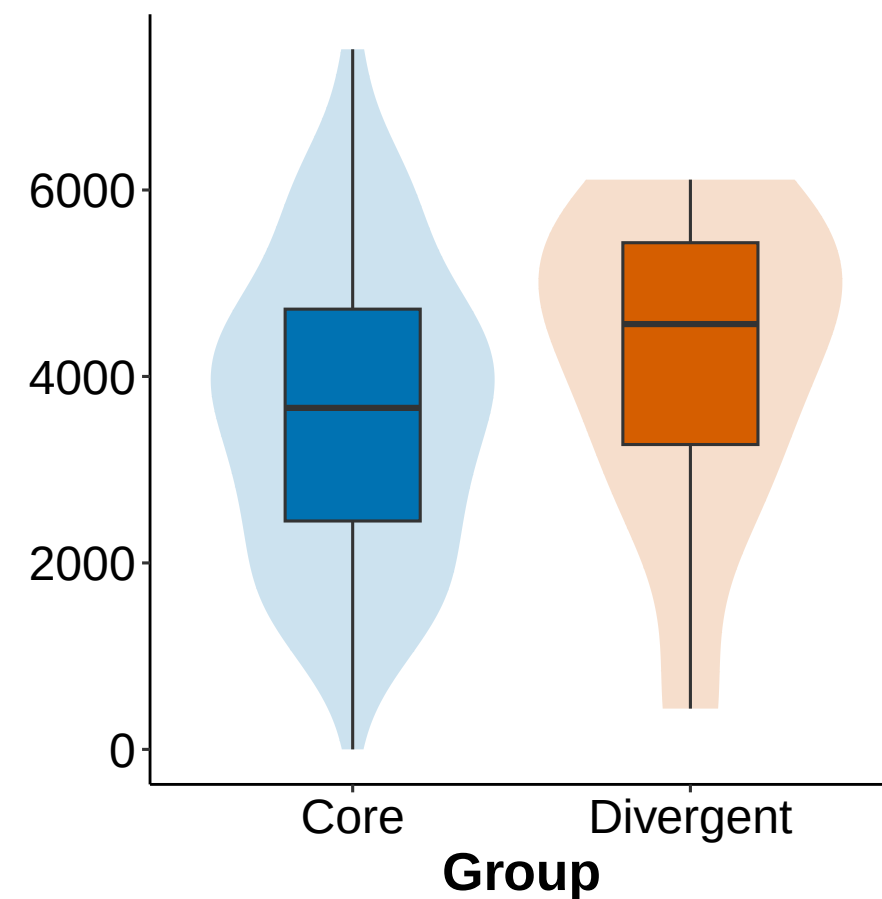
SA_Galactose_metabolism_L1w

p-adj: 0.992 | .=0 (Negligible) | .%: 69.1



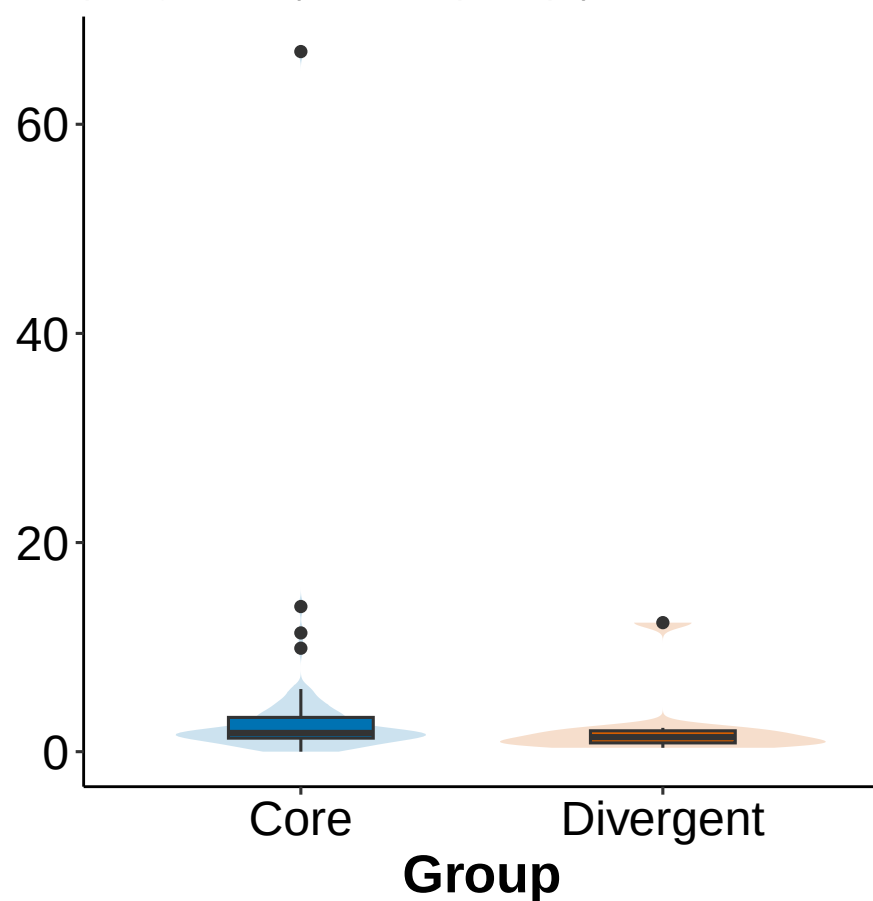
SA_Glutamate_metabolism_FBA

p-adj: 0.885 | .=0.24 (Small) | .%: 24.5



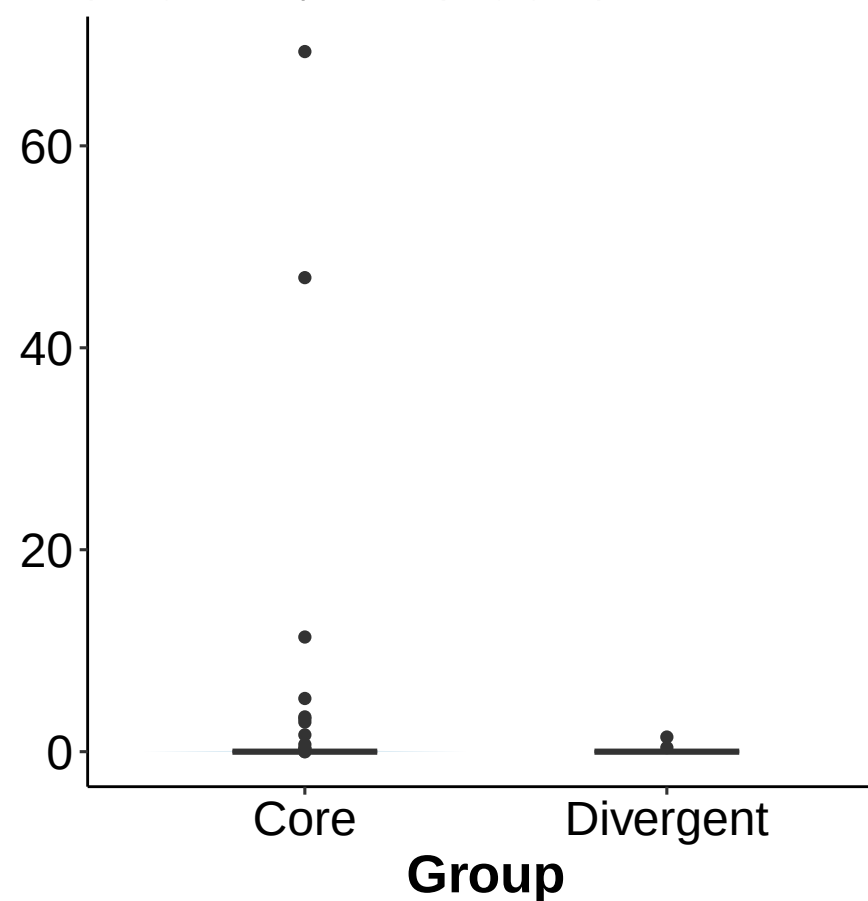
SA_Glutamate_metabolism_L1w

p-adj: 0.872 | .=-0.26 (Small) | .%: -22.4



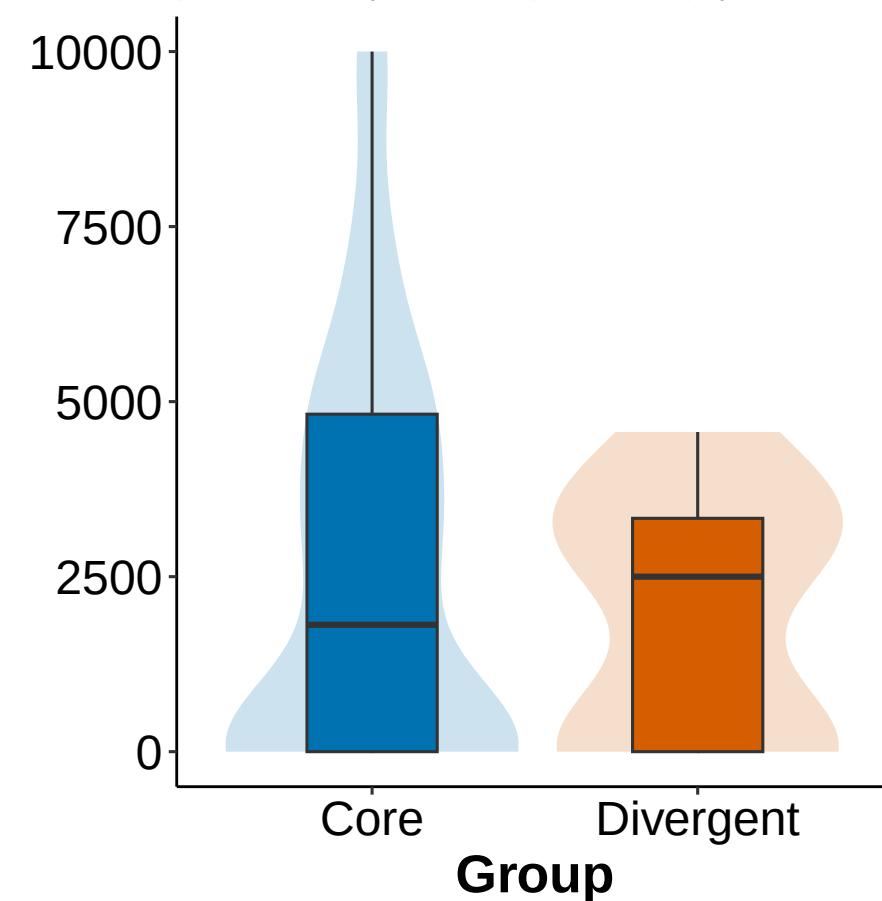
SA_Glutamate_metabolism_pFBA

p-adj: 0.885 | .=0.15 (Negligible)

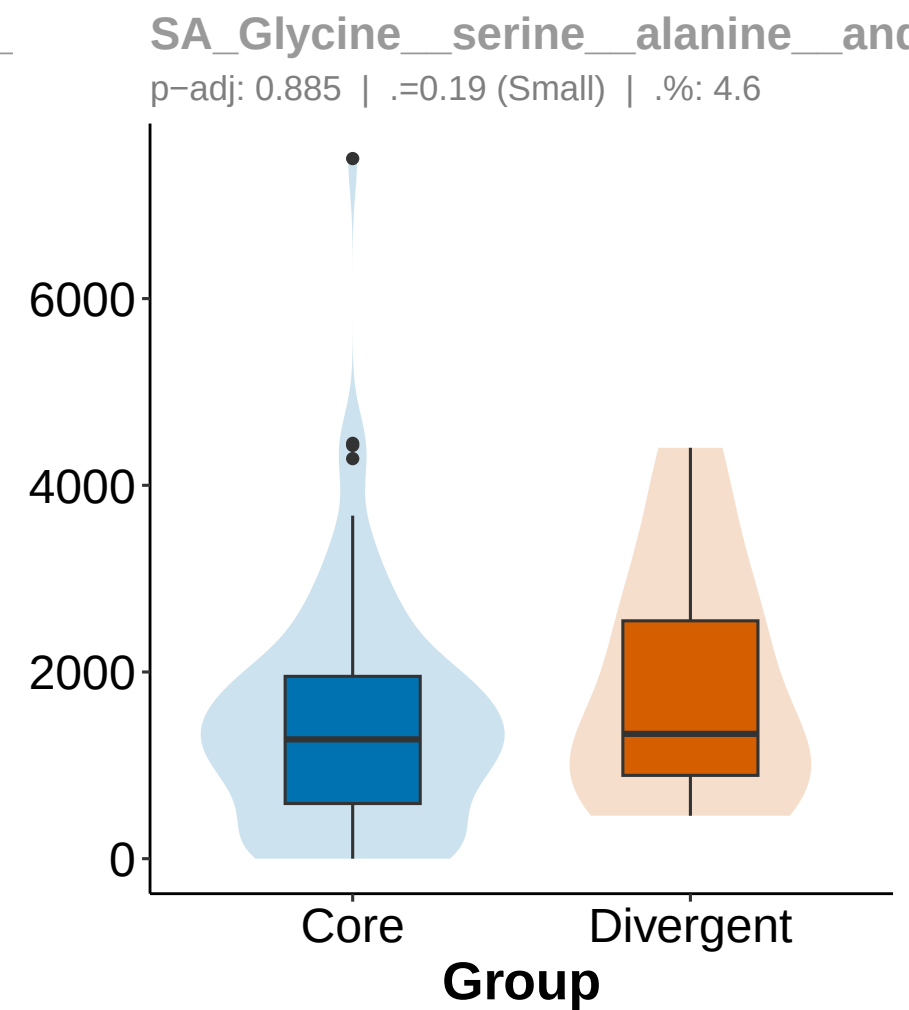
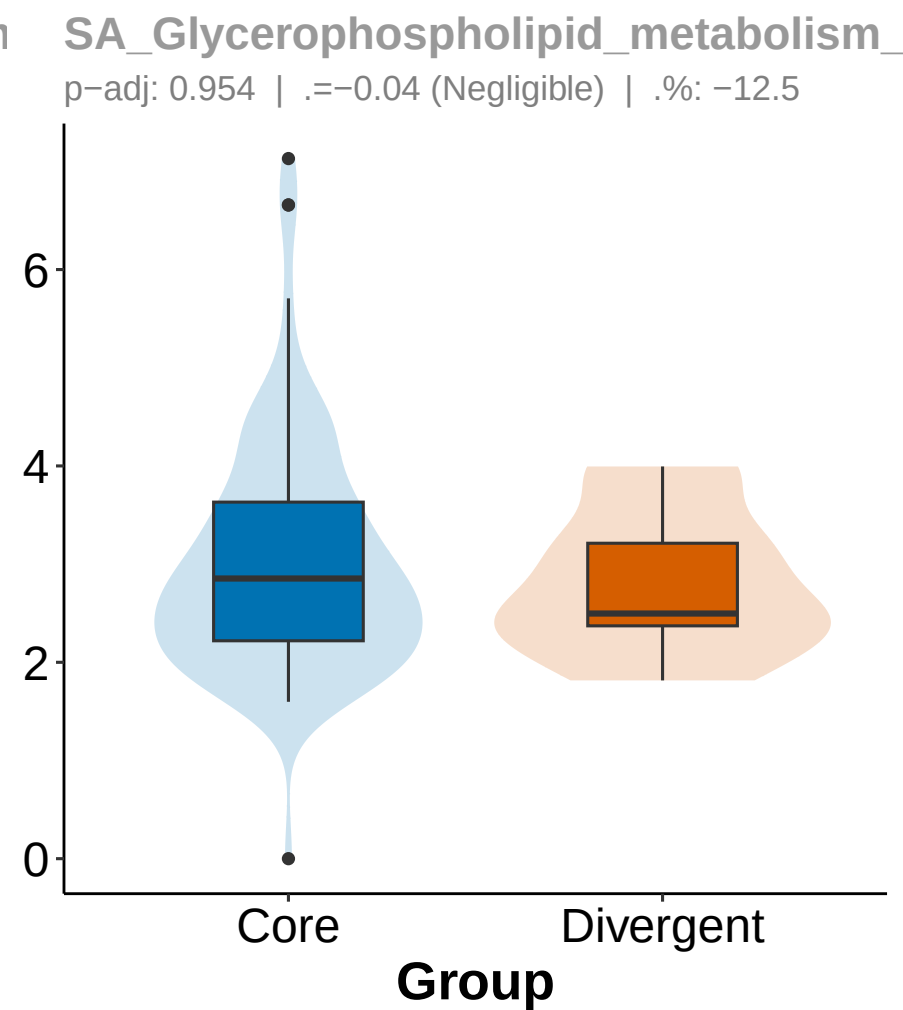
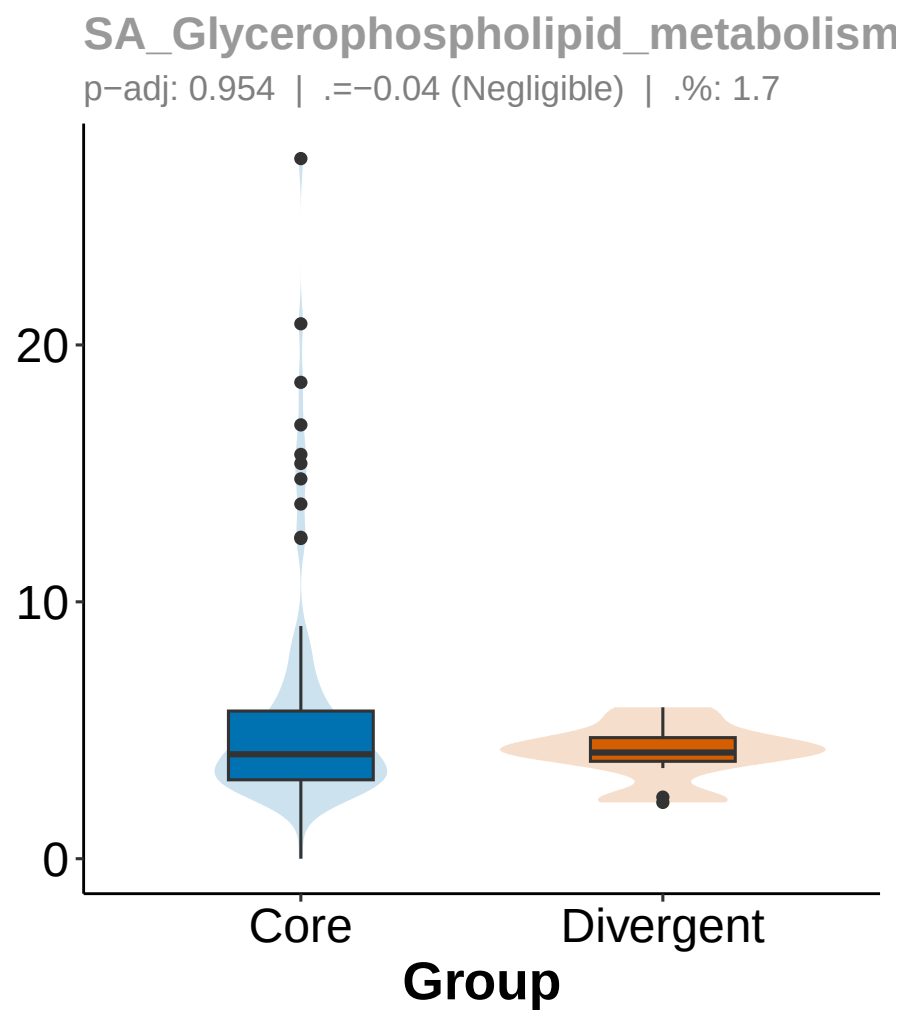
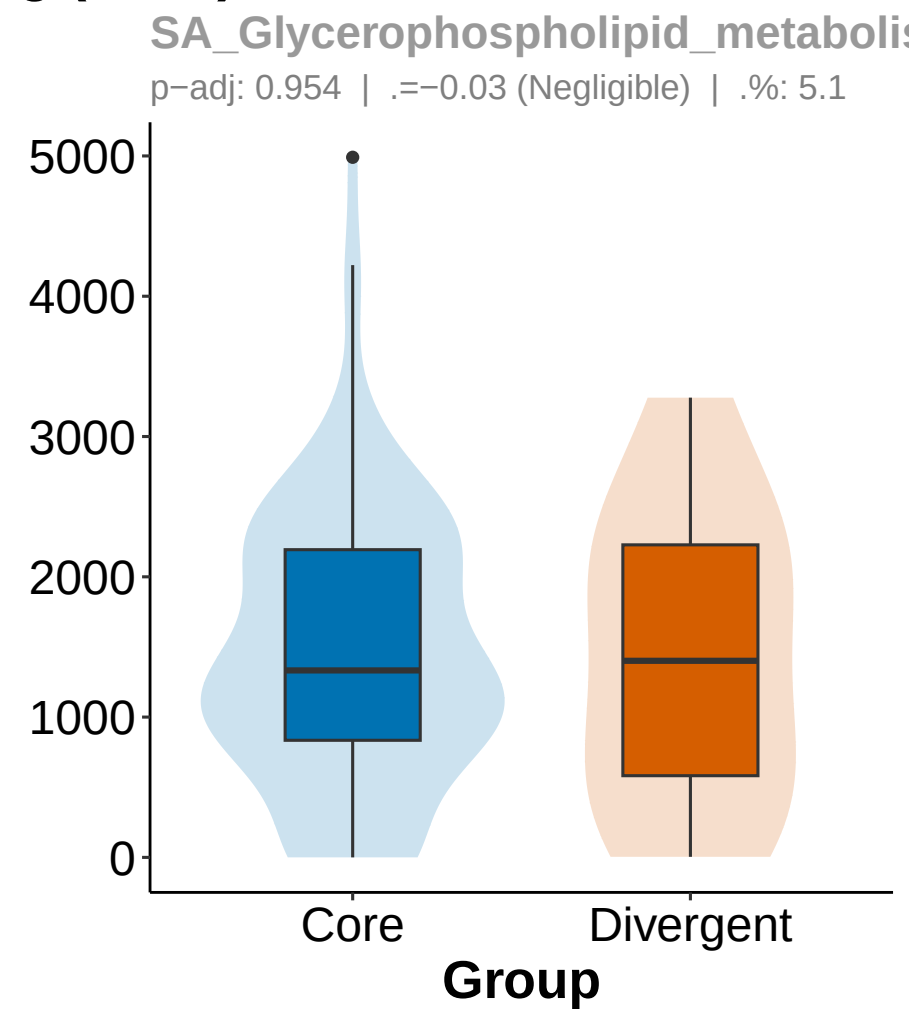
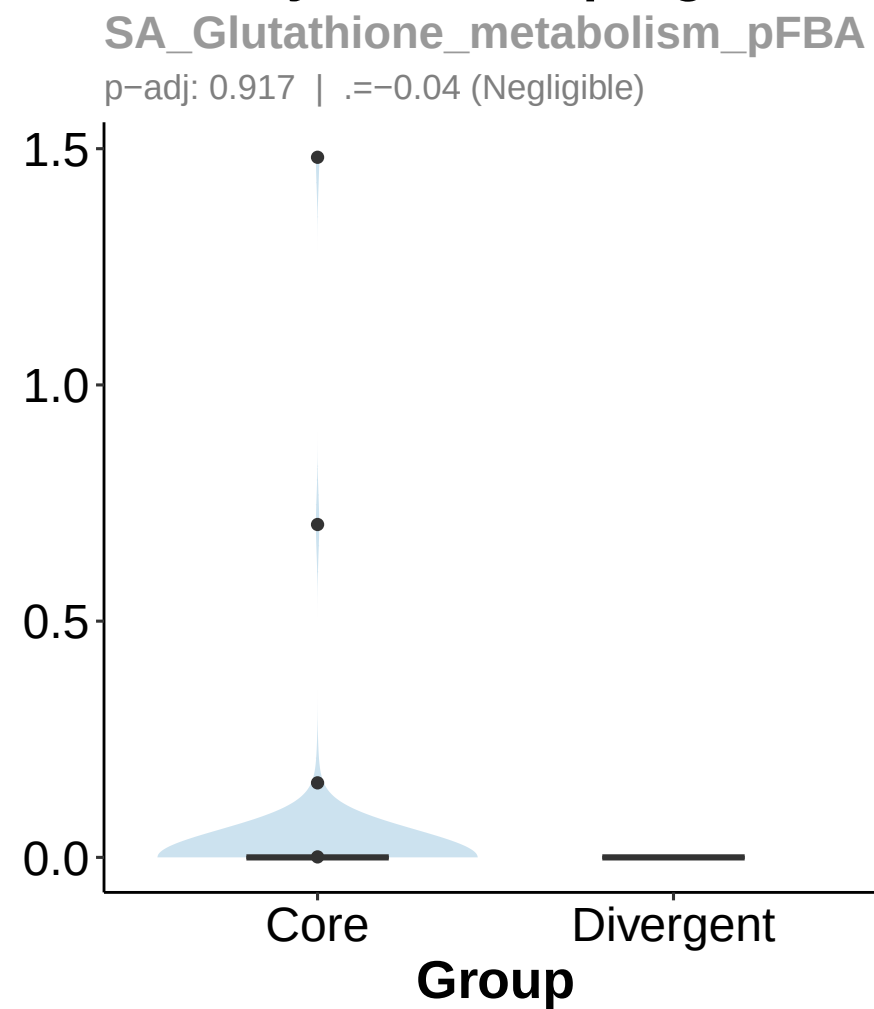
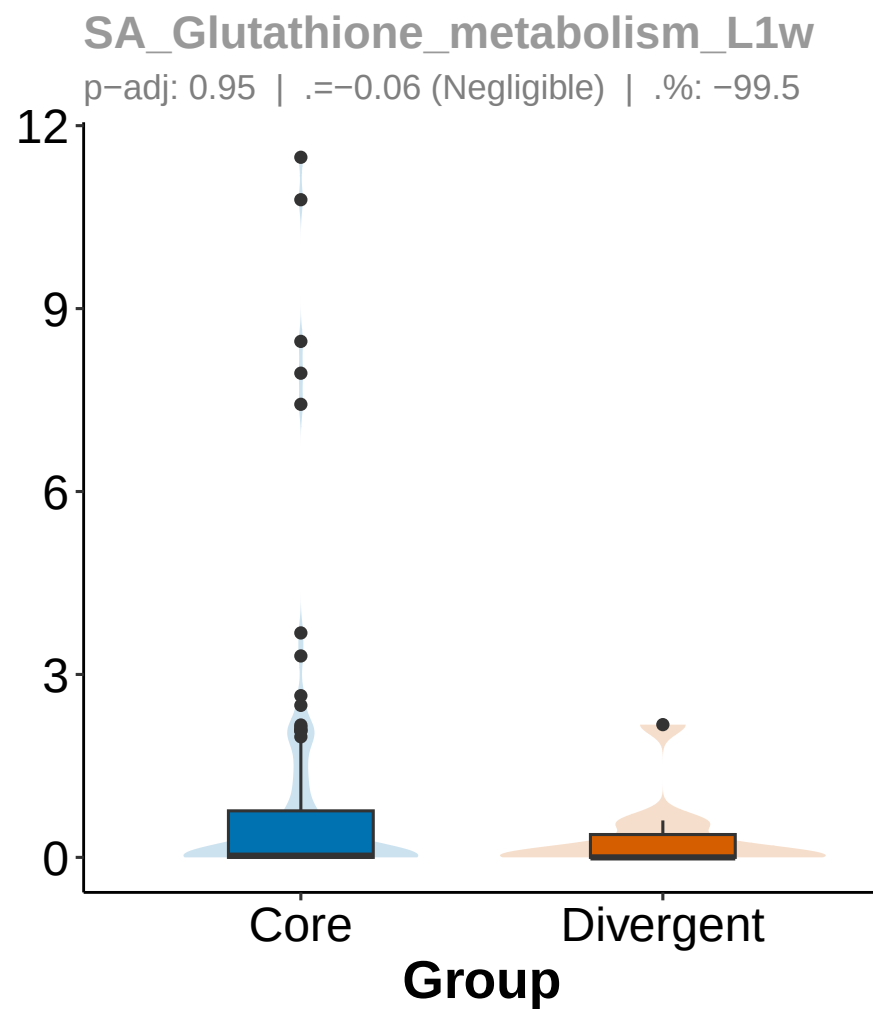


SA_Glutathione_metabolism_FBA

p-adj: 0.95 | .=-0.07 (Negligible) | .%: 37.9



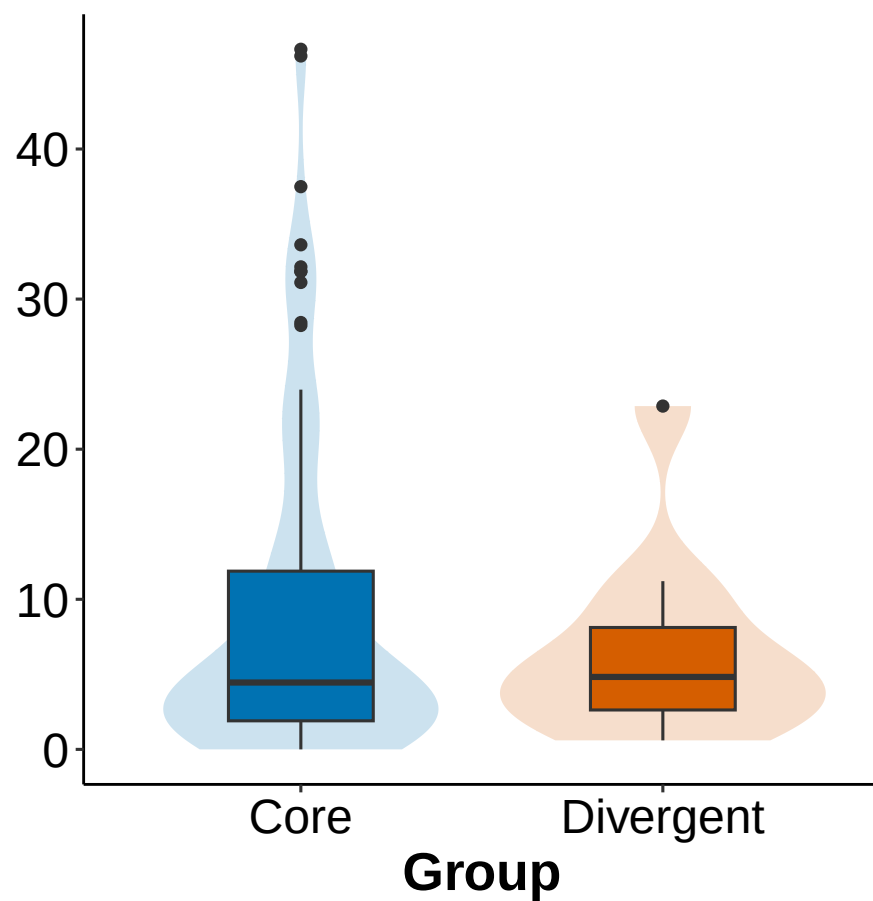
Metabolism – . 5. Systemic Reprogramming (SA_*)



Metabolism – . 5. Systemic Reprogramming (SA_*)

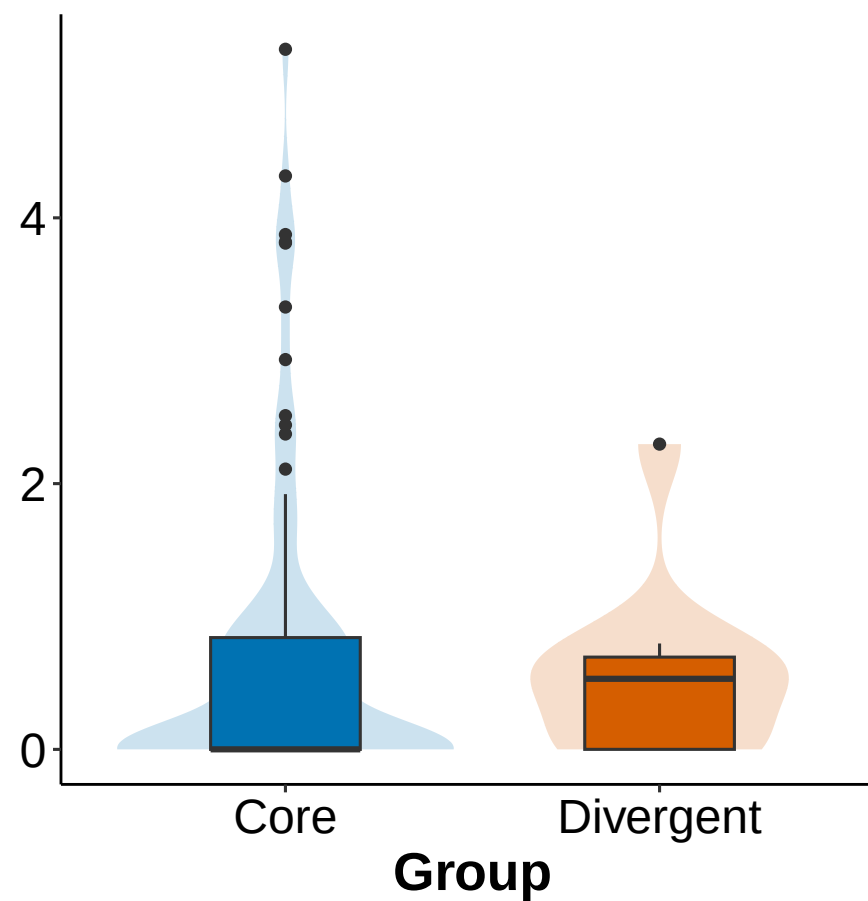
SA_Glycine__serine__alanine__and_t

p-adj: 0.992 | .=0 (Negligible) | .%: 8.4



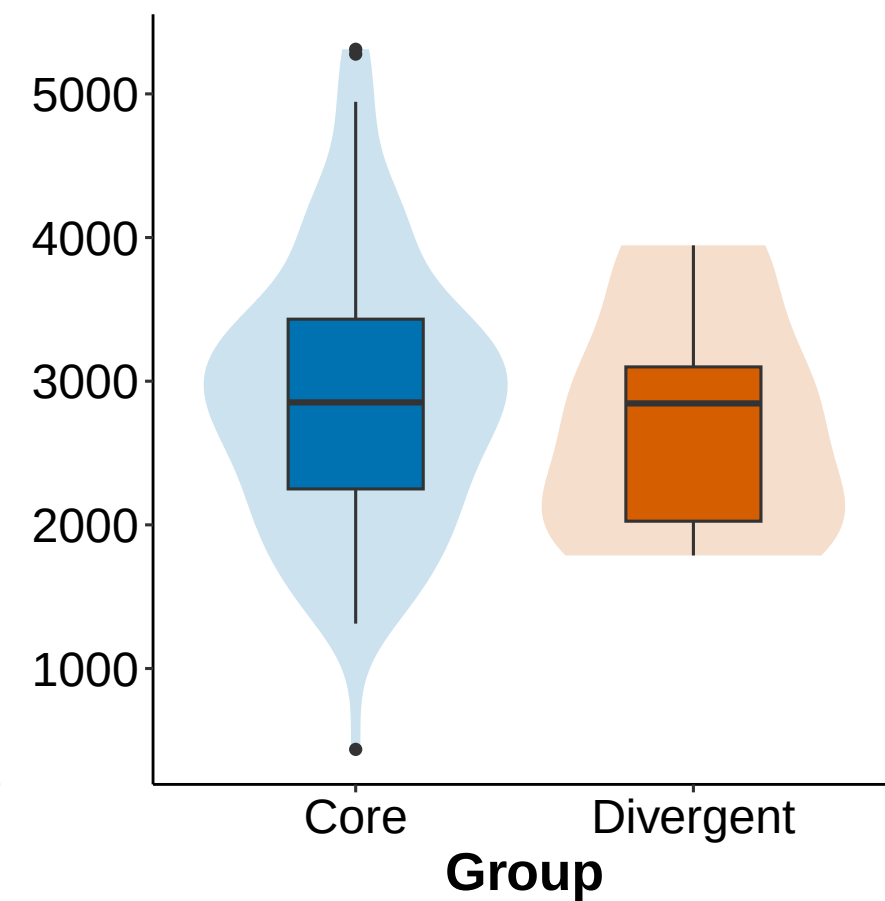
SA_Glycine__serine__alanine__and_th

p-adj: 0.95 | .=0.08 (Negligible) | .%: 140163.2



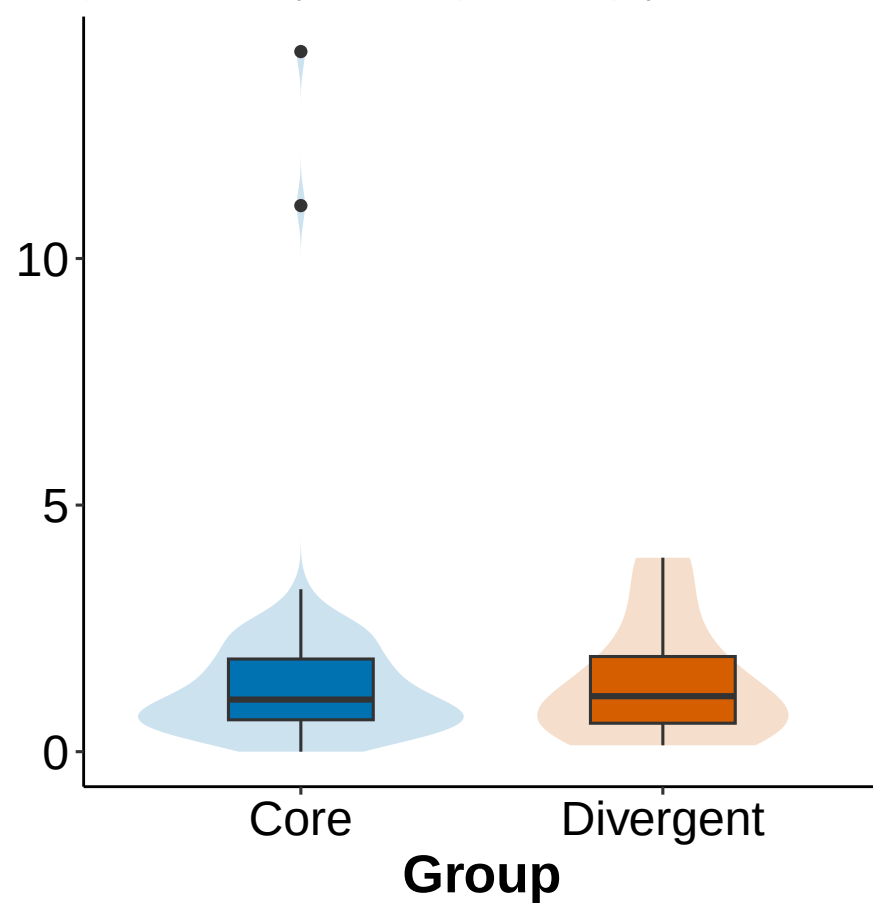
SA_Glycolysis_gluconeogenesis_F

p-adj: 0.917 | .=-0.11 (Negligible) | .%: -0.2



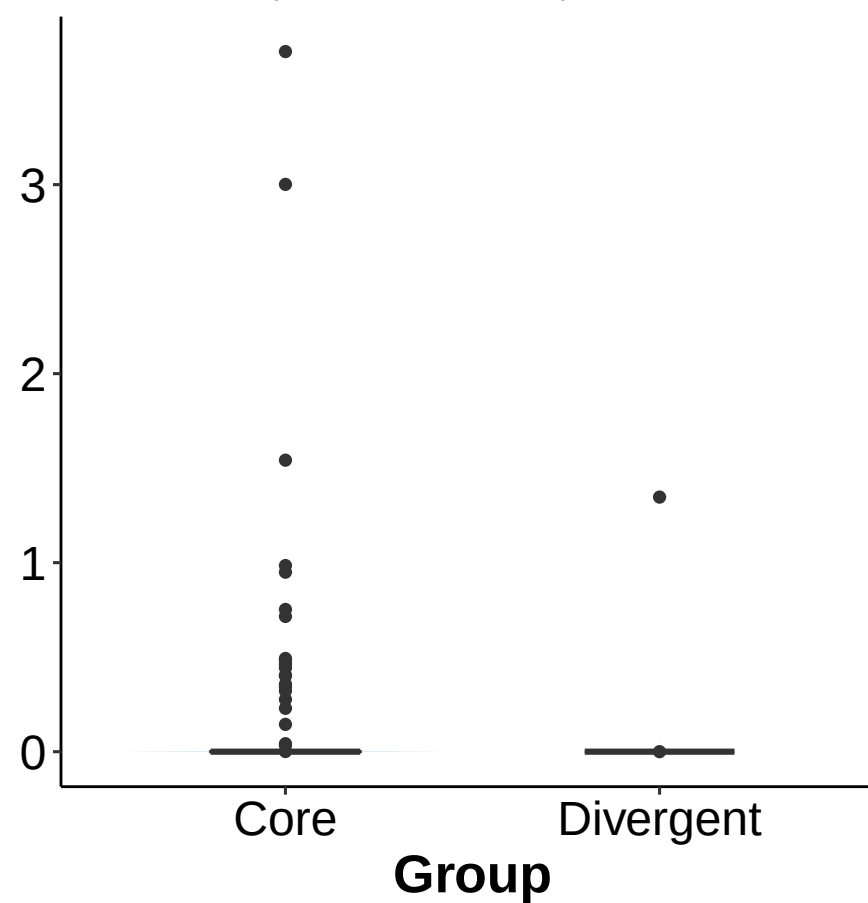
SA_Glycolysis_gluconeogenesis_L1w

p-adj: 0.992 | .=-0.01 (Negligible) | .%: 6.5



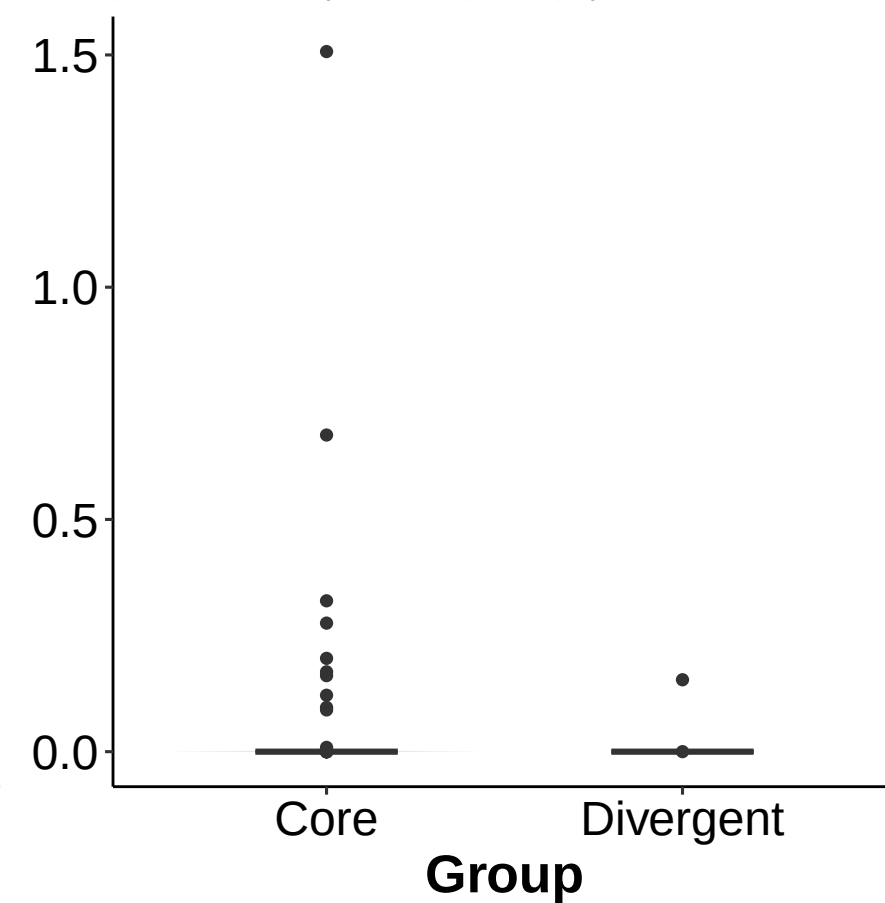
SA_Glycolysis_gluconeogenesis_pFB/

p-adj: 0.749 | .=-0.48 (Large) | .%: -100



SA_Glycosphingolipid_metabolism_L

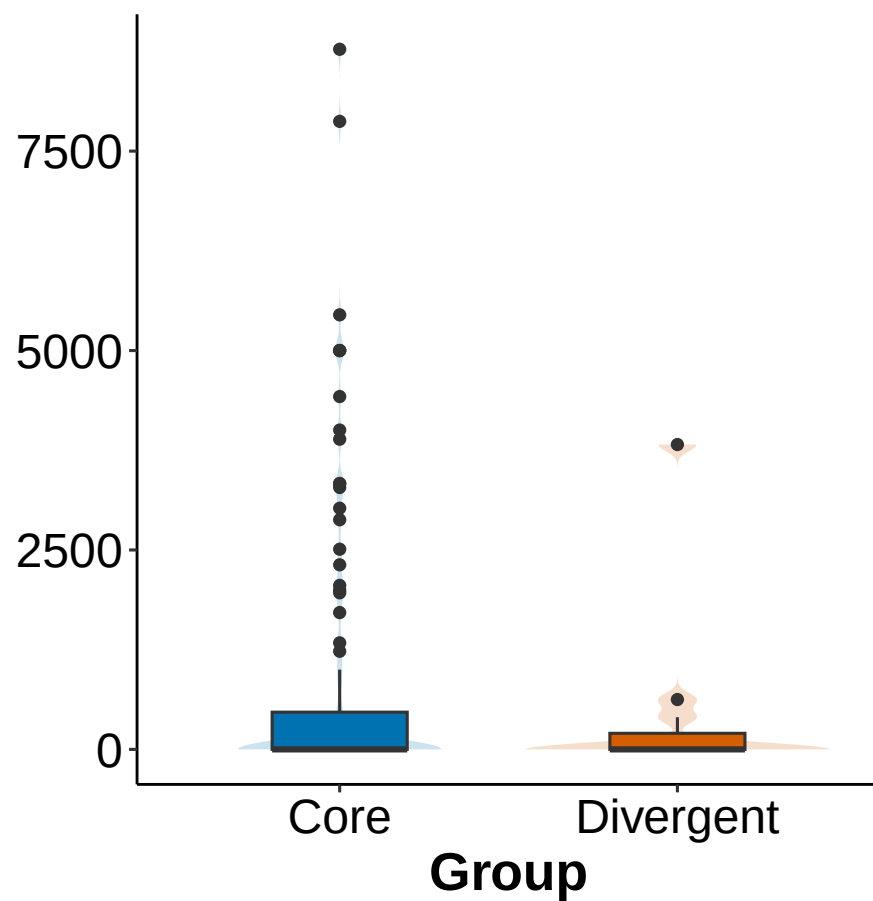
p-adj: 0.885 | .=0.15 (Small) | .%: 50.8



Metabolism – . 5. Systemic Reprogramming (SA_*)

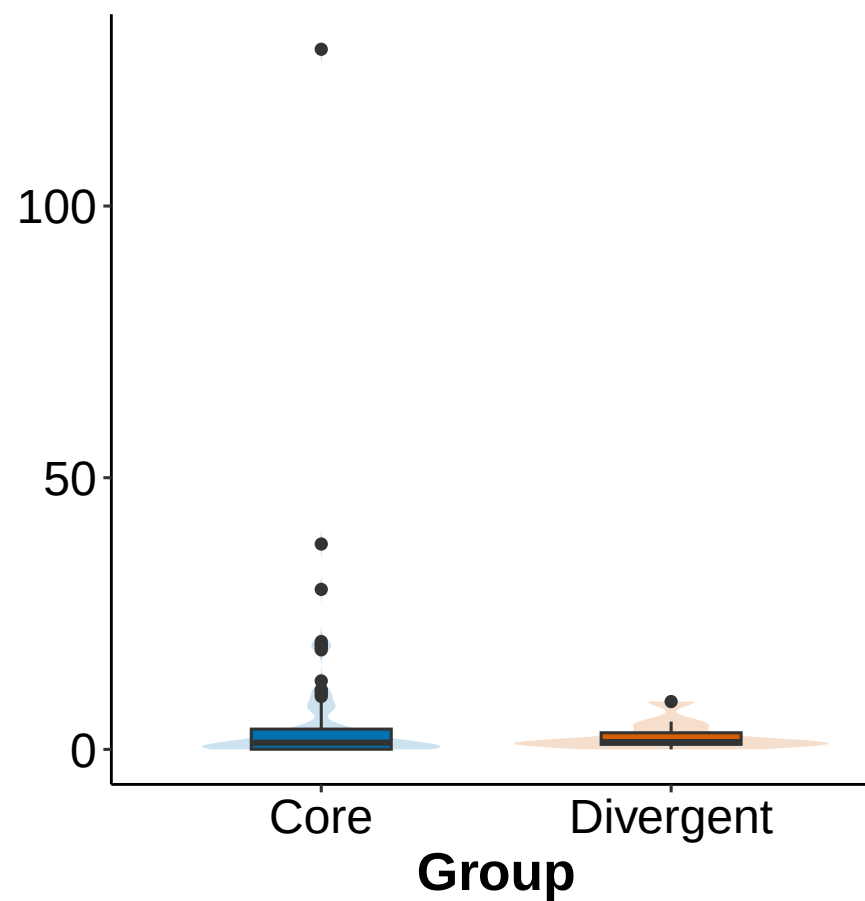
SA_Glyoxylate_and_dicarboxylate_

p-adj: 0.95 | .=-0.06 (Negligible)



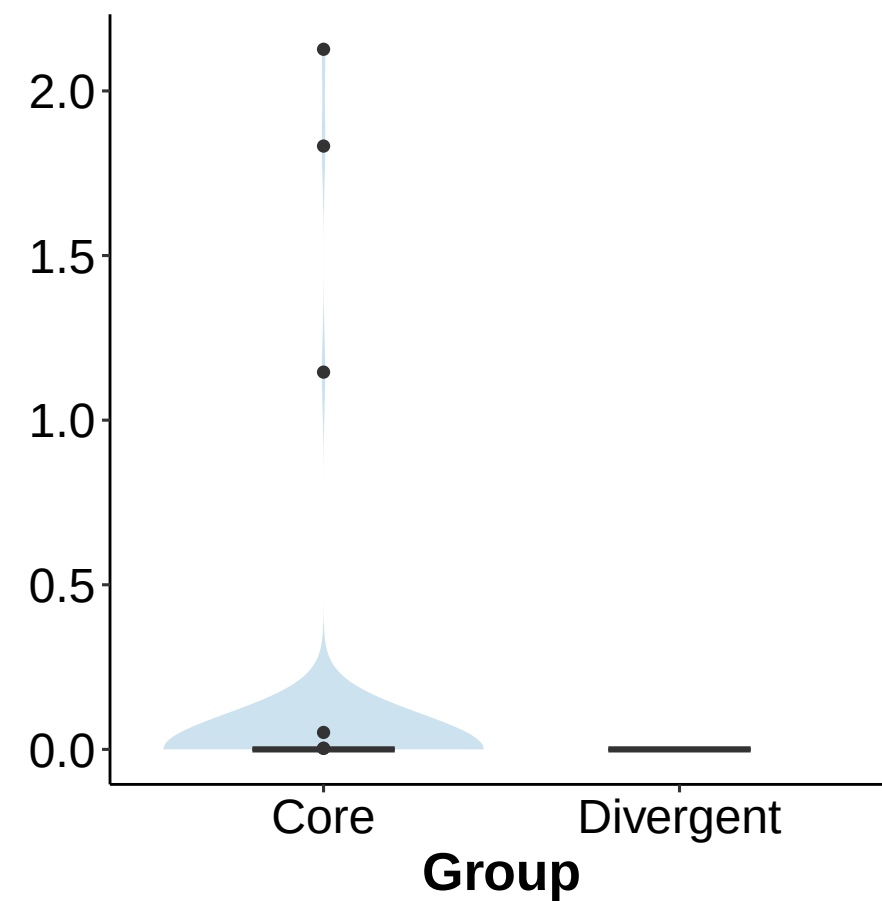
SA_Glyoxylate_and_dicarboxylate_n

p-adj: 0.917 | .:=0.11 (Negligible) | .%: 8.9



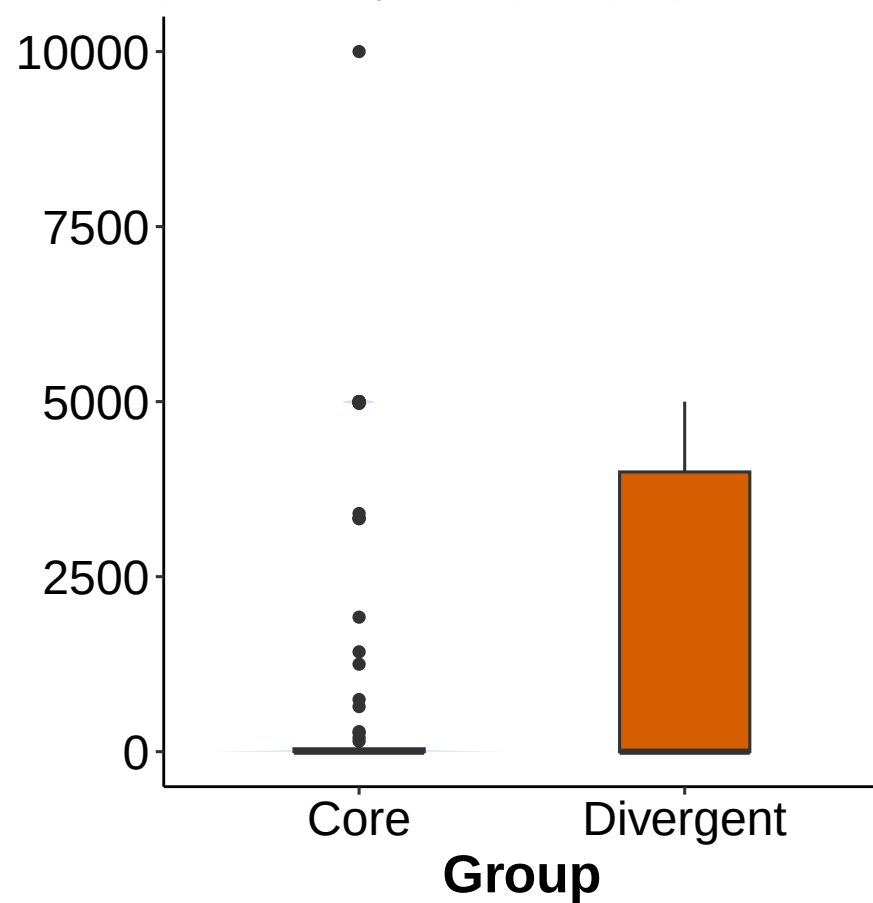
SA_Glyoxylate_and_dicarboxylate_m

p-adj: 0.885 | .=-0.06 (Negligible)



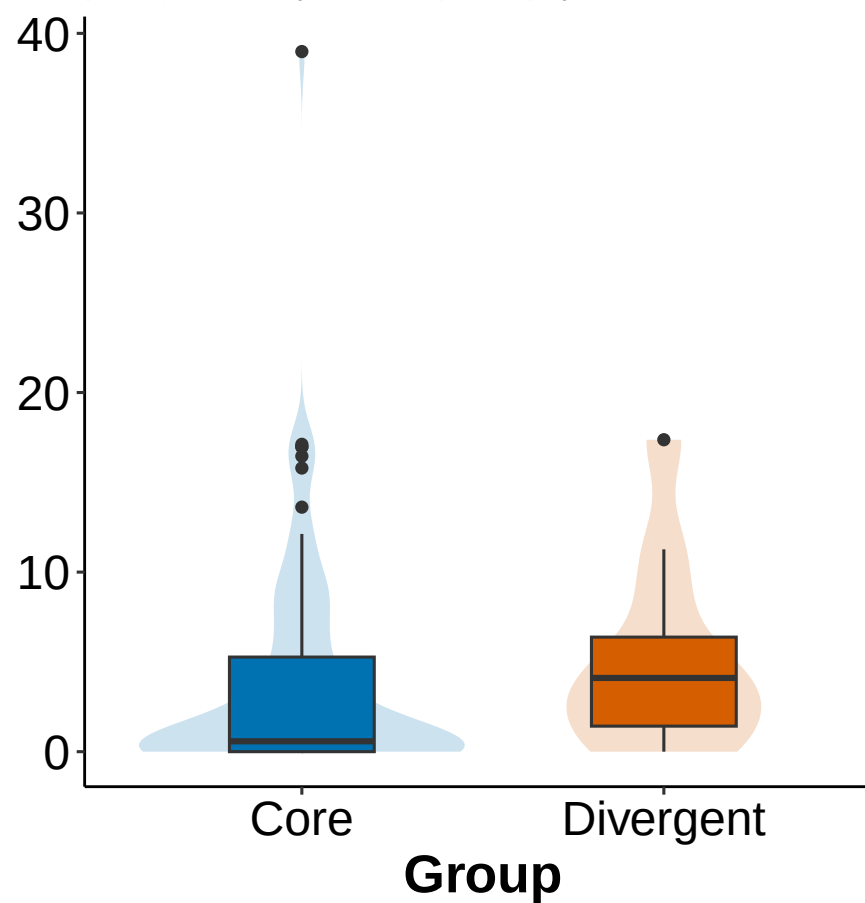
SA_Heme_degradation_FBA

p-adj: 0.885 | .:=0.14 (Negligible)



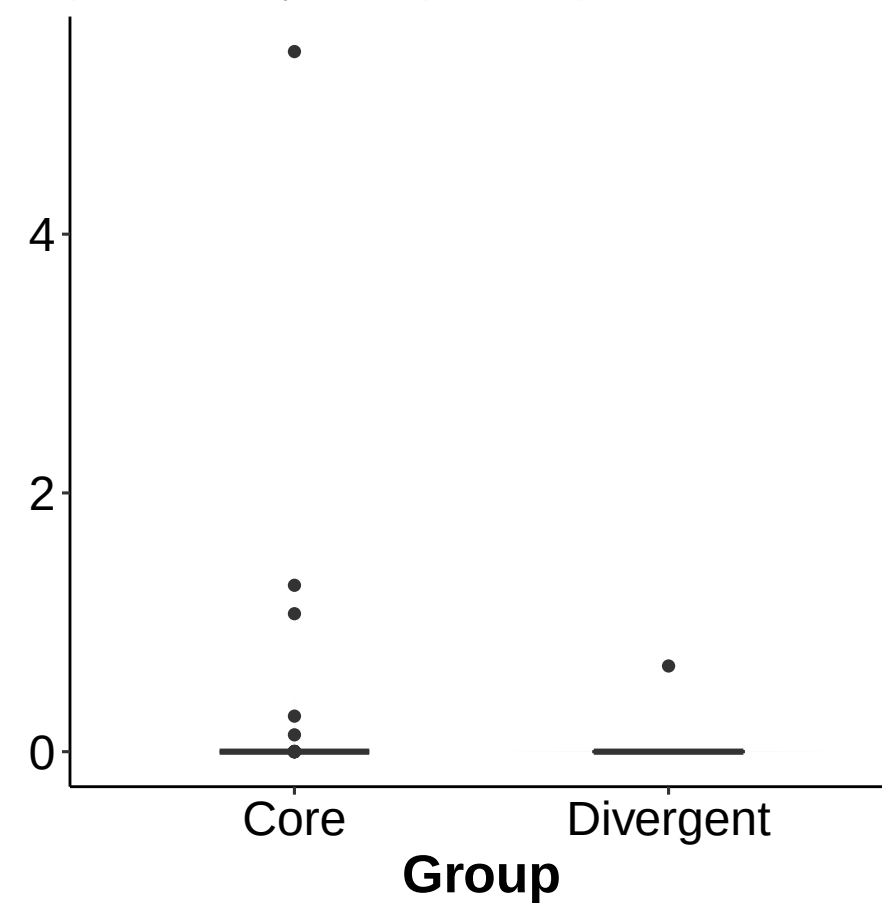
SA_Heme_degradation_L1w

p-adj: 0.872 | .:=0.29 (Small) | .%: 610.2



SA_Heme_degradation_pFBA

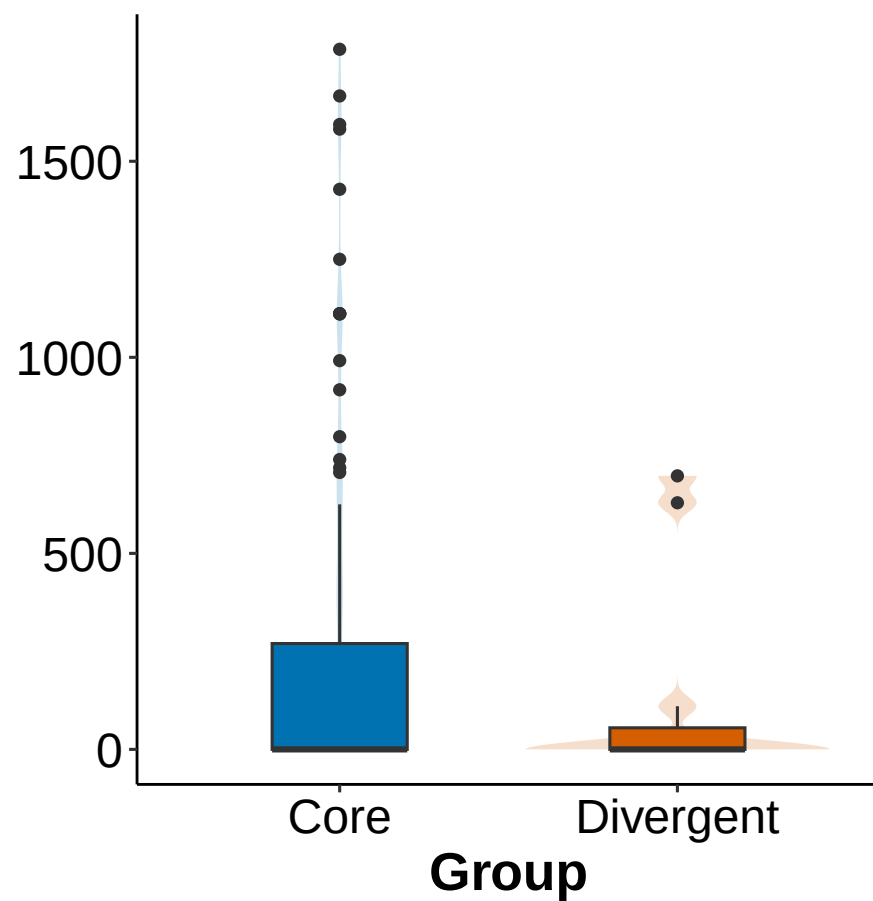
p-adj: 0.885 | .:=0.14 (Negligible)



Metabolism – . 5. Systemic Reprogramming (SA_*)

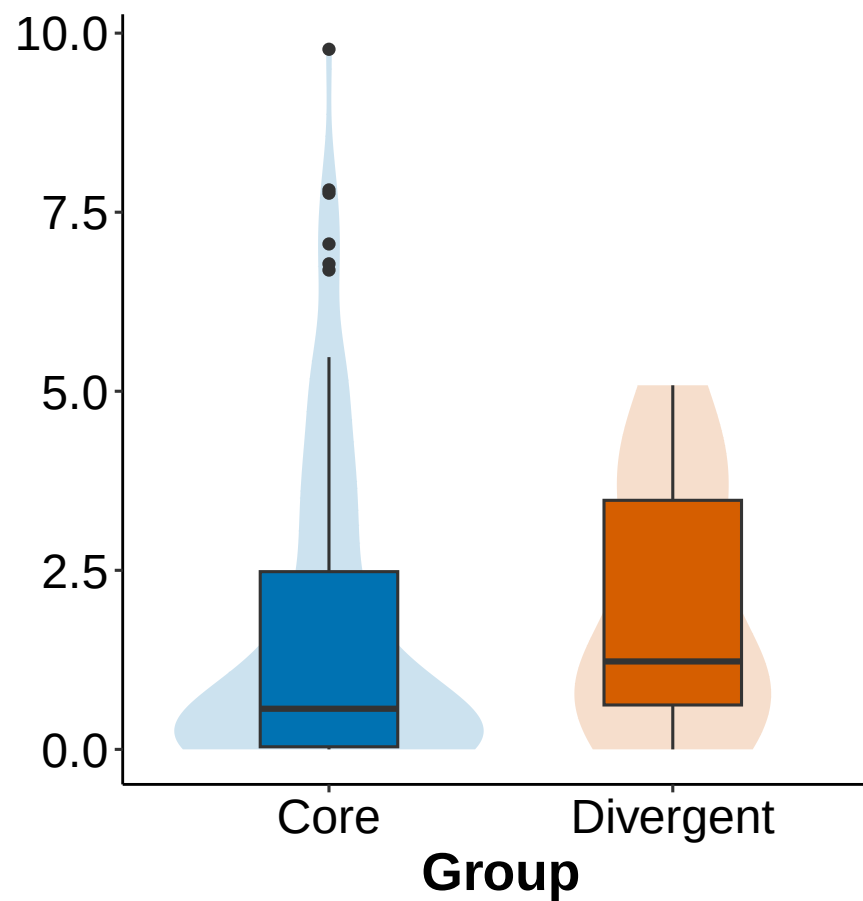
SA_Heme_synthesis_FBA

p-adj: 0.992 | . = 0 (Negligible)



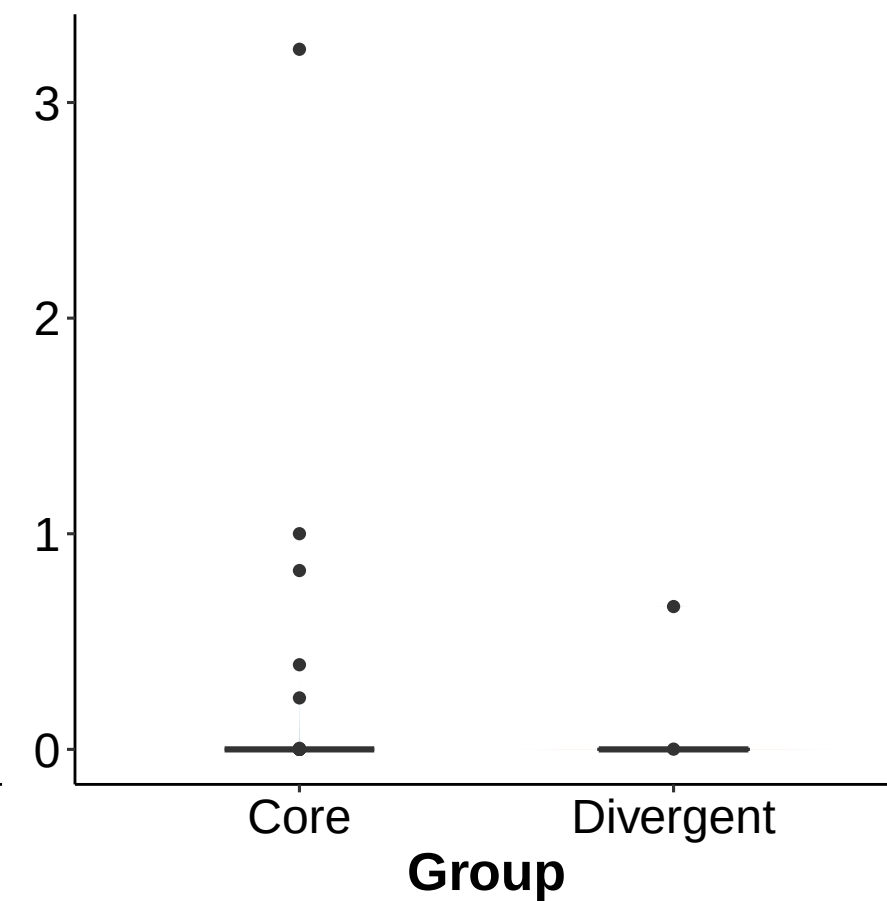
SA_Heme_synthesis_L1w

p-adj: 0.885 | . = 0.22 (Small) | . %: 116.4



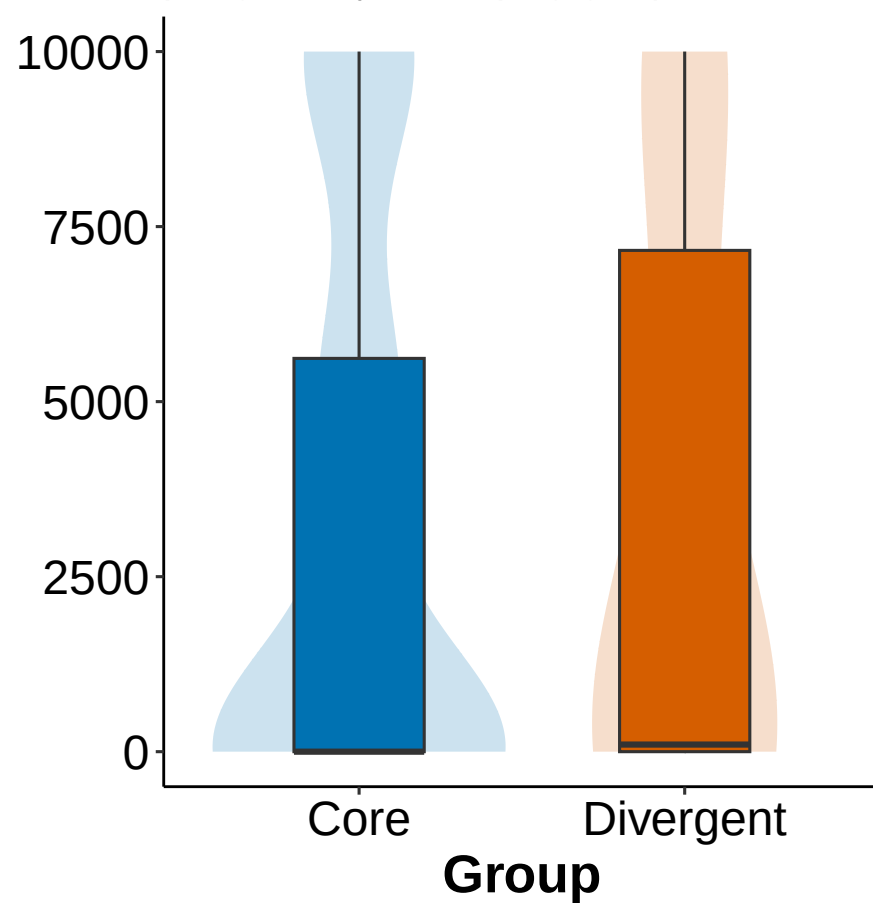
SA_Heme_synthesis_pFBA

p-adj: 0.885 | . = 0.14 (Negligible)



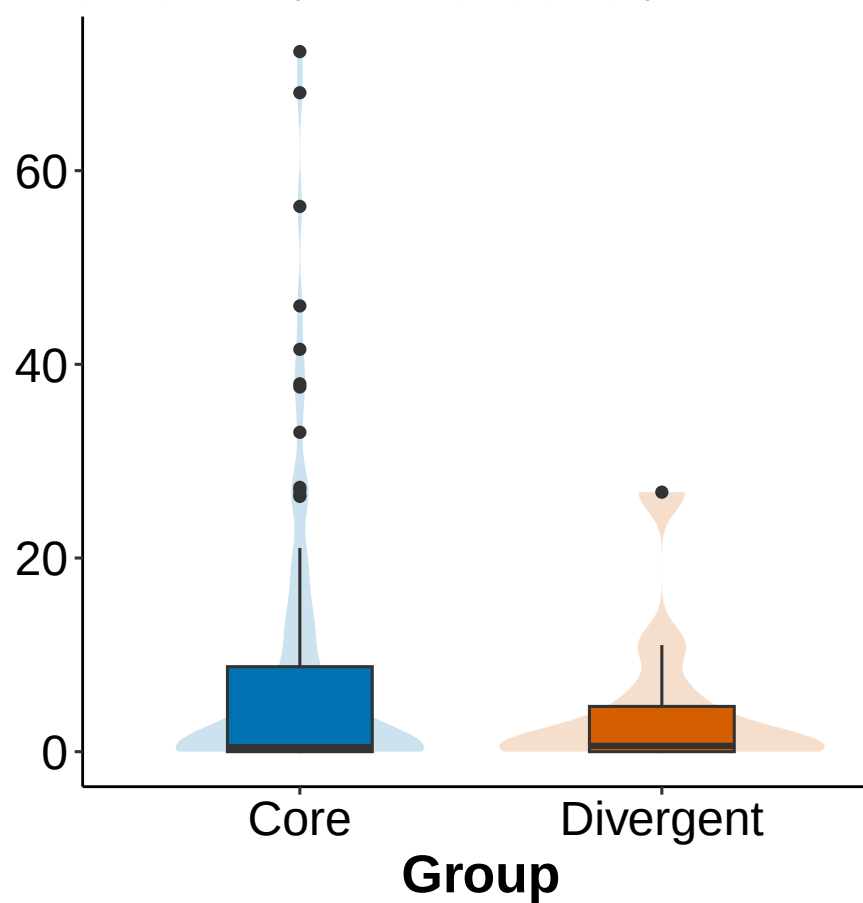
SA_Histidine_metabolism_FBA

p-adj: 0.95 | . = 0.06 (Negligible)



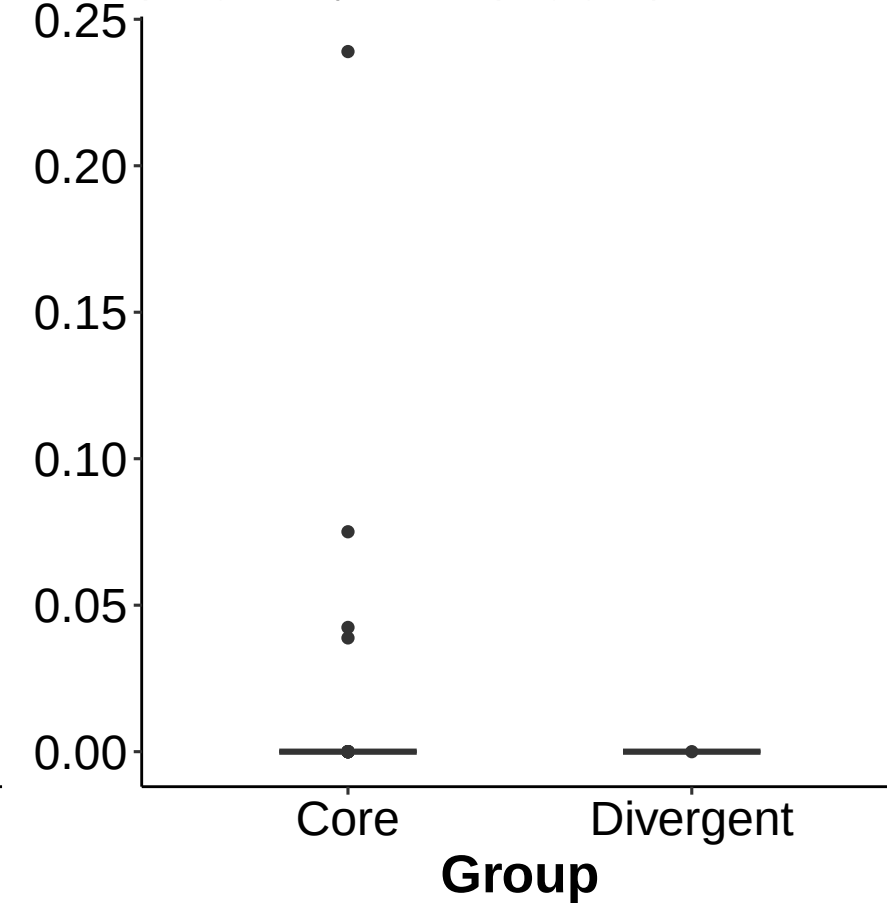
SA_Histidine_metabolism_L1w

p-adj: 0.954 | . = -0.05 (Negligible) | . %: 28.1



SA_Hyaluronan_metabolism_L1w

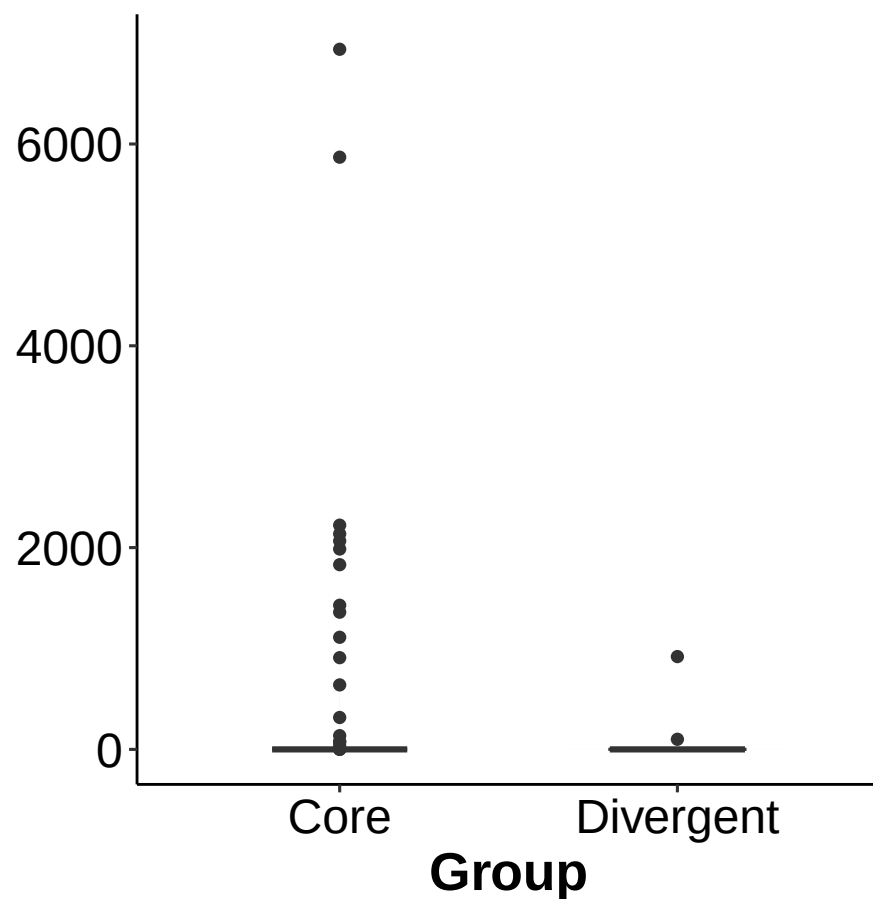
p-adj: 0.95 | . = -0.04 (Negligible)



Metabolism – . 5. Systemic Reprogramming (SA_*)

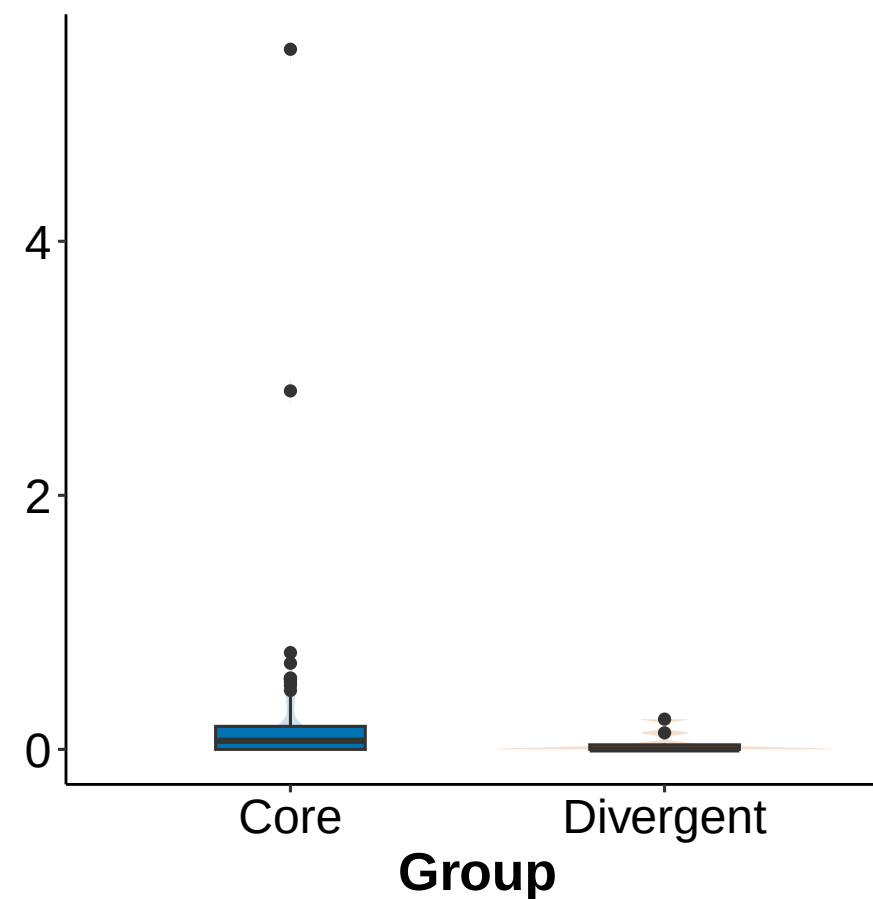
SA_Inositol_phosphate_metabolism_L1w

p-adj: 0.885 | .:=0.12 (Negligible)



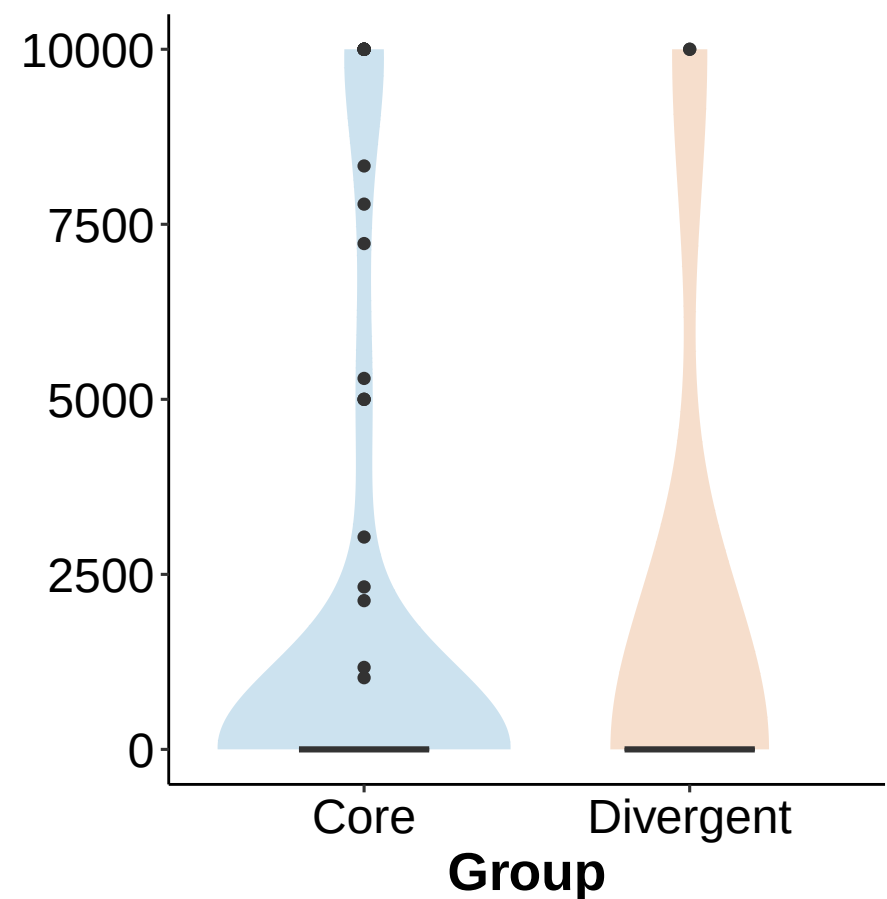
SA_Inositol_phosphate_metabolism_L1w

p-adj: 0.872 | .:=−0.31 (Small) | .%: −99.8



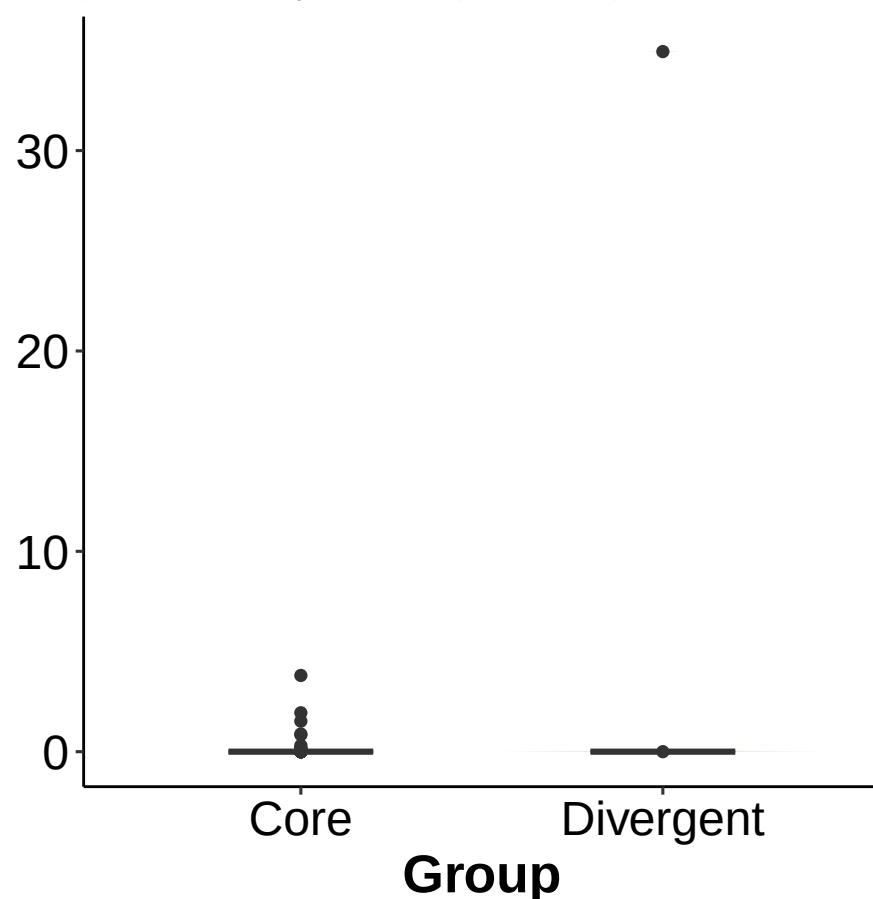
SA_Leukotriene_metabolism_FBA

p-adj: 0.957 | .:=−0.02 (Negligible)



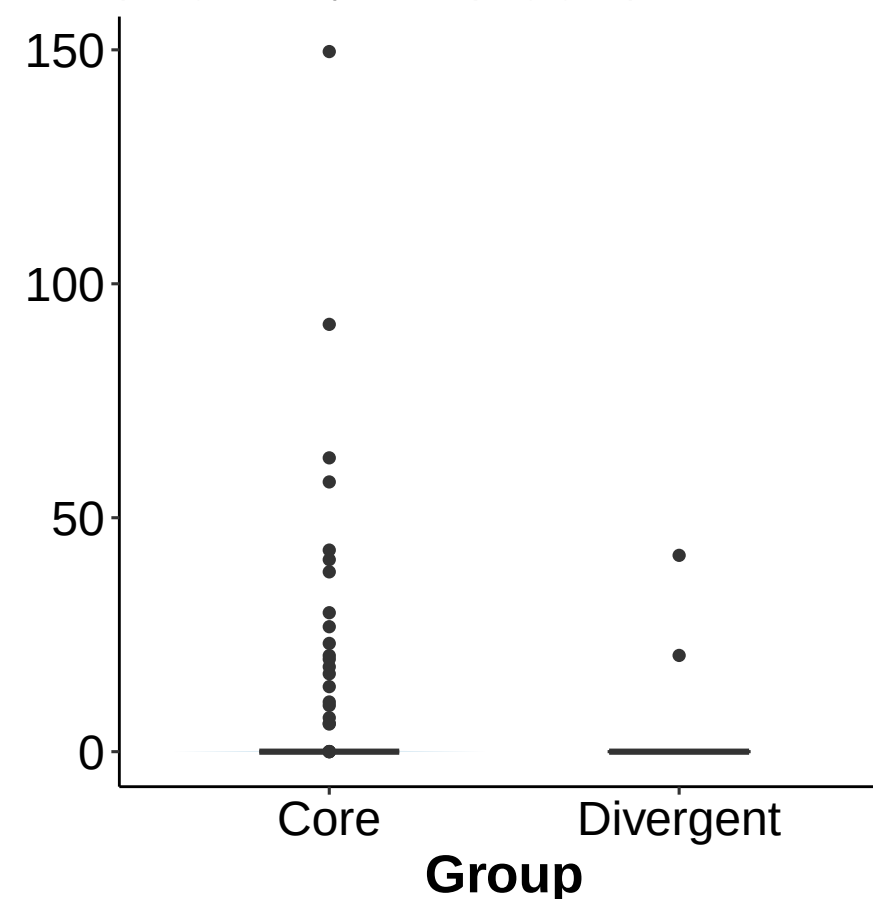
SA_Leukotriene_metabolism_L1w

p-adj: 0.917 | .:=−0.11 (Negligible)



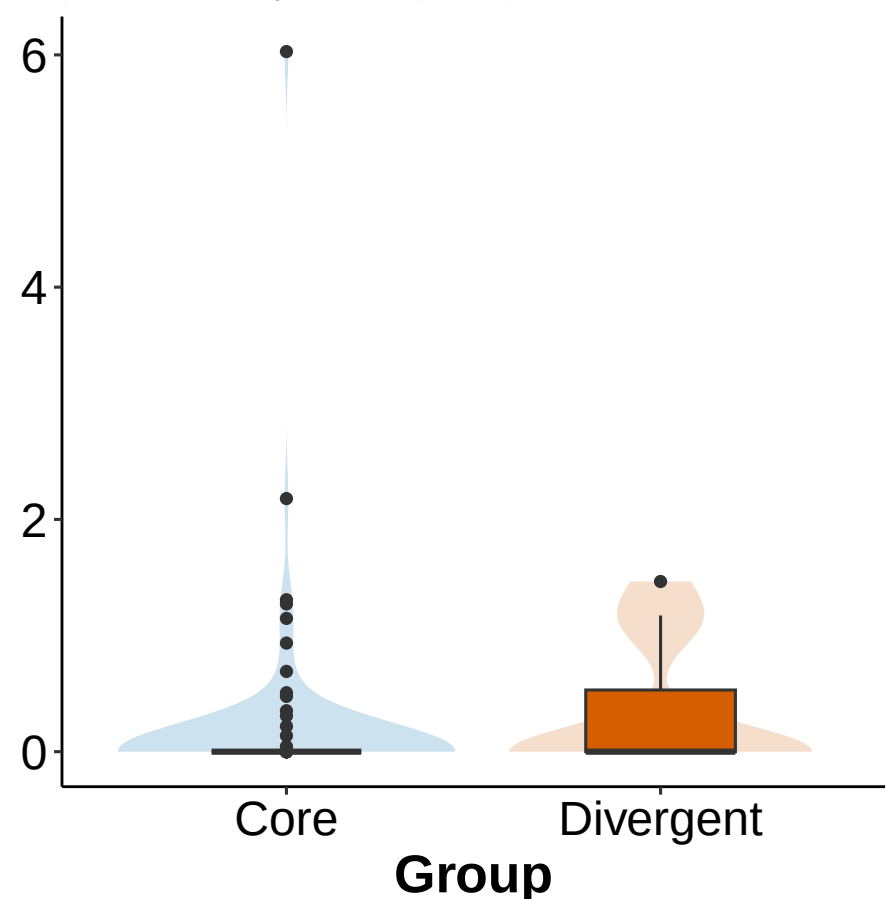
SA_Limonene_and_pinene_degradation_L1w

p-adj: 0.885 | .:=0.14 (Negligible)

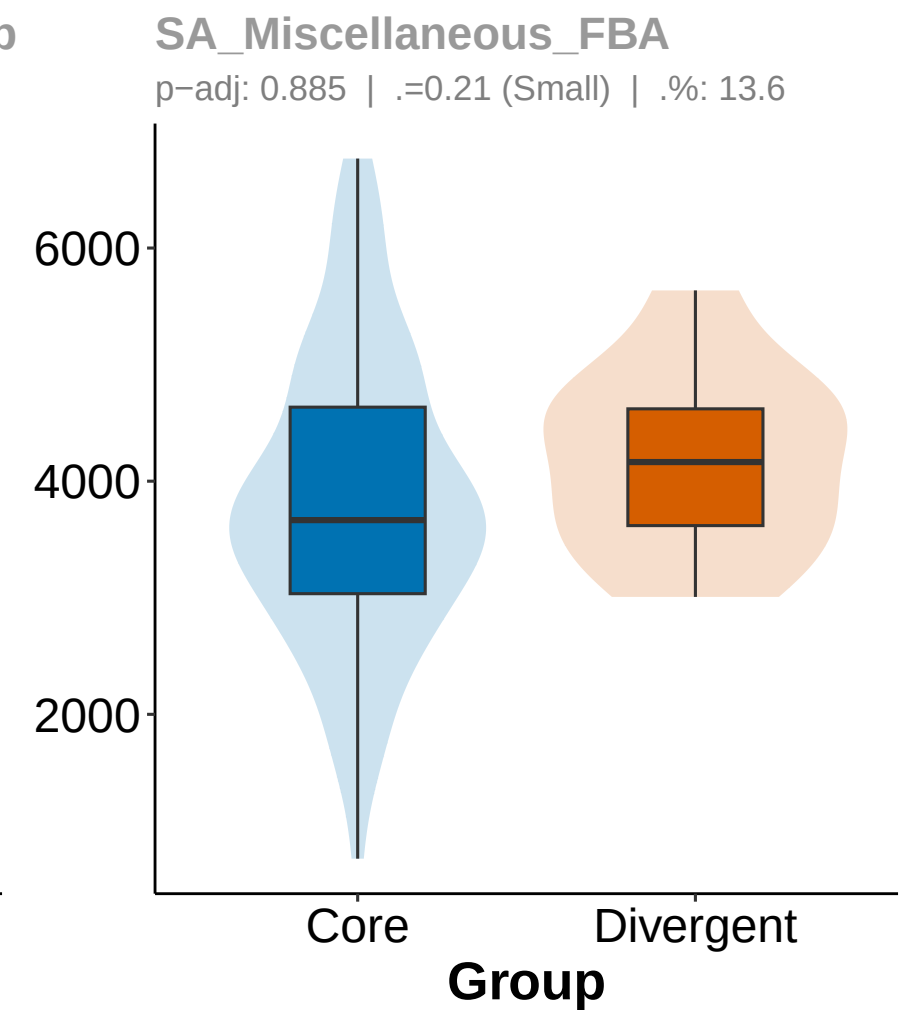
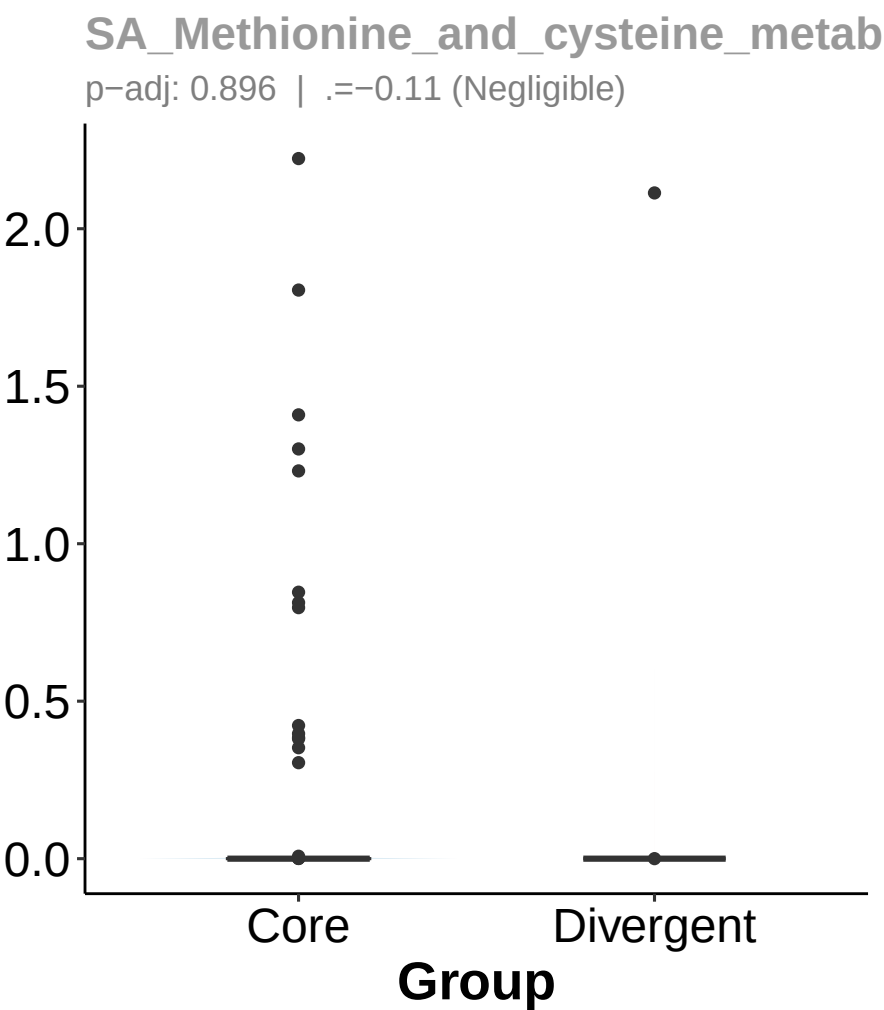
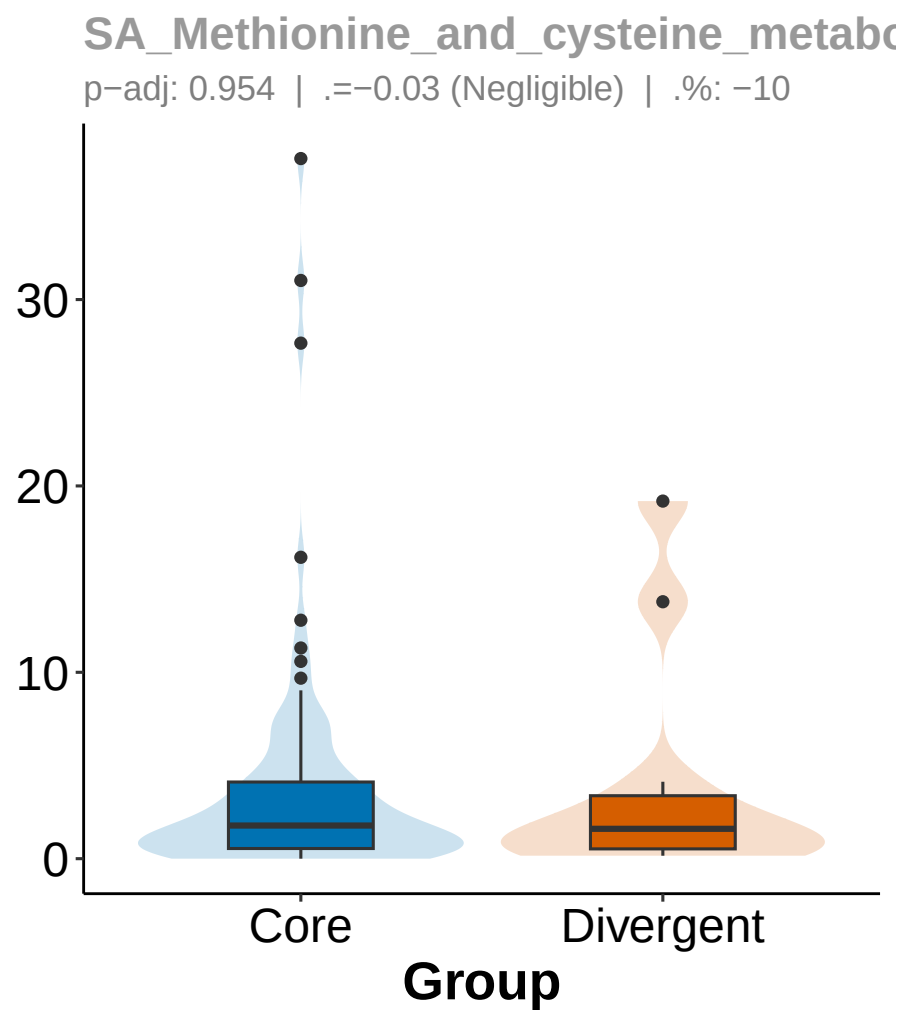
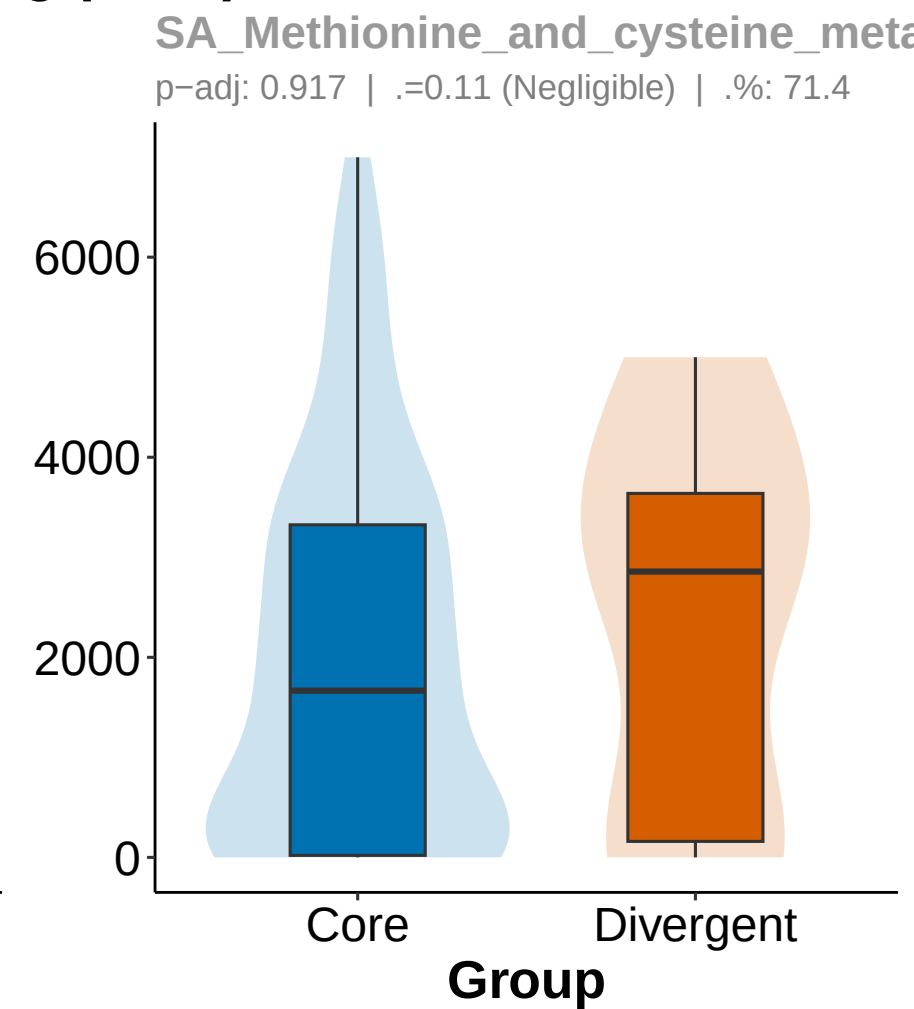
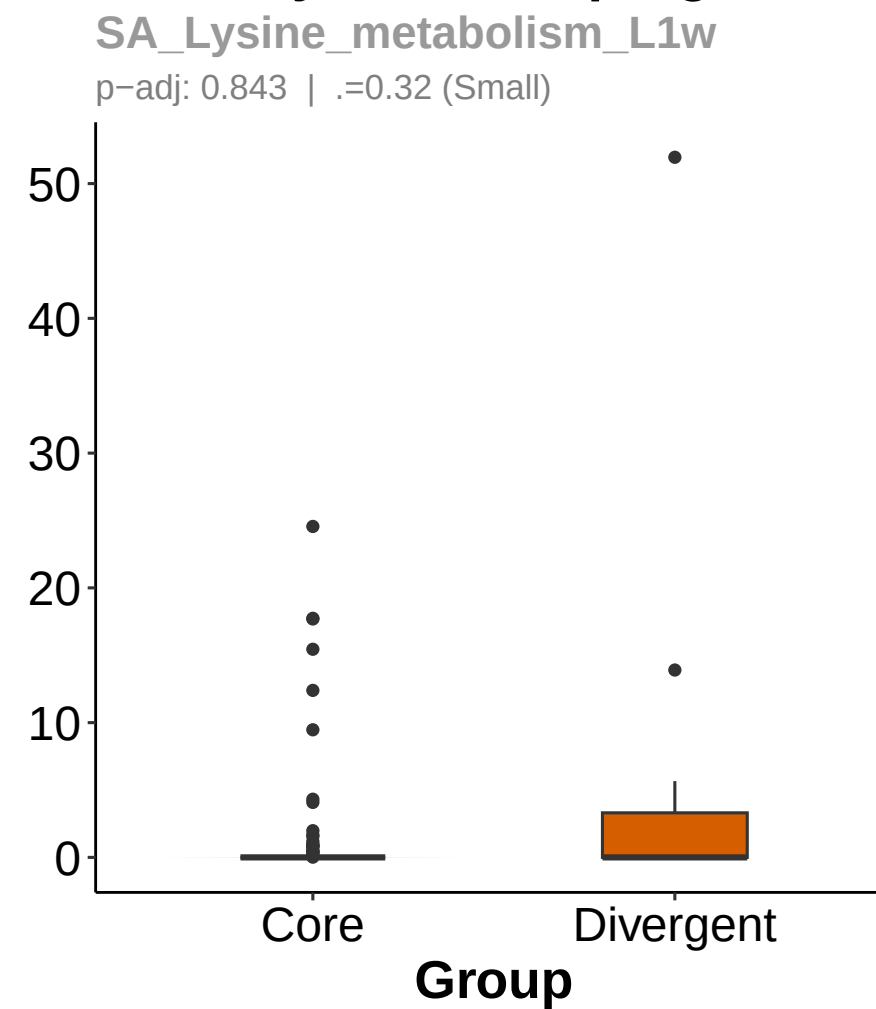
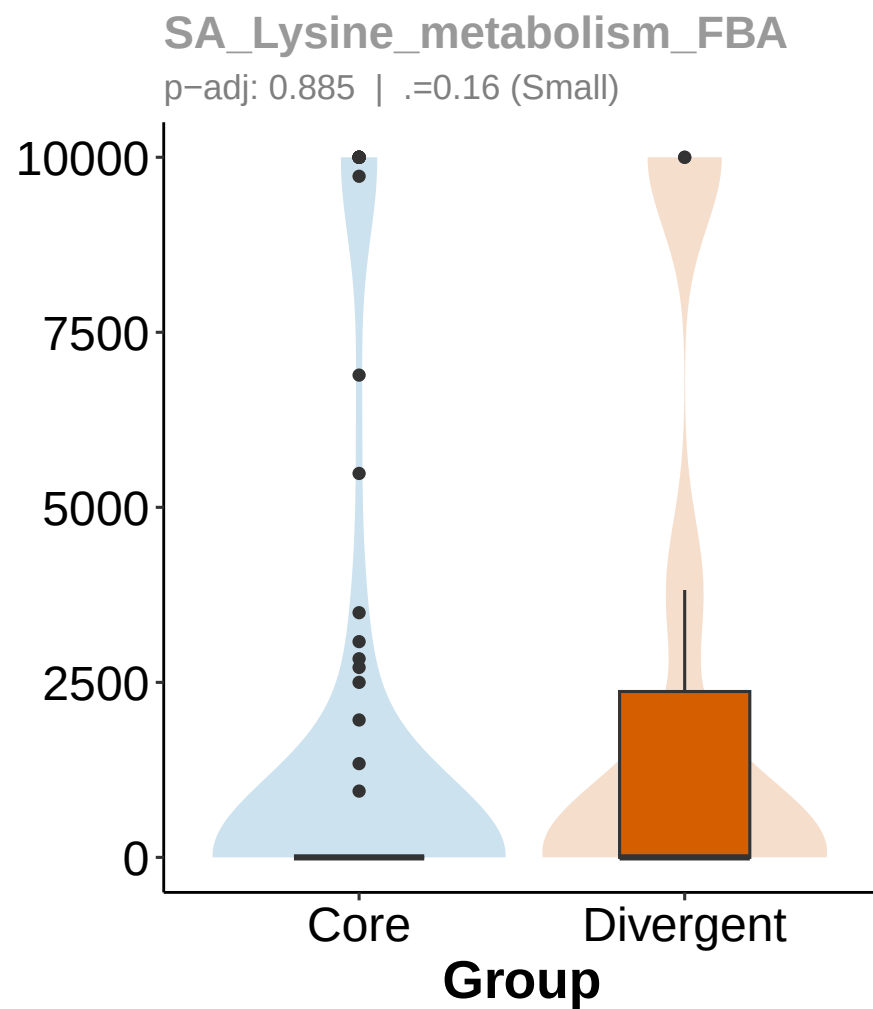


SA_Linoleate_metabolism_L1w

p-adj: 0.885 | .:=0.17 (Small)



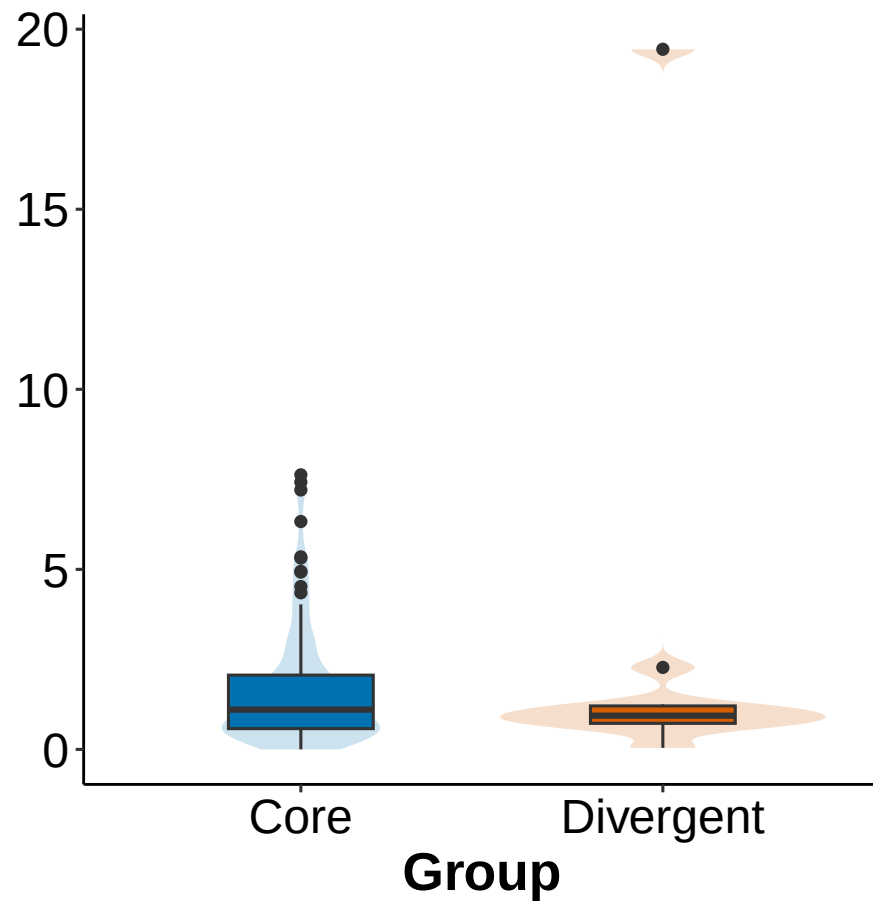
Metabolism – . 5. Systemic Reprogramming (SA_*)



Metabolism – . 5. Systemic Reprogramming (SA_*)

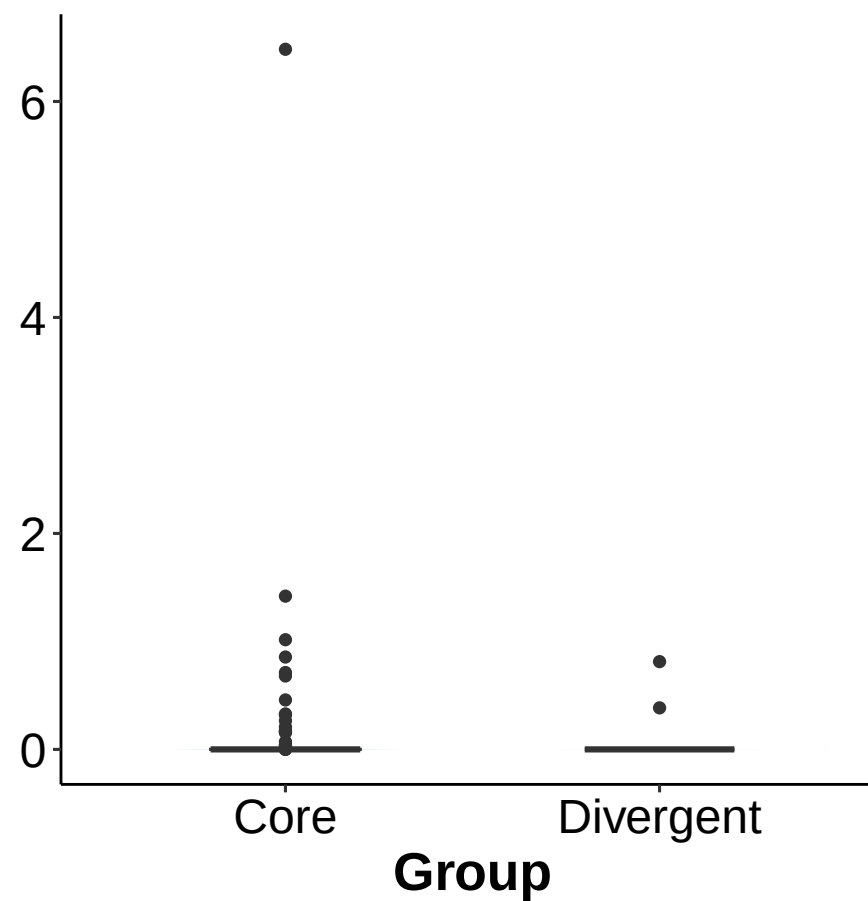
SA_Miscellaneous_L1w

p-adj: 0.954 | .=-0.04 (Negligible) | .%: -15.2



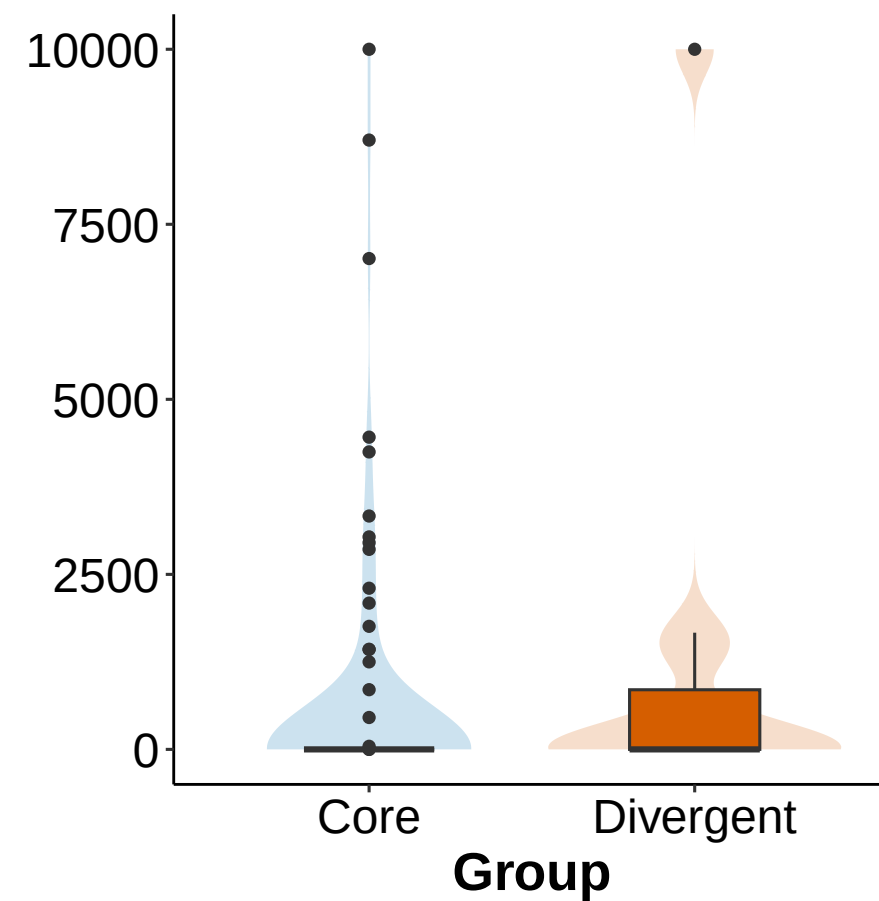
SA_Miscellaneous_pFBA

p-adj: 0.885 | .=-0.18 (Small) | .%: -57.2



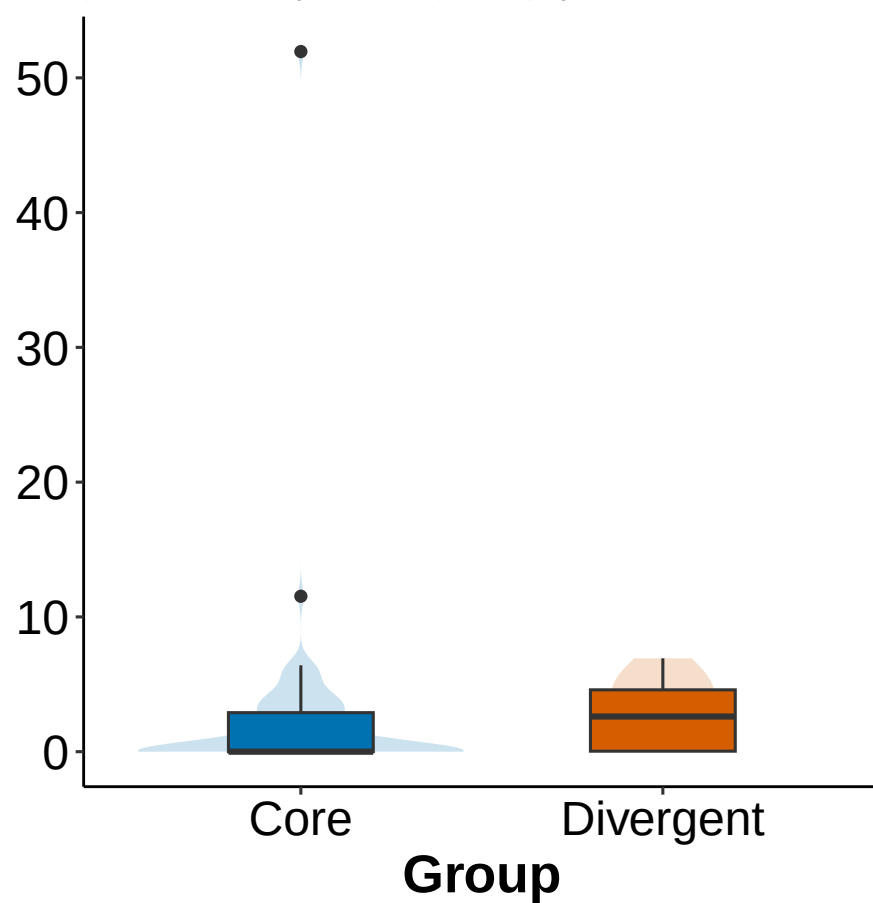
SA_NAD_metabolism_FBA

p-adj: 0.885 | .=0.15 (Small)



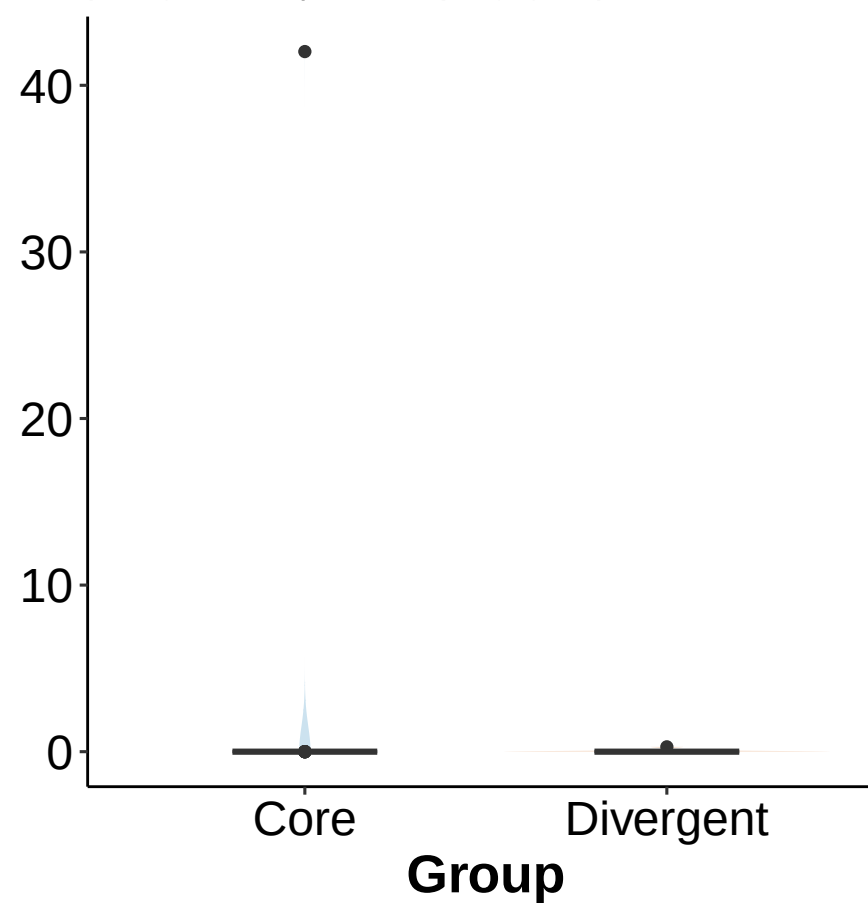
SA_NAD_metabolism_L1w

p-adj: 0.872 | .=0.27 (Small) | .%: 7933817.9



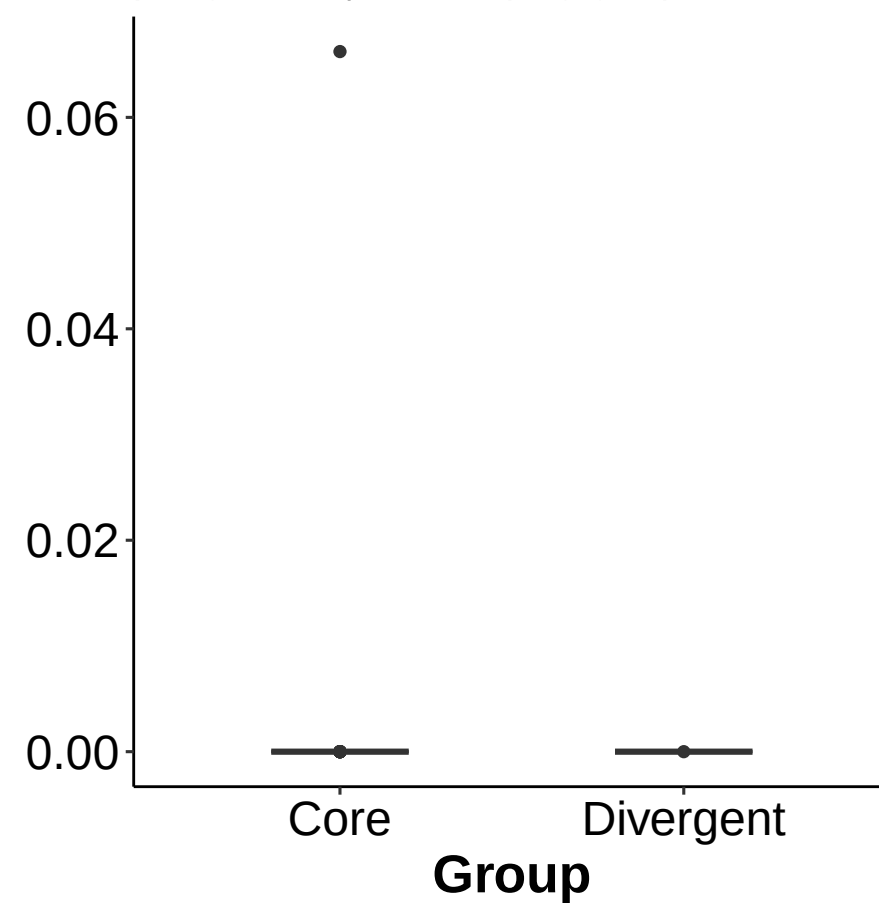
SA_NAD_metabolism_pFBA

p-adj: 0.954 | .=0.02 (Negligible)



SA_N_glycan_synthesis_L1w

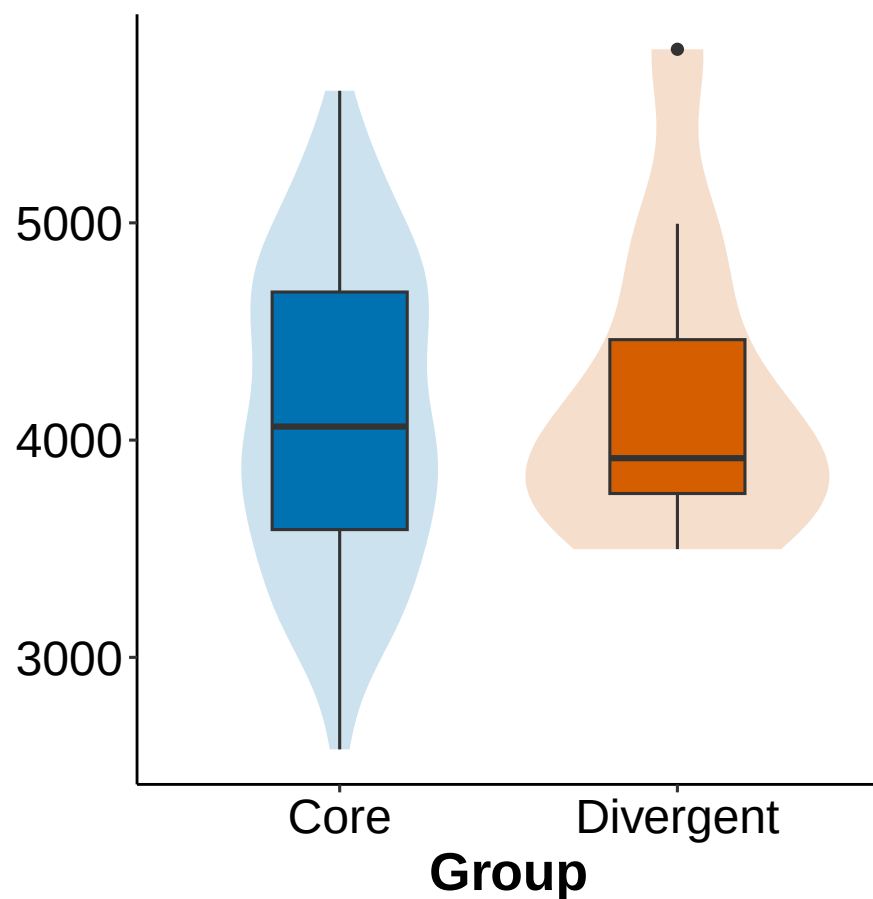
p-adj: 0.974 | .=-0.01 (Negligible)



Metabolism – . 5. Systemic Reprogramming (SA_*)

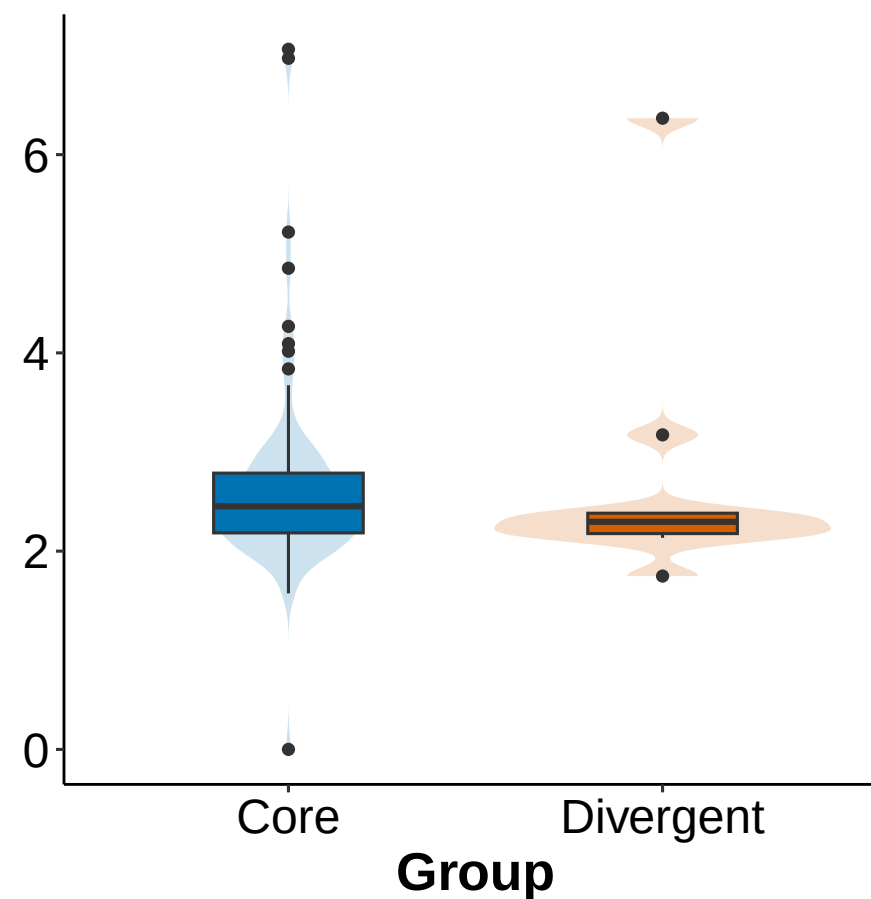
SA_Nucleotide_interconversion_FB

p-adj: 0.954 | . = 0.03 (Negligible) | .%: -3.6



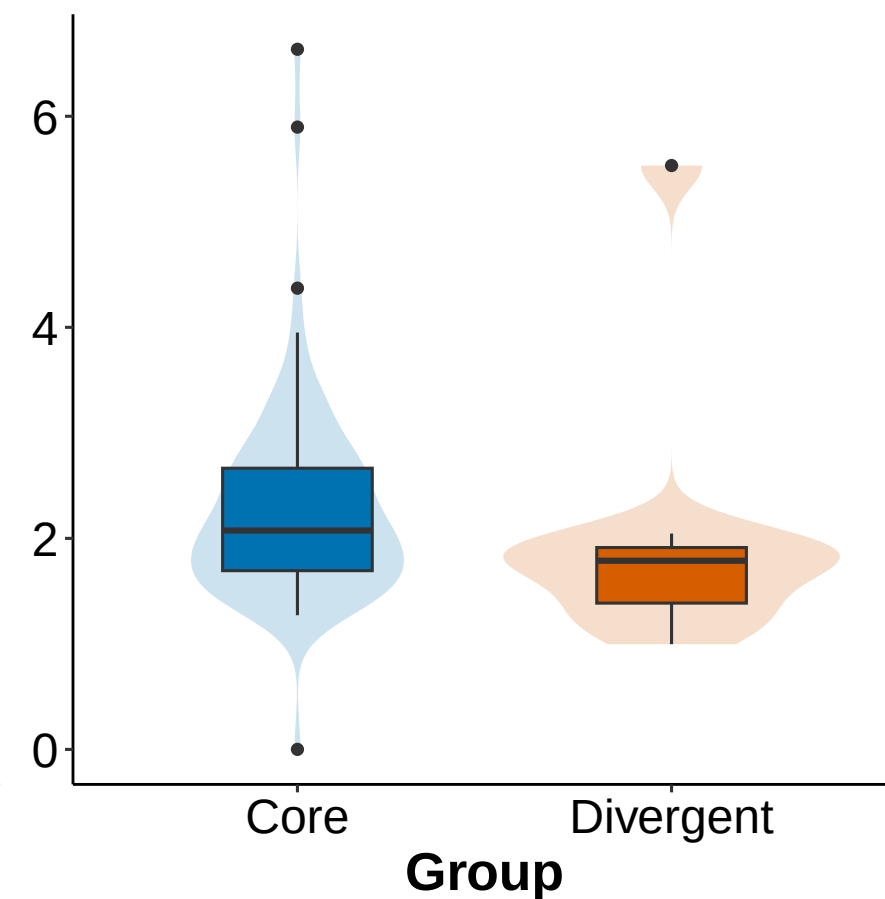
SA_Nucleotide_interconversion_L1w

p-adj: 0.885 | . = -0.18 (Small) | .%: -6.4



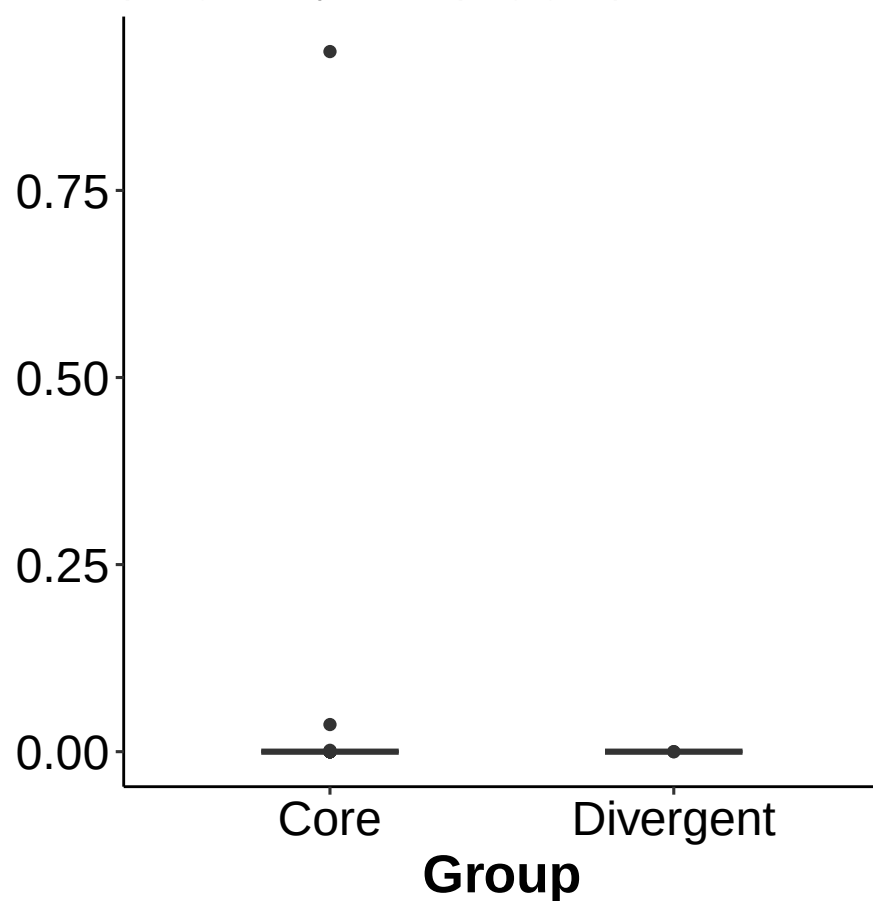
SA_Nucleotide_interconversion_pFBA

p-adj: 0.843 | . = -0.39 (Medium) | .%: -13.8



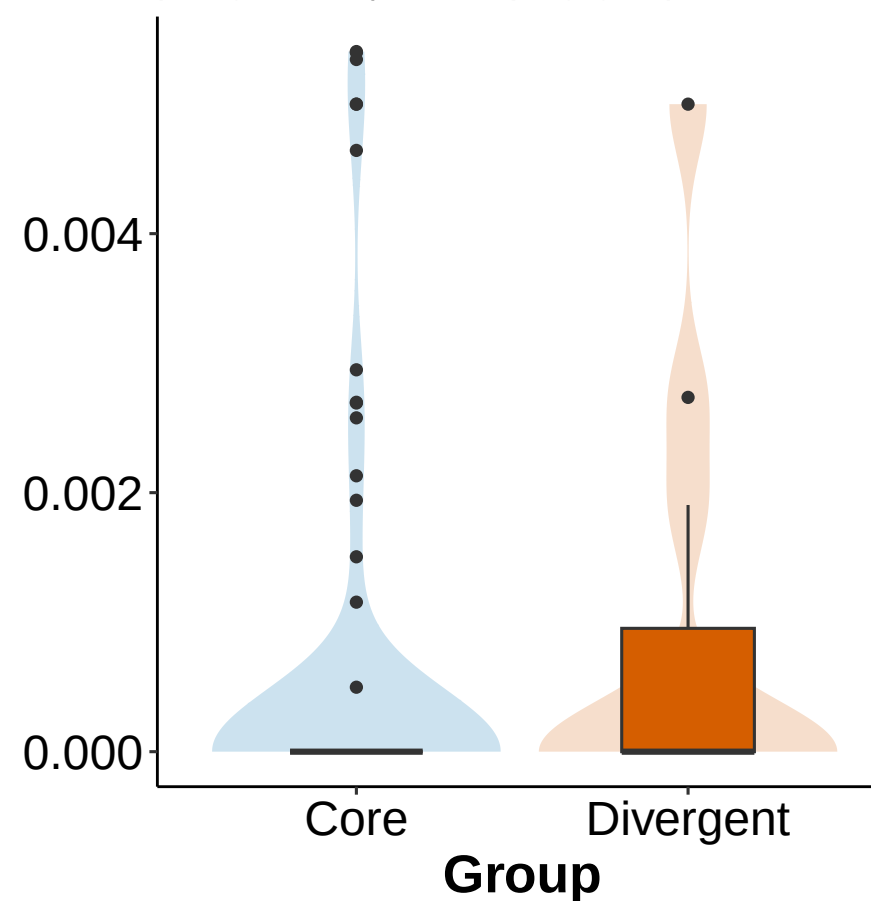
SA_Nucleotide_metabolism_L1w

p-adj: 0.95 | . = 0.06 (Negligible)



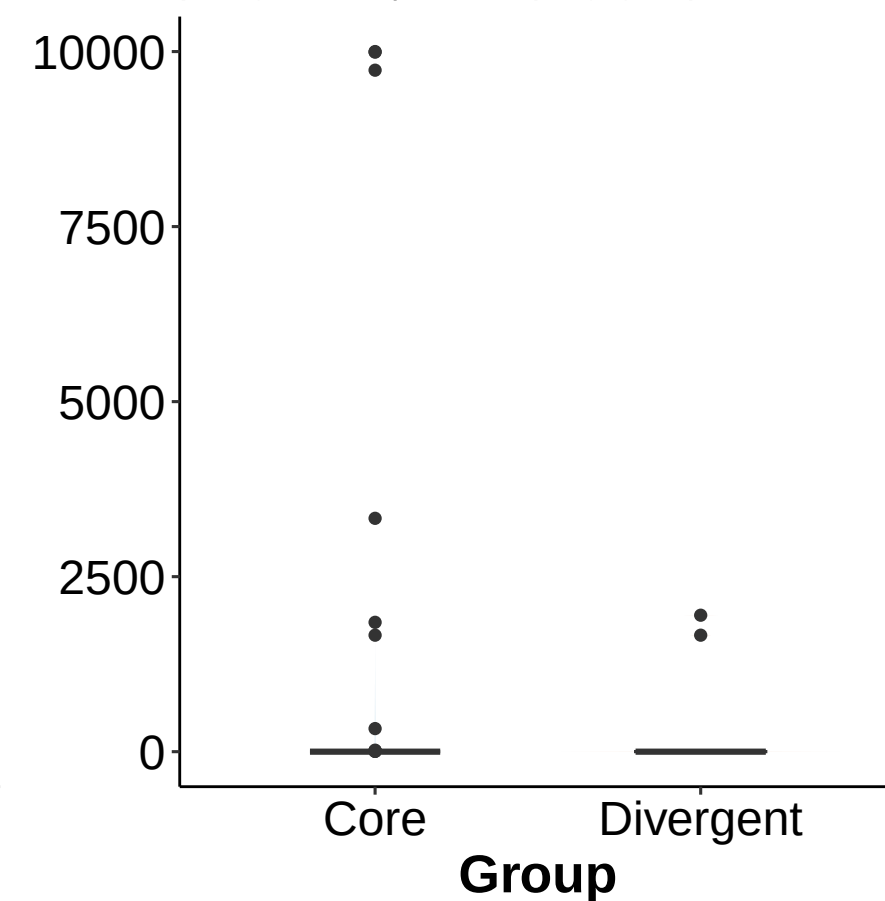
SA_Nucleotide_metabolism_pFBA

p-adj: 0.885 | . = 0.12 (Negligible)



SA_Nucleotide_salvage_pathway_pFBA

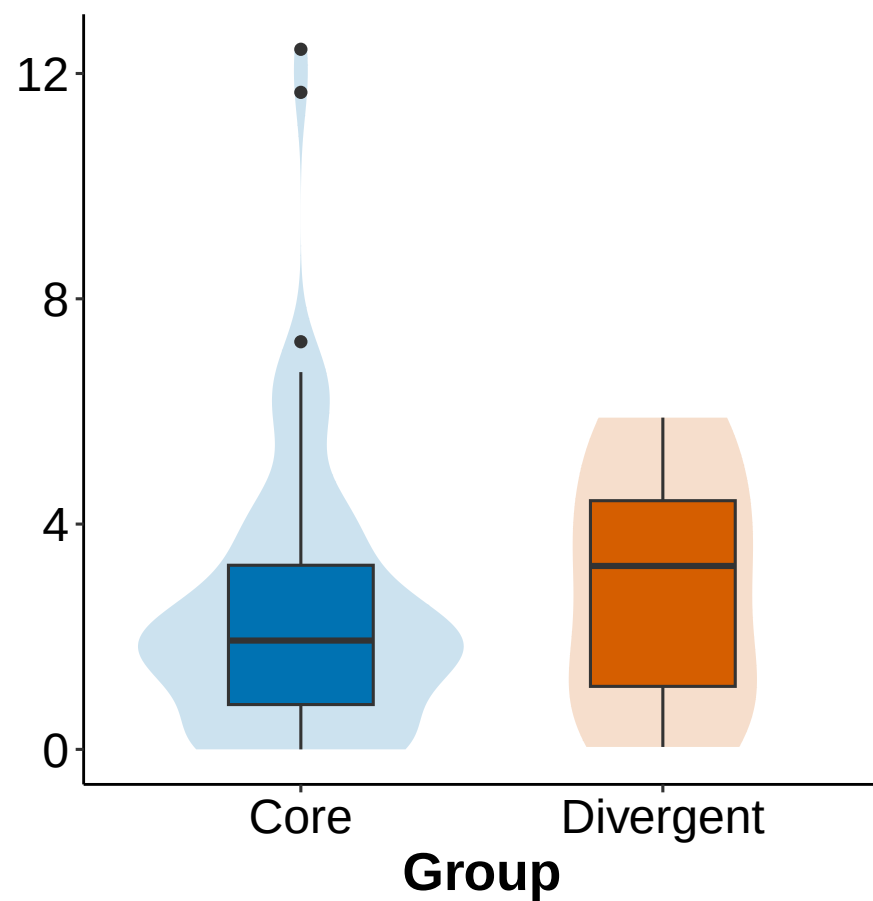
p-adj: 0.885 | . = 0.13 (Negligible)



Metabolism – . 5. Systemic Reprogramming (SA_*)

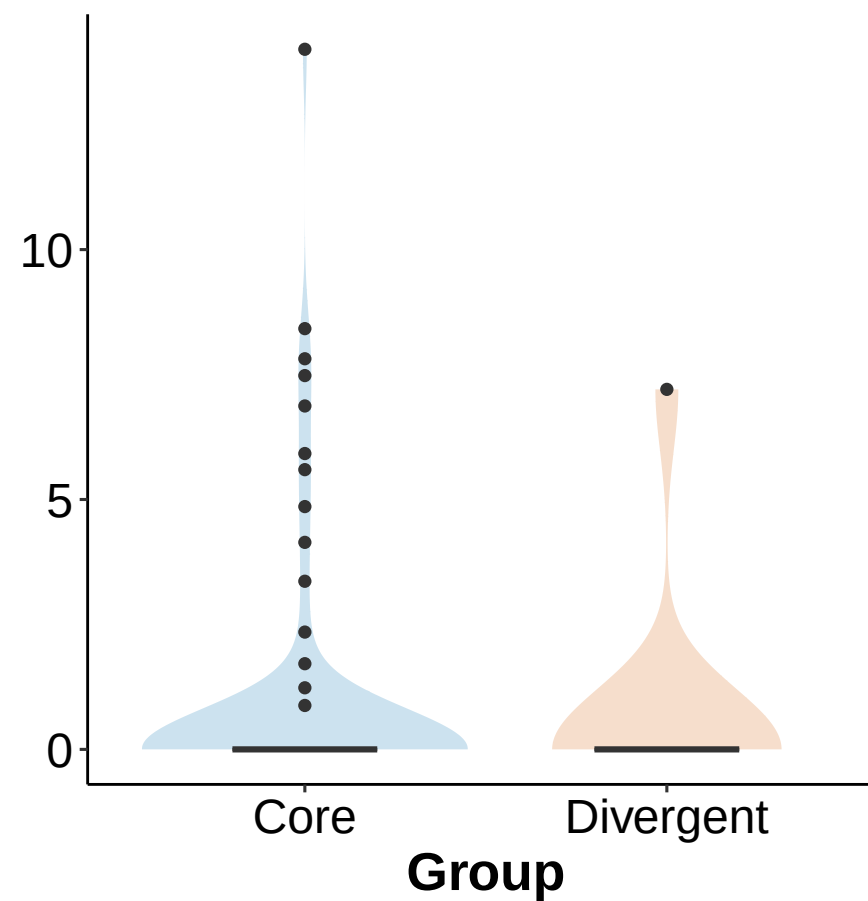
SA_Nucleotide_salvage_pathway_L1v

p-adj: 0.885 | .:=0.17 (Small) | .%: 68.5



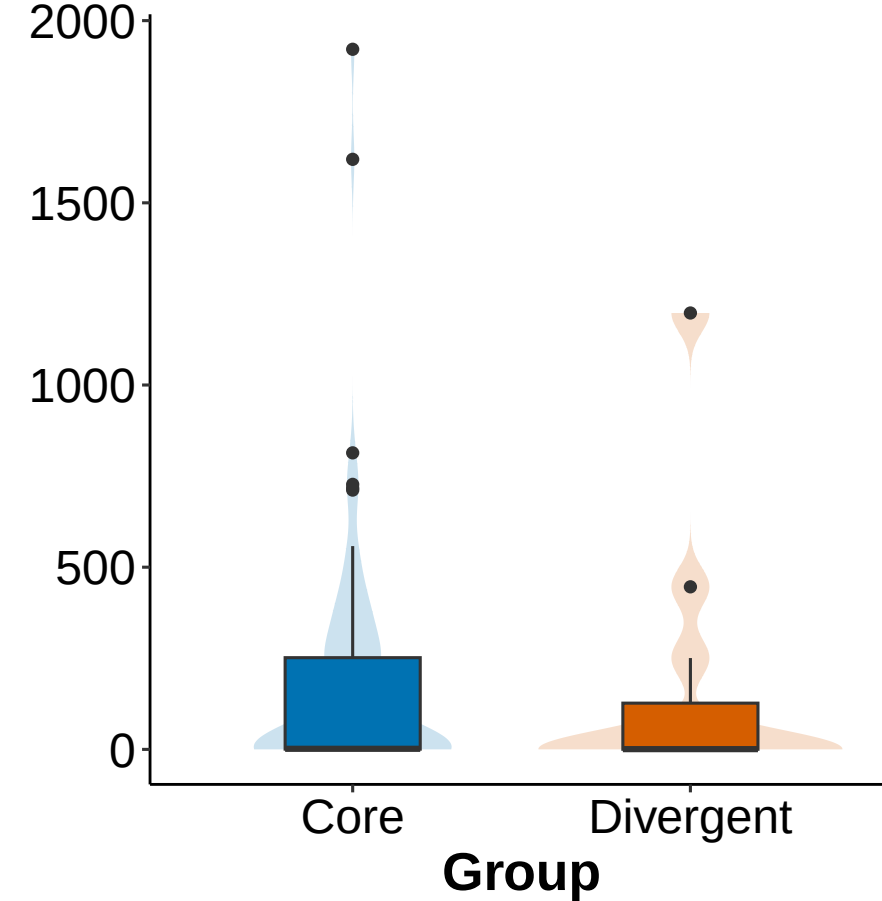
SA_Nucleotide_salvage_pathway_pFE

p-adj: 0.95 | .:=−0.04 (Negligible)



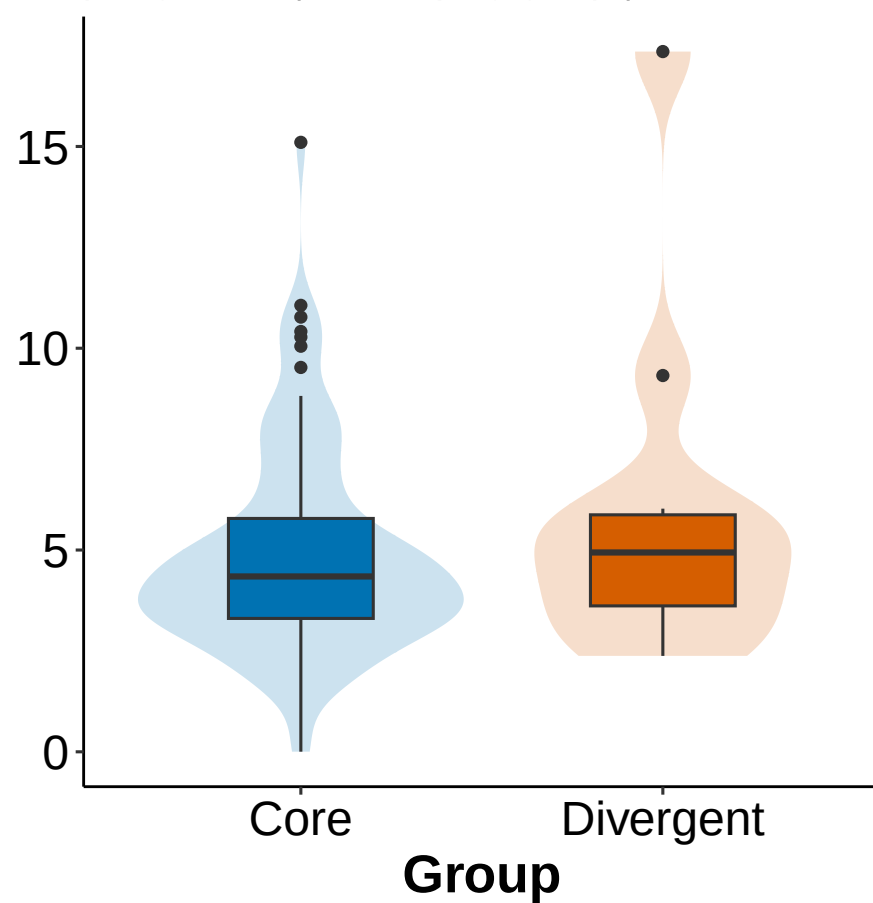
SA_Oxidative_phosphorylation_FB

p-adj: 0.885 | .:=−0.16 (Small) | .%: −59.2



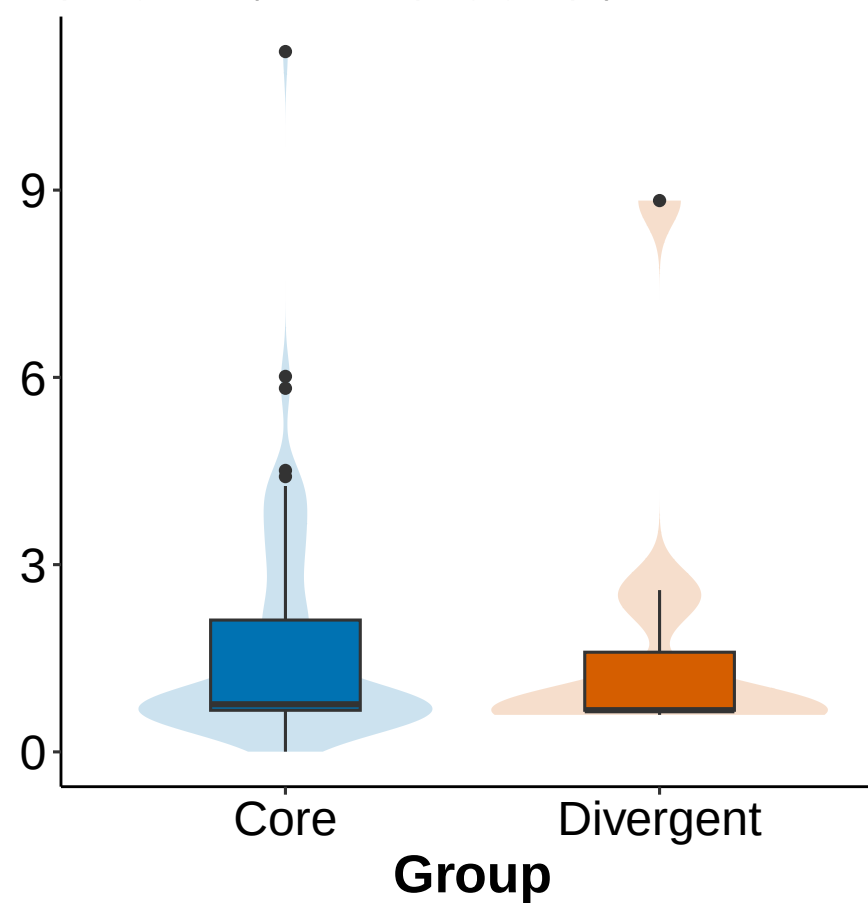
SA_Oxidative_phosphorylation_L1w

p-adj: 0.903 | .:=0.13 (Negligible) | .%: 13.7



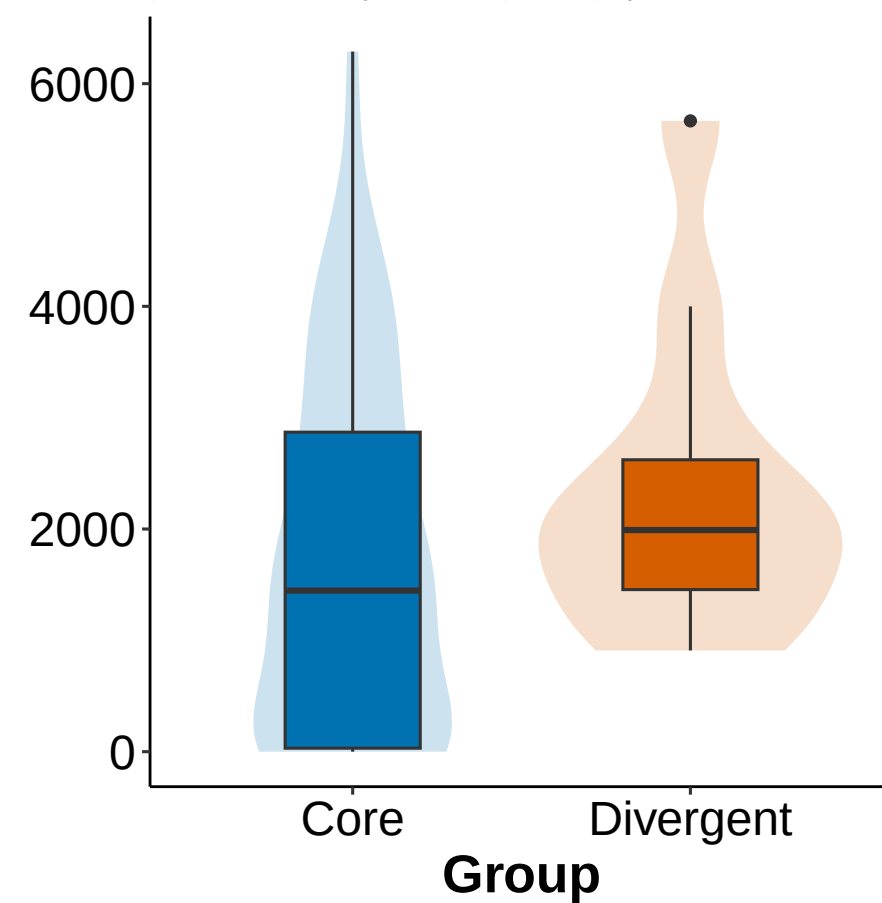
SA_Oxidative_phosphorylation_pFBA

p-adj: 0.95 | .:=−0.09 (Negligible) | .%: −12.4



SA_Pentose_phosphate_pathway_L1w

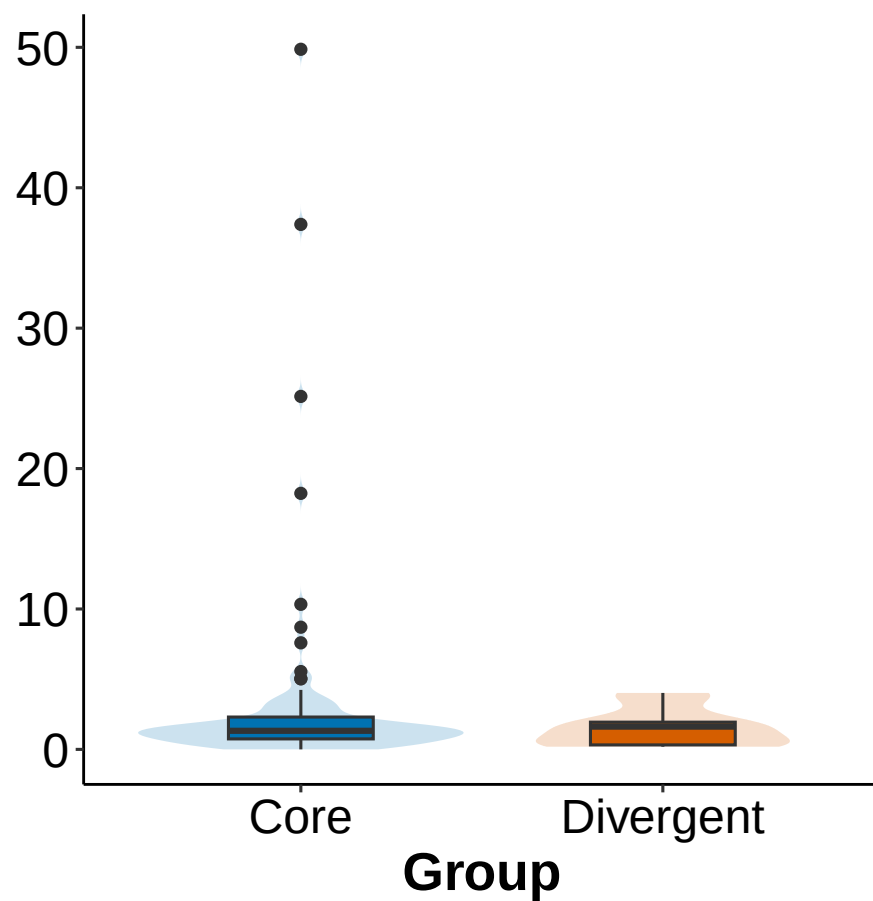
p-adj: 0.872 | .:=0.26 (Small) | .%: 37.6



Metabolism – . 5. Systemic Reprogramming (SA_*)

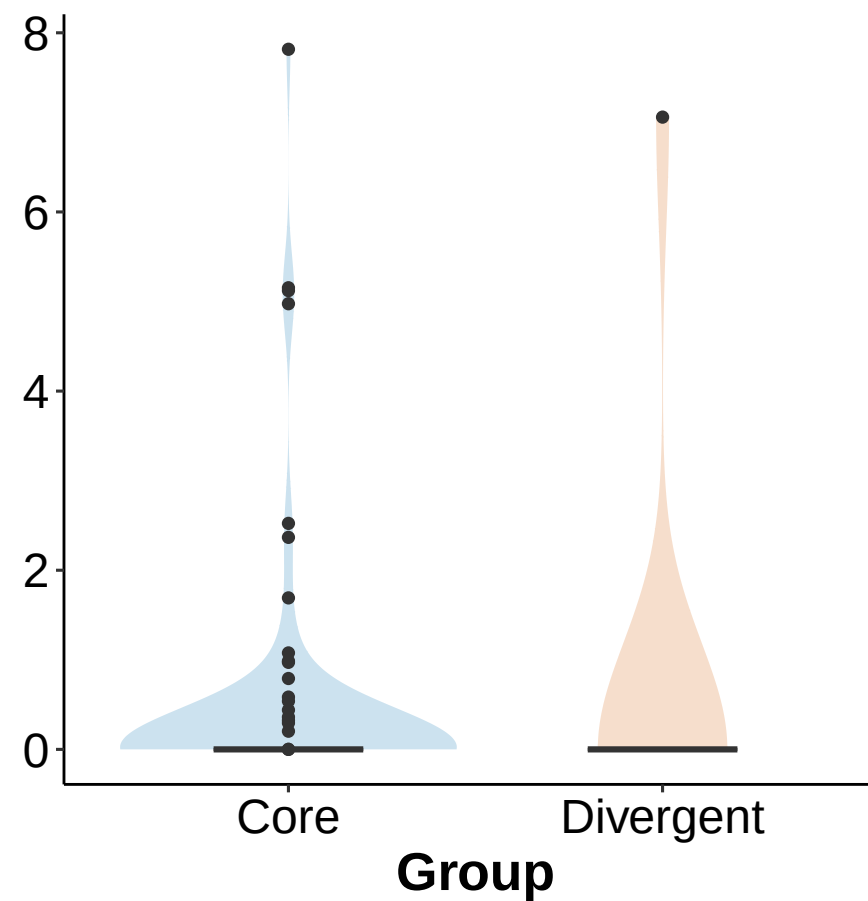
SA_Pentose_phosphate_pathway_L1w

p-adj: 0.928 | .=-0.1 (Negligible) | .%: 22.6



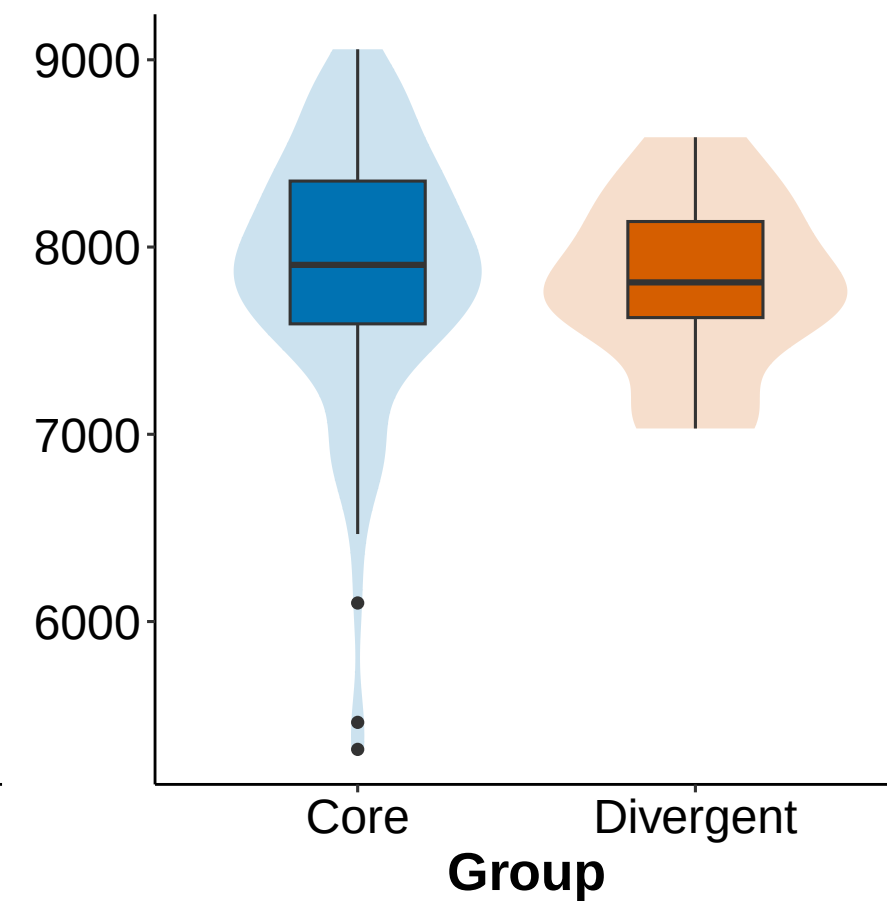
SA_Pentose_phosphate_pathway_pFB

p-adj: 0.896 | .=-0.09 (Negligible)



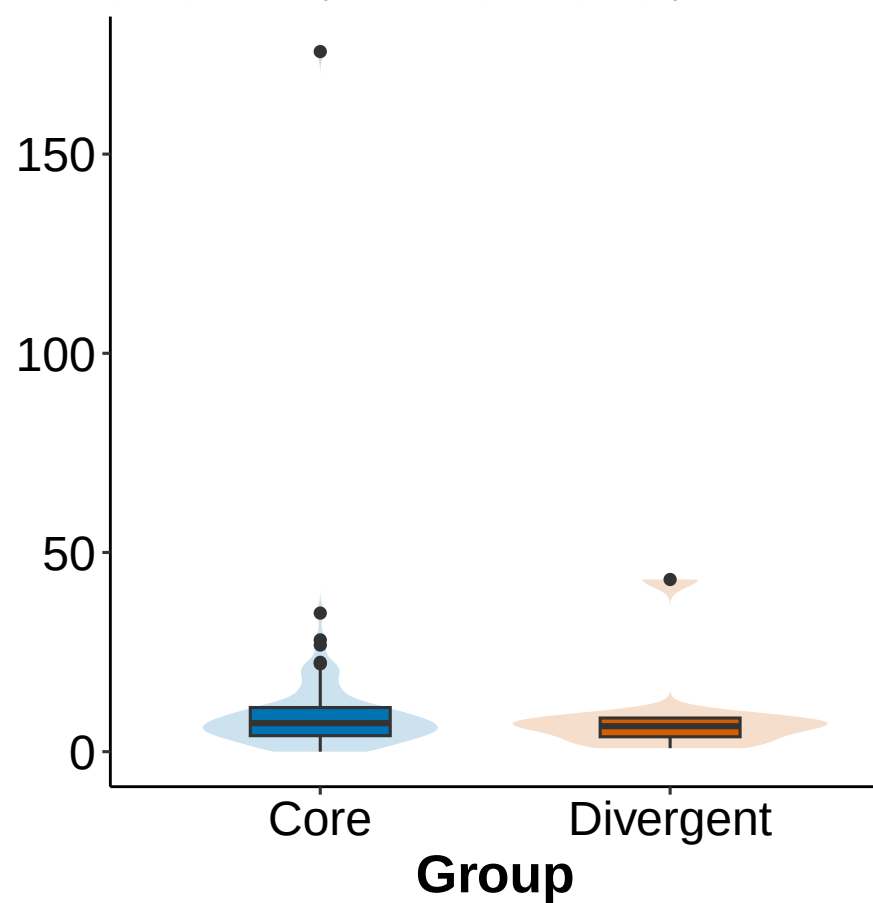
SA_Peptide_metabolism_FBA

p-adj: 0.917 | .=-0.13 (Negligible) | .%: -1.2



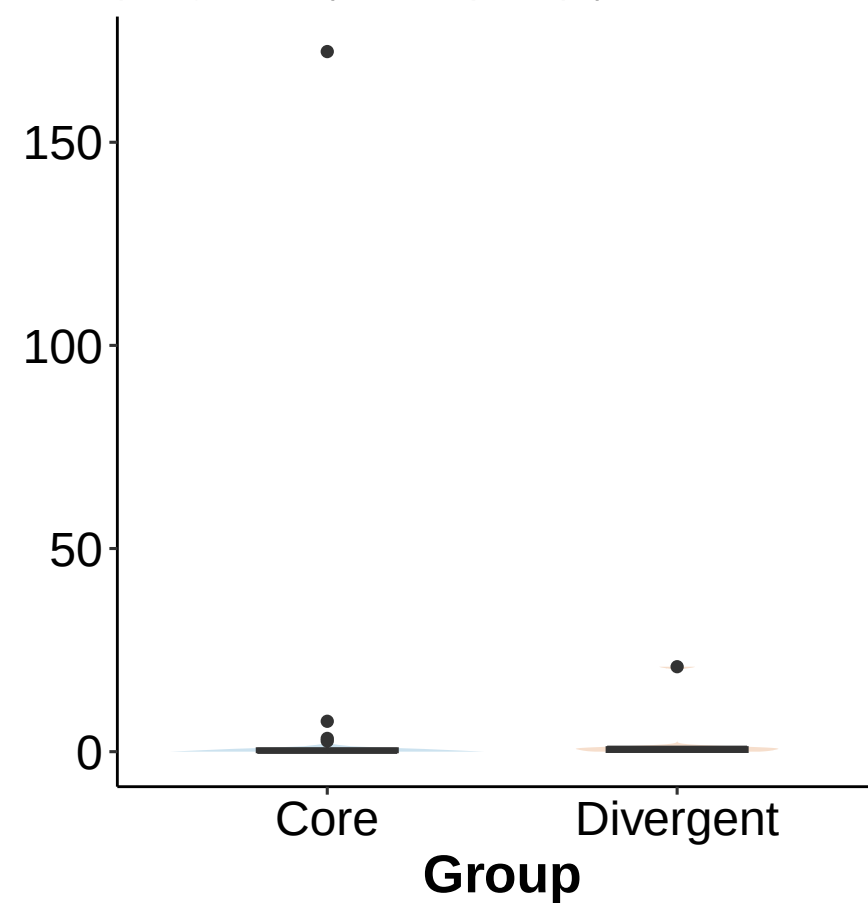
SA_Peptide_metabolism_L1w

p-adj: 0.917 | .=-0.11 (Negligible) | .%: -10.7



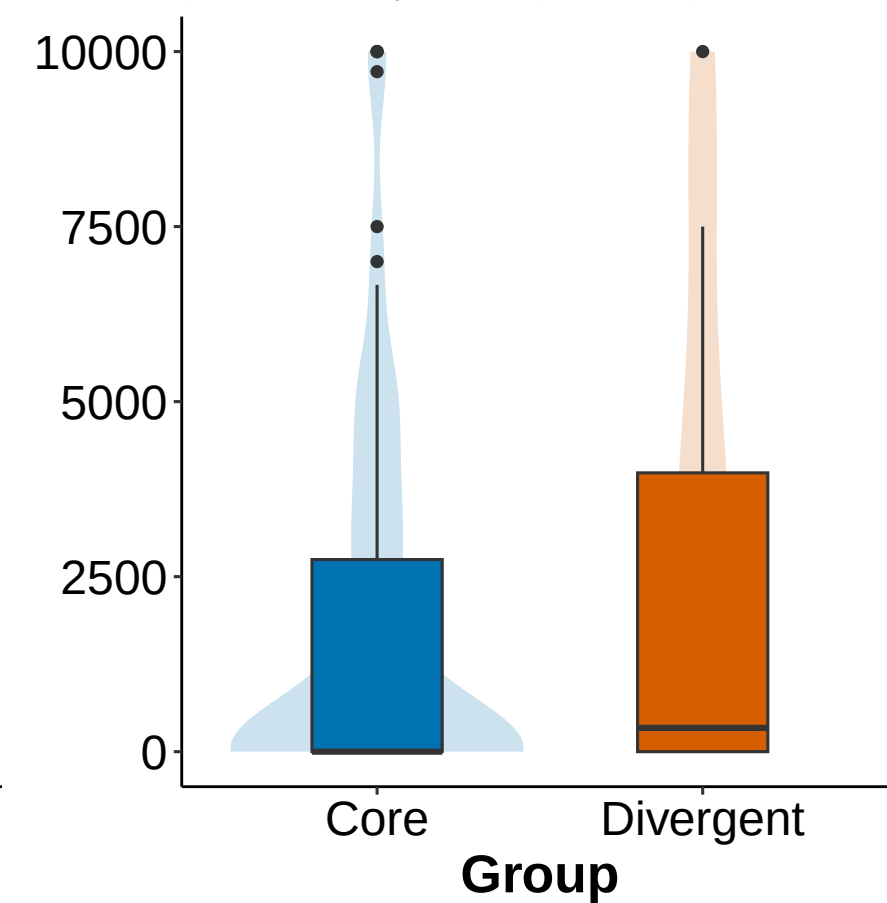
SA_Peptide_metabolism_pFBA

p-adj: 0.843 | .=-0.31 (Small) | .%: 233.7



SA_Phenylalanine_metabolism_FBA

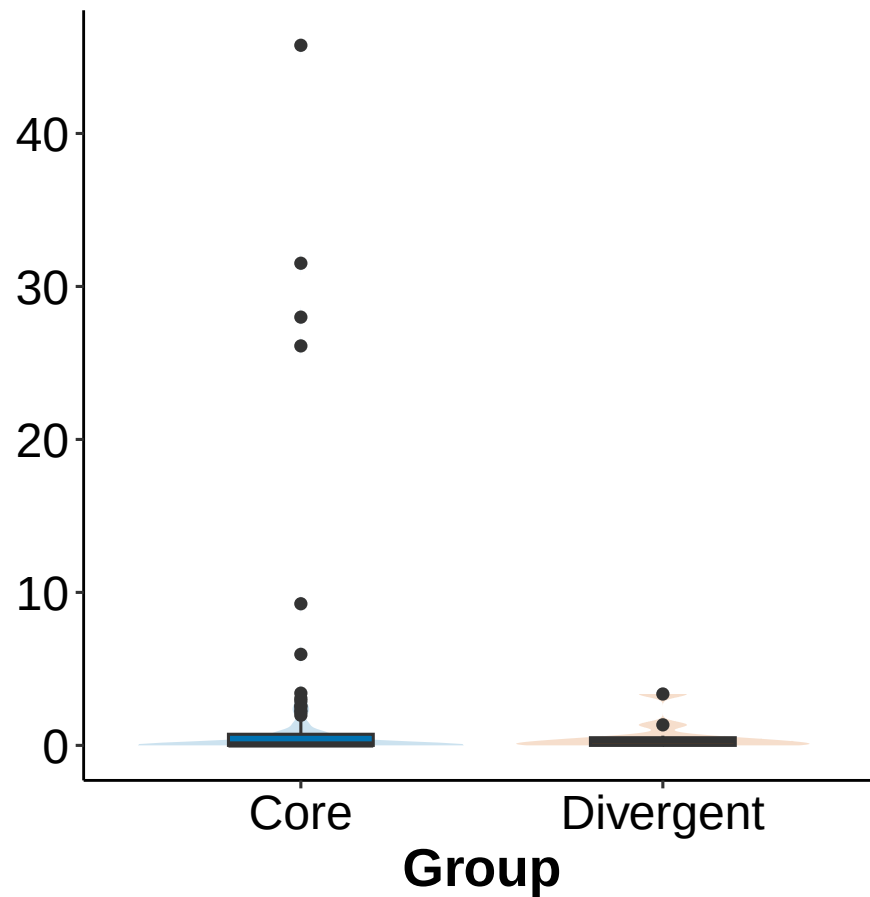
p-adj: 0.885 | .=-0.14 (Negligible)



Metabolism – . 5. Systemic Reprogramming (SA_*)

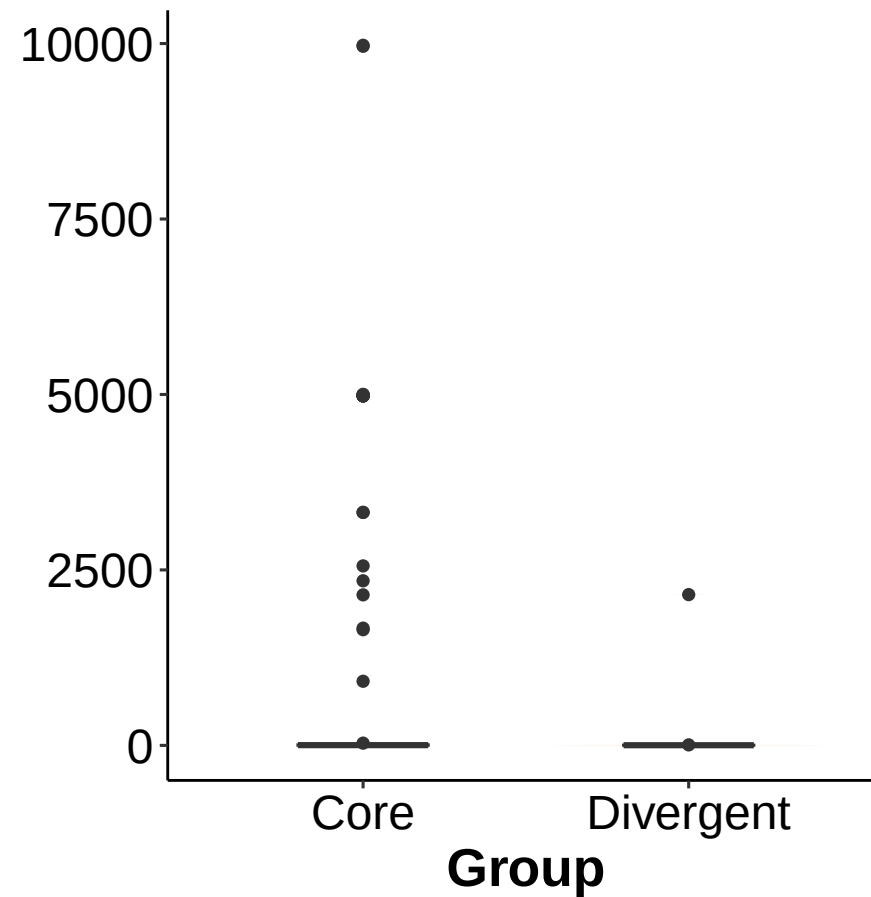
SA_Phenylalanine_metabolism_L1w

p-adj: 0.896 | .:=0.14 (Negligible) | .%: 390.3



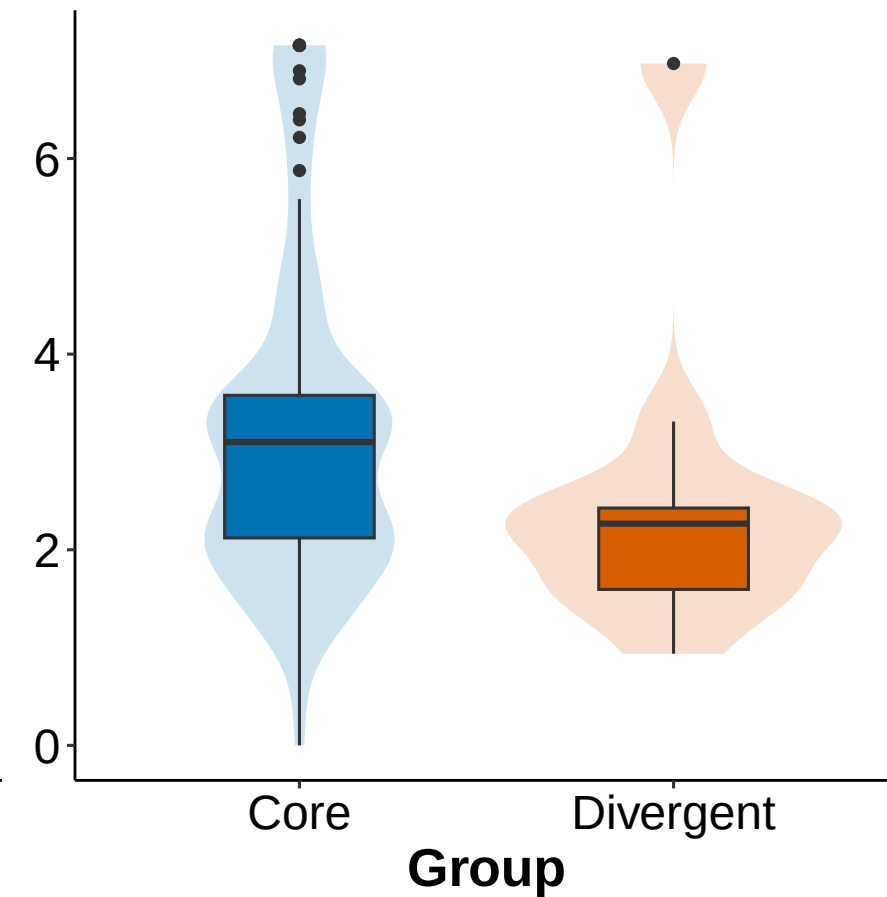
SA_Phosphatidylinositol_phosphatidylcholine_m

p-adj: 0.843 | .:=−0.36 (Medium) | .%: −33.3



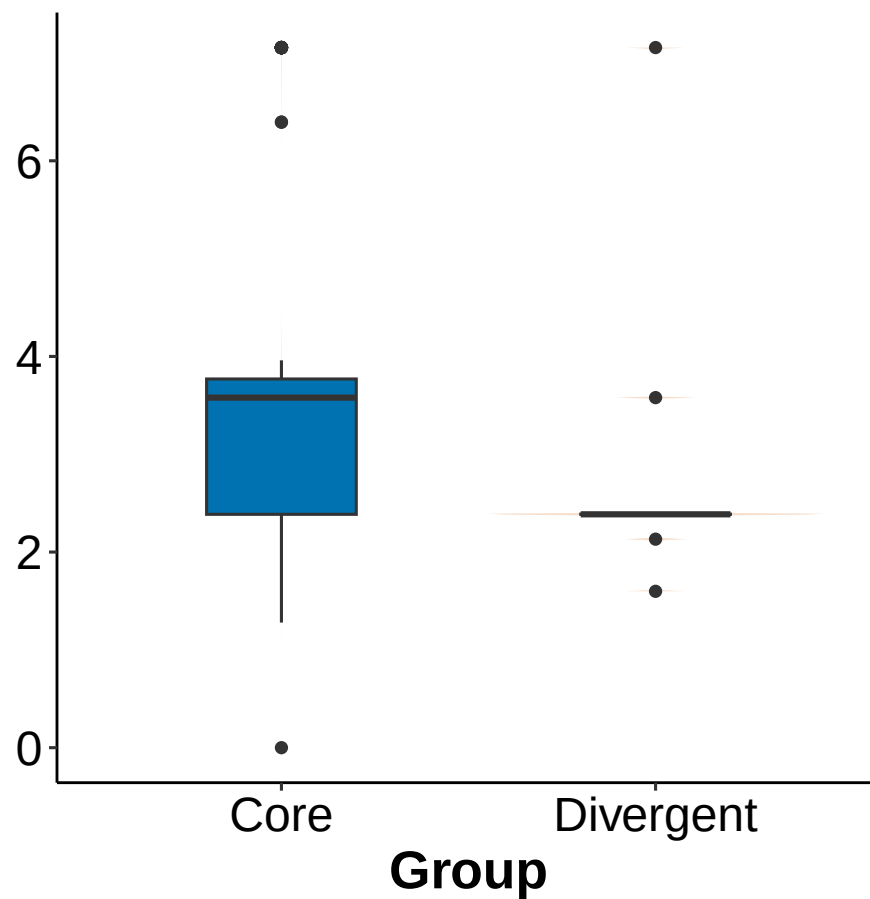
SA_Phosphatidylinositol_phosphate_m

p-adj: 0.872 | .:=−0.31 (Small) | .%: −26.9



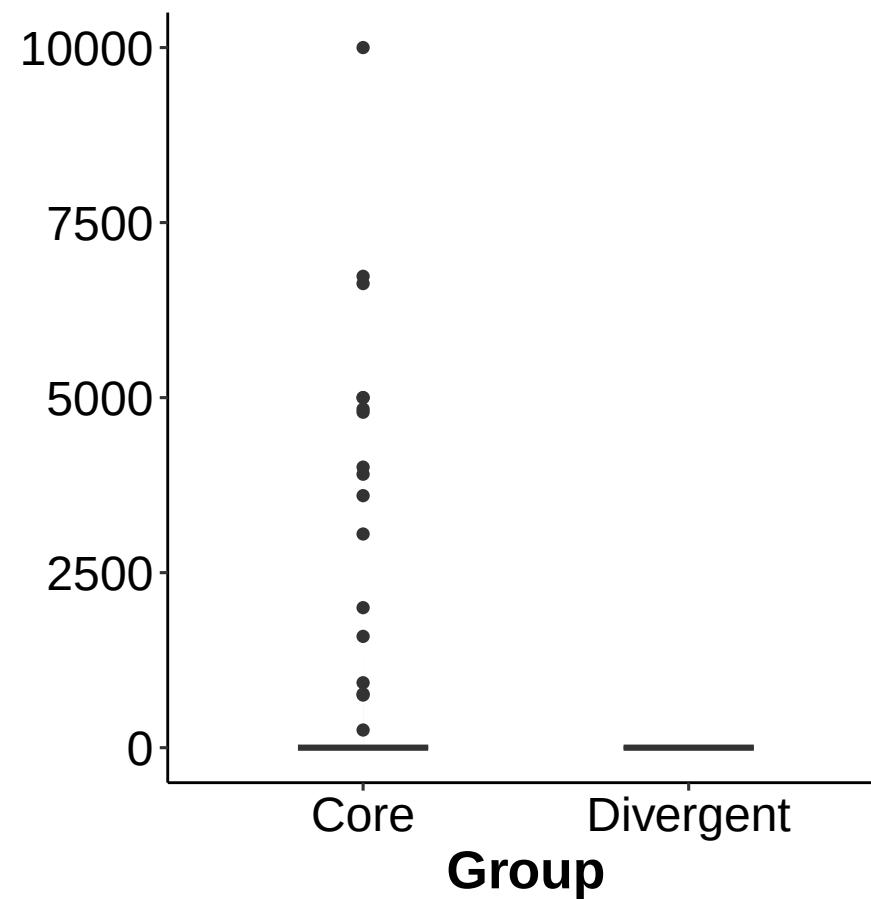
SA_Phosphatidylinositol_phosphate_n

p-adj: 0.843 | .:=−0.39 (Medium) | .%: −33.3



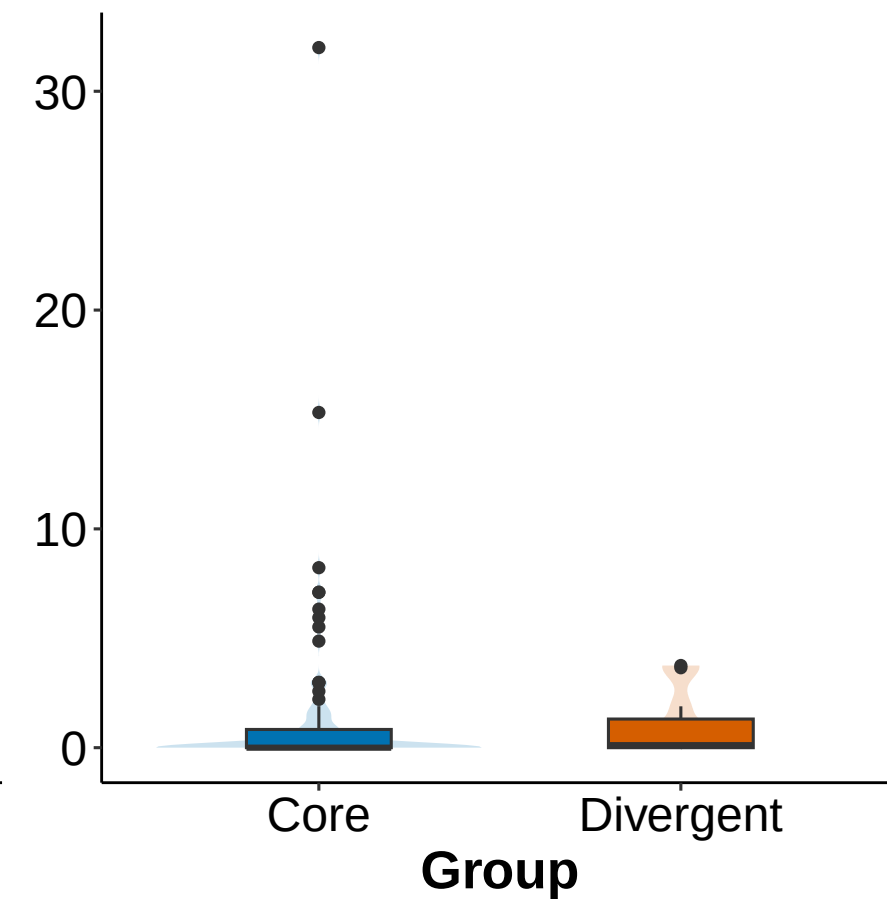
SA_Propanoate_metabolism_FBA

p-adj: 0.872 | .:=−0.17 (Small)



SA_Propanoate_metabolism_L1w

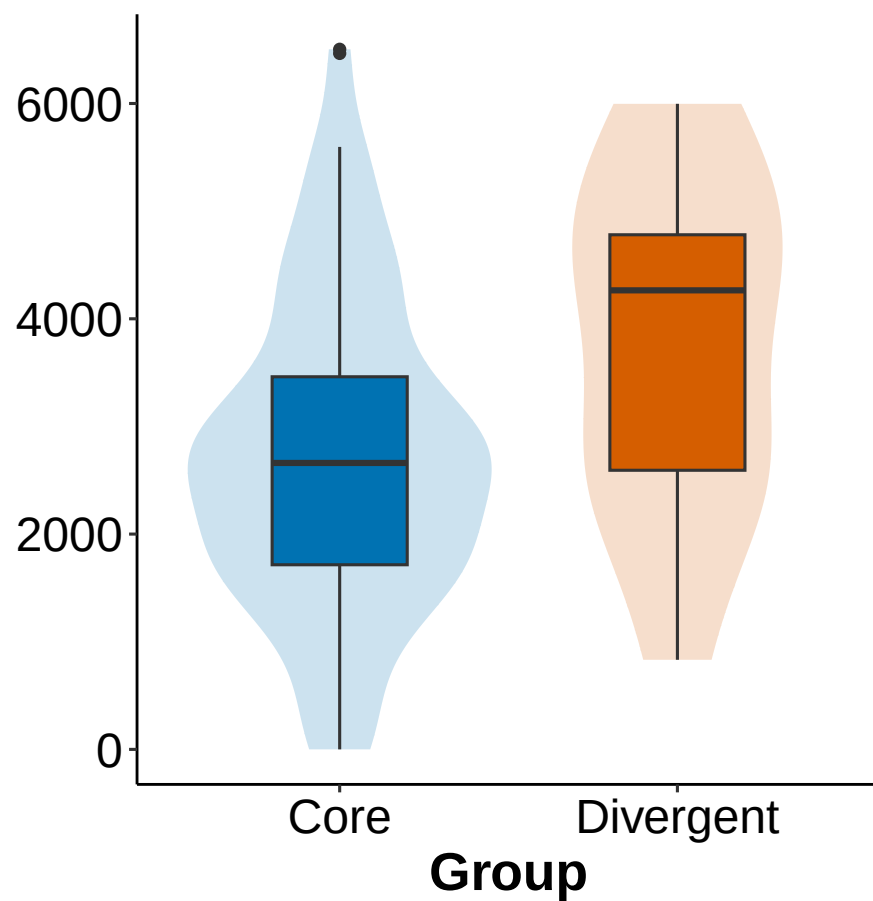
p-adj: 0.885 | .:=0.15 (Small) | .%: 1578181497152



Metabolism – . 5. Systemic Reprogramming (SA_*)

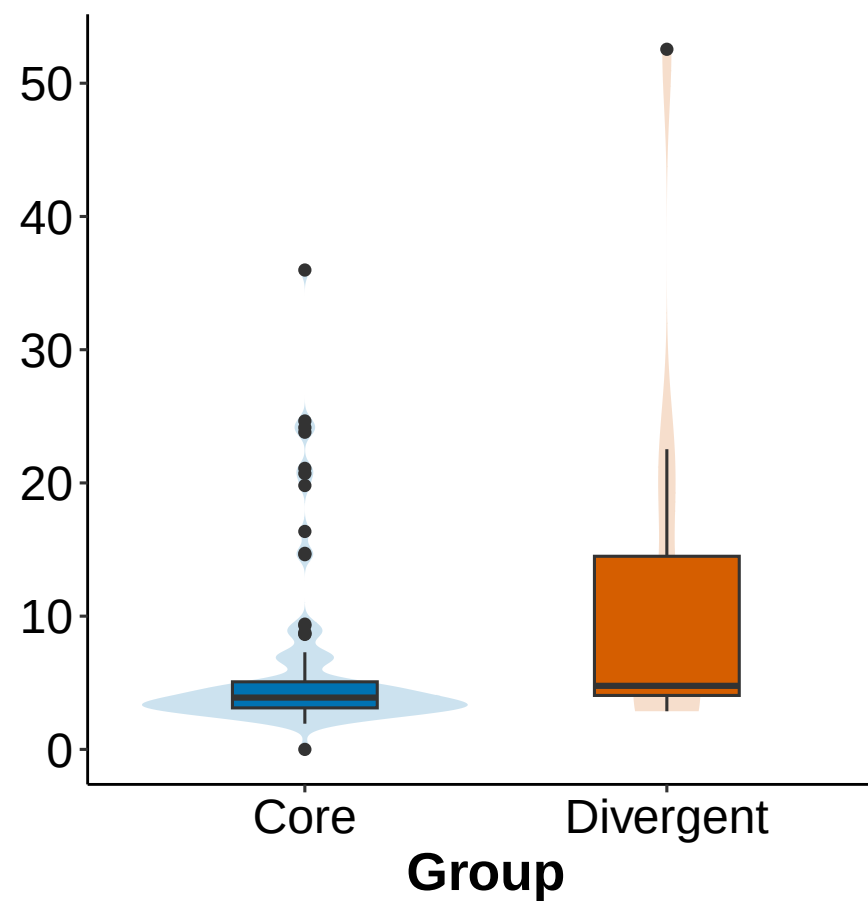
SA_Purine_catabolism_FBA

p-adj: 0.843 | .=0.36 (Medium) | .%: 60.3



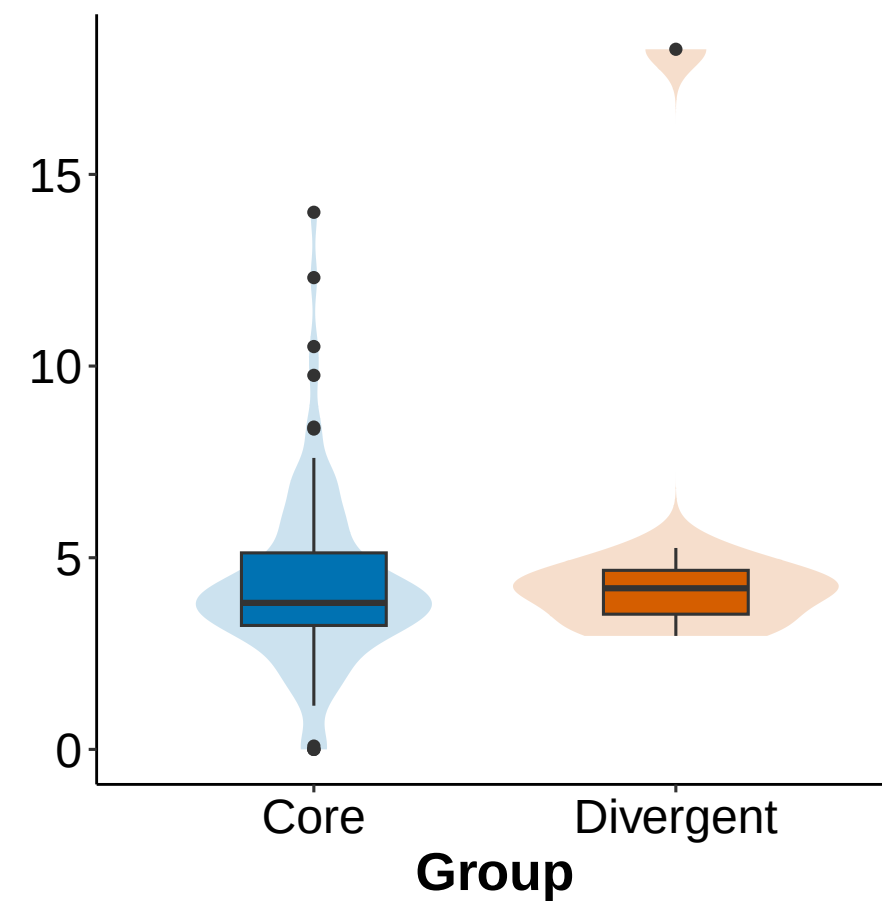
SA_Purine_catabolism_L1w

p-adj: 0.843 | .=0.35 (Medium) | .%: 22.7



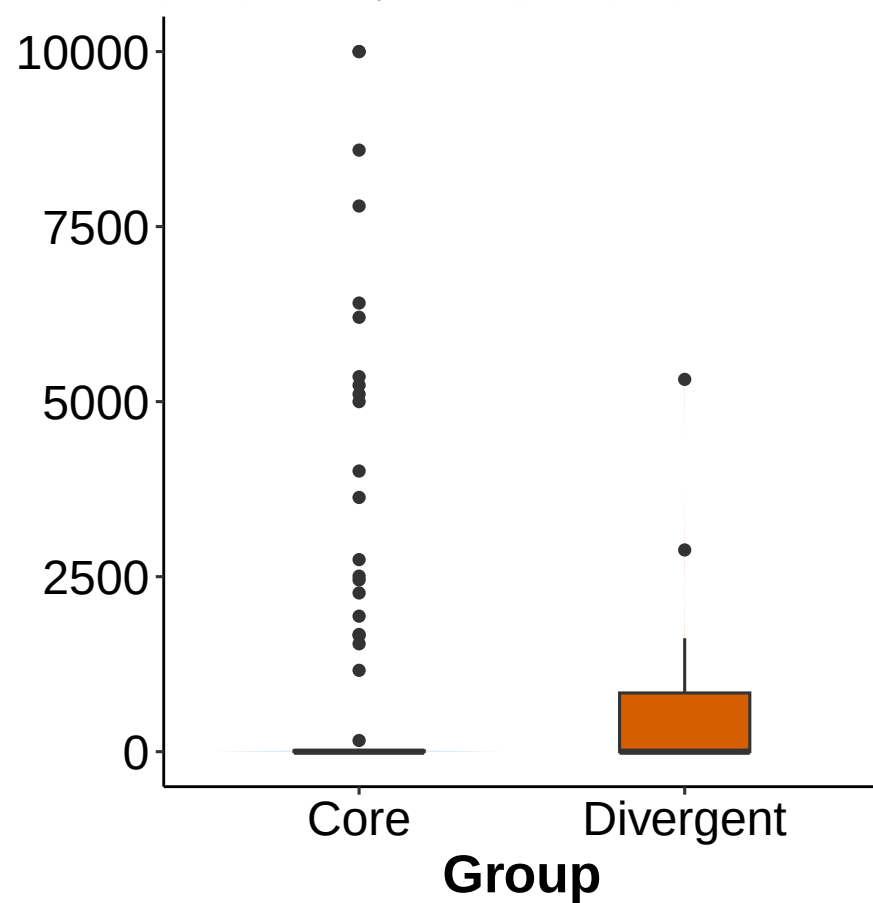
SA_Purine_catabolism_pFBA

p-adj: 0.917 | .=0.12 (Negligible) | .%: 10



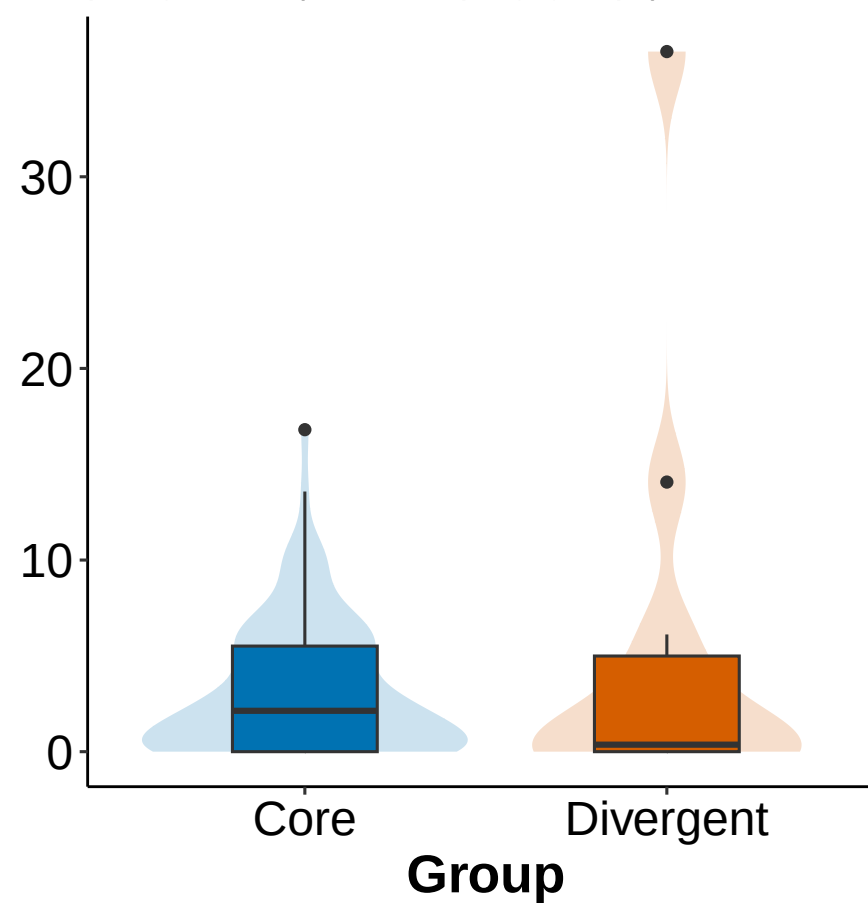
SA_Purine_synthesis_FBA

p-adj: 0.945 | .=0.07 (Negligible)



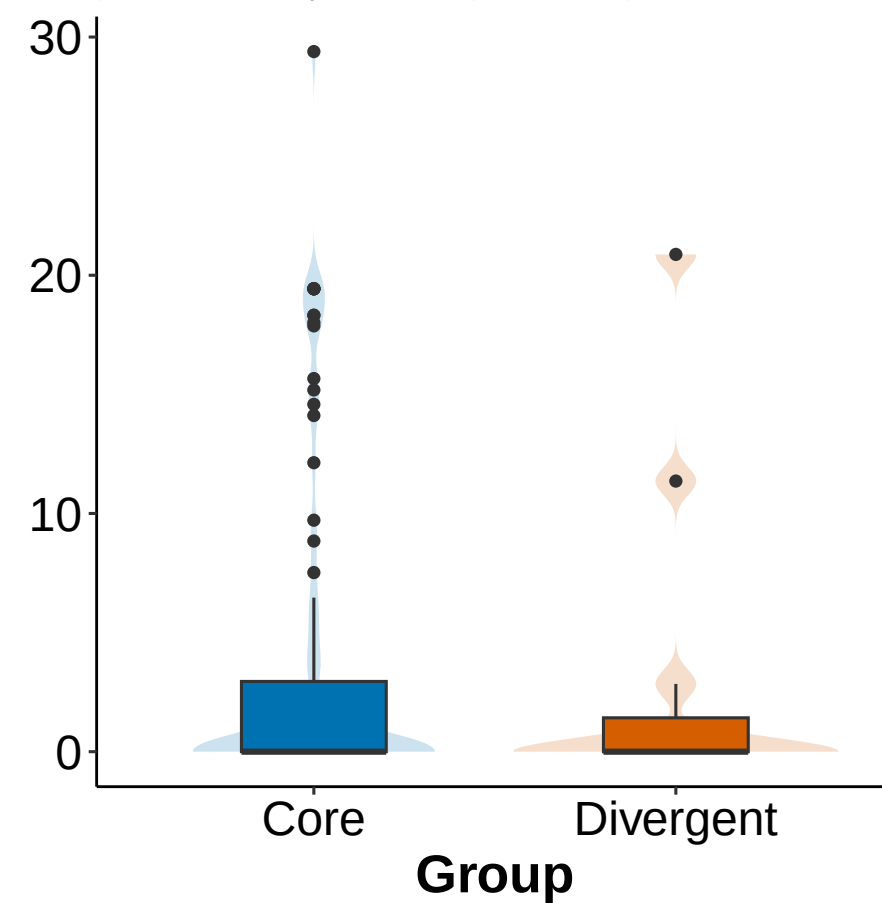
SA_Purine_synthesis_L1w

p-adj: 0.927 | .=-0.11 (Negligible) | .%: -82.8



SA_Purine_synthesis_pFBA

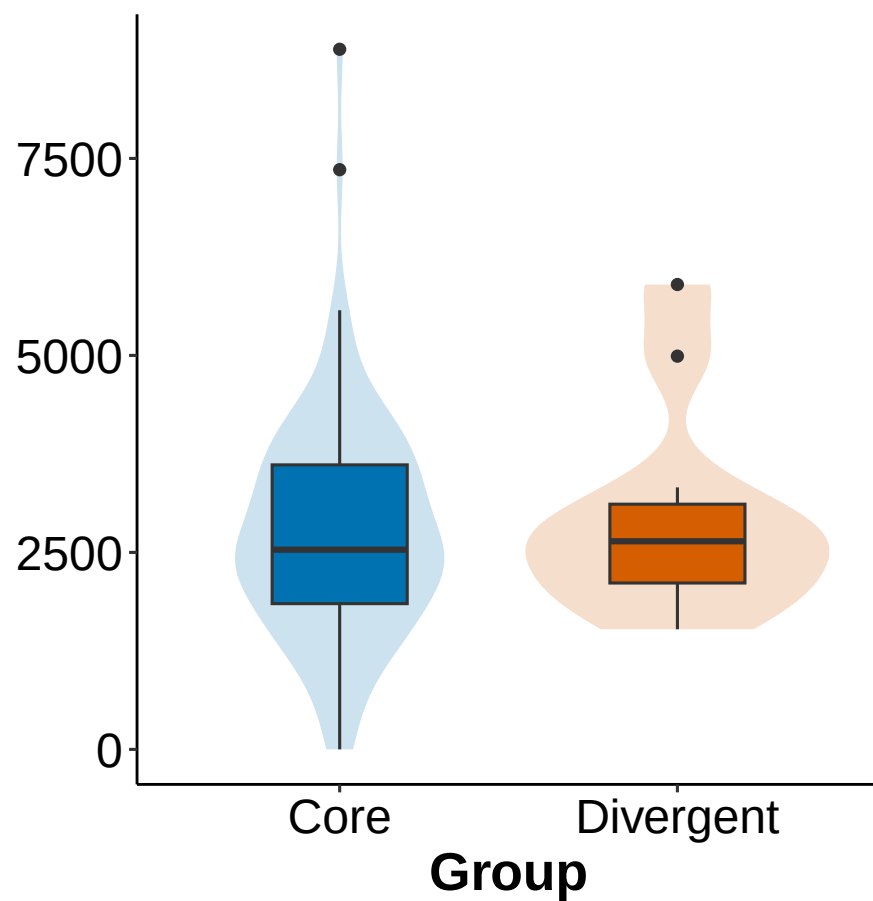
p-adj: 0.954 | .=-0.04 (Negligible)



Metabolism – . 5. Systemic Reprogramming (SA_*)

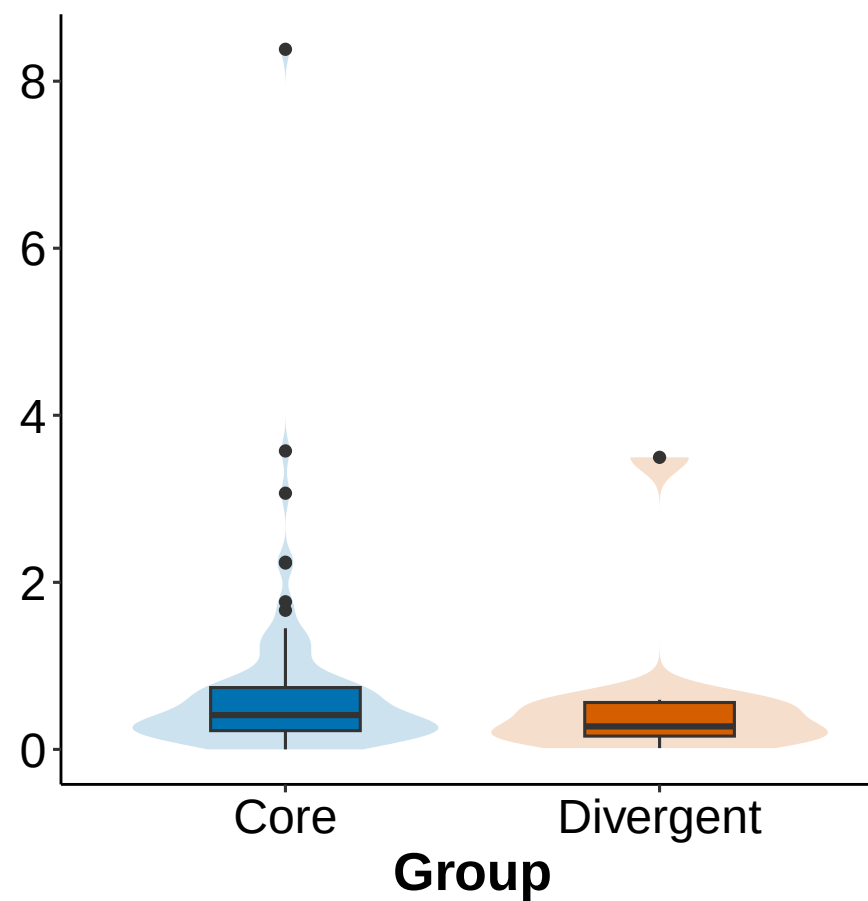
SA_Pyrimidine_catabolism_FBA

p-adj: 0.954 | .>=0.04 (Negligible) | .%: 4.2



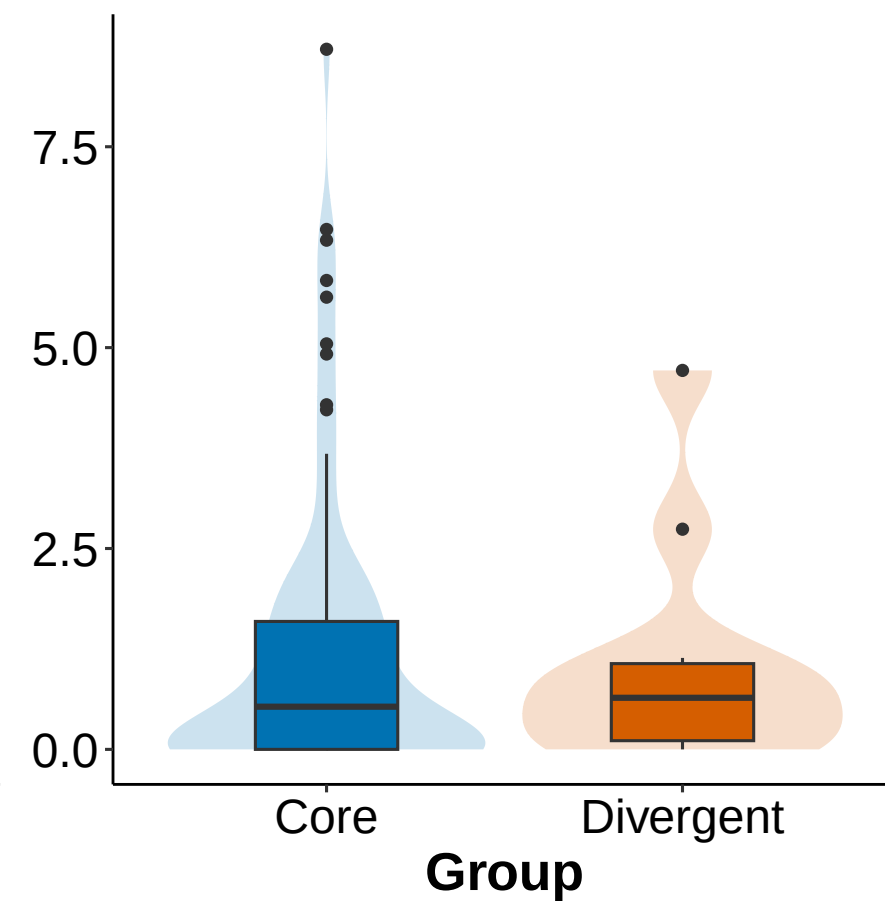
SA_Pyrimidine_catabolism_L1w

p-adj: 0.885 | .=-0.19 (Small) | .%: -32.9



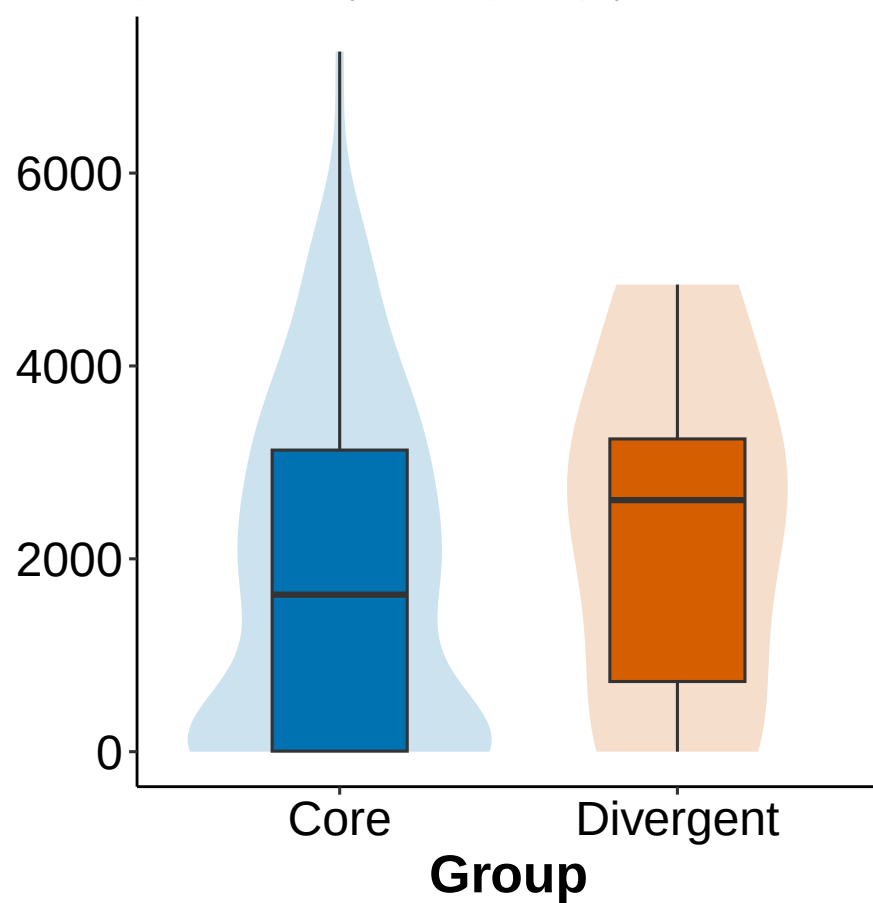
SA_Pyrimidine_catabolism_pFBA

p-adj: 0.95 | .>=0.06 (Negligible) | .%: 20.9



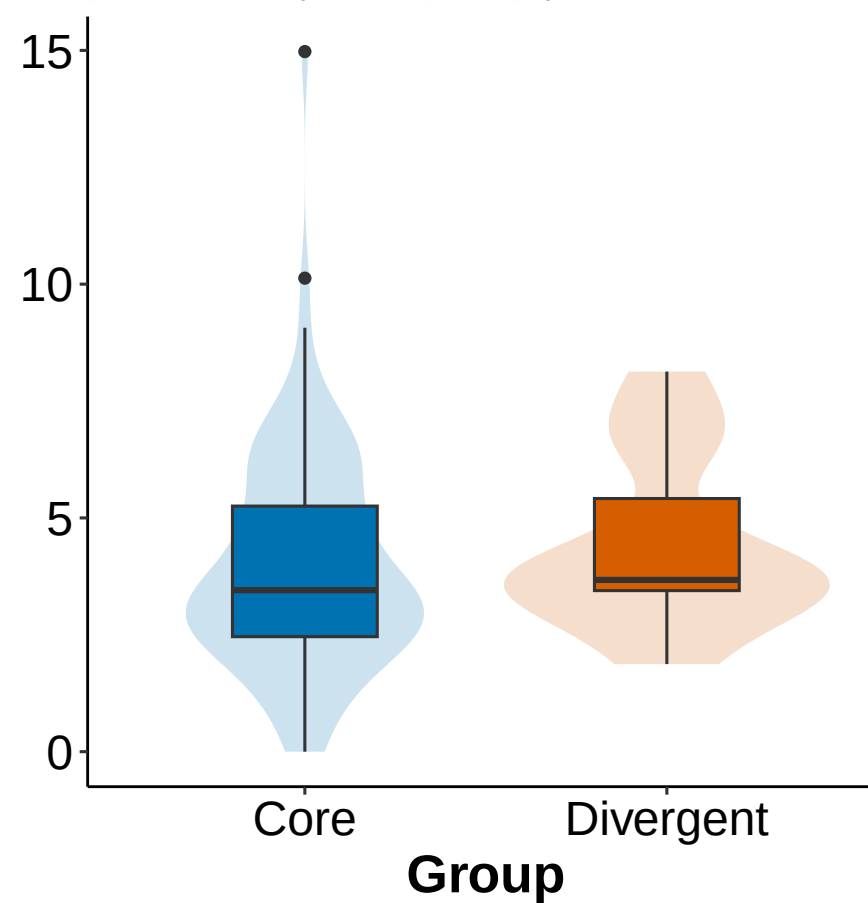
SA_Pyrimidine_synthesis_FBA

p-adj: 0.885 | .>=0.18 (Small) | .%: 60.2



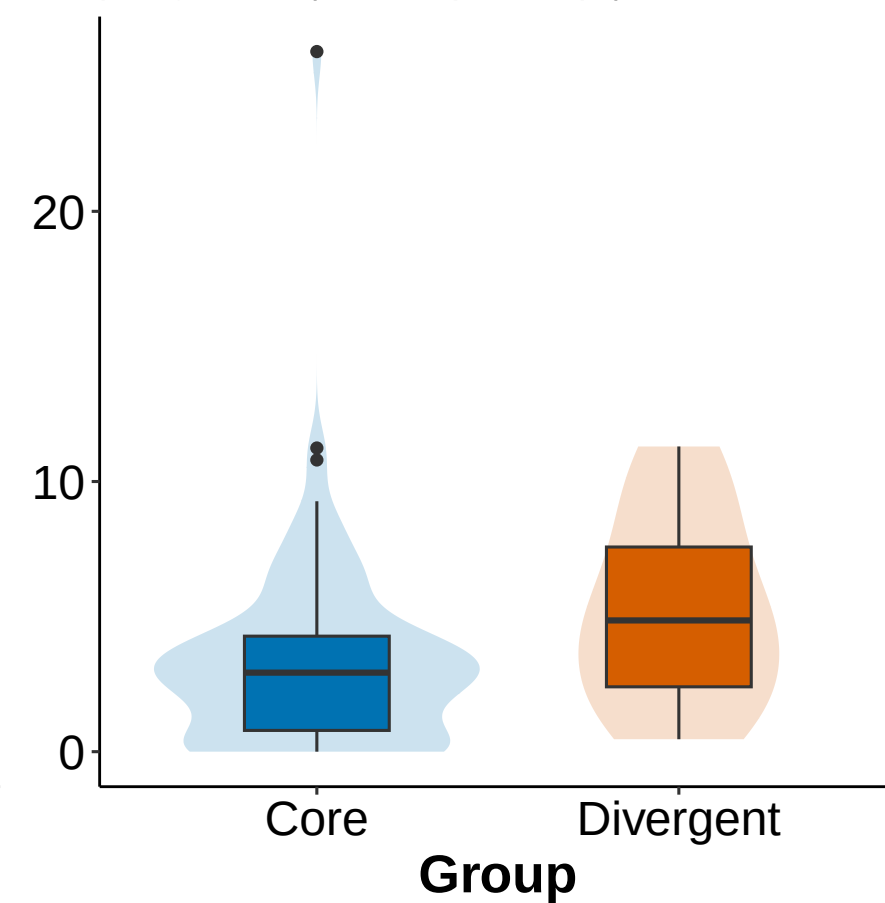
SA_Pyrimidine_synthesis_L1w

p-adj: 0.885 | .>=0.2 (Small) | .%: 6.3



SA_Pyrimidine_synthesis_pFBA

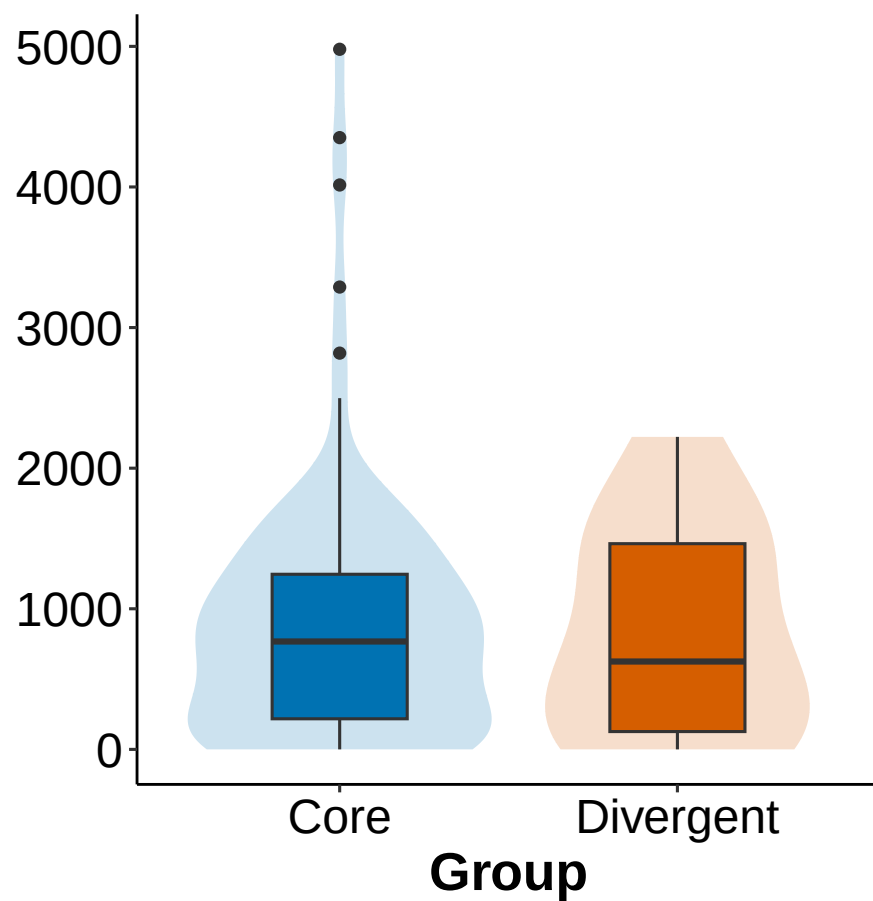
p-adj: 0.843 | .>=0.37 (Medium) | .%: 66



Metabolism – . 5. Systemic Reprogramming (SA_*)

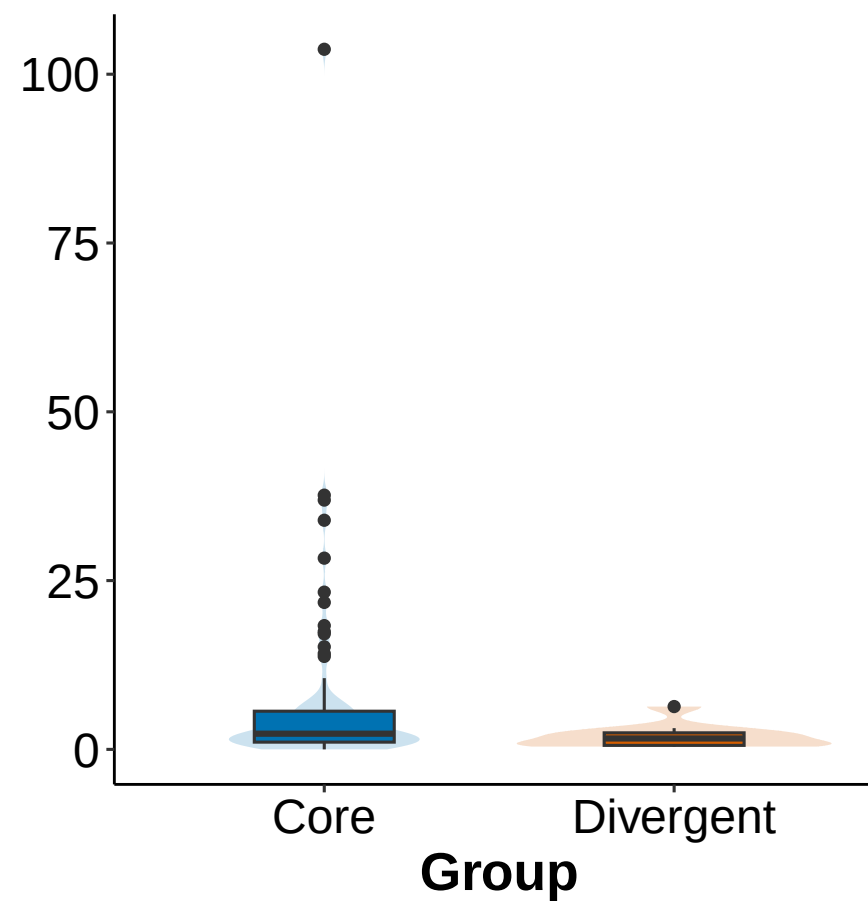
SA_Pyruvate_metabolism_FBA

p-adj: 0.957 | .=-0.02 (Negligible) | .%: -18.5



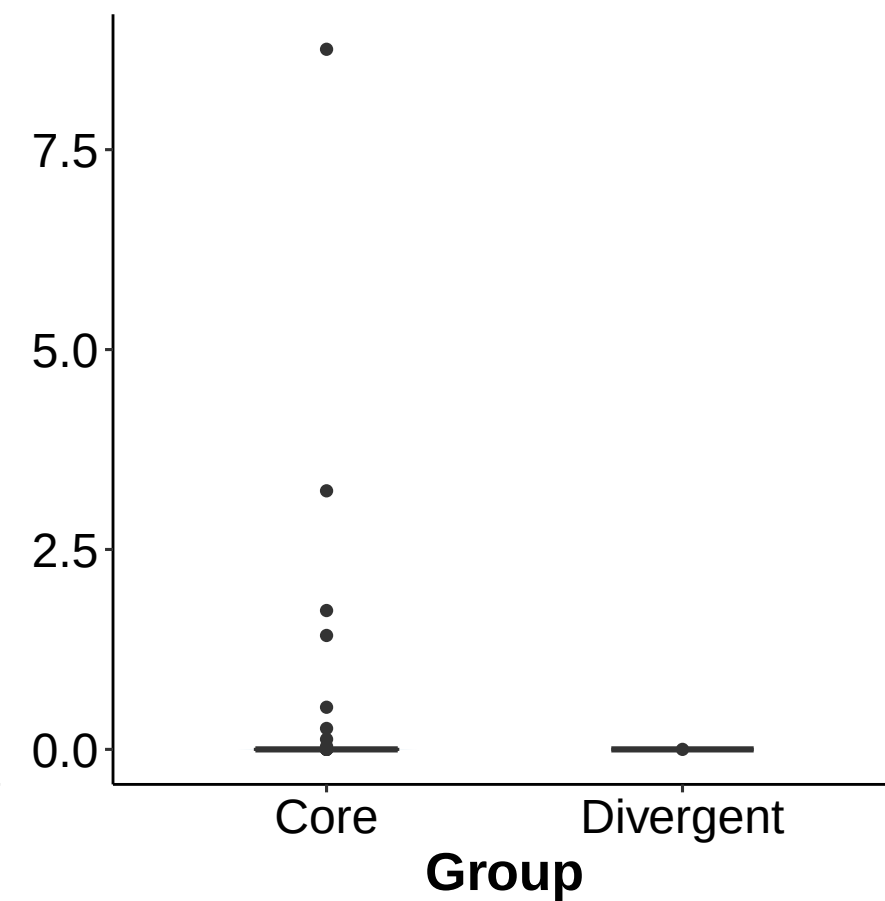
SA_Pyruvate_metabolism_L1w

p-adj: 0.872 | .=-0.26 (Small) | .%: -30.4



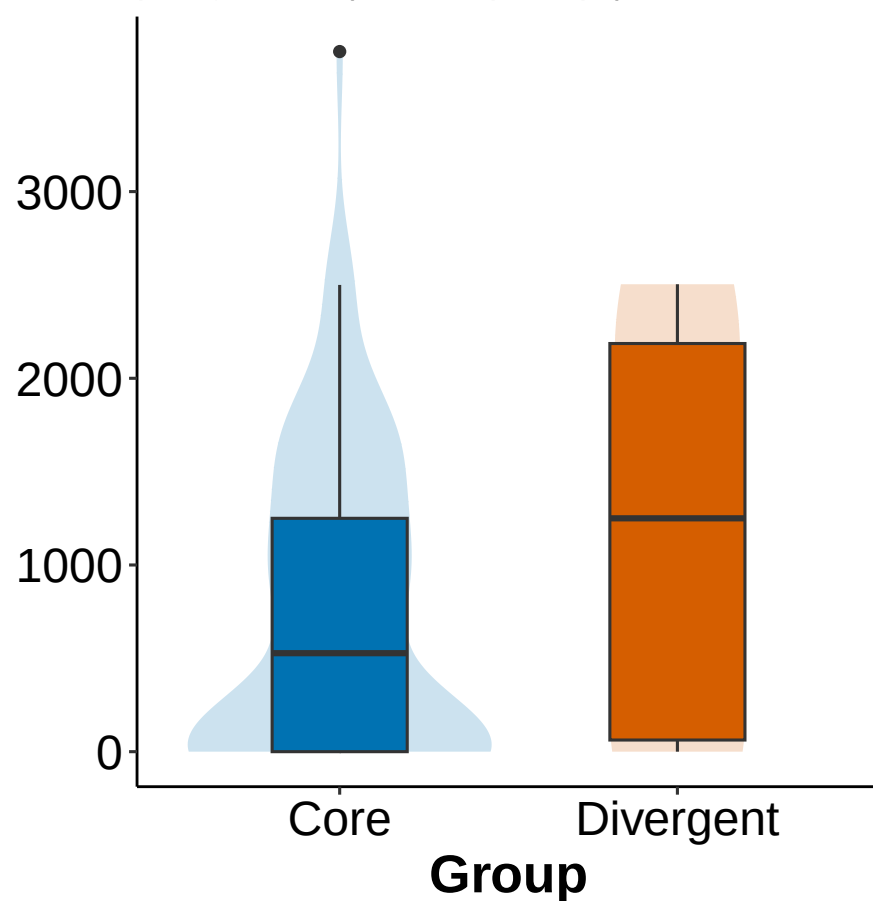
SA_Pyruvate_metabolism_pFBA

p-adj: 0.843 | .=-0.41 (Medium) | .%: -100



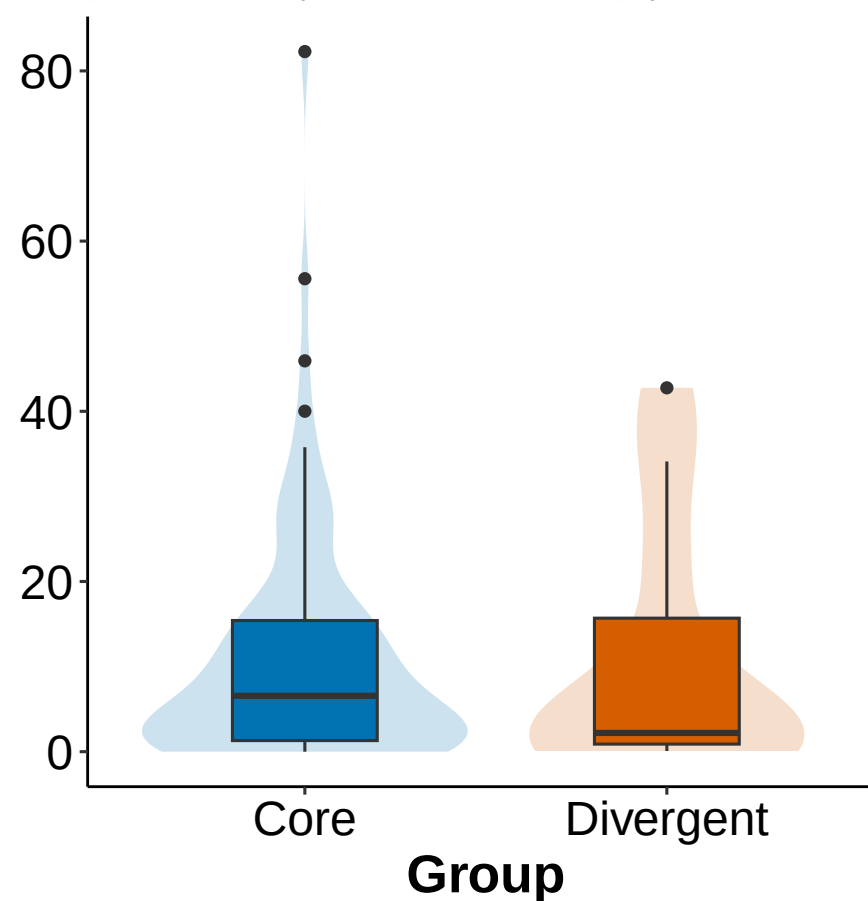
SA_ROS_detoxification_FBA

p-adj: 0.872 | .=-0.26 (Small) | .%: 137



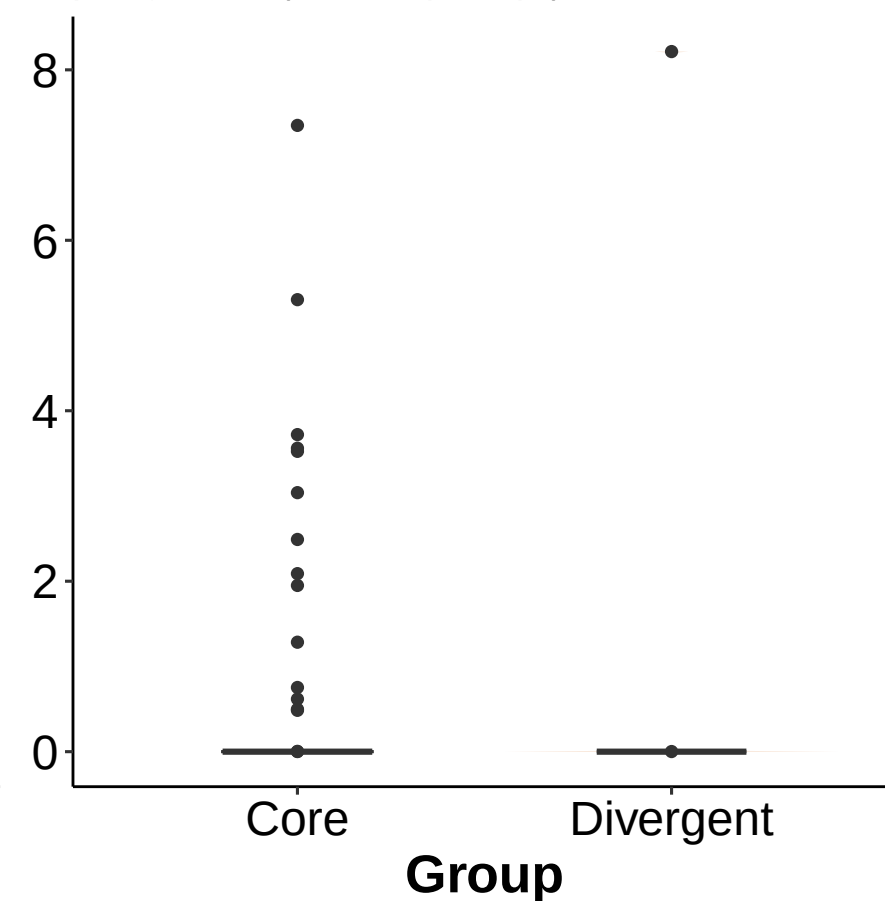
SA_ROS_detoxification_L1w

p-adj: 0.945 | .=-0.09 (Negligible) | .%: -66.4



SA_ROS_detoxification_pFBA

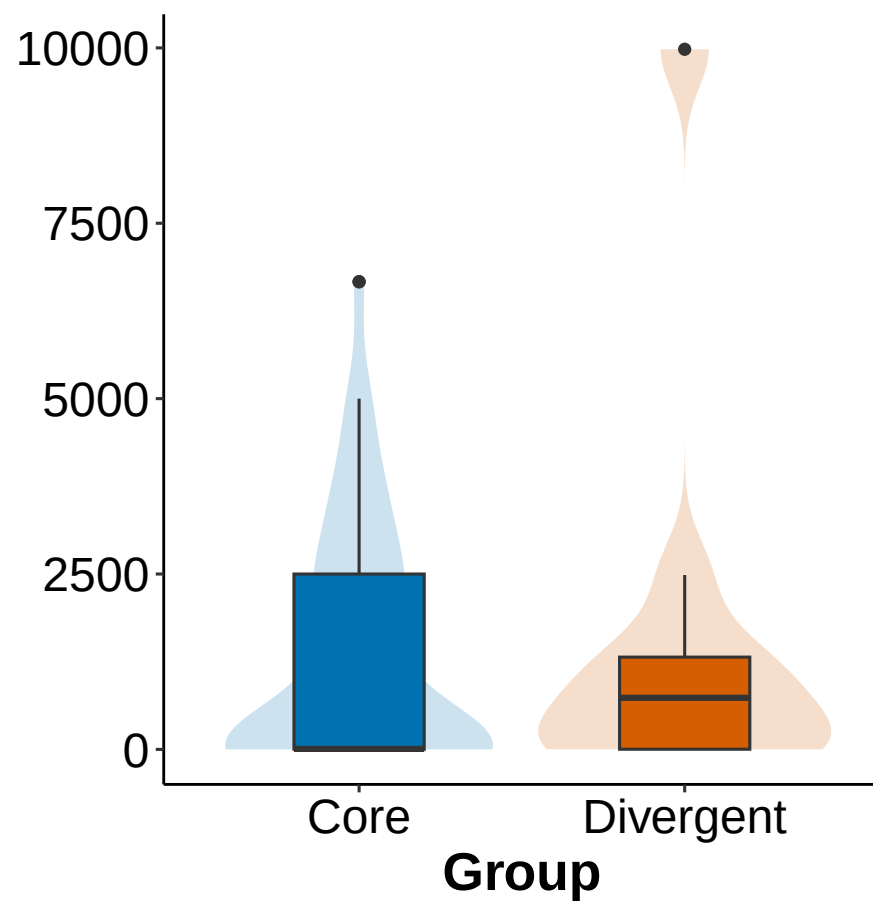
p-adj: 0.885 | .=-0.2 (Small) | .%: -100



Metabolism – . 5. Systemic Reprogramming (SA_*)

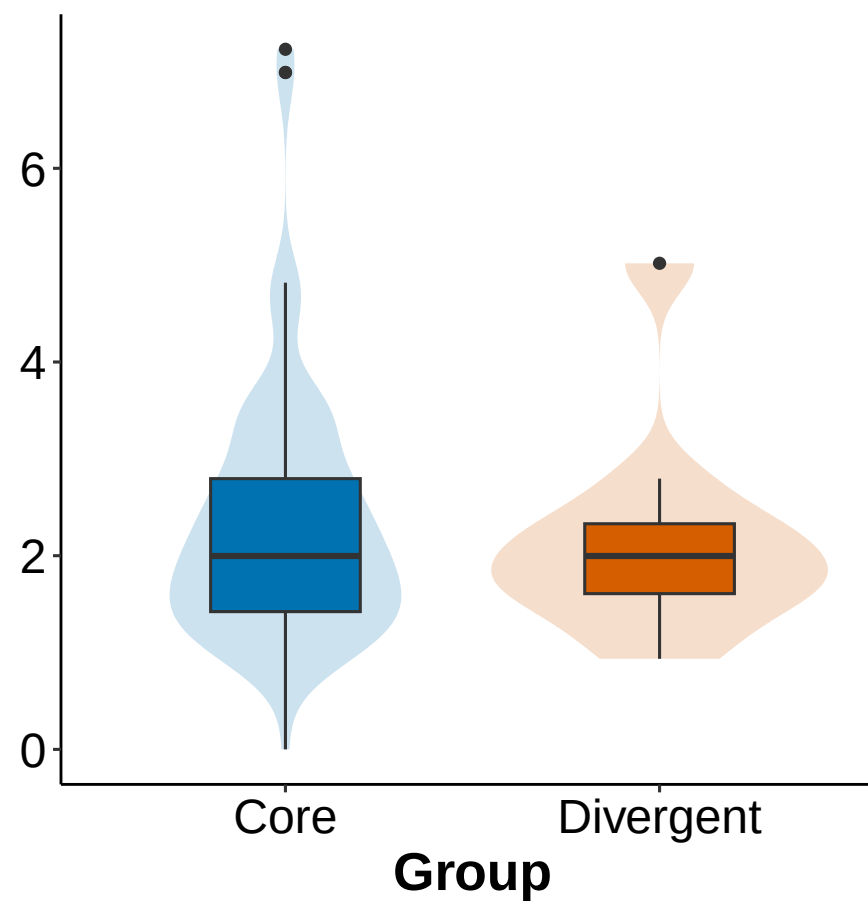
SA_Sphingolipid_metabolism_FB

p-adj: 0.954 | .=-0.03 (Negligible) | .%: 10307



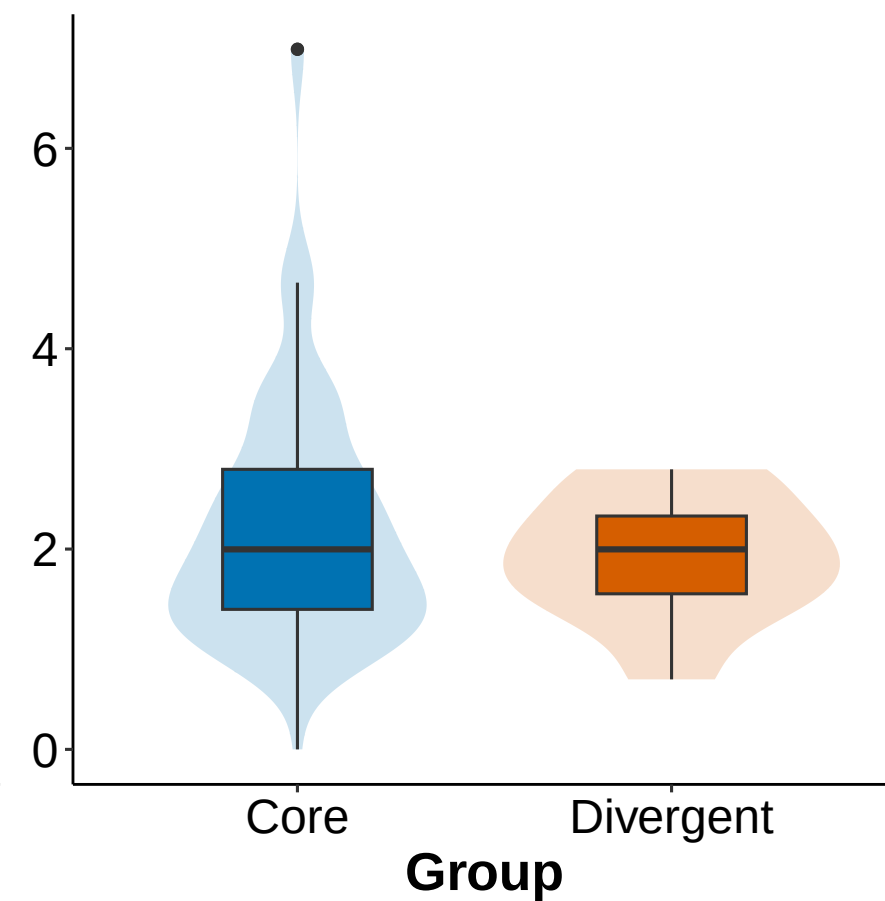
SA_Sphingolipid_metabolism_L1w

p-adj: 0.95 | .=-0.08 (Negligible) | .%: 0



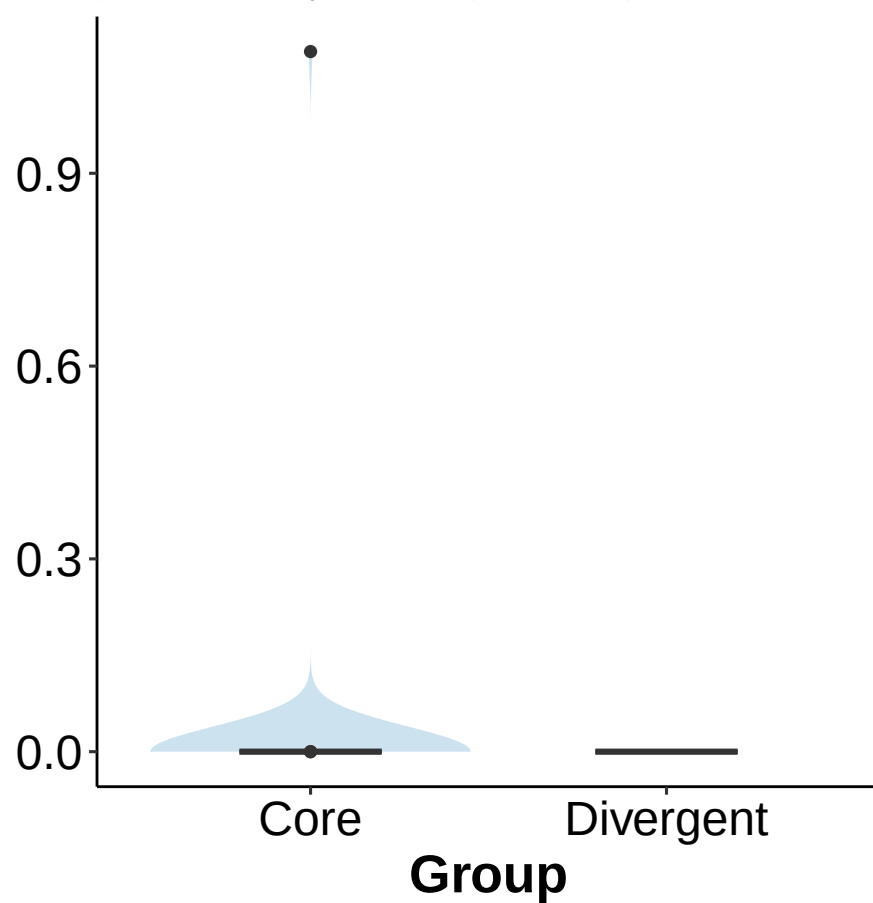
SA_Sphingolipid_metabolism_pFBA

p-adj: 0.954 | .=-0.05 (Negligible) | .%: 0



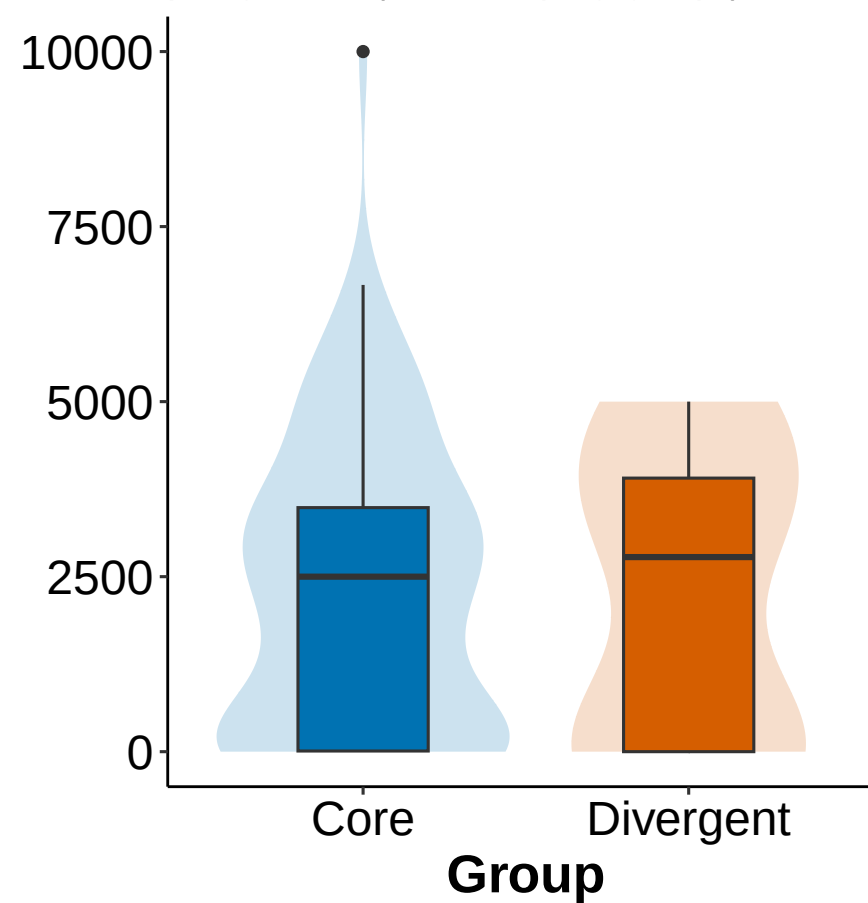
SA_Squalene_and_cholesterol_synth

p-adj: 0.927 | .=-0.03 (Negligible)



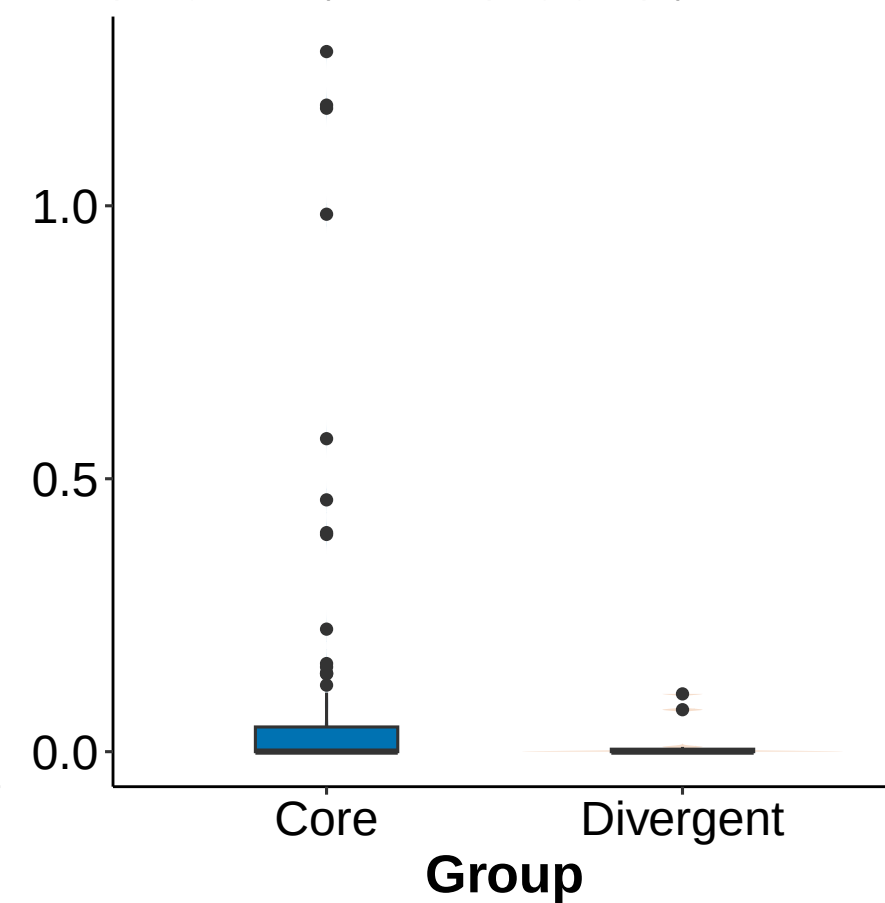
SA_Starch_and_sucrose_metabol

p-adj: 0.979 | .=-0.01 (Negligible) | .%: 11.2

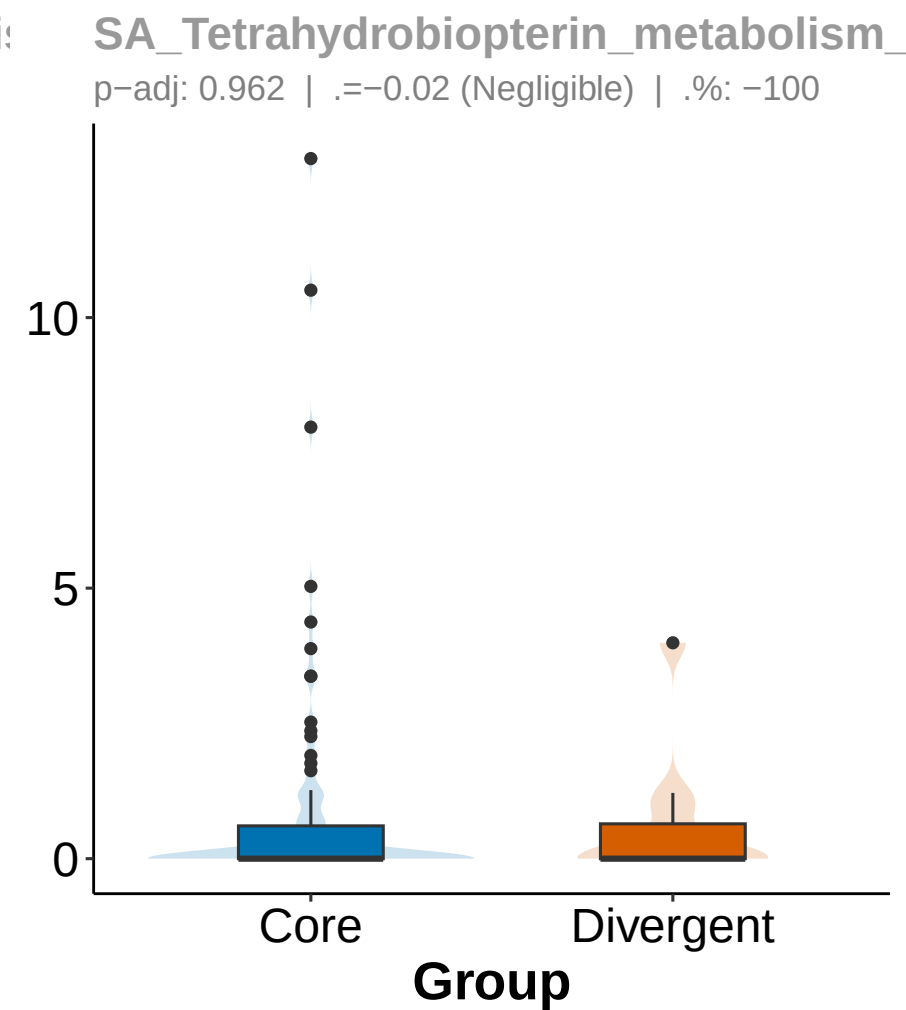
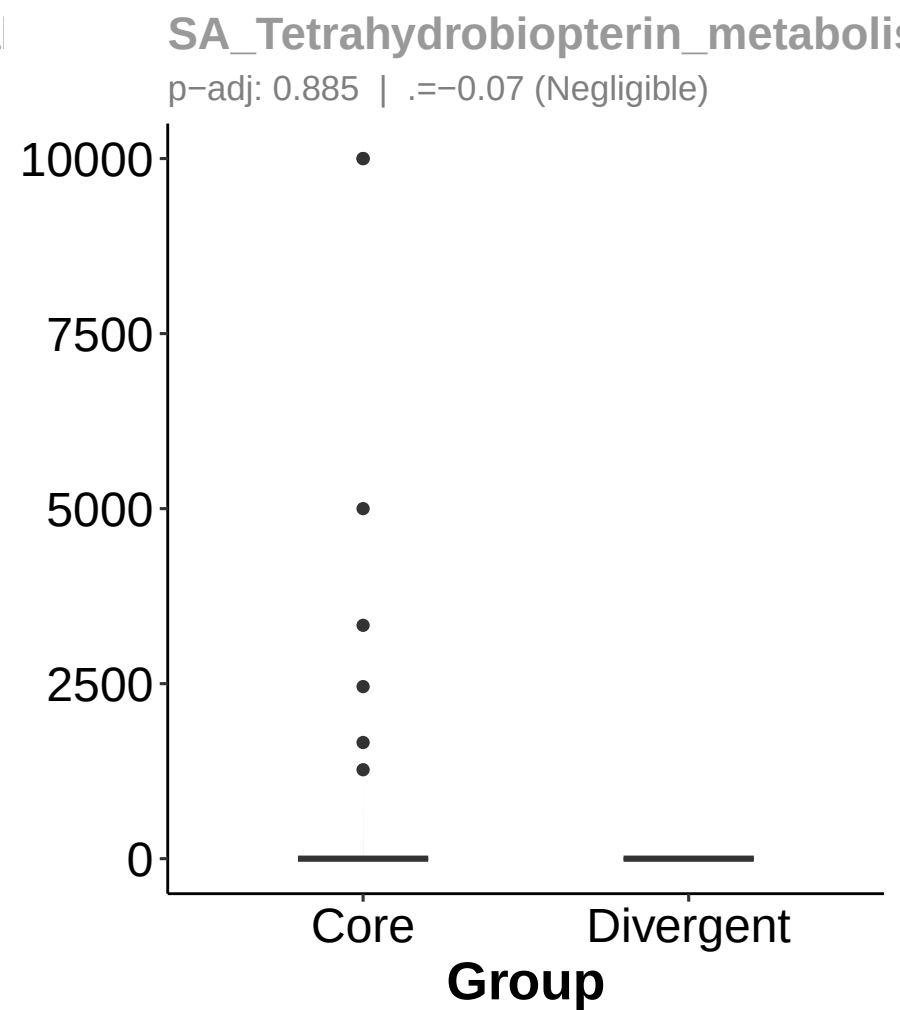
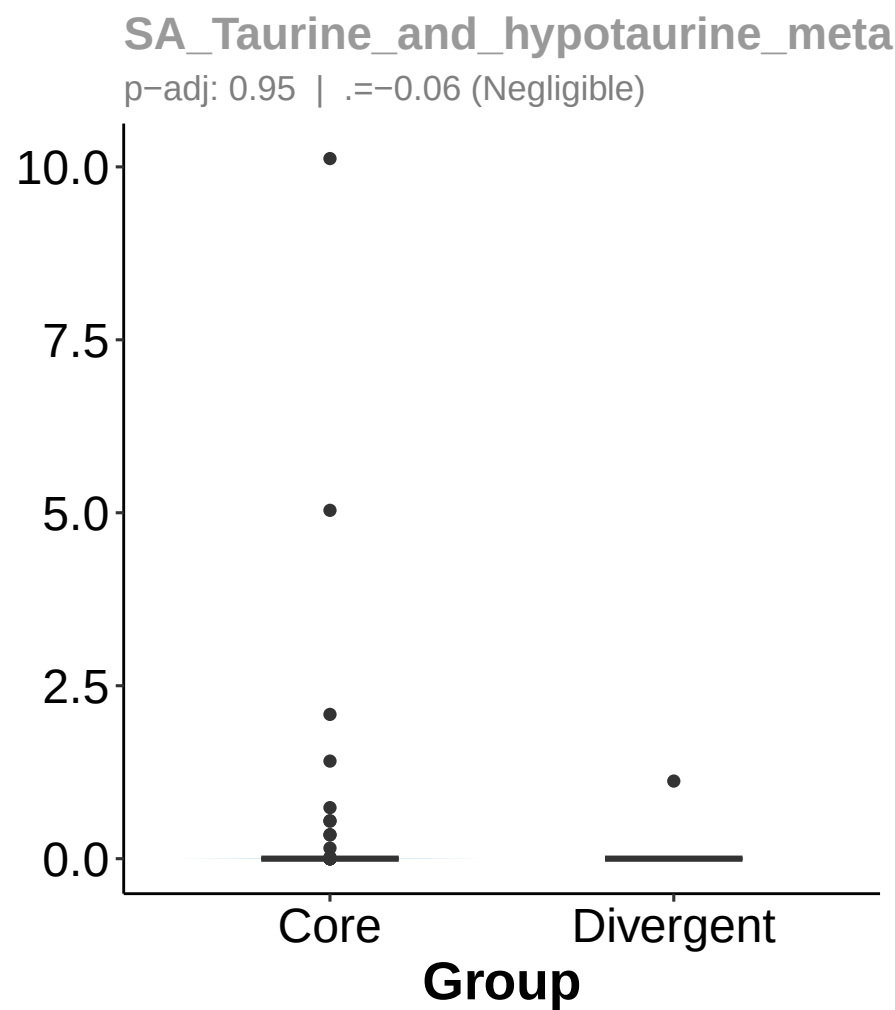
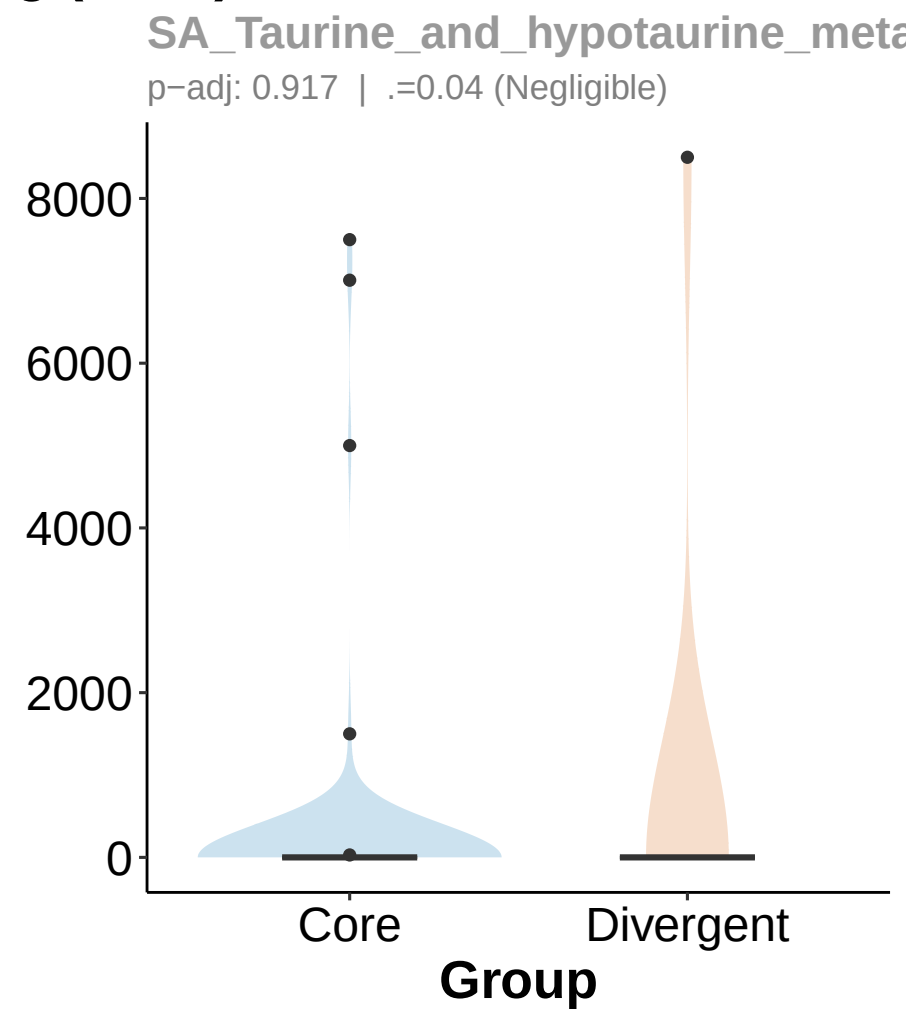
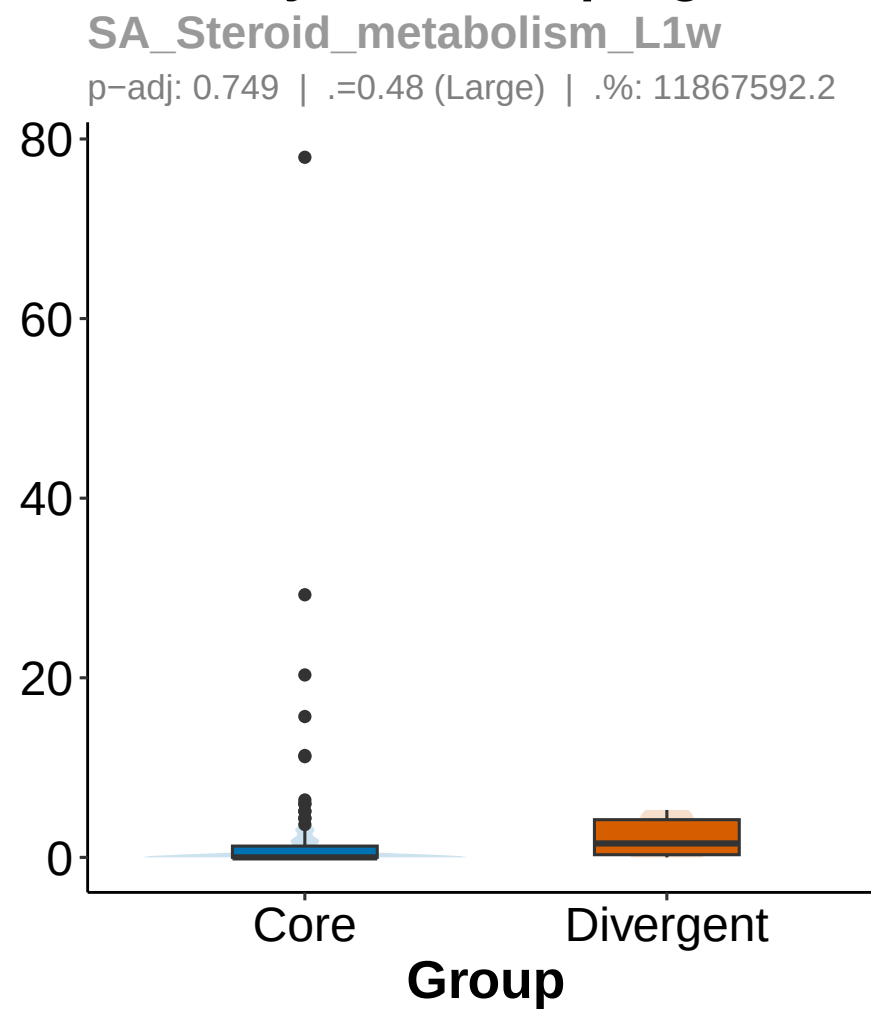
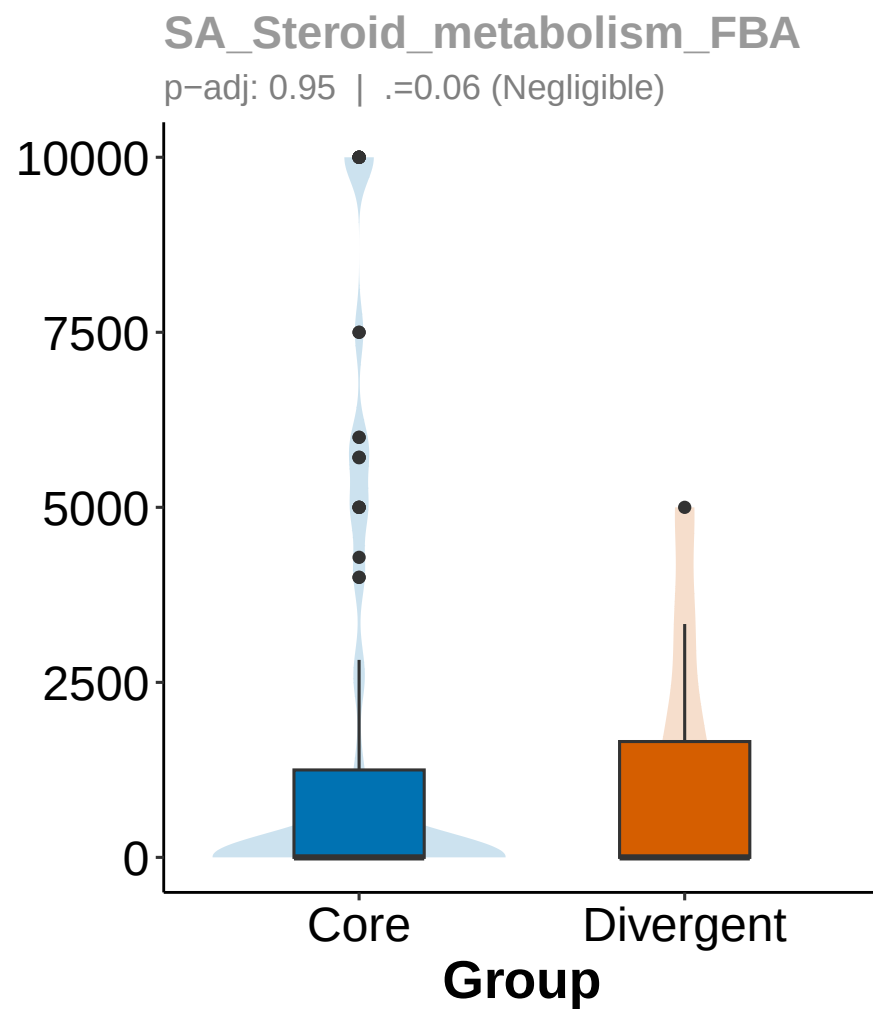


SA_Starch_and_sucrose_metabolism

p-adj: 0.917 | .=-0.11 (Negligible) | .%: -30.9



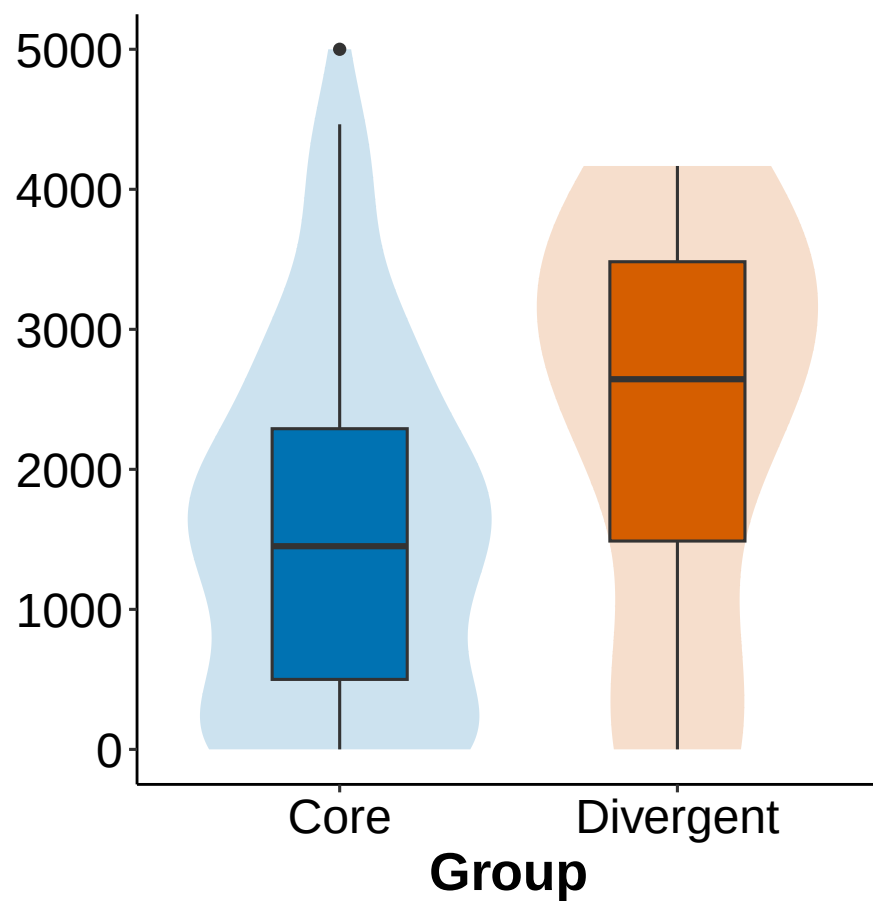
Metabolism – . 5. Systemic Reprogramming (SA_*)



Metabolism – . 5. Systemic Reprogramming (SA_*)

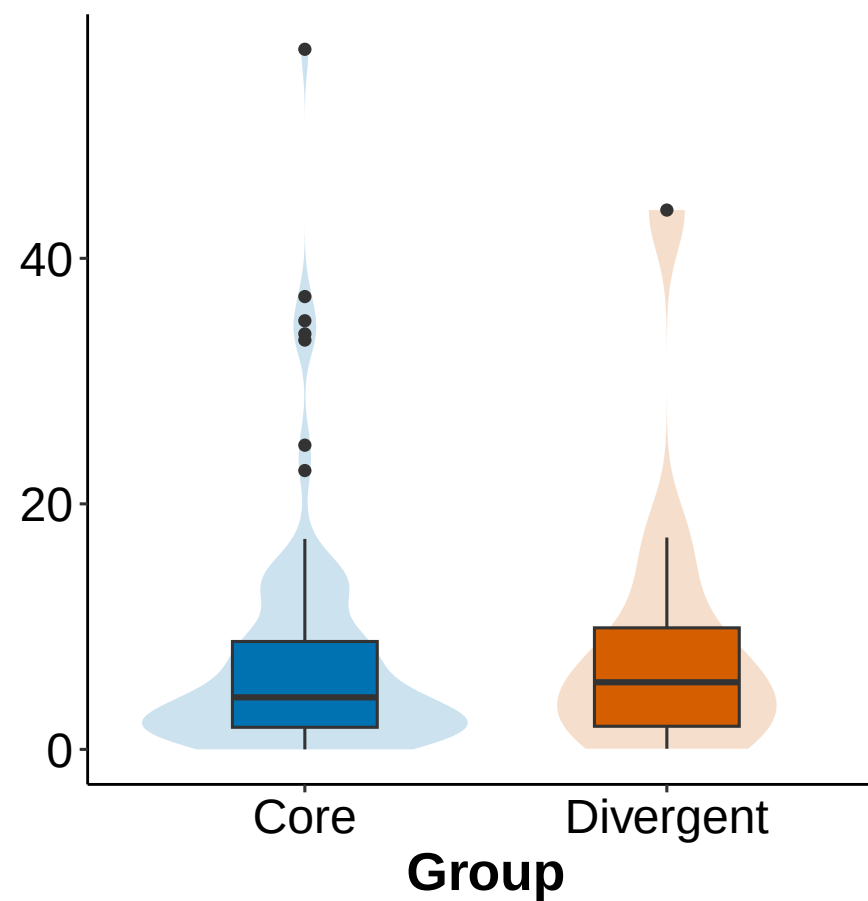
SA_Transport__endoplasmic_reticu

p-adj: 0.843 | .>=0.34 (Medium) | .%: 82.2



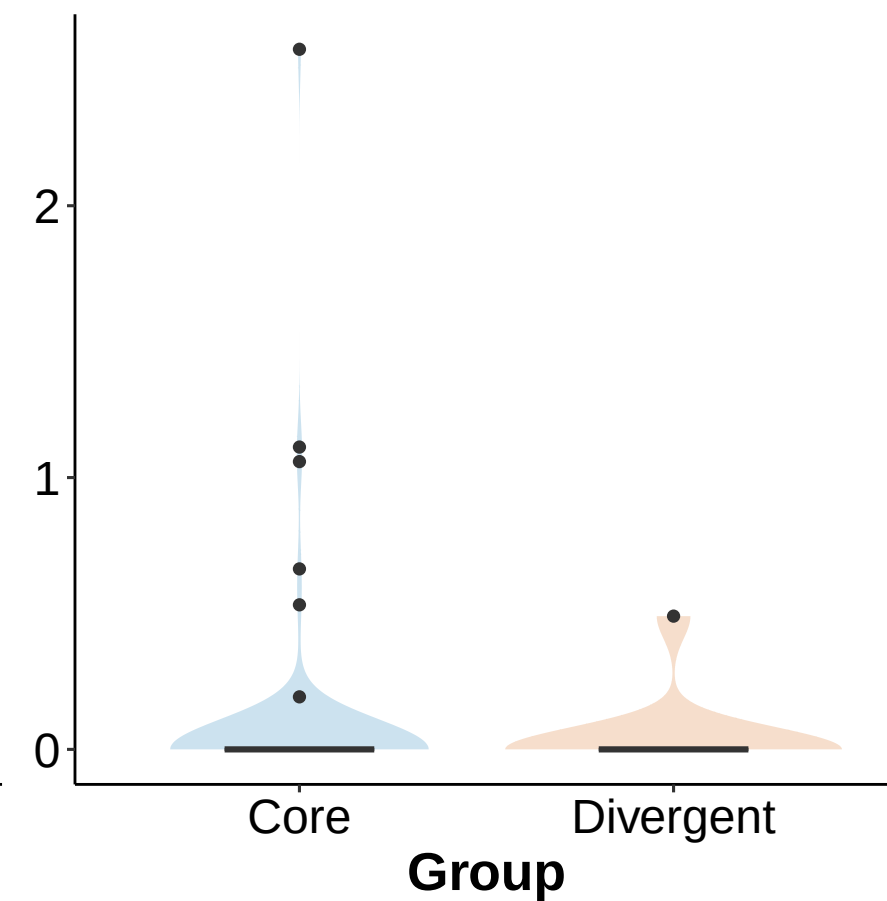
SA_Transport__endoplasmic_reticula

p-adj: 0.95 | .>=0.06 (Negligible) | .%: 29



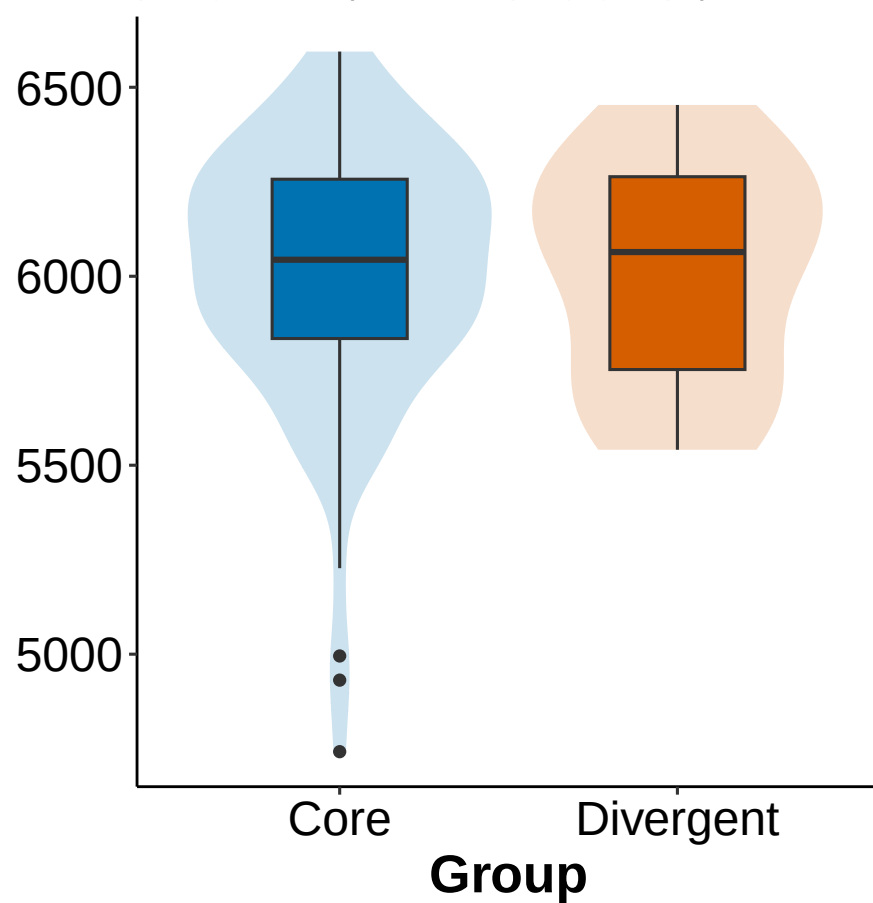
SA_Transport__endoplasmic_reticular_

p-adj: 0.95 | .>=0.03 (Negligible)



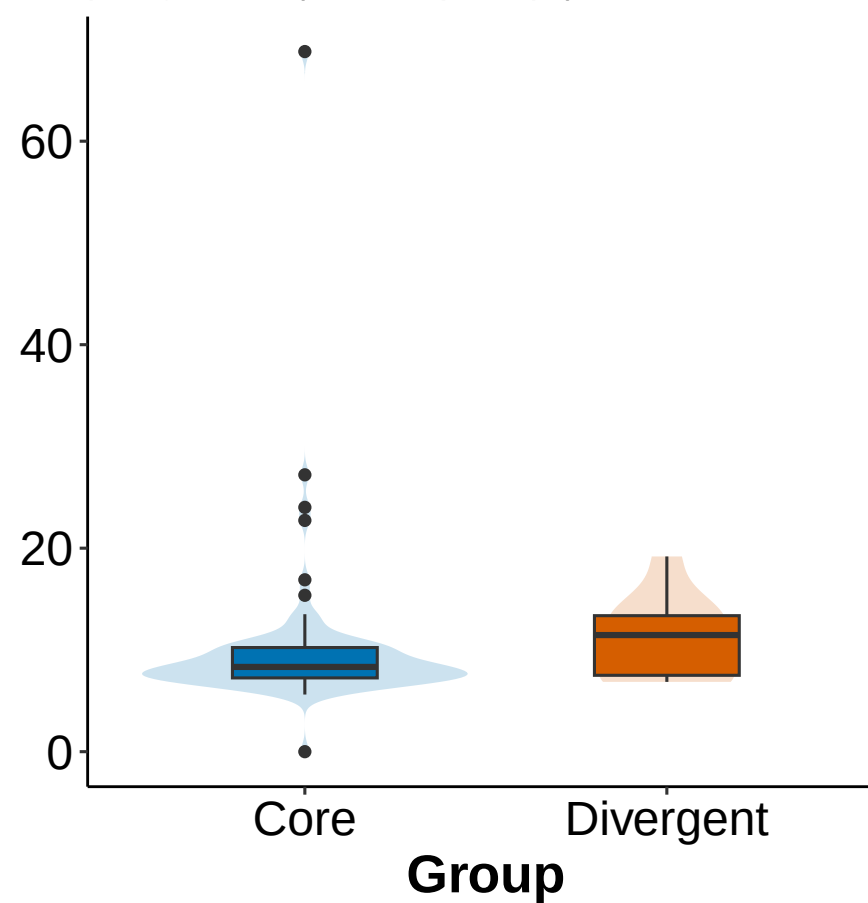
SA_Transport__extracellular_FBA

p-adj: 0.954 | .>=-0.03 (Negligible) | .%: 0.3



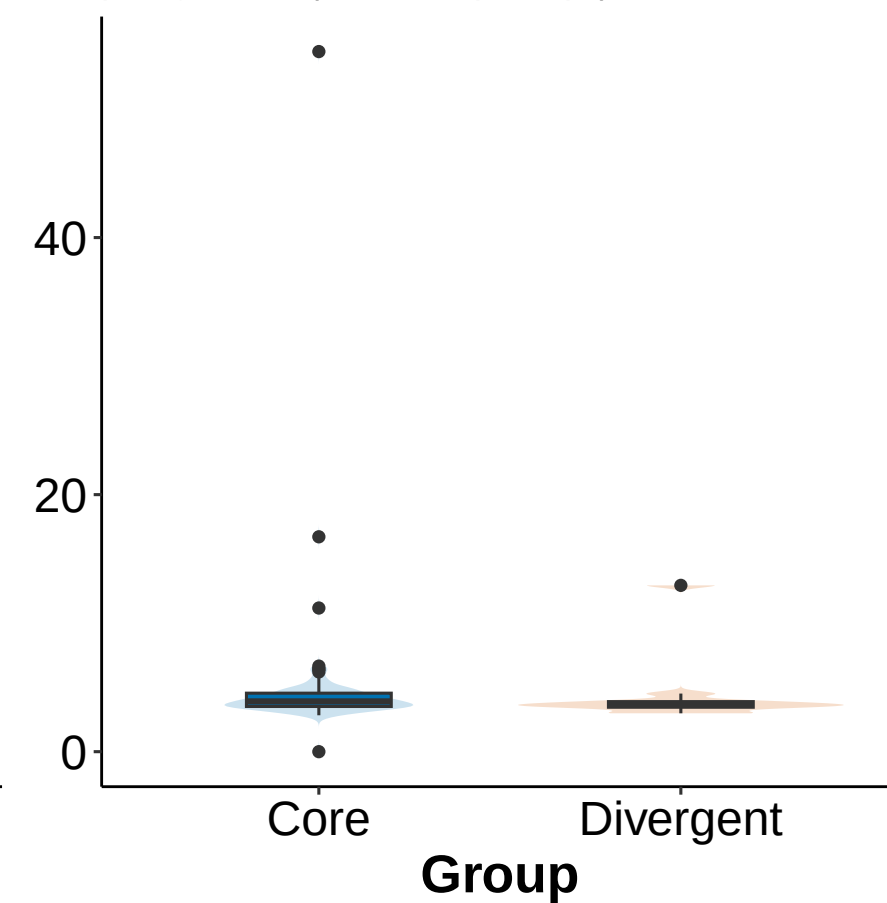
SA_Transport__extracellular_L1w

p-adj: 0.885 | .>=0.25 (Small) | .%: 37.6



SA_Transport__extracellular_pFBA

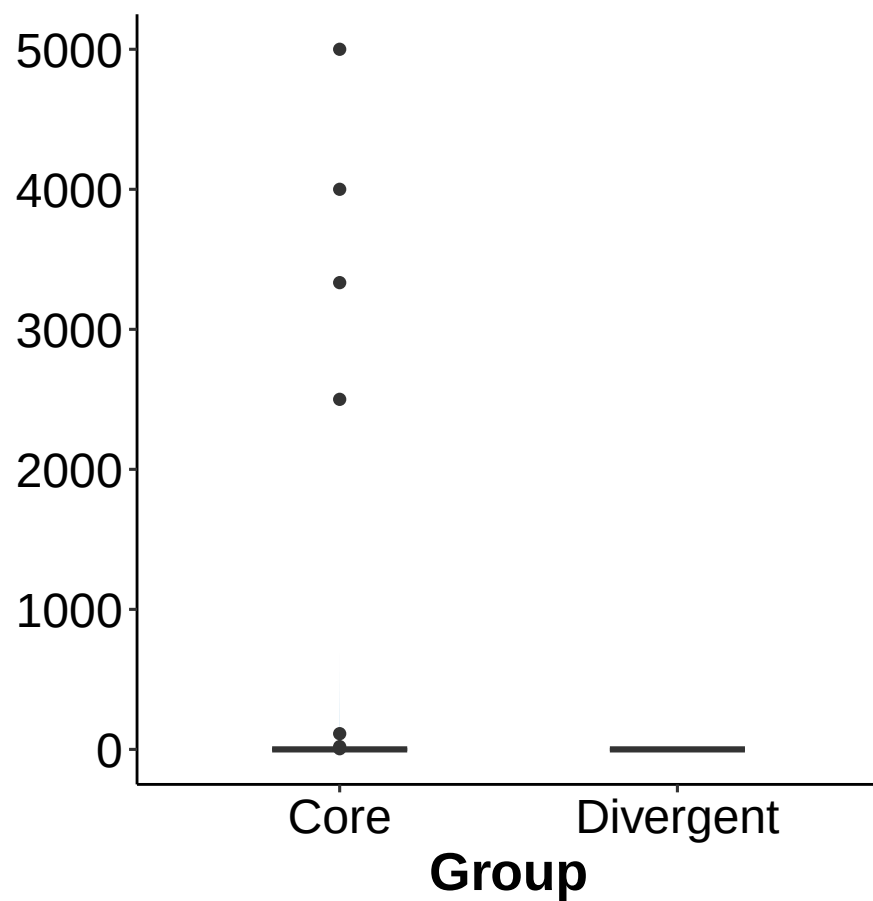
p-adj: 0.885 | .>=-0.21 (Small) | .%: -7



Metabolism – . 5. Systemic Reprogramming (SA_*)

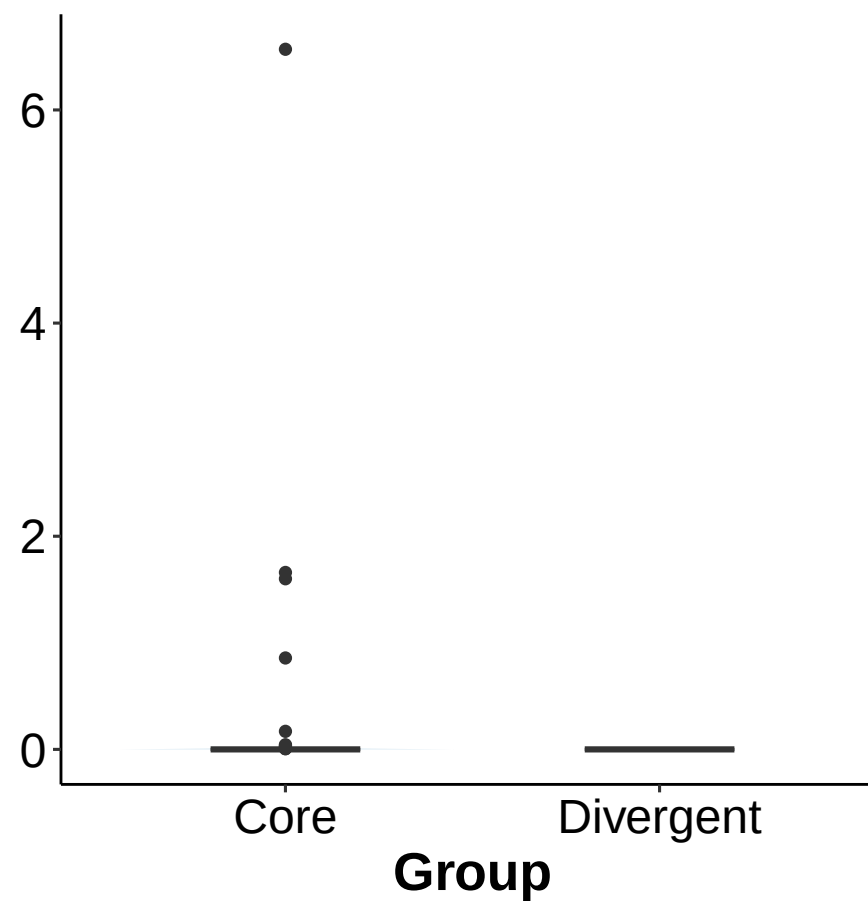
SA_Transport__golgi_apparatus_Fl

p-adj: 0.885 | .=-0.07 (Negligible)



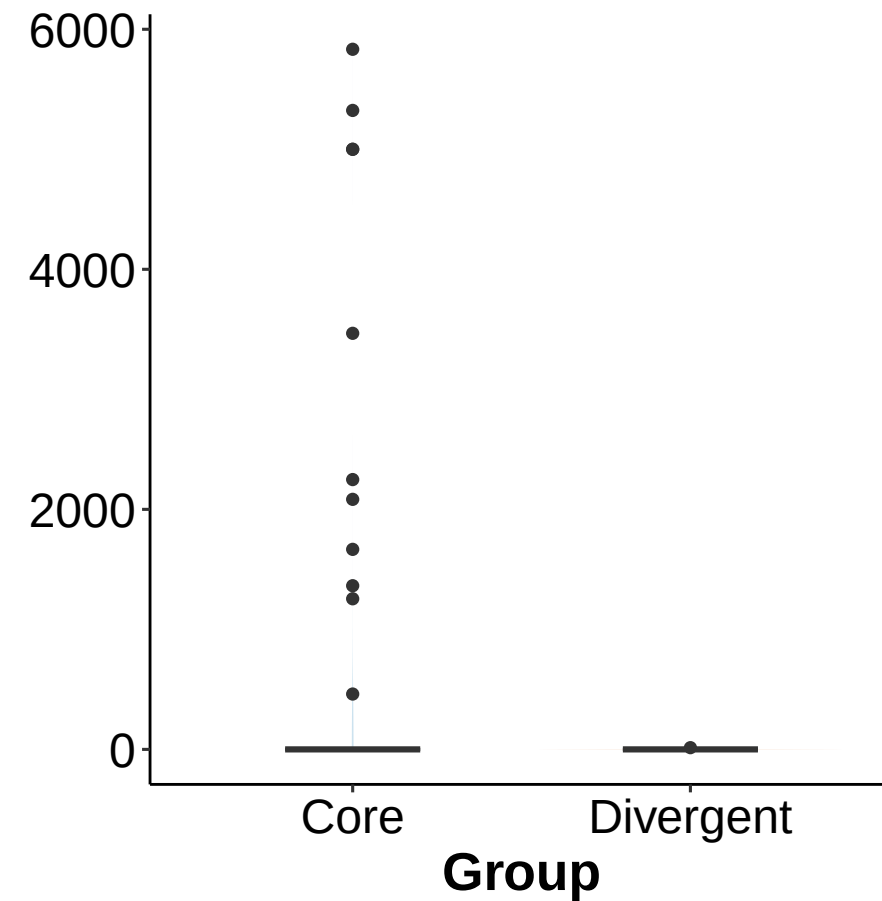
SA_Transport__golgi_apparatus_L1w

p-adj: 0.885 | .=-0.15 (Small) | .%: -28.4



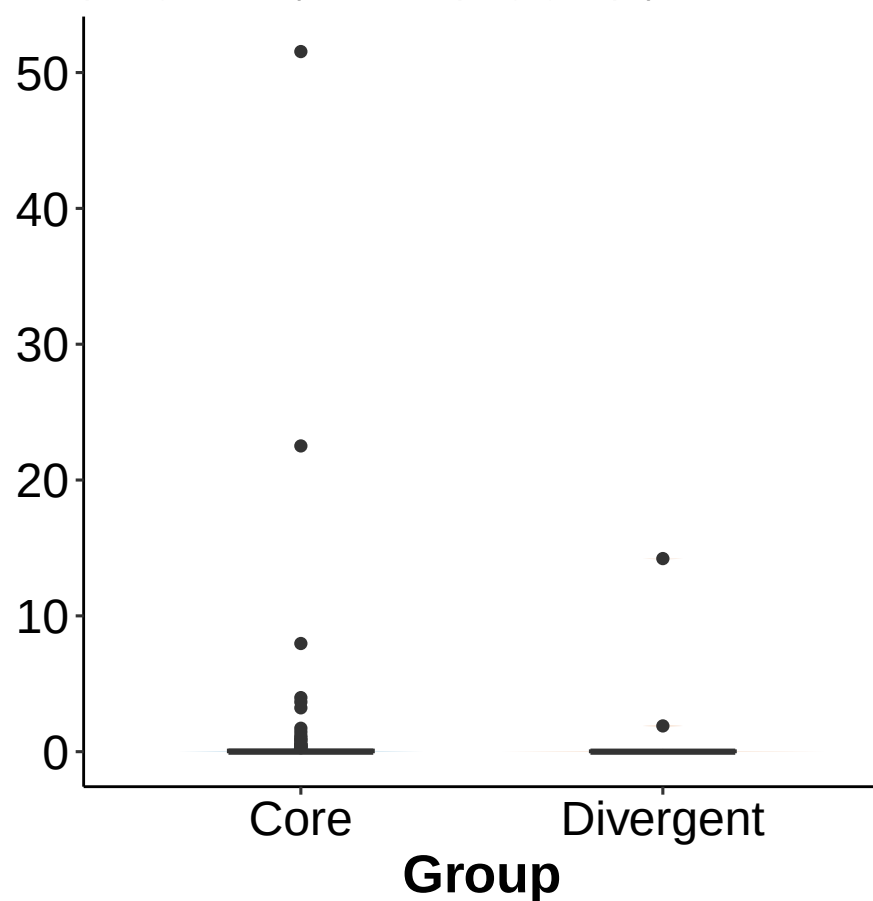
SA_Transport__lysosomal_FBA

p-adj: 0.95 | .=-0.03 (Negligible)



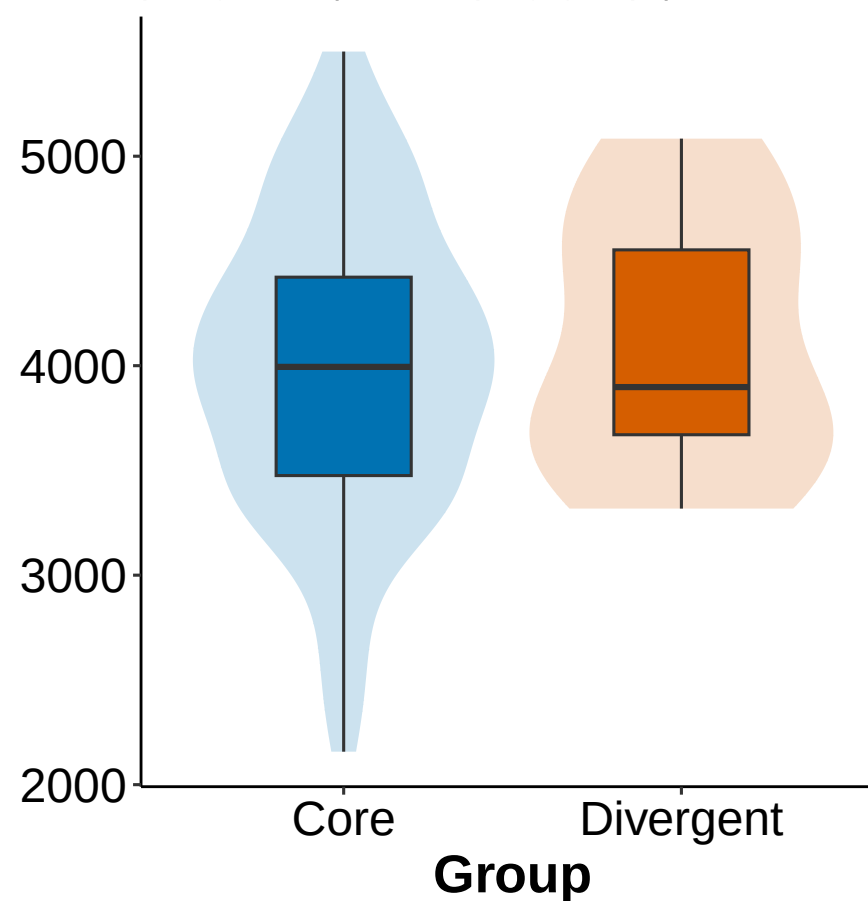
SA_Transport__lysosomal_L1w

p-adj: 0.896 | .=-0.14 (Negligible) | .%: -99.6



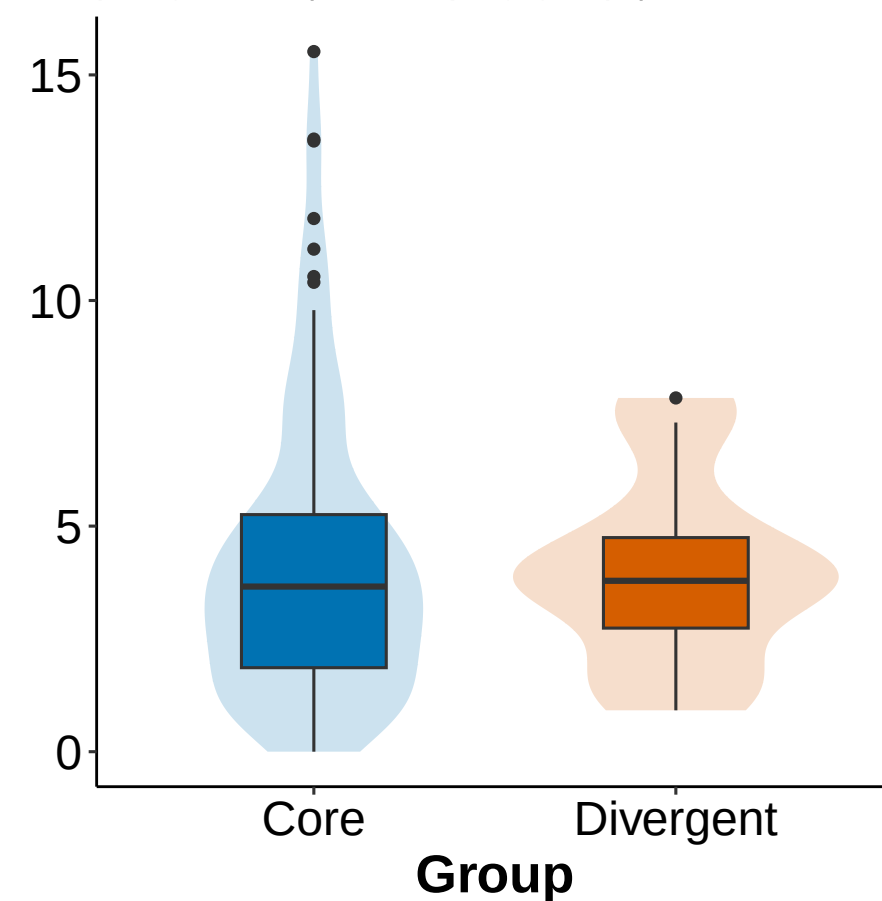
SA_Transport__mitochondrial_FBA

p-adj: 0.95 | .=0.08 (Negligible) | .%: -2.4



SA_Transport__mitochondrial_L1w

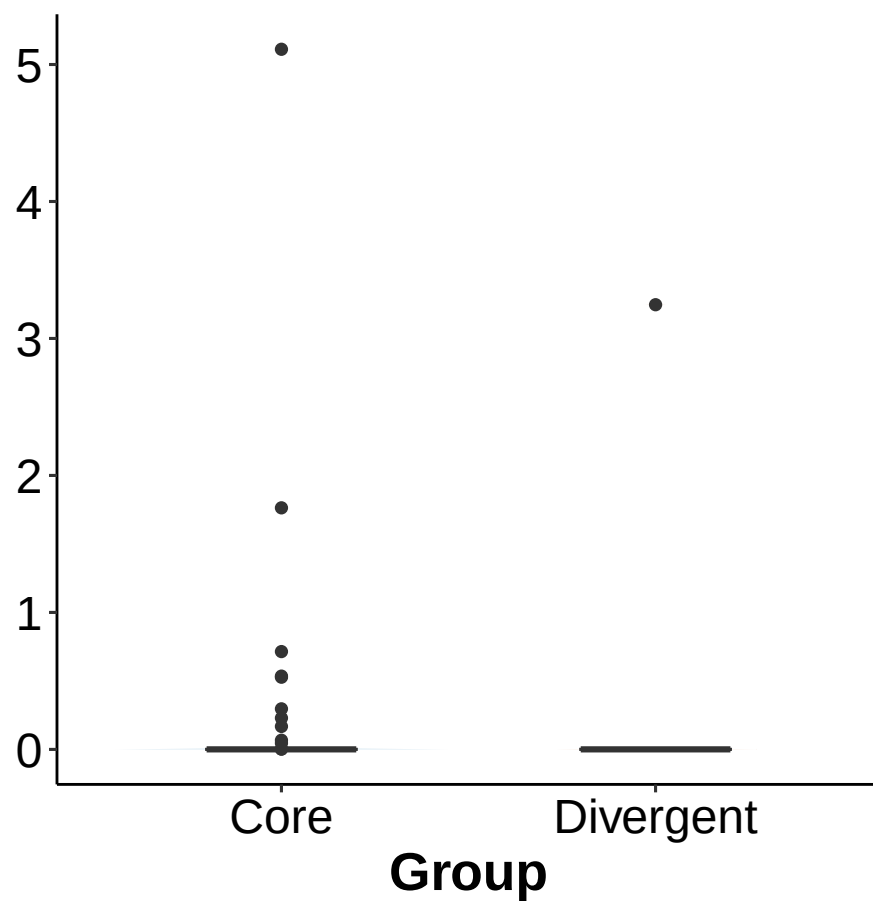
p-adj: 0.954 | .=0.05 (Negligible) | .%: 3.5



Metabolism – . 5. Systemic Reprogramming (SA_*)

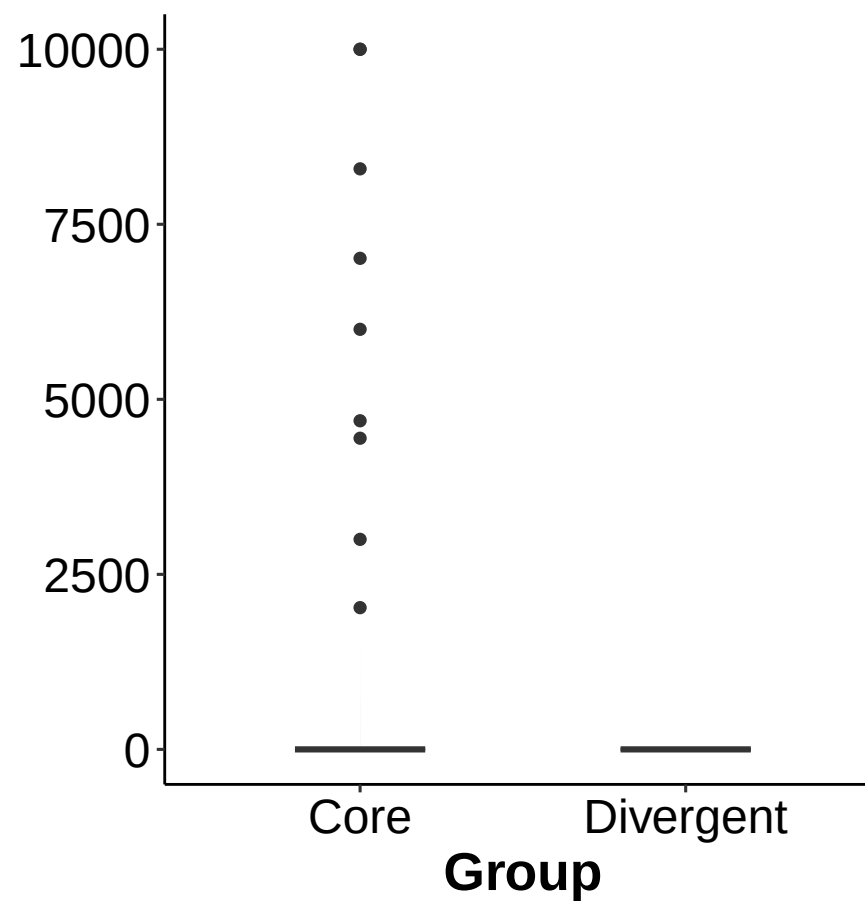
SA_Transport__mitochondrial_pFBA

p-adj: 0.885 | .=-0.16 (Small) | .%: -9.9



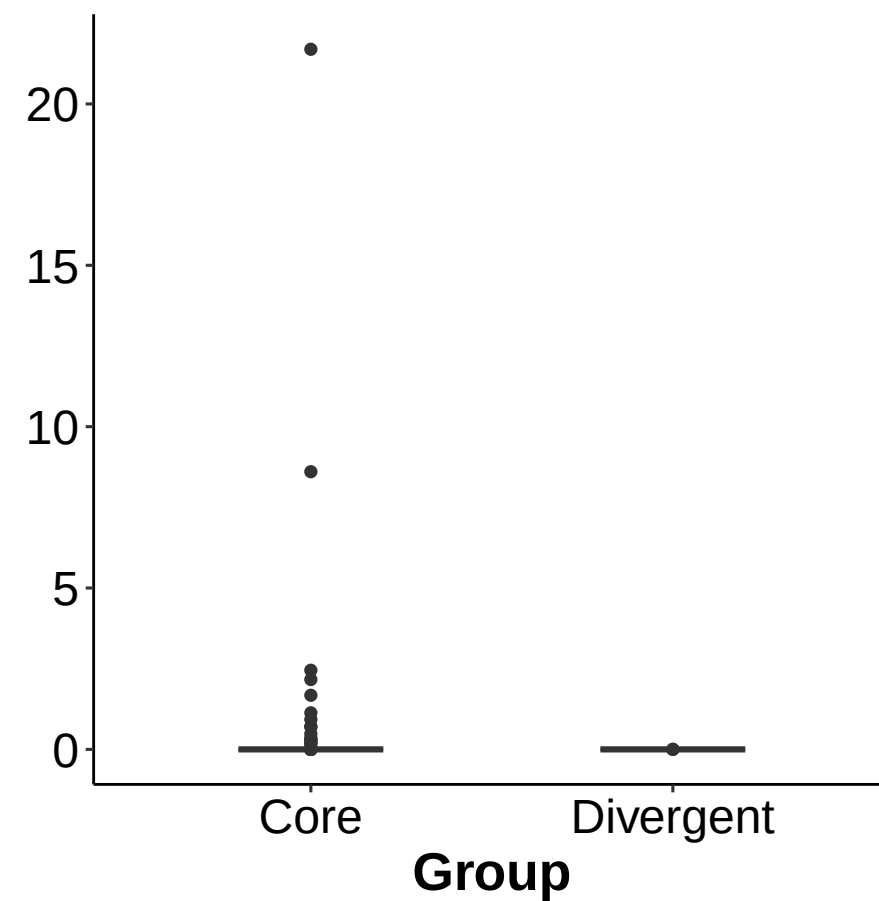
SA_Transport__nuclear_FBA

p-adj: 0.885 | .=-0.09 (Negligible)



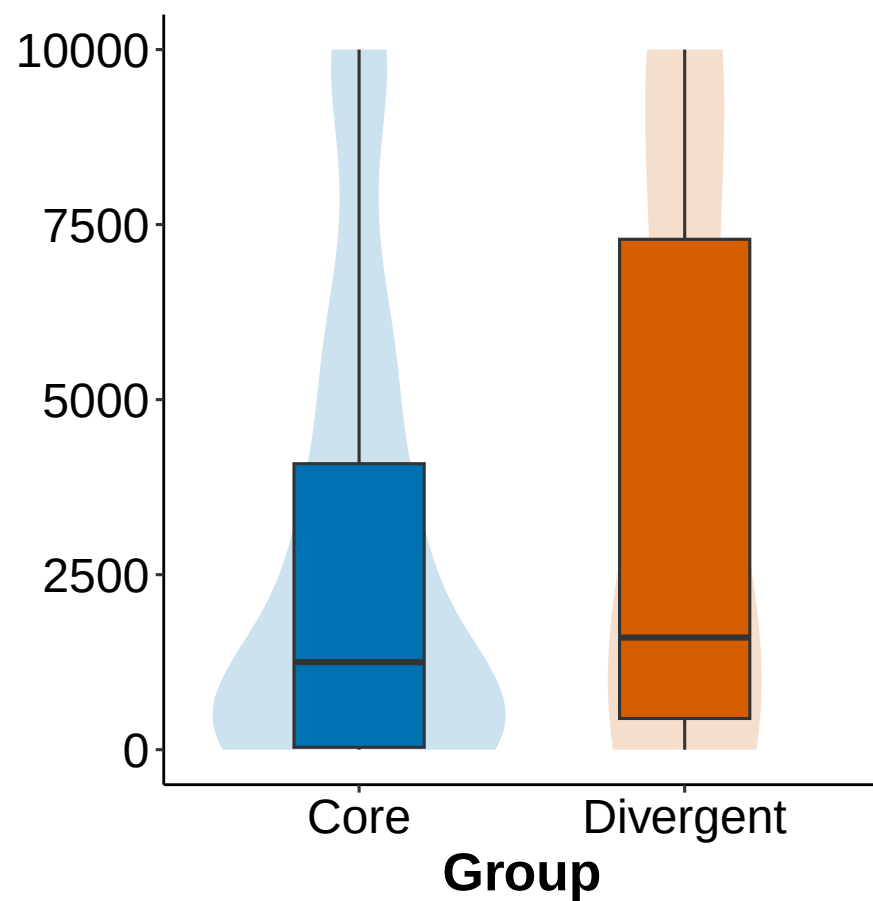
SA_Transport__nuclear_L1w

p-adj: 0.885 | .=-0.18 (Small)



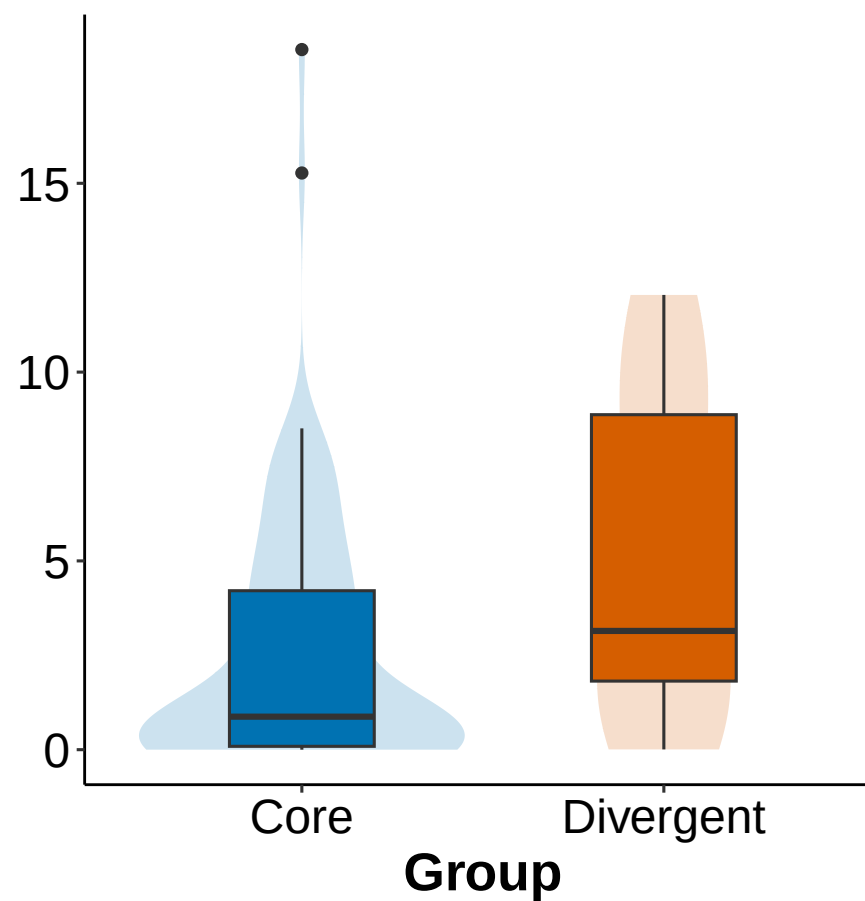
SA_Transport__peroxisomal_FBA

p-adj: 0.917 | .=0.12 (Negligible) | .%: 28.2



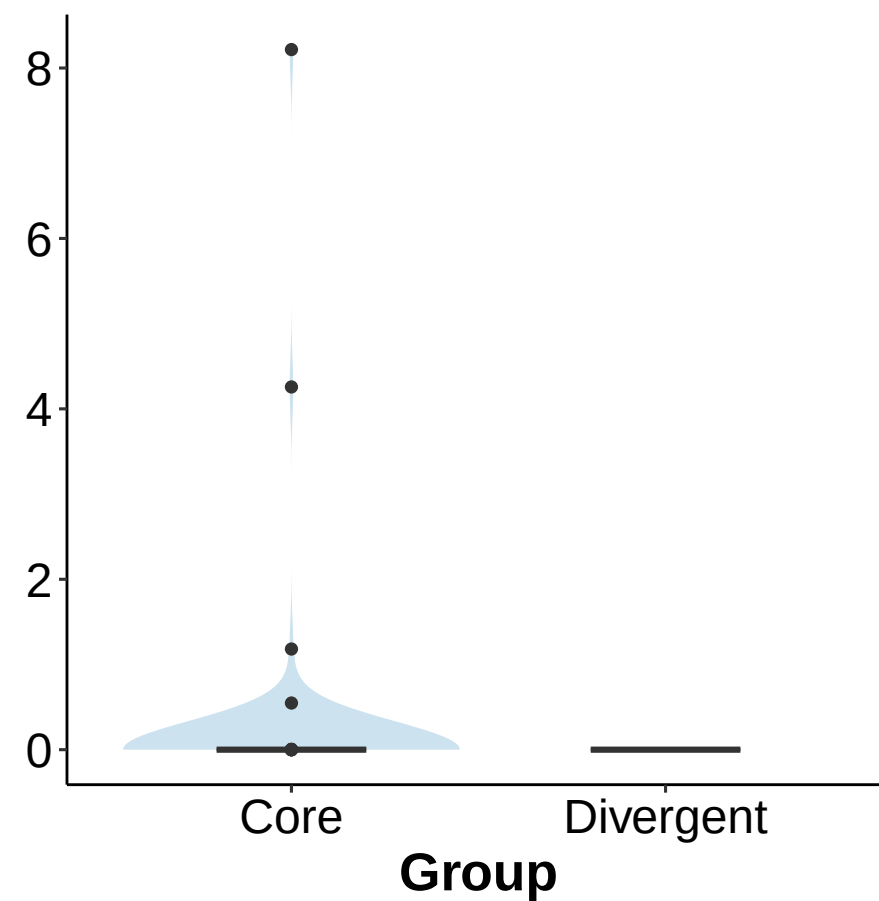
SA_Transport__peroxisomal_L1w

p-adj: 0.843 | .=0.39 (Medium) | .%: 259.5

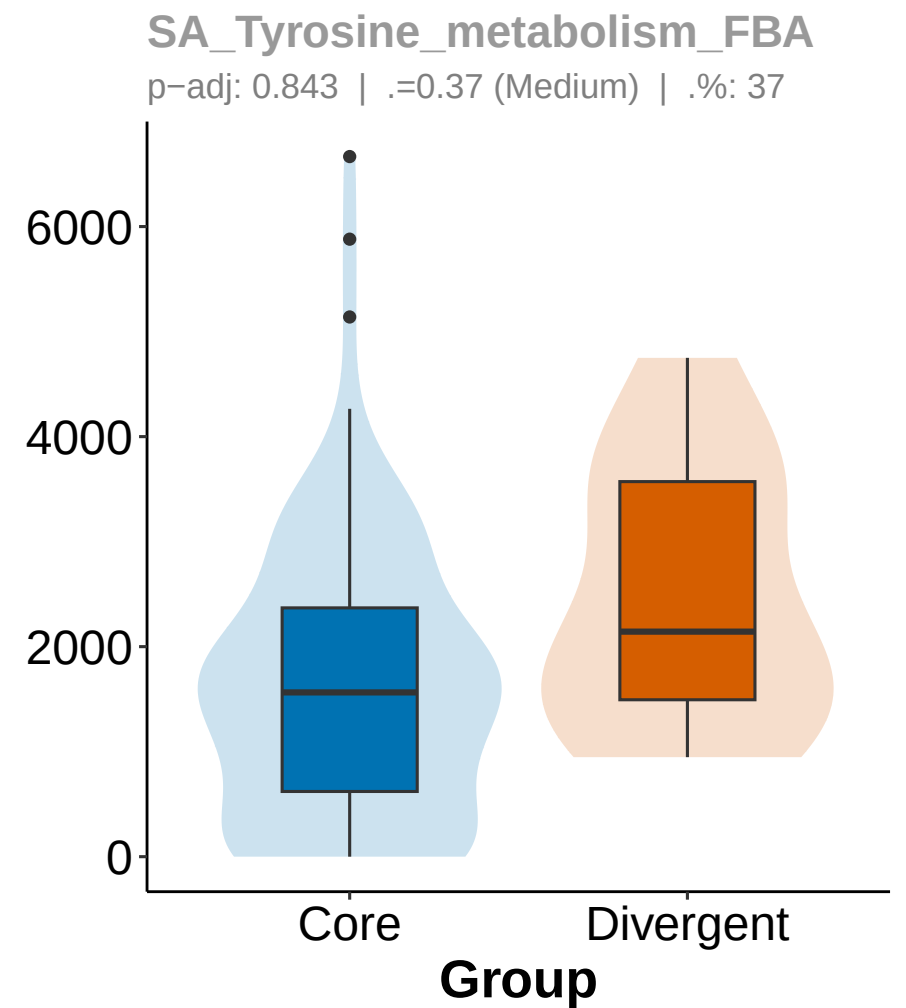
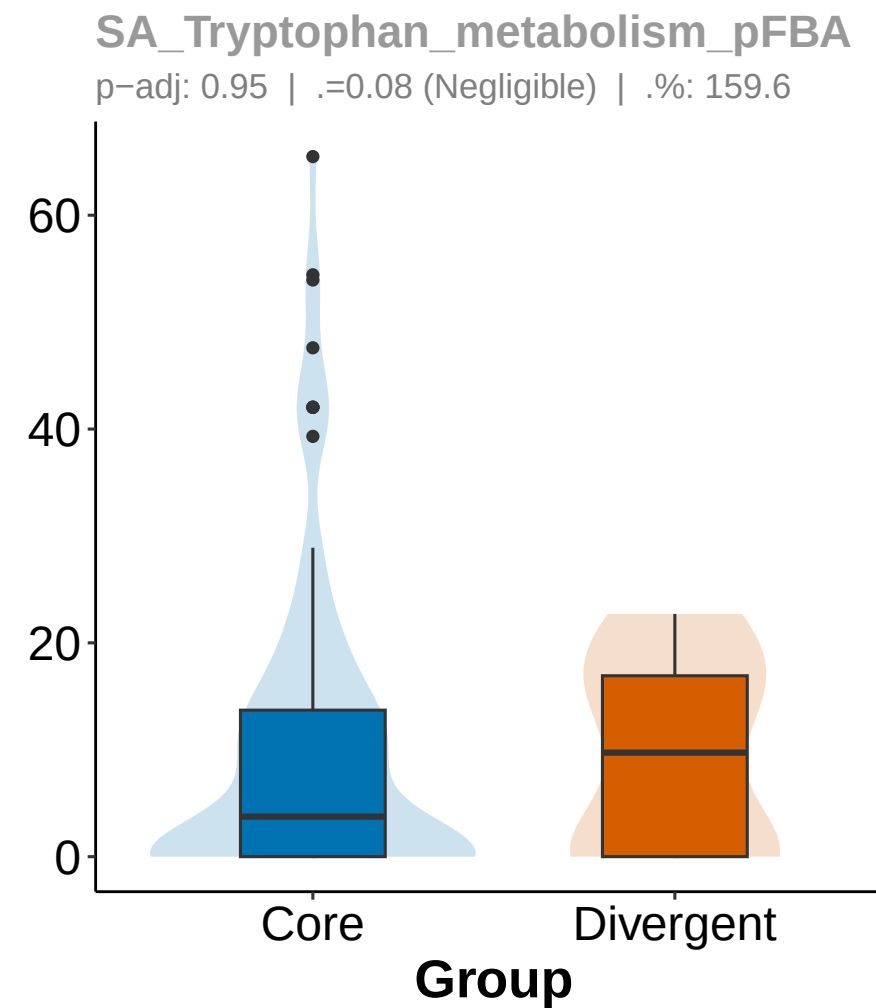
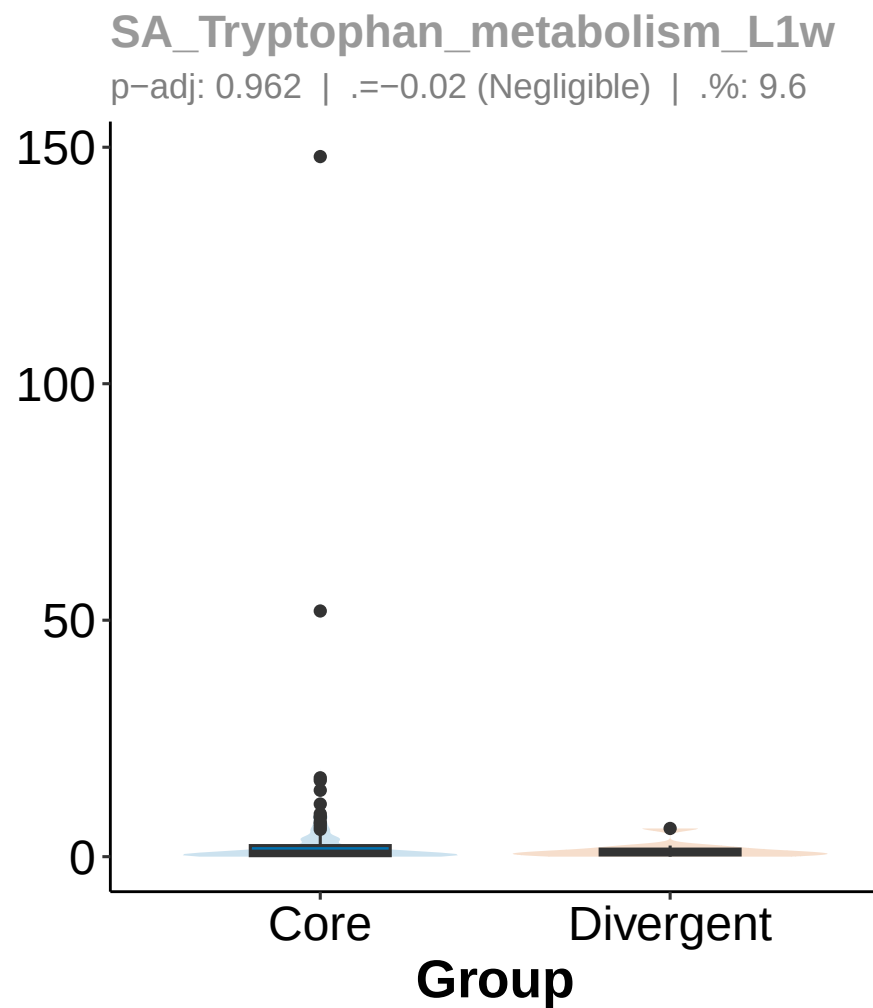
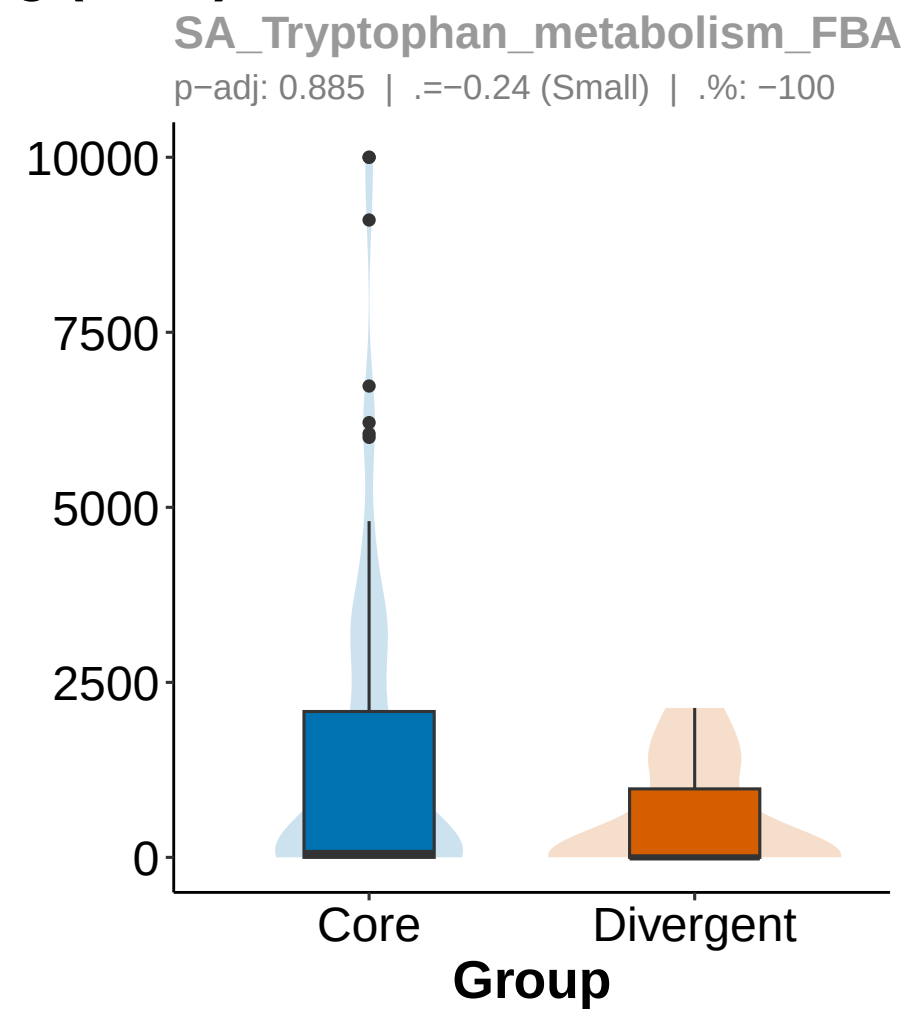
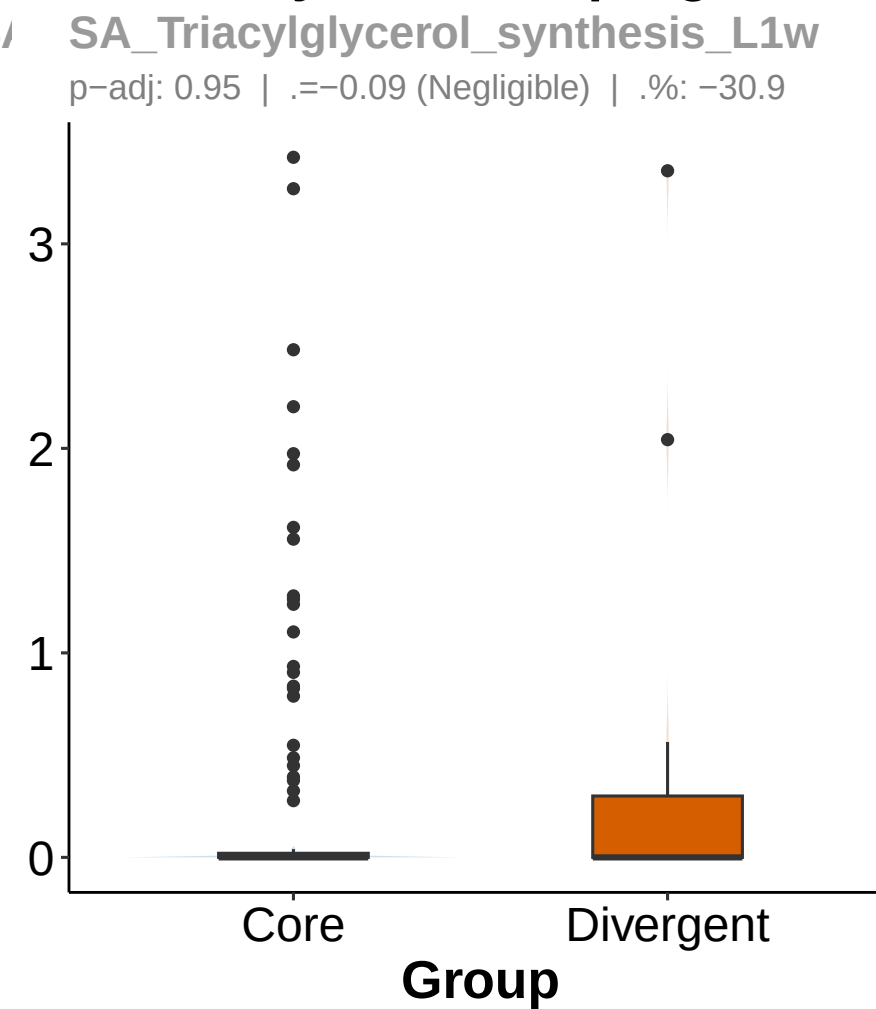
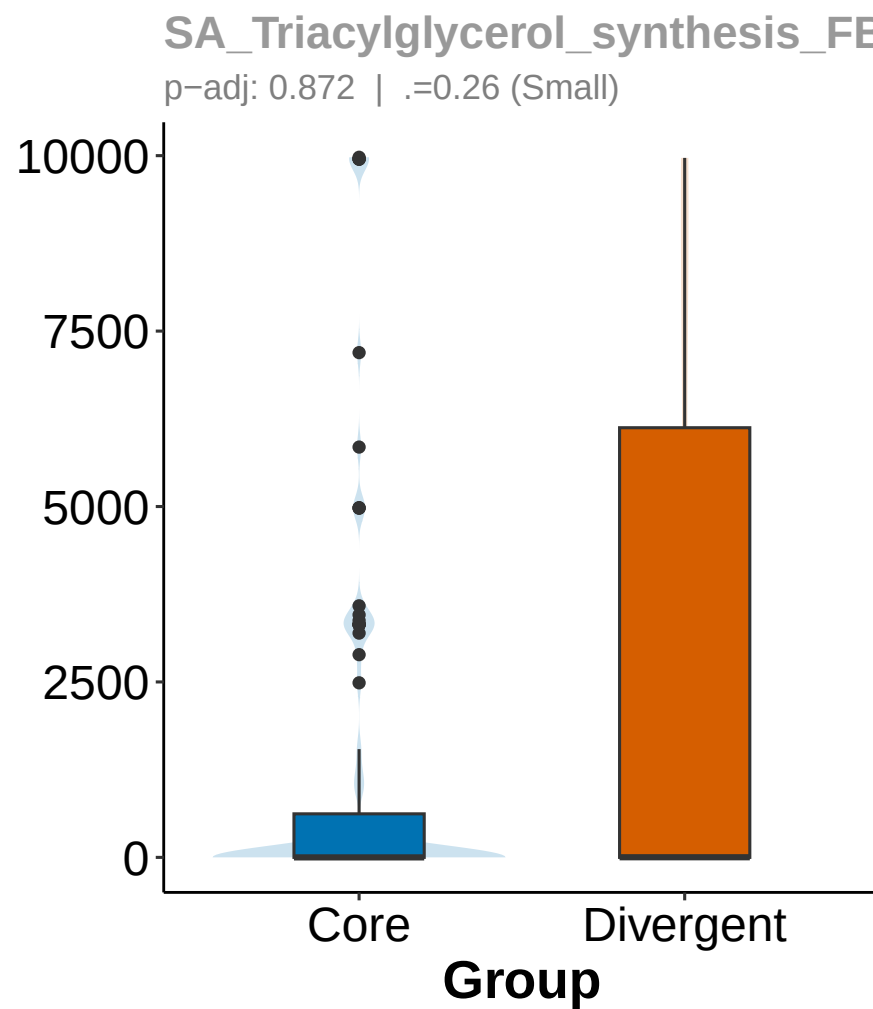


SA_Transport__peroxisomal_pFBA

p-adj: 0.885 | .=-0.1 (Negligible)



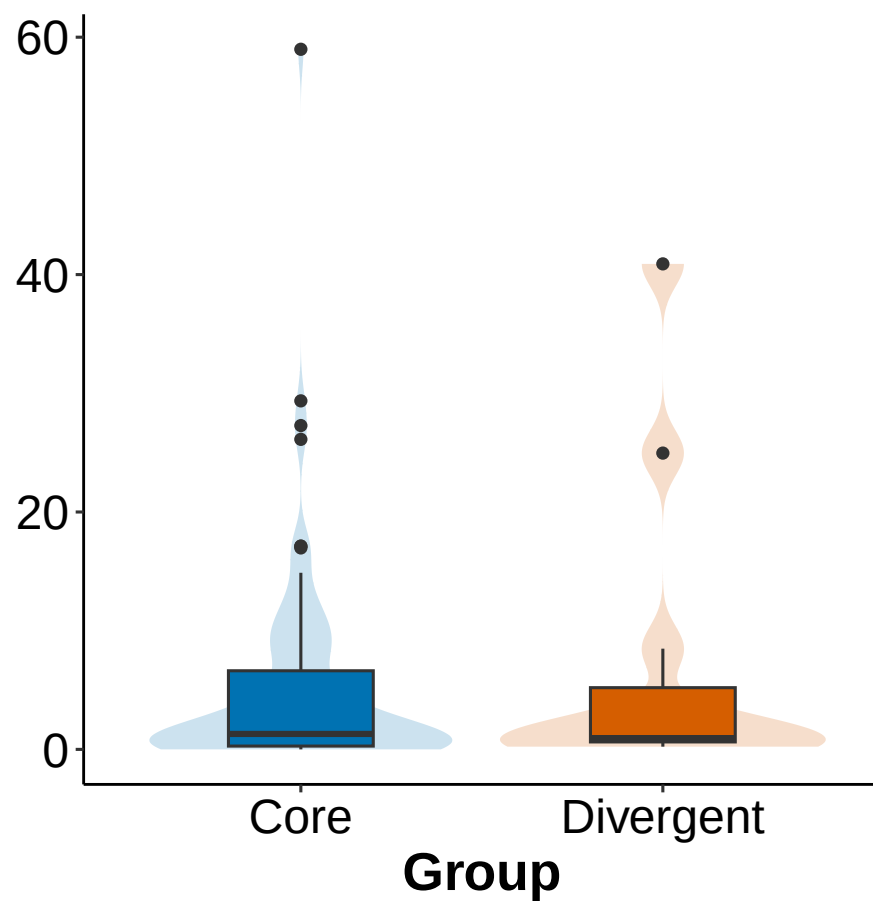
Metabolism – . 5. Systemic Reprogramming (SA_*)



Metabolism – . 5. Systemic Reprogramming (SA_*)

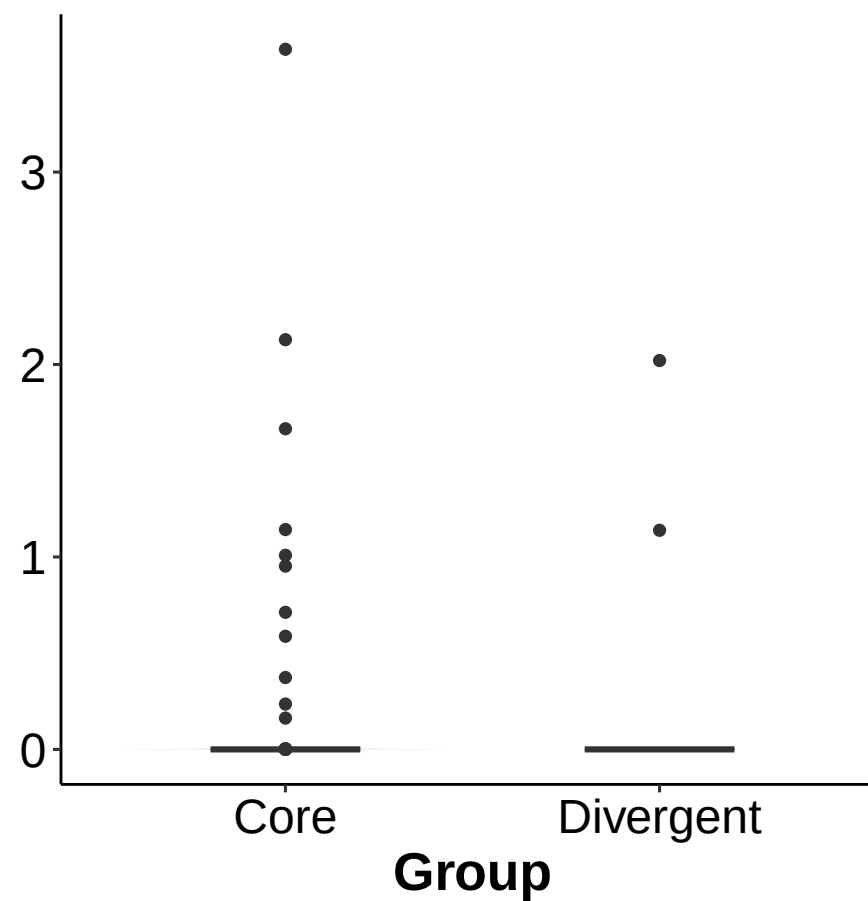
SA_Tyrosine_metabolism_L1w

p-adj: 0.95 | . = 0.06 (Negligible) | .%: -26.9



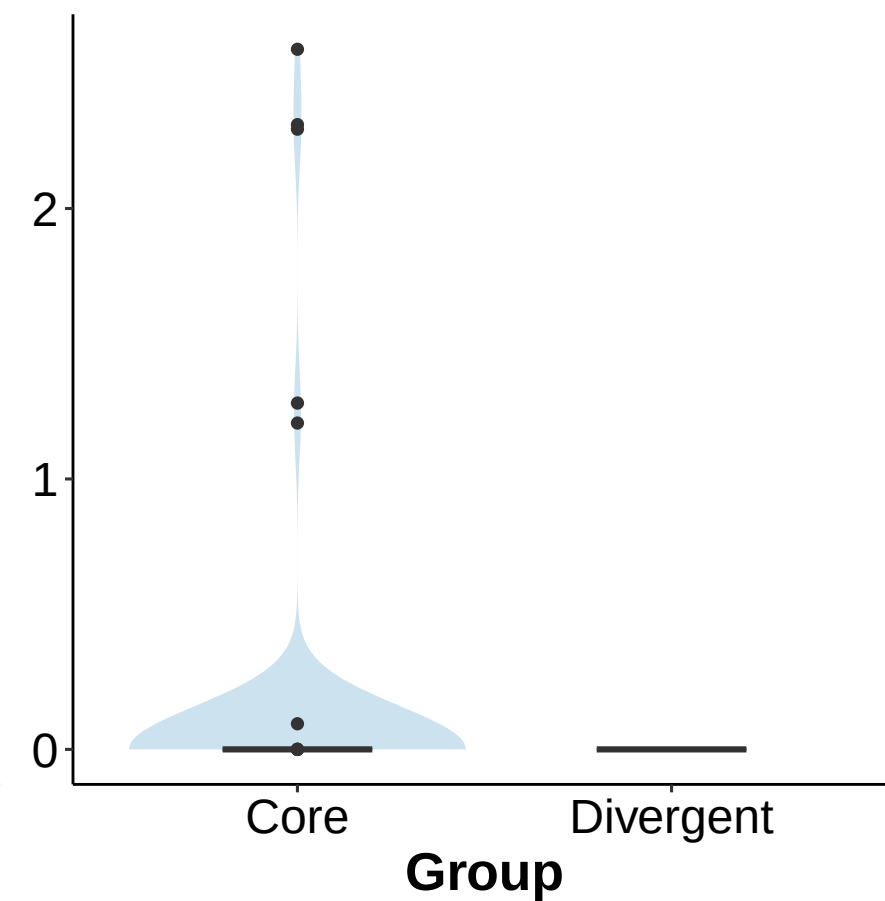
SA_Tyrosine_metabolism_pFBA

p-adj: 0.954 | . = -0.04 (Negligible)



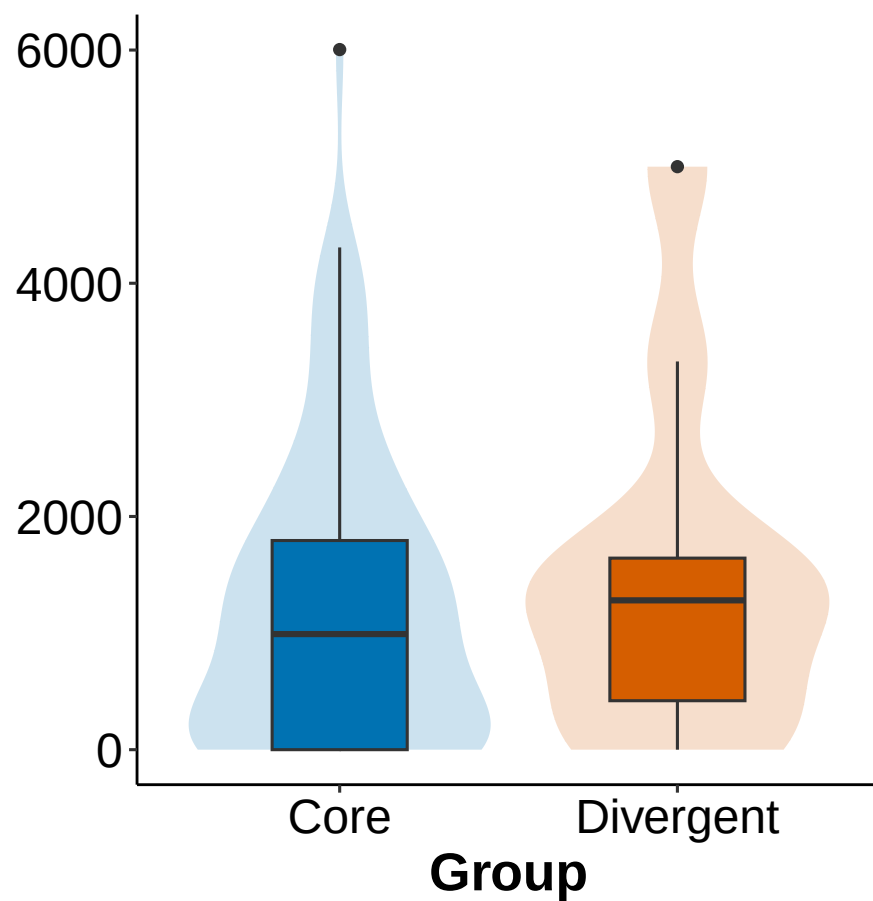
SA_Ubiquinone_synthesis_L1w

p-adj: 0.885 | . = -0.1 (Negligible)



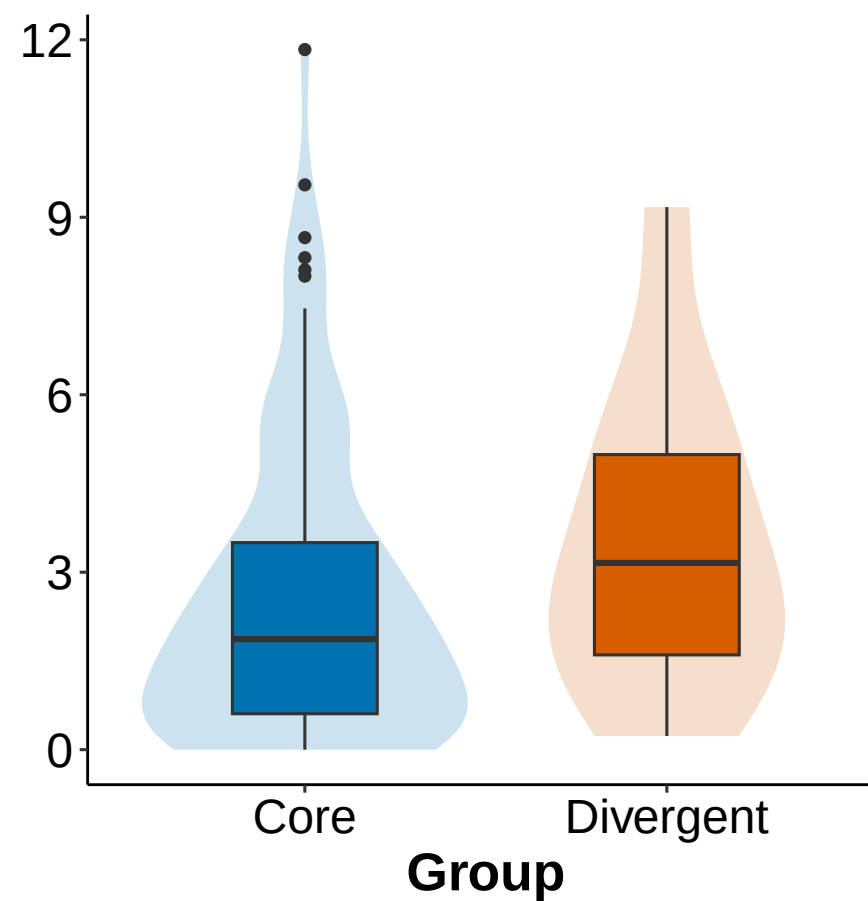
SA_Urea_cycle_FBA

p-adj: 0.945 | . = 0.09 (Negligible) | .%: 29.2



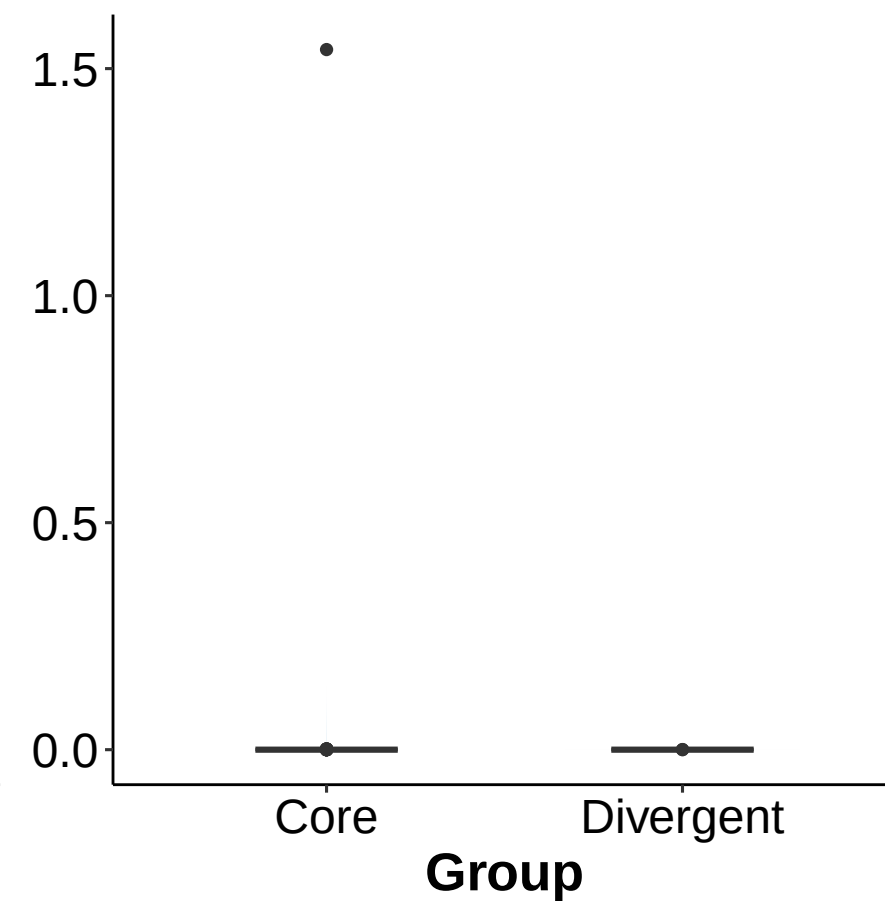
SA_Urea_cycle_L1w

p-adj: 0.872 | . = 0.28 (Small) | .%: 68.8



SA_Urea_cycle_pFBA

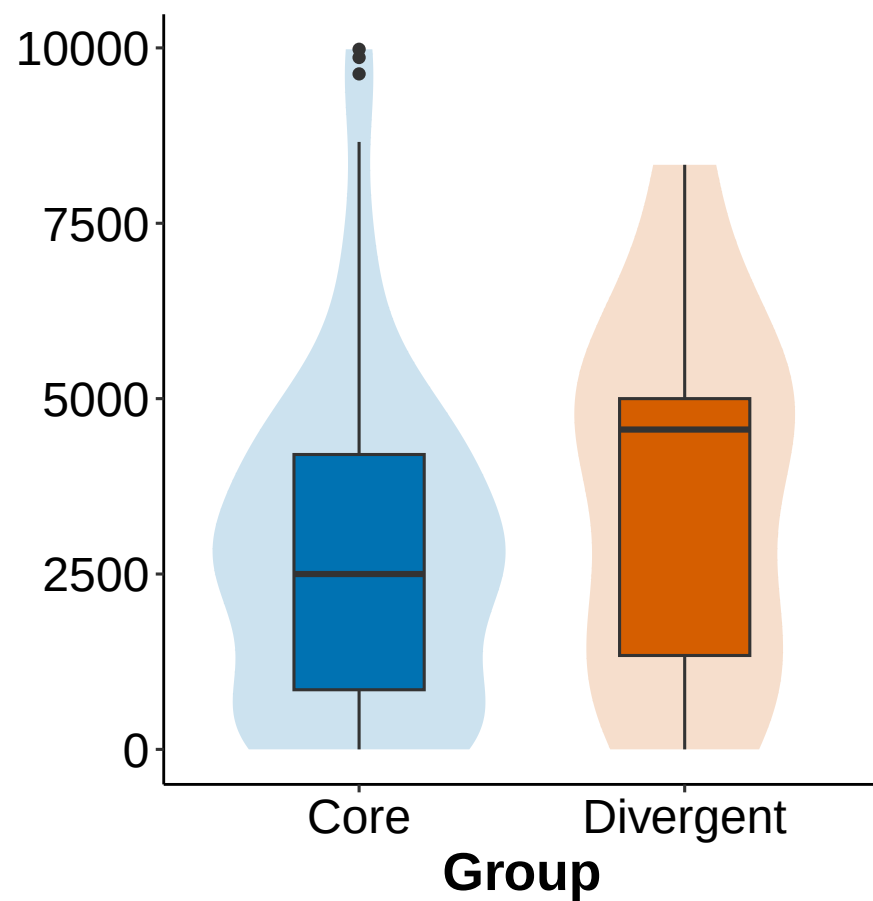
p-adj: 0.917 | . = -0.08 (Negligible)



Metabolism – . 5. Systemic Reprogramming (SA_*)

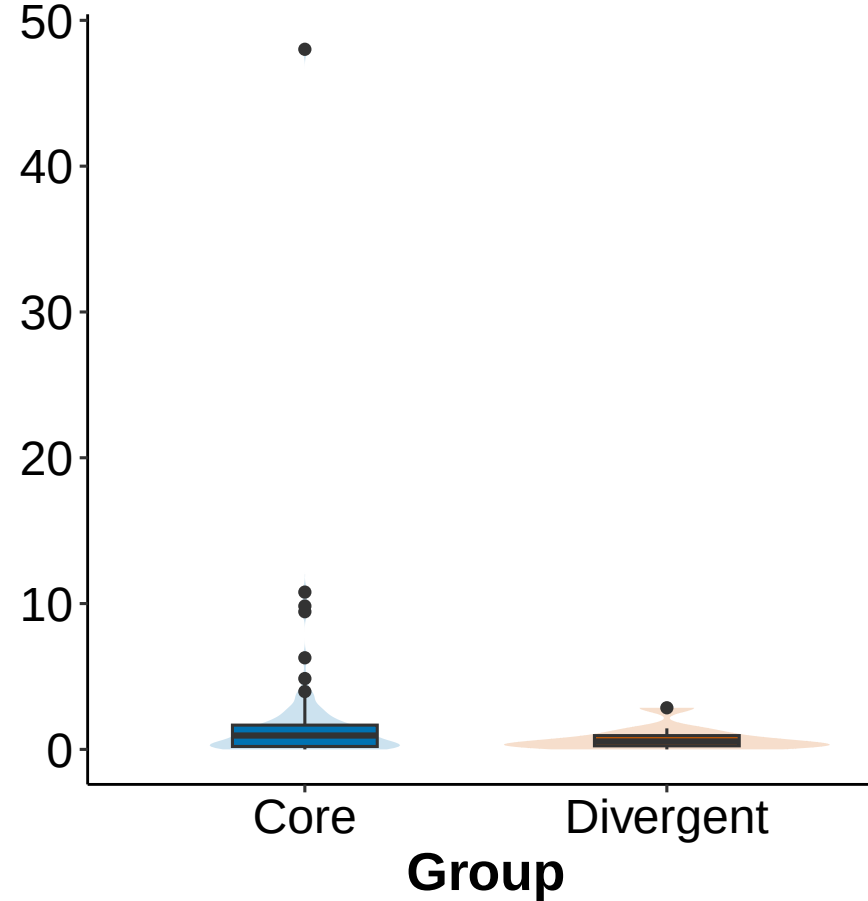
SA_Valine__leucine__and_isoleuc

p-adj: 0.885 | .:=0.21 (Small) | .%: 82.4



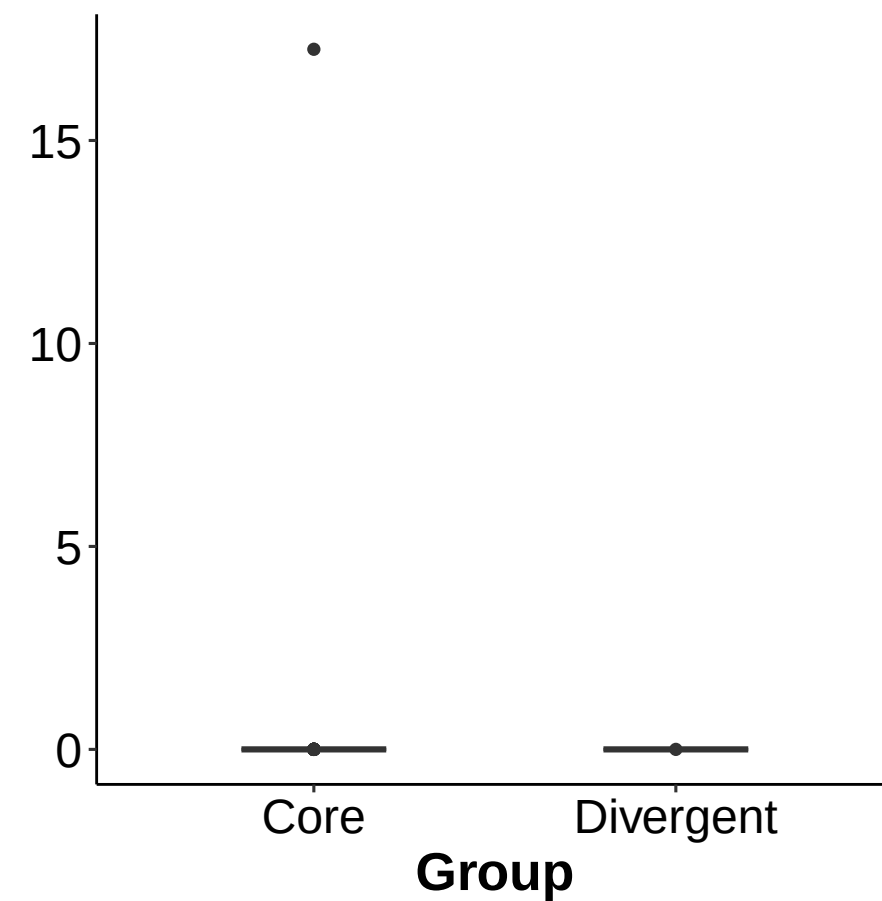
SA_Valine__leucine__and_isoleucine

p-adj: 0.885 | .:=−0.18 (Small) | .%: −42.5



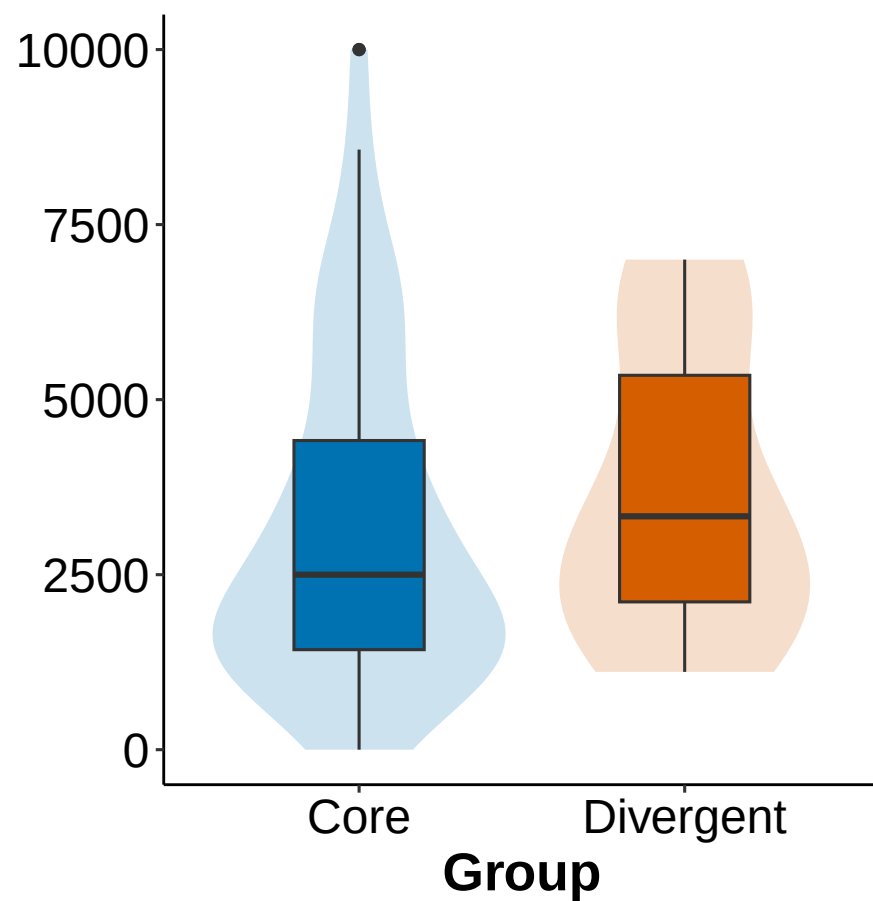
SA_Valine__leucine__and_isoleucine

p-adj: 0.885 | .:=−0.1 (Negligible)



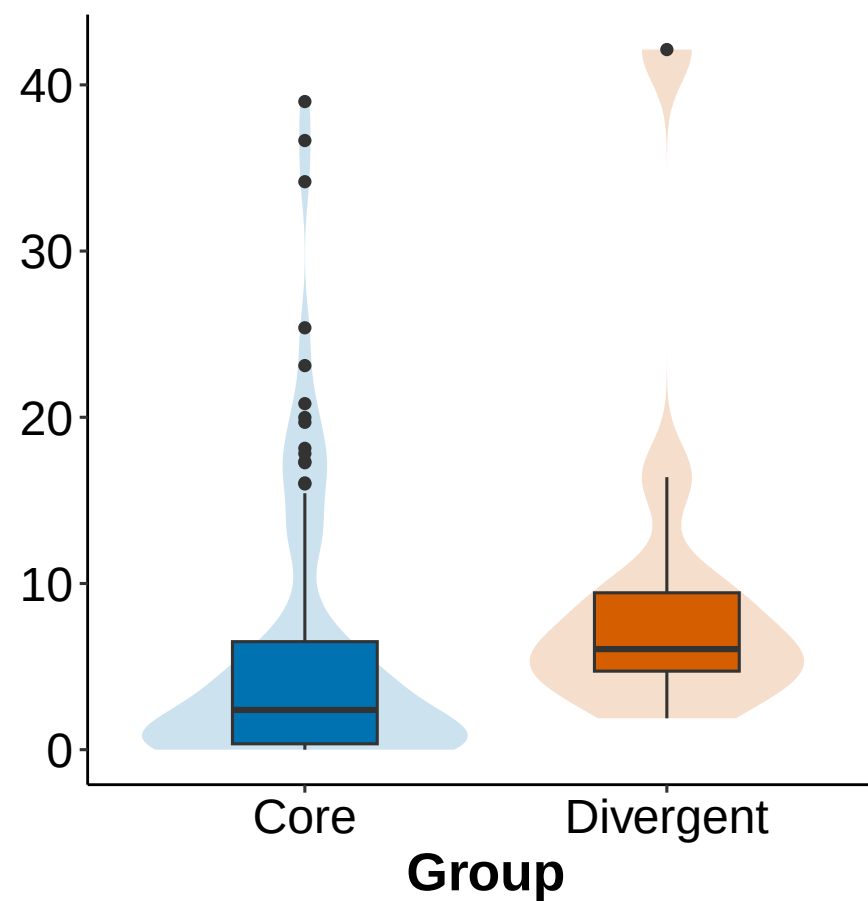
SA_Vitamin_A_metabolism_FBA

p-adj: 0.885 | .:=0.19 (Small) | .%: 33.3



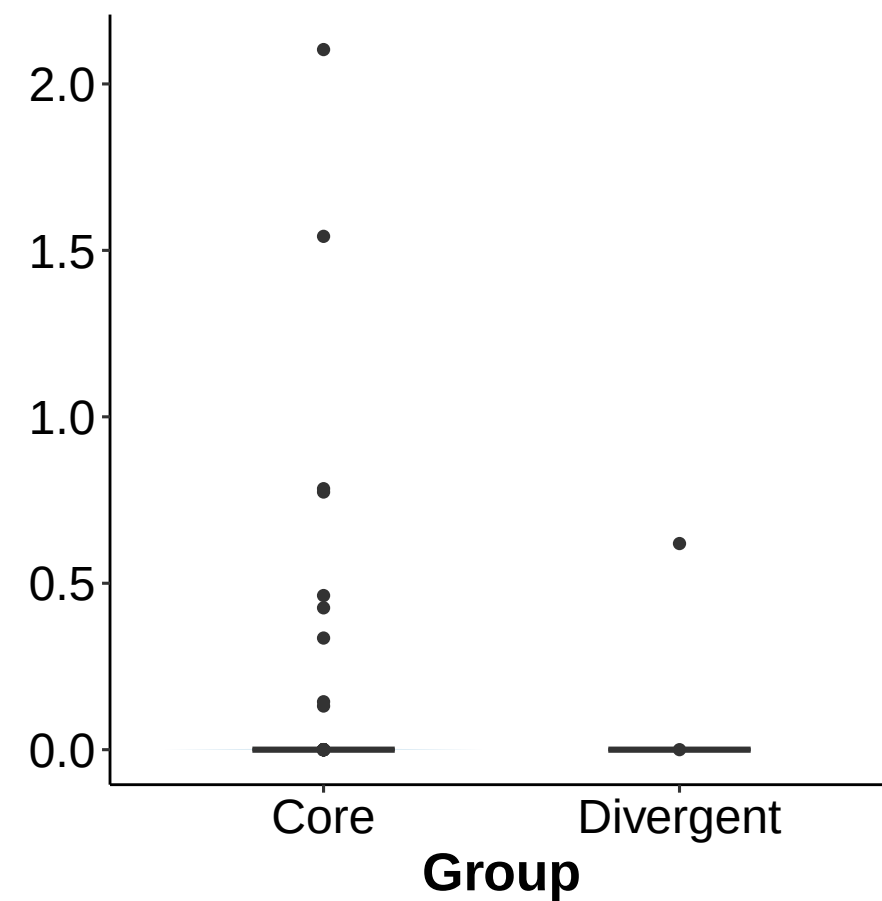
SA_Vitamin_A_metabolism_L1w

p-adj: 0.79 | .:=0.46 (Medium) | .%: 153



SA_Vitamin_B2_metabolism_L1w

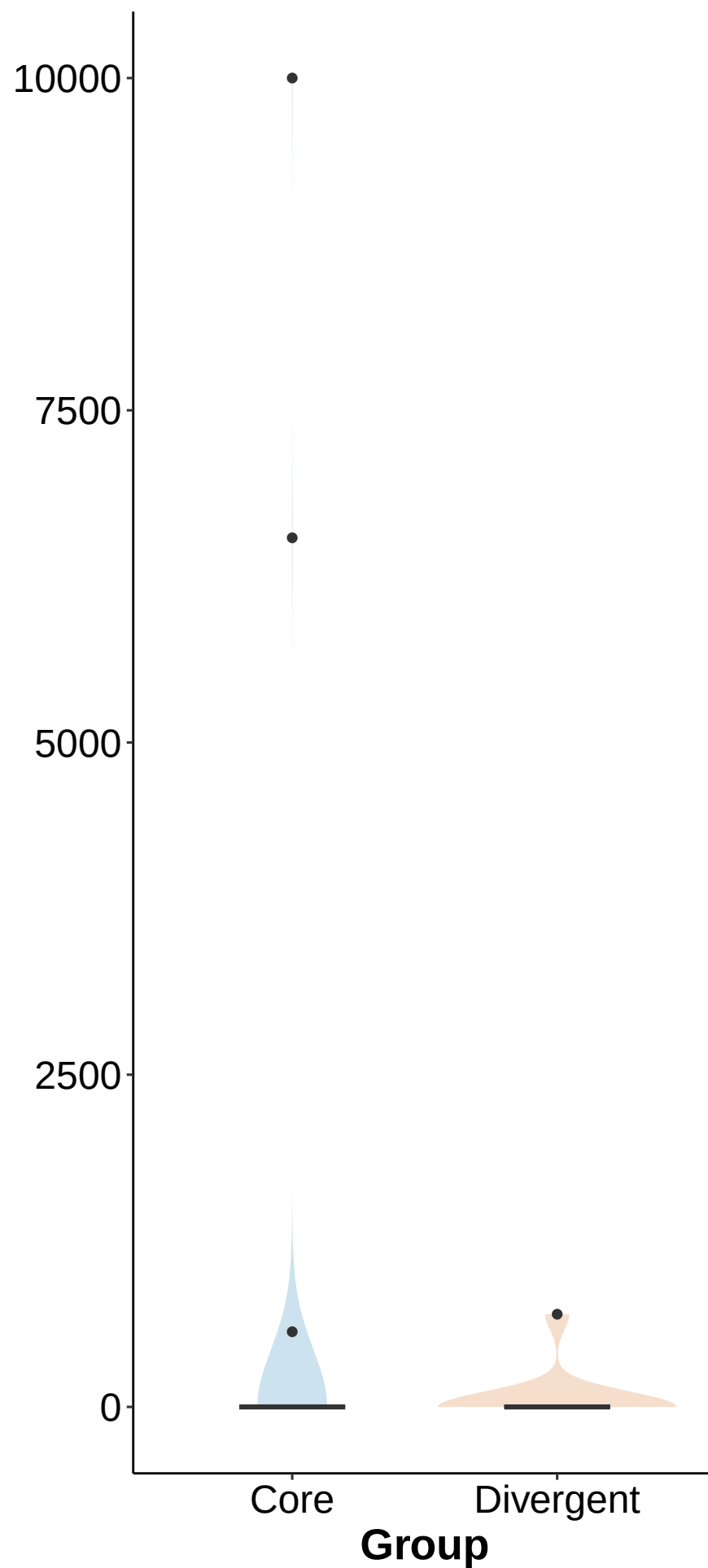
p-adj: 0.95 | .:=0.05 (Negligible)



Metabolism – . 5. Systemic Reprogramming (SA_*)

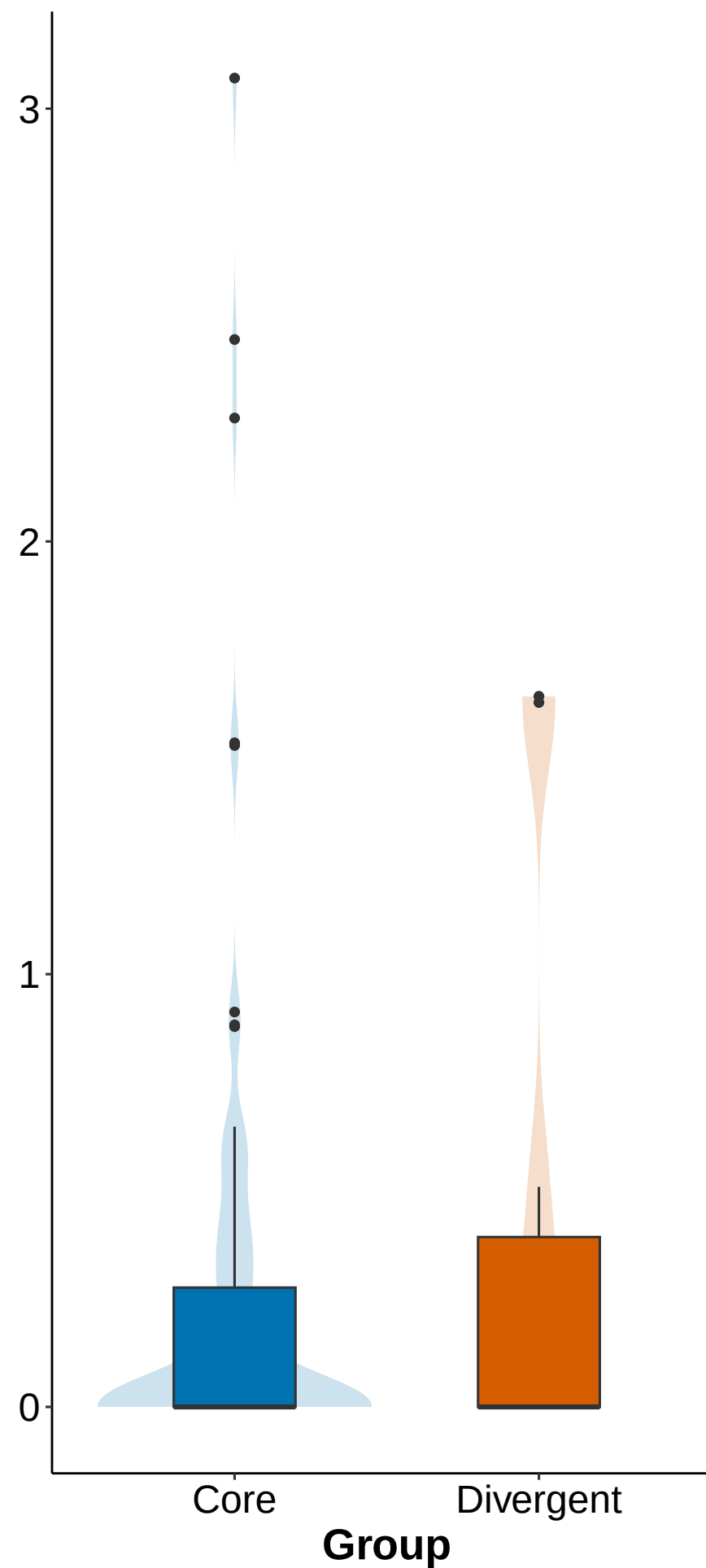
SA_Vitamin_C_metabolism_FBA

p-adj: 0.885 | . = 0.06 (Negligible)



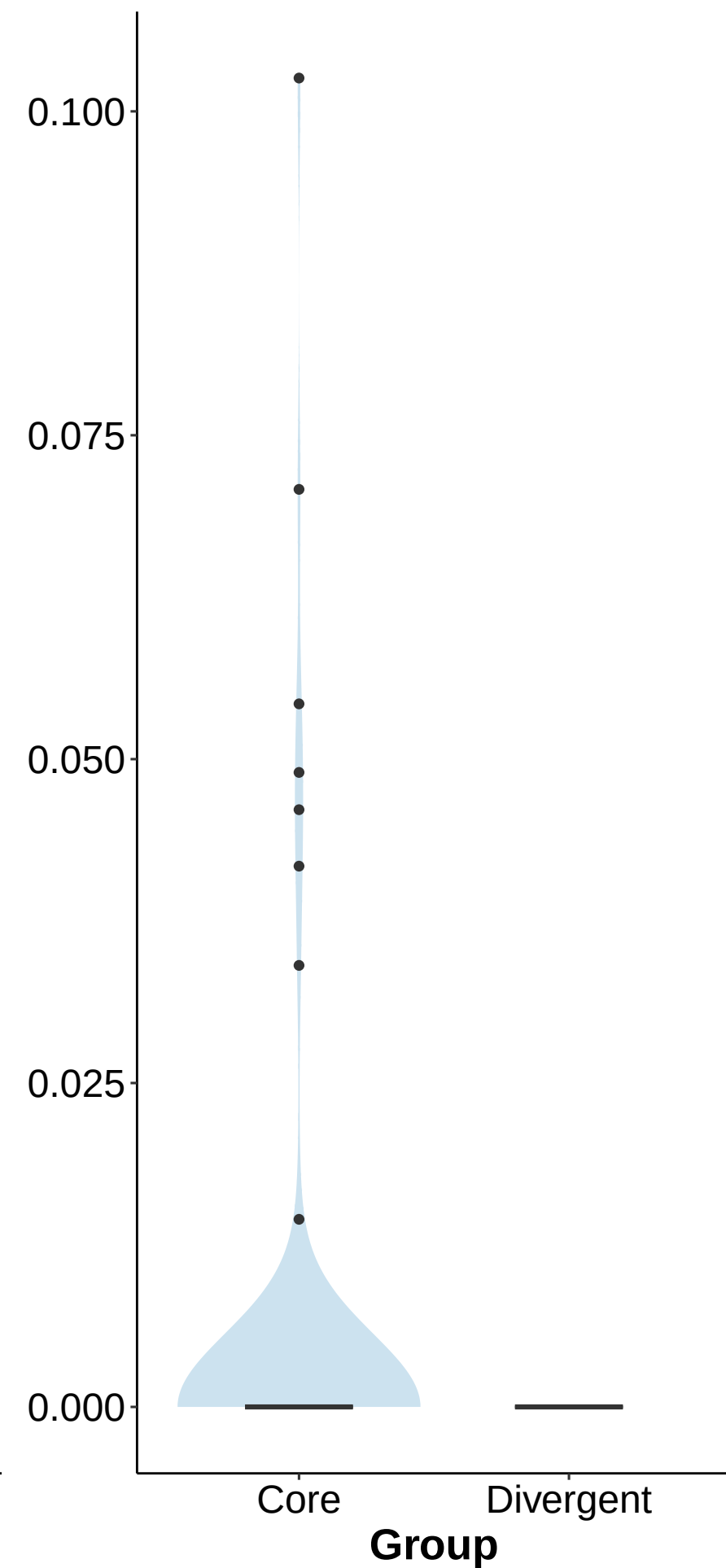
SA_Vitamin_C_metabolism_L1w

p-adj: 0.95 | . = 0.06 (Negligible) | .%: -22.6



SA_Vitamin_D_metabolism_L1w

p-adj: 0.885 | . = -0.08 (Negligible)



Cliff's .: Top 12 Variables

Blue = Core > Divergent | Orange = Divergent > Core | * p_{adj}<0.05

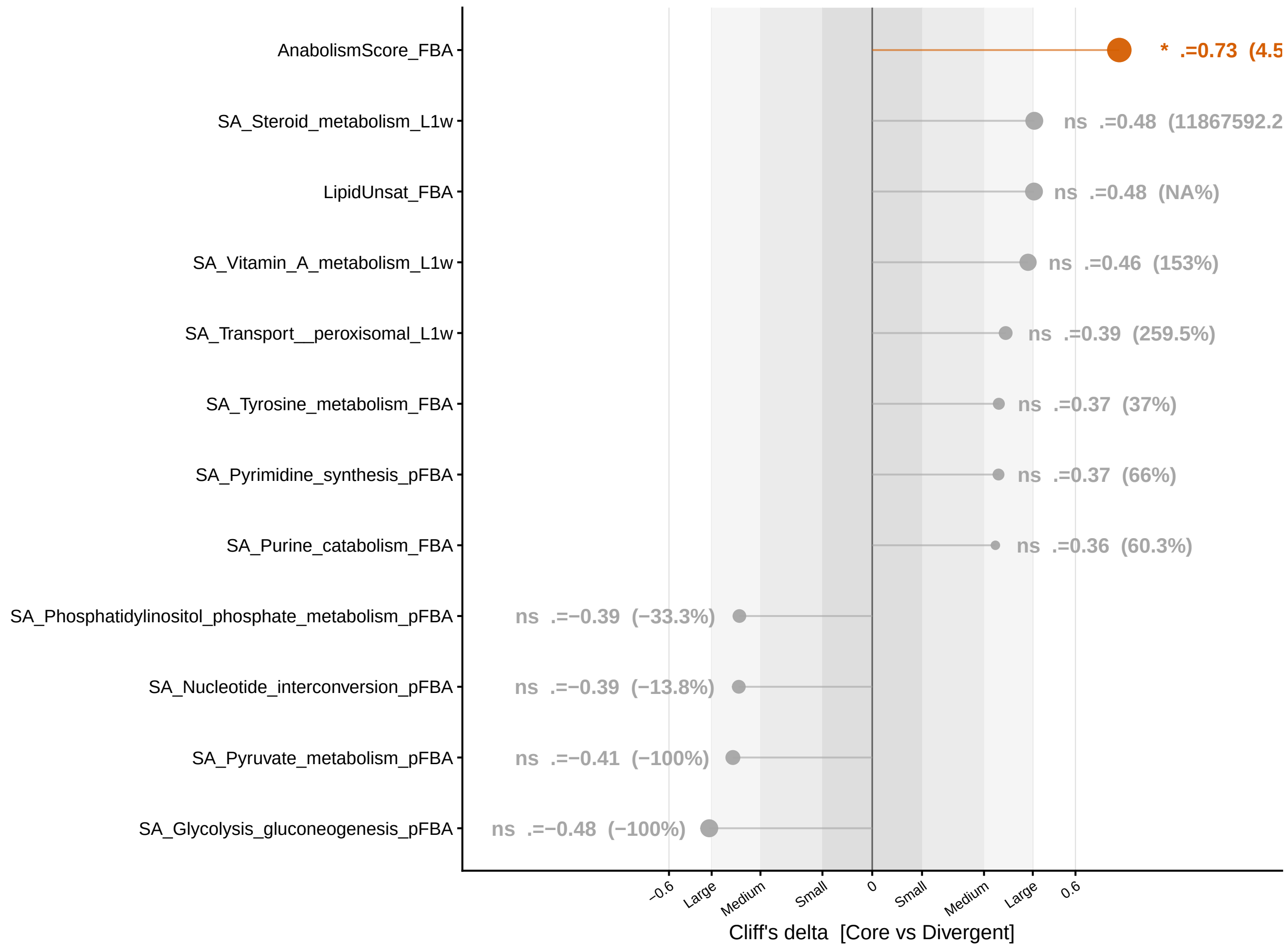
Cliff's .
Div vs Core



AnabolismScore_FBA	* .=0.73
SA_Steroid_metabolism_L1w	ns .=0.48
LipidUnsat_FBA	ns .=0.48
SA_Vitamin_A_metabolism_L1w	ns .=0.46
SA_Transport__peroxisomal_L1w	ns .=0.39
SA_Tyrosine_metabolism_FBA	ns .=0.37
SA_Pyrimidine_synthesis_pFBA	ns .=0.37
SA_Purine_catabolism_FBA	ns .=0.36
SA_Phosphatidylinositol_phosphate_metabolism_pFBA	ns .=-0.39
SA_Nucleotide_interconversion_pFBA	ns .=-0.39
SA_Pyruvate_metabolism_pFBA	ns .=-0.41
SA_Glycolysis_gluconeogenesis_pFBA	ns .=-0.48

. Top 12 Variables – Cliff's .

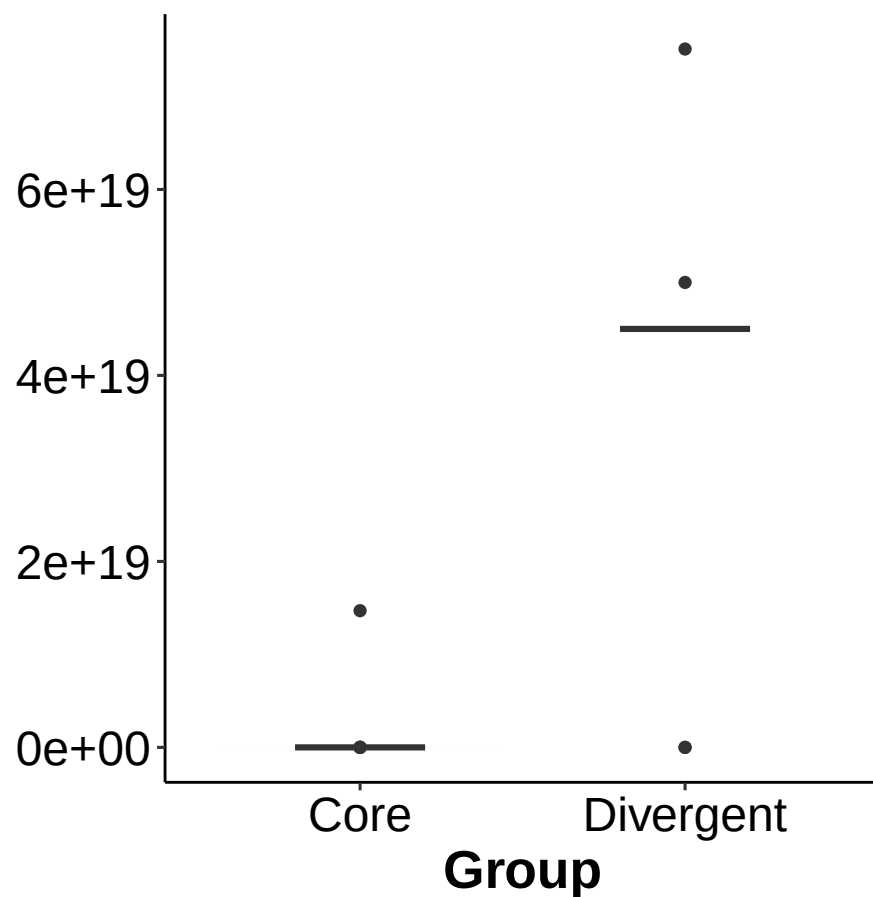
Cliff's . + .% median | . higher in Core . higher in Divergent



. Top 12 Metabolic Variables (p_adj + Cliff's .)

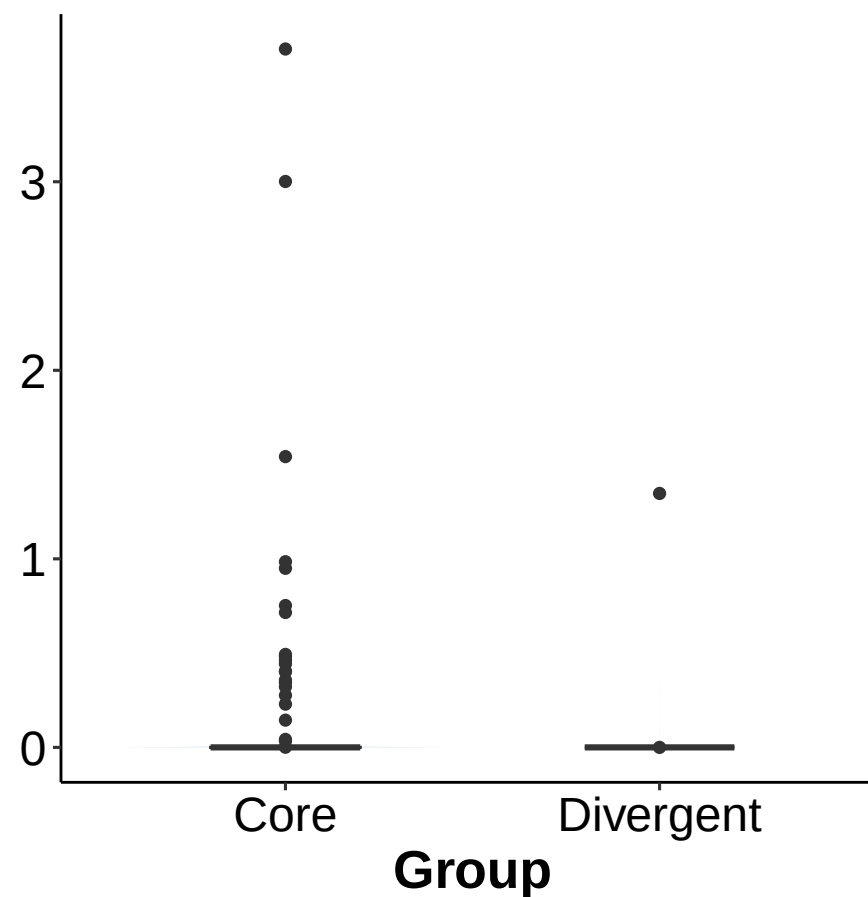
AnabolismScore_FBA

p-adj: 0.0481 | . = 0.73 (Large) | .%: 4.562543



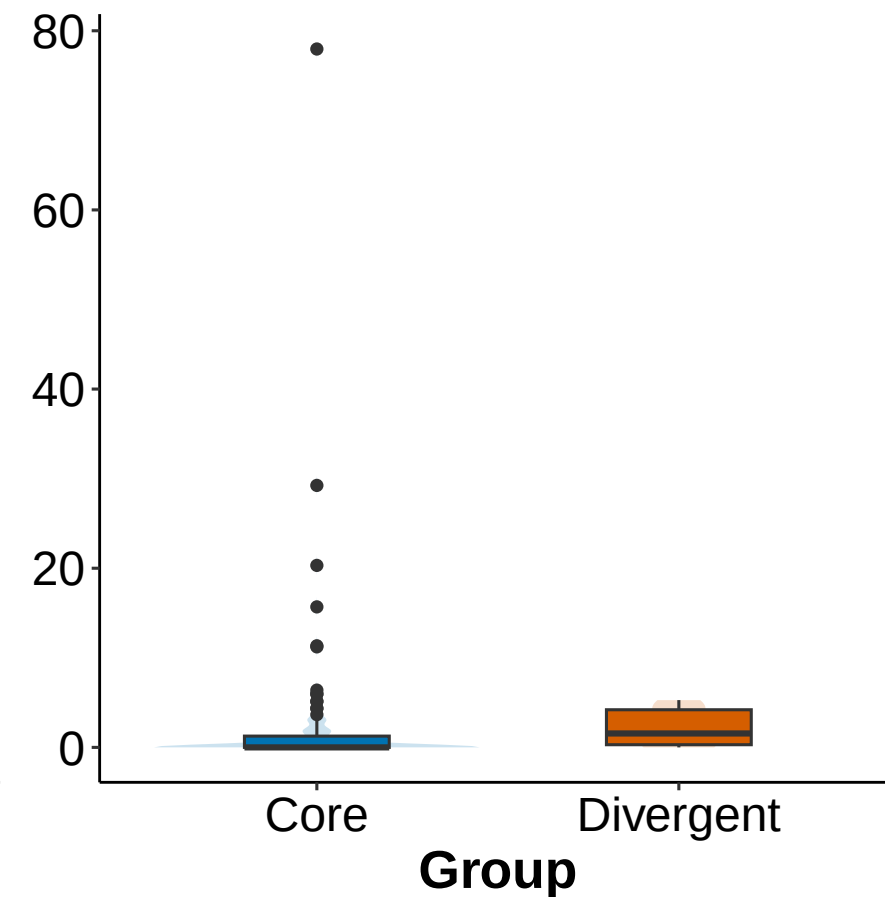
SA_Glycolysis_gluconeogenesis_pFBA

p-adj: 0.749 | . = -0.48 (Large) | .%: -100



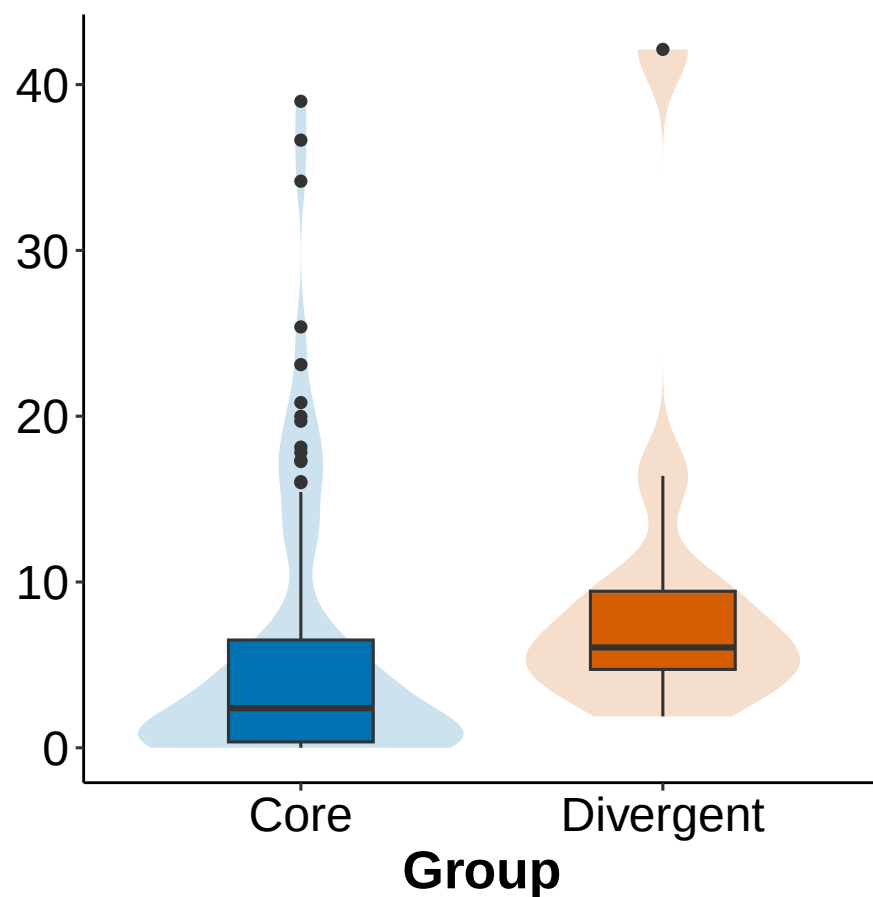
SA_Steroid_metabolism_L1w

p-adj: 0.749 | . = 0.48 (Large) | .%: 11867592.2



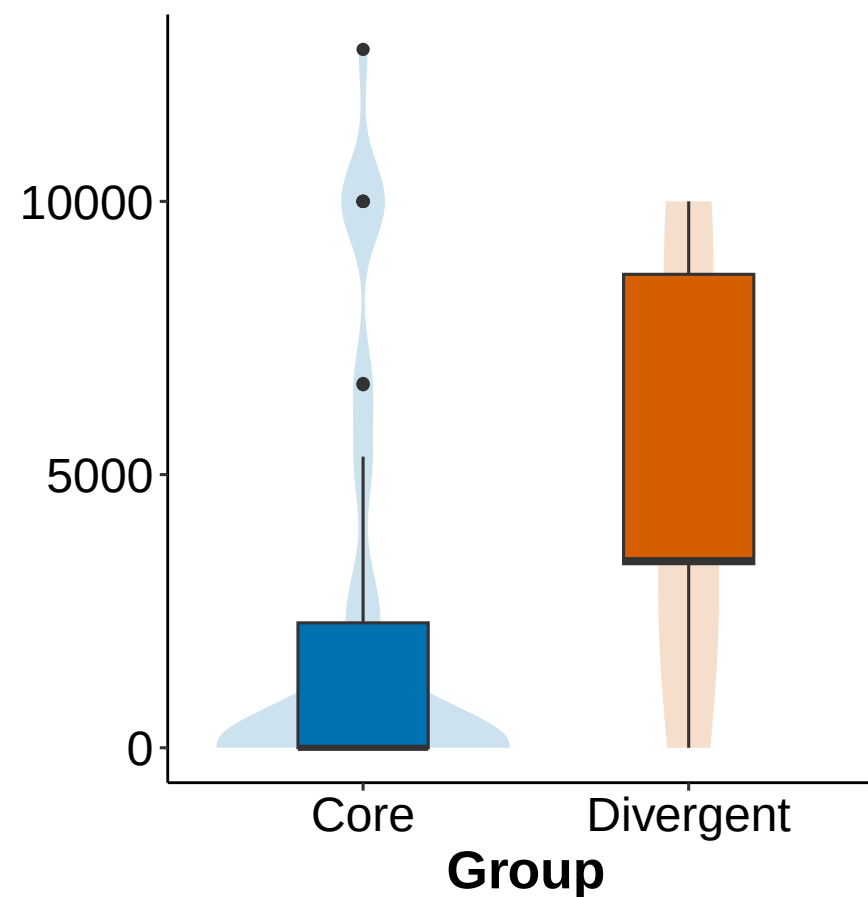
SA_Vitamin_A_metabolism_L1w

p-adj: 0.79 | . = 0.46 (Medium) | .%: 153



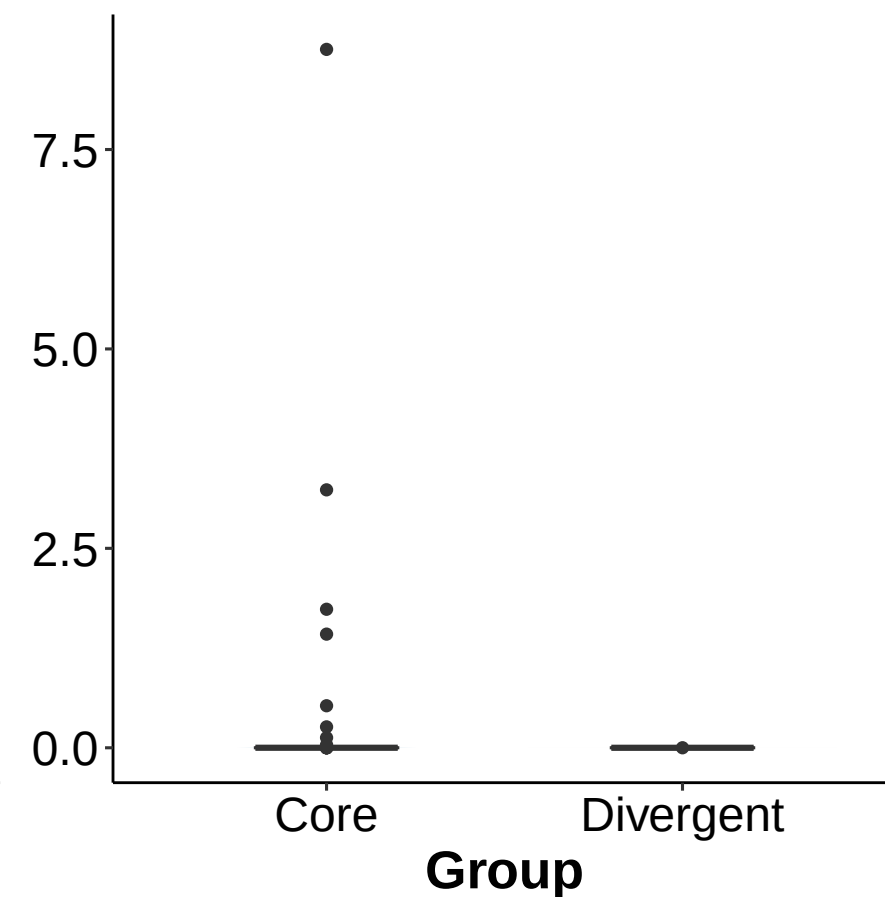
LipidUnsat_FBA

p-adj: 0.843 | . = 0.48 (Large)



SA_Pyruvate_metabolism_pFBA

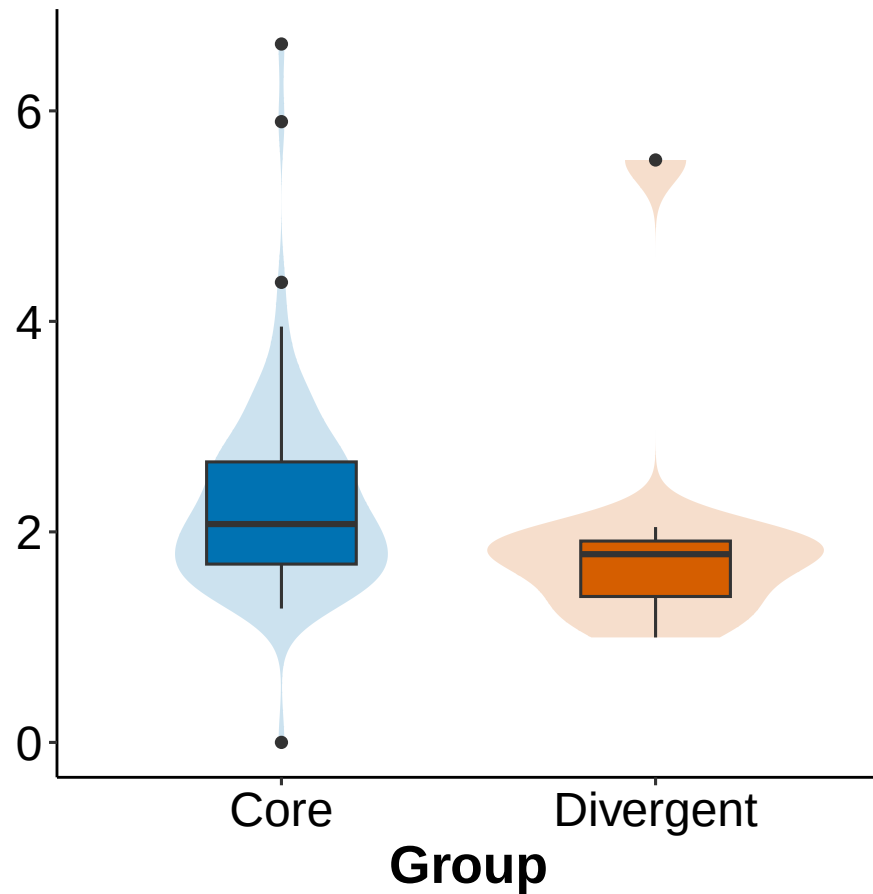
p-adj: 0.843 | . = -0.41 (Medium) | .%: -100



. Top 12 Metabolic Variables (p_adj + Cliff's .)

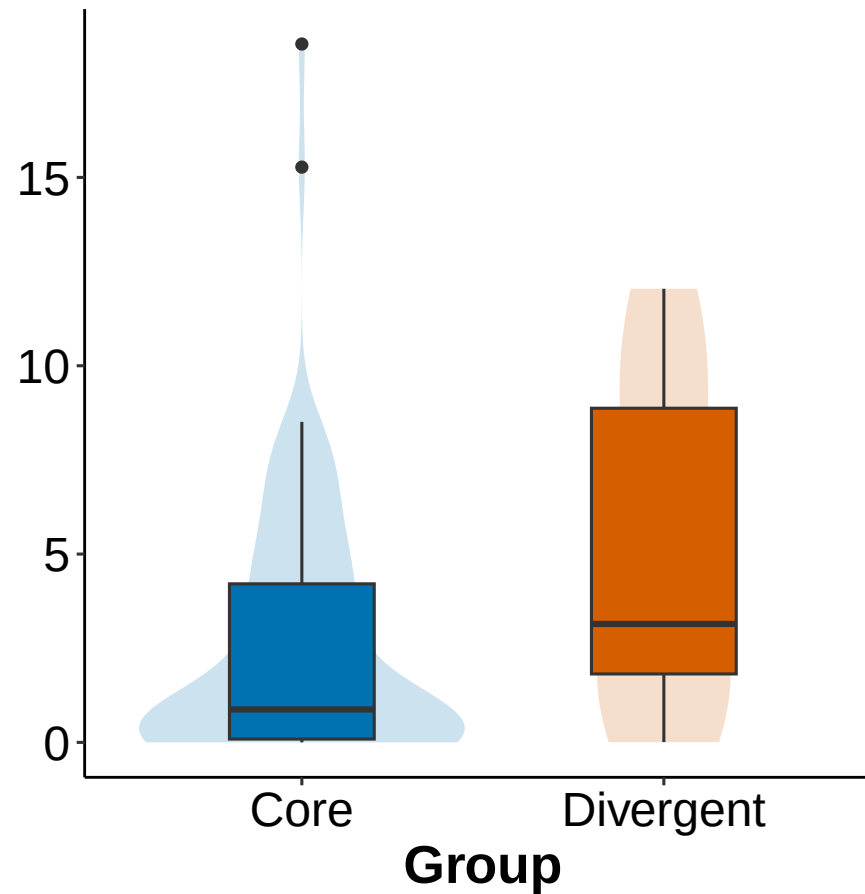
SA_Nucleotide_interconversion_pFBA

p-adj: 0.843 | .=-0.39 (Medium) | .%: -13.8



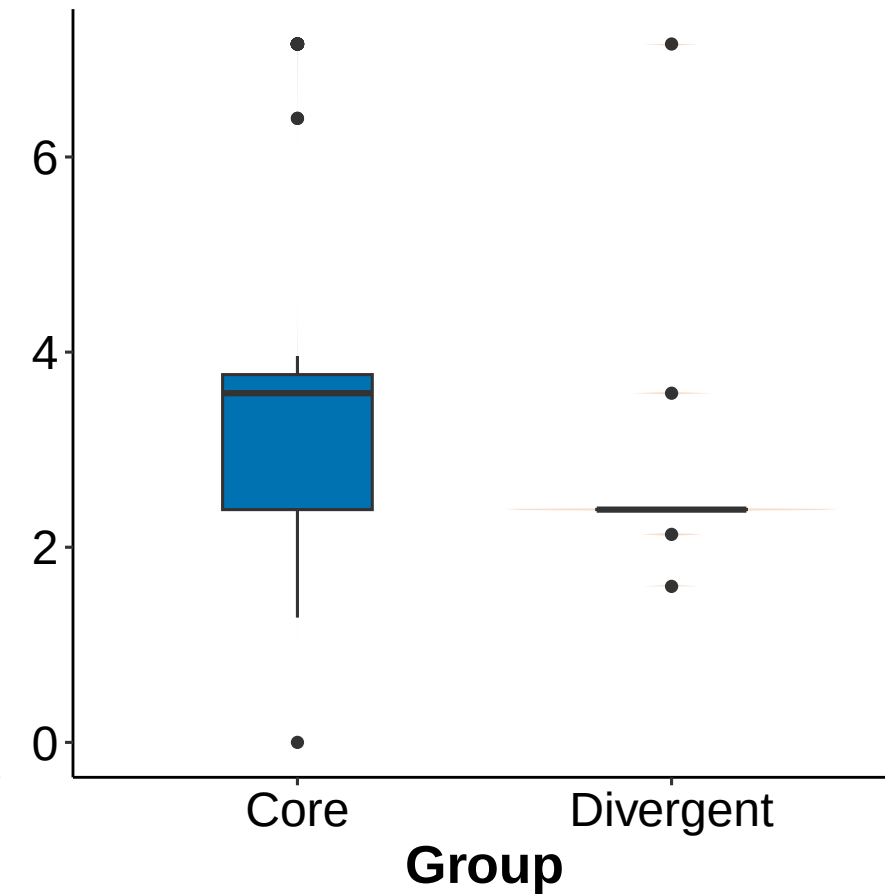
SA_Transport__peroxisomal_L1w

p-adj: 0.843 | .=-0.39 (Medium) | .%: 259.5



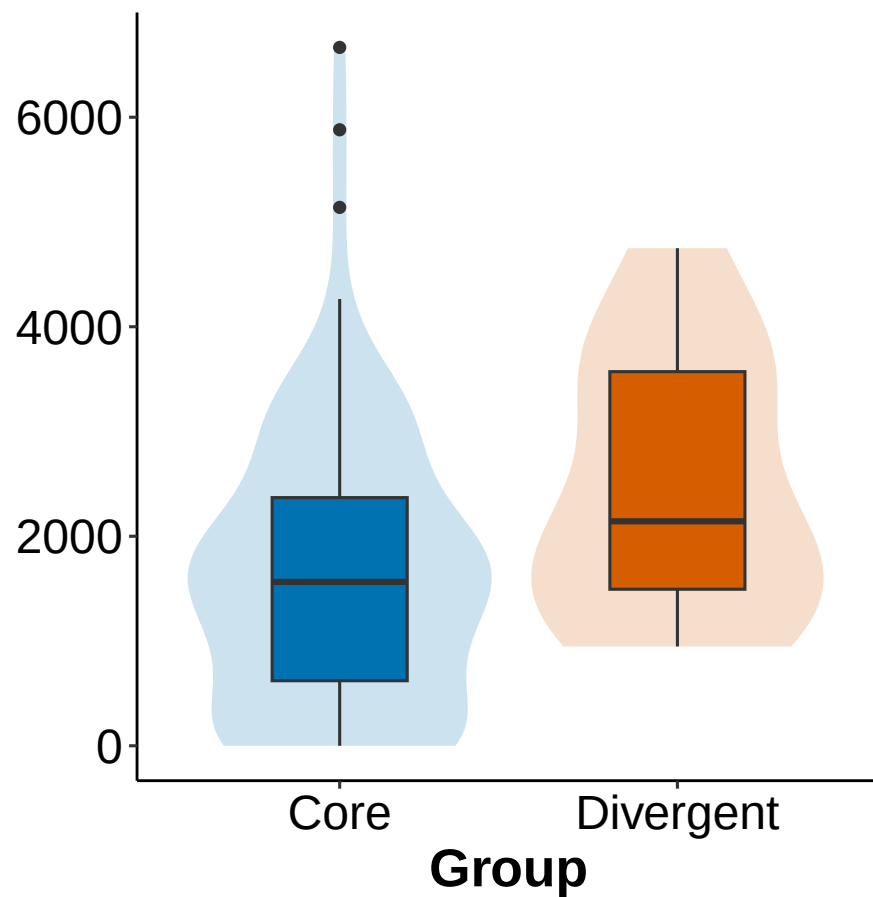
SA_Phosphatidylinositol_phosphate_m

p-adj: 0.843 | .=-0.39 (Medium) | .%: -33.3



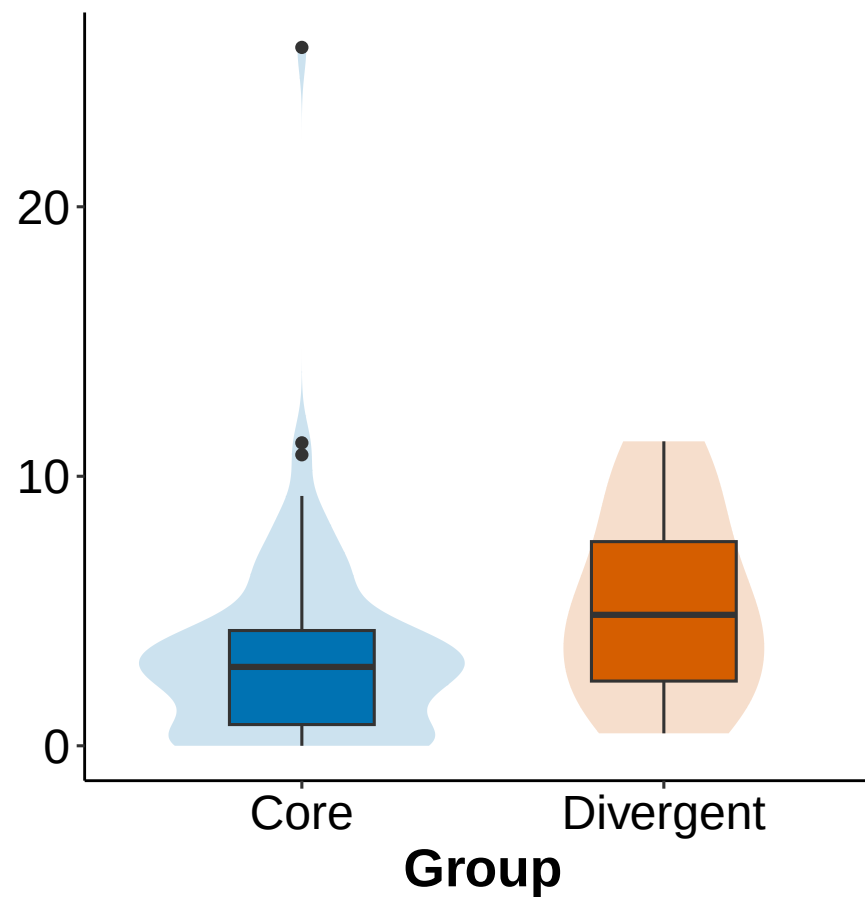
SA_Tyrosine_metabolism_FBA

p-adj: 0.843 | .=-0.37 (Medium) | .%: 37



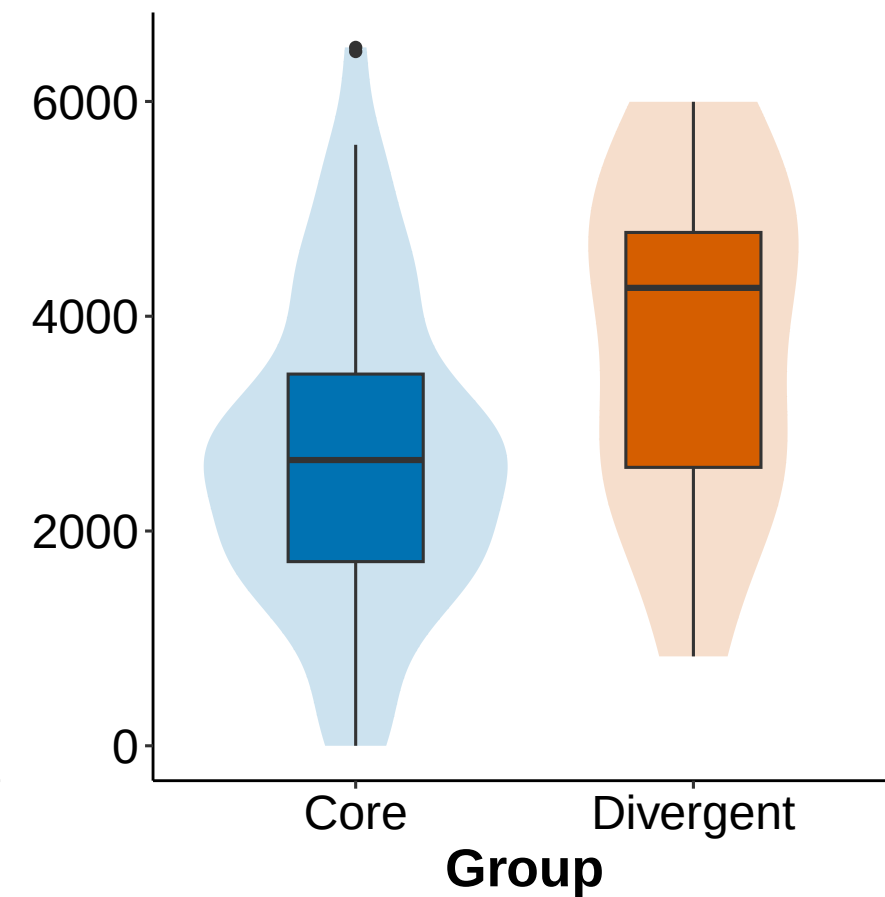
SA_Pyrimidine_synthesis_pFBA

p-adj: 0.843 | .=-0.37 (Medium) | .%: 66



SA_Purine_catabolism_FBA

p-adj: 0.843 | .=-0.36 (Medium) | .%: 60.3



SECTION III

Integrated Analysis: Clinical <—> Metabolism

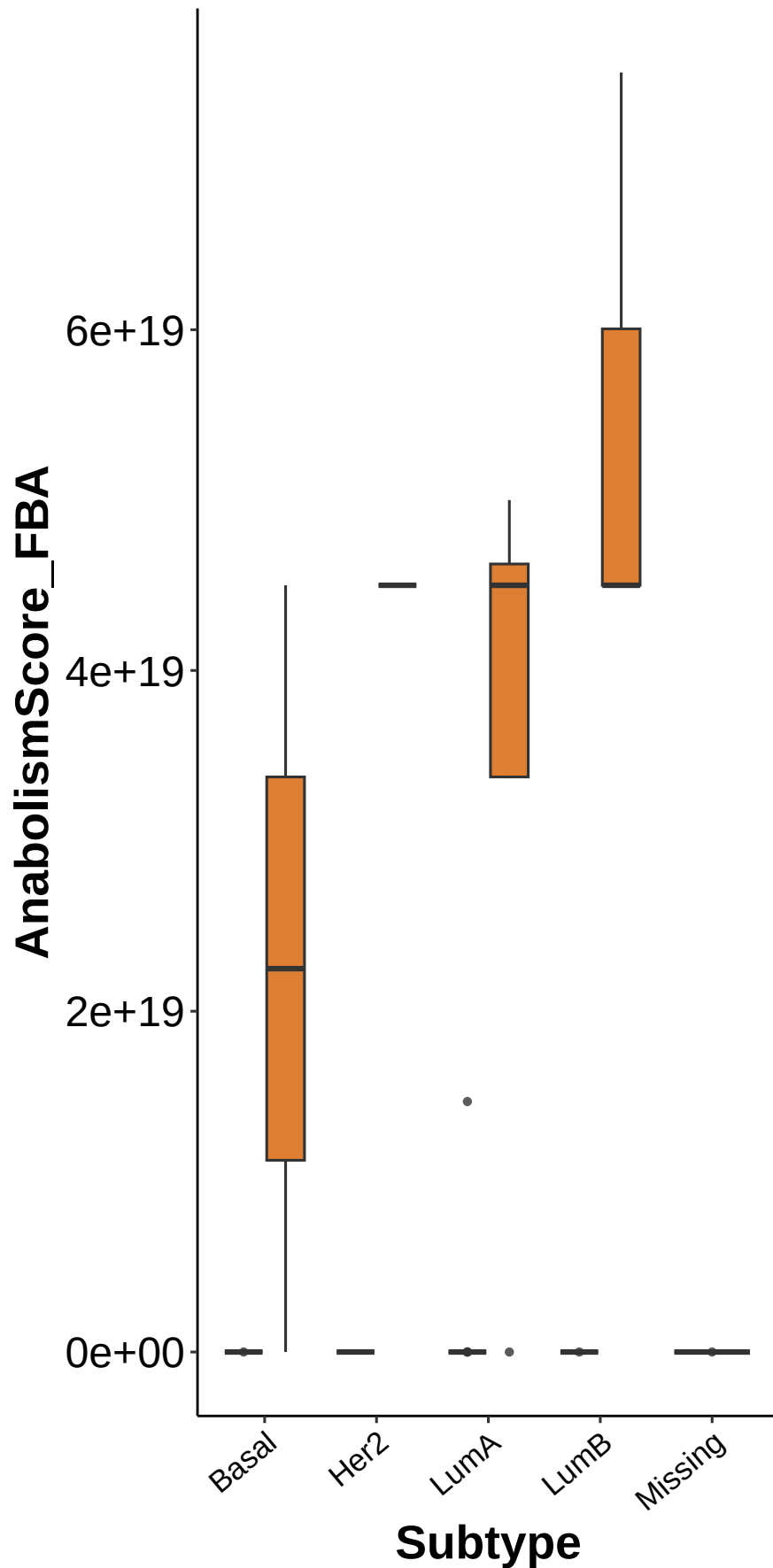
Is Core vs Divergent a reflection of subtype? Or an independent metabolic reprogramming?

Integration: Top metabolic × Subtype

AnabolismScore_F

Is the Core/Divergent difference

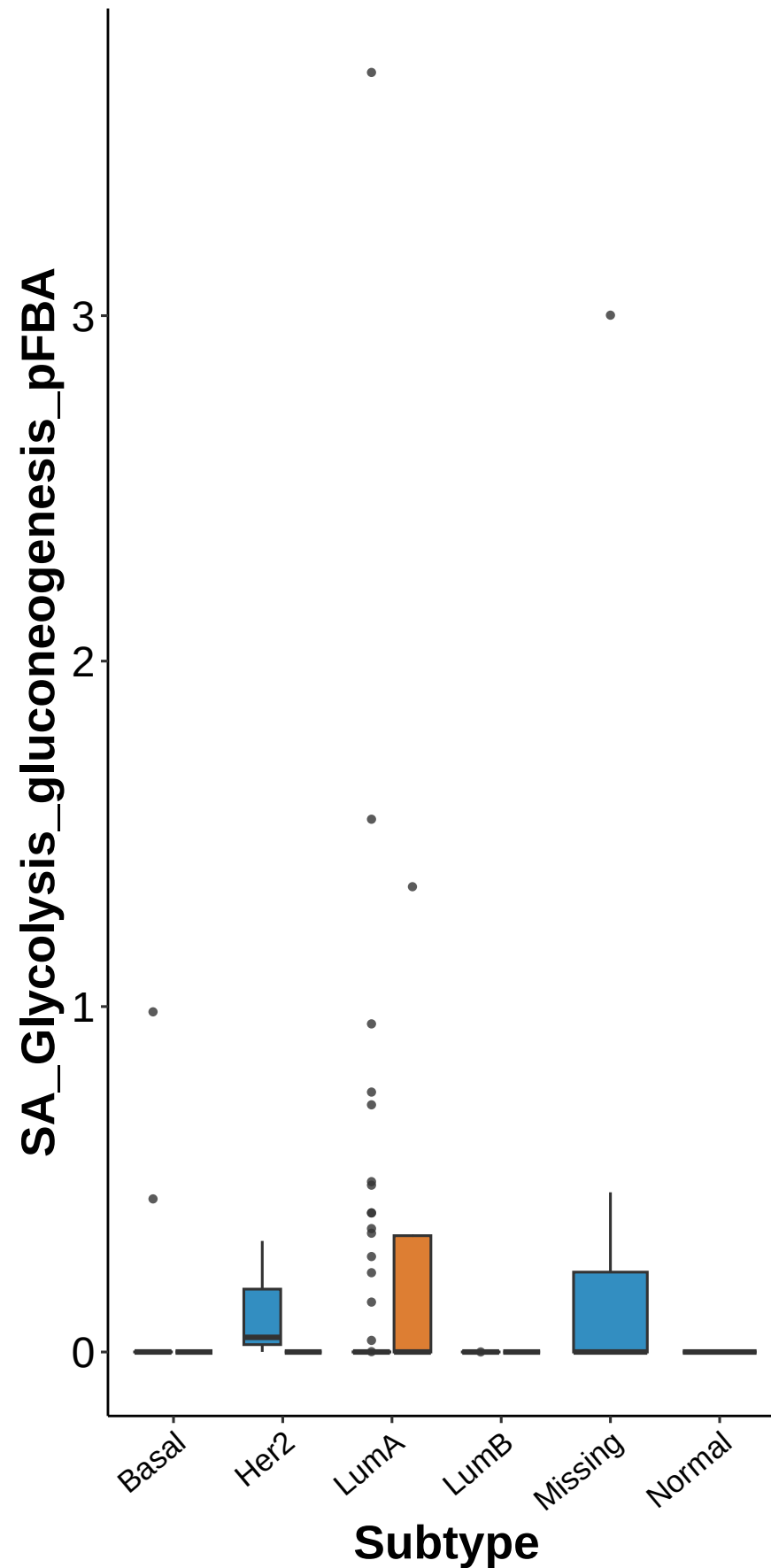
Group  Core  Divergent



SA_Glycolysis_gluco

Is the Core/Divergent difference inde

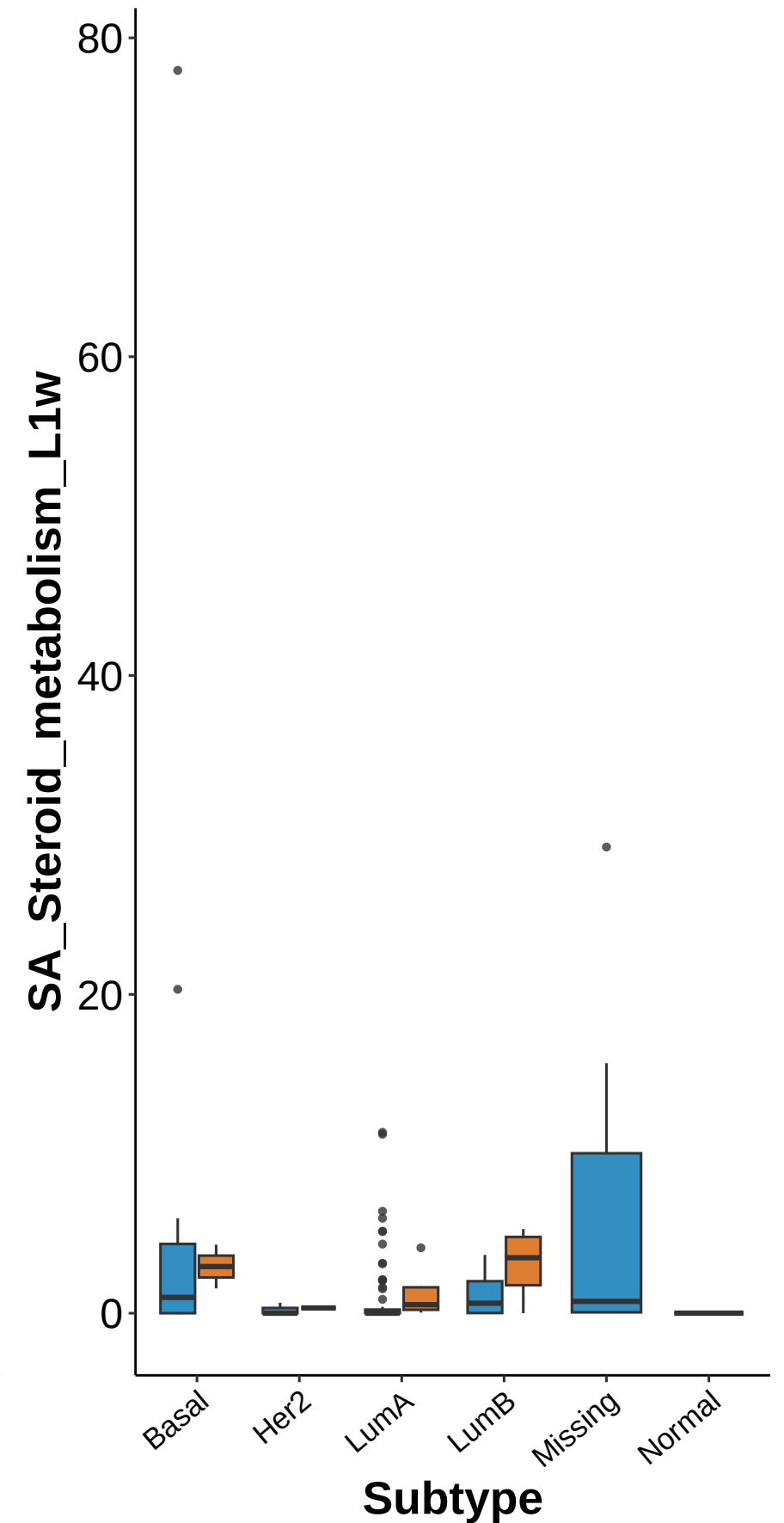
Group  Core  Divergent



SA_Steroid_metaboli

Is the Core/Divergent difference ind

Group  Core  Divergent

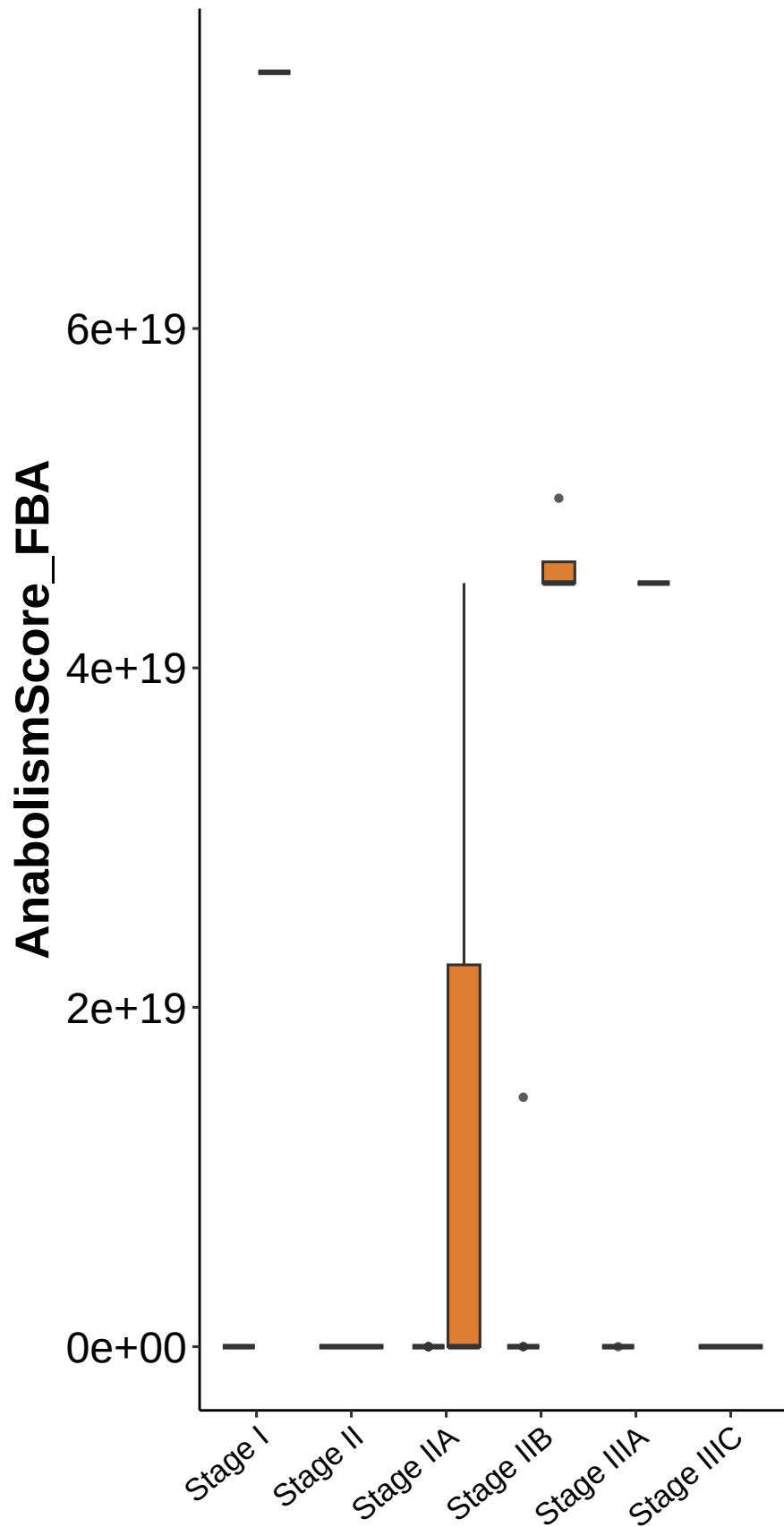


Integration: Top metabolic × ajcc_pathologic_stage.diagnoses

AnabolismScore_F

Is the Core/Divergent difference

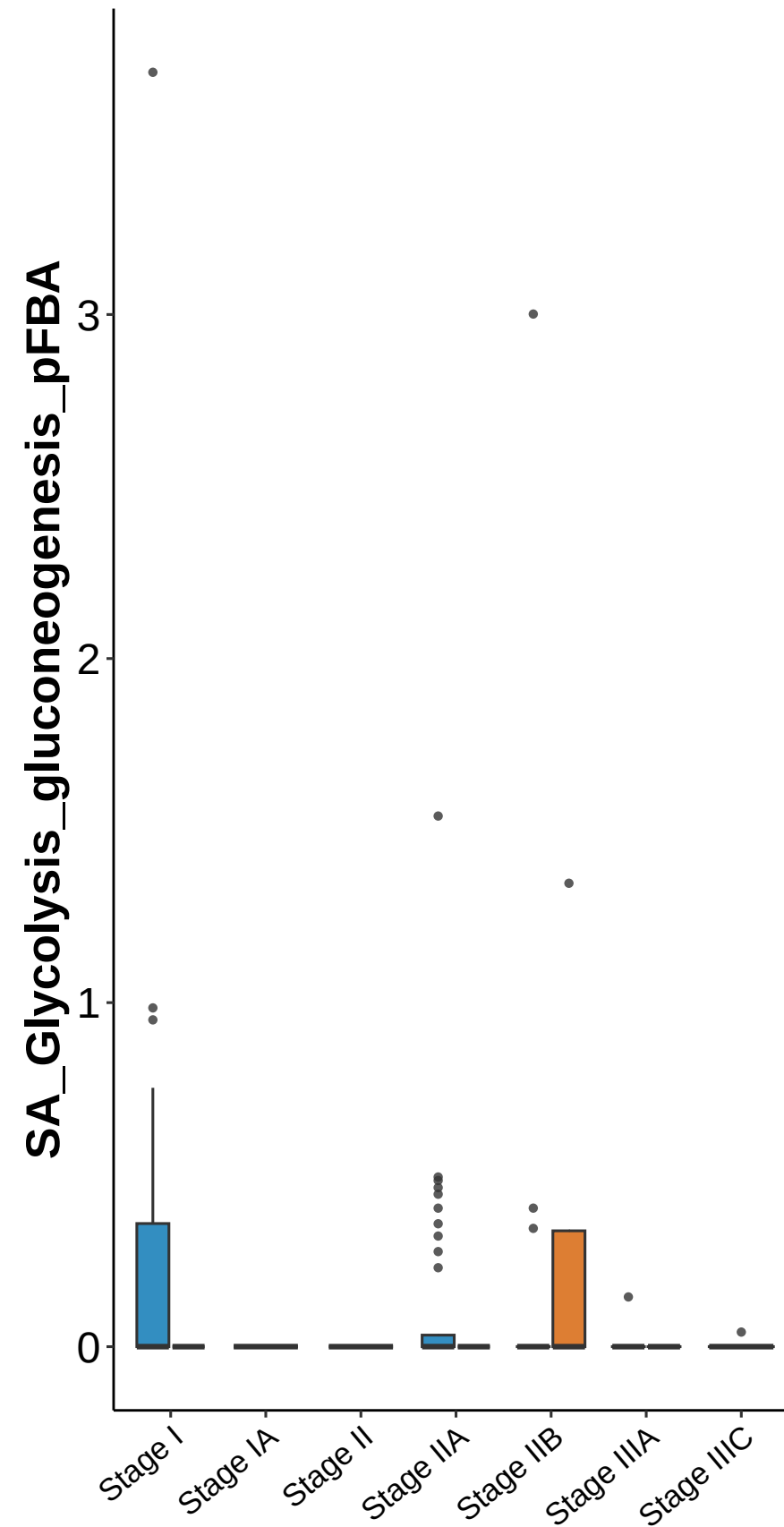
Group Core Divergent



SA_Glycolysis_gluco

Is the Core/Divergent difference inde

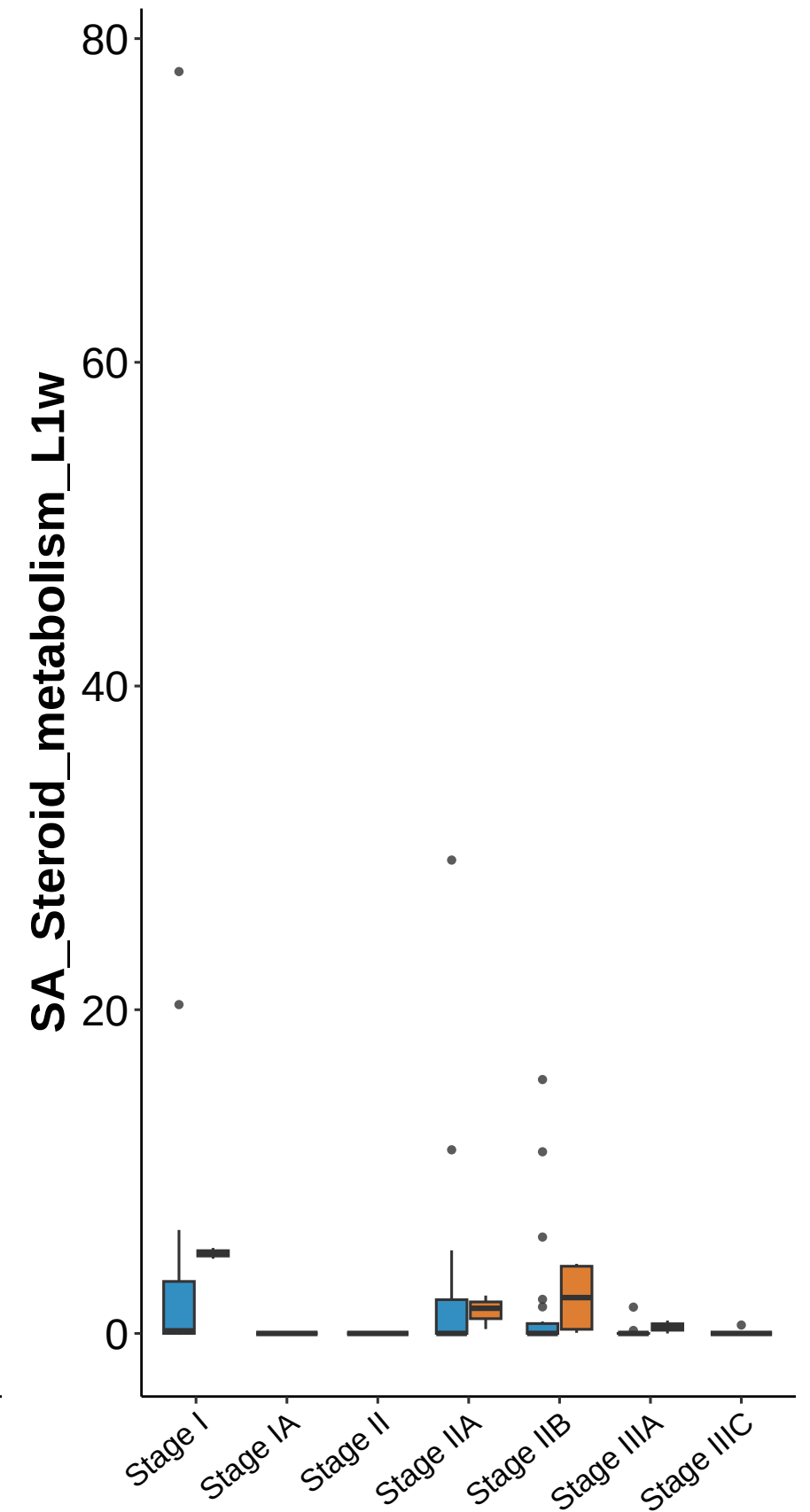
Group Core Divergent



SA_Steroid_metaboli

Is the Core/Divergent difference ind

Group Core Divergent

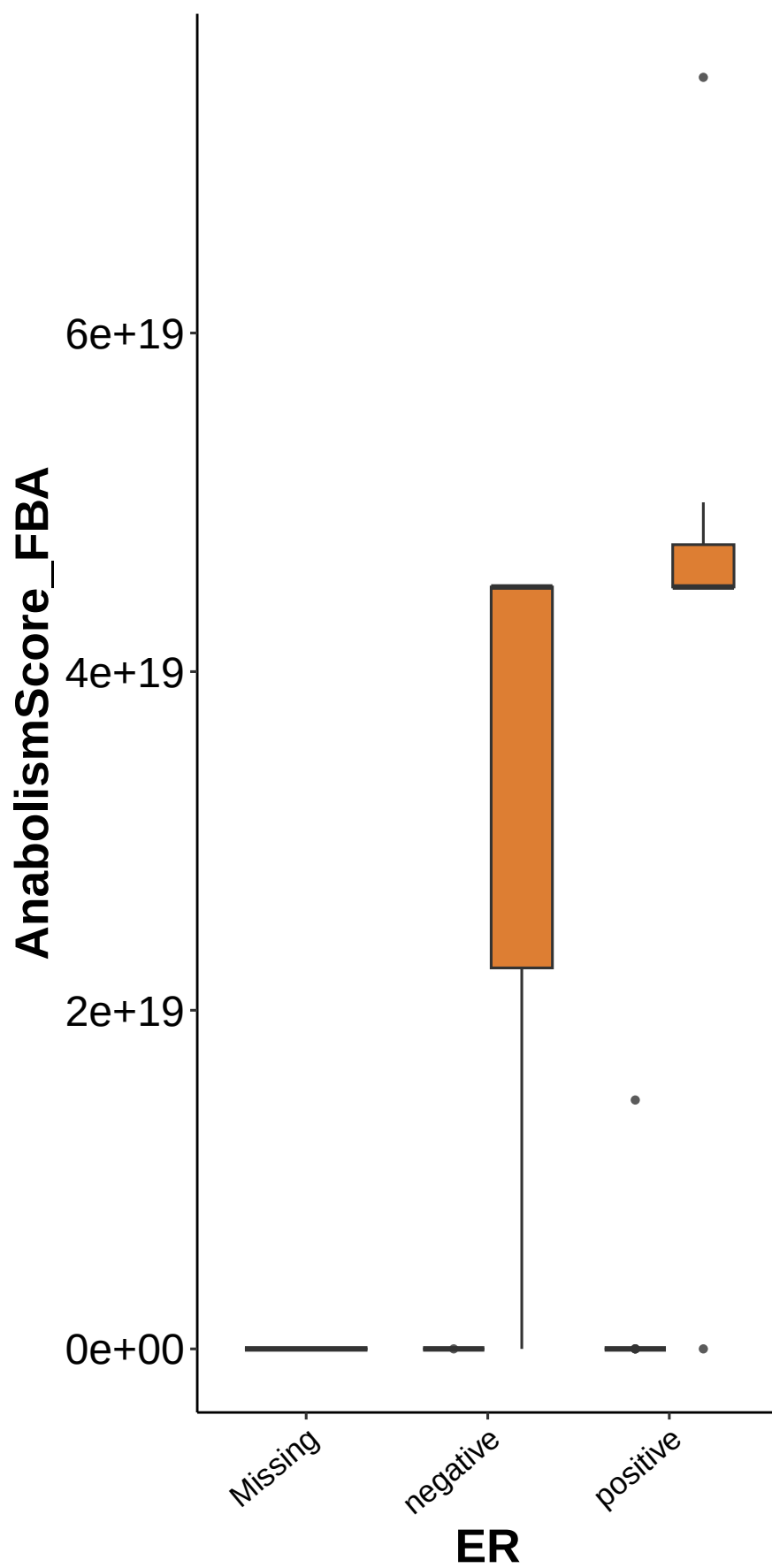


Integration: Top metabolic × ER

AnabolismScore_F

Is the Core/Divergent difference

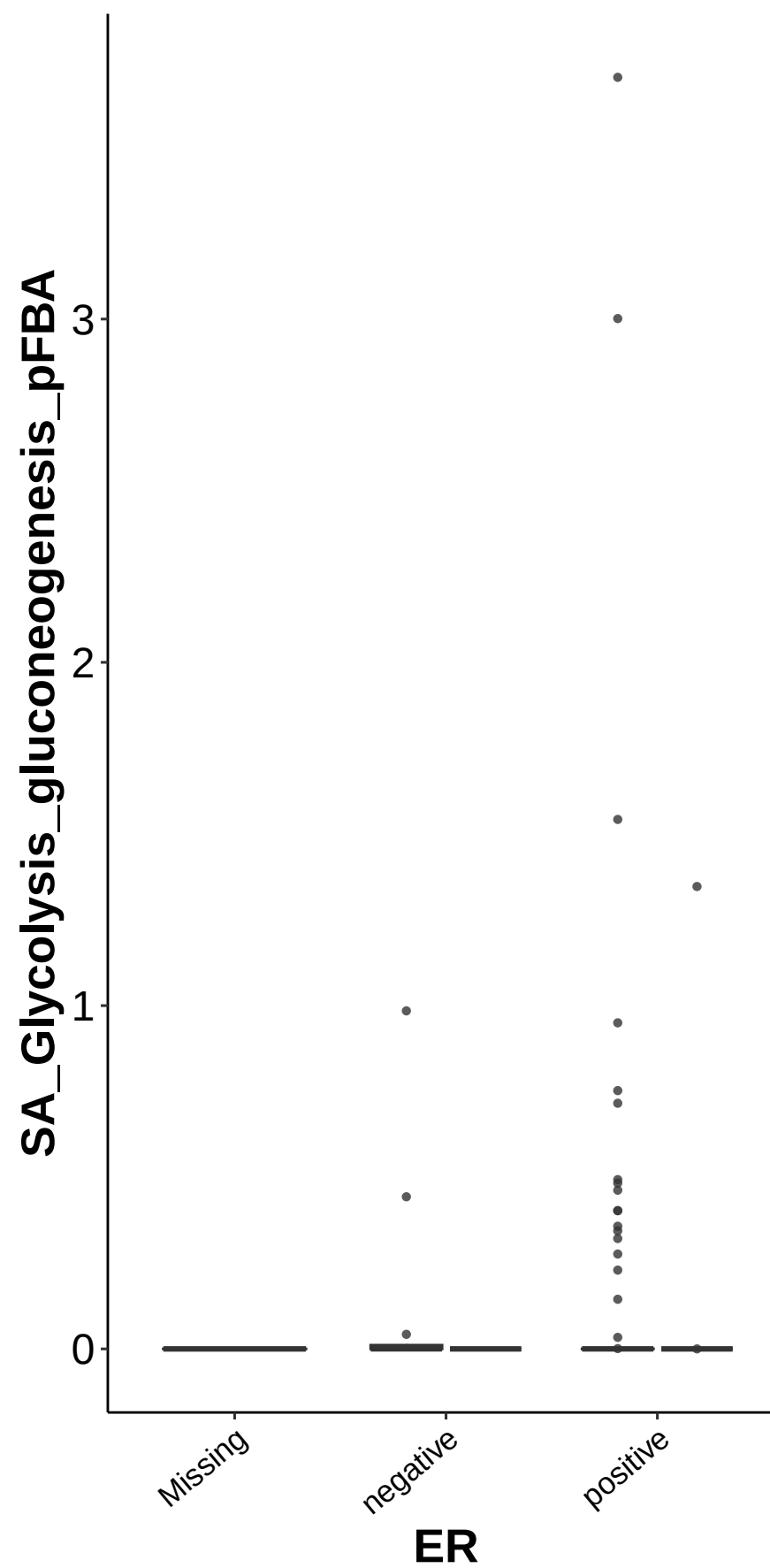
Group  Core  Divergent



SA_Glycolysis_glucon

Is the Core/Divergent difference inde

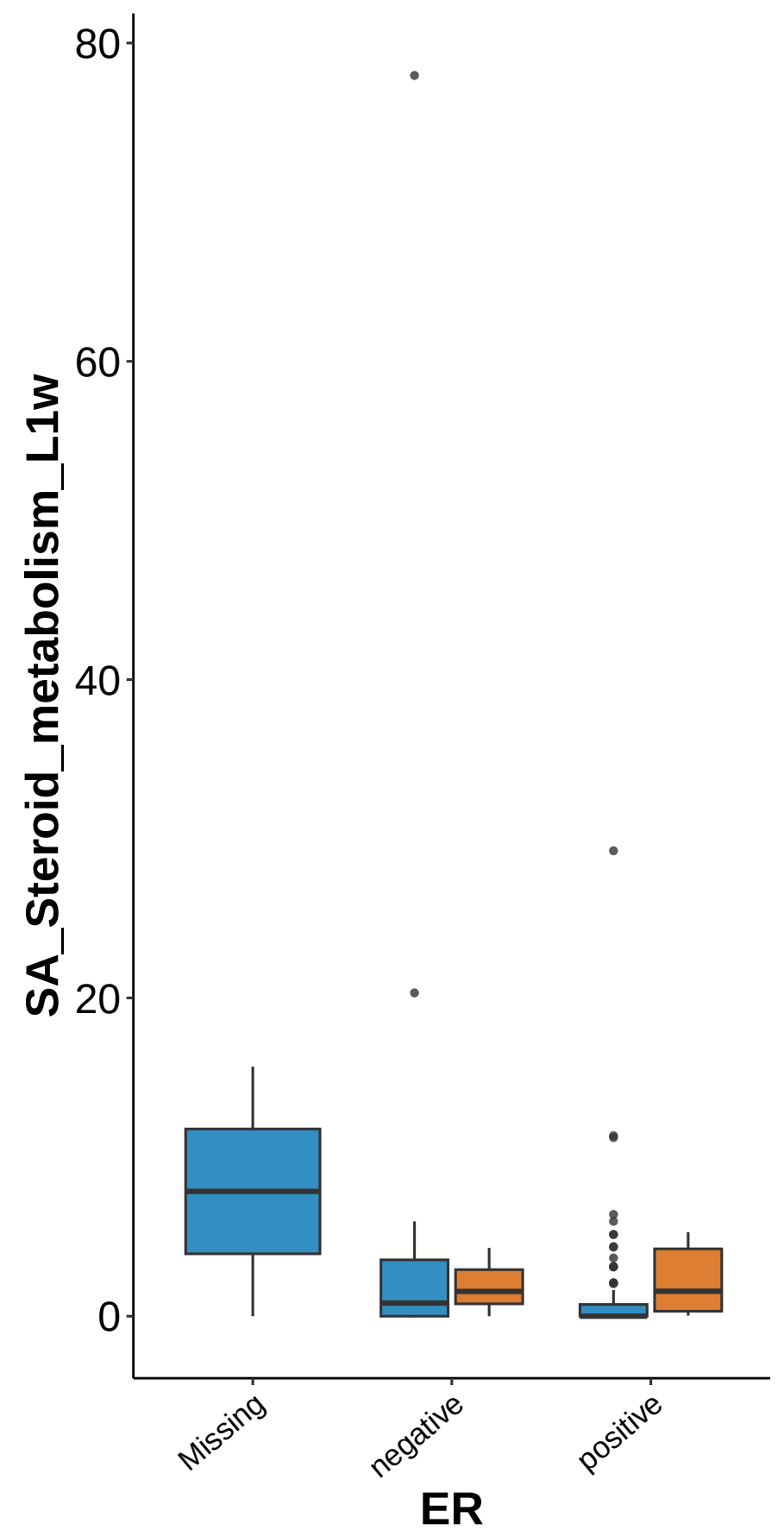
Group  Core  Divergent



SA_Steroid_metaboli

Is the Core/Divergent difference ind

Group  Core  Divergent

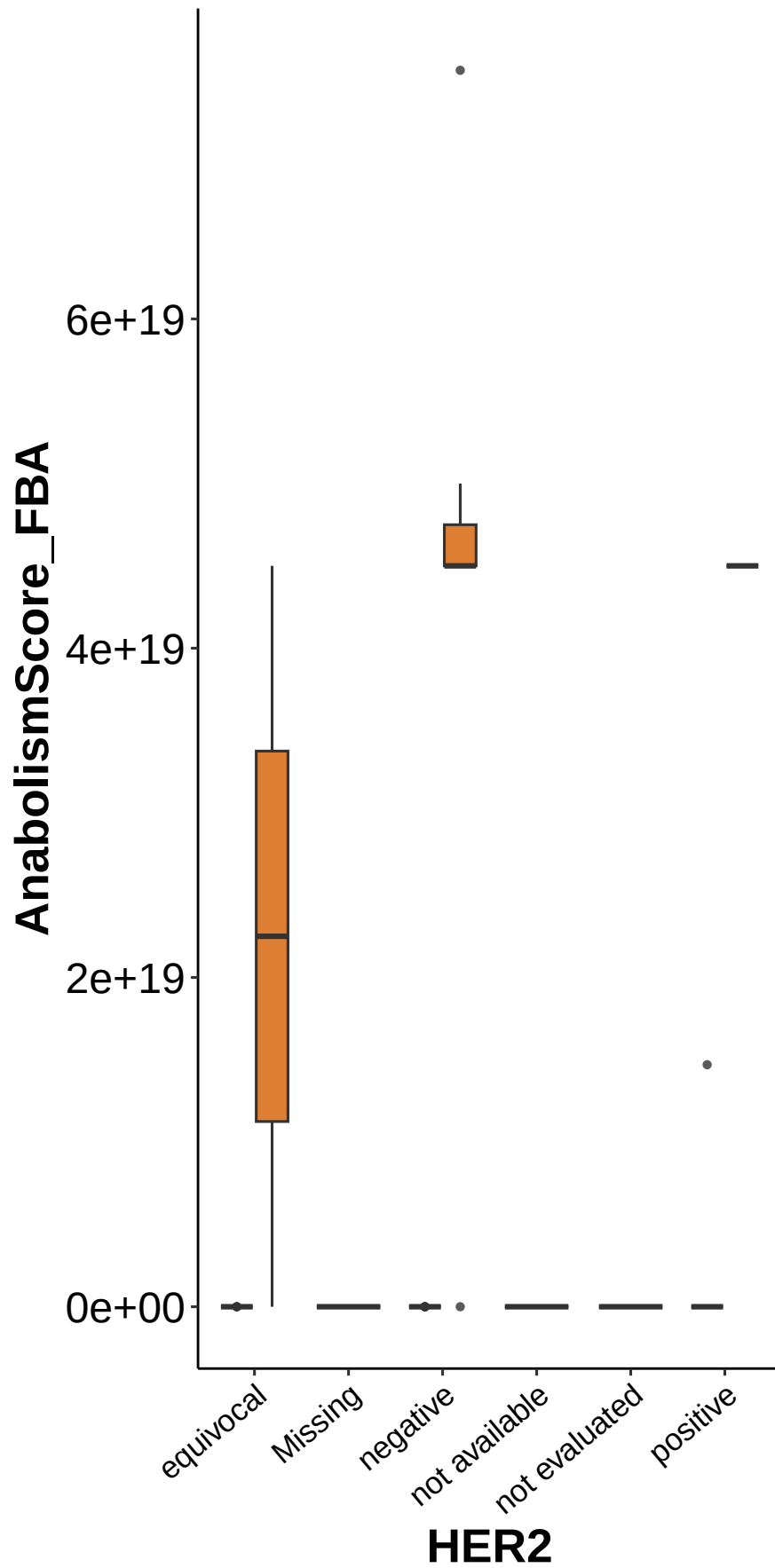


Integration: Top metabolic × HER2

AnabolismScore_F

Is the Core/Divergent difference

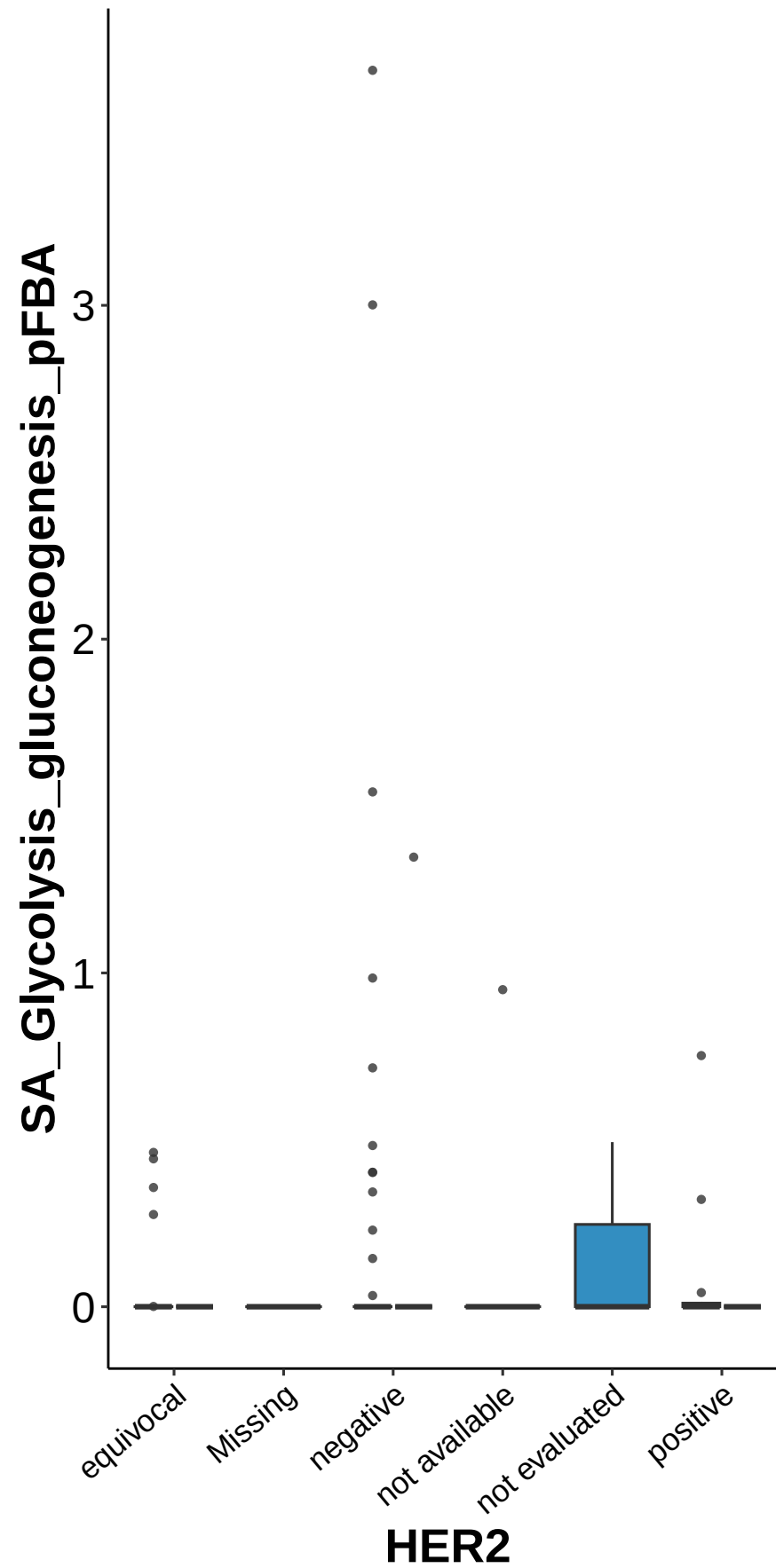
Group Core Divergent



SA_Glycolysis_gluco

Is the Core/Divergent difference inde

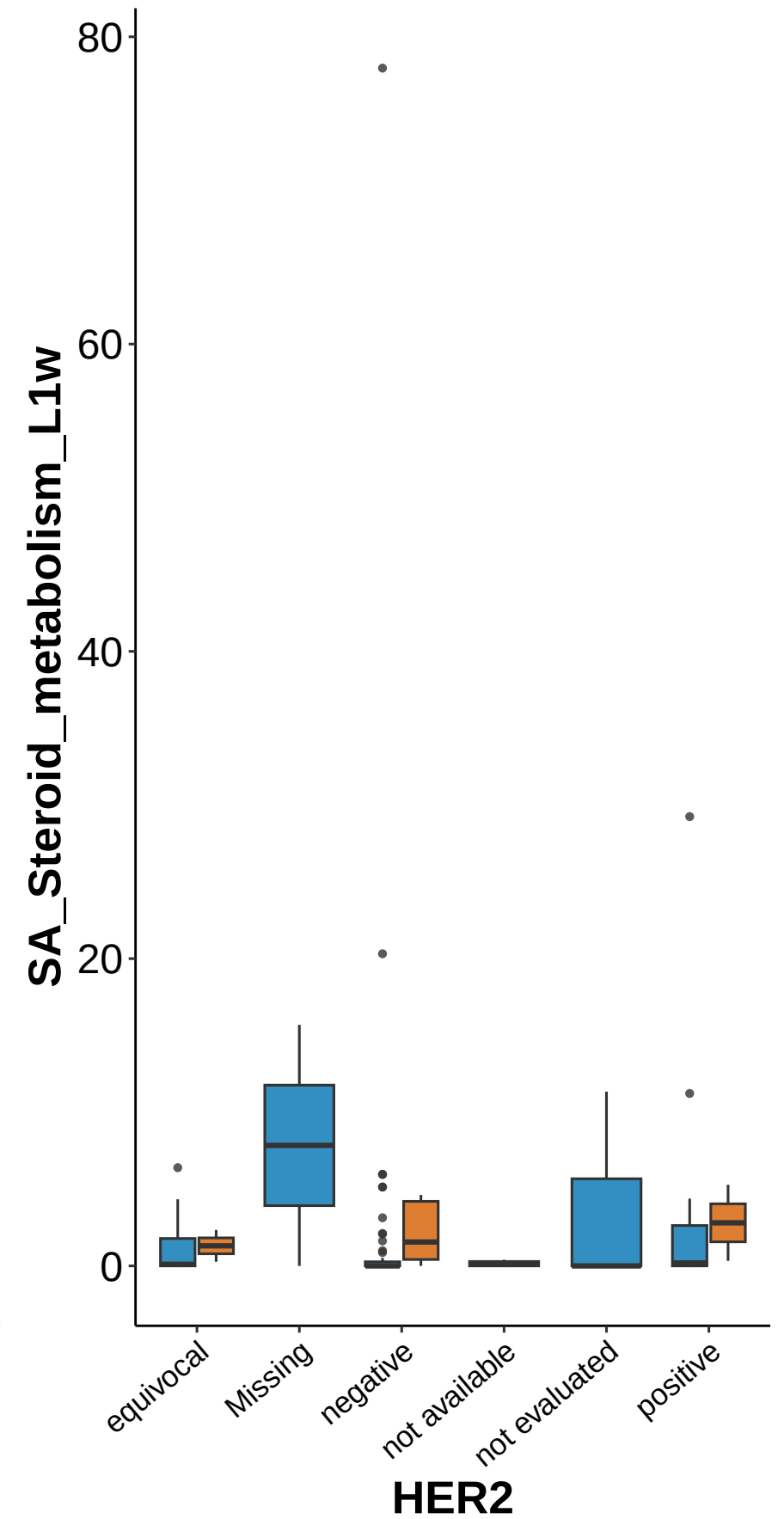
Group Core Divergent



SA_Steroid_metaboli

Is the Core/Divergent difference inde

Group Core Divergent

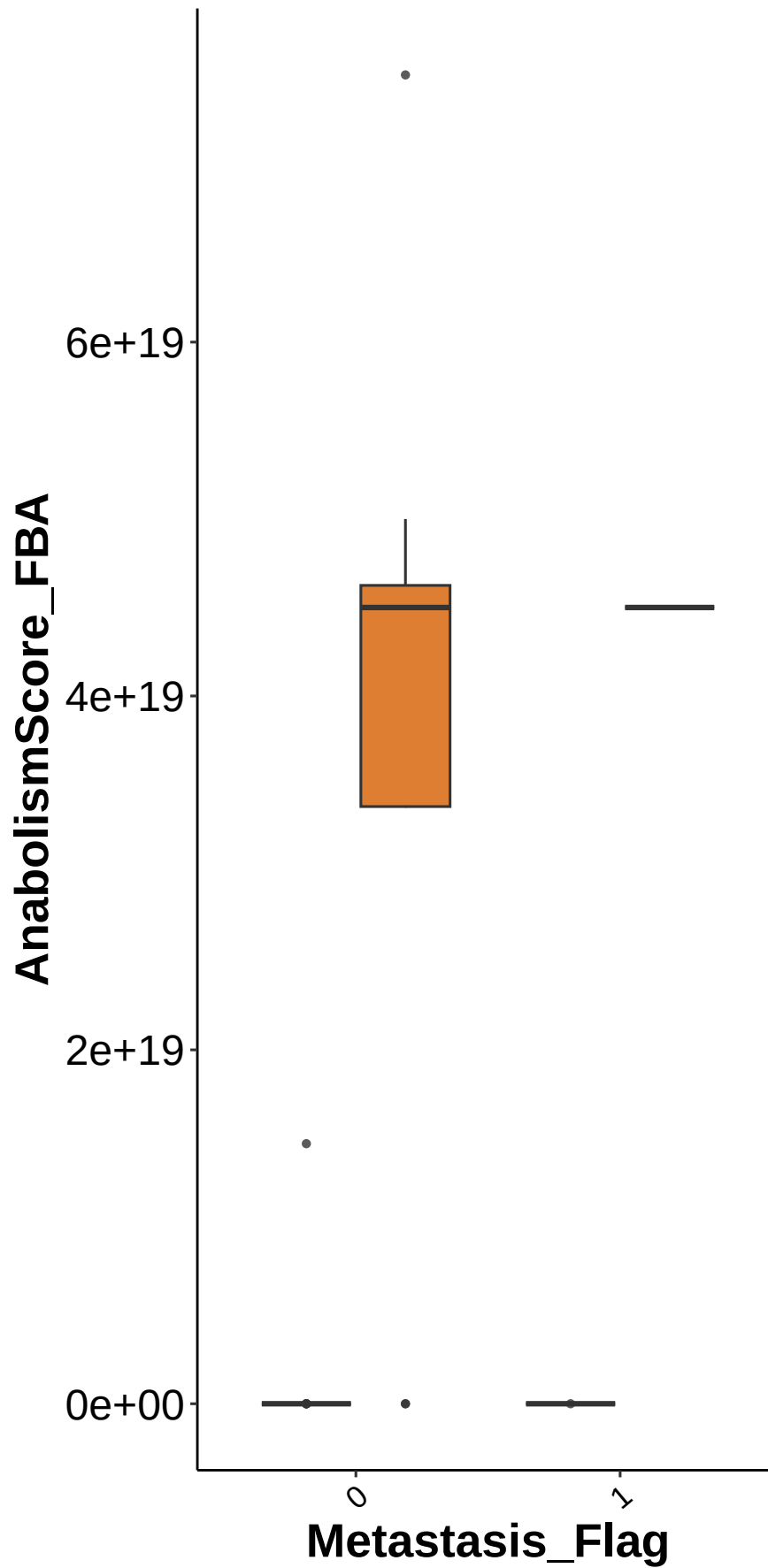


Integration: Top metabolic × Metastasis_Flag

AnabolismScore_F

Is the Core/Divergent difference

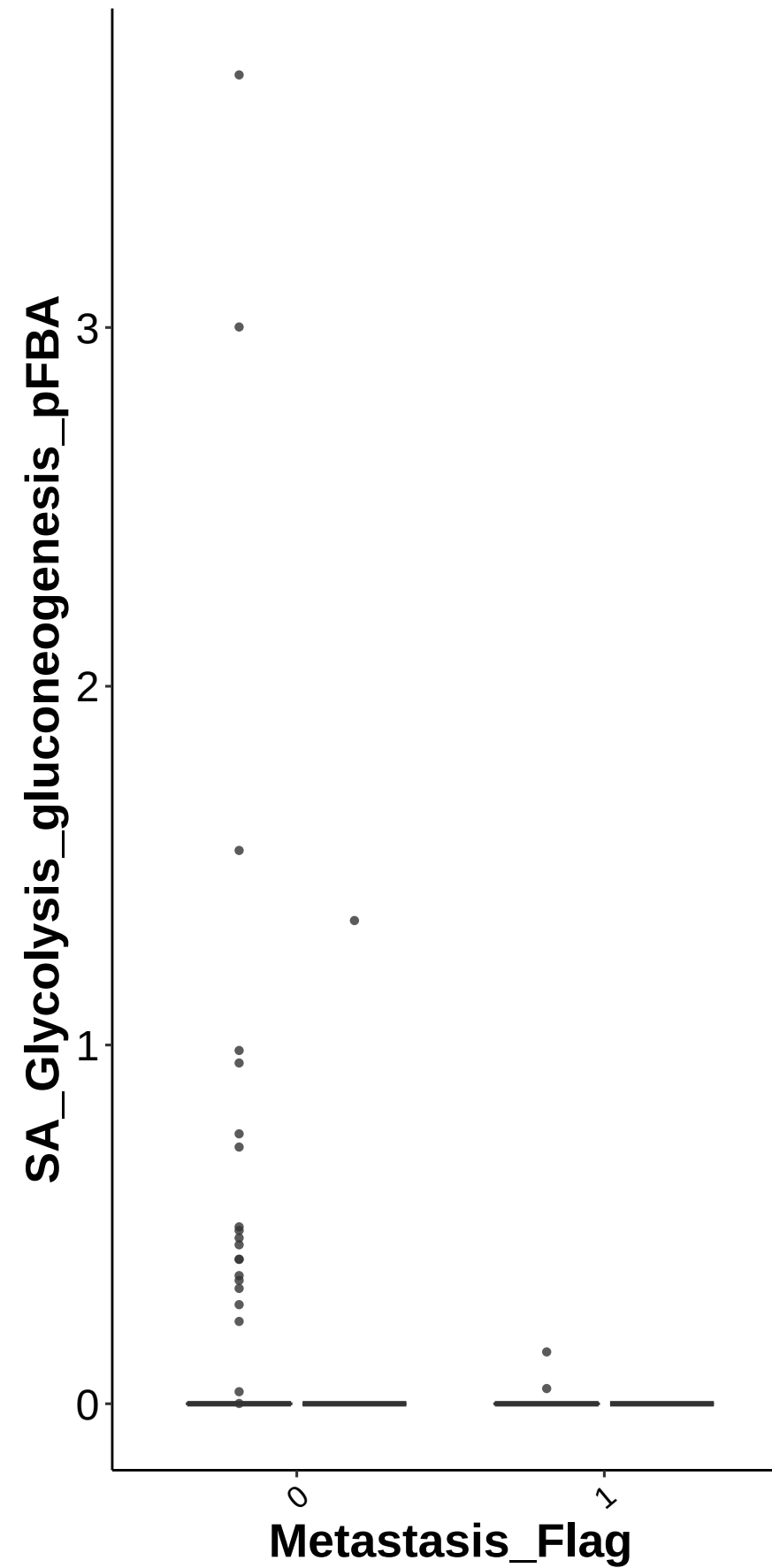
Group Core Divergent



SA_Glycolysis_gluco

Is the Core/Divergent difference inde

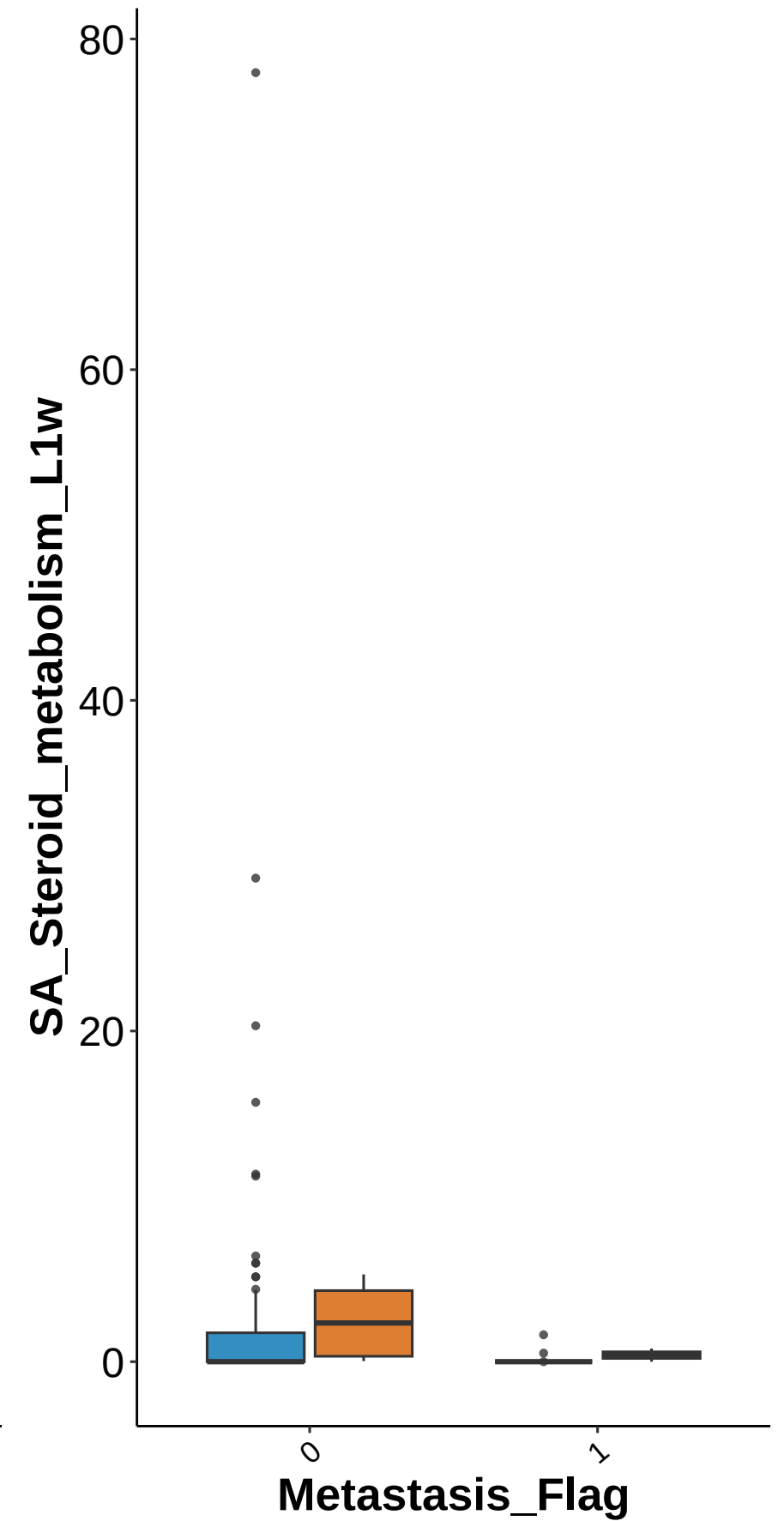
Group Core Divergent



SA_Steroid_metaboli

Is the Core/Divergent difference ind

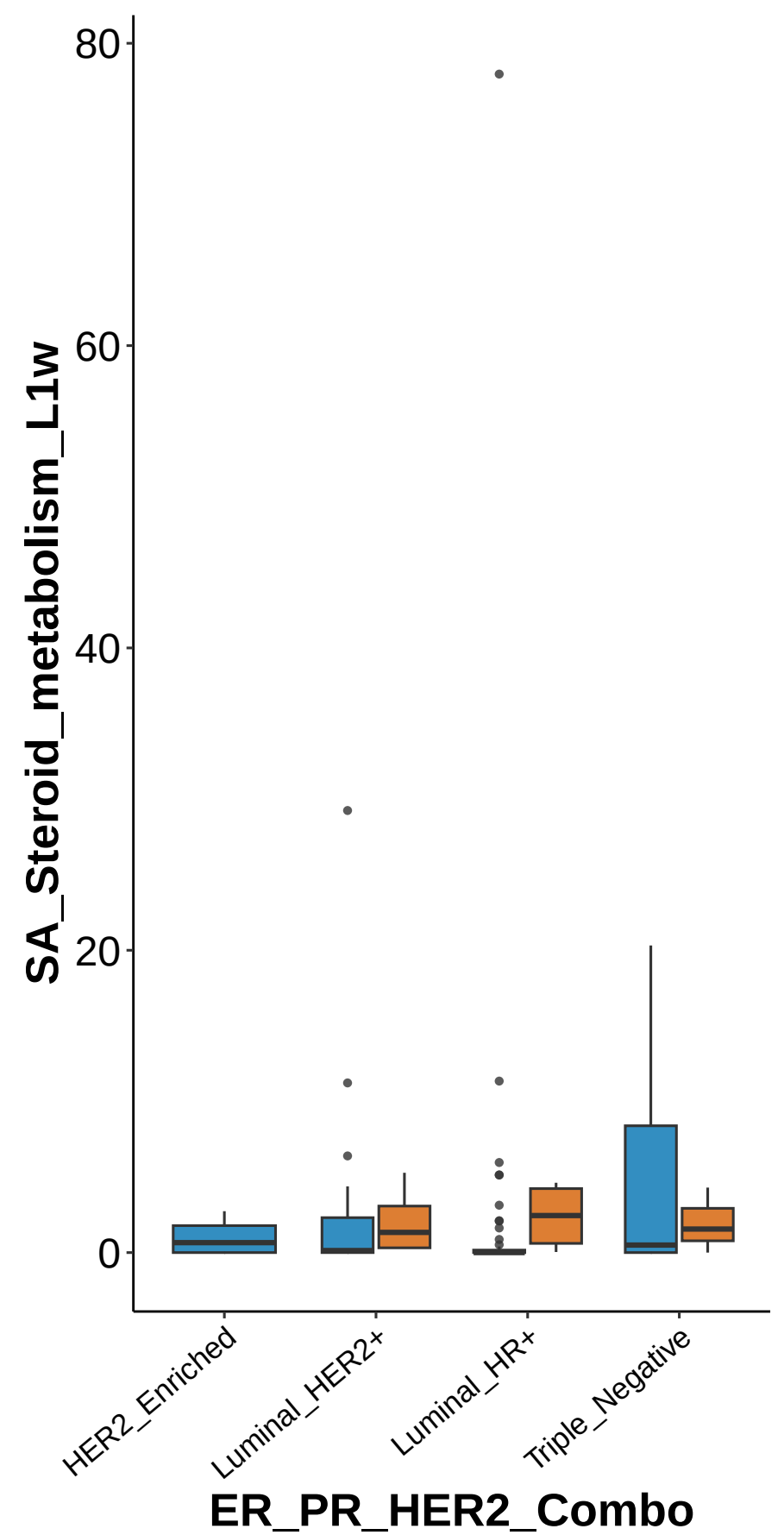
Group Core Divergent



SA_Steroid_metaboli

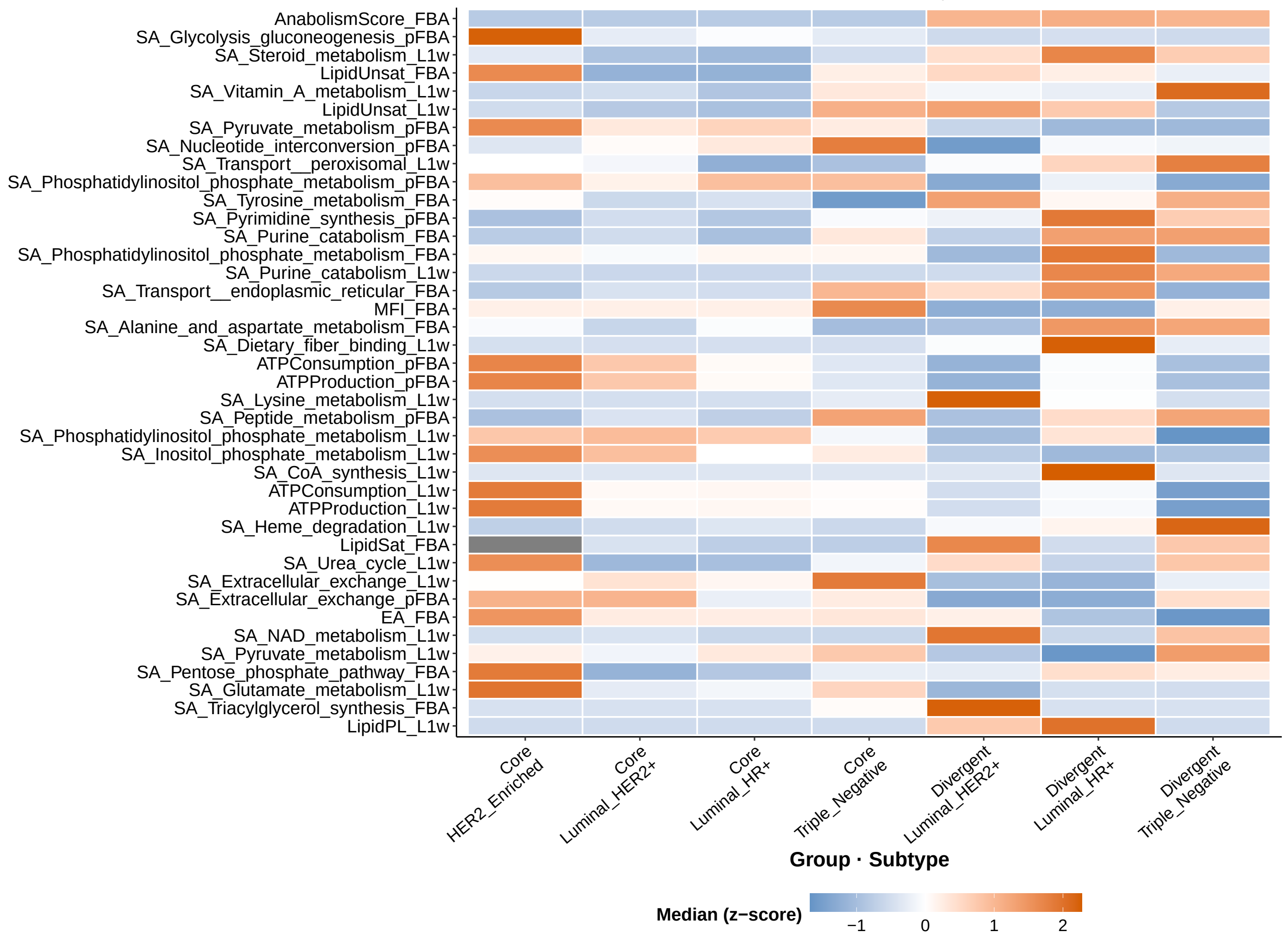
Is the Core/Divergent difference ind

Group  **Core**  **Divergent**



Metabolic Signature by Molecular Subtype (ER_PR_HER2_Combo)

Variables ordered by Cliff's . | FDR < 0.05



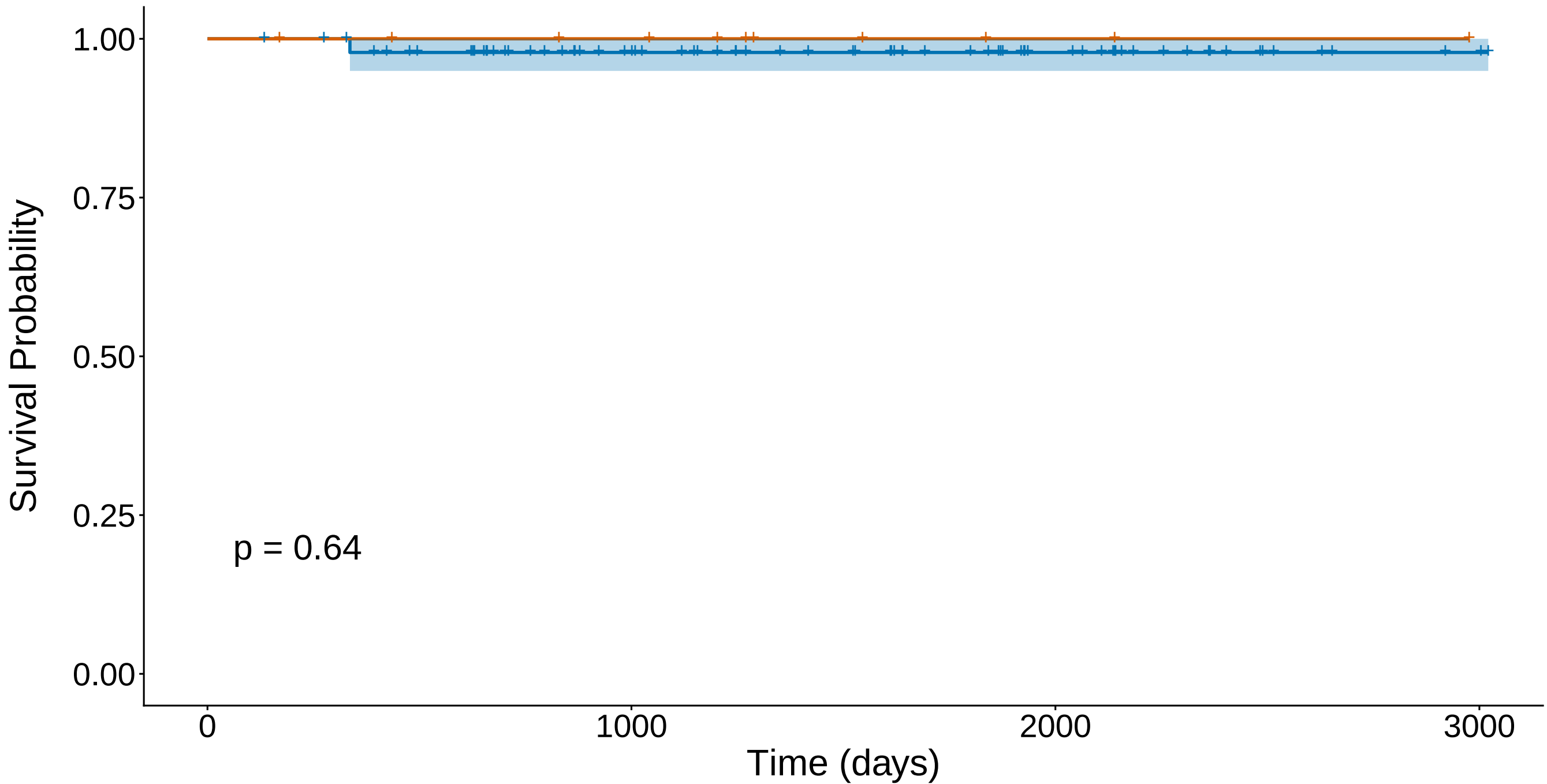
SECTION IV

Survival Analysis

Does the metabolic phenotype Core/Divergent predict prognosis?

Overall Survival Curve: Core vs Divergent

Strata + Core + Divergent



Number at risk

Core

99

63

27

2

Divergent

11

8

2

0

0

1000

2000

3000

SECTION IV–B

Survival by Molecular Subtype (ER_PR_HER2_Combo)

Each subtype: Core vs Divergent | min. 5 patients per group

SECTION IV–C

Survival by Molecular Subtype (Subtype / PAM50)

Each subtype: Core vs Divergent | min. 5 patients per group

Survival Summary

Core: N=99 | Events=2 | Median OS (events)=336 days | Median OS (all)=1417 days

Divergent: N=11 | Events=0 | Median OS (events)=NA days | Median OS (all)=1270 days

N by Molecular Subtype:

HER2_Enriched	Core	N= 5	Events=0
Luminal_HER2+	Core	N=32	Events=0
Luminal_HER2+	Divergent	N= 4	Events=0
Luminal_HR+	Core	N=54	Events=2
Luminal_HR+	Divergent	N= 4	Events=0
Triple_Negative	Core	N= 8	Events=0
Triple_Negative	Divergent	N= 3	Events=0

Metabolic Signature by Molecular Subtype – Top 40

Variables ordered by Cliff's . | FDR < 0.05

